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Peter Shiu

Number Theory with Computations

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Peter Shiu

Number Theory with Computations

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Preface

In fond memory of my teacher Heini Halberstam (1926–2014).

Number theory had been considered to be the 'purest' branch of mathematics, with hardly any significant applications to other sciences. This was so up to about 1975, when an impetus to human activity came with the rapid development of the computer, heralding the era of digital communication. For mathematics, and particularly for number theory, such a development is a two-way process: the new subject of computational number theory is now an active area of research, the fruits of which have become indispensable in science in general and communications in particular.

Computational experiments in mathematics, especially in number theory, help the investigator to test hypotheses, construct counter examples, and even pinpoint special cases in an argument, so that a suitable statement can be formulated before a proof is contemplated. Indeed, even the verification of an established result can be useful because computation clarifies an argument used and sometimes offers additional insight to the solved problem. This has always been the case, perhaps even in ancient times, but very much so nowadays because machines can be used to gather and process data much more accurately and efficiently than human beings can.

There being no shortage of number theory texts at the undergraduate level, some justification is required to add another one. Computational number theory is rapidly becoming important, but hardly any of the existing texts cater for the beginner who wants to know about algorithms and their implementations to deliver results of experiments. With the ubiquity of computing machines, number theory is the most suitable subject for students to develop and explore algorithms for themselves, thereby having a much more concrete grasp of the subject.

The book contains a basic course in number theory, with emphasis placed on algorithms and their implementations wherever it is suitable. Much of the content can be considered an introduction to computational number theory, but our main aim is to encourage readers to use the results of computations to enhance their understanding of the theory of numbers. Some proofs may appear terse, but only when the essence of the argument can be captured in a few words or sentences. Lengthier arguments are usually accompanied by concrete examples. Algorithms are given either with their implementations as ready to run computer programs,

or in pseudo codes. It is often the implementation that beginners find challenging, and we have chosen to use the open-source general purpose programming language `Python`, which is popular and freely available, because the language has the virtue of being easy to interpret pseudo codes. Although the codes are set out in `Python3`, the required modification for `Python2` is trivial.

There is very little prerequisite to make a gainful study of the text, which is written mainly for university students intending to study number theory as a final year option. The book is divided into two parts: *Elementary number theory* and *Analytic number theory*. The basics in elementary number theory are covered in the first part; it may help, but not absolutely necessary, if the reader has come across some algebraic notions, such as *integral domains, groups, rings* and *fields*. Continued fractions is an important topic in number theory, which has been neglected over the years, mainly because of the tediousness of the calculations involved without a computer. In Chap. 7, we give an account on such fractions, together with results of computer experiments to back up the theory.

Part II of the book is an introduction to analytic number theory, and the reader is expected to be familiar with topics in real and complex analysis, and also linear algebra. There is some depth in the chosen topics, including *Diophantine approximations*, *Dirichlet's theorem on the existence of primes in arithmetic progressions*, *the prime number theorem*, and *smooth numbers*. The last chapter is meant to be a guided tour for the more advanced students who may wish to know something about the esoteric *circle method* of Hardy, Littlewood and Ramanujan; much of the details of the intricate proofs are omitted.

The learning process in any subject, including science and mathematics, is not linear. A topic, or a theorem, may be invoked before being considered properly, and certain important results are mentioned more than once or twice; as long as the argument is not circular, such a process usually just reinforces the significance of the result. Indeed, the book is suitable for 'fun-read' at any place of the reader's choice; just delve in and enjoy learning some mathematics.

There are examples throughout the text, and also exercises in each chapter which includes a *Summary and Notes*. Mathematics is not a spectator sport, and the reader is strongly encouraged to tackle the exercises, most of which have given answers, hints or solutions.

The text is not written to be followed in a course on number theory, but a lecturer can use Part I as the basis for such a course, with selected topics in Part II for a second course for third year students. We have chosen to leave out algebraic number theory in the text. Besides the lack of space, it is our belief that the subject matter requires a certain amount of mathematical maturity, and it is more suitable as a course for graduate students.

With the slow, and then rapid, development of computing machines, there was not an optimal time for such a text to be written, but why now? The answer is: Covid-19. As a rule octogenarians do not write books for students because, being long retired, such a writer is no longer in touch with the current teaching scene and activities. However, rather than idling and simply wasting time during the long lockdown period, I decided to make a positive stand on the dreaded virus. The

outcome is an unusual text in that it lacks the injection of wisdom from friends and colleagues, including some who might advise '*Don't do it!*' Anyway, having produced the manuscript, I am brave enough to let it see the light of day.

It remains for me to thank Dr Remi Lodh of Springer who enthusiastically took up the project from the moment I approached him. He and his team of experts have been very reassuring in my being able to produce a suitable manuscript.

Sheffield, UK Peter Shiu
April, 2024

Notation

The following notation and conventions are used in the text.

$\mathbb{Z}$	The set of integers
$\mathbb{N}$	The set of positive integers
$\mathbb{P}$	The set of prime numbers
$\mathbb{W}$	The set of numbers representable as sums of two squares
$\mathbb{U}$	The set of products of distinct primes $p \equiv 1 \pmod 4$
$\mathbb{Q}$	The set of rational numbers
$\mathbb{R}$	The set of real numbers
$\mathbb{C}$	The set of complex numbers
$\mathbb{H}$	The subset of $\mathbb{C}$ in the upper-half plane
$(\mathbb{Z}/m\mathbb{Z})^+$	The additive group of order m
$(\mathbb{Z}/m\mathbb{Z})^\times$	The multiplicative group of reduced residue modulo m
$\mathbb{F}_p$	The finite field with p (prime) elements
$\|\mathrm{A}\|$	The number of elements of the finite set A

$a \pmod m$	The residue class a modulo m
$a \mid b$	means a divides b
$\gcd(a, b)$	The greatest common divisor of a and b
LCM	The least common multiple of the numbers concerned
PSP, EPSP	Pseudoprime, Euler pseudoprime (page 78)
$\left(\frac{p}{q}\right)$, $\left(\frac{a}{m}\right)$	Legendre symbol, Jacobi symbol (page 96, 103)
$\mathrm{e}(x)$	$= e^{2\pi i x}$ (page 59)
$\lfloor\lambda\rfloor$, $\lceil\lambda\rceil$, $\{\lambda\}$	The floor, the ceiling, the fractional part, of λ
$\|\lambda\|$	The distance between λ and the nearest integer
B	Landau-Ramanujan constant (page 151)
K	Khinchin's constant (page 190)
γ	Euler's constant (page 53, 239)
π	= Circumference/diameter of a circle (page 172, 208, 219)

The letter p denotes a prime, but there are obvious exceptions such as it being used in the form p/q for a fraction, or to denote the value of a probability.

Both Landau's notation $f(x) = O(g(x))$ and Vinogradov's notation $f(x) \ll g(x)$ are used to estimate the size of $|f(x)|$. They mean that there is a positive constant C such that $|f(x)| \le Cg(x)$ for all relevant x. Note that $f(x)$ may be complex-valued, but $g(x)$ has to be non-negative; on some rare occasions we may write $f(x) \gg g(x)$, with an obvious meaning and $f(x), g(x) > 0$. Also, $f(x) = o(g(x))$ and $f(x) \sim g(x)$ mean that

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0 \quad \text{and} \quad \lim_{x\to\infty} \frac{f(x)}{g(x)} = 1, \quad \text{respectively.}$$

CCH is my own convention—it stands for 'cows (may) come home' to indicate that the process may take a very long time before a solution emerges.

Function	Meaning	Page number
$c_q(n)$	Ramanujan sum	60
$d(n)$	Number of divisors of n	9, 50
$f*g$	Dirichlet convolution	54
$g(k)$, $G(k)$	See Waring's problem	377
$G(p)$	Quadratic Gauss sum	116
$\mathrm{li}(x)$	Logarithmic integral $\int_2^x \frac{dt}{\log t}$	309
$L(s,\chi)$	Dirichlet L-function	256
$M(x)$	Mertens function $\sum_{n\le x} \mu(n)$	287
$\mathrm{ord}_m(a)$	Order of $a \pmod m$	43
$p(q,a)$	Least Prime $p \equiv a \pmod q$	329
$P(n)$	Largest Prime divisor of n	349
$Q(x)$	Counting function for squarefree numbers	281
$s(h,k)$	Dedekind sum	375
$W(x)$	Counting function for sums of two squares	151
$\phi(n)$	Euler totient function	38
$\Gamma(n)$	Gamma function	351
$\lambda(n)$	Liouville function	49
$\lambda(n)$	Carmichael function	133
$\Lambda(n)$	Mangoldt function	55
$\mu(n)$	Möbius function	35
$\nu(p)$	Exponent of the prime p	21
$\pi(x)$	prime counting function	8, 65, 235
$\pi(x;q,a)$	prime counting function over $p \equiv a \pmod q$	328
$\rho(u)$	Dickman function	354

(continued)

Function	Meaning	Page number
$\psi(x)$	$\sum_{n\leq x}\Lambda(n)$	236
$\psi(x;q,a)$	$\sum_{n\leq x,\ n\equiv a\pmod q}\Lambda(n)$	328
$\Psi(x,y)$	Counting function of y-smooth numbers	349
$\theta(x)$	$\sum_{p\leq x}\log p$	236
$\theta(x;q,a)$	$\sum_{p\leq x,\ p\equiv a\pmod q}\log p$	328
$\tau(\chi)$	Gauss sum of χ	341
$\omega(n)$	Number of distinct prime divisors of n	8
$\Omega(n)$	Total number of prime divisors of n	8
$\zeta(s)$	Riemann zeta-function	66

Contents

Part I Elementary Number Theory

1 Basics 3
- 1.1 Well Ordering, Induction and Pigeonhole Principle 3
- 1.2 Division Process, Quotient and Remainder 7
- 1.3 Common Divisors, gcd and Euclidean Algorithm 10
- 1.4 Linear Indeterminate Equations 14
- 1.5 Chinese Remainder Theorem 19
- 1.6 Fundamental Theorem of Arithmetic 20
- 1.7 Theorems of Dirichlet, Thue and Aubry 22
- 1.8 Lagrange's Theorem 25
- 1.9 Evaluation of $a^b \pmod{n}$ 27
- 1.10 Summary and Notes 28
- 1.11 Exercises 30

2 Arithmetic Functions I 33
- 2.1 Additive and Multiplicative Functions 33
- 2.2 Möbius Function 35
- 2.3 Euler Totient Function 38
- 2.4 Fermat-Euler Theorem 42
- 2.5 Order of $a \pmod{n}$ 43
- 2.6 Decimals and Order of 10 $\pmod{n}$ 43
- 2.7 Liouville Function $\lambda(n)$ 49
- 2.8 Divisor Function $d(n)$ 50
- 2.9 Dirichlet Convolution 54
- 2.10 Mangoldt Function $\Lambda(n)$ 55
- 2.11 Residue Classes and Additive Characters 56
- 2.12 Gauss Sum and Ramanujan Sum 59
- 2.13 Summary and Notes 62
- 2.14 Exercises 63

3 Prime Numbers: Euclid and Eratosthenes 65
3.1 Some Landmarks 66
3.2 Sieve of Eratosthenes 67
3.3 Counting Primes 69
3.4 Gauss' Dream 75
3.5 Factorisation: Fermat's Method, and Pollard's ρ Method 80
3.6 One-Way Functions and Cryptography 83
3.7 RSA Public-Key Cryptography 85
3.8 AKS Primality Test 87
3.9 Summary and Notes 89
3.10 Exercises 93

4 Quadratic Residues and Congruences 95
4.1 Legendre Symbol and Euler's Criterion 95
4.2 Law of Quadratic Reciprocity 98
4.3 Solving Quadratic Congruences (mod p) 109
4.4 Solving Quadratic Congruences (mod m) 115
4.5 Gauss Sum and its Evaluation 116
4.6 Summary and Notes 121
4.7 Exercises 122

5 Primitive Roots 125
5.1 Primitive Roots (mod p) 126
5.2 Primitive Roots Theorem 127
5.3 Carmichael Function and Carmichael Numbers 133
5.4 Discrete Logarithms 136
5.5 Key-Exchange Protocols 138
5.6 Summary and Notes 142
5.7 Exercises 143

6 Sums of Squares 145
6.1 Sums of Two Squares 146
6.2 Fermat's Theorem for Primes $p \equiv 1 \pmod 4$ 147
6.3 Lagrange's Theorem 153
6.4 Three-Squares Theorem 154
6.5 Finding Representations 159
6.6 Summary and Notes 162
6.7 Exercises 165

7 Continued Fractions 167
7.1 Basic Properties 169
7.2 Continuants 174
7.3 Periodic Continued Fractions 176
7.4 Factorisation by Continued Fractions 186
7.5 Computing Continued Fractions 190
7.6 Summary and Notes 201
7.7 Exercises 202

Part II Analytic Number Theory

8 Diophantine Approximations 207
8.1 Accurate But Bad Approximations to π 208
8.2 Good and Best Approximations 210
8.3 Conditions for Being Convergents 212
8.4 Fractional Parts of $n\theta$ 215
8.5 Liouville's Theorem and Diophantine Equations 216
8.6 Transcendence of e and π 219
8.7 Irrationality of $\zeta(3)$ 227
8.8 Summary and Notes 232
8.9 Exercises 233

9 Distribution of Prime Numbers 235
9.1 Chebyshev's Theorem 236
9.2 Euler Summation Formula 238
9.3 Sums Related to the Distribution of Primes 243
9.4 Arithmetic Progressions and Dirichlet Characters 248
9.5 Euler Products 254
9.6 Dirichlet Series and Riemann Zeta-Function 255
9.7 Poisson, Mellin and Perron 262
9.8 Summary and Notes 266
9.9 Exercises 268

10 Arithmetic Functions II 271
10.1 Moment Sums 271
10.2 Highly Composite Numbers 272
10.3 Euler Totient Function II 276
10.4 Sparsely Totient Numbers 278
10.5 Squarefree Numbers 281
10.6 Prime Divisor Functions 288
10.7 Normal Order 289
10.8 Summary and Notes 295
10.9 Exercises 296

11 Prime Number Theorem 299
11.1 Brief Sketch 299
11.2 Newman's Simplification 301
11.3 Proof of the Prime Number Theorem 305
11.4 Error Term of the Prime Number Theorem 309
11.5 Zero-Free Region and the Riemann Hypothesis 310
11.6 Prime Gaps 311
11.7 Summary Notes 313
11.8 Exercises 316

12 Primes in Arithmetic Progressions 321
12.1 Plan for the Proof 322
12.2 Replacing Sums Over Primes with Sums Over Integers 323
12.3 Nonvanishing of $L(1, \chi)$ for Characters $\chi \neq \chi_0$ 325
12.4 Prime Number Theorem in Arithmetic Progressions 328
12.5 Titchmarsh-Linnik Theorem 331
12.6 Prime Number Races 335
12.7 Primitive Characters 339
12.8 Summary and Notes 344
12.9 Exercises 346

13 Smooth Numbers 349
13.1 Introduction 350
13.2 Asymptotic Formula for $\Psi(x, p_k)$ 352
13.3 Table of Values for $\Psi(x, y)$ 353
13.4 Dickman Function 354
13.5 Rankin's Trick Applied to $\Psi(x, y)$ 358
13.6 Application to the Discrete Logarithm Problem 361
13.7 Summary and Notes 367
13.8 Exercises 369

14 Circle Method 371
14.1 Partition Function: Rademacher Series 372
14.2 Sums of Powers: Waring's Problem 377
14.3 Circle Method for $G(k)$ 382
14.4 Sums of Three Primes: Vinogradov's Theorem 396
14.5 Summary and Notes 406
14.6 Exercises 408

Answers/Hints/Solutions to Selected Exercises 411

References 433

Index 437

Part I
Elementary Number Theory

Chapter 1
Basics



Abstract After the introduction of the well-ordering axiom and the principle of mathematical induction, the topics being covered include the Euclidean algorithm, the Chinese remainder theorem, and the fundamental theorem of arithmetic. There are also the theorems of Lagrange, Dirichlet, Thue, and Aubry, and an explanation on how $a^b \pmod{n}$ can be evaluated efficiently.

Being able to count is what distinguishes human beings from other species and, as an activity beyond what is immediately necessary for survival, it dates back to millenniums across different civilisations. However, as a scientific discipline, number theory is the creation of relatively modern times, dating from mid-seventeenth century in Europe, particularly from the discoveries of P. Fermat (1601–1665). More specifically, his invention of the method of infinite descent to solve various problems can be considered the point when number theory became a deductive science.

1.1 Well Ordering, Induction and Pigeonhole Principle

Counting involves the 'well ordering' property associated with the set

$$\mathbb{N} = \{1, 2, 3, \ldots\}$$

of natural numbers, or positive integers. There are two basic arithmetic operations, namely *addition* and *multiplication* on the set. For closure with respect to the operation of subtraction, the inverse operation of addition, we need to extend $\mathbb{N}$ to include 0 and the negative numbers to the set

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$$

of integers. Together with the operations of addition and multiplication, the set forms the *ring of integers*.

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Instead of recalling the definitions of *rings, groups,* and *fields*, which the reader may have come across, we mention the following three axioms, or principles, concerning $\mathbb{N}$ which are inter-related, and also to Fermat's method of descent:

(i) Well ordering axiom: *Any non-empty subset of* $\mathbb{N}$ *has a least member.*
(ii) Principle of mathematical induction: *Let* $P(n)$ *be a statement associated with* $n \in \mathbb{N}$. *Suppose that* $P(1)$ *is true, and if* $P(k)$ *is true then* $P(k+1)$ *is also true. Then* $P(n)$ *is true for every* $n \in \mathbb{N}$.
(iii) Pigeonhole principle: *If there are more objects than boxes which contain them then one of the boxes contains more than one object.*

Many propositions in number theory are statements concerning every $n \in \mathbb{N}$, the proof of which will then require the well ordering axiom (i). However, instead of appealing to the axiom directly, we often use its equivalent formulation (ii). Naively, the principle of mathematical induction amounts to saying the following: We are given that $P(1)$ is true, from which we deduce that $P(2)$ is true; then $P(3)$ being true also follows, and so on ..., so that $P(n)$ is true for every $n \in \mathbb{N}$. Indeed, the principle is just a license for 'and so on ...' being a legitimate deduction, and the well ordering axiom can be considered an endorsement of the scheme.

For the principle, the assumption that $P(k)$ is true is called the *induction hypothesis*, and the argument used to deduce that $P(k+1)$ then follows is called the *induction step*. According to the principle, we need to verify that $P(1)$ is true and to complete the inductive step so that $P(n)$ is then true for all $n \in \mathbb{N}$.

Example 1.1 Show that, for every $n \in \mathbb{N}$, we have

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \Big(\frac{n(n+1)}{2}\Big)^2.$$

Solution Let $P(n)$ be the proposition that the displayed equation is true. We first note that $P(1)$ is true, because $1^3 = (1{\cdot}2/2)^2$. Next, suppose, as induction hypothesis, that $P(k)$ is true for an arbitrary $k \in \mathbb{N}$, that is

$$1^3 + 2^3 + 3^3 + \cdots + k^3 = \Big(\frac{k(k+1)}{2}\Big)^2.$$

We proceed to establish the inductive step that $P(k+1)$ is true. For this purpose, we consider

$$1^3 + 2^3 + 3^3 + \cdots + k^3 + (k+1)^3 = \Big(\frac{k(k+1)}{2}\Big)^2 + (k+1)^3,$$

in which we have made used of the induction hypothesis. Now, the right-hand side of the equation is $(k+1)^2(k^2/4 + (k+1)) = (k+1)^2(k+2)^2/2^2$, which delivers $P(k+1)$. Therefore $P(n)$ is true for every $n \in \mathbb{N}$. □

It will be relevant to make the following remarks. First, it is important to show that $P(1)$ is true, so that the induction argument can kick off. Under some situations,

it may be that $P(1)$ is false, but $P(n)$ and true for all $n > n_0$, for some $n_0 \in \mathbb{N}$; in such cases, we need only set $P'(n) = P(n - n_0)$, and apply the argument to $P'(n)$ instead. Next, the induction hypothesis is vital in a proof by induction and, in a complicated argument, it may be useful to write it out explicitly for its application in the deduction of the inductive step—after all, if the step could be established without the induction hypothesis, then it is not a proof by induction.

Novices may wonder whether there is a sleight of hand involved, because the correct answer has been supplied, and some may even suspect that all we are asked to do is to establish what we assume. The advice is to do the abstract exercise of showing that the principle is equivalent to the well ordering axiom. For many problems, including the example above, there are direct proofs without involving induction. Nevertheless, it is important for the novice to accept that, given the well ordering of $\mathbb{N}$, a proof by induction is logically sound. Moreover, in various abstract situations, a concrete direct proof may become available only after much experiments following a proof by induction.

Mathematical induction can be used to establish the colourfully named principle (iii), which is particularly useful in arguments involving counting. We illustrate the principle with two of the earliest nontrivial applications in Sect. 1.7.

Fermat's method of descent is a kind of proof by contradiction: in order to show that a given equation has no solution in $\mathbb{N}$, the argument shows that the existence of a solution implies that of a smaller solution, leading to an infinite descent, which is impossible according to the well ordering axiom. The argument can sometimes be applied 'in reverse' to become an ascent one, which then shows that if a solution does exist then there are infinitely many solutions in $\mathbb{N}$. For example, readers will be familiar with the Pythagorean school's proof that $\sqrt{2}$ is an irrational number—an examination of the proof shows the argument depends on the well ordering axiom. A more direct argument is to show that the set $\{\sqrt{2}q : q, \sqrt{2}q \in \mathbb{N}\}$ has no least member, and is therefore empty.

Example 1.2 Use Fermat's method of descent to prove that $\sqrt{2}$ is not a fraction, and show that there are infinitely many fractions p/q such that

$$0 < \frac{p}{q} - \sqrt{2} < \frac{1}{2\sqrt{2}q^2}.$$

Solution Suppose, if possible, that $\sqrt{2} = p/q$, with $p, q \in \mathbb{N}$. Then $\sqrt{2}q = p$ and $\sqrt{2}p = 2q$, so that $\sqrt{2}(p - q) = 2q - p$. From $1 < \sqrt{2} < 2$ we deduce that $1 < q < p < 2q$, so that we may write

$$\sqrt{2} = \frac{2q - p}{p - q}, \qquad 0 < p - q < q.$$

This means that $\sqrt{2}$ can be represented as a fraction with a positive denominator strictly less than q, the denominator of the supposed representation. Applying the argument repeatedly, there is then a sequence of fractions, each representing $\sqrt{2}$,

with strictly decreasing positive denominators. This is impossible, so that the assumption that $\sqrt{2} = p/q$ is false, and $\sqrt{2}$ cannot be a fraction.

The 'descending' map is $(p, q) \mapsto (2q - p, p - q)$, that is

$$A\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 2q - p \\ p - q \end{pmatrix}, \quad \text{where} \quad A = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}, \quad |A| = -1.$$

Let $B = A^{-1} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$, with $|B| = -1$, and

$$\begin{pmatrix} p_{n+1} \\ q_{n+1} \end{pmatrix} = B\begin{pmatrix} p_n \\ q_n \end{pmatrix} = \begin{pmatrix} p_n + 2q_n \\ p_n + q_n \end{pmatrix}.$$

This is now an 'ascending' map, with (q_n) strictly increasing and $p_{n+1}^2 - 2q_{n+1}^2 = -(p_n^2 - 2q_n^2)$. Starting with $p_0^2 - 2q_0^2 = 1$, we then have $p_n^2 - 2q_n^2 = (-1)^n$, that is $(p_n + \sqrt{2}q_n)(p_n - \sqrt{2}q_n) = (-1)^n$, and hence

$$\frac{p_n}{q_n} - \sqrt{2} = \frac{(-1)^n}{(p_n/q_n + \sqrt{2})q_n^2}, \qquad n = 0, 1, 2 \ldots .$$

Setting $\binom{p_0}{q_0} = \binom{3}{2}$, so that $p_0^2 - 2q_0^2 = 9 - 8 = 1$, we then find that $p_n/q_n > \sqrt{2}$ for each even n, and therefore

$$0 < \frac{p_n}{q_n} - \sqrt{2} < \frac{1}{2\sqrt{2}q_n^2}, \qquad n = 0, 2, 4 \ldots .$$

Taking $n = 50$, say, we have

$$p_n = 40114893348711941777, \quad q_n = 28365513113449345692,$$

and p_n/q_n is an approximation to $\sqrt{2}$ from above with an error less that 10^{-39}.

Long before Fermat, there were isolated topics, such as Pythagorean triples, the Euclidean algorithm, and the Chinese remainder theorem, being discovered. The most important is the work of Diophantus[1] (200–284) who made significant contributions to a topic which now bears his name: the study of equations with solutions restricted to integers or rational numbers. Indeed, Fermat had made copious study of the Latin translation[2] of his *Arithmetica*, writing relevant notes and comments in the margins.

[1] Only a small fraction of his *Arithmetica* survived.

[2] By Claude-Gaspar Bachet in 1621.

1.2 Division Process, Quotient and Remainder

The ring $\mathbb{Z}$ is not closed under division: when 'long division' is applied to numbers in $\mathbb{N}$, there is usually a remainder as well as a quotient. For example, when 107 is divided by 14, the quotient is 7, and the remainder is $107 - 7{\cdot}14 = 9$. How is the quotient to be specified? We take the largest multiple of 14 which does not exceed 107, so that the remainder is less than the divisor 14. In general, the well ordering axiom ensures the existence of such a quotient, and the remainder then follows.

Definition 1.1 (Division Process) Let $n \geq d \geq 1$, and set $q = \lfloor n/d \rfloor$, the *floor* of the fraction n/d, or the greatest integer not exceeding n/d. Then $qd \leq n < (q+1)d$ so that, on writing $r = n - qd$,

$$n = qd + r, \qquad 0 \leq r < d.$$

We call this the *division process* applied to n and d; the numbers q and r are the *quotient* and *remainder*, respectively. The process can be extended to allow $n \in \mathbb{Z}$, but $d \in \mathbb{N}$, so that the quotient $q \in \mathbb{Z}$.

By a *multiple* of $d \in \mathbb{N}$ we mean any number md, where $m \in \mathbb{Z}$. If n is a multiple of d, say $n = md$, then the remainder is 0, and we call d a *divisor* of n, indicated by $d|n$; note that $m = n/d$ is also a divisor, called the *companion divisor*, which may coincide with d when $n = d^2$. A divisor d satisfying $1 < d < |n|$ is called a *proper divisor* of n.

In `Python`, we write `q = n//d` for integer division to deliver the quotient, and `r = n%d` is the remainder.

Some care is required in interpretation when n is negative.

Primes, Composites and Factorisations

The set $\mathbb{N}$ is partitioned into three subsets: the singleton set $\{1\}$ whose sole member is called the *unit*, or the *multiplicative identity*, the set

$$\mathbb{P} = \{2, 3, 5, 7, 11, 13, 17, \ldots\}$$

of *prime numbers*, each with no proper divisors, so that there are exactly two divisors, namely 1 and the number itself; and the set

$$\{4, 6, 8, 9, 10, 12, 14, 15, 16, \ldots\}$$

of *composite numbers*, each with at least one proper divisor.

If n is composite then its least proper divisor is necessarily a prime, because otherwise when the argument is applied to it there would be a smaller proper divisor of n. Therefore every $n > 1$ has a prime divisor, possibly n itself.

Even numbers exceeding 2 have the proper divisor 2, and are therefore composite. However, there is no known explicit expression for an arbitrarily large prime so that it is not immediately obvious that there are also infinitely primes, a fact first proved by Euclid (300 BCE). Being considered the best example of Greek mathematics his famous argument is worth repeating.

Let $\mathcal{P}$ denote any finite collection of primes, and consider

$$Q = \prod_{p \in \mathcal{P}} p + 1,$$

where the product is over the primes $p \in \mathcal{P}$. Since $Q > 1$, it has a prime divisor which cannot belong to $\mathcal{P}$, because Q has the remainder 1 when divided by each $p \in \mathcal{P}$. Therefore no finite collection of primes can include all the primes. We let $\pi(x)$ be the counting function for the set $\mathbb{P}$, that is

$$\pi(x) = \sum_{p \le x,\, p \in \mathbb{P}} 1,$$

so that, by Euclid's theorem, $\pi(x)$ is an increasing function which is unbounded. For example,

$$\pi(10) = 4, \quad \pi(100) = 25, \quad \pi(1000) = 168, \quad \pi(10000) = 1229.$$

By *trial division* we mean the process of determining whether a given $n > 1$ is a prime, or a composite, by taking $d = 2, 3, \ldots, n$ successively and checking whether $d|n$; the least such d has to be a prime divisor p (because otherwise d would not be the least). Now if $p = n$ then n is prime, of course; if not then p is a proper divisor, and n is composite. If n is composite then its least prime divisor p is at most the value of its companion divisor, that is $p \le n/p$, so that $p^2 \le n$, and thus $p \le \sqrt{n}$.

On applying trial division repeatedly to the resulting companion divisor, all the prime divisors of n are revealed, thereby showing that n has the *standard prime factorisation* representation:

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}, \qquad p_1 < p_2 < \cdots < p_k, \qquad p_i \in \mathbb{P}, \quad a_i \in \mathbb{N}. \tag{1.1}$$

For example, we find that $22869 = 3^3 7 \cdot 11^2$.

We let $\omega(n)$ denote the number of distinct prime divisors of n, and $\Omega(n)$ the total number of such divisors, that is

$$\omega(n) = k, \qquad \Omega(n) = a_1 + a_2 + \cdots + a_k; \tag{1.2}$$

we say that n is *squarefree* if $\omega(n) = \Omega(n)$, so that n is not divisible by the square of a prime. A divisor m of n then takes the form

$$m = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}, \qquad 0 \le b_i \le a_i.$$

There being $a_i + 1$ choices for each of the exponents b_i it follows that

$$d(n) = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1), \tag{1.3}$$

the number of divisors of n; we call $d(n)$ the *divisor function*. In particular,

$$d(p^a) = a + 1 \quad \text{and} \quad d(p_1 p_2 \cdots p_k) = 2^k,$$

and that $d(n)$ is an odd number if and only if n is a square.

Actually, for the reason to be given in Sect. 1.6, the definitions for these functions $\omega(n)$, $\Omega(n)$ and $d(n)$, are not yet 'properly given'.

If n is small then trial division can be applied to deliver the standard factorisation. However, if $n > 10^{40}$, say, so that $\sqrt{n} > 10^{20}$, then cows may come home (see Note 4 on CCH) before the least prime divisor p emerges. Nevertheless trial division should not be ruled out because a small p may appear before the cows do.

Congruence Classes and Modular Arithmetic

L. Euler was the first mathematician who became aware that, in number theory, he was dealing not just with integers as such, but whole classes of them represented by the individual integers. It will therefore be useful and important to become familiar with the notion of classes of numbers and their arithmetic. In the classic text *The Higher Arithmetic* [21] H. Davenport wrote

> The fact that there are only a finite number of essentially different numbers in arithmetic modulo m means that there are algebraic relations which are satisfied by every number in that arithmetic. There is nothing analogous to these relations in ordinary arithmetic.

Let $m > 1$ be fixed. The set $\mathbb{Z}$ is partitioned into the m subsets

$$C_r = \{n : n = qm + r,\ q \in \mathbb{Z}\}, \qquad r = 0, 1, 2, \ldots, m - 1,$$

called the *congruence classes* $r \pmod m$. Members of C_r differ from each other only by multiples of m; they are called *residues* $r \pmod m$, and any such residue is said to represent its class, which may also be called a *residue class*. A *complete residue system* $\pmod m$ is a set of m residues taken from the m distinct congruence classes; in particular, $\{0, 1, 2, \ldots, m - 1\}$ is a complete residue system $\pmod m$, its members being the least non-negative residues $\pmod m$.

C. F. Gauss introduced the notion of a *congruence*, together with an appropriate notation which help enormously in the arithmetic of congruence classes. Write

$$a \equiv b \pmod{m}$$

to mean that $m|a-b$, and say that a and b are *equivalent* (mod m); thus the class $C_r \pmod{m}$ is the set of numbers equivalent to $r \pmod{m}$. For example, $22 \equiv 7 \pmod{15}$ and $C_7 \pmod{15}$ is the set of numbers of the form $n = 15k + 7$ where $k \in \mathbb{Z}$. It can be verified that being equivalent (mod m) is an *equivalence relation*: the usual *reflexive, symmetric*, and *transitive* rules hold. Indeed, since the time of Gauss, number theory deals with congruences as much as with equations; we omit listing the familiar properties of arithmetic modulo m, except to mention the cancellation rule, which states that if $ac \equiv bc \pmod{m}$, then $a \equiv b \pmod{m}$, provided that c satisfies a certain condition which is to be considered in due course.

To summarise, the set $\mathbb{Z}$ is partitioned into m equivalent classes C_r whose members are numbers which are equivalent modulo m. A class C_r may be represented by one of its members, and any set of m incongruous representatives form a complete residue system. In algebraic terminology the congruent classes C_r form an (additive) cyclic group of order m; the group is denoted by $(\mathbb{Z}/m\mathbb{Z})^+$, or simply $(\mathbb{Z}/m\mathbb{Z})$, and it can then be represented by a complete residue system, such as $\{0, 1, 2, \ldots, m-1\}$.

1.3 Common Divisors, gcd and Euclidean Algorithm

A *common divisor* of $m, n \geq 1$ is a number which divides both m and n; the set of common divisors is non-empty because 1 divides every number, and each such divisor is at most $\min(m, n)$, so that there is a *greatest common divisor* (gcd); we shall see later that the gcd is a multiple of every common divisor. We write $\gcd(m, n)$ for the gcd—although sometimes (m, n) is used when the context is clear; we also let $\gcd(0, m) = m$ for $m \geq 1$. If $(m, n) = 1$ then we say that m and n are *relatively prime*, or *coprime*.

Similarly, a common multiple of m, n is a multiple of both m and n, so that the product mn is a common multiple. The least such multiples is called the *least common multiple* and is denoted by $\mathrm{LCM}(m, n)$, which is at least $\max(m, n)$. The notions of gcd and LCM generalises to any finite collection of numbers. If $m = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$ and $n = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}$, where p_i are distinct primes, and $a_i, b_i \geq 0$, then

$$\gcd(m, n) = p_1^{\min(a_1,b_1)} p_2^{\min(a_2,b_2)} \cdots p_k^{\min(a_k,b_k)},$$
$$\mathrm{LCM}(m, n) = p_1^{\max(a_1,b_1)} p_2^{\max(a_2,b_2)} \cdots p_k^{\max(a_k,b_k)}.$$

Because $\min(a,b)+\max(a,b)=a+b$, it follows that $\gcd(m,n)\cdot\mathrm{LCM}(m,n)=mn$, but $\gcd(m,n,k)\cdot\mathrm{LCM}(m,n,k)$ need not be mnk.

Example 1.3 Let $a=12$, $b=15$, $c=21$, so that $a=2^2 3$, $b=3\cdot 5$, $c=3\cdot 7$. Then

$$\gcd(a,b,c)=2^0 3^1 5^0 7^0=3 \quad \text{and} \quad \mathrm{LCM}(a,b,c)=2^2 3^1 5^1 7^1=420,$$

so that $\gcd(a,b,c)\cdot\mathrm{LCM}(a,b,c)=1260$, and $abc=3780=3\times 1260$.

The gcd can be computed without involving the factorisation of the numbers concerned. The process is the best known, and indeed the grandfather of all algorithms namely, the *Euclidean algorithm*. Suppose that $a>b$, and that $b\nmid a$; the idea is to apply the division process to (a,b), that is $a=qb+r$, with $0<r<b$, and make the observation that a divisor d of any two of the three numbers $\{a,b,r\}$ has to divide the third one, which then shows that $\gcd(a,b)=\gcd(b,r)$. Thus, replacing (a,b) with (b,r), and applying the process repeatedly, delivers a strictly decreasing sequence of non-negative remainders r, so that the process has to terminate, with the last positive remainder $r=\gcd(a,b)$. As a byproduct of the algorithm for $\gcd(a,b)$ it also delivers the solutions x, y to the equation

$$ax+by=d, \qquad d=\gcd(a,b); \tag{1.4}$$

the equation, known as the *Bézout identity* for the gcd, shows that any common divisor of a, b is a divisor of $\gcd(a,b)$. We give the `Python` code for the algorithm, together with the output of an example, which delivers $\gcd(a,b)$ and x, y.

```
# Program: The Euclidean algorithm for gcd(a,b).
# The algorithm returns a triple [x,y,g] such that
# ax + by = g = gcd(a,b).
def euclid(a,b):
  x=1; y=0; g=a; u=0; v=1; w=b
  while w>0:
    q=g//w; r=g-w*q
    print ((g,w,q,r))
    oldx=x; oldy=y; oldg=g
    x=u; y=v; g=w;
    u=oldx-q*u; v=oldy-q*v; w=r
  return [x,y,g]
```

The following are the outputs for `euclid(a,b)` with $a=456888$, $b=123777$; the intermediate entries in (g,w,q,r) are the numbers in the division process applied to the updated values of $(a,b)\to(g,w)$.

```
(456888, 123777, 3, 85557)
(123777, 85557, 1, 38220)
(85557, 38220, 2, 9117)
(38220, 9117, 4, 1752)
(9117, 1752, 5, 357)
(1752, 357, 4, 324)
(357, 324, 1, 33)
(324, 33, 9, 27)
(33, 27, 1, 6)
(27, 6, 4, 3)
(6, 3, 2, 0)
[-18722, 69107, 3]
```

The equations corresponding to the output are

$$\begin{aligned} 456888 &= 3\times 123777 + 85557 \\ 123777 &= 1\times 85557 + 38220 \\ &\vdots \\ 27 &= 4\times 6 + 3 \\ 6 &= 2\times 3 + 0 \end{aligned}$$

The last positive remainder is $\gcd(a, b) = 3$, and $-18722a + 69107b = 3$.

The gcd algorithm is super-efficient, because the number of steps involved depends only linearly on the number of digits of the input numbers. By a step, we mean the production of a line obtained from the division process to deliver the quotient and the remainder, before we replace the numbers concerned for the next step. Let $a > b$ be the pair of input numbers to the Euclidean algorithm. In applying the algorithm, the remainders in successive lines decrease strictly, so that the number of lines has the trivial bound b. The following argument given by G. Lamé (1795–1870) shows that the much smaller bound $O(\log b)$ suffices.

We may suppose that the 'worst case' occurs when all the quotients in the division process take the value 1, so that each remainder is just the difference between the two numbers concerned. In other words the worst case is when b, a are two successive terms F_n, F_{n+1} of the *Fibonacci sequence*, defined by

$$F_0 = 1, \quad F_1 = 1, \qquad F_n = F_{n-1} + F_{n-2}, \qquad n = 2, 3, 4, \ldots ,$$

so that $(F_n) = (1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots)$. Let $\alpha > \beta$ be the two roots of the associated auxiliary equation $\lambda^2 = \lambda + 1$, so that F_n satisfies *Binet's formula*:

$$F_{n-1} = \frac{\alpha^n - \beta^n}{\alpha - \beta} \sim \frac{\alpha^n}{\sqrt{5}}, \quad n \to \infty, \qquad \alpha = \frac{1+\sqrt{5}}{2}, \quad \beta = \frac{1-\sqrt{5}}{2};$$

this can be proved by induction, using $\alpha^k + \alpha^{k-1} = \alpha^{k+1}$ and $\beta^k + \beta^{k-1} = \beta^{k+1}$, for the induction hypothesis. The number of steps involved is n, so that from $b \geq F_n$ we deduce that $n \ll \log F_n \leq \log b$. A brief explanation of the notion of the complexity of an algorithm is given in Note 1.3.

By a *linear congruence* we mean

$$ax \equiv b \pmod{n};$$

this is equivalent to the linear equation $ax + ny = b$; so that, by (1.4), it is soluble if $d = \gcd(a, n)$ divides b. In particular, if a and n are coprime then there is a unique solution $\bar{a} \pmod{n}$ so that

$$a\bar{a} \equiv 1 \pmod{n};$$

we call $\bar{a}$ the reciprocal of $a \pmod{n}$, and the solution to the congruence is given by $x \equiv \bar{a}b \pmod{n}$. The reciprocals $\bar{a} \pmod{n}$, for $n = 11, 15$, when $\bar{a}$ exists, are given in Table 1.1.

Returning to the cancellation rule in the arithmetic of congruence, we may now state that if $ca \equiv cb \pmod{n}$, and c is coprime with n, then, on multiplication of the reciprocal $\bar{c} \pmod{n}$, we find that $a \equiv b \pmod{n}$.

By a *reduced residue system* $\pmod{n}$, we means the system of congruence classes C_r, where r is coprime with n. Following our earlier remark, we may represent the system by $\{r_1, r_2, \ldots, r_h\}$, where h is the number of classes in the system, and each r_i is coprime with n. Note that 0 does not belong to the system, so that $h < n$. More important to note is that if a is any number coprime with n, then $\{ar_1, ar_2, \ldots, ar_h\}$ also represent the same reduced residue system. This is because the h numbers are incongruent to each other, because $ar_i \equiv ar_j \pmod{n}$ implies $r_i \equiv r_j \pmod{n}$. That each r has a reciprocal $\bar{r} \pmod{m}$ shows that the system forms a multiplicative *group*, usually denoted by $(\mathbb{Z}/m\mathbb{Z})^\times$; indeed, if $m = p$ is a prime then the complete residue classes form a *finite field* $\mathbb{F}_p$ with p elements.

Table 1.1 Reciprocals $\pmod{n}$

$n = 11$	a	1	2	3	4	5	6	7	8	9	10	11
	$\bar{a}$	1	6	4	3	9	2	8	7	5	10	*

$n = 15$	a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	$\bar{a}$	1	8	*	4	*	*	13	2	*	*	11	*	7	14	*

Since any number equivalent (mod n) to a solution of a congruence is also a solution, it may be more helpful to refer the whole congruence class C_x as *the* solution to the congruence. Actually, once the notion of congruence classes and the concept of equivalence have become familiar, it does not matter what we mean by a solution, unless clarification is necessary. Thus we may consider a complete residue system (mod n) to be numbers in $\mathbb{Z}/n\mathbb{Z}$, a reduced residue system to be numbers in $(\mathbb{Z}/n\mathbb{Z})^\times$, and $\mathbb{F}_p$ to be the field of numbers $\{0, 1, \ldots, p-1\}$. This is common practice in mathematics; for another example of such practice, once the notion of a function f has become familiar we often abuse the language and call the value $f(n)$ that it takes as the function.

1.4 Linear Indeterminate Equations

There being two unknowns x, y in the linear equation (1.4) we may wish to find all the solutions, by means of an integer parameter.

Theorem 1.1 *Let $(a, b) = 1$, and x_0, y_0 be a solution to $ax + by = n$. Then the general solution is given by*

$$x = x_0 + bt, \quad y = y_0 - at, \qquad t \in \mathbb{Z}.$$

Proof From $ax + by = n$ and $ax_0 + by_0 = n$ we have $a(x - x_0) + b(y - y_0) = 0$. Since a, b are coprime we deduce that $a|y - y_0$. Let $y = y_0 - at$, so that $x = x_0 + bt$ for any $t \in \mathbb{Z}$. Then x, y is a solution. □

Theorem 1.2 *Let $(a, b) = 1$ and $a > 0$, $b > 0$. Then every integer greater than $ab - a - b$ is representable as $ax + by$ with $x \geq 0$, $y \geq 0$. Moreover, $ab - a - b$ is not representable as such.*

Proof Consider the equation $ax + by = n$, where $n > ab - a - b$. We know that there is a solution $(x, y) = (x_0, y_0)$ and, by the previous theorem, the general solution takes the form $x = x_0 + bt$, $y = y_0 - at$, with $t \in \mathbb{Z}$. Our task is to show that t may be so chosen that both x, y are non-negative. Since $\{0, 1, 2, \ldots, a-1\}$ forms a complete residue system (mod a), there is a unique t such that $0 \leq y_0 - at \leq a - 1$. It then follows that

$$(x_0 + bt)a = n - (y_0 - at)b > ab - a - b - (a-1)b = -a,$$

which implies $x_0 + bt \geq 0$, as required.

Finally suppose, if possible, that $ax + by = ab - a - b$ for some $x, y \geq 0$. Then $ab = (x+1)a + (y+1)b$. Since a, b are coprime, it follows that $a|y+1$ and $b|x+1$, so that $y + 1 \geq a$ and $x + 1 \geq b$, leading to $ab \geq 2ab$, which is impossible. □

Example 1.4 (Chicken Nugget Problem) What is the largest number of chicken nuggets that *cannot* be purchased if such nuggets are only sold in packs of 6, 9, 20?

Solution The answer is 43. Since $6 \equiv 9 \equiv 0 \pmod 3$ and $20 \equiv 2 \pmod 3$, we need at least two packs of 20 in order to deliver $m \equiv 1 \pmod 3$, from which it follows that 43 is not representable as a non-negative linear combination of 6, 9, 20. Moreover, it can be readily verified that the next 6 consecutive numbers $m = 44, 45, 46, 47, 48, 49$ are representable as such. Now we are done, because if $m \geq 50$ then there is a number $u \geq 0$ such that $m - 6u$ is one of the 6 numbers.

Example 1.5 Let a, b be coprime positive integers. The number of non-negative solutions to $ax + by = n$ is either $\lfloor n/ab \rfloor$ or $\lfloor n/ab \rfloor + 1$.

Solution Rewrite the equation $ax + by = n$ as the congruence $ax \equiv n \pmod b$. Let $\bar{a}$ be defined by $\bar{a}a \equiv 1 \pmod b$, $1 \leq \bar{a} < b$, so that the solution to the congruence is $x \equiv \bar{a}n \pmod b$. The solutions x to the equation $ax + by = n$ are now given by $x = \bar{a}n + hb$, where h is an integer satisfying the condition that $0 \leq x \leq n/a$. Thus

$$0 \leq \bar{a}n + bh \leq \frac{n}{a}, \quad \text{that is} \quad -\theta \leq h \leq \frac{n}{ab} - \theta, \qquad \text{where} \quad \theta = \frac{\bar{a}n}{b}.$$

The length of the closed interval for h is n/ab, so that the number of possible values for h is at least $\lfloor n/ab \rfloor$, and at most $\lfloor n/ab \rfloor + 1$. That both cases may happen is illustrated in Exercise 1.12.

The Case with Three Unknowns

Let a, b, c be positive integers with $\gcd(a, b, c) = 1$. The determination of the largest integer not representable as $ax + by + cz$, with $x, y, z \geq 0$ is a very much more difficult problem which has only been solved in recent times; see Note 5. However, there is a simple solution to the following special case.

Theorem 1.3 *Let a, b, c be pairwise coprime positive integers, and*

$$A = bc, \ B = ca, \ C = ab.$$

Then the largest integer not representable as $Ax + By + Cz$, with $x, y, z \geq 0$, is $2abc - bc - ca - ab$.

Proof Write

$$M = 2abc - bc - ca - ab, \qquad K = ab - a - b.$$

Since C, c are coprime, the reciprocal $\bar{C}$ of $C \pmod c$ exists, and it satisfies $\bar{C}C \equiv 1 \pmod c$, $1 \leq \bar{C} < c$. For $m \geq 0$, we may now define

$$z \equiv \bar{C}m \pmod c, \qquad 0 \leq z \leq c - 1,$$

so that $Cz = m - ck$ for some $k \in \mathbb{Z}$. The equation $Ax + By + Cz = m$ then becomes $Ax + By = ck$, that is $bx + ay = k$, which is soluble with $x, y \in \mathbb{Z}$ because a, b are coprime. Moreover, by Theorem 1.2, there are non-negative solutions (x, y) if $k > K$, and no such solution if $k = K$. Now if $m \geq M$ then

$$ck = m - Cz \geq M - abz = cK + ab(c - 1 - z) \geq cK,$$

so that $k \geq K$; and $k = K$ if $m = M$. This means that there are non-negative solutions to $Ax + By + Cz = m$ if $m > M$, and no such solution if $m = M$. The theorem is proved. □

The proof of the theorem amounts to reducing the three variables problem to a two variables one, and the discernible relationship between m and k is not revealed. It will be instructive to consider the solution to the following example which retains the symmetry of the three variables concerned; for this, we make use of the Chinese remainder theorem to be given in Sect. 1.5. We illustrate this by plotting k against m in Fig. 1.1.

Example 1.6 Find the 4 largest positive integers not representable as

$$35x + 21y + 15z, \quad x, y, z \geq 0.$$

Solution Take $(a, b, c) = (3, 5, 7)$, so that $abc = 105$, $M = 139$, $K = 7$ in the proof of the theorem. The values of m which are representable as $35x + 21y + 15z$, correspond to the dots which are above the horizontal line $K = 7$ in the diagram.

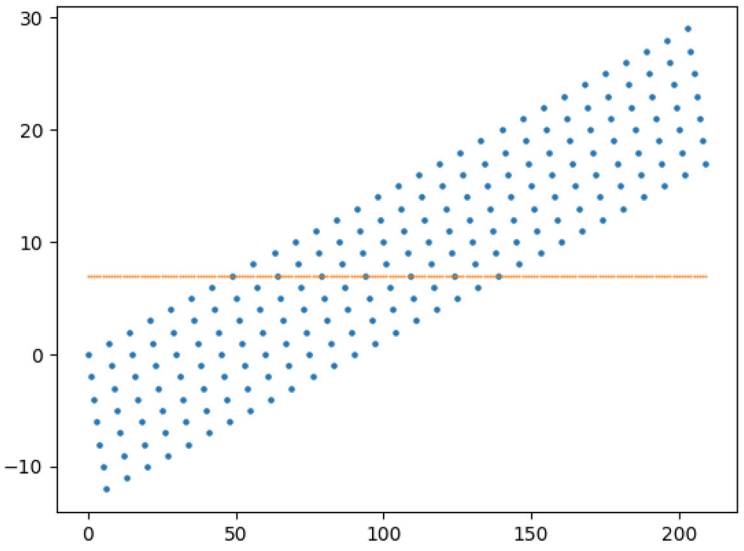


Fig. 1.1 k versus m, $1 \leq m \leq 210$

By a *negative* solution, we mean a solution (x, y, z) with at least one of x, y, z being negative, and call m *exceptional* if all the solutions are negative. Define $m_0 \equiv m \pmod N$, with $0 \le m_0 < N$, where

$$N = \mathrm{LCM}(35, 21, 15) = 105 = 3{\times}5{\times}7.$$

Note that $\gcd(21, 15) = 3,\ \gcd(35, 15) = 5,\ \gcd(35, 21) = 7$, and further define

$$m_3 \equiv m(\mathrm{mod}\ 3), \quad m_5 \equiv m(\mathrm{mod}\ 5), \quad m_7 \equiv m(\mathrm{mod}\ 7),$$

with $0 \le m_3 < 3, 0 \le m_5 < 5, 0 \le m_7 < 7$. Now, let

$$X = 70m_3 + 21m_5 + 15m_7,$$

so that $X \equiv m_0 \pmod N$, which can be checked by considering the congruence modulo 3, 5, 7 individually. Note that $0 \le X \le 70{\cdot}2 + 21{\cdot}4 + 15{\cdot}6 = 244 < 3N$, so that we may write

$$m = uN + m_0, \quad X = vN + m_0, \qquad u \ge 0,\ \ v = 0, 1, 2.$$

The required general solution of the equation should contain two independent integer parameters; keeping symmetry, we use three parameters together with a constraint on them. Anyway, the general solution can be given by

$$(x, y, z) = (2m_3 + 3r,\ m_5 + 5s,\ m_7 + 7t), \qquad r, s, t \in \mathbb{Z},$$

with the constraint $r + s + t = u - v$, because

$$\begin{aligned} 35x + 21y + 15z &= 70m_3 + 21m_5 + 15m_7 + N(r + s + t) \\ &= X + N(r + s + t) \\ &= m_0 + N(v + r + s + t) \\ &= m + N(r + s + t - (u - v)) = m. \end{aligned}$$

If $m \ge 2N$, then $u \ge 2$, $r + s + t \ge 2 - v \ge 0$, and we may set $r, s, t \ge 0$, delivering solutions (x, y, z) with $x, y, z \ge 0$, so that m is not exceptional. We proceed to identify the exceptional m, coprime with N, in $N \le m < 2N$.

Such an exceptional m has $u = 1$, and we need to have $v = 2$, to deliver $r + s + t = -1$, so that at least one of r, s, t is negative. A key observation is that we cannot have $m_3 = 2$, because it would then allow us to set $r = -1$, and we still have $x = 2m_3 + 3r = 4 - 3 = 1 > 0$. Therefore $m_3 = 1$ for exceptional numbers m, coprime with N, in $N \le m < 2N$. We may identify m by the Chinese remainder theorem from $(m_3, m_5, m_7) = (1, m_5, m_7)$, so that there are at most $4{\times}6 = 24$ of them. However, there is no need to go through all the cases, because the condition

$v = 2$ implies $X = 70+21m_5+15m_7 \geq 2N$, so that $7m_5+5m_7 \geq (2N-70)/3 = 140/3 > 46$. There are then only 4 cases for which this is satisfied, namely

$$(1, m_5, m_7) = (1, 3, 6),\ (1, 4, 4),\ (1, 4, 5), (1, 4, 6),$$

and the Chinese remainder theorem delivers the 4 largest exceptional numbers coprime with 105:

$$m = N + (13,\ 4,\ 19,\ 34) = 118,\ 109,\ 124,\ 139.$$

Note that, in agreement with Theorem 1.3, $2N - 5{\cdot}7 - 3{\cdot}7 - 3{\cdot}5 = 139$. □

Application to Round-Robin Tournaments

Linear congruences can be applied to schedule round-robin tournaments, with N competing teams. We need N to be even, so that every team participates in each round of the tournament, with every team playing every other team exactly once in $(N-1)$ rounds of the tournament. If N is odd we expedite the situation by having a dummy team—when a team is paired with the dummy team in a particular round, the team concerned is given a 'bye' and does not play in that round.

Label the teams $1, 2, \ldots, N$. In the kth round we let Team N play with Team i, where i is the unique solution of $2x \equiv k \pmod{N-1}$, which is soluble because $N-1$ is odd. For the rest of the teams, we let Team i play Team j if $i + j \equiv k \pmod{N-1}$, and $i \neq j$, of course. It is readily checked that each team then plays every other team exactly once in $(N-1)$ rounds.

The schedule for $N = 12$ teams is shown in Table 1.2.

Table 1.2 Round-robin schedule

	Team											
Round	1	2	3	4	5	6	7	8	9	10	11	12
1	11	10	9	8	7	12	5	4	3	2	1	6
2	12	11	10	9	8	7	6	5	4	3	2	1
3	2	1	11	10	9	8	12	6	5	4	3	7
4	3	12	1	11	10	9	8	7	6	5	4	2
5	4	3	2	1	11	10	9	12	7	6	5	8
6	5	4	12	2	1	11	10	9	8	7	6	3
7	6	5	4	3	2	1	11	10	12	8	7	9
8	7	6	5	12	3	2	1	11	10	9	8	4
9	8	7	6	5	3	3	2	1	11	12	9	10
10	9	8	7	6	12	4	3	2	1	11	10	5
11	10	9	8	7	6	5	4	3	2	1	12	11

1.5 Chinese Remainder Theorem

The *Chinese remainder theorem* asserts that, under suitable conditions, a system of simultaneous linear congruences has a solution.

Theorem 1.4 (Chinese Remainder Theorem) *Let $m_1, m_2, \ldots, m_k$ be pairwise coprime, and $b_1, b_2, \ldots, b_k \in \mathbb{Z}$. Then the system of simultaneous congruences*

$$x \equiv b_i \pmod{m_i} \qquad i = 1, 2, \ldots, k,$$

has a unique solution modulo $M = m_1 m_2 \cdots m_k$, given by

$$x \equiv \sum_{i=1}^{k} b_i M_i \bar{M}_i \pmod{M}, \quad \text{where} \quad M_i = \frac{M}{m_i}, \quad M_i \bar{M}_i \equiv 1 \pmod{m_i}.$$

Proof That x satisfies each of the congruences follows from $M_i \bar{M}_i \equiv 1$ or $0 \pmod{m_j}$, depending on whether $i = j$ or not. Also, any two solutions to the system have to be congruent (mod m_i), and hence congruent (mod M), because $\{m_i\}$ are pairwise coprime. □

Let $\mathcal{V}$ denote the set of vectors $\mathbf{v} = (b_1, b_2, \ldots, b_k)$, with $0 \le b_i < m_i$. The mapping f of the interval $0 \le x < M$ to $\mathcal{V}$ by the Chinese remainder theorem is a bijection, but there is much algebraic structure involved; in particular, we find that $f(x + y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$ if sums and products are interpreted appropriately in terms of the elements being considered as congruence classes. Abstractly, the Chinese remainder theorem says that the map

$$x \pmod{M} \mapsto (x \pmod{m_1},\ x \pmod{m_2},\ \ldots,\ x \pmod{m_k})$$

is a ring isomorphism $\mathbb{Z}_M \cong \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_k}$.

Note that $\omega(M) \ge k$, and that we may assume that b_i are distinct, since otherwise k can be reduced by combining the congruences having the same b_i with the product of their moduli.

Armed with `Python` codes for the Euclidean algorithm, and thus the reciprocal $\bar{a}$ of a (mod m), the reader can easily construct the codes for the Chinese remainder theorem. It is only necessary to deal with $k = 2$, and apply the result repeatedly for the case $k > 2$. However, it is not difficult to have a useful one that cater for $k = 4$ (say) directly. There is a more sophisticated, and more efficient, algorithm for the Chinese remainder theorem; see Algorithm 2.1.7 in [19].

The following frequently used fact is a consequence of the Chinese remainder theorem. Let N_1 and N_2 be coprime numbers. Then, as x_1 and x_2 run over complete residue systems (mod N_1) and (mod N_1), respectively, $N_2 x_1 + N_1 x_2$ runs over a complete residue system (mod $N_1 N_2$).

There are three types of applications of the Chinese remainder theorem. The first type can be considered mundane: they may involve the organisation of tournament schedules, the construction of hashing functions, the distribution of cypher keys, or the breaking up of large numbers into smaller parts for the purpose of efficient storage, transmission and manipulation. In this type of applications, a substantial portion, if not all, of the numbers in the interval $0 \le x < M$ are involved. The second type of applications is to use the theorem to establish the existence of a number with certain properties by the imposition of congruence conditions that the number has to satisfy. It is the nature of the method that there is then little control of (the size of) the resulting number, apart from the fact that it lies in an interval $0 \le x < M$, where the bound M is usually far too large comparing to the actual solution; see Example 10.2 for an illustration.

The third type of applications involve some abstract notions, particularly those in algebraic number theory. In any case, the Chinese remainder theorem, together with its more elderly relative the Euclidean algorithm, are by far the most frequently used procedures in computer programs.

If the moduli $m_1, m_2, \ldots, m_k$ are not pairwise coprime, then some condition on $b_1, b_2, \ldots, b_k$ is required to ensure that a simultaneous solution to the system $x \equiv b_i \pmod{m_i}$, $i = 1, 2, \ldots, k$, exists. For example, the system $x \equiv 1 \pmod 6$, $x \equiv 2 \pmod{15}$ does not have a simultaneous solution. The appropriate condition is that $b_i \equiv b_j \pmod{\gcd(m_i, m_j)}$, and there will then be a unique simultaneous solution modulo $\mathrm{LCM}(m_1, m_2, \ldots, m_k)$. To see this we need only consider the case $k = 2$, and repeat the argument inductively.

Consider then the system $x \equiv a \pmod m$ and $x \equiv b \pmod n$. Writing $d = \gcd(m, n)$, so that $\ell = (a-b)/d$ is an integer by the condition. We may choose u, v so that $mu + nv = d$, and thus $mu\ell + nv\ell = a - b$. Setting $g = a - mu\ell = b + nv\ell$, and $L = \mathrm{LCM}(m, n)$, we find that $x \equiv g \pmod L$ is the solution to the system.

1.6 Fundamental Theorem of Arithmetic

We have shown that each $n > 1$ has a standard prime factorisation representation. It had been considered obvious[3] that such a representation is unique, until it was pointed out by Gauss that the uniqueness assertion, now known as the *fundamental theorem of arithmetic*, required a proof. Indeed, sometimes we might not even have realised that the theorem had been invoked in an argument. For example, our definitions for the prime divisor functions $\omega(n)$, $\Omega(n)$, and the divisor function $d(n)$, introduced in Sect. 1.2 are properly defined only if there is uniqueness in the standard prime factorisation of n. The importance of the fundamental theorem will surface properly in the study of algebraic number theory, in which the theorem may not apply in some new situations.

[3] Apparently a certain author wrote that it follows from 'the law of thought', whatever that was!

Lemma 1.1 *Let a prime $p|m_1 m_2 \cdots m_k$. Then $p|m_i$ for some i.*

Proof Suppose first that $k = 2$, $m_1 = a$, $m_2 = b$, and that $p \nmid a$, so that $\gcd(p, a) = 1$. By the Bézout identity we may write $ax + py = 1$ for some $x, y \in \mathbb{Z}$, and hence $abx + pby = b$. By hypothesis $p|ab$, so that p divides the left-hand side of the equation, and hence $p|b$. We omit the inductive argument for $k > 2$. □

Theorem 1.5 (Fundamental Theorem of Arithmetic) *The representation of $n > 1$ as a product of primes is unique, apart from the ordering of the factors.*

Proof We use induction on n; the case $n = 2$ is true. Take, as induction hypothesis, that the result is true for all m in $1 < m < n$; the induction step is to show that the result is true also for n.

If n is prime then we are done. Suppose then that n is composite and that there are two distinct prime factorisations

$$n = p_1 p_2 \cdots p_s = p'_1 p'_2 \cdots p'_t, \qquad p_i, p'_j \in \mathbb{P}, \quad s, t \geq 2;$$

we have to show that $s = t$, and that each p is some p'. Since $p_1 | p'_1 p'_2 \cdots p'_t$, by the lemma, $p_1 | p'_j$ for some j. That p_1, p'_j are primes imply $p_1 = p'_j$, and, on relabelling if necessary, we may take $j = 1$, so that $p_1 = p'_1$. Set $m = n/p_1$ so that $1 < m < n$ and, by the induction hypothesis, the two factorisations of $m = p_2 p_3 \cdots p_s = p'_2 p'_3 \cdots p'_t$ are the same apart from the ordering. Therefore $s = t$, and the two factorisations of n are also the same apart from the ordering. The inductive step is established, and the theorem is proved. □

Prime Power in a Factorial

For each prime p, we let $v_p(n)$ denote the exact exponent of the prime power p^j which divides n, so that $p^j | n$ and $p^{j+1} \nmid n$. The arithmetic function v_p is called the p-adic valuation.

Theorem 1.6 *For $n \in \mathbb{N}$ and primes p, we have*

$$v_p(n!) = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots .$$

Proof This follows from the observation that, in the defining product for $n!$, there are $\lfloor n/p \rfloor$ multiples of p, and $\lfloor n/p^2 \rfloor$ multiples of p^2, and so on. Note that there are only finitely many nonzero terms in the series. □

Corollary 1.1 *We have, for $n > 1$,*

$$\frac{n}{p} - 1 < v_p(n!) < \frac{n}{p} + \frac{n}{p(p-1)}.$$

Proof Since $\lfloor x \rfloor > x - 1$ for $x \in \mathbb{R}$, the first inequality follows. From $\lfloor x \rfloor \leq x$, we also have, on summing a geometric progression,

$$v_n(p) < \frac{n}{p} + n \sum_{j=2}^{\infty} \frac{1}{p^j} = \frac{n}{p} + \frac{n}{p^2} \cdot \frac{1}{1 - 1/p} = \frac{n}{p} + \frac{n}{p(p-1)}.$$

□

Corollary 1.2 *The binomial coefficient* $\binom{n}{r} = \frac{n!}{r!\,(n-r)!}$ *is an integer.*

Proof The exact power of a prime p in the factorisation of $\binom{n}{r}$ is given by

$$v_p(n!) - v_p(r!) - v_p((n-r)!) = \sum_{m \geq 1} \left(\left\lfloor \frac{n}{p^m} \right\rfloor - \left\lfloor \frac{r}{p^m} \right\rfloor - \left\lfloor \frac{n-r}{p^m} \right\rfloor \right),$$

which is a non-negative integer, because $\lfloor \alpha \rfloor - \lfloor \beta \rfloor$ is either $\lfloor \alpha - \beta \rfloor$ or $\lfloor \alpha - \beta \rfloor + 1$. □

It is easy to show that if $a, b \in \mathbb{N}$ then $\lfloor n/ab \rfloor = \lfloor \lfloor n/a \rfloor / b \rfloor$, which is useful for integer arithmetic computations.

Example 1.7 In the evaluation of $v_7(1000!)$, we need only consider

$$\left\lfloor \frac{1000}{7} \right\rfloor = 142, \quad \left\lfloor \frac{1000}{7^2} \right\rfloor = \left\lfloor \frac{142}{7} \right\rfloor = 20, \quad \left\lfloor \frac{1000}{7^3} \right\rfloor = \left\lfloor \frac{20}{7} \right\rfloor = 2,$$

to deliver $v_7(1000!) = 142 + 20 + 2 = 164$.

1.7 Theorems of Dirichlet, Thue and Aubry

The following two theorems are useful applications of the pigeonhole principle.

Theorem 1.7 (Dirichlet Approximation Theorem) *Let* $\theta \in \mathbb{R}$, *and* $Q > 1$. *Then there are coprime integers* p, q, *with* $1 \leq q \leq Q$ *such that* $|q\theta - p| < 1/Q$.

Proof For $\lambda \in \mathbb{R}$, write $\{\lambda\} = \lambda - \lfloor \lambda \rfloor$, the 'fractional part' of λ, and consider

$$\{r\theta\}, \qquad r = 0, 1, 2, \ldots, Q.$$

These $Q+1$ fractional parts lie in the interval $0 \leq t < 1$, which is to be divided into Q subintervals of the form $a/Q \leq t < (a+1)/Q$, for $a = 0, 1, 2, \ldots, Q-1$, each with length $1/Q$. By the pigeonhole principle, there are two fractional parts $\{r\theta\}$ lying in some subinterval, so that they differ by less than $1/Q$. Let the two numbers concerned be $\{r_1\theta\}, \{r_2\theta\}$, where $r_1 > r_2$, and set $q = r_1 - r_2$, $p = \lfloor r_1\theta \rfloor - \lfloor r_2\theta \rfloor$. Then $|q\theta - p| < 1/Q$, with $1 \leq q \leq Q$. Finally, divide by $\gcd(p, q)$ if necessary to ensure that p, q are coprime. □

Corollary 1.3 *Let θ be an irrational number. There are infinitely many fractions p/q, with $\gcd(p, q) = 1$ such that $|\theta - p/q| < 1/q^2$.*

Proof Suppose that there were only n fractions $p_1/q_1, p_2/q_2, \ldots, p_n/q_n$ satisfying the corollary. Since θ is irrational, the minimum $\min |\theta - p_i/q_i|$ over i is positive, so that there exists $Q \in \mathbb{N}$ such that $|\theta - p_i/q_i| > 1/Q$ for each $i = 1, 2, \ldots, n$. By the theorem, there is a fraction p/q, with $1 \leq q \leq Q$, such that

$$\left|\theta - \frac{p}{q}\right| < \frac{1}{qQ} \leq \frac{1}{Q} < \left|\theta - \frac{p_i}{q_i}\right|, \qquad i = 1, 2, \ldots, n.$$

This shows that $p/q \neq p_i/q_i$, and also $|\theta - p/q| < 1/q^2$, contradicting the assumption that p_i/q_i were the only n fractions satisfying the corollary. □

The following is a useful criterion for irrationality:

Corollary 1.4 *A real number $\theta > 0$ is irrational if, and only if, there is a sequence of distinct rationals (p_k/q_k) such that $p_k - q_k\theta \to 0$ as $k \to \infty$.*

Proof If θ is rational, say $\theta = m/n$, then $|p - q\theta| = |pn - qm|/n \geq 1/n$ for all $p/q \neq \theta$, so that the condition cannot be satisfied.

If θ is irrational we can apply the previous corollary so that there are infinitely many p/q with $|p - q\theta| < 1/q \to 0$ as $q \to \infty$. □

Theorem 1.8 (Thue) *Let $m > 1$, $(a, m) = 1$, and $M = \lfloor\sqrt{m}\rfloor$. Then there are integers x, y in $1 \leq x, y < M$ such that $ax \pm y \equiv 0 \pmod{m}$.*

Proof Note that $M \leq \sqrt{m} < M + 1$, and consider the set of residues $ax - y \pmod{m}$ corresponding to pairs (x, y) in $0 \leq x, y \leq M$. Since the number of pairs is $(M + 1)^2 > m$, by the pigeonhole principle, there are two distinct pairs (x_1, y_1), (x_2, y_2) such that $ax_1 - y_1 \equiv ax_2 - y_2 \pmod{m}$, so that

$$ax_1 - y_1 - (ax_2 - y_2) \equiv a(x_1 - x_2) - (y_1 - y_2) \equiv 0 \pmod{m}.$$

We further deduce that $x_1 \neq x_2$ and $y_1 \neq y_2$ because equality for one implies equality for both, which is not possible for two distinct pairs. We may assume without loss that $x_1 > x_2$, and the required result follows by setting $x = x_1 - x_2$ and $y = |y_1 - y_2|$. □

The next theorem by L. Aubry [7], is a good illustration of Fermat's descent in action. Although we use vectors in the proof, the equations concerned involve mostly rational numbers, so only vectors are set in bold. Thus we shall write xy for the inner product of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{Q}^3$, and x^2 is then the square of the length of $\mathbf{x}$; in this context, x by itself is not defined, and we write (xy) instead of xy for clarity sake. By $\mathrm{lcm}(\mathbf{x})$ of a vector $\mathbf{x} \in \mathbb{Q}^3$ we mean the *least common multiple* of the three denominators of its components, and $\mathbf{x} \in \mathbb{Z}^3$ is called an integer vector.

Theorem 1.9 (Aubry) *An integer representable as a sum of three squares of rational numbers is representable as a sum of three squares of integers.*

We first give a sketch of the proof, which involves Diophantine consideration. Let $\mathbf{x} \in \mathbb{Q}^3$, with $d = \mathrm{lcm}(\mathbf{x})$. If $\|\mathbf{x}\|^2 \in \mathbb{N}$, then we say that $\mathbf{x}$ is admissible with respect to d, or simply that the integer vector $d\mathbf{x} \in \mathbb{Z}^3$ is *admissible*. Aubry's clever idea is to use $d = \mathrm{lcm}(\mathbf{x})$ as the integer parameter for the descent argument. The proof shows that, given an admissible vector $\mathbf{x} \in \mathbb{Q}^3$ with respect to some $d > 1$, there is an admissible $\mathbf{y} \in \mathbb{Q}^3$ with respect to some $d' < d$, and also $\|\mathbf{y}\| = \|\mathbf{x}\|$. Repeated application of the process then leads to $d' = 1$, and thus an integer vector, thereby giving a representation of x^2 as a sum of three squares of integers. More precisely, the admissible $d\mathbf{x}$ is the integer vector 'stretched' out from $\mathbf{x} \in \mathbb{Q}^3$, by the factor $d = \mathrm{lcm}(\mathbf{x}) > 1$, which is to be 'shrunk' back, not to $\mathbf{x}$ itself, but to an *integer* vector $\mathbf{q} \in \mathbb{Z}^3$ nearest to $\mathbf{x}$; this means that $\mathbf{q}$ is a lattice point inside the three dimensional sphere centred $\mathbf{x}$, radius $\frac{1}{2}$. There is then a vector $\mathbf{y} \in \mathbb{Q}^3$ in the neighbourhood of $\mathbf{q}$, given by $\mathbf{y} = \mathbf{q} - \kappa\mathbf{r}$, where κ is a fraction chosen so that $y^2 = x^2$. It follows that $\mathbf{y}$ is an admissible vector with respect to $\mathrm{lcm}(\mathbf{y})$, the denominator of κ, which turns out to be at most $3d/4$, thereby delivering the descent argument.

The proof is particularly suitable for implementation by an integer arithmetic algorithm for the delivery of the three squares.

Proof Let $d\mathbf{x}$, with $d > 1$, be admissible, and write

$$\mathbf{x} = \Big(\frac{m_1}{d}, \frac{m_2}{d}, \frac{m_3}{d}\Big) \in \mathbb{Q}^3, \qquad m_i \in \mathbb{Z},\ m_i \geq 0; \quad x^2 \in \mathbb{N},$$

so that $d\mathbf{x} = (m_1, m_2, m_3) \in \mathbb{Z}^3$. Apply the division process to each m_i with the divisor d, taking the quotient q_i so that the remainder r_i is least in absolute value, that is we write $m_i = q_i d + r_i$ with $|r_i| \leq d/2$. Set

$$\mathbf{q} = (q_1, q_2, q_3) \in \mathbb{Z}^3, \quad \mathbf{r} = (r_1, r_2, r_3) \in \mathbb{Z}^3,$$

so that $d\mathbf{x} = d\mathbf{q} + \mathbf{r}$, and

$$1 \leq r^2 = r_1^2 + r_2^2 + r_3^2 \leq 3\Big(\frac{d}{2}\Big)^2 = \frac{3d^2}{4}.$$

From $d^2x^2 = d^2q^2 + 2d(qr) + r^2$, we find that

$$x^2 = q^2 + \frac{2qr}{d} + \frac{r^2}{d^2} = q^2 + \frac{2qr + r^2/d}{d} \in \mathbb{N},$$

which implies that $d|r^2$, so that $r^2 = dd'$, where $1 \leq d' \leq 3d/4$. Thus

$$x^2 = q^2 + K, \qquad K = \frac{2qr + d'}{d} \in \mathbb{N}.$$

Now the vector

$$\mathbf{y} = \mathbf{q} - \kappa \mathbf{r} \in \mathbb{Q}^3, \qquad \kappa = \frac{K}{d'},$$

satisfies

$$\begin{aligned} y^2 &= q^2 - 2\kappa(qr) + \kappa^2 r^2 = q^2 - \kappa(2qr + \kappa r^2) \\ &= (x^2 - K) - \kappa(2qr - Kd) = (x^2 - K) - \kappa(-d') \\ &= (x^2 - K) + K = x^2. \end{aligned}$$

Moreover, $\operatorname{lcm}(\mathbf{y}) \leq d' \leq 3d/4 < d$, so that the descent argument is complete, and the theorem is proved. □

1.8 Lagrange's Theorem

The fundamental theorem of algebra states that a polynomial of degree n in the complex field $\mathbb{C}$ has exactly n zeros. Apart from the use of analysis to establish the existence of a zero, the fact that there are exactly n zeros can be obtained by mathematical induction, using the 'remainder theorem' to extract a linear factor for the inductive step. The same method can be used to establish the corresponding theorem for polynomial congruences modulo a prime, but we need the division process applied to polynomials; see Note 1.5.

Theorem 1.10 (Lagrange) *Let $f(x)$ be a polynomial with degree n, modulo a prime p. Then there are at most n solutions to the congruence $f(x) \equiv 0 \pmod{p}$.*

Proof We use mathematical induction on n, the case $n = 1$ being true. Suppose, as induction hypothesis, that the theorem is true for polynomials of degree $n - 1$, and let $f(n)$ be a polynomial with degree n. If $f(x)$ has no zero in $0 \leq x < p$ then we are done; otherwise let $f(x_0) \equiv 0 \pmod{p}$ for some x_0. Taking $g(x) = x - x_0$, by the division algorithm for polynomials, we have $f(x) = (x - x_0)q(x)$, where $\deg(q) = n - 1$. The induction step follows from the induction hypothesis there are at most $n - 1$ solutions to $q(x) \equiv 0 \pmod{p}$. □

It is important that the modulus p in the theorem is a prime—the (nonzero) residue classes corresponding to a composite modulus m do not form a field, because there are 'zero divisors'. Take for example, $m = 8$; we then have $2 \times 4 \equiv 0 \pmod{8}$, with $2, 4 \not\equiv 0 \pmod{8}$. Indeed, there are four solutions to the quadratic congruence $x^2 \equiv 1 \pmod{8}$, namely $x = 1, 3, 5, 7$.

For an odd prime p, there are exactly two solutions to $x^2 \equiv 1 \pmod{p}$, namely $x \equiv \pm 1 \pmod{p}$. We jump the gun by making use of *Fermat's Little Theorem* (Sect. 2.4) to deliver:

Corollary 1.5 *Let p be a prime, and $d|p-1$. Then there are exactly d solutions to the congruence $x^d - 1 \equiv 0 \pmod{p}$.*

Proof Writing $p - 1 = dk$ we find that

$$x^{p-1} - 1 = (x^d - 1)(x^{d(k-1)} + x^{d(k-2)} + \cdots + x^d + 1).$$

The second factor on the right-hand side of the equation is a polynomial with degree $d(k-1)$, so that there are at most $d(k-1)$ zeros (mod p). By Fermat's Little Theorem, each x in $1 \le x < p$ is a solution to $x^{p-1} - 1 \equiv 0 \pmod{p}$. Therefore, there are at least $(p-1) - d(k-1) = d$ solutions to $x^d - 1 \equiv 0 \pmod{p}$, and hence exactly d solutions by applying the theorem again. □

Theorem 1.11 (Wilson) *Every prime p satisfies $(p-1)! \equiv -1 \pmod{p}$.*

The theorem, together with its converse, was published in 1770 by E. Waring, who ascribed it to his student John Wilson, and it was proved by Lagrange in the same year. The proof of the converse is simpler: If m is composite then its least proper divisor d is less than $m-1$, so that $(m-1)! \equiv 0 \pmod{d}$, which implies $(m-1)! \equiv 0 \pmod{m}$, contradicting $(m-1)! \equiv -1 \pmod{m}$. The theorem therefore gives a necessary and sufficient condition for m to be a prime; however, it is useless as a primality test, because the evaluation of $(m-1)! \pmod{m}$ is more onerous than trial division.

Proof The cases $p = 2, 3$ are trivial, so we suppose that $p > 3$ and consider the reduced residue system $\{1, 2, \ldots, p-1\}$. These numbers form pairs $(m, \bar{m})$, where $\bar{m}$ is the reciprocal of $m \pmod{p}$, that is $m\bar{m} \equiv 1 \pmod{p}$. Moreover, by Lagrange's theorem, there are exactly two self-pairs, namely $(1, 1)$ and $(p-1, p-1)$, which correspond to the two solutions of $x^2 \equiv 1 \pmod{p}$. Discarding these two pairs we then find that

$$2 \cdot 3 \cdot 4 \cdots (p-2) \equiv \prod (m\bar{m}) \equiv 1 \pmod{p},$$

and hence the required result. □

Corollary 1.6 (Lagrange) *Let $p = 4k + 1$ be a prime. Then the solutions to $u^2 + 1 \equiv 0 \pmod{p}$ are given by $u \equiv \pm(2k)! \pmod{p}$.*

Proof We have, by the theorem, $(4k)! + 1 \equiv 0 \pmod{p}$, and

$$4k \equiv -1, \quad 4k - 1 \equiv -2, \quad \ldots, \quad 2k + 1 \equiv -2k \pmod{p},$$

so that $(1 \cdot 2 \cdot 3 \cdots 2k)^2 + 1 \equiv 0 \pmod{p}$, because the number of negative signs in the product is the even number $2k$. The required result follows. □

The formula $u \equiv \pm(2k)! \pmod{p}$ in the corollary, though explicit, is practically useless for computation unless p is small. In fact, the efficient evaluation of u poses a rather interesting problem which will be considered later.

1.9 Evaluation of $a^b \pmod n$

We often need to perform arithmetic operations on the residue classes (mod n), which are identified by the numbers in $\{0, 1, 2, \ldots, n-1\}$. The problem is trivial for addition, subtraction, and multiplication; and division amounts to multiplication of the reciprocal of the number (mod n) when it exists. We also need to deal with exponentiation: the determination of the residue class to which $a^b \pmod n$ belongs, supposing that $1 < a, b < n$.

Even when n is of modest size, the number a^b itself is enormous if $b > n/2$. However, in arithmetic (mod n), we are only interested in the residue class to which a^b belongs, and this can be determined by repeated multiplication by a in ordinary arithmetic, with reduction (mod n) after each multiplication, so that the numbers involved need not exceed n^2. Nevertheless, the number of multiplications involved is b, which is an unacceptably large number of operations. There is, instead, a much more efficient procedure to determine $a^b \pmod n$, called the 'square and conquer' method which requires only $O(\log b)$ (see Notation for the Big-O) multiplications. The idea is to apply

$$a^{2^{k+1}} = \left(a^{2^k}\right)^2, \quad \text{for} \quad k = 0, 1, 2, \ldots,$$

in the computation, and an example will be enough for the explanation.

Example 1.8 Determine $a^b \pmod n$ for $n = 777777$, $a = 11$, $b = 765432$.

Solution Writing $b = 765432$ in its binary representation, we have

$$\begin{aligned} b &= (1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0)_2 \\ &= 2^{19} + 2^{17} + 2^{16} + 2^{15} + 2^{13} + 2^{11} + 2^{10} + 2^8 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3, \\ a^b &= a^{2^{19}} a^{2^{17}} a^{2^{16}} a^{2^{15}} a^{2^{13}} a^{2^{11}} a^{2^{10}} a^{2^8} a^{2^7} a^{2^6} a^{2^5} a^{2^4} a^{2^3}. \end{aligned}$$

Since each a^{2^k} can be obtained by repeated squaring, the required residue (mod n) can be delivered from $a \pmod n$, starting form the right of the product, and by squaring numbers 20 times, together with the same number of multiplications, reducing (mod n) at the end of each such operation. Apart from the conversion of b into its binary representation, the number of (mod n) multiplications involved is bounded $(2 \log b)/(\log 2)$.

For $a = 11$ and $n = 777777$, we have $a^b \equiv 292930 \pmod n$. □

1.10 Summary and Notes

Summary

- Problems concerned with the whole of $\mathbb{N}$ will require the *well-ordering axiom*, or the principle of *mathematical induction*.
  Fermat's descent method can be used to show that some equations have no solution, or that there are infinitely many solutions which may be delivered by an algorithm.
- For $n, d \in \mathbb{N}$, with $d < n$, the *division process* enables us to write $n = dq + r$, where $q = \lfloor n/d \rfloor$ is the *quotient* and $0 \leq r < d$ is the *remainder*.
  The *Euclidean algorithm* is efficient in the computation of $\gcd(a, b)$, and can be used to solve the congruence $ax \equiv c \pmod{b}$ if a, b are coprime. The congruence is equivalent to the linear equation $ax + by = c$.
- The notion of *congruence classes*, and their arithmetic, enable us to use congruences instead of equations, which often simplify matters in finding solutions.
- The set $\mathbb{P}$ of primes is infinite, and the *fundamental theorem of arithmetic* states that the factorisation of $n \in \mathbb{N}$ into $p \in \mathbb{P}$ is essentially unique.
- The *Chinese remainder theorem* enables us to find a unique solution to a system of simultaneous congruences.
- The very important and useful *Dirichlet approximation theorem* can be established by the *pigeonhole principle*, as can Thue's theorem on the existence of a small solution to a linear congruence.
- Lagrange's theorem states that the number of solution of the congruence $f(x) \equiv 0 \pmod{p}$, where $f(x)$ is a polynomial with degree n, and p is a prime, is at most n.
- The value of $a^b \pmod{n}$ can be evaluated efficiently by the 'square and conquer' method.

Notes

1. The more general pigeonhole principle states that if there are n objects in m boxes, then there is a box which contains at least $\lfloor n/m \rfloor$ objects. The principle is also related to the *birthday problem*; see Note 3.2.
2. The argument used in Example 1.2 was first put into print by T. Estermann [31]. It can be extended to show that $N^{1/k}$ is either an integer or an irrational number; see Exercise 1.5.

   There is an algorithm which, corresponding to an irrational θ as input, delivers a sequence of fractions (p_n/q_n) satisfying Corollary 1.3. However, except for the case when θ is a quadratic irrational, such as $\sqrt{2}$ in Example 1.2, there is no known simple relationship between such successive fractions p_n/q_n; see Chap. 7 on continued fractions.

3. The complexity of an algorithm is a topic in computer science. It involves the study of the number of bit-operations involved in each step of the algorithm, which usually amounts to the consideration of the number of bit-operations in the division of two numbers to obtain the quotient, and then multiplication and subtraction to obtain the remainder.

   Denote by $M(N)$ the number of bit-operations needed to multiply two N-bit integers. It can be shown that if b is an N-bit number and a has at most $2N$-bits, then $\lfloor a/b \rfloor$ can be obtained using $O(M(N))$ bit-operations. Therefore, corresponding to the input integers $a > b$, the complexity of the Euclidean algorithm is $O(N \cdot M(N))$, where $N = \log a$.

   What can we say about $M(N)$? By the usual digit-by-digit method used in 'long-multiplication', and taking care of the 'positioning' of the various products before summing them up, we find that $M(N) = O(N^2)$ is certainly correct. However, the use of more sophisticated methods of multiplying large numbers delivers $M(N) = O(N^{1+\epsilon})$ for any $\epsilon > 0$.
4. When the computing time involved is exceedingly long, because the algorithm involved is inefficient, I write CCH to mean that "cows may come home before any result emerges!"
5. The solution for the largest integer not representable as $ax + by + cz$, where $\gcd(a, b, c) = 1$ and $x, y, z \geq 0$ is given in [99] by A. Tripathi.
6. Chinese remainder theorem: The Chinese were the first to consider such a system; more specifically, the mathematician Sun Zi (c 200 CE) wrote an arithmetic text which contains practical problems and their solutions, after which many authors wrote stories, novels, and songs including puzzles which involve such systems and their solutions. Mathematics was not 'theorem and proof' for them, although some of their methods for the solutions amounted to algorithms from which proofs might be constructed; however, the more definitive versions were given only in 1247 by Qin Jiushao in his *Mathematical Treatise in Nine Sections*. Take, for example, one of the simplest systems:

$$x \equiv a \pmod 3, \qquad x \equiv b \pmod 5, \qquad x \equiv c \pmod 7.$$

   The solution was given in a well known enigmatic verse in the year 1593:

> *Three people walking together, 'tis rare that one be seventy;*
> *Five cherry blossoms trees, twenty-one branches bearing flowers;*
> *Seven disciples reunite for the half-moon;*
> *Take away multiples of one-hundred-and-five and you shall know.*

   In short, the verse says that the solution is

$$x \equiv 70a + 21b + 15c \pmod{105},$$

   not that it tells us that 70, 21, 15, corresponding to the moduli 3, 5, 7, came from an algorithm of theirs to solve linear equations.

There is some confusion in the use of the name 'Sun Zi' for the Chinese remainder theorem, and also for the famous military treatise *The Art of War*. The authors of the latter were members of the Sun family, including the distinguished general Sun Wu (500 BCE). The mathematician Sun lived centuries later, but very little is known about his life. The Chinese character for 'Zi' is an honorary title bestowed on a learned person, rather than being a personal name. Thus 'Sun Zi, Kong Zi, Meng Zi' are 'Masters Sun, Kong (Confucius), Meng (Mencius)', respectively.

7. Division process for polynomials: The set $F[x]$ of polynomials with the variable x over a field F form an integral domain (having no 'zero divisors') with a division algorithm. If $f, g \in F[x]$ and g not a constant, then there are $q, r \in F[x]$ such that

$$f(x) = g(x)q(x) + r(x), \qquad \deg(r) < \deg(g),$$

or else r is identically zero; here $\deg(f)$ is the degree of f. For our purpose, we take the congruence classes modulo a prime p to be the field. The remainder theorem then amounts to saying that if we take $g(x) = x - x_0$, where $f(x_0) \equiv 0 \pmod{p}$, then there exists $q(x)$ such that $f(x) = (x - x_0)q(x)$, because r is identically zero.

1.11 Exercises

1. Show that the *well ordering axiom* is equivalent to the *principle of mathematical induction*.
2. Establish the formula for the sum $1^3+2^3+3^3+\cdots+n^3$ in Example 1.1 without using induction.
3. Show that, for $n > 1$, the value of the fraction $1 + \frac{1}{2} + \cdots + \frac{1}{n}$ is never an integer.
4. Find all the solutions to the equation $m^n = n^m$ in positive integers.
5. Apply Estermann's argument to establish that $\sqrt[5]{3}$ is irrational.
6. Prove that there are infinitely many primes $p \equiv 2 \pmod 3$.
7. Let $0 < \theta < 1$ be an irrational number. Show that there are infinitely many Pythagorean triples (a, b, c) such that

$$0 < \frac{a}{b} - \theta < \frac{7}{c}.$$

8. *Fermat numbers* are defined by $F_n = 2^{2^n} + 1$ for $n \geq 0$. Prove, by induction, that

$$F_0F_1F_2\cdots F_n = F_{n+1} - 2, \qquad n = 1, 2, \ldots .$$

Deduce that Fermat numbers are pairwise coprime, and apply the argument to show that $\pi(x) > \log\log x$ for $x > 5$.

9. Blankets, towels and handkerchieves are sold at a market store at £20, £1 and 25p for each, respectively. Suppose that the storekeeper collects £200 by selling 100 articles. How many blankets were sold?
10. Show that $2^{\omega(n)} \le d(n) \le 2^{\Omega(n)}$ for $n \in \mathbb{N}$.
11. Write a `Python` program which, from an input value n, delivers the reciprocal $\bar{a} \pmod n$ for each a in $1 \le a \le n$ with $\gcd(a, n) = 1$.
12. Find the non-negative solutions to $3x + 7y = n$ when $n = 95$, and $n = 96$.
13. Let p be an odd prime not dividing m. Prove that the least positive solution of $mx \equiv \pm 2 \pmod p$ satisfies $1 \le x < p/2$.
14. Let $n > 3$ be an odd number, and let k and ℓ be the least positive integers such that $kn + 1$ and ℓn are squares. Prove that n is prime if, and only if, $k, \ell > n/4$.
15. Find, for a fixed $m \ge 0$, a formula for the general solution of

$$40x + 24y + 15z = m, \qquad x, y, z \in \mathbb{Z},$$

and use it to determine the four largest values of m, coprime with 120, for which there is no solution (x, y, z) with $x, y, z \ge 0$.

16. Construct a schedule for a round robin tournament for 11 teams with the following conditions.

(i) there are 11 rounds, in each of which one team takes a 'bye';
(ii) two teams in a game can be considered a 'home' or 'away' tie;
(iii) each team plays 5 home games, and 5 away games in the tournament.

17. Let p_n denote the nth prime. Determine the smallest solution to the system of 6 simultaneous congruences

$$p_k x \equiv p_{k+1} \pmod{p_{k+2}}, \qquad k = 1, 2, 3, 4, 5, 6.$$

18. Three children in a family measure the length of the dining room using their feet which are 5, 7 and 9 inches long. Although there is a dispute on the numbers of steps taken in their measurements they agree that there are 1, 4 and 6 inches left over respectively. Determine, as their mathematician guest, the length of their dining room and the numbers of steps taken.
19. Use the fact that $(\sqrt{2} - 1)^n \to 0$, as $n \to \infty$, to deduce that $\sqrt{2}$ is irrational.
20. Verify that $N = (m_1/d)^2 + (m_2/d)^2 + (m_3/d)^2$ is an integer for $d = 9949$ and

$$(m_1, m_2, m_3) = (10998058553422, 1222006144467945,$$
$$12559804947610504).$$

Use Aubry's theorem to find integers (a_1, a_2, a_3) with $N = a_1^2 + a_2^2 + a_3^2$.

Chapter 2
Arithmetic Functions I



Abstract Besides the notions of additive and multiplicative functions, and examples thereof, there is the Fermat–Euler theorem, and a description of the decimal system. Other topics include Dirichlet convolutions, additive characters, Gauss sum and Ramanujan sum.

A function $f : \mathbb{N} \mapsto \mathbb{C}$ is called an *arithmetic function*. The definition is perhaps too wide ranging, and we may prefer to add that such functions should carry some information on the arithmetical property of either the range of f, or the image $f(n)$, but it is not easy to set such down in a satisfactory way. Thus an arithmetic function is just a sequence of complex numbers, not that such a statement adds anything to the notion that it is a mapping from $n \in \mathbb{N}$ to an image $f(n) \in \mathbb{C}$.

The more basic arithmetic functions, such as $\omega(n)$, $\Omega(n)$ and $d(n)$, have the range restricted to $\mathbb{N}$, or $\mathbb{Z}$.

2.1 Additive and Multiplicative Functions

There are two important classes of arithmetic functions. A function f is *additive* if

$$f(mn) = f(m) + f(n) \quad \text{for all } m, n \text{ with } \quad \gcd(m, n) = 1;$$

and it is *multiplicative* if

$$f(mn) = f(m)f(n) \quad \text{for all } m, n \text{ with } \quad \gcd(m, n) = 1.$$

Being *completely additive*, and *completely multiplicative*, mean that the coprime condition can be dispensed with. For example, it follows from the definitions that $\omega(n)$ is additive, $\Omega(n)$ is completely additive, the divisor function $d(n)$ is multiplicative, and that $\sqrt{n}$ is completely multiplicative.

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Besides the zero function, a multiplicative function f satisfies $f(1) = 1$, and $f(n)$ is then determined by the values it takes at the prime powers:

$$f(p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}) = f(p_1^{a_1}) f(p_2^{a_2}) \cdots f(p_k^{a_k}), \qquad p_i \in \mathbb{P}, \quad a_i \in \mathbb{N}.$$

Let $\sum_{d|n}$ denote summation over the divisors of n, so that, for a given arithmetic function $f(n)$, we have a new function, namely the sum function $F(n)$ defined by

$$F(n) = \sum_{d|n} f(d). \tag{2.1}$$

Theorem 2.1 *The sum function of a multiplicative function is multiplicative.*

Proof Let $F(n)$ be the sum function of $f(n)$, and let $\gcd(m, n) = 1$. A divisor d of mn can be uniquely expressed as $d_1 d_2$ where $d_1 | m$ and $d_2 | n$, and then $(d_1, d_2) = 1$. Thus, for such m, n we have

$$\begin{aligned} F(mn) &= \sum_{d|mn} f(d) = \sum_{d_1|m,\, d_2|n} f(d_1 d_2) = \sum_{d_1|m,\, d_2|n} f(d_1) f(d_2) \\ &= \sum_{d_1|m} f(d_1) \sum_{d_2|n} f(d_2) = F(m) F(n). \end{aligned}$$

□

Since $d(n)$ counts the divisors of n, it is the number of pairs (a, d) satisfying $n = ad$, so that

$$d(n) = \sum_{d|n} 1 = \sum_{ad=n} 1. \tag{2.2}$$

It is not difficult to construct n so that $d(n)$ is large; indeed, $d(2^k) = k + 1$, and $d(p_1 p_2 \cdots p_k) = 2^k$ for distinct primes p_i. However, we find that $d(n) < n^{\epsilon}$, for any fixed ϵ and all large n.

Theorem 2.2 *Corresponding to each $K \in \mathbb{N}$, there are infinitely many n such that $d(n) > \log^K n$.*

Proof Let $p_1, p_2, \ldots, p_{K+1}$ be distinct primes, and set $n = (p_1 p_2 \cdots p_{K+1})^m$. Then

$$\begin{aligned} d(n) &= (m+1)^{K+1} > m^{K+1} = m \cdot m^K = m \Big(\frac{\log n}{\log(p_1 p_2 \cdots p_{K+1})} \Big)^K \\ &= \frac{m}{\log(p_1 p_2 \cdots p_{K+1})^K} (\log n)^K \geq (\log n)^K, \end{aligned}$$

for each such n if $m \geq \log(p_1 p_2 \cdots p_{K+1})^K$. □

Theorem 2.3 *For each fixed $\delta > 0$, we have $d(n)/n^{\delta} \to 0$ as $n \to \infty$.*

Proof It suffices to prove that $d(n)/n^{2\delta} \to 0$ as $n \to \infty$, for any fixed $\delta > 0$. Let $n = \prod_{p|n} p^a$, so that $d(n) = \prod_{p|n}(a+1)$, and

$$\frac{d(n)}{n^{\delta}} = \prod_{p|n} \frac{a+1}{p^{a\delta}}.$$

First, if $p^{\delta} \geq 2$ then

$$\frac{a+1}{p^{a\delta}} \leq \frac{a+1}{2^a} \leq 1.$$

Next, since $p^{a\delta} \geq 2^{a\delta} = \exp(a\delta(\log 2)) > a\delta \log 2$, it follows that

$$\frac{a+1}{p^{a\delta}} < 1 + \frac{a}{p^{a\delta}} < 1 + \frac{1}{\delta \log 2} < \exp\Big(\frac{1}{\delta \log 2}\Big),$$

so that

$$\frac{d(n)}{n^{\delta}} \leq \prod_{p \leq 2^{1/\delta}} \exp\Big(\frac{1}{\delta \log 2}\Big) < \exp\Big(\frac{2^{1/\delta}}{\delta \log 2}\Big),$$

a bound that does not depend on n. This implies $d(n)/n^{2\delta} \to 0$ as $n \to \infty$. □

2.2 Möbius Function

The Möbius function $\mu(n)$ is defined as follows:

$$\mu(1) = 1; \quad \mu(n) = \begin{cases} (-1)^k & \text{if} \quad n = p_1 p_2 \cdots p_k, \quad p_i \text{ distinct primes,} \\ 0 & \text{otherwise.} \end{cases} \tag{2.3}$$

Note that if n is squarefree then $|\mu(n)| = \mu^2(n) = 1$. The values for $\mu(n)$, $1 \leq n \leq 30$, are given in Table 2.1:

Table 2.1 Möbius function $\mu(n)$

$\mu(n) = -1$	2	3	5	7	11	13	17	19	23	29	30
$\mu(n) = +1$	1	6	10	14	15	21	22	26			
$\mu(n) = 0$	4	8	9	12	16	18	20	24	25	27	28

It is clear that if $\gcd(m,n) = 1$ then $\mu(mn) = \mu(m)\mu(n)$, so that $\mu(n)$ is multiplicative. The function comes up often in number theory, particularly in counting arguments, because of the following useful and important property:

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n > 1. \end{cases} \tag{2.4}$$

For if $F(n) = \sum_{d|n} \mu(d)$, so that $F(n)$ is multiplicative, and $F(1) = 1$; also, for prime powers p^a, we find that

$$F(p^a) = \mu(1) + \mu(p) + \mu(p^2) + \cdots + \mu(p^a) = 1 - 1 + 0 + \cdots + 0 = 0.$$

Möbius Inversion

Theorem 2.4 (Möbius Inversion) *Let $F(n)$ be the sum function of $f(n)$. Then*

$$f(n) = \sum_{d|n} \mu\Big(\frac{n}{d}\Big) F(d) = \sum_{d|n} F\Big(\frac{n}{d}\Big)\mu(d).$$

Corollary 2.1 *If $F(n)$ in the multiplicative then so is $f(n)$.*

Proof If $d|n$ then we may write $n = dm$, so that

$$\sum_{d|n} \mu\Big(\frac{n}{d}\Big) F(d) = \sum_{dm=n} \mu(m) F(d) = \sum_{dm=n} \mu(m) \sum_{a|d} f(a).$$

Write $d = ab$, so that $n = abm$; the double sum does not involve b, so that

$$\sum_{d|n} \mu\Big(\frac{n}{d}\Big) F(d) = \sum_{am=n} \mu(m) f(a) = \sum_{a|n} f(a) \sum_{m|n/a} \mu(a) = f(n),$$

because the value of the inner sum is 1 when $a = n$, and 0 otherwise. The second equation in the theorem is the 'conjugate' sum over the companion divisors. □

For the corollary, let $(m,n) = 1$, take $a|m$, $b|n$ with $(a,b) = 1$, so that

$$f(mn) = \sum_{a|m\ b|n} \mu\Big(\frac{mn}{ab}\Big) F(ab) = \sum_{a|m} \mu\Big(\frac{m}{a}\Big) F(a) \sum_{b|n} \mu\Big(\frac{n}{b}\Big) F(b) = f(m) f(n);$$

we have used the fact that $\mu(n)$ is multiplicative, and also that $F(n)$ is multiplicative by hypothesis. □

Suppose that $f(x)$ has domain $\mathbb{R}$, so that we may define

$$g(x) = \sum_{n \le x} f\Big(\frac{x}{n}\Big).$$

In this case, we also have an extension of the Möbius inversion formula, namely that $f(x)$ is then given by

$$f(x) = \sum_{n \le x} \mu(n) g\Big(\frac{x}{n}\Big), \tag{2.5}$$

and the converse also holds. Thus, making use of $\sum_{n|k} \mu(n) = 1$ or 0, depending on $k = 1$ or $k > 1$, we have

$$\sum_{n \le x} \mu(n) g\Big(\frac{x}{n}\Big) = \sum_{n \le x} \mu(n) \sum_{m \le x/n} f\Big(\frac{x}{mn}\Big) = \sum_{k \le x} f\Big(\frac{x}{k}\Big) \sum_{n|k} \mu(n) = f(x).$$

The proof of the converse is similar.

Indeed, there are even more general and useful inversion formulae, and the following is particularly important when applied to series.

Theorem 2.5 (Generalised Möbius Inversion Formula) *Let $h(n)$ be a completely multiplicative arithmetic function. Then*

$$G(x) = \sum_{n \le x} h(n) F\Big(\frac{x}{n}\Big) \quad \text{if, and only if,} \quad F(x) = \sum_{n \le x} \mu(n) h(n) G\Big(\frac{x}{n}\Big).$$

Proof Suppose that $G(x) = \sum_{n \le x} h(n) F(x/n)$. Then, on writing $mn = d$, so that $h(m)h(n) = h(d)$, with $n|d$, we find that

$$\begin{aligned} \sum_{n \le x} \mu(n) h(n) G\Big(\frac{x}{n}\Big) &= \sum_{n \le x} \mu(n) h(n) \sum_{m \le x/n} h(m) F\Big(\frac{x}{mn}\Big) \\ &= \sum_{d \le x} h(d) F\Big(\frac{x}{d}\Big) \sum_{n|d} \mu(n) = F(x), \end{aligned}$$

because the inner sum is 1, or 0, corresponding to $d = 1$, or $d > 1$. The proof of the converse is similar. □

Example 2.1 Show that

$$\sum_{1 \le n \le x} \mu(n) \Big\lfloor \frac{x}{n} \Big\rfloor = 1 \quad \text{and} \quad \Big| \sum_{1 \le n \le x} \frac{\mu(n)}{n} \Big| \le 1, \qquad \text{for} \quad x \ge 1.$$

Solution Setting $h(n) = 1$ and $F(x) = 1$ in the theorem, so that $G(x) = \lfloor x \rfloor$, the required equation follows. We may assume that $x \geq 2$ is an integer. Writing $\{z\}$ for the factional part of z, we then have, by what has just been established,

$$\left| x \sum_{1 \leq n \leq x} \frac{\mu(n)}{n} - 1 \right| = \left| \sum_{1 \leq n \leq x} \mu(n) \left(\frac{x}{n} - \left\lfloor \frac{x}{n} \right\rfloor \right) \right|$$

$$= \left| \sum_{1 \leq n \leq x} \mu(n) \left\{ \frac{x}{n} \right\} \right| = \left| \sum_{2 \leq n \leq x} \mu(n) \left\{ \frac{x}{n} \right\} \right| \leq x - 1.$$

By the triangle inequality, we now have

$$x \left| \sum_{1 \leq n \leq x} \frac{\mu(n)}{n} \right| \leq 1 + (x - 1) = x,$$

and the required result follows on division by x. □

Remark The series $\sum_{n \geq 1} \mu(n)/n$ converges to 0 but, as we remark in Sect. 9.3, the proof is difficult.

2.3 Euler Totient Function

In 1749 Euler introduced his eponymous totient function $\phi(n)$, which counts the numbers m in $1 \leq m \leq n$ that are coprime with n, that is

$$\phi(n) = \sum_{1 \leq m \leq n,\ (m,n)=1} 1.$$

A set of $h = \phi(n)$ numbers, each of which is coprime with n, is said to form a *reduced system of residues* (mod n). It is useful and important to note that if $a_1, a_2, \ldots, a_h$ form a reduced system of residues (mod n), and $\gcd(k, n) = 1$, then $ka_1, ka_2, \ldots, ka_h$ also form such a system. This follows from the observation that each member of the second set is coprime with n and, by multiplication of the reciprocal $\bar{k}$ (mod n), they are distinct modulo n. Thus, the summation condition in the definition of $\phi(n)$ may be replaced by m belonging to any reduced residue system (mod n). The values for $\phi(n)$ in $1 \leq n \leq 16$ are set out in Table 2.2.

Table 2.2 Euler totient function $\phi(n)$

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$\phi(n)$	1	1	2	2	4	2	6	4	6	4	10	4	12	6	8	8

For example, the 6 numbers $\{1, 2, 4, 5, 7, 8\}$ form a reduced residue system (mod 9).

Our first task is to show that $\phi(n)$ is multiplicative. For if $k|mn$, where mn are coprime, then $k = ab$, with $a|m$, $b|n$ and $\gcd(a, b) = 1$, so that

$$\phi(mn) = \sum_{1\le k\le mn,\ (k,mn)=1} 1 = \sum_{1\le a\le m,\ (a,m)=1} \sum_{0\le b<n,\ (b,n)=1} 1 = \phi(m)\phi(n).$$

The result is also a consequence of our remark following the Chinese remainder theorem, for which we give another proof. Suppose that m, n are coprime, and a runs through a complete residue system (mod m), and b runs through a complete residue system (mod n). Then $an + bm$ runs through a complete residue system (mod mn). To see this we note that there are mn numbers being considered and $a_1n + b_1m \equiv a_2n + b_2m \pmod{mn}$ implies $a_1n \equiv a_2n \pmod{m}$, so that $a_1 \equiv a_2 \pmod{m}$, and similarly $b_1 \equiv b_2 \pmod{n}$. Thus the mn numbers are all incongruent and form a complete residue system. The same argument shows that as a runs through a reduced residue system (mod m), and b runs through a reduced residue system (mod n), then $an + bm$ runs through a reduced residue system (mod mn). This shows that $\phi(mn) = \phi(m)\phi(n)$ if m, n are coprime.

For another explanation consider the following $m\times n$ rectangular array:

$$\begin{Bmatrix} 1 & m+1 & 2m+1 & \cdots & (n-1)m+1 \\ 2 & m+2 & 2m+2 & \cdots & (n-1)m+2 \\ 3 & m+3 & 2m+3 & \cdots & (n-1)m+3 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ m & 2m & 3m & \cdots & nm \end{Bmatrix}$$

Each column is a run of m consecutive numbers, so that there are $\phi(m)$ numbers in a column which are coprime with m. Next, consider the ath row, where $1 \le a \le m$. Since m, n are coprime, the n numbers in the row, which have the form $bm+a$, with $0 \le b < n$, form a complete residue system (mod n), so that there are $\phi(n)$ numbers in the row which are coprime with n; moreover, if $\gcd(a, m) > 1$, then each $bm + a$ cannot be coprime with m. Therefore in counting the numbers in the array which are coprime with mn, such numbers have to lie in $\phi(m)$ rows with $(a, m) = 1$, each of which contains $\phi(n)$ numbers being coprime with n. There being $\phi(mn)$ numbers in the array to be counted, it follows that $\phi(mn) = \phi(m)\phi(n)$.

In other words, we break up the interval $1 \le k \le mn$ into m arithmetic progressions, with common difference m, that is $k = bm + a$, for $0 \le b < n$ and $1 \le a \le m$. Such k form a complete residue modulo n, and k is coprime with m only if a is coprime with m.

Theorem 2.6 *For all $n \geq 1$, we have*

$$\sum_{d|n} \phi(d) = n.$$

Proof Partition the set of numbers in the interval $\{1 \leq m < n\}$ into sets according to the value of $d = \gcd(m, n)$, by defining

$$A(d) = \{m : 1 \leq m < n, d = (m, n)\}, \qquad d|n.$$

For $d = (m, n)$, we may write $m = ad$, $n = bd$, where $(a, b) = 1$. It then follows that $A(d) = \{a : 1 \leq a < b, (a, b) = 1\}$ and hence $|A(d)| = \phi(b) = \phi(n/d)$. Therefore

$$n = \sum_{d|n} |A(d)| = \sum_{d|n} \phi\Big(\frac{n}{d}\Big) = \sum_{d|n} \phi(d).$$

□

Apparently Euler was unaware of the theorem, from which it also follows that $\phi(n)$ is multiplicative by considering Möbius inversion.

Theorem 2.7 *Euler's totient function $\phi(n)$ is multiplicative, and*

$$\phi(p^a) = p^a - p^{a-1} = p^a\Big(1 - \frac{1}{p}\Big), \qquad p \in \mathbb{P}, \quad a \in \mathbb{N}.$$

Also

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right) = n \sum_{d|n} \frac{\mu(d)}{d}.$$

Proof We have already established that $\phi(n)$ is multiplicative. For a prime p, there are p^{a-1} multiples of p in the interval $1 \leq m < p^a$, so that $\phi(p^a) = p^a - p^{a-1}$. Next, if $n = \prod p^a$ then $\phi(n) = \prod \phi(p^a) = \prod p^a \prod(1 - 1/p) = n \prod(1 - 1/p)$. The last equation in the theorem follows from applying Möbius inversion. □

From the theorem it follows that $\phi(n)/n$ is small when the product $\prod_{p|n}(1-1/p)$ is small. We shall see in Corollary 9.5 that

$$\frac{n}{\phi(n)} \ll \log\log n \quad \text{as} \quad n \to \infty. \tag{2.6}$$

For $m = p$, a prime, the residue classes $\mathbb{Z}/p\mathbb{Z}$, with the complete residue system $\{0, 1, 2, \ldots, p-1\}$ as representatives, form a *finite field* denoted by $\mathbb{F}_p$; there is no divisors of zero in the sense that $ab \equiv 0 \pmod{p}$ implies either $a \equiv 0 \pmod{p}$ or $b \equiv 0 \pmod{p}$. We shall see in Chap. 5 that the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ of the system of reduced residues $\{1, 2, \ldots, p-1\}$ is *cyclic*, with a certain generator.

Suppose that we wish to compute $\phi(n)$ for $1 \le n \le x$. It will not be efficient to evaluate $\phi(n)$ systematically and individually for each n. If n is composite then it has a prime divisor $p < n$, so that $\phi(n) < n - 1$; therefore $\phi(n) \le n - 1$ with equality if and only if n is a prime. Indeed, if $p|n$ then $\phi(pn) = p\phi(n)$, and if $p \nmid n$ then $\phi(pn) = \phi(p)\phi(n) = (p-1)\phi(n)$. We may therefore use the reduction scheme

$$\phi(pn) = \begin{cases} p\phi(n) & \text{if} \quad p \mid n, \\ (p-1)\phi(n) & \text{if} \quad p \nmid n; \end{cases} \tag{2.7}$$

the `Python` code, together with the plot of $\phi(n)$ for $1 \le n \le 1000$, are given here and in Fig. 2.1.

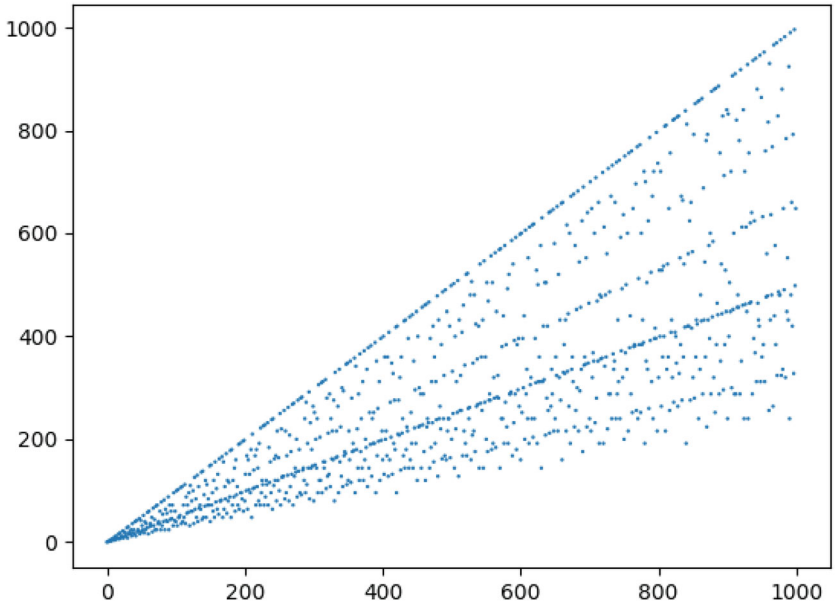


Fig. 2.1 $\phi(n)$, $1 \le n \le 1000$

```
# Program: Listing φ(n) for n < 1000.
bound=1000+1
phi=[]; phi.append(0); phi.append(1)
def f(n):
  if n==1: return n
  p=2
  while n%p>0 and p*p<=n: p+=1
  if p*p<=n:
    if n%(p*p)==0: return p*f(n//p)
    else: return (p-1)*f(n//p)
  return n-1
for n in range(2,bound+1): phi.append(f(n))
# print (phi)
```

2.4 Fermat-Euler Theorem

In a letter on 18 October 1640, Fermat wrote that a prime p divides $a^p - a$ for every a, that is $a^p \equiv a \pmod{p}$. The theorem (usually called Fermat's Little Theorem) was discovered from 'additive' considerations—each of the binomial coefficients

$$\binom{p}{k} = \frac{p!}{k!\,(p-k)!}, \qquad 1 \le k < p$$

is a multiple of p, so that $(x+1)^p \equiv x^p + 1 \pmod{p}$, and the required result follows by summing over $0 \le x < a$. The theorem was generalised by Euler in 1760:

Theorem 2.8 (Fermat-Euler) *Let $\phi(n)$ be Euler's totient function. Then*

$$a^{\phi(n)} \equiv 1 \pmod{n}, \quad \text{for all } a \text{ coprime with } n.$$

Proof Write $h = \phi(n)$; let $\gcd(a, n) = 1$ and $\{a_1, a_2, \ldots, a_h\}$ be a reduced residue system. Then $\{aa_1, aa_2, \ldots, aa_h\}$ also form a reduced system because $aa_i \equiv aa_j \pmod{n}$ implies $a_i \equiv a_j \pmod{n}$ by the multiplication of the reciprocal $\bar{a}$. Consequently $a^h a_1 a_2 \cdots a_h \equiv a_1 a_2 \cdots a_h \pmod{n}$, and the required result follows from the multiplication of the reciprocal of the factor $a_1 a_2 \ldots a_h$. □

The multiplicative consideration in the proof is more natural than the additive version considered by Fermat; indeed, it is said that the theorem and the proof mark the birth of group theory. However, the additive version has an important extension which will be considered in Sect. 3.8.

Table 2.3 Order of x (mod 13)

x	1	12	3, 9	5, 8	4, 10	2, 6, 7, 11
$\mathrm{ord}_p(x)$	1	2	3	4	6	12

2.5 Order of a (mod n)

Let a be coprime with n. By the *order* of a (mod n), we mean the least positive integer d such that $a^d \equiv 1 \pmod{n}$; it is denoted by $\mathrm{ord}_n(a)$. We also say that a belongs to the exponent d (mod n).

Let $k \in \mathbb{N}$ satisfies $a^k \equiv 1 \pmod{n}$. From the division process applied to k and $d = \mathrm{ord}_n(a)$, we may write $k = qd + r$, $0 \le r < d$, so that

$$a^r \equiv a^{k-qd} \equiv a^{-qd} \equiv (a^d)^{-q} \equiv 1 \pmod{n},$$

which implies $r = 0$, that is $d|k$; in particular $\phi(n)$ is a multiple of $\mathrm{ord}_n(a)$.

Example 2.2 Take $p = 13$, so that $\phi(p) = 12$. Then, each $x \in \{1, 2, \ldots, 12\}$, is a solution to $x^{12} \equiv 1 \pmod{13}$, and its order $\mathrm{ord}_p(x)$ is a divisor of 12.

Note that, in Table 2.3, the number of x with $\mathrm{ord}_p(x) = d$ is $\phi(d)$; in particular, for $d = \phi(p) = 12$, there are $\phi(d) = \phi(12) = 4$ values x, namely $x = 2, 6, 7, 11$, with $\mathrm{ord}_p(x) = 12$.

2.6 Decimals and Order of 10 (mod n)

Every $N \in \mathbb{N}$ has the representation $N = (d_0 d_1 \cdots d_k)_b$ in base $b > 1$, meaning that

$$N = d_0 b^k + d_1 b^{k-1} + \cdots + d_{k-1} b + d_k, \qquad d_i \in \{0, 1, 2 \ldots, b\}.$$

If x is a real number in $0 \le x < 1$ then we may express it by writing

$$x = (0{\cdot}a_1 a_2 \cdots)_b, \qquad a_n \in \{0, 1, 2 \ldots, b\},$$

defined to be the sum of the series $\sum_{n \ge 1} a_n / b^n$. Then any $x > 0$ can be given the representation

$$x = (d_0 d_1 \cdots d_k{\cdot}a_1 a_2 \cdots)_b = (d_0 d_1 \cdots d_k)_b + (0{\cdot}a_1 a_2 \cdots)_b.$$

The integers d_i, a_n are called the *digits* of x in base b, and the usual decimal representation corresponds to $b = 10$, for which we simply write

$$x = d_0 \cdots d_k{\cdot}a_1 a_2 \cdots .$$

If $x < 10^k$ it may be convenient to allow paddings with the digit 0 from the front in order to preserve $k+1$ digits for $\lfloor x \rfloor$ during manipulations of the number.

A decimal representation may have only finitely many nonzero digits, so that it has a terminating representation such as $x = 0{\cdot}a_1a_2\cdots a_n$, with a_n being the last nonzero digit, in which case x also has the infinite representation

$$x = 0{\cdot}a_1a_2\cdots a_{n-1}(b_n)999\cdots, \qquad b_n = a_n - 1,$$

because

$$\sum_{m>n} \frac{9}{10^m} = \frac{9}{10^{n+1}} \sum_{m\geq 0} \frac{1}{10^m} = \frac{9}{10^{n+1}} \cdot \frac{1}{1-1/10} = \frac{1}{10^n}.$$

We are familiar with the result of the operation of multiplication, or division, by 10^k amounting to shifting the 'decimal point' to the right, or to the left, by k places. When the context is clear, it will be convenient to use a_i for the digits both before and after the decimal point. Every real $x > 0$ has a representation which is unique if terminating representations are converted to infinite ones ending in the repeated digit 9, and padding with 0s from the front is removed—we omit the proof which involves the completeness axiom of $\mathbb{R}$.

From long division, we are familiar with the representations

$$\frac{1}{2} = 0{\cdot}5, \quad \frac{1}{4} = 0{\cdot}25, \quad \frac{1}{5} = 0{\cdot}2, \quad \frac{1}{8} = 0{\cdot}125,$$

whereas

$$\frac{1}{3} = 0{\cdot}333333\cdots = 0{\cdot}\bar{3}, \qquad \frac{1}{6} = 0{\cdot}166666\cdots = 0{\cdot}1\bar{6},$$

and many of us will also be familiar with

$$\frac{1}{7} = 0{\cdot}\overline{142857}, \quad \frac{1}{11} = 0{\cdot}\overline{09}, \quad \text{and} \quad \frac{1}{13} = 0{\cdot}\overline{076923};$$

here the bar denotes the *period* which repeats indefinitely. The use of long division that does not terminate may need justification, which is based on the definition of a periodic representation being given by the sum of a geometric progression; by this we mean that the value of $0{\cdot}\overline{00\cdots001}$, with ℓ digit under the bar, is given by

$$\sum_{m=0}^{\infty} \frac{1}{10^{m\ell}} = \frac{1}{1-1/10^\ell} = \frac{10^\ell}{10^\ell - 1}, \qquad \ell = 1, 2, \ldots.$$

Thus, we have

$$\frac{1}{7} = \frac{142857}{10^6 - 1} = \frac{142857}{10^6} \cdot \frac{10^6}{10^6 - 1} = \frac{1}{10^6} \sum_{n \geq 0} \frac{142857}{10^{6n}} = 0{\cdot}\overline{142857}.$$

A period need not start immediately after the decimal point—for example $\frac{1}{6} = 0{\cdot}1\bar{6}$; so periodic means being eventually periodic. By the period *length*, we mean the number ℓ of digits under the bar—however, any multiple of ℓ is also such a length, so we take the least one, of course. If the context is clear, the length is also called the period.

Lemma 2.1

(i) *Let $n = 2^k 5^\ell$. Then $1/n$ has a terminating representation.*
(ii) *Let $1 < m < n$ and $\gcd(10m, n) = 1$. Then the representation of m/n is periodic with period length the order of* 10 (mod n).

Proof

(i) If $n = 2^k$ then we write $5^k = a_1 a_2 \cdots a_k$, so that

$$\frac{1}{n} = \frac{5^k}{10^k} = \frac{a_1 a_2 \cdots a_k}{10^k} = 0{\cdot}a_1 a_2 \cdots a_k;$$

similarly if $n = 5^\ell$ and $2^\ell = b_1 b_2 \cdots b_\ell$ then

$$\frac{1}{n} = \frac{2^\ell}{10^\ell} = \frac{b_1 b_2 \cdots b_\ell}{10^\ell} = 0{\cdot}b_1 b_2 \cdots b_\ell.$$

If $n = 2^k 5^\ell$, then let $h = \min(k, \ell)$, so that n is either $10^h 2^{k-h}$ or $10^h 5^{k-\ell}$, and the required result follows from the previous two cases.

(ii) Let ℓ be the order of 10 (mod n), so that $10^\ell - 1 = qn$, $q \in \mathbb{N}$. Then $qm < qn < 10^\ell$, and we may write $qm = a_1 a_2 \cdots a_\ell$. We then have

$$\frac{m}{n} = \frac{qm}{10^\ell - 1} = \frac{a_1 a_2 \cdots a_d}{10^d} \cdot \frac{10^\ell}{10^\ell - 1} = \frac{1}{10^\ell} \sum_{k \geq 0} \frac{a_1 a_2 \cdots a_\ell}{10^{k\ell}} = 0{\cdot}\overline{a_1 a_2 \cdots a_\ell},$$

the required periodic representation with period length ℓ.

□

Theorem 2.9 *A necessary and sufficient condition for the representation of x in $0 < x < 1$ to be either terminating, or periodic, is that x be rational.*

Proof Necessity. It is clear that a terminating representation is rational. Suppose next that $x = 0{\cdot}a_1a_2\cdots a_k\overline{a_{k+1}a_{k+2}\cdots a_{k+\ell}}$, that is $x = X + Y_1 + Y_2 + \cdots + Y_\ell$, where

$$X = \sum_{m=1}^{k} \frac{a_m}{10^m}, \quad Y_j = \frac{a_{k+j}}{10^{k+j}} \sum_{m=0}^{\infty} \frac{1}{10^{m\ell}} = \frac{a_{k+j}}{10^{k+j}} \cdot \frac{10^\ell}{10^\ell - 1}, \quad j = 1, 2, \ldots, \ell.$$

Thus X and each Y_j is rational, so that x is rational.

Sufficiency. Suppose that $x = m/n$ is rational, where $\gcd(m, n) = 1$. Let $n = 2^k 5^\ell N$, with $\gcd(10, N) = 1$, and write $2^k 5^\ell m = a_1a_2\cdots a_j$, so that

$$x = \frac{M}{N}, \qquad \text{where} \quad M = \frac{2^\ell 5^k m}{10^{k+\ell}} = \frac{a_1a_2\cdots a_j}{10^{k+\ell}}.$$

It follows from the lemma that the representation of x is periodic, with the period length being the order of 10 (mod N). □

Example 2.3 Let (i) $(0{\cdot}011010100010\ldots)_{10}$ where the digit a_n is 1 if n is prime and 0 otherwise, and (ii) $(0{\cdot}2357111317192329\ldots)_{10}$ where the sequence of digits is formed by the increasing sequence of primes. Then both numbers are irrational, because there is no periodicity in the expansions.

Example 2.4 Determine the decimal representation of $x = 47/1400$.

Long division gives $47/1400 = 0{\cdot}033\overline{571428}$, but we can also follow the proof of the theorem to deliver

$$\frac{47}{1400} = \frac{1}{1000} \cdot \frac{5\times 47}{7} = \frac{1}{1000}\left(33 + \frac{4}{7}\right) = \frac{1}{1000}(33{\cdot}\overline{571428}) = 0{\cdot}033\overline{571428}.$$

Periods and Divisibility Tests

Let $p > 5$ be a prime number, and d the order of 10 (mod p), so that $d|p - 1$. Then, for $1 \le m < p$, the period length of the decimal representation of m/p is d. If $d = p - 1$, the representation of m/p has 'full-period'; for example there is full-period for $p = 7$, but not for $p = 13$. It is not known if there are infinitely primes p for which there is full-period for the representation of m/p; see Chap. 5 on primitive roots.

We say something about the relationship between the various periods of m/p. Taking $p = 7$, we find that

$$\frac{1}{7} = 0{\cdot}\overline{142857}, \quad \frac{2}{7} = 0{\cdot}\overline{285714}, \quad \frac{3}{7} = 0{\cdot}\overline{428571},$$
$$\frac{4}{7} = 0{\cdot}\overline{571428}, \quad \frac{5}{7} = 0{\cdot}\overline{714285}, \quad \frac{6}{7} = 0{\cdot}\overline{857142},$$

which shows that each period is a cyclic permutations of $(1, 4, 2, 8, 5, 7)$. For $p = 13$, we find that

$$\frac{1}{13} = 0{\cdot}\overline{076923}, \quad \frac{3}{13} = 0{\cdot}\overline{230769}, \quad \frac{4}{13} = 0{\cdot}\overline{307692},$$
$$\frac{9}{13} = 0{\cdot}\overline{692307}, \quad \frac{10}{13} = 0{\cdot}\overline{769230}, \quad \frac{12}{13} = 0{\cdot}\overline{923076},$$
$$\frac{2}{13} = 0{\cdot}\overline{153846}, \quad \frac{5}{13} = 0{\cdot}\overline{348615}, \quad \frac{6}{13} = 0{\cdot}\overline{461538},$$
$$\frac{7}{13} = 0{\cdot}\overline{538641}, \quad \frac{8}{13} = 0{\cdot}\overline{615384}, \quad \frac{11}{13} = 0{\cdot}\overline{846153},$$

which shows that there are two distinct periods, both with length 6; the 12 fractions m/p are separated into the two classes $m = 1, 3, 4, 9, 10, 12$ and $m = 2, 5, 6, 7, 8, 11$, and within a class, the periods are cyclic permutations of each other. This is true in general: if 10 is a primitive root (mod p) then all $p - 1$ fractions m/p share the same period, with cyclic permutations of each other. If $p - 1 = \ell k$, where the proper divisor ℓ is the order of 10 (mod p), then there are k distinct periods, and the $p - 1$ fractions m/p fall into k classes with ℓ members in each class, and the period is a cyclic permutation of other periods in the class. The proof is not difficult, but it belongs properly to group theory, wherein useful tools are available.

The division process, applied to n and d, delivers the quotient $q = \lfloor n/d \rfloor$ and the remainder $r = n - qd$. In decimals, there are familiar party tricks that people know, but perhaps not all of them can explain why such tricks work, or how they can be extended.

Let N be a modestly large number given in its decimal representation $a_1 a_2 \cdots a_n$. Divisibility by 2 or 5 is trivial, and many people are aware of the fact that N is divisible by 3 if and only if the sum of its digits a_i is divisible by 3, and indeed the same also applies to divisibility by 9. The reason behind this is simple: from $10^k \equiv 1 \pmod 9$, and hence also (mod 3), it follows that

$$N \equiv a_1 10^n + a_2 10^{n-1} + \cdots + a_n \equiv a_1 + a_2 + \cdots + a_n \pmod 9,$$

so that we can say even more, namely that the remainder of N when divided by 9 is the same as that of $\sum a_i \pmod 9$. Indeed, before the advent of computing machines, the trick was often used to check sums.

Fewer people seem to know about divisibility by 7, by 11 and by 13, but the extension of the trick is not that much more complicated. We make use of the interesting fact that $7\times11\times13 = 1001$, so that

$$10^{3k} \equiv 1000^k \equiv (-1)^k \,(\mathrm{mod}\ p), \qquad p = 7, 11, 13, \quad k = 1, 2, \ldots .$$

An example will be enough to explain the extended trick.

Example 2.5 Find $N \pmod p$ when $N = 746665091182$ and $p = 7, 11, 13$.

We consider

$$\begin{aligned} N &= 746\times10^9 + 665\times10^6 + 91\times10^3 + 182 \\ &\equiv -746 + 665 - 91 + 182 \equiv 10 \,(\mathrm{mod}\ p), \end{aligned}$$

so that $N \equiv 3, 10, 10 \pmod p$, for $p = 7, 11, 13$.

If one is good with mental arithmetic, and wants to show off at parties, the further extension to $p = 17$ depends on $10^{8k} \equiv (-1)^k \pmod{17}$.

Missing Digit Problem

Consider the set of numbers with the digit 7 missing in each of their decimal representations, that is the set

$$\mathcal{M}_7 = \{n : n \in \mathbb{N},\ n = (d_0 d_1 \cdots d_k)_{10},\ d_i \neq 7\}.$$

Intuition may suggest that some $\frac{1}{10}$ of the numbers are missing in $\mathcal{M}_7$, so that some $\frac{9}{10}$ of the numbers are left in $\mathcal{M}_7$. Thus, letting

$$M_7(N) = \sum_{1\le n\le N,\ n\in\mathcal{M}_7} 1$$

count the numbers in $\mathcal{M}_7$ with $n \le N$, a careless consideration leads one to expect that $M_7(N) \approx \frac{9}{10}N$ for large N. Well, such intuition is completely wrong, because almost all the numbers are missing in $\mathcal{M}_7$ in the sense of the following theorem.

Theorem 2.10 *Let $\mathcal{M}_d$ denote the set of numbers with the digit $d \in \{0, 1, 2, \ldots, 9\}$ missing in their decimal representations. Then the counting function of $\mathcal{M}_d$ satisfies $M_d(N) = o(N)$ as $N \to \infty$, and the series $\sum_{n\in\mathcal{M}_d} 1/n$ is convergent.*

Proof Fix $d \in \{1, 2, \ldots, 9\}$, and count the numbers $n \in \mathcal{M}_d$ in the interval $10^{\ell-1} \le n < 10^\ell$. Each such number has the representation $n = a_1 a_2 \cdots a_\ell$, with $a_i \neq d$ and $a_1 > 0$, so that there are $8\times9^{\ell-1}$ such numbers. If $d = 0$ then there are 9^k of them.

For $N > 1$, let $10^{k-1} \le N < 10^k$. Then, as $N \to \infty$, so that $k \to \infty$, we have

$$M_d(N) \le 9 + 9^2 + \cdots 9^k < 9^{k+1} \le 9^2\Big(\frac{9}{10}\Big)^{k-1} N = o(N),$$

and

$$\sum_{n\in\mathcal{M}_d} \frac{1}{n} = \sum_{k\ge 1} \sum_{10^{k-1}\le n<10^k,\, n\in\mathcal{M}_d} \frac{1}{n} < \sum_{k\ge 1} \frac{9^k}{10^{k-1}} = 90.$$

□

2.7 Liouville Function $\lambda(n)$

The Liouville lambda function, denoted by $\lambda(n)$, is defined by

$$\lambda(n) = (-1)^{\Omega(n)}.$$

Since $\Omega(n)$ is completely additive, $\lambda(n)$ is completely multiplicative. It is closely related to the Möbius function $\mu(n)$; thus, on writing $n = a^2 b$, where b is squarefree, we find that $\lambda(n) = \mu(b)$. Also,

$$\sum_{d|n} \lambda(d) = \begin{cases} 1 & \text{if } n = m^2 \text{ is a square,} \\ 0 & \text{otherwise,} \end{cases}$$

the characteristic function on the set of squares. The proof amounts to checking that the value of the sum $1 + \lambda(p) + \lambda(p^2) + \cdots + \lambda(p^a)$ is 1 or 0, depending whether a is even or odd. On applying Möbius inversion, we find that

$$\lambda(n) = \sum_{d^2|n} \mu\Big(\frac{n}{d^2}\Big).$$

For large N, the behaviour of

$$\sum_{n\le N} \mu(n) \quad \text{and} \quad \sum_{n\le N} \lambda(n)$$

poses some interesting questions which we shall consider later. In particular, it has been established that either of $\sum_{n\le N} \mu(n) = o(N)$ or $\sum_{n\le N} \lambda(n) = o(N)$, as

$N \to \infty$, is 'equivalent' to the *prime number theorem*, which states that

$$\pi(x) \sim \frac{x}{\log x} \quad \text{as} \quad x \to \infty.$$

Indeed, there is a recent elementary proof [81] of $\sum_{n \le N} \lambda(n) = o(N)$ as $N \to \infty$, which constitutes a new elementary proof of the prime number theorem.

2.8 Divisor Function $d(n)$

We say more on the divisor function $d(n)$. For a prime p, we have $d(p^a) = a + 1$, so that the range of $d(n)$ is the whole of $\mathbb{N}$. Note that $d(n)$ depends only on the exponents of the primes $p|n$, and not on p; we recall that if $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, where p_i are distinct primes, then

$$\omega(n) = k, \quad \Omega(n) = a_1 + a_2 + \cdots + a_k, \quad d(n) = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1).$$

It follows from $2 \le a + 1 \le 2^a$ that $2^{\omega(n)} \le d(n) \le 2^{\Omega(n)}$, and that $d(n)$ is odd if and only if n is a square. The values for $d(n)$ in $1 \le n \le 16$ are set out in Table 2.4.

A longer list can be obtained by a scheme similar to the sieve of Eratosthenes. Choose a suitable value for N and create a list `d[n]` of zeros for $1 \le n \le N$. For each such n, run `sift` through with `sift+=1`, and update `d[sift]+=1`. The resulting list `d[n]` will be the values of the divisor function up to N. The following is the `Python` code for the scheme, and the plot for $d(n)$ with $1 \le n \le 1000$ in Fig. 2.2.

```
N=10000; d=[]
for n in range(N+1): d.append(0)
for n in range(1,N+1):
    sift=0
    while sift+n<N+1:
        sift+=n; d[sift]+=1
```

For a chosen N, we can determine the largest value $K = K(N)$ taken by $d(n)$ by finding the number $s_k = s_k(N)$ of solutions to $d(n) = k$ for each k in $1 \le k \le K$. We can also evaluate the average value of $d(n)$ over the interval.

For $N = 10000$, we find that $K = 64$, taken by $d(n)$ at $n = 7560 = 2^3 3^3 5 \cdot 7$, and the values for s_k $(1 \le k \le 24)$ are given in Table 2.5. The determination of the

Table 2.4 The divisor function $d(n)$

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$d(n)$	1	2	2	3	2	4	2	4	3	4	2	6	2	4	4	5

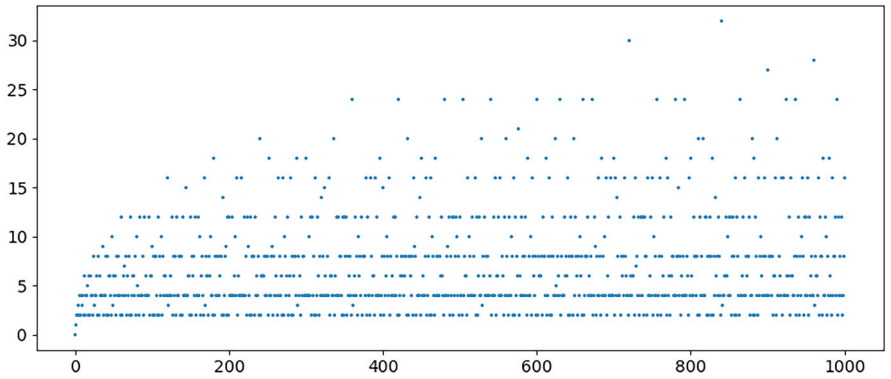


Fig. 2.2 $d(n)$, $1 \le n \le 1000$

Table 2.5 The number of solutions to $d(n) = k$ in $1 \le k \le 24$

k	1	2	3	4	5	6	7	8	9	10	11	12
s_k	1	1229	25	2608	4	764	2	2114	32	150	1	1040
k	13	14	15	16	17	18	19	20	21	22	23	24
s_k	1	41	15	800	0	159	0	157	5	3	0	449

average value of $d(n)$ over $1 \le n \le N$ amounts to the evaluation of the sum $\sum d(n)$ over the interval, which can be obtained from the list; for $N = 10000$, the average value is $93668/10000 = 9{\cdot}3668$.

The small sample already offers some clues to what happens when N is large. For example, the values of s_q, with q being a prime, can be determined without the list. This is because if $d(n) = q$ then $n = p^{q-1}$ for primes p, so that $s_q = \pi(N^{1/(q-1)})$, which explains the entries $s_q = 1229, 25, 4, 2, 1, 1$ for $2 \le q \le 13$ in the table, and that $s_q = 0$ for $q = 17, 19, 23$ because $2^{16} > N$.

Dirichlet's Divisor Problem

Let $x \in \mathbb{N}$, and consider the sum of $d(n)$ over $1 \le n \le x$ by setting

$$D(x) = \sum_{n \le x} d(n) = d(1) + d(2) + \cdots + d(x).$$

Since $d(n) = \sum_{a|n} 1 = \sum_{ab=n} 1$ is the number of ways of writing n as a product ab, we have

$$D(x) = \sum_{ab \le x} 1,$$

a double sum over $a, b \in \mathbb{N}$ with $ab \le x$. Summing over a first, we find that

$$D(x) = \sum_{a \le x} \sum_{b \le x/a} 1 = \sum_{a \le x} \left\lfloor \frac{x}{a} \right\rfloor,$$

which shows that $D(x)$ can be evaluated without knowing the values for $d(n)$. The expression is much easier to compute, and it can be used to check the results of our more elaborate listing method.

The expression for $D(x)$ also gives an inkling of the size of the sum. Replacing the integer part $\lfloor x/a \rfloor$ by x/a will incur an error which is at most 1, so that

$$D(x) = \sum_{a \le x} \frac{x}{a} + O(x) = x \sum_{a \le x} \frac{1}{a} + O(x) = x \log x + O(x).$$

We may say that, for large x, the average value of $d(n)$ over $1 \le n \le x$ is $\log x$.

As Dirichlet showed in 1849, we can do much better exploiting the symmetry of the sum $\sum_{ab \le x} 1$ by combining the two cases $a < b$ and $b < a$ together, and to take the case $a = b$ separately. Setting $M = \lfloor \sqrt{x} \rfloor$, we then have

$$\begin{aligned} D(x) &= 2 \sum_{ab \le x,\, a < b} 1 + \sum_{a^2 \le x} 1 = 2 \sum_{a < \sqrt{x}} \sum_{a < b \le x/a} 1 + \sum_{a \le \sqrt{x}} 1 \\ &= 2 \sum_{a \le M} \left(\left\lfloor \frac{x}{a} \right\rfloor - a \right) + M = 2 \sum_{a \le M} \left\lfloor \frac{x}{a} \right\rfloor - M^2, \end{aligned} \tag{2.8}$$

since $2 \sum_{a \le M} a = M(M+1)$; an alert reader will have noticed that, in replacing the condition $a < \sqrt{x}$ with $a \le M$, there is an extra term corresponding to $a = M$ in the case $x = M^2$, but the term concerned is $M^2/M - M = 0$.

The new formula gives a much more efficient method to evaluate $D(x)$, because the new sum is over the much shorter interval $1 \le a \le M$. Again, it can be used to check the previous calculations, but more to the point is that $D(x)$ can now be evaluated with a much larger x. Indeed, the new expression should enable us to have an asymptotic formula for $D(x)$ with a much smaller error term, for which we need a good estimate for the partial sum of the harmonic series. We shall see later (Sect. 9.2) that there is a constant, called Euler's constant γ, such that

$$H(n) = \sum_{1 \le m \le n} \frac{1}{m} = \log n + \gamma + O\Big(\frac{1}{n}\Big), \qquad n \to \infty.$$

Theorem 2.11 (Dirichlet) *As $x \to \infty$, we have*

$$\sum_{n \le x} d(n) = x \log x + (2\gamma - 1)x + O(\sqrt{x}).$$

Proof Writing $M = \lfloor\sqrt{x}\rfloor = \sqrt{x} - \theta = \sqrt{x}(1 - \theta/\sqrt{x})$, where $0 \le \theta < 1$, we find that, as $x \to \infty$,

$$\log M = \log\sqrt{x} + \log\Big(1 - \frac{\theta}{\sqrt{x}}\Big) = \frac{1}{2}\log x + O\Big(\frac{1}{\sqrt{x}}\Big)$$

and that $M^2 = x + O(\sqrt{x})$. We then have

$$\begin{aligned} D(x) &= 2\sum_{n\le M}\Big(\frac{x}{a} - \Big\{\frac{x}{a}\Big\}\Big) - x + O(\sqrt{x}) \\ &= 2xH_M + O(M) - x + O(\sqrt{x}) \\ &= 2x\left(\log M + \gamma + O\Big(\frac{1}{M}\Big)\right) - x + O(\sqrt{x}) \\ &= x\log x + (2\gamma - 1)x + O(\sqrt{x}), \end{aligned}$$

as required. □

So, a more accurate approximation to the average value of $d(n)$ over $1 \le n \le x$ is $\log x + 2\gamma - 1$; at $x = 10000$, the approximation is 9.36, which is quite good compare to the exact value 9.3668.

Averages need not be restricted to the interval $1 \le n \le x$, and we say that the average value of $d(n)$ is $\log n$, meaning that the average is over a reasonably long interval containing n. For example, by considering $D(x + h) - D(x)$, the theorem can be used to show that the average value of $d(n)$ in $x \le n \le x + h$ is $\log(x + h)$, provided that $h \ge \sqrt{x}$.

The geometric interpretation for the sum $D(x) = \sum_{ab\le x} 1$ is the number of lattice points (u, v) in the region on or below the rectangular hyperbola $uv = x$ in the first quadrant of the uv-plane. The arithmetic in the above amounts to pairing the lattice points in the two symmetric regions being separated by the line $v = u$.

Writing $\Delta(x) = D(x) - x\log x - (2\gamma - 1)x$, the *Dirichlet divisor problem* is the determination of the infimum θ_0 of the set of numbers θ for which $\Delta(x) = O(x^\theta)$ is admissible. Thus, the result $\theta_0 \le \frac{1}{2}$ was established by Dirichlet, and $\theta_0 \le \frac{1}{3}$ by Voronoi (1903) and Sierpiński (1906). After many improvements, the current best result is $\theta_0 \le \frac{517}{1648} = 0{\cdot}3137\ldots$ by Bourgain and Watt (2017). The lower bound $\theta_0 \ge \frac{1}{4}$ was established in 1916 independently by Hardy and Landau, and it is conjectured that $\theta_0 = \frac{1}{4}$. See Note 3, and also Note 10.6.

We use (2.8) to compute $D(x)$ for $x = 10^k$, with $k = 15, 16, 17, 18$. The results are given in Table 2.6, with $\theta = (\log|\Delta(x)|)/(\log x)$ being truncated to two decimal places, and the elapsed computing time given in seconds. It is known that $\Delta(x)$ changes signs infinitely often as $x \to \infty$, and it will be interesting to find a value for x_0 with $\Delta(x_0) < 0$.

Table 2.6 Value of $D(x)$

x	$D(x)$	$\lfloor \Delta(x) \rfloor$	θ	Time
10^{15}	34693207724724246	10496	0·26	10
10^{16}	369957928177109416	31424	0·28	31
10^{17}	3929837791070240368	55808	0·27	99
10^{18}	41600963003695964400	73728	0·27	323

2.9 Dirichlet Convolution

The result in Theorem 2.7, namely that

$$\phi(n) = \sum_{d|n} \mu(d)\frac{n}{d},$$

suggests that we can have a more general method of defining a new function from that given in (2.1).

Definition 2.1 Let f and g be arithmetic functions. The *Dirichlet convolution* $f * g$ is the function h defined by the equation

$$h(n) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right).$$

By (2.2) the equation can also be written as $h(n) = \sum_{ab=n} f(a)g(b)$, which shows that the operation $*$ on the class of arithmetic functions is commutative. That it is also associative can be shown by observing that both $((f * g) * k)(n)$ and $(f * (g * k))(n)$ are given by $\sum_{abc=n} f(a)g(b)k(c)$.

It will be useful to introduce the following three functions:

Definition 2.2 The *unit, identity*, and *fixed*, functions u, I, and N are defined by

$$u(n) = 1, \qquad \text{for all } n \in \mathbb{N};$$

$$i(n) = \begin{cases} 1, & \text{if } n = 1, \\ 0, & \text{if } n > 1; \end{cases}$$

$$N(n) = n, \qquad \text{for all } n \in \mathbb{N}.$$

We call i the identity function because $f * i = f$, whereas $f * u$ is the usual sum function of f; in particular, the divisor function is just $i * i$, and Euler's function is $\phi = \mu * N$. Also, (2.4) and Theorem 2.6 become $\mu * u = i$ and $\phi * u = N$. Moreover, Möbius inversion, that is, Theorem 2.4 becomes: If $F = f * u$ then $f = \mu * F$, and the argument in the proof of Theorem 2.1 can be used to show that the convolution of multiplicative functions is multiplicative. Nothing new has been

added here—the introduction of u, i, N only shows the relevance of the notion of Dirichlet convolution.

If f is an arithmetic function with $f(1) \neq 0$ then there is a unique arithmetic function f^{-1}, called the *Dirichlet inverse* of f, such that $f * f^{-1} = i$. We omit the proof, which uses induction on n in the construction of the values of $f^{-1}(n)$.

If f and g are multiplicative then so is $f * g$; indeed, if f and $f * g$ are multiplicative then so is g. In particular, if f is multiplicative then so is f^{-1}. Thus, under the operation $*$ for Dirichlet convolution, the set of multiplicative functions form a subgroup of the group of multiplicative functions; we omit the proofs, which are not difficult.

2.10 Mangoldt Function $\Lambda(n)$

The Mangoldt function $\Lambda(n)$ is defined by

$$\Lambda(n) = \begin{cases} \log p, & \text{if } n = p^m \text{ for some prime } p \text{ and some } m \geq 1, \\ 0, & \text{otherwise.} \end{cases} \tag{2.9}$$

The first few values are set out in Table 2.7.

Table 2.7 Mangoldt function $\Lambda(n)$

n	1	2	3	4	5	6	7	8	9	10
$\Lambda(n)$	0	$\log 2$	$\log 3$	$\log 2$	$\log 5$	0	$\log 7$	$\log 2$	$\log 3$	0

Theorem 2.12 *We have* $\log = \Lambda * i$, *that is*

$$\log n = \sum_{d|n} \Lambda(d).$$

Corollary 2.2 *We have* $\Lambda = \mu * \log$, *so that*

$$\Lambda(n) = \sum_{d|n} \mu(d) \log\left(\frac{n}{d}\right) = -\sum_{d|n} \mu(d) \log d.$$

Proof As it is usual, an empty sum, and an empty product, are given the values 0, and 1, respectively. If $n = \prod_{1 \leq k \leq \ell} p^{a_k}$ is given by its standard factorisation, then

$$\log n = \sum_{1 \leq k \leq \ell} a_k \log p_k.$$

On the other hand, from the definition of $\Lambda(n)$, we have

$$\sum_{d|n} \Lambda(d) = \sum_{1\le k\le \ell} \sum_{1\le m\le a_k} \Lambda(p_k^m) = \sum_{1\le k\le \ell} \sum_{1\le m\le a_k} \log p_k = \sum_{1\le k\le \ell} a_k \log p_k.$$

The theorem is established. On application of Möbius inversion we find that

$$\Lambda(n) = \sum_{d|n} \mu(d) \log\left(\frac{n}{d}\right) = \log n \sum_{d|n} \mu(d) - \sum_{d|n} \mu(d) \log d,$$

where $\sum_{d|n} \mu(d) = 0$, except when $n = 1$, for which $\log n = 0$. The corollary therefore follows. □

The function $\Lambda(n)$ plays an important role in the study of the distribution of primes. In sums over prime numbers p, we sometimes replace $\log p$ by $\Lambda(n)$, and consider the more 'convenient and smoother' sum over the integers n instead. The reader may ask: Why not just take $f(n) = \log n$ when $n = p$ is a prime, and 0 otherwise? Well, that would just be relabelling, which is hardly 'smoothing' things over. At this early stage, we can only give a rather unsatisfactory answer, namely that $\Lambda(n)$ possesses an interesting analytic property which can be exploited in the modified sum—see Sect. 11.1. There is, of course, a price to be paid for the manoeuvre, namely the difference between the two sums. Fortunately the price is often relatively low because prime powers are literally 'few and far between' among the integers. We shall see such action when tackling problems involving prime numbers.

2.11 Residue Classes and Additive Characters

The additive group of residue classes (mod m) can be given a concrete representation so that actions on the group may become more tangible. Let $\omega = e^{2\pi i/m}$, a complex mth root of unity, that is a solution of $\omega^m = 1$. Consider the map

$$C_r \mapsto \omega_r; \quad \omega_r = \omega^r = e^{2\pi ir/m}, \qquad r = 0, 1, 2, \ldots, m.$$

Since $e^{2\pi in} = 1$ for $n \in \mathbb{Z}$, integers belonging to the same residue class are represented by the same root of unity, and if $r + s \equiv t \pmod m$ then $\omega_r\omega_s = \omega_t$, so that the abstract notion of the addition of residue classes has the more concrete interpretation of the product of complex roots of unity. Results on congruences can now be interpreted in, or derived from, those on roots of unities, the underlying reason for the importance of exponential, or trigonometric, sums in the study of number theory.

Besides taking $\omega = e^{2\pi i/m}$, we may also take any fixed a coprime with m and use $\omega^a = e^{2\pi ia/m}$ instead, that is $C_r \mapsto e^{2\pi ira/m}$; this shows that there are $\phi(m)$

different representations of the residue classes. There are no other representations with complex numbers that have the required properties in the above. For if we take any $\eta_r \in \mathbb{C}$ to represent C_r then, from $mr \equiv 0 \pmod{m}$, we need to have $\eta_r^m = \eta_0$. Now $\eta_0^2 = \eta_0$, and if $\eta_0 = 0$ then $\eta_r = 0$, and all the representatives are zero. Thus $\eta_0 = 1$ so that η_r must be an mth root of unity, that is $\eta_r = \omega^a$ for some a.

The useful result

$$\int_0^1 e^{2\pi i n x} dx = \begin{cases} 1, & \text{if } n = 0, \\ 0 & \text{if } n \neq 0, \end{cases}$$

which is related to the orthogonal relations associated with the trigonometric functions $\sin nx$ and $\cos nx$, also has the following discrete version. The sum of the arithmetic progression $1 + z + z^2 + \cdots + z^{m-1}$ is m, or $(1 - z^m)/(1 - z)$, depending on whether $z = 1$, or not. Thus if $\omega_n = e^{2\pi i n/m}$ then

$$\sum_{0 \le a < m} \omega_n^a = \begin{cases} m, & \text{if } m \mid n, \\ 0 & \text{if } m \nmid n. \end{cases}$$

Now let $w_0, w_1, \ldots, w_{q-1}$ be any q distinct complex numbers. Then there are uniquely determined complex numbers $a_0, a_1, \ldots, a_{q-1}$ such that

$$w_m = \sum_{0 \le n < q} a_n e^{2\pi i m n/q}, \qquad m = 0, 1, \ldots, q-1. \tag{2.10}$$

This follows from the application of the interpolation formula to the polynomial

$$P(z) = \sum_{0 \le n < q} a_n z^n,$$

with degree $q - 1$, at the points $(z_m, P(z_m)) = (z_m, w_m)$, for $m = 0, 1, \ldots, q - 1$. Indeed, the values for a_n are given by the formulae

$$a_n = \frac{1}{q} \sum_{0 \le m < q} w_m e^{-2\pi i m n/q}, \qquad n = 0, 1, \ldots, q-1. \tag{2.11}$$

This is because, from formula (2.10) for w_m, the sum in (2.11) is given by

$$\sum_{0 \le m < q} \left(\sum_{0 \le j < q} a_j e^{2\pi i m j/q} \right) e^{-2\pi m n/q} = \sum_{0 \le j < q} a_j \sum_{0 \le m < q} e^{2\pi i m (j-n)/q},$$

and the value of the inner sum is q or 0, corresponding to $j = n$ or $j \neq n$.

In the language of the *finite Fourier transform* of a periodic function $f(n)$ with period q, the transform $\hat{f}$ is given by

$$\hat{f}(m) = \frac{1}{q} \sum_{0 \le n < q} f(n) e^{-2\pi i m n/q}.$$

To recover $f(n)$ we make use of

$$\sum_{0 \le a < q} e^{2\pi a(m-n)/q} = \begin{cases} q, & \text{if} \quad m = n, \\ 0, & \text{otherwise,} \end{cases}$$

so that

$$\begin{aligned} f(n) &= \sum_{0 \le m < q} \frac{f(m)}{q} \sum_{0 \le a < q} e^{2\pi i a(m-n)/q} \\ &= \sum_{0 \le a < q} \frac{e^{-2\pi i a n/q}}{q} \sum_{0 \le m < q} f(m) e^{2\pi i a m/q} \\ &= \sum_{0 \le a < q} e^{-2\pi i a n/q} \hat{f}(-a) \\ &= \sum_{0 \le a < q} \hat{f}(a) e^{2\pi i a n/q}. \end{aligned}$$

The orthogonality property of $e^{2\pi i a/q}$ for $a = 0, 1, \ldots, q-1$ also delivers the following useful relation between $f(n)$ and its transform $\hat{f}(m)$:

$$\begin{aligned} \sum_{0 \le n < q} |f(n)|^2 &= \sum_{0 \le n < q} f(n) \overline{f(n)} \\ &= \sum_{0 \le n < q} \sum_{0 \le a < q} \hat{f}(a) e^{2\pi i a n/q} \sum_{0 \le b < q} \overline{\hat{f}(b)} e^{-2\pi i b n/q} \\ &= \sum_{0 \le a, b < q} \hat{f}(a) \overline{\hat{f}(b)} \sum_{0 \le n < q} e^{2\pi i n(a-b)/q} \\ &= q \sum_{0 \le a < q} \hat{f}(a) \overline{\hat{f}(a)} \\ &= q \sum_{0 \le m < q} |\hat{f}(m)|^2. \end{aligned}$$

We have shown that a characteristic function of an arithmetic progression (mod q) can be expressed as a linear combination of the orthogonal sequence

$e^{2\pi ia/q}$; for this reason such functions are sometimes called the *additive characters* (mod q): they can be considered as homomorphisms from the additive group $(\mathbb{Z}/q\mathbb{Z})^+$ of integers (mod q) to the multiplicative group $\mathbb{C}^\times$ of non-zero numbers in $\mathbb{C}$.

In problems of a multiplicative nature, particularly those concerned with the distribution of prime numbers in arithmetic progressions, we shall require a similar characteristic function of the reduced residue class (mod q). The corresponding functions are called *Dirichlet characters* which are completely multiplicative functions; their construction in Chap. 12 is much more complicated.

2.12 Gauss Sum and Ramanujan Sum

For a proper study of number theory, sooner or later, we will need to be familiar with the notion of a Gauss sum—a finite periodic sum of exponentials. We shall consider three types of such sums, although two of them are actually the same, and the third (see Sect. 12.7) one is just a generalisation, or extension.
For convenience, we write

$$\mathrm{e}(t) = \exp(2\pi it) = e^{2\pi it}, \quad t \in \mathbb{R};$$

for the present, we consider only for $t \in \mathbb{Q}$, so that $t = m/q$, where $m, q \in \mathbb{Z}$ and $q > 0$. By periodicity of the exponential function we have

$$\mathrm{e}\Big(\frac{m}{q}\Big) = \mathrm{e}\Big(\frac{m'}{q}\Big), \qquad \text{if} \quad m \equiv m' \pmod{q}.$$

A *quadratic Gauss sum* (see also Sect. 4.5) is the expression

$$S(m,q) = \sum_{0 \le h < q} \mathrm{e}\Big(\frac{h^2 m}{q}\Big); \tag{2.12}$$

note that the summation condition may be replaced by h running over any complete set of residues (mod n). We shall consider Gauss sums in more detail later; meanwhile we remark that if q and q' are coprime then

$$S(m, qq') = S(mq', q)S(mq, q').$$

To see this, we let $H = hq' + h'q$, where q, q' run through complete residue systems (mod q), (mod q'), respectively, so that H runs through a complete residue system (mod qq'). From $H^2 = (hq' + h'q)^2 \equiv h^2q'^2 + h'^2q^2 \pmod{qq'}$, we find that

$$\begin{aligned} S(mq', q)S(mq, q') &= \sum_{0\le h<m} \mathrm{e}\Big(\frac{h^2mq'}{q}\Big) \sum_{0\le h'<q} \mathrm{e}\Big(\frac{h'^2mq}{q'}\Big) \\ &= \sum_{0\le h<m} \sum_{0\le h'<q} \mathrm{e}\Big(\frac{m(h^2q'^2 + h'^2q^2)}{qq'}\Big) \\ &= \sum_{0\le H<qq'} \mathrm{e}\Big(\frac{mH^2}{qq'}\Big) = S(m, qq'). \end{aligned}$$

By a *Ramanujan sum* we mean the expression

$$c_q(m) = \sum_{\substack{0\le h<q \\ \gcd(h,q)=1}} \mathrm{e}\Big(\frac{hm}{q}\Big);$$

again, the summation condition may be replaced by h running over any reduced set of residues (mod q), and the same argument used in the above shows that

$$c_{qq'}(m) = c_q(m)c_{q'}(m), \qquad \text{if} \quad \gcd(q, q') = 1.$$

Thus we find that

$$\begin{aligned} c_q(m)c_{q'}(m) &= \sum_{\substack{0\le h<q \\ \gcd(h,q)=1}} \sum_{\substack{0\le h'<q' \\ \gcd(h',q')=1}} \mathrm{e}\Big(m\Big(\frac{h}{q} + \frac{h'}{q'}\Big)\Big) \\ &= \sum_{\substack{0\le h<q \\ \gcd(h,q)=1}} \sum_{\substack{0\le h'<q' \\ \gcd(h',q')=1}} \mathrm{e}\Big(\frac{m(hq' + h'q)}{qq'}\Big) = c_{qq'}(m). \end{aligned}$$

One useful and important fact concerning the Ramanujan sum is that

$$c_q(1) = \mu(q),$$

the Möbius function. More generally, the sum is given by

$$c_q(m) = \sum_{d\,|\,\gcd(m,q)} d\,\mu\Big(\frac{q}{d}\Big). \tag{2.13}$$

To see this we rewrite the sum for

$$J(q) = \sum_{0 \le h < q} \mathrm{e}\Big(\frac{hm}{q}\Big),$$

as a double sum, that is

$$J(q) = \sum_{d|q} \sum_{\substack{0 \le h < q \\ \gcd(h,q)=d}} \mathrm{e}\Big(\frac{hm}{q}\Big) = \sum_{d|q} \sum_{\substack{0 \le h' < q' \\ \gcd(h',q')=1}} \mathrm{e}\Big(\frac{h'm}{q'}\Big),$$

where we have taken $h = dh'$, $q = dq'$, with $\gcd(h'q') = 1$. Now the inner sum is just $c_{q'}(m) = c_{q/d}(m)$, so that

$$J(q) = \sum_{d|q} c_{q/d}(m) = \sum_{d|q} c_d(m).$$

On applying Möbius inversion, we find that $c_q(m) = \sum_{d|q} \mu(d) J(q/d)$. Since $J(q) = q$ or 0 depending on whether $q|m$ or not, we have arrived at (2.13).

Another formula for the Ramanujan sum that we shall use later is

$$c_q(m) = \frac{\phi(q)\mu(N)}{\phi(N)}, \qquad N = \frac{q}{\gcd(m,q)}, \tag{2.14}$$

For the proof, we let $a = \gcd(m,q)$, so that $q = Na$ and, by (2.13)

$$c_q(m) = \sum_{d|a} d\,\mu\Big(\frac{Na}{d}\Big) = \sum_{cd=a} \frac{a}{c}\,\mu(Nc) = a \sum_{c|a} \frac{\mu(Nc)}{c}.$$

Now $\mu(Nc) = \mu(N)\mu(c)$ or 0, according to $\gcd(N,c) = 1$ or not, so that

$$c_q(m) = a\mu(N) \sum_{\substack{c|a \\ \gcd(c,N)=1}} \frac{\mu(c)}{c} = a\mu(N) \prod_{p|a\ p\nmid N} \Big(1 - \frac{1}{p}\Big),$$

and (2.14) follows from

$$\frac{\phi(q)}{\phi(N)} = \frac{q}{N} \prod_{p|q\ p\nmid N} \Big(1 - \frac{1}{p}\Big) = a \prod_{p|q\ p\nmid N} \Big(1 - \frac{1}{p}\Big).$$

2.13 Summary and Notes

Summary

- There are two important classes of arithmetic functions: The *additive* functions, and the *multiplicative* functions. Their definitions require the condition that the numbers concerned are coprime; if the coprime condition can be dispensed with then *completely* is added to the definitions.
- The *sum function* $F(n) = \sum_{d|n} f(d)$ of a multiplicative function $f(n)$ is multiplicative. The *Möbius function* $\mu(n)$ has the important property that the value of its sum function is 1 or 0, depending on whether the parameter is 1 or otherwise. Moreover, *Möbius inversion* recovers a function from its sum function.
- *Euler's totient function* $\phi(n)$ counts the number $a \leq n$ which are coprime with n; it is a multiplicative function. The *Fermat-Euler theorem* states that $a^{\phi(n)} \equiv 1 \pmod{n}$ for all a coprime with n.
- For $\gcd(a, n) = 1$, the *order* of $a \pmod{n}$ is the least positive integer d such that $a^d \equiv 1 \pmod{n}$.
- The *divisor function* $d(n)$ has an average value $\log x$ over the range $1 \leq n \leq x$; such a result can best be dealt with using Dirichlet's *hyperbolic method*.
  For any K, there are values of n for which $d(n) > \log^K n$; nevertheless, $d(n) < n^\epsilon$ for any $\epsilon > 0$, and all large n.
- The *Dirichlet convolution* $h = f * g$ is defined by

$$h(n) = \sum_{d|n} f(d)g(n/d) = \sum_{ab=n} f(a)g(b).$$

- The *Mangoldt function* $\Lambda(n)$ takes the value $\log p$ when $n = p^k$, the power of a prime p, and 0 otherwise. We then have

$$\log n = \sum_{d|n} \Lambda(d) \quad \text{and} \quad \Lambda(n) = -\sum_{d|n} \mu(d) \log d.$$

- The additive group of residue classes $(\bmod d)$ can be mapped into the set of complex roots of unity, so that the addition of such classes becomes the multiplication of such roots. The *orthogonality property* of the exponential function enables a periodic function f with period q to be given a finite Fourier transform $\hat{f}$; there is then the relationship $q \sum |f(n)|^2 = \sum |\hat{f}(m)|^2$ when summed over a period.
- The sums

$$S(m, q) = \sum_{0 \leq h < q} e^{2\pi i h^2 m/q} \quad \text{and} \quad c_q(m) = \sum_{0 \leq h < q,\, \gcd(h,q)=1} e^{2\pi i h m/q}$$

are called *quadratic Gauss sums* and *Ramanujan sums*, respectively. They have the properties that, for coprime q, q',

$$S(m, qq') = S(mq, q')S(mq', q) \quad \text{and} \quad c_{qq'}(m) = c_q(m)c_{q'}(m).$$

A useful fact is that $c_q(1) = \mu(q)$, and in general there is the formula (2.14).

Notes

1. The elegant proof of the Fermat-Euler theorem seemed to have eluded even Gauss; it was given by J. Ivory in 1806.
2. The method of dealing with the sum in the Dirichlet divisor problem (Theorem 2.11), and its generalisation or extension, is called the *hyperbola method*.
3. The Dirichlet divisor problem is similar, and closely related, to the *circle problem* of Gauss to be considered in Theorem 6.5.

2.14 Exercises

1. Find the 20 smallest positive integers k for which $\phi(n) = k$ has no solution.
2. Find all the solutions of $\phi(n) = 24$.
   Show that, corresponding to any $k > 1$, there exists m_k such that there are at least k solutions of $\phi(n) = m_k$.
3. Find the smallest $n > 0$ for which $\phi(6n + 6) > \phi(6n + 2)$. Explain why such values n are rare, and show that there are infinitely many of them.
4. Prove that if n is composite, then $\phi(n) \le n - \sqrt{n}$.
5. In 1980, D. W. Masser said the following to me: *There is a simple argument which shows that, for any positive real x, the digit 7 appears in the decimal representation of one of the numbers $x, 2x, \dots, 79x$. I spent a wet Saturday afternoon showing that 79 can be reduced to 42, which is best possible.*
6. Find the primes p in $5 < p < 100$ for which the order of 10 (mod p) is $p - 1$.
   Determine the periods of the decimal expansion of $1/p$ for $p = 37, 41, 43, 47$.
7. Find the digital representations of 10^{20} in bases 2, 3, 5, 6, 7.
8. Find the last decimal digit of 13^{2345}.
9. Show that if $\gcd(m, n) = 1$ then $m^{\phi(n)} + n^{\phi(m)} \equiv 1 \pmod{mn}$.
10. Find numbers m, n satisfying

$$m < 10^{100}, \quad d(m) > 10^8; \qquad n < 10^{1000}, \quad d\big(d(n)\big) > 10^6.$$

11. Take a suitable x of your choice and evaluate $d(1) + d(2) + \cdots + d(x)$ in the three ways described in Sect. 2.8, and compare the computing times involved.
12. Show that if n is the product of the first $k > 1$ primes then $d(n \pm 1) \le 2^{k-1}$.

13. It has been established by D. R. Heath-Brown that there are infinitely many n such that $d(n) = d(n+1)$. Find the first 10 such n, and the number of such $n < 10000$.
14. It is conjectured that there are infinitely many n such that $\phi(n) = \phi(n+1)$. Find 10 such n.
15. Show that, for any fixed $m > 1$, and $x \to \infty$,

$$\sum_{\substack{n \le x \\ \gcd(n,m)=1}} n \sim \frac{\phi(m)x^2}{2m}.$$

16. Construct lists of values of $\omega(n)$, $\Omega(n)$, $\mu(n)$, $\lambda(n)$ for $n \le 10000$. Discover a simple formula for

$$\sum_{n \le N} \lambda(n) \left\lfloor \frac{N}{n} \right\rfloor$$

by evaluating the sum for various chosen values of N. Use Example 2.1 to prove the formula.
17. Show that, for $n \in \mathbb{N}$,

$$\sum_{a|n} (d(a))^3 = \Big(\sum_{a|n} d(a) \Big)^2.$$

18. Let $\psi(x) = \sum_{n \le x} \Lambda(n)$. Show that, for $m \in \mathbb{N}$,

$$\sum_{1 \le n \le m} \psi\Big(\frac{m}{n}\Big) = \log m!.$$

Chapter 3
Prime Numbers: Euclid and Eratosthenes



Abstract After Euclid's theorem on the infinitude of primes, the sieve of Eratosthenes is explained, and backed up with concrete results from a Pythod program. There is also a detailed description of the Meissel–Lehmer partial sieving method for the counting of primes, and the factorisation methods of Fermat and Pollard. Probabilistic primality tests, including the AKS test, one-way function and RSA public-key encryption are also covered.

We already mentioned that Euclid proved that there are infinitely many primes. Let $\pi(x)$ count the primes $p \leq x$, that is

$$\pi(x) = \sum_{p \leq x} 1, \tag{3.1}$$

so that $\pi(x) \to \infty$ as $x \to \infty$, by Euclid's theorem. His argument can be used to deliver an explicit lower bound for $\pi(x)$. Thus, if p_n denotes the nth prime, then $p_{n+1} \leq p_1 p_2 \cdots p_n + 1$, from which it follows by induction that $p_{n+1} < 2^{2^n}$, and hence $\log\log p_{n+1} < n \log 2 + \log\log 2$. For $x \geq 3$, define n by $p_n \leq x < p_{n+1}$; then

$$\pi(x) = n > \log\log p_{n+1} > \log\log x.$$

We are not, of course, suggesting that Euclid could have done this because he was some two millenniums before the notions of exponents and logarithms, never mind double exponentiations and repeated logarithms. The distribution of primes among the positive integers is one of the most interesting problems in number theory, and the following are some famous landmarks after Euclid and before the year 1900.

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3.1 Some Landmarks

In 1737 L. Euler found that the reciprocal sum $\sum 1/p$ of all the primes p is a divergent series, thereby giving a new proof of Euclid's theorem. His argument shows that

$$\sum_{p \le x} \frac{1}{p} \approx \log\log x \qquad \text{as} \qquad x \to \infty,$$

a result which imposes both an upper bound and a lower bound for $\pi(x)$. For example, on one hand, it can be used to obtain a much better lower bound for $\pi(x)$ than that from Euclid's argument. On the other hand, it can also be used to deduce that 'almost all numbers are composite', in the sense that $\pi(x)/x \to 0$ as $x \to \infty$, the first nontrivial upper bound for $\pi(x)$. The correct order of magnitude for $\pi(x)$ was establishes in 1851 by P. L. Chebyshev, who showed that there are fixed numbers a, b, with $0 < a < 1 < b$, such that $ax/\log x < \pi(x) < bx/\log x$ for all large x. The theorems of Euler and Chebyshev are given in Chap. 9.

Meanwhile, in an important paper in 1837, P. G. L. Dirichlet proved that if $\gcd(a, q) = 1$ then the series $\sum 1/p$, summed over primes p in the arithmetic progression $p \equiv a \pmod{q}$, is divergent, so that there are infinitely many such primes; see Sect. 12.1. Next, in a famous paper in 1859, B. Riemann showed that problems of the distribution of primes are closely tied up with the positions of the complex zeros of his eponymous zeta-function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \qquad s \in \mathbb{C}.$$

These two seminal pieces of work heralded the new subject of *analytic number theory*, the use of complex analysis to study problems in number theory; indeed much of function theory itself was developed with the aim of solving the problems.

In 1896, following ideas in Riemann's paper, J. Hadamard and C. de la Vallée Poussin independently proved the *prime number theorem*, which states that

$$\pi(x) \sim \frac{x}{\log x} \qquad \text{as} \qquad x \to \infty. \tag{3.2}$$

The asymptotic formula was conjectured by Legendre, and a rather young Gauss, almost exactly one hundred years earlier; an explanation on how they had arrived at the formula is given in Chap. 11, where a proof of the theorem is also presented.

We already mentioned that the use of trial division to find a proper divisor of a composite n, or to establish that n is prime, is extremely laborious, especially so if n is large. Although there are much better procedures to find a proper divisor

of a composite, none of them can be described as being efficient. However, there are efficient methods by which a given $n > 1$ can be determined to be prime or composite. In particular, there is the more recently invented AKS test which we describe in Sect. 3.8. Note that a number can be proved to be composite without delivering a proper divisor.

We shall require plenty of primes for experiments, so our first task is the listing of all the primes up to a chosen bound x. Instead of applying a primality test to each individual number to build the required list we apply a process called the *sieve of Eratosthenes*. With such a list, the value of $\pi(x)$ essentially comes for free, but if we just want to know $\pi(x)$ then we need not list all the primes $p \le x$, we need only count them.

3.2 Sieve of Eratosthenes

Each composite n has a prime divisor $p \le \sqrt{n}$. The *sieve of Eratosthenes* amounts to the observation that if we delete, or sieve out, the numbers in the list $\{2, 3, 4, \ldots, x\}$ which are multiples of a prime $p \le \sqrt{x}$, then the remaining, or unsifted, numbers in the list are *all* the primes p satisfying $\sqrt{x} < p \le x$.

Take, for example, $x = 1000$, so that $\lfloor\sqrt{1000}\rfloor = 31$. We first create a list '`prime`' by setting

$$\texttt{prime} = [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31].$$

Meanwhile a '`dummy`' list is also created by setting $\texttt{dummy}[n] = n$ for $0 \le n < 1000$. A double loop is then set up for the sieving process: Run the primes $p \in \texttt{prime}$ over the outer loop, run the numbers $n = mp < 1000$ over the inner loop, and reset $\texttt{dummy}[n] = 0$. At the end of the execution of the double loop, the unsifted elements of `dummy` are positive integers—in fact, apart from $\texttt{dummy}[1] = 1$, they are the primes in $31 < p < 1000$ which can be appended to the list `prime`. The following is the `Python` code, together with the output of the 78 primes $p < 400$, and the count 168 of the primes $p < 1000$.

```
# Program: Listing the primes p < 1000 by the
# Sieve of Eratosthenes.
bound=1000
prime=[2,3,5,7,11,13,17,19,23,29,31]
dummy=[]
for d in range(bound): dummy.append(d)
for p in prime:
  sift=0
  while sift+p<bound:
    sift+=p
    dummy[sift]=0
for d in dummy:
  if d>1: prime.append(d)
print (prime[0:78],len(prime))
```

[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337, 347, 349, 353, 359, 367, 373, 379, 383, 389, 397] 168

We may, of course, dispense with entering the set of primes by performing an initial sieve instead. Note that `prime[0]` $= 2$; if we want `prime[n]` to be the nth prime, we need to make the adjustment `prime.insert(0, 0)`, so that `prime[0]` $= 0$, `prime[1]` $= 2$. The set `dummy` can be converted to a list for the prime counting function $\pi(z)$, a step function with jumps by a unit when z is a prime.

Anyway, with the 168 primes $p < 1000$ in `prime`, we can have a second run of the sieve by resetting the bound $x = 1000^2 = 10^6$, thereby having the 78498 primes $p < 10^6$ in `prime`, and a table of values for $\pi(z)$ up to $z = 10^6$. In this way we can apply the sieve to deliver all the primes $p \leq x$ for any x, subject to the limit on the time and the storage capacity of the machine used. There is a simple trick which can be deployed to reduce the storage requirement: perform a preliminary sieve of a few small primes and enter only the unsifted numbers in `dummy`. For example, because the primes $p > 3$ take the form $6k \pm 1$, only such numbers need to be stored, although some adjustment is needed to land on the right store in `dummy` when sieving with $p > 3$.

There is a much better way to overcome the storage space problem when x is large, because the sieve can be applied to numbers in an interval, say $m < n \leq m+h$ with length h. Divide the interval $0 < n \leq x$ into $k = \lfloor x/h \rfloor$ subintervals, or 'blocks', with length h, together with a remaining last block $kh < n \leq x$. Take, for example, $x = 10^9$; we may set $h = 10^5$ and divide $0 < n \leq x$ into 10^4 blocks. Apply the sieve to a block, collect and process the required information on the primes in the block before updating `dummy` for the next block to be sieved. It is a good exercise to practise one's programming skill because care has to be taken in identifying the initial point to be sieved in the next block.

3.3 Counting Primes

In applying the sieve of Eratosthenes, some numbers are deleted more than once, and an expression for $\pi(x)$ can be constructed by keeping note of the number of times each number has been deleted.

Inclusion-Exclusion Principle

Let $C(1), C(2), \ldots, C(k)$ be clubs in a city with population N. For $1 \le h \le k$, and for each set of integers $a_1, a_2, \ldots, a_h$ satisfying $1 \le a_1 < a_2 < \cdots < a_h \le k$, denote by $N(a_1, a_2, \ldots, a_h)$ the number of people belonging to each one of the clubs $C(a_1), C(a_2), \ldots, C(a_h)$. Write

$$S_0 = N, \qquad S_h = \sum N(a_1, a_2, \ldots, a_h), \qquad 1 \le h \le k,$$

the summation being over all possible $a_1, a_2, \ldots, a_h$ satisfying $1 \le a_1 < a_2 < \cdots < a_h \le k$. The *inclusion-exclusion principle* asserts that the number of people *not* belonging to any club is given by

$$S_0 - S_1 + S_2 - S_3 + \cdots + (-1)^k S_k.$$

To see this, we note that a person which belongs to exactly ℓ clubs is counted once in S_0, $\binom{\ell}{1}$ times in S_1, $\binom{\ell}{2}$ in S_2, $\binom{\ell}{3}$ in S_3, and so on, where $\binom{\ell}{m}$ is a binomial coefficient, so that the total number of times the person being counted is

$$1 - \binom{\ell}{1} + \binom{\ell}{2} - \binom{\ell}{3} + \cdots = (1-1)^\ell = \begin{cases} 1 \text{ if } \ell = 0, \\ 0 \text{ if } \ell > 0, \end{cases}$$

and the principle is established.

Sieve of Eratosthenes-Legendre

Not long before the year 1800 Legendre applied the inclusion-exclusion principle to the sieve of Eratosthenes, which results in a formula for $\pi(x)$.

The number of integers in the interval $1 \le m \le x$ which are multiples of distinct primes $q_1, q_2, \ldots, q_h$ is $\lfloor x/q_1 q_2 \cdots q_h \rfloor$. Therefore, by the inclusion-exclusion principle, the number of integers in the interval which are not divisible by a prime $q \le \sqrt{x}$ is $S_0 - S_1 + S_2 - S_3 + \cdots + (-1)^k S_k$, where

$$S_0 = \lfloor x \rfloor, \qquad S_h = \sum_{q_1 < q_2 < \cdots < q_h \le \sqrt{x}} \left\lfloor \frac{x}{q_1 q_2 \cdots q_h} \right\rfloor, \quad h = 1, 2, \ldots, k = \pi(\sqrt{x}\,);$$

an h-fold sum corresponding to products of h distinct primes $q \le \sqrt{x}$. Recalling that the number concerned is $\pi(x) - \pi(\sqrt{x}) + 1$, and the definition of the Möbius function, we have established the *sieve of Eratosthenes-Legendre*:

$$\pi(x) = \sum_{d|P} \mu(d) \left\lfloor \frac{x}{d} \right\rfloor + \pi(\sqrt{x}) - 1, \qquad P = \prod_{p \le \sqrt{x}} p. \tag{3.3}$$

Example 3.1 Use the Eratosthenes-Legendre sieve formula to evaluate $\pi(400)$.

Solution For $x = 400$, we have $\sqrt{x} = 20$, $\pi(20) = 8$, and $q = 19$ is the largest prime to be used in the sieve. Also, from $2{\cdot}3{\cdot}5{\cdot}7 < x < 2{\cdot}3{\cdot}5{\cdot}7{\cdot}11$, we deduce that $S_h = 0$ if $h > 4$, and we find that

$$S_0 = \lfloor x \rfloor = 400,$$

$$S_1 = \sum_{q<20} \left\lfloor \frac{x}{q} \right\rfloor = \left\lfloor \frac{400}{2} \right\rfloor + \left\lfloor \frac{400}{3} \right\rfloor + \cdots + \left\lfloor \frac{400}{19} \right\rfloor = 580,$$

$$S_2 = \sum_{q_1<q_2<20} \left\lfloor \frac{x}{q_1 q_2} \right\rfloor = \left\lfloor \frac{400}{2{\cdot}3} \right\rfloor + \cdots + \left\lfloor \frac{400}{3{\cdot}5} \right\rfloor + \cdots + \left\lfloor \frac{400}{17{\cdot}19} \right\rfloor = 324,$$

$$S_3 = \sum_{q_1<q_2<q_3<20} \left\lfloor \frac{x}{q_1 q_2 q_3} \right\rfloor = \left\lfloor \frac{400}{2{\cdot}3{\cdot}5} \right\rfloor + \cdots + \left\lfloor \frac{400}{13{\cdot}17{\cdot}19} \right\rfloor = 76,$$

$$S_4 = \sum_{q_1<q_2<q_3<q_4<20} \left\lfloor \frac{x}{q_1 q_2 q_3 q_4} \right\rfloor = \left\lfloor \frac{400}{2{\cdot}3{\cdot}5{\cdot}7} \right\rfloor + \cdots + \left\lfloor \frac{400}{11{\cdot}13{\cdot}17{\cdot}19} \right\rfloor = 3.$$

For $h = 0, 1, 2, 3, 4$, the number of terms in the sum S_h is $\binom{8}{h} = 1, 8, 28, 56, 70$, where we note that 29 of the terms in S_3, and 67 of the terms in S_4, are zero. From the sieve formula (3.3), we now have

$$\pi(400) = (400 - 580 + 324 - 76 + 3) + (8 - 1) = 78.$$

□

It is a laborious way to evaluate $\pi(400)$, and the use of pencil and paper on a 20×20 grid in the original sieve of Eratosthenes for $\pi(400)$ may be simpler. In any case it is clear that, for larger values of x, say $x > 10^9$, the proliferation of the large number of small terms among the various sums S_h will become unmanageable even for a large machine. As it stands, the sieve of Eratosthenes-Legendre (3.3) is almost useless, and it seems hopeless trying to estimate the contribution from the large number of small terms with seemingly uncertain attached signs. We should come to grief if we force the issue by trying to derive a theoretical estimate for $\pi(x)$ by simply replacing $\lfloor x/d \rfloor$ with $x/d + O(1)$ in the formula; see Note 1 for confirmation. Some clever adjustment is necessary, both in theory and in practice for actual evaluation; we deal only with the latter here.

Meissel-Lehmer Method

In 1870, E. D. F. Meissel[1] came up with the idea of a 'partial sieving' process: instead of sieving with primes $p \leq \sqrt{x}$, apply it only for $p \leq \sqrt[3]{x}$. There is then a dramatic reduction of the number of terms in the 'Legendre sums' but, of course, there is a price to be paid because the unsifted numbers in the interval $\sqrt[3]{x} < m \leq x$ include composites besides the primes. However, for the purpose of counting, instead of listing, the primes, we need only come up with a formula for the number of unsifted composites. The important point to note is that such composites must have exactly two (possibly equal) prime divisors, because such divisors exceed $\sqrt[3]{x}$. Meissel derived a formula for the count of such unsifted composites, thereby succeeded in the evaluations of $\pi(10^7)$, $\pi(10^8)$ and $\pi(10^9)$.

In 1959, D. H. Lehmer much clarified and simplified Meissel's method, and he extended Meissel's idea by sieving only up to $p \leq \sqrt[4]{x}$. There is, of course, further reduction of the number of terms in the Legendre sum, but there are also unsifted composite numbers in $\sqrt[4]{x} < m \leq x$ which are products of three (not necessarily distinct) primes $p > \sqrt[4]{x}$, so he derived a formula for the count of such numbers. By this time computing machines had become available and Lehmer remarked in his paper that "Meissel's calculation of $\pi(10^9)$ ... must be regarded as one of the outstanding single calculations of the nineteenth century, even though his value is slightly in error." Lehmer discovered that the often quoted value of $\pi(10^9)$ was too small by 56; by an ironic twist of fate, Lehmer's value for $\pi(10^{10})$ was too large by 1!

Meanwhile, there was to be a general reappraisal of the notion of a sieve, with more sophisticated ideas from combinatorics. Indeed, there are now various types of sieves, such as Brun's sieve and Selberg's sieve, which have become indispensable tools in the study of number theory.

Legendre Sums $\phi(x, a)$

For flexibility, we introduce a parameter $a < \pi(x)$, and consider partial sieving for $p \leq p(a)$, where $p(a)$ is the ath prime. Thus, $a = \pi(\sqrt{x})$, $a = \pi(\sqrt[3]{x})$ and $a = \pi(\sqrt[4]{x})$ correspond to the schemes of Legendre, Meissel and Lehmer, respectively.

Let $\phi(x, a)$ count the numbers in the interval $1 \leq m \leq x$ which are *not* divisible by a prime $p \leq p(a)$, and let $P_r(x, a)$ count the same numbers which are products of r primes exceeding $p(a)$. By our discussion in the above we then have

$$\pi(x) = \begin{cases} \phi(x,a) + a - 1, & \pi(\sqrt{x}) \leq a < \pi(x), \\ \phi(x,a) - P_2(x,a) + a - 1, & \pi(\sqrt[3]{x}) \leq a < \pi(\sqrt{x}), \\ \phi(x,a) - P_2(x,a) - P_3(x,a) + a - 1, & \pi(\sqrt[4]{x}) \leq a < \pi(\sqrt[3]{x}), \end{cases} \tag{3.4}$$

[1] It is interesting that both Eratosthenes and Meissel are better known for their works in astronomy.

and the task now is to devise suitable methods to evaluate the expressions $\phi(x, a)$, $P_2(x, a)$ and $P_3(x, a)$; we allow x to be a rational number in the following.

The count $\phi(x, a)$ is called the *Legendre sum*; by the inclusion-exclusion principle we have

$$\phi(x, a) = S_0 - S_1 + S_2 - S_3 + \cdots + (-1)^k S_k,$$

where

$$S_0 = \lfloor x \rfloor, \qquad S_h = \sum_{q_1 < q_2 < \cdots < q_h \le p(a)} \left\lfloor \frac{x}{q_1 q_2 \cdots q_h} \right\rfloor, \qquad h = 1, 2, \ldots, k; \tag{3.5}$$

the q_i are primes, and k satisfies

$$X_k \le x < X_{k+1}, \qquad \text{where} \quad X_k = \prod_{p \le p(k)} p.$$

The computation of the Legendre sum $\phi(x, a)$ by the formula is still unwieldy, so we apply the following adjustment. By definition, the Legendre sum satisfies the reduction formula:[2]

$$\phi(x, a) = \phi(x, a-1) - \phi\left(\frac{x}{p(a)}, a-1\right), \tag{3.6}$$

because numbers not divisible by any prime $q \le p(a)$ are precisely those integers not divisible by any prime $q \le p(a-1)$, with the exception of those which are not divisible by $p(a)$. We shall make particular use of

$$X_6 = 2.3.5.7.11.13 = 30030,$$

mainly because it is about the right size for our purpose. The count $\phi(z, 6)$ increases with z, and there are precisely

$$F_6 = \phi(X_6) = 5760 \qquad \text{(Euler's totient } \phi\text{)}$$

numbers in the interval $1 \le r < X_6$ which are coprime with X_6. Moreover, since each interval of length X_6 contains F_6 numbers coprime with X_6, by dividing the interval $1 \le m < z$ into $Q = \lfloor z/X_6 \rfloor$ blocks with length X_6, the value for $\phi(z, 6)$ can be obtained from

$$\phi(z, h) = F_6 Q + \phi(r, 6), \quad \text{where} \quad Q = \lfloor z/X_6 \rfloor,\ \lfloor z \rfloor = X_6 Q + r,\ 0 \le r < X_6,$$

[2] A reduction formula of this type is named after A. A. Buchstab (see also Sect. 13.1).

together with a table for $\phi(r, 6)$, which can be constructed by counting the numbers $m \le X_6$ which are coprime with X_6.

Returning to the reduction formula for $\phi(x, a)$, its repeated application then brings its value down to a combination of $\phi(z, 6)$ for various z. More precisely, the inclusion-exclusion argument delivers $\phi(x, a) = S'_0 - S'_1 + S'_2 - S'_3 + \cdots (-1)^{k'} S'_k$, where

$$S'_0 = \phi(x, 6), \quad S'_h = \sum_{p(6)<q_1<q_2<\cdots<q_h\le p(a)} \phi\left(\frac{x}{q_1 q_2 \cdots q_h}, 6\right), \quad h = 1, 2, \ldots, k,$$

and k now satisfies

$$Y_k \le x < Y_{k+1}, \qquad Y_k = \prod_{p(6)<p\le p(6+k)} p,$$

so that, for example, $Y_7 = p(7)p(8)\cdots p(13) = 17{\cdot}19{\cdot}23{\cdot}29{\cdot}31{\cdot}37{\cdot}41$.

With $z = x/q_1 q_2 \cdots q_h$, the sum S'_h can now be evaluated in a nested loop of h layers. Indeed, the evaluation of all S'_h can be done together in a single nested loop of k layers, and we may break out of an inner loop when $z = 0$. Some computing time can be saved if multiplication by F_6, and reading of $\phi(r, 6)$ from the said table, are done outside the loop. Since Y_k increases rapidly, the size of k is rather modest; for example, $k = 6, 7, 8$ correspond to $x = 10^{10}, 10^{11}, 10^{12}$.

Meissel and Lehmer found, respectively, that

$$\phi(10^9, 168) = 81515102, \qquad \phi(10^{10}, 65) = 964916390,$$

where the corrections of $+57$ and -1, respectively, have been made.

Evaluations of $P_2(x, a)$ and $P_3(x, a)$

The formulae for $P_2(x, a)$ and $P_3(x, a)$ are

$$P_2(x, a) = \sum_{a<i\le b} \pi\left(\frac{x}{p(i)}\right) - \frac{1}{2}(a + b - 1)(b - a),$$

$$P_3(x, a) = \sum_{i>a} P_2\left(\frac{x}{p(i)}, a\right) = \sum_{a<i\le c} \sum_{i\le j\le b(i)} \left(\pi\left(\frac{x}{p(i)p(j)}\right) - (j - 1)\right),$$

where

$$b = \pi\left(\sqrt{x}\,\right), \qquad c = \pi\left(\sqrt[3]{x}\,\right), \qquad b(i) = \pi\left(\sqrt{x/p(i)}\,\right). \tag{3.7}$$

Their derivations are not difficult, and they serve as useful exercises for the readers before they apply their programming skill. Instead, we describe the method for their evaluations.

Note that $b = \pi(\sqrt{x})$ means $p(b) \le \sqrt{x} < p(b+1)$, so that $x/p(b) \ge \sqrt{x}$. On writing $z_i = x/p(i)$, where $a < i \le b$, we have a decreasing sequence:

$$\sqrt{x} \le \frac{x}{p(b)} = z_b < z_{b-1} < \cdots < z_{a+1} = \frac{x}{p(a+1)}.$$

The evaluation of the sum in $P_2(x, a)$ then amounts to reading, and summing, the prime counts $\pi(z_i)$ from the table for $\pi(z)$ in the interval $\sqrt{x} \le z < x/p(a+1)$. The summation should be done in the reverse order of the stated sum so that the required values for $\pi(z)$ can be read in increasing order. If $x/p(a+1)$ is large the table for $\pi(z)$ can be supplied in blocks as it was described earlier. The double sum for $P_3(x, a)$ can be evaluated similarly—the required table for $\pi(z)$ being much shorter, its evaluation is easier; in fact it can be nested inside the loop for the evaluation of $P_2(x, a)$.

Meissel set $x = 10^9$ and $a = 168$, so that the interval for is $31622 \le z < 991080$, and he worked out that (the correction of $+1$ has been made)

$$P_2(10^9, 168) = 30667735,$$

so that, together with the earlier calculated value for $\phi(10^9, 168)$,

$$\pi(10^9) = \phi(10^9, 168) - P_2(10^9, 168) + 168 - 1 = 50847534. \tag{3.8}$$

Lehmer set $x = 10^{10}$ and $a = 65$, so that the interval is $10^5 \le z < 31545741$, and he found that

$$P_2(10^{10}, 65) = 463026862, \qquad P_3(10^{10}, 65) = \ \ 46837081,$$

so that, together with the earlier calculated value for $\phi(10^{10}, 65)$,

$$\pi(10^{10}) = \phi(10^{10}, 65) - P_2(10^{10}, 65) - P_3(10^{10}, 65) + 65 - 1 = 455052511. \tag{3.9}$$

For a fixed x, the computing times for $\phi(x, a)$ and $P_2(x, a)$ are increasing and decreasing, respectively, with respect to a, and the combined time is near the minimum when the two times are about the same—we already hinted that the time for $P_3(x, a)$ is relatively short. The value of $\pi(x)$ can be evaluated more than once by making different choices of a, which is useful for the purpose of

checking the calculations. The following are the results for our evaluations of $\pi(10^{11})$ and $\pi(10^{12})$.

$$\begin{aligned}
\phi(10^{11}, 220) &= 7\,741\,200\,576 & \phi(10^{11}, 250) &= 7\,589\,266\,196\\
P_2(10^{11}, 220) &= 3\,506\,484\,668 & P_2(10^{11}, 250) &= 3\,387\,392\,900\\
P_3(10^{11}, 220) &= \quad 116\,661\,314 & P_3(10^{11}, 250) &= \quad 83\,818\,732\\
\phi(10^{12}, 300) &= 73\,723\,884\,081 & \phi(10^{12}, 360) &= 71\,750\,032\,273\\
P_2(10^{12}, 300) &= 32\,294\,725\,048 & P_2(10^{12}, 360) &= 32\,850\,710\,547\\
P_3(10^{12}, 300) &= \quad 1\,821\,247\,314 & P_3(10^{12}, 360) &= \quad 1\,291\,410\,067
\end{aligned}$$

The four sets of results, with four choices of a, and $x = 10^{11}, 10^{12}$, give

$$\pi(10^{11}) = 4\,118\,054\,813, \qquad \pi(10^{12}) = 37\,607\,912\,018.$$

Using a desktop machine with a `Python` program, the two computing times are about 6 and 30 minutes, respectively.

We also found that, for $x = 10^{13}$ and $a = 700$:

$$\phi(x, a) = 657389071477,\ P_2(x, a) = 297595323135,\ \ P_3(x, a) = 10128212202,$$

giving $\pi(10^{13}) = 346065536839$. A description of the Meissel-Lehmer method, and an analytic method, for the computation of $\pi(x)$ can be found in [52], and [51], respectively.

3.4 Gauss' Dream

The decomposition of a number into its prime factors has always fascinated people, from enthusiastic students to seasoned professionals. In Article 329 of his groundbreaking *Disquisitiones Arithmeticae* the young Gauss wrote passionately:

> The problem of distinguishing prime numbers from composite numbers, and to resolve the latter into their prime factors in known to be one of the most important and useful in arithmetic ... [T]he dignity of the science itself seems to require that every possible means be explored for the solution of a problem so elegant and so celebrated.

The two problems have become known as Gauss' dream: An efficient primality test, and an efficient factorisation algorithm. Almost exactly 200 years later, the first part of his dream came true when three computer scientists, Manindra Agrawal, Neeraj Kayal, and Nitin Saxena, created a deterministic polynomial time algorithm (AKS test, see Sect. 3.8) which establishes the primality of an input. However, the second part of Gauss' dream may never be fulfilled, because there seems to be good reasons that efficient factorisation algorithms may not exist.

The factorisation of a modestly large number is an art, and there are plenty of books from the elementary to research monographs devoted to the subject; see for

example [82] and [19]. The trial division method of finding the least prime divisor of a given number is fine, provided such a divisor is small. Here we consider two of the simplest factorisation methods: Fermat's method and Pollard's ρ-method. Another method based on the use of continued fractions is given in Chap. 7.

Factorisation methods are time-consuming, and the worst case scenario of a chosen factorisation method happens if the input is a prime. Therefore some effort should be made to see if the number concerned is known, or at least very likely, to be composite. For most composite n, there is a simple procedure described in the next subsection which ensures that the compositeness claim is correct. A randomly chosen number n is likely to have small prime divisors, so trial division should be applied first, and it is easy to check whether a number is a perfect power.

Probabilistic Primality Tests

There is some confusion in the use of the term 'test' to determine whether a given number is prime or composite. First, a definitive test is an *efficient* procedure applied to an input n, with an output which declares whether n is a prime or a composite. For such tests, besides the input n, there are additional parameters which can be chosen and altered so that a successful outcome is delivered. Note that a number can be shown to be composite without the delivery of a proper divisor. Trial division requires only the input n, delivering its least prime divisor, but it is not considered a proper test because it is not efficient in application.

A *primality test* should then be derived from a theorem which gives a *sufficient* condition for n to be a prime; conversely, a test corresponding to a theorem which gives only a *necessary* condition for n to be prime should be called a *compositeness test*. As we already remarked, the first part of Gauss' dream has been realised, because the AKS test is efficient theoretically. Nevertheless, there is still much interest in the subject because there are other more efficient tests, although their results need not always be definitive.

Since almost all numbers are composite, and compositeness tests are easier to apply, we always opt for a compositeness test first. If a number n passes a compositeness test with respect to some parameter a, then n is composite, and we may call a a *witness* for its compositeness. If n fails the test, then n cannot be declared composite, nor can it be claimed to be a prime. Rather than switching to a more elaborate primality test, we may choose to change the parameter for the test, or change to another compositeness test, hoping that n now passes the test with a newly found witness.

Suppose that n fails a compositeness test repeatedly even with changes of parameters, so that we suspect that the reason for the failures is that n is prime. For this reason, we sometime call a compositeness test a *probabilistic primality test*, perhaps to emphasise that if a number has failed the test, it is suspected to be prime, but the dubious claim of primality has to include a caveat that it can be wrong. The confusion comes when some writers chose to drop the word 'probabilistic', and

also omit mentioning the caveat. For these reasons, some rather clumsy and ugly nomenclature will be necessary to clarify the situation.

Since a compositeness test is also called a probabilistic primality test, if n fails such a test, it is called a *probable prime*, with respect to the (failed) test and the parameters; the status for such n is currently not known—it can be a prime, but it has not been ruled out that n can be composite. If such n can be established to be composite, either by a more adroit choice of parameter, or by a different test, then n is called a *pseudoprime*, with respect to the (failed) compositeness test and the chosen parameters. In particular, a pseudoprime is composite, by definition, whereas a probable prime may turn out to be composite, or prime, by another test, say the definitive AKS test, for example.

Fermat Test

Fermat's Little Theorem in Sect. 2.4 gives a necessary condition for n to be a prime, but not a sufficient one, so that the corresponding test is a compositeness one. Taking the contrapositive of the theorem, it states that if $a^n \not\equiv a \pmod{n}$ for some $a < n$ then n is composite. We may as well assume that $\gcd(a, n) = 1$, since otherwise its value is a proper divisor of n.

Theorem 3.1 (Fermat Test) *Suppose that* $1 < a < n$, *and* $\gcd(a, n) = 1$.

$$\text{If} \quad a^{n-1} \not\equiv 1 \pmod{n}, \quad \text{then } n \text{ is composite.}$$

□

Example 3.2 Verify that $n = (10^{59} - 1)/9 = 111 \cdots 111$ is composite.

Solution We can take $a = 2$, or a random $a < n$; there is no need to check that $\gcd(a, n) = 1$. Thus, for $a = 748056155141604309$, we find that

$$a^{n-1} \equiv 44469930257769197 \pmod{n},$$

so that n is composite; the computing time taken for the test is trivial (under one milliseconds). Note that we have established that n is composite without finding a proper divisor—if trial division is used, the least prime divisor $p = 2559647034361$ will not emerge before CCH; however, see Pollard's ρ-method below. □

If n passes[3] the test, that is, if there exists a such that $a^{n-1} \not\equiv 1 \pmod{n}$, then a is a witness to n being composite. If n fails the test, that is, $a^{n-1} \equiv 1 \pmod{n}$ for some a coprime with n, then we call n a *probable prime* with respect to the

[3] Unfortunately, to add to the confusion, considering it as a probabilistic primality test, the use of 'pass/fail' by some authors is exactly the opposite to what we use here.

parameter a, written PRP(a) for short. If, on a change of parameter or a change of test, n is subsequently established to be composite, then we call n a *pseudoprime* with respect to the parameter a, written PSP(a) for short; thus a PSP is, by definition, composite.

Even when n fails Fermat test for all possible values of the parameter a, we can still only say that n is a probable prime. In other words, the following statement, which is the converse of the theorem, is a

FALSE statement: If $a^{n-1} \equiv 1 \pmod{n}$ for all a coprime with n, then n is prime.

Numbers n for which the statement is false are called *Carmichael numbers*, and $n = 561, 1105, 1729$, are the first three such numbers; more information on them are given in Sect. 5.3.

The first ten PSP(2) are

$$341, 561, 645, 1105, 1387, 1729, 1905, 2047, 2465, 2701.$$

Thus $n = 341 = 11 \times 31$ is composite, and $2^{340} \equiv 1 \pmod{341}$. It has been found that there are 118 968 378 PSP(2) up to 2^{64} and $\pi(2^{64}) = 425\,656\,284\,035\,217\,742$, so that, of the number of $n \leq 2^{64}$ satisfying $2^{n-1} \equiv 1 \pmod{n}$, some 99·99999997% of them are primes; this explains the term 'probable prime' being used.

Can a number be both PSP(2) and PSP(3)? The answer is yes: we need only find a Carmichael number which is coprime with 6, so 1105 and 1729 are both PSP(2) and PSP(3). There is significant overlap among lists of pseudoprimes to different bases, for which reason once a number has been shown to be PSP(a) it is more likely than on average to be PSP(a') for many a'. For example, it is known that of the 21852 PSP(2) up to $25 \cdot 10^9$, over 20% of them are also PSP(3), and over 8% are PSP(a) for $a = 2, 3, 5, 7$.

Example 3.3 Find the first twelve $a > 1$ such that 341 is PSP(a).

Solution For $n = 341$, we find that

$$a^{n-1} \equiv 1 \pmod{n} \quad \text{when } a = 2, 4, 8, 15, 16, 23, 27, 29, 30, 32, 35, 39, \ldots,$$

so that 341 is PSP(a) for these values of a. □

For a general odd n, there is the problem of what should be done if the test keeps failing, that is, we keep getting $a^{n-1} \equiv 1 \pmod{n}$, for many random choices of a? There seems to be good reasons that probable primality tests with different bases are not independent, and it may not be wise to apply the test endlessly for more bases hoping to find one that the number will pass. Perhaps the test should be abandoned after taking four bases, and some experts even suggest that if it fails for $a = 2$ then switch immediately to other 'stronger' tests. We shall describe one of these tests in due course; see Sect. 4.2.

Meanwhile, if $n < 2^{64}$ passes Fermat test with $a = 2$, then we need only check if n is one of the 118 968 378 found PSP(2); in other words a probable prime which is not in the list has to be a prime. For this reason, Fermat test with parameter $a = 2$ is a deterministic primality test for such n.

Theorems of Lucas, Pocklington and Lehmer

A deterministic primality test requires a theorem which gives a suitable sufficient condition for primality. Some additional hypotheses can be attached to the Fermat-Euler theorem to obtain such a sufficient condition, and the earliest such considerations were given by E. Lucas (1876) and H. C. Pocklington (1914); their results were clarified and much simplified by D. H. Lehmer in 1927.

Theorem 3.2 (Lucas-Lehmer) *Let $a, n > 1$ satisfy $a^{n-1} \equiv 1 \pmod{n}$, and*

$$a^{(n-1)/q} \not\equiv 1 \pmod{n} \quad \text{for every prime } q \mid n-1.$$

Then n is prime.

Proof Let $d = \operatorname{ord}_n(a)$, the order (see Sect. 2.4) of $a \pmod{n}$. The hypothesis $a^{n-1} \equiv 1 \pmod{n}$ implies $d \mid n-1$, and the remaining hypothesis implies that d is not a proper divisor of $n-1$, so that $d = n-1$. It follows that $n-1 \mid \phi(n)$, so that $n-1 \le \phi(n) \le n-1$. Therefore $\phi(n) = n-1$, and n is a prime. □

The implementation of the Lucas-Lehmer theorem as a definitive primality test requires the complete prime factorisation of $n-1$. Sometimes, even a partial factorisation is enough.

Theorem 3.3 (Pocklington-Lehmer) *Suppose that $n - 1 = FR$, where $F \ge \sqrt{n}$, and that its complete prime factorisation is known. Suppose further that there exists a, coprime with n, such that*

$$a^{n-1} \equiv 1 \pmod{n}, \quad \text{and} \quad \gcd(a^{(n-1)/q} - 1, n) = 1 \text{ for each prime } q \mid F.$$

Then n is prime. □

The proof is essentially an elaboration of the argument used in the proof of the Lucas-Lehmer theorem. More specifically, making use of $F = (n-1)/R$, the hypotheses imply that F divides $p-1$ for every prime divisor p of n. Thus $p > F \ge \sqrt{n}$ for every such p, so that n has to be a prime.

Besides being a primality test, the theorem can also be used to construct special types of large primes.

Example 3.4 Use Theorem 3.3 to construct a few primes exceeding 10^{100}.

Solution There is much flexibility in the following. On application of the Fermat test with base 2 to $123456789101 + i$ for $i = 0, 2, 4, \ldots$, we find that

$$p = 123456789133 > 10^{11}$$

is a PRP(2), and trial division confirms that p is prime. Next, let

$$F = 2p^5, \qquad R_k = 10^{50} + k, \qquad n_k = FR_k + 1,$$

where $0 \le k < 1000$. Then $F > 10^{55}$, so that $10^{105} < FR_k < n_k < 2FR_k < F^2$.

By applying the Fermat test with base 2 to n_k we find that n_k is a PRP(2), for

$$k = 101, 131, 136, 298, 406, 491, 620, 647, 868, 938, 985.$$

Moreover, for $k = 101, 298, 406, 938, 985$, we find that

$$\gcd(2^{(n_k-1)/2} - 1, n_k) = \gcd(2^{(n_k-1)/p} - 1, n_k) = 1,$$

so that n_k is prime for each such k according to Theorem 3.3. □

3.5 Factorisation: Fermat's Method, and Pollard's ρ Method

Fermat's Method

Fermat was the first person to consider the factorisation of numbers besides using trial division. His idea was that every odd composite number with the representation $n = ab$, where $a > b > 2$, can be rewritten as a difference of two squares:

$$n = x^2 - y^2 = (x - y)(x + y), \qquad \text{with} \quad x = \frac{a+b}{2}, \qquad y = \frac{a-b}{2},$$

so that the decomposition always captures a proper divisor of such n. The method then relies on testing whether $z = x^2 - n$ is a perfect square, by taking successively values of x, starting from $x = \lceil \sqrt{n} \rceil$. It is useful to note that the next value to be tested is $(x + 1)^2 - n = z + 2x + 1$, so that the iterative step may take the form:

$$\begin{cases} \mathtt{z} += 2 * \mathtt{x} + 1 \\ \mathtt{x} += 1 \end{cases}$$

We need an efficient method to detect which z is a perfect square—we can evaluate $\lfloor \sqrt{z} \rfloor$ by Newton's method, and then check whether $\lfloor \sqrt{z} \rfloor^2 = z$. However, since only a few values of z to be found are squares, we should employ a simpler preliminary process by which those z which are clearly not squares can first be

disposed of, so that the number of testing is reduced significantly. For example, since the last decimal digit of a square is restricted to the values 0, 1, 4, 5, 6, 9, we know that z cannot be a square if $z \equiv 2, 3, 7, 8 \pmod{10}$. The idea can be extended using other moduli besides 10. We shall consider the problem again in Chap. 4 on the theory of quadratic residues.

If the companion divisors a, b are near $\sqrt{n}$ then Fermat's method works well, because $(a + b)/2 \approx \sqrt{n}$ will soon be reached from the starting value for x. On the other hand, if a, b are primes which differ significantly in size, say $a \approx n^{2/3}$, $b \approx n^{1/3}$ then the number of required steps to reach factorisation is $(a + b)/2 \approx \frac{1}{2}a \approx \frac{1}{2}n^{2/3}$, so that the running time is worse than trial division.

Example 3.5 Use Fermat's method to factorise

$$n = 70481903 \qquad \text{and} \qquad n = 28959000082152109.$$

Solution For $n = 70481903$, we take $x = \lceil \sqrt{n} \rceil = 8396$ and $z = x^2 - n = 10913$. Iterating $x \mapsto x + 1$, we then find that at $x = 8412$, $z = 279841 = 529^2$, so that $(x, y) = (8412, 529)$, and $n = (8412 - 529)(8412 + 529) = 7883 \cdot 8941$.

We apply the same procedure for $n = 28959000082152109$, taking $x = \lceil \sqrt{n} \rceil = 170173442$ and $z = x^2 - n = 279975255$. After 8838903 iterations, we arrive at $(x, y) = (179012345, 55555554)$, and the factorisation

$$n = (x - y)(x + y) = 123456791 \cdot 234567899,$$

in 230 seconds computing time. In Sect. 4.2 we show how the number of applications to check whether $\lfloor \sqrt{z} \rfloor^2 = z$ can be reduced significantly. □

Although Fermat's method is rarely used nowadays, it is the basis of more recent methods of factorisation. Thus, in 1798, Legendre had suggested that there was extra mileage if Fermat's decomposition equation was converted to a congruence. In the 1920s M. Kraitchik offered the idea of finding pairs (x, y) such that $x^2 \equiv y^2 \pmod{n}$, so that $n | (x - y)(x + y)$, and a proper divisor of n could be delivered from a suitable pair (x, y) by the application of the Euclidean algorithm for $\gcd(n, x - y)$. Indeed, some very powerful methods of factorisation are based on such a consideration.

Pollard's ρ-Method

Let n be an odd composite number with the least prime divisor p, which is yet to be found. In 1975 J. M. Pollard [74] came up with the fabulous idea of using a sequence of random numbers (x_i), hoping that p may emerge from the evaluation of $\gcd(n, x_i - x_j)$ by the Euclidean algorithm with various pairs of x_i, x_j.

The idea is liken to saying "Never mind the ball, just get on with the game!" The ball being p, of course, and the game is to devise a method by which the large

number of pairs of x_i, x_j to be considered is so reduced that the scheme might become viable.

The construction of the sequence (x_i) of random numbers in $0 \le x < n$ is only a very minor part of the game; for concreteness, let $f(x) = x^2 + 1$ and define

$$x_0 = 2, \qquad x_i \equiv f(x_{i-1}) \pmod{n}, \qquad 0 \le x_i < n, \quad i = 1, 2, \ldots .$$

There being only p congruence classes (mod p), the residues $x_i \pmod{p}$ must eventually repeat a term, and then become cyclic, with a certain period $\ell \le p$. An alert reader may interrupt here with the remark:

Hey, we do not know p, so we cannot find $x_i \pmod{p}$, never mind ℓ.

True, but we did say, never mind the ball, just get on with the game!

A reader with knowledge of the so-called 'birthday problem' (see Note 2), will know that a pair x_i, x_j such that $x_i \equiv x_j \pmod{p}$ is likely to exist if $\ell \approx \sqrt{p}$. Thus the number of pairs i, j to be considered is about $\ell^2 \approx p \le \sqrt{n}$, and the expected number of gcd computations for a successful delivery of p is about $\sqrt{n}$. Unfortunately this is worse than the trial division method, which guarantees the delivery of p after at most $\sqrt{n}$ trials.

The game changer is the application of the very nice *Floyd cycle-finding* method which asserts that we may set $j = 2i$, so that we need only deal with ℓ pairs x_i, x_{2i}. Armed with such a scheme we are then likely to succeed in about $\ell \le \sqrt{p} \le n^{1/4}$ steps, which makes the method viable, provided that n is not too large; indeed, the method should work well as long as p is relatively small.

Theorem 3.4 (Floyd Cycle-Finding) *Let $f : S \to S$ be a mapping from a finite set S to itself. Choose any $x_0 \in S$, and define $x_i = f(x_{i-1})$ for $i = 1, 2, \ldots$. Then there exists i such that $x_i = x_{2i}$.*

Proof Since S is a finite set, the sequence (x_i) is eventually cyclic, with a period ℓ, say. Let x_j be the first term of the sequence which belongs to the cycle. Then $x_i = x_{i+\ell} = x_{i+2\ell} = \cdots$ for every $i \ge j$. Take a multiple of ℓ which is at least j, say $i = h\ell \ge j$. Then $x_i = x_{i+h\ell} = x_{2i}$. □

Inside the loop of the factorisation algorithm, only the values of the current pair (x_i, x_{2i}) need to be retained because the next pair (x_{i+1}, x_{2i+2}) can then be evaluated by $x_{i+1} = f(x_i)$ and $x_{2i+2} = f(f(x_{2i}))$. To save running time, the gcd calculation can be delayed and done in batches by forming the product $X \pmod{n}$ of a batch of values $x_i - x_{2i}$ for the evaluation of $\gcd(n, X)$.

With randomness being involved in Pollard's ρ-method, we may as well allow the initial seed x_0, or even the function $f(x)$, to be changed for whatever reason, including abortion due to impatience with the computed $\gcd(n, x_i - x_{2i})$ keeps failing to deliver the goods. We may reset x_0 in $0 \le x_0 < n$, and/or take $f(x) = x^2 + a$ for any new a in $1 \le a < n - 2$.

We can forget about the ball p, because the game has moved on to the evaluation of $\gcd(n, x_i - x_{2i})$ for the delivery of a proper divisor of n. Indeed, if or when it does, it may not always deliver p. After all, in a game involving chance, the favourite

can be beaten by an outsider, so that the algorithm may occasionally even deliver a composite proper divisor!

```
# Program: Pollard's ρ-method of factorisation.
def f(x,a,n): return (x*x+a)%n
n=(10**59-1)//9
a=random.randint(1,n-3)
s=random.randint(0,n-1)
x=y=s
found=False
count=0
while not found:
  count+=1
  if count%100000==0: print (count,"gcd done")
  x=f(x,a,n)
  y=f(y,a,n); y=f(y,a,n)
  g=gcd(n,x-y)
  if g>1 and g<n:
    print ("Factors found for",n)
    print ("Factors",g,n//g)
    found=True
  if count**4>n:
    found=True
    print ("Fail; change function and/or seed")
```

Let $N = N(n) = (10^n - 1)/9 = 111\cdots111$. It is easy to extract a factor of N when n is composite. However, there is no simple way to determine whether N is prime or composite when n is a prime. Using the program here the prime factor $p = 2559647034361$ of $N = N(59)$ was delivered in about 70 seconds after some 2 million gcd evaluations.

What has 'ρ' got to do with the method? Because of the random process involved, Pollard called his factorisation algorithm the Monte Carlo method. The values of the sequence $(x_i) \pmod p$ may be divided into two parts: the cyclic part, and the initial part before the terms enter the cycle. It is suggested that the cyclic part forming the oval shape of the Greek letter ρ, and the initial part forming the 'tail' of the letter, so practitioners call the algorithm Pollard's ρ-method.

3.6 One-Way Functions and Cryptography

Arithmetic operations in $\mathbb{Z}$ and in $\mathbb{R}$ are quite different, both in theory and in practice. Thus, from input values in $\mathbb{Z}$, an algorithm on them will deliver the output of the arithmetic operation, whereas there is no finite process which can take numbers in $\mathbb{R}$ as input values, let alone developing algorithms to perform arithmetic operations on them.

Arithmetic operations in $\mathbb{Z}$ amounts to such operations (mod n) with a suitably chosen n. However, there are still practical difficulties concerned with the efficiency of the operations, and often some ingenuity is required to overcome them. Indeed, only since the development of computers are such problems being considered. For example, it was found that there was a very significant improvement even in the usual basic operation of the multiplication of numbers, after which computer programs that often require products of very large numbers can perform much more efficiently.

For another example, we saw in Sect. 1.9 that, given $a, b > 1$, the value of $m \equiv a^b \pmod{n}$ can be delivered in $O(\log b) = O(\log n)$ multiplications if we use the 'square-and-conquer' method. However, with a, m as inputs, there is no known efficient method which can deliver b such that $a^b \equiv m \pmod{n}$, even with the assumption that b exists. We can, of course, repeatedly multiply a by itself until m emerges, but it may take $O(b)$ multiplications, and CCH (see Note 1.4.) is likely to happen. In other words, we have a rather interesting situation in arithmetic (mod n): It is easy to exponentiate, but it is difficult to take the inverse, i.e., the evaluation of the 'discrete algorithm' of m with respect to the base a. We shall study the phenomenon and the problem arisen in more detail in later chapters.

Indeed, we already remarked in Sect. 1.2 that CCH may happen if we apply trial division to find the least proper divisor of a number. In other words, we are still waiting for someone to deliver the second half of Gauss' dream: An efficient factorisation algorithm. These two seemingly intractable problems lead us to the following somewhat vague notion.

Definition 3.1 A bijection between two finite sets $\mathcal{N}$ and $\mathcal{M}$ may be called a *one-way* function if the computation of the image $m \in \mathcal{M}$ under $n \in \mathcal{N}$ is 'easy', but the computation of n from m (that is, the inverse mapping $m \mapsto n$) is 'difficult'.

The terms being 'easy' and 'difficult' are deliberately left unquantified. The reason for this is that, up to the present, if such terms are defined, no one has been able to show the existence of a one-way function according to the definition.

It has always been necessary to have a cypher scheme by which a sender uses an encoding key e to scramble a secret massage to be read only by the receiver, who then has to unscramble the cyphered message with the decoding key d. The two keys e and d are, of course, closely related to each other—they can be considered as reciprocal keys, and sometimes there is an obvious 'symmetry' between them.

Up to 1975 the only reliable, or unbreakable, cypher is the so called *one-time-pad*, in which both the encoding and decoding keys take the form of the same stream of random numbers, shared between the sender and the receiver, who therefore had to meet and agree on the construction and delivery of the random numbers for their keys. The idea is that if the message from the sender is merged with the random numbers before transmission through an open channel, only the receiver can decode the message by the removal of the random numbers in the received message. The major problem with the scheme is that the sender and the receiver had to meet to agree on the stream of random numbers, which can only be used once for a single transmission; note also that the key has to be at least as long as the message. The

most famous case for such a scheme is the Moscow-Washington hotline established in 1963 that allows direct communication between leaders of the Russian Federation and the United States using one-time pad encryption.

The cryptographic scene changed forever in 1975 when W. Diffie, M. Hellman and R. Merkle proposed new ideas and procedures by which secret messages can be exchanged between the sender and the receiver not having to meet in order to agree on a secret key; see also Note 3.4. The solution can be either in the form of two asymmetric keys, or a certain protocol for key-exchange.

Modern cryptographic schemes are based on what are perceived to be one-way functions although, as we remarked earlier, there is no function which has been *proved* to satisfy the one-way property when being easy or difficult are properly defined. Nevertheless, the factorisation problem and the discrete logarithm problem are deemed intractable enough for multiplication and exponentiation to be considered one-way functions by industry for business and commerce.

3.7 RSA Public-Key Cryptography

In 1977 R. Rivest, A. Shamir, and L. Adleman invented their famous RSA public-key-code, which is based on the assumption that factorisation of large numbers is difficult, so that multiplication of such numbers is a one-way function. Two asymmetric numerical keys, the encoding key e and the decoding key d, are involved; by asymmetric we mean that knowledge of the exact value of e offers little or no help for the construction of d.

Suppose that Axl wishes to send a secret plaintext message m, in the form of a large number with about 400 decimal digits, to Beatrice whom he need not have met before. In such a scenario, they go through the following procedure in which Beatrice the receiver has to make the first move:

- *Beatrice: Beatrice constructs two distinct primes p, q, each with some 200 decimal digits. She then evaluates $r = pq$ and $\phi(pq) = (p-1)(q-1)$. Next, she selects a random number e, coprime with $\phi(r)$, and sends Axl the pair of numbers (r, e) through an open channel. Meanwhile she computes d, the reciprocal of e (mod $\phi(r)$), that is $de \equiv 1 \pmod{\phi(r)}$, for herself.*

  ∘ *Axl: On receiving (r, e), Axl scrambles his message m by computing $M \equiv m^e \pmod{r}$, which he then sends to Beatrice through the channel.*

- *Beatrice: On receiving the scrambled message M, Beatrice uses d as the magic decoding key to unscramble M by computing $M^d \equiv m \pmod{r}$.*

The values of $\{p, q, \phi(pq)\}$ must be kept secret by Beatrice, the legitimate receiver of the message. In this scenario, a nosy eavesdropper Eve can intercept (r, e) and M, and we need to explain why Beatrice can decode M, whereas Eve cannot.

First, the decoding key d does decode because $de \equiv 1 \pmod{\phi(r)}$ so that, by the Fermat-Euler theorem, Beatrice finds that

$$M^d \equiv (m^e)^d \equiv m^{de} \equiv m^1 \equiv m \pmod{r}.$$

The interception of (r, e) and M by Eve is of no real help to her for want of the decoding key d, which requires $\phi(r)$. We now show that she cannot know $\phi(r)$.

Proposition *Let p, q be large distinct primes. Then their values can be obtained from knowledge of $r = pq$ and $\phi(r) = (p-1)(q-1)$ without applying factorisation.*

Proof Write $f = \phi(r) = (p-1)(q-1) = pq - (p+q) + 1$, so that $p + q = r - f + 1$. From $(p+q)^2 - (p-q)^2 = 4pq = 4r$, we deduce that $|p-q| = \sqrt{(r-f+1)^2 - 4r}$, so that

$$\{p, q\} = \frac{A \pm \sqrt{A^2 - 4r}}{2}, \quad A = r - \phi(r) + 1.$$

□

Thus, if Eve had knowledge of $\phi(r)$ then she would be able to find the factors p and q, contradicting the assumption that factorisation of r is difficult. Without the value of $\phi(r)$, Eve cannot compute d to decode M. Indeed, after scrambling m, even Axl himself cannot unscramble it!

The pair (r, e) is called the *public key* set up by Beatrice: Any member of the public can use the encoding key e, together with the modulus r, to scramble a message m, and only Beatrice can unscramble it with her private key d.

It is beyond our scope to consider how such a scheme might be attacked by cryptanalysts, and how extra procedures are needed in choosing the primes p and q. Except for top secret communications, large primes constructed by various methods, such as the Pocklington-Lehmer theorem will be enough. Indeed, for ephemeral communications, even large probable primes p, q can be used instead. See also Note 4.

Example 3.6 Let

$$m = 1234567890987654321$$

be the plaintext that Axl wishes to send Beatrice. Help Beatrice to find the least primes p and q, with

$$p > 1111111222222200,$$

$$q > 9999999888888800.$$

Compute $r = pq$ and $\phi(r)$, and select a random e, coprime with $\phi(r)$, for her, so that she can send the public key (r, e) to Axl.
Next, help Axl to scramble his message m by computing $M \equiv m^e \pmod{r}$.
Finally, help Beatrice to unscramble M by computing $m \equiv M^d \pmod{r}$.

Solution Let $n = 1111111222222200$. On applying the Fermat test to $n + i$ successively for $i = 1, 3, 5, \ldots$ we find that $p = n + 23 = 1111111222222223$ is a PRP(2), and trial division confirms that p is a prime. Similarly, we find that

$q = 9999999888888811$ is a prime. The relevant numbers for the scheme are:

$$\begin{aligned}
p &= 1111111222222223, \text{ (Secret kept by Beatrice)}\\
q &= 9999999888888811, \text{ (Secret kept by Beatrice)}\\
\phi(pq) &= 11111112098765329876534469135820; \text{ (Secret kept by Beatrice)}\\
&- - - - - - - - - -\\
m &= 1234567890987654321, \text{ (Axl's secret message)}\\
r = pq &= 11111112098765340987645580246853, \text{ (Publicly available modulus)}\\
e &= 88273732901308392123569253775 87; \text{ (Publicly available key)}\\
&- - - - - - - - - -\\
d &= 71748606101029600571840483947 03, \text{ (Beatrice's secret decoding key)}\\
M &= 26447036237898425760034957046 9, \text{ (Scrambled message)}\\
M^d &\equiv 1234567890987654321 (\text{mod } r). \text{ (Decoded message)}
\end{aligned}$$

□

3.8 AKS Primality Test

Corresponding to an input value $n > 1$, a polynomial-time algorithm delivering an output is one in which the number of basic arithmetic, or bit, operations involved is at most $\log^{\kappa} n$, where κ is a constant not depending on n. For a definitive primality test the output is either 'n is a prime', or 'n is composite', and the test amounts to the verification of a condition which is necessary and sufficient for n to be prime. The holy grail of such tests is a polynomial-time algorithm which is deterministic in the sense that it does not depend on probability considerations, such as making guesses. As we mentioned earlier, it is the first part of Gauss' dream, and the grail was found in 2002. The algorithm is now called the AKS test, the basis of which is the following generalisation of Fermat's little theorem:

Theorem 3.5 *Let $n > 1$, $a \in \mathbb{Z}$ and $\gcd(a, n) = 1$. Then n is prime if and only if*

$$(x + a)^n \equiv x^n + a \pmod{n}, \qquad \textit{for every } x \in \mathbb{Z}_n.$$

Proof By the binomial theorem, for $0 < i < n$, the coefficient of x^i in $(x + a)^n - (x^n + a)$ is $\binom{n}{i}a^{n-i}$. If $n = p$ is a prime, then $v_p\left(\binom{p}{i}\right) = 1$ so that $p|\binom{p}{i}$, for each such i. If n is composite then it has a prime divisor q. Let $v_q(n) = k$; then $q^k \nmid \binom{n}{q}$, and $\gcd(q, a) = 1$. Thus the coefficient of x^q is not 0 (mod n), so that $(x + a)^n - (x^n + a)$ is not identically zero over $\mathbb{Z}_n$. □

To apply the theorem, given n, we need to choose an appropriate a, or a set of values for a, so that the characterisation of n is not too onerous, but how? The clever idea is to reduce the size of the set of values for a by the introduction of an auxiliary polynomial of the form $x^r - 1$, with an appropriately chosen small r. In other words, for a coprime with n, we consider instead

$$(x + a)^n \equiv x^n + a \pmod{x^r - 1, n}; \tag{3.10}$$

here we need to explain that $f(x) \equiv 0 \pmod{h(x), n}$, with the two moduli $h(x), n$ means that $f(x) = nu(x) + h(x)v(x)$ for polynomials u, v with integer coefficients. The point of the auxiliary polynomial is that $(x + a)^n$ can now be computed quickly because, for the congruence (3.10), all polynomials can be reduced to degrees less than r, which is small relative to n.

Note that all prime values of n satisfy the new congruence (3.10), for all a coprime with n, because Theorem 3.5 still applies. However, some composite n may also satisfy the congruence for some values of r and a. The really clever part is to show that, for a judicious choice of r, there are relatively few values of a that allow composite n to satisfy (3.10). Indeed, the characterisation of n is 'almost' restored by such a scheme; almost here means that we cannot rule out the possibility that n may have a small prime divisor, or that it is a prime power. But small prime divisors can be found, and prime powers can be identified, both easily, so that characterisation *is* completely restored. The difficult part of the argument is to show that the correct output is delivered when the chosen value for r, and the resulting number of a, are both bounded by a polynomial in $\log n$.

The AKS algorithm itself is not at all complicated. In the following $\phi(n)$ is Euler's totient function, $\log_2$ means logarithm to the base 2, and $\mathrm{ord}_r(n)$ is the order of $n \pmod r$, that is the least positive integer k such that $n^k \equiv 1 \pmod r$.

AKS primality test algorithm. `Input: integer` $n > 1$.

1. (Power test) `If` $n = m^k$ `for` $m, k \in \mathbb{N}$, $k > 1$, `return composite.`
2. `Find the smallest` r `such that` $\mathrm{ord}_r(n) > \log_2^2 n$; `if` n `has a divisor in` $2 \le m \le \sqrt{\phi(r)}\,\log n$, `return composite.`
3. `For` $1 \le a \le \sqrt{\phi(r)}\,\log n$ `do:`
   `if` $(x + a)^n \not\equiv x^n + a \pmod{x^r - 1, n}$, `return composite;`

   `Return prime.`

Theorem 3.6 (Agrawal, Kayal, Saxena) *The AKS algorithm returns* `prime` *if and only if* n *is prime.*

Remarks

(i) The proof involves a fair amount of algebra (cyclotomy) and analysis (sieves); the interested reader can consult [1].

(ii) Newton's method can be used for the power test; note that $k \le \log_2 n$.

(iii) In Step 2, the integer r is to be found by sequential search, starting with $\lceil \log_2^2 n \rceil$, and it will emerge before $\lceil \log_2^5 n \rceil$ according to Theorem 3.7 below. Once r is found, we can check if $\gcd(r, n) > 1$ for a divisor of n, and the

interval of search for such divisors is reduced to $2 \le m \le \log_2^2 n$. This is because the search for r itself recognises those n with a proper divisor in the range $\log_2^2 n < m \le r$.

(iv) If n is prime the time-consuming part of the algorithm is Step 3. It is established in [1] that the asymptotic time complexity of the algorithm is $O(\log^\kappa n)$, where $\kappa > 21/2$. This has been improved to $\kappa > 6$ in [58].

Theorem 3.7 *Let $n \ge 4$, and r be the least integer with $\mathrm{ord}_r(n) > \log_2^2 n$. Then $r \le \log_2^5 n$.*

Proof Let r_0 be the least prime number that does *not* divide

$$N = n(n-1)(n^2-1)\cdots(n^\ell - 1), \qquad \ell = \lfloor \log_2^2 n \rfloor, \quad N > 31.$$

Then r_0 does not divide any of the factors $n^i - 1$ for $1 \le i \le \ell$, so that r_0 is the least prime such that $\mathrm{ord}_r(n) > \ell$, and hence $r \le r_0$.

The product of the primes dividing N is a divisor of N, so that such a product is at most

$$N < n^{1+1+2+\cdots+\ell} = n^{\frac{\ell^2}{2}+\frac{\ell}{2}+1} < n^{\log_2^4 n} = 2^{\log_2^5 n},$$

whereas the product of *all* the primes $p \le N$ exceeds 2^N when $N > 31$ (see Sect. 11.3). It follows from $r_0 \nmid N$ that $r \le r_0 \le \log_2^5 n$ when $\log_2^5 n \ge 31$, that is when $n \ge 4$. □

The AKS test sensationally settled the theoretical issue of primality testing. The proper understanding of the method involves a fair amount of algebraic number theory, particularly in the arithmetic in quotients of $(\mathbb{Z}/n\mathbb{Z})[x]$. The algorithm is rarely implemented in practice because it remains to be seen whether the test is competitive for input values n which are currently considered to be large. Thus, at present, if it is required to establish the primality of a large number, say with several hundred decimal digits, there are much speedier tests, some of which can also deliver 'primality certificates' which can be used for independent and rapid verification of the results.

3.9 Summary and Notes

Summary

- Euler discovered that the sum $\sum_{p \le x} 1/p$ over primes p is approximately $\log\log x$; the argument used gives new bounds for the prime counting function $\pi(x)$.

- There was much activity on the problems of the distribution of primes after Euler. Some important highlights are that Legendre and Gauss (independently) conjectured the *prime number theorem*, which states that

$$\pi(x) \sim \frac{x}{\log x} \quad \text{as} \quad x \to \infty.$$

  Dirichlet proved that the series $\sum 1/p$ summed over primes $p \equiv a \pmod{m}$ diverges when a, m are coprime. Chebyshev established that $\pi(x)$ does indeed have the order $x/\log x$; and Riemann showed that the secrets of $\pi(x)$ are tied up with the complex zeros of his function $\zeta(s) = \sum_{n\geq 1} 1/n^s$, where $s = \sigma + it$, with real σ, t. The prime number theorem was proved independently by Hadamard and de la Vallée Poussin in 1896.
- The *sieve of Eratosthenes* is a device for the listing of the primes $p \leq x$, using primes $p \leq \sqrt{x}$. When combined with the principle of inclusion-exclusion we have the *Eratosthenes-Legendre formula* for $\pi(x)$ which, by itself, is not much use.
- The *Meissel-Lehmer method* of counting primes is a 'partial-sieving' process of sieving only up to a prime $p(a) \leq \sqrt{x}$; it then involves the evaluation of a much shortened Legendre sum $\phi(x, a)$, together with formulae for the number of composite numbers left over as the result of partial sieving.
- Given a large n, we want an efficient method to find out if it is a prime or a composite number. The problem has been solved, at least in theory, recently by the AKS test, and there are also more efficient, but not always definitive, tests for the task. However, the problem of finding an efficient deterministic factorisation algorithm still eludes us, and such an algorithm may not exist.
- Pollard's ρ-method is a useful probabilistic scheme that can factorise a composite number of very modest size. It is based on the detection of a short cycle in a randomly chosen sequence.
- The solution to the 'birthday problem' indicates that, for randomly chosen numbers up to N, there is likely to be a collision when there are about $\sqrt{N}$ terms.
- Assuming that factorisation in intrinsically difficult, the RSA public-key cryptographic scheme enables two users to communicate with each other over an open channel with a pair of asymmetric keys.

Notes

1. *Sieve methods*. The number of terms in the sum in the sieve of Eratosthenes-Legendre (3.3) is of order $2^{\pi(\sqrt{x})}$. If we try to derive an estimate for $\pi(x)$ by simply replacing $\lfloor x/d \rfloor$ with $x/d + O(1)$ in the formula, we arrive at

$$\pi(x) = x \prod_{p \leq \sqrt{x}} \left(1 - \frac{1}{p}\right) + O(2^{\pi(\sqrt{x})}), \qquad x \to \infty.$$

There is no hope that this can be a genuine asymptotic formula, because the first term of the expression on the right-hand side is of a much smaller order of magnitude than that of the error term. As we shall see later, by Mertens' theorem (Theorem 9.3 (ii)), the main term here is asymptotic to $2e^{-\gamma}x/\log x$, where γ is Euler's constant, whereas the expected result is $\pi(x) \sim x/\log x$, according to the prime number theorem, which will be proved in due course. In any case $2^{\pi(\sqrt{x})}$ is far too large to be an error term; for example, it exceeds any fixed power of x. The stated result is therefore completely useless, and something sophisticated is required to retrieve the situation.

In about 1920, the Norwegian mathematician V. Brun created his theory of the combinatorial sieve, which is based on the observation that truncated sums of the binomial expansion $(1-1)^n$ deliver upper and lower estimates for sums of the Möbius function. The method can be used to overcome the main difficulty here, and Brun applied it to show, among other things, that the sum of the reciprocal of prime twins form a convergent series. Later (1940s) an elegant upper bound sieve method was developed by A. Selberg. The new idea is based on the observation that a Legendre-type sum can be estimated from above by the square of another sum with a free choice of an auxiliary function, so that it becomes a matter of optimising the auxiliary function. Although no one has been able to do this in general, the idea is the basis of the useful Selberg's sieve. The theory of sieves is now a thriving and important subject in analytic number theory, with many exciting new developments and some sensational results. For example, a spectacular success is the theorem of J. R. Chen which states that every large even number is the sum of a prime and a number that is a product of at most two primes. The reader interested in sieves should first consult the book [38] for an expository account.

2. *Birthday problem.* The problem asks for the probability that, in a set of n randomly chosen people, at least two of them share a birthday. It seems counterintuitive that such a probability exceeds $\frac{1}{2}$ when n is as small as 23. Apparently H. Davenport was aware of this fact in the 1920s, but he did not put it in print because he assumed that "someone must have done so already".

   We may as well assume that $n \le 365$, the number of possible birthdays in an ordinary year, and that each person is equally likely to have a birthday on any one of the days. It will be simpler to consider the probability of all such persons having distinct birthdays. Let the first person have a specific birthday from the choice of 365 days; the second person may then have 364 possible choices, and the third person then has 363 choices, and so on, so that the probability of all of them having distinct birthdays is

$$\begin{aligned} p(n) &= \frac{365}{365} \times \frac{364}{365} \times \frac{363}{365} \times \cdots \times \frac{365-n+1}{365} \\ &= \Big(1-\frac{1}{365}\Big)\Big(1-\frac{2}{365}\Big)\cdots\Big(1-\frac{n-1}{365}\Big). \end{aligned}$$

Thus $p(n)$ decreases from 1 to 0, as n increases in $1 \le n \le 366$, and it is readily checked that $p(22) > \frac{1}{2} > p(23)$.

Extension. The birthday problem is related to the pigeonhole principle: We now consider many pigeons in various holes in some random manner, and we ask for the likelihood that some hole should contain at least two pigeons.

Let (x_i) be a sequence of n integers in the interval $1 \le x \le N$, where N is large. The terms x_i may be all distinct if $n \le N$; for example, we may take $x_i = i$. Also, by the pigeonhole principle, if $n > N$, then there are two indices $j < k$ with $x_j = x_k$; we call this a matching pair, or a *collision*, when it occurs.

Suppose that there is some semblance of randomness among the numbers x_i. Then, by the solution to the birthday problem, we may expect that there is already a collision with indices much smaller than N. To see this, let

$$p(N,n) = \Big(1 - \frac{1}{N}\Big)\Big(1 - \frac{2}{N}\Big) \cdots \Big(1 - \frac{n}{N}\Big), \qquad 1 \le n < N,$$

so that

$$\log p(N,n) = \sum_{1 \le r \le n} \log\Big(1 - \frac{r}{N}\Big) \approx - \sum_{1 \le r \le n} \frac{r}{N} \approx -\frac{n^2}{N},$$

and thus $p(N,n) \approx \exp(-n^2/N)$. This means that if $n \gg \sqrt{N}$ then $p(N,n)$ is small, and we should expect collisions to occur.

3. *Fermat's method of factorisation*. Suppose that $n = pq$, where $p > q > 2$ are odd primes. Let $x_0 = \lceil \sqrt{n} \rceil$ and x_1 be the initial and final values of the iterative scheme; then $x_1^2 - z^2 = n = pq$, and hence $x_1 + z = p$, $x_1 - z = q$, giving $x_1 = (p+q)/2$. The number of steps involved in the scheme is then

$$x_1 - x_0 = \frac{p+q}{2} - \lceil \sqrt{pq} \rceil,$$

which is essentially the excess of the arithmetic-mean over the geometric-mean of the two primes p and q.

4. It has been acknowledged that public key cryptography (PKC) was first discovered inside the British secret establishment GCHQ a few years before similar discoveries being made outside. More specifically, James Ellis proposed the possibility of PKC in 1970, and devised in 1973 a system based on the difficulty of factoring. This was soon followed by the work of Malcolm Williamson and Clifford Cocks who designed protocols to be implemented. The three individuals are *Historical Figures* on the *2021 Hall of Honor Inductees* of the American secret establishment NSA.

5. Obviously size matters the most when it comes to the breaking of the RSA scheme. We need only mention the following: The RSA-768 challenge, a number with 232 decimal digits, was factorised on 12 December 2009, using

the *number field sieve*, the asymptotically fastest known algorithm for factoring large numbers that do not have particular special forms.

There is also *quantum computing* on the horizon, about which the author knows next to nothing!

3.10 Exercises

1. Write a program using the sieve of Eratosthenes in blocks to list the primes in the 10 interval $1000k < p \le 1000(k+1)$ for $k = 0, 1, 2, \ldots, 9$.
2. Let $U(x)$ count the primes $p \le x$ such that $p+2$ is also a prime, and $V(x)$ the primes $p \le x$ such that $p+2$, $p+6$, $p+8$ are also primes. Determine $U(10^k)$ for $k = 1, 2, \ldots, 6$, and $V(10^6)$.
3. Find the largest prime gap for primes $p < 10^6$.
4. Use Meissel's method to evaluate $\pi(10^9)$, by sieving through the 168 primes $p < 1000$.
5. Use the Meissel-Lehmer method to evaluate $\pi(10^{10})$, by sieving through primes up to a bound of your choice, and check the calculation with the choice of a different bound.
6. Derive the formulae for $P_2(x, a)$ and $P_3(x, a)$ in the text.
7. Show that, corresponding to each integer $k > 1$, there is a number $n = n_k$ such that $k\pi(n) = n$. Show further that there are 9 solutions of $10\pi(n) = n$, and only one solution of $11\pi(n) = n$.
8. Let $a = 3$, take various random numbers n of modest size, and evaluate $a^{n-1} \pmod n$ until the value turns out to be 1 for some $n = N$. Then change the value of a and evaluate $a^{N-1} \pmod N$. If this turns out to be 1 again for several different values of a then stop and try to find out if N is prime or not.
9. Apply the Pocklington-Lehmer theorem to construct a prime $p > 10^{150}$.
10. Apply Fermat's method to factorise 490191327, noting the number of steps in the iterative scheme. Compare the number with the formula in the Note 3.3.
11. Apply Pollard's ρ-method to factorise various chosen numbers $n \approx 10^{20}$.
12. Compute the smallest numbers n_1, n_2 such that

$$\prod_{1 \le k < n_1} \frac{365-k}{365} < \frac{2}{3}, \qquad \prod_{1 \le k < n_2} \frac{3000-k}{3000} < \frac{1}{2}.$$

13. Use any method to construct two primes $p, q \approx 10^{20}$. Take an $m < pq$ and carry out the arithmetic for the RSA scheme.
14. Give the reason why it is not safe to use the same on-time pad more than once.

Chapter 4
Quadratic Residues and Congruences



Abstract After the introduction of the Legendre's symbol and Euler's criterion, the law of quadratic reciprocity is proved using Gauss' lemma. There is a detailed explanation of how to solve a quadratic congruence, including the Tonelli–Shanks algorithm and the accompanying pseudo codes.

Consider the quadratic congruence $ax^2 + bx + c \equiv 0 \pmod{p}$, where p is an odd prime. We may assume that p does not divide a, since otherwise the congruence becomes a linear one. On multiplication by the reciprocal $\bar{a} \pmod{p}$, together with the usual method of completing a square and reassigning of symbols, the general congruence becomes

$$x^2 \equiv a \pmod{p}, \tag{4.1}$$

where we may assume again that p does not divide a. If the congruence is soluble then there are two solutions $\pm x \pmod{p}$, and either of them may be called *the* solution.

There are now two problems associated with the quadratic congruence (4.1). The first is to develop a simple procedure by which (4.1) can be determined to be soluble or not. The problem is first solved by *Euler's criterion*, but the *law of quadratic reciprocity* is much simpler, and thus more efficient, in application.

When the congruence (4.1) is known to be soluble, we have the further problem of finding the solution efficiently.

4.1 Legendre Symbol and Euler's Criterion

We say that a is a *quadratic residue*, or *nonresidue*, depending on whether (4.1) is soluble, or not. Such residues, for $p = 3, 5, 7, 11, 13$, are set out in Table 4.1.

The number $(p-1)/2$ comes up frequently in investigations, and we shall denote it by P whenever it is convenient to do so; note that P is even, or odd, depending on

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Table 4.1 Quadratic residues and nonresidues

	$p = 3$	$p = 5$	$p = 7$	$p = 11$	$p = 13$
Residues	1	1, 4	1, 2, 4	1, 3, 4, 5, 9	1, 3, 4, 9, 10, 12
Nonresidues	2	2, 3	3, 5, 6	2, 6, 7, 8, 10	2, 5, 6, 7, 8, 11

whether $p \equiv 1$, or 3, (mod 4). Also, we say 'residue' instead of quadratic residue when the context is clear.

Theorem 4.1 *Let p be an odd prime. Then there are P quadratic residues and P nonresidues; the residues are given by*

$$1^2, 2^2, \ldots, P^2 \pmod{p}, \qquad P = (p-1)/2.$$

Proof All the quadratic residues are given by $x^2 \pmod{p}$ as x runs through $1 \le x < p$; in fact the residues are run over exactly twice because $(p-x)^2 \equiv x^2 \pmod{p}$, and if $x^2 \equiv y^2 \pmod{p}$ for $1 \le x, y \le P$ then $p|x-y$ so that $x = y$. The theorem is proved. □

Legendre Symbol

The *Legendre symbol* for an odd prime p not dividing a is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} +1, & \text{if } a \text{ is a residue,} \\ -1, & \text{if } a \text{ is a nonresidue.} \end{cases}$$

It follows at once that

$$\left(\frac{a^2}{p}\right) = 1, \quad \text{and} \quad \left(\frac{a}{p}\right) = \left(\frac{b}{p}\right) \quad \text{if } a \equiv b \pmod{p}.$$

The values of the symbol for $p = 17$ are set out in Table 4.2. Whether the quadratic congruence (4.1) is soluble, or not, then amounts to the determination of the value of the Legendre symbol $\left(\frac{a}{p}\right)$. As we shall see, the symbol has the useful property of being a completely multiplicative function of a corresponding to a fixed prime p; that is

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right).$$

Table 4.2 Legendre symbol

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$\left(\frac{a}{17}\right)$	$+1$	$+1$	-1	$+1$	-1	-1	-1	$+1$	$+1$	-1	-1	-1	$+1$	-1	$+1$	$+1$

Thus, given $p > 2$, the problem becomes one of the determination of the symbols

$$\left(\frac{-1}{p}\right), \quad \left(\frac{2}{p}\right), \quad \left(\frac{q}{p}\right), \qquad q \text{ an odd prime.}$$

Experiments show that

$$\left(\frac{-1}{p}\right) = \begin{cases} +1, & \text{if } p \equiv 1 \pmod 4, \\ -1, & \text{if } p \equiv 3 \pmod 4, \end{cases}$$

which was first established in 1749 by Euler after repeated failures; later (1755) he discovered his eponymous criterion from which the result follows at once. It was pointed out by Lagrange in 1773 that it also follows from the corollary to Wilson's theorem (Theorem 1.11). Anyway, the result shows that if $p \equiv 1 \pmod 4$ then the quadratic residues come in pairs $\pm x$, whereas if $p \equiv 3 \pmod 4$ then x is a residue if and only if $-x$ is a nonresidue; see, for example, the entries in the table in Sect. 4.

Euler's Criterion

Theorem 4.2 (Euler's Criterion) *Let p be an odd prime not dividing a. Then*

$$a^P \equiv \left(\frac{a}{p}\right) \pmod p, \qquad P = \frac{p-1}{2}.$$

Proof By Fermat's Little Theorem, $a^{p-1} \equiv 1 \pmod p$, so that $a^P \equiv \pm 1 \pmod p$. If a is a quadratic residue then there exists x such that $x^2 \equiv a \pmod p$, and hence $a^P \equiv x^{p-1} \equiv 1 \pmod p$. Moreover, by Lagrange's theorem (Theorem 1.10), there can be no other solution to the polynomial congruence $x^P \equiv 1 \pmod p$ but the P quadratic residues. Consequently, if a is a nonresidue then $a^P \equiv -1 \pmod p$. The criterion is established. □

Corollary 4.1 *Let p be an odd prime. Then $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$.*

Proof By the criterion, and the range of the Legendre symbol being $\{\pm 1\}$,

$$\left(\frac{ab}{p}\right) \equiv (ab)^P \equiv a^P b^P \equiv \left(\frac{a}{p}\right)\left(\frac{b}{p}\right) \pmod p.$$

□

We already mentioned that the result for $\left(\frac{-1}{p}\right)$ follows from the criterion. For a fixed p, the problem of whether $x^2 \equiv a \pmod{p}$ is soluble or not can be considered solved, because the evaluation of $a^{(p-1)/2} \pmod{p}$ is not onerous in modular arithmetic. Nevertheless, for a fixed $a \neq \pm 1$, it is not easy to determine the set of primes p for which a is a residue; for example, it is not easy to use the criterion to evaluate $\left(\frac{2}{p}\right)$ and $\left(\frac{3}{p}\right)$, although both Euler and Legendre managed to find the answers. Indeed, from their investigations, they essentially formulated the general problem concerned, and they even came close to finding the solution.

4.2 Law of Quadratic Reciprocity

The much simpler method to determine $\left(\frac{q}{p}\right)$ is given by the following theorem.

Theorem 4.3 (Law of Quadratic Reciprocity) *Let p and q be distinct odd primes. Then*

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{(p-1)(q-1)/4}.$$

This famous result was first proved by the nineteen year old Gauss in 1796 and published in 1801 in his *Disquisitiones Arithmeticae*. The law is by far the most useful and most important result in elementary and algebraic number theories—there are dozens of themes, ideas and methods on which some 200 proofs (8 of which are by Gauss) of the law are given in the book [57]. The law and its generalisation and extension to other reciprocity laws are central to the development of algebraic number theory.

For our purpose we only consider the most basic meaning of the law and its implications. First, we note that $(p-1)(q-1)/4$ is even, unless $p \equiv q \equiv 3 \pmod{4}$, in which case it is odd. Consequently, an alternative formulation of the law is that $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$, unless $p \equiv q \equiv 3 \pmod{4}$, in which case $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$. Relating this back to quadratic congruences, the law then states that the two congruences

$$x^2 \equiv p \,(\mathrm{mod}\ q) \quad \text{and} \quad x^2 \equiv q \,(\mathrm{mod}\ p)$$

are either both soluble, or both insoluble, unless $p \equiv q \equiv 3 \pmod{4}$, in which case exactly one of them is soluble.

In science, an important discovery often contains some element of simplicity or surprise. Here, for such qualities, H. Davenport wrote in [21]:

> The law of quadratic reciprocity is one of the most famous theorems in the whole of the theory of numbers. It reveals a simple and striking relationship between the solubility of the two congruences . . ., a relationship which is by no means obvious.

Gauss' complicated first proof of the law is by induction. In 1808 he discovered a simple lemma on which his third proof was based, but the derivation of the law

from the lemma is not easy, even after much simplification. We give this proof here, and mention only that Gauss sum has been used in several proofs.

Recall that a complete residue system (mod p) is a set of p incongruent numbers modulo p, and that the integers in the interval $0 \le x < p$ form the system of least non-negative residues. Similarly, the integers in the interval $-p/2 < x < p/2$ form the system of *absolute least residues*. We delete 0 for the reduced system, so that the remaining $p-1$ residues are $\pm 1, \pm 2, \ldots, \pm P$, where $P = (p-1)/2$; note that 'absolute' here refers to being 'least' rather than being 'non-negative'.

Lemma 4.1 (Gauss' Lemma) *Let p be an odd prime not dividing a, and $P = (p-1)/2$. Denote by n the number of integers in the set $\{a, 2a, \ldots, Pa\}$ whose absolute least residues are negative. Then $\left(\frac{a}{p}\right) = (-1)^n$.*

Example 4.1 Let $p = 11$ and $a = 2$. Then $P = 5$ and the set concerned is $\{2, 4, 6, 8, 10\}$. Thus 2, 4 are the positive absolute least residues, and the absolute least residues of 6, 8, 10 are $-5, -3, -1$, so that $n = 3$, and $\left(\frac{2}{11}\right) = -1$.

Remarks We need only the parity of n for the result. Note, en passant, in this example that $\{2, 4, |-5|, |-3|, |-1|\} = \{1, 2, 3, 4, 5\}$. That this property holds in general is used in the proof of the lemma.

Proof Let $-r_1, -r_2, \ldots, -r_n$ denote the negative absolute least residues of the members of $A = \{a, 2a, \ldots, Pa\}$, and let $s_1, s_2, \ldots, s_m$ denote the remaining positive absolute least residues, so that $m + n = P$ and $1 \le r_i, s_j \le P$. Taking the product of the members of A, we then find that

$$a^P P! \equiv (-1)^n (r_1 r_2 \cdots r_n)(s_1 s_2 \cdots s_m) \pmod{p}.$$

We proceed to show that members of $B = \{r_1, r_2, \ldots, r_n, s_1, s_2, \ldots, s_m\}$ are incongruent (mod p). Since members of A are distinct (mod p), so are $r_1, r_2, \ldots, r_n$ and $s_1, s_2, \ldots, s_m$. Suppose, if possible, that $r_i \equiv s_j \pmod{p}$ for some pair i, j. Identifying $-r_i, s_j$ with their respective members of A, we may write $-r_i \equiv ga$, $s_j \equiv ha \pmod{p}$, where $1 \le g, h \le P$. Then $(g+h)a \equiv -r_i + s_j \equiv 0 \pmod{p}$, which is impossible for a coprime with p and $2 \le g + h \le 2P < p$. Therefore members of B are incongruent (mod p), so that $B = \{1, 2, \ldots, P\}$, and that

$$P! \equiv (r_1 r_2 \cdots r_n)(s_1 s_2 \cdots s_m) \pmod{p}.$$

Therefore $a^P \equiv (-1)^n \pmod{p}$, and the required result follows from Euler's criterion. □

Corollary 4.2 *We have $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$; in other words,*

$$\left(\frac{2}{p}\right) = \begin{cases} 1, & \text{if } p \equiv 1, 7 \pmod{8}, \\ -1, & \text{if } p \equiv 3, 5 \pmod{8}. \end{cases}$$

Proof Take $a = 2$ in the lemma, so that $A = \{2, 4, \ldots, 2P\}$. Since members of A are less than p, the number n in the lemma is the number of x satisfying $p/2 < 2x < p$, that is $p/4 < x \leq P$. Thus $n = P - \lfloor p/4 \rfloor$, and the required result is obtained by considering the parity of n corresponding to $p \equiv 1, 3, 5, 7 \pmod 8$. □

Corollary 4.3 *Suppose that a is odd. Then the number n in Gauss' lemma satisfies*

$$n \equiv \sum_{1 \leq x \leq P} \left\lfloor \frac{ax}{p} \right\rfloor \pmod 2.$$

Proof For $1 \leq x \leq P$, apply the division process to ax with the divisor p:

$$ax = q_x p + u_x, \qquad q_x = \left\lfloor \frac{ax}{p} \right\rfloor \quad 1 \leq u_x < p.$$

Summing over $1 \leq x \leq P$, we have

$$\frac{p^2 - 1}{8} a = p \sum_{1 \leq x \leq P} \left\lfloor \frac{ax}{p} \right\rfloor + \sum_{1 \leq x \leq P} u_x.$$

The number n is the number of x for which $u_x > p/2$. In fact, recalling the numbers r_i, s_j in the proof of the lemma, we see that

$$\{u_1, u_2, \ldots, u_P\} = \{s_1, s_2, \ldots, s_m, p - r_1, p - r_2, \ldots, p - r_n\}$$

and that $\{1, 2, \ldots, P\} = \{s_1, s_2, \ldots, s_m, r_1, r_2, \ldots, r_n\}$. Summing the terms in these two sets, and cancelling out $\sum r_i$, we find that

$$\sum_{1 \leq x \leq P} u_x + \frac{p^2 - 1}{8} = 2 \sum_{1 \leq j \leq m} s_j + np.$$

The sum $\sum u_x$ can now be eliminated in the two displayed equations involving sums to deliver

$$\frac{p^2 - 1}{8}(a + 1) = p\left(\sum_{1 \leq x \leq P} \left\lfloor \frac{ax}{p} \right\rfloor + n \right) + 2 \sum_{1 \leq j \leq m} s_j.$$

Reducing the equation (mod 2), the required result follows. □

There are various ways to simplify Gauss' third proof of the law; one of the more elegant ones is due to F. G. Eisenstein. A point (x, y) on the plane is called a *lattice point* if $x, y \in \mathbb{Z}$.

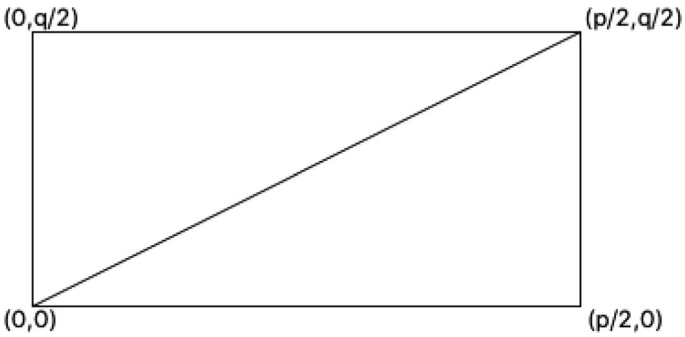


Fig. 4.1 Rectangle $\mathcal{R}$ in the proof

Lemma 4.2 (Eisenstein's Lemma) *Let p, q be distinct odd primes. Then*

$$\sum_{1\le x\le P}\left\lfloor\frac{qx}{p}\right\rfloor+\sum_{1\le x\le Q}\left\lfloor\frac{px}{q}\right\rfloor=PQ,\qquad P=\frac{p-1}{2},\quad Q=\frac{q-1}{2}.$$

Proof The number of lattice points in the rectangle

$$\mathcal{R}=\{(u,v):0<u<p/2,\ 0<v<q/2\},\qquad u,v\in\mathbb{R},$$

in Fig. 4.1 is PQ. If a lattice point (x, y) lies on the diagonal of $\mathcal{R}$ through the origin then $xq - yp = 0$ so that $p|x, q|y$, which shows that (x, y) has to be outside $\mathcal{R}$.

For $1 \le x \le P$, the number of lattice points (x, y) in $\mathcal{R}$ which lies below the said diagonal is $\lfloor qx/p\rfloor$. Similarly, for $1 \le y \le Q$, the number of lattice points on the horizontal line through y lying in the triangular region above the diagonal is $\lfloor py/q\rfloor$. Therefore the two sums in the lemma are the counts of the lattice points in the triangular regions of $\mathcal{R}$ being below and being above the diagonal, so that the required result is established. □

Proof of the Law of Quadratic Reciprocity Let P, Q be defined in Eisenstein's lemma. Set $a = q$ in Gauss' lemma, and let $m = \sum_{1\le y\le Q}\lfloor py/q\rfloor$, so that, by Corollary 2 to Gauss' lemma,

$$\left(\frac{p}{q}\right)=(-1)^m\quad\text{and}\quad\left(\frac{q}{p}\right)=(-1)^n.$$

Also, by Eisenstein's lemma, we have $m + n \equiv PQ \pmod 2$, so that

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right)=(-1)^{m+n}=(-1)^{PQ},$$

as required. □

Example 4.2 Let the prime $p = 887$, and $a = 777$. Determine whether $x^2 \equiv a \pmod{p}$ is soluble.

First Solution We find that $a^{(p-1)/2} \equiv 777^{443} \equiv 1 \pmod{p}$. By Euler's criterion, $\left(\frac{a}{p}\right) = 1$, so that the congruence is soluble.

Second Solution We consider instead that $p \equiv 3 \pmod{4}$ and $777 = 3{\cdot}7{\cdot}37$. By the law of quadratic reciprocity, we have

$$\left(\frac{3}{887}\right) = -\left(\frac{887}{3}\right) = -\left(\frac{2}{3}\right) = (-1)(-1) = +1,$$
$$\left(\frac{7}{887}\right) = -\left(\frac{887}{7}\right) = -\left(\frac{5}{7}\right) = -\left(\frac{7}{5}\right) = -\left(\frac{2}{5}\right) = (-1)(-1) = +1,$$
$$\left(\frac{37}{887}\right) = \left(\frac{887}{37}\right) = \left(\frac{36}{37}\right) = \left(\frac{-1}{37}\right) = +1$$

so that $\left(\frac{777}{877}\right) = (+1)(+1)(+1) = +1$, and the congruence is soluble. □

Example 4.3 Determine the odd primes p for which $x^2 \equiv 7 \pmod{p}$ is soluble.

Solution First, let $p \equiv 1 \pmod{4}$. Then

$$\left(\frac{7}{p}\right) = \left(\frac{p}{7}\right) = \begin{cases} +1 & \text{if} \quad p \equiv 1, 2, 4 \pmod{7}, \\ -1 & \text{if} \quad p \equiv 3, 5, 6 \pmod{7}. \end{cases}$$

Next, let $p \equiv 3 \pmod{4}$. Then

$$\left(\frac{7}{p}\right) = -\left(\frac{p}{7}\right) = \begin{cases} +1 & \text{if} \quad p \equiv 3, 5, 6 \pmod{7}, \\ -1 & \text{if} \quad p \equiv 1, 2, 4 \pmod{7}, \end{cases}$$

We therefore consider the residue classes modulo $4{\times}7 = 28$ to which p belongs, and apply the Chinese remainder theorem to give

$$\left(\frac{7}{p}\right) = \begin{cases} +1 & \text{if} \quad p \equiv 1, 3, 9, 19, 25, 27 \pmod{28}, \\ -1 & \text{if} \quad p \equiv 5, 11, 13, 15, 17, 23 \pmod{28}. \end{cases}$$

□

Fermat's Method: A Revisit

We return briefly to Fermat's method of factorisation in Sect. 3.5. A more efficient way to find a perfect square among a set of numbers z is to compute the Legendre

symbol $\left(\frac{z}{p}\right)$ for the first few primes p in a chosen list. If $\left(\frac{z}{p}\right) = -1$ for some p then z cannot be a square, so that we need only apply Newton's method to compute $\lfloor\sqrt{z}\rfloor$ for those rare z such that $\left(\frac{z}{p}\right) = +1$ for every prime p in the list.

Example 4.4 Factorise $n = 28959000082152109$ by Fermat's method, with the enhancement of using the Legendre symbol $\left(\frac{z}{p}\right)$ for primes $p \leq 43$.

Solution This is the larger n in Example 3.5, where we took $x = \lceil\sqrt{n}\,\rceil = 170173442$ and $z = x^2 - n = 279975255$. After 8838903 iterations $z \mapsto z + 2x + 1$, $x \to x + 1$ we found the factorisation $n = 123456791 \cdot 234567899$ in 230 seconds computing time.

When going through the 8838903 iterations for z, we now insert 'Legendre hurdles' with primes $p \leq 43$, breaking for the next z if $\left(\frac{z}{p}\right) = -1$. In the table below, the entry $|\{z\}|$ for a prime $p \leq 43$ is the count for z which survives the previous hurdles but now falls at the present one. The last entry is then the number of z with $\left(\frac{z}{p}\right) = +1$ for every $p \leq 43$. As it is shown in Table 4.3, there are then only 1589 surviving z requiring testing for a square, and the computing time is reduced down to 20 seconds.

Jacobi Symbol

We are not yet done in the search for a simple procedure (without applying Euler's criterion directly) to evaluate the Legendre symbol $\left(\frac{a}{p}\right)$, because we will hit a snag when we fail to determine the standard factorisation of a in order to proceed. For this reason, we need to extend the definition of the Legendre symbol to include composite numbers in the 'denominator' of the symbol; meanwhile, it will be convenient to define $\left(\frac{a}{p}\right) = 0$ when $p|a$.

Let $m = p_1 p_2 \cdots p_k$ be a product of (not necessarily distinct) odd primes, and $a \in \mathbb{Z}$. We define the *Jacobi symbol* $\left(\frac{a}{m}\right)$ by

$$\left(\frac{a}{m}\right) = \left(\frac{a}{p_1}\right)\left(\frac{a}{p_2}\right)\cdots\left(\frac{a}{p_k}\right),$$

Table 4.3 Number of z surviving prime hurdles

p	3	5	7	11	13	17	19
$\lvert\{z\}\rvert$	2946301	2357041	1515240	1101993	494479	199454	106309
p	23	29	31	37	41	43	≥ 47
$\lvert\{z\}\rvert$	61594	29239	14015	6809	3308	1532	1589

where the symbols on the right-hand side are Legendre symbols. There is consistency in that if m is prime then the two symbols are the same; note that if a, m are not coprime then $\left(\frac{a}{m}\right) = 0$.

Some care is needed when we try to relate the value of the new symbol to a quadratic congruence $x^2 \equiv a \pmod{m}$. The congruence is soluble if and only if $x^2 \equiv a \pmod{p}$ is soluble for each $p|m$. Thus if $x^2 \equiv a \pmod{m}$ is soluble then $\left(\frac{a}{m}\right) = 1$; but not necessarily conversely, because it is possible that there are precisely two p_i, p_j such that $\left(\frac{a}{p_i}\right) = \left(\frac{a}{p_j}\right) = -1$. Apart for this the two symbols enjoy essentially the same properties.

Theorem 4.4 *We have*

(i) $\left(\dfrac{-1}{m}\right) = 1$ *if and only if* $m \equiv 1$ *(mod 4);*

(ii) $\left(\dfrac{2}{m}\right) = 1$ *if and only if* $m \equiv 1, 7$ *(mod 8);*

(iii) $\left(\dfrac{m}{n}\right)\left(\dfrac{n}{m}\right) = (-1)^{\frac{(m-1)(n-1)}{4}}$ *if m, n are distinct odd numbers.*

Proof For (i) and (ii), we let $m = p_1 p_2 \cdots p_k$, where $p_i \equiv 3 \pmod{4}$ for exactly r values i. Then $\left(\frac{-1}{m}\right) = \prod_i \left(\frac{-1}{p_i}\right)$, which takes the value $+1$ if and only if r is even, that is if and only if $m \equiv 1 \pmod{4}$. A similar argument applies to $\left(\frac{2}{m}\right)$.

(iii) Let $m = p_1 \cdots p_k$, $n = q_1 \cdots q_\ell$, where p_i and q_j are odd primes, so that

$$\left(\frac{m}{n}\right) = \left(\frac{p_1 \cdots p_k}{q_1 \cdots q_\ell}\right) = \prod_{i,j} \left(\frac{p_i}{q_j}\right),$$

where there are $k\ell$ terms in the product. By the law of quadratic reciprocity (for Legendre symbols) $\left(\frac{p_i}{q_j}\right) = \left(\frac{q_j}{p_i}\right)$, unless $p_i \equiv q_j \equiv 3 \pmod{4}$, in which case we need to attach a negative sign. It follows that

$$\left(\frac{m}{n}\right) = (-1)^{rs} \prod_{i,j} \left(\frac{q_j}{p_i}\right) = (-1)^{rs} \left(\frac{n}{m}\right),$$

where r, s are the numbers of $p_i \equiv 3$, $q_j \equiv 3 \pmod{4}$, respectively. If rs is even then $\left(\frac{m}{n}\right) = \left(\frac{n}{m}\right)$, which is in agreement with (iii) because then either $m \equiv 1$ or $n \equiv 1 \pmod{4}$. If rs is odd then $\left(\frac{m}{n}\right) = -\left(\frac{n}{m}\right)$, which is also in agreement with (iii) because then $m \equiv n \equiv 3 \pmod{4}$. □

There is now no need for factorisation in the computation of Jacobi symbols, apart from the extraction of the factor 2^k. The evaluation is similar to the use of the Euclidean algorithm in the determination of the gcd without factorisation. Indeed, the complexity of the two algorithms are essentially the same, both $O(N^3)$, where N is the number of digits of the input numbers.

Example 4.5 Evaluate the Legendre symbol $\left(\frac{12345}{65537}\right)$.

Solution Using Legendre symbols, we consider $12345 = 3{\cdot}5{\cdot}823$, and

$$\left(\frac{3}{65537}\right) = \left(\frac{65537}{3}\right) = \left(\frac{2}{3}\right) = -1,$$
$$\left(\frac{5}{65537}\right) = \left(\frac{65537}{5}\right) = \left(\frac{2}{5}\right) = -1,$$
$$\left(\frac{823}{65537}\right) = \left(\frac{65537}{823}\right) = \left(\frac{520}{823}\right) = \left(\frac{2^3 5{\cdot}13}{823}\right) = (+1)^3\left(\frac{823}{5}\right)\left(\frac{823}{13}\right)$$
$$= (+1)\left(\frac{3}{5}\right)\left(\frac{2^2}{13}\right) = (+1)(-1)(+1) = -1,$$

so that $\left(\frac{12345}{65537}\right) = (-1)(-1)(-1) = -1$.

Using Jacobi symbols, we have

$$\left(\frac{12345}{65537}\right) = \left(\frac{65537}{12345}\right) = \left(\frac{3812}{12345}\right) = \left(\frac{2^2 953}{12345}\right) = (+1)\left(\frac{909}{953}\right)$$
$$= \left(\frac{2^2 11}{909}\right) = (+1)\left(\frac{909}{11}\right) = \left(\frac{2^3}{11}\right) = (-1)^3 = -1.$$

□

The following is the `Python` code for Jacobi symbols.

```
# Program: Jacobi symbol
def jacobi(a,m):
     aa=a%m; mm=m; t=1
     while aa!=0:
         while aa%2==0:
             aa=aa//2
             d=mm%8
             if d==3 or d==5: t=-t
         olda==aa
         aa=mm; mm=olda
         if aa%4==3 and mm%==3: t=-t
         aa=aa%mm
     if mm==1: return t
     return 0
```

Theorem 4.5 (M. Hall [39]) *A number $n \in \mathbb{Z}$ is a square if and only if n is a quadratic residue of every prime.*

Proof If $n = a^2$ then, for every prime p, we have $\left(\frac{n}{p}\right) = \left(\frac{a^2}{p}\right) = \left(\frac{a}{p}\right)^2 = 1$.

Suppose that n is not a square; we then need to show that there exists a prime p such that $\left(\frac{n}{p}\right) = -1$, but it is enough to show that there exists some $k > 1$ such that the Jacobi symbol $\left(\frac{n}{k}\right) = -1$. For this purpose, we consider some appropriate

conditions depending on n and apply the Chinese remainder theorem to show that there exists k satisfying such conditions.

If $n = -a^2$ then choose $k \equiv 3 \pmod 4$ and $\gcd(k, a) = 1$, so that

$$\left(\frac{n}{k}\right) = \left(\frac{-1}{k}\right)\left(\frac{a^2}{k}\right) = (-1)(+1) = -1.$$

Next, suppose that $n = \pm 2^s t$, where both s, t are odd and positive. If $t = 1$ then set $k = 5$; otherwise choose k satisfying

$$k \equiv 5 \pmod 8 \quad \text{and} \quad k \equiv 1 \pmod t.$$

Then $\left(\frac{\pm 1}{k}\right) = 1$, $\left(\frac{2}{k}\right) = -1$ and $\left(\frac{t}{k}\right) = \left(\frac{k}{t}\right) = 1$ by reciprocity, so that $\left(\frac{n}{k}\right) = (+1)(-1)^s(+1) = -1$.

The remaining case is $n = \pm 2^{2u} t$, where $t > 1$ is odd and is not a square; we may then suppose that there is an odd prime q which divides t an exact odd number of times. Thus, $n = \pm 2^{2u} q^s v$, where both s, v are odd and positive.

Let r be a quadratic nonresidue (mod q), and choose k so that

$$k \equiv 1 \pmod{4v} \quad \text{and} \quad k \equiv r \pmod q.$$

Then, by reciprocity,

$$\left(\frac{n}{k}\right) = (+1)(+1)\left(\frac{q}{k}\right)^s \left(\frac{v}{k}\right) = \left(\frac{k}{q}\right)^s \left(\frac{k}{v}\right) = \left(\frac{r}{q}\right)^s (+1) = (-1)^s = -1.$$

□

The following theorem is a much stronger result. The proof makes use of Dirichlet's theorem on the existence of primes in arithmetic progressions, which will be given in Chap. 12.

Theorem 4.6 *A number which is not a square is a quadratic nonresidue of infinitely many primes congruent to* 3 (mod 4).

Proof A number which is not a square has the form QR^2, where $Q > 1$ is squarefree. Define P by

$$P = \begin{cases} 1 + 2Q & \text{if} \quad Q \equiv 3 \pmod 4, \\ 1 + Q & \text{if} \quad Q \equiv 2 \pmod 4, \\ n & \text{if} \quad Q \equiv 1 \pmod 4, \end{cases}$$

where $n \equiv 3 \pmod 4$ is chosen with Jacobi symbol $\left(\frac{n}{Q}\right) = -1$. By Dirichlet's theorem for the existence of primes in arithmetic progressions, there are infinitely many primes $p \equiv P \pmod{4Q}$. We proceed to show that $\left(\frac{Q}{p}\right) = -1$ for each

such p, which is congruent to 3 (mod 4). Thus, if $Q \equiv 1 \pmod 4$ then, by the law of quadratic reciprocity,

$$\left(\frac{Q}{p}\right) = \left(\frac{p}{Q}\right) = \left(\frac{n}{Q}\right) = -1,$$

and if $Q \equiv 3 \pmod 4$ then

$$\left(\frac{Q}{p}\right) = -\left(\frac{p}{Q}\right) = -\left(\frac{1}{Q}\right) = -1.$$

If Q is even then, being squarefree, it is twice an odd number, so that we may consider $Q = 2(Q/2)$, and recall that $\left(\frac{2}{p}\right) = 1$ if and only if $p \equiv \pm 1 \pmod 8$. Moreover, from

$$\frac{p^2-1}{8} + \frac{Q/2-1}{2} \equiv \frac{(1+Q)^2 - 1 + 2Q - 4}{8} \equiv \frac{Q^2+4Q-4}{8} \equiv 1 \pmod 2,$$

we see that the two numbers on the left-hand side are of opposite parity, so that

$$\left(\frac{Q}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{Q/2}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{p}{Q/2}\right)(-1)^{(Q/2-1)/2} = -\left(\frac{p}{Q/2}\right)$$
$$= -\left(\frac{1}{Q/2}\right) = -1.$$

Therefore $\left(\frac{QR^2}{p}\right) = -1$ for the infinitely many primes $p \equiv P \pmod{4Q}$ which are coprime with R. □

Euler Test and Strong Pseudoprimes

The Jacobi symbol can be used to strengthen the Fermat test. If $a^{n-1} \equiv 1 \pmod n$, and n is a prime, then $a^{(n-1)/2} \equiv \pm 1 \pmod n$, although there may be other possibilities if n is composite. In any case, by Euler's criterion, we now have:

Theorem 4.7 (Euler Test) *Suppose that* $\gcd(a, n) = 1$.

$$\textit{If} \quad a^{(n-1)/2} \not\equiv \left(\frac{a}{n}\right) \pmod n, \quad \textit{then } n \textit{ is composite.}$$

□

Recall that there are pseudoprimes and Carmichael numbers which fail Fermat test. With the Euler test, many such numbers are revealed as composites. An odd composite n for which $a^{(n-1)/2} \equiv \left(\frac{a}{n}\right) \pmod n$ with $\gcd(a, n) = 1$ is called an

Euler pseudoprime for the base a, EPSP(a) for short, because its compositeness is not revealed by such a in the test. The first ten EPSP(2) are

$$561, 1105, 1729, 1905, 2047, 2465, 3277, 4033, 4681, 6601;$$

the set of EPSP(2) is a proper subset of PSP(2); indeed, about half of the PSP(a) are EPSP(a) so that, as a probabilistic primality test, Euler test is slightly stronger than Fermat test.

Example 4.6 Apply the Euler test to $n = 341$, and $n = 561$.

Solution For $n = 341$, computation shows that $2^{n-1} \equiv 1 \pmod{n}$, so that n is a Fermat probable prime. However, $2^{(n-1)/2} \equiv 1 \pmod{n}$, whereas $\left(\frac{2}{n}\right) = -1$, so that n is composite; it is a Fermat PSP(2), but not an EPSP(2).

For the Carmichael number $n = 561 \in$ EPSP(2), we can take $a = 13$, and find that $a^{(n-1)/2} \equiv 1 \pmod{n}$, whereas $\left(\frac{a}{n}\right) = -1$. Therefore n is composite. □

The idea of using Euler test instead of Fermat test can be taken further, thereby giving rise to the following strengthening of the notion of a pseudoprime.

Recall that if there exists a coprime with n such that $a^{n-1} \equiv 1 \pmod{n}$ then we say that n is a probable prime to the base a. Now let $n = 2^s t + 1$, where t is odd; if

$$a^t \equiv 1 \,(\text{mod } n) \quad \text{or} \quad a^{t2^r} \equiv -1 \,(\text{mod } n), \quad \text{for some } r \text{ in } 0 \le r < s,$$

(which implies $a^{n-1} \equiv a^{2^s t} \equiv 1 \pmod{n}$) then we say that n is a *strong probable prime* to the base a, SPRP(a) for short.

Example 4.7 Find 10 numbers $n > 10^{50}$ which are SPRP(2).

Solution Take a random $m \approx 10^{50}$, say

$$m = 235863671125473301922070344236774838417646222194 80,$$

and, for $k = 0, 1, 2, \ldots$, set $n = 2^{10}(m + k) + 1 = 2^s t + 1$, where $s \ge 10$ and t is odd. Then n is SPRP(2) for $k = 54, 192, 203, 248, 255, 339, 362, 405, 429, 459$. For example, when $k = 248$ we find that $n = 2^{14}t + 1$ and $2^{t2^{12}} \equiv -1 \pmod{n}$. □

A SPRP(a) is likely to be prime, but if it is shown to be composite, by whatever means, then we call it a *strong pseudoprime*, SPSP(a) for short. The set of SPSP(a) is proper subset of the EPSP(a), so that, as a probabilistic primality test, it is stronger than the Euler test. The first ten SPSP(2) are

$$2047, 3277, 4033, 4681, 8321, 15841, 29341, 42799, 49141, 52633.$$

Suppose that all SPSP(2) up to a bound N have been found and stored. Then, any SPRP(2) $n \le N$ which is not a SPSP(2) has to be a prime so that, by making such a check, the probabilistic primality test becomes a deterministic one for such n.

A Carmichael number may be a SPSP(a) for some a; for example, 561 is SPSP(50); but such numbers cannot be SPSP(a) for all a coprime with n.

Example 4.8 Verify that $n = 65281$ is SPSP(2).

Verification We note that $n = 255\times 2^8 + 1$, so that $s = 8$ and $t = 255$. We find that $2^{t2^r} \equiv 32768, 65217, 4096, -1, 1 \pmod{n}$, for $r = 0, 1, 2, 3, 4$, respectively, so that n is SPRP(2). Since $n = 97\times 673$ is composite, it is SPSP(2). □

4.3 Solving Quadratic Congruences (mod p)

We already remarked that a general quadratic congruence modulo a prime p can be reduced to (4.1). Let a be a quadratic residue (mod p), so that $x^2 \equiv a \pmod{p}$ has two solutions $\pm x$; we may call either one *the* solution, and we show how it can be found efficiently. Euler's criterion will be heavily involved, and the procedure is complicated for primes $p \equiv 1 \pmod 8$, but the resulting algorithm is still very fast in execution. In this section the unspecified use of the equivalence sign '$\equiv$' is meant for the modulus p.

Cases $p \equiv 3, 5, 7 \pmod 8$

The case $p \equiv 3 \pmod 4$ is simplest, because $x \equiv a^{(p+1)/4}$ is the solution. Thus, by Euler's criterion, $a^{(p-1)/2} \equiv 1$, so that

$$x^2 \equiv a^{(p+1)/2} \equiv a^{(p-1)/2}a \equiv a.$$

The case $p \equiv 5 \pmod 8$ is only slightly more complicated; since $(p-1)/2$ is even, Euler's criterion now gives $a^{(p-1)/4} \equiv \pm 1$, and there are two possibilities to consider. If $a^{(p-1)/4} \equiv 1$ then we set $x \equiv a^{(p+3)/8}$ to give $x^2 \equiv a^{(p+3)/4} \equiv a$. If $a^{(p-1)/4} \equiv -1$ then we set $x \equiv ua^{(p+3)/8}$, where $u^2 + 1 \equiv 0$, so that $x^2 \equiv -a^{(p+3)/4} \equiv a$. The congruence for u has already appeared in the corollary to Wilson's theorem (Theorem 1.11); for our present purpose, we take $u \equiv n^{(p-1)/4}$, where n is any quadratic nonresidue, because then $u^2 \equiv -1$ by Euler's criterion. In particular, $n = 2$ will do for $p \equiv 5 \pmod 8$. Therefore, for $p \equiv 5 \pmod 8$, the solution is given by

$$x \equiv \begin{cases} a^{(p+3)/8} & \text{if} \quad a^{(p-1)/4} \equiv +1, \\ 2^{(p-1)/4}a^{(p+3)/8} & \text{if} \quad a^{(p-1)/4} \equiv -1. \end{cases} \tag{4.2}$$

We have made use of the fact that -1 is a nonresidue for primes $p \equiv 3 \pmod 4$, and 2 is a nonresidue for primes $p \equiv 5 \pmod 8$. The following is the pseudo code for the solution of $x^2 \equiv a$ with $p \equiv 3, 5, 7 \pmod 8$, and a being a residue:

```
If p ≡ 3,7 (mod 8): return a^((p+1)/4).
If p ≡ 5 (mod 8):
    x ← a^((p+3)/8) (mod p); y ← x^2;
    if y ≢ a: x ← x2^((p−1)/4);
    return x.
```

Case $p \equiv 1 \pmod 8$

One of the reasons for the complication here is that we can no longer specify a nonresidue for such p. Write

$$p = 2^s t + 1, \qquad s \geq 3, \quad t \text{ odd},$$

and let $A \equiv a^t$, so that

$$A^{2^{s-1}} \equiv a^{2^{s-1}t} \equiv a^{(p-1)/2} \equiv 1$$

by Euler's criterion, which shows that $\mathrm{ord}_p(A) = 2^r < 2^s$. Moreover, if $x_0 \equiv a^{(t+1)/2}$ then $x_0^2 \equiv a^{t+1} \equiv Aa$, which shows that A is also a residue.

We now switch our attention to solving $z^2 \equiv A$ instead, so that $x^2 \equiv a$ then has the solution $x \equiv x_0 \bar{z}$. An alert reader may ask:

But $z^2 \equiv A$ is just another quadratic congruence! Is it easier to solve?

The answer is 'Yes!', because the restriction of $\mathrm{ord}_p(A)$ to being a power of 2 is the key to a much simplified procedure.

We proceed to construct a number N with order 2^s, so that any number with order $2^r < 2^s$ must be one of the numbers in $\{N, N^2, N^3, \ldots, N^{2^s}\}$. In particular, $A \equiv N^m$ for some m, which has to be even, because $1 \equiv A^{2^r} \equiv N^{m2^r}$, so that $2^s | m2^r$, and thus $2|m$. Write $m = 2\mu$, so that $z \equiv N^\mu$, and we will then have the sought after solution $x \equiv x_0 \bar{z}$. The two outstanding tasks are:

(i) Find a number N with order $2^s \pmod p$;
(ii) Find the number μ, so that $A \equiv N^{2\mu}$, and we may then set $z \equiv N^\mu$.

Lest we forget, these two tasks have to be done efficiently. Well, the first task is not difficult: we just take *any* nonresidue n; then $N \equiv n^t$ has order 2^s. This is because $N^{2^s} \equiv n^{2^s t} \equiv n^{p-1} \equiv 1$, by Fermat's Little Theorem, so that $\mathrm{ord}_p(N) | 2^s$, and indeed $\mathrm{ord}_p(N) = 2^s$ because, by Euler's criterion, $N^{2^{s-1}} \equiv n^{(p-1)/2} \equiv -1$.

Example 4.9 Solve the congruence $x^2 \equiv 13 \pmod{113}$.

Solution Here the prime $p = 113 = 2^4 7 + 1$, so that $s = 4$ and $t = 7$. Evaluating the Legendre symbol $\left(\frac{n}{p}\right)$ for successive prime values of n, we find that $n = 3, 5, 17, 19, 23, \ldots$ are nonresidues, noting en passant that $a = 13$ is a residue. We may as well choose the least $n = 3$, so that $N \equiv 3^7 \equiv 40$ should have order 16, and in fact the set $\{N, N^2, N^3, \ldots, N^{16}\}$ is equivalent (mod 113) to

$$\{40, 18, 42, 98, 78, 69, 48, 112, 73, 95, 71, 15, 35, 44, 65, 1\}.$$

Next, we have $A \equiv a^t \equiv 13^7 \equiv 69 \equiv N^6$, so that $\mu = 3$, giving $z \equiv N^\mu \equiv N^3 \equiv 42$. Meanwhile, we also compute $x_0 \equiv a^{(t+1)/2} \equiv 13^4 \equiv 85$, delivering the solution $x \equiv x_0 \bar{z} \equiv 85 \cdot \overline{42} \equiv 37$. □

With a different choice of the nonresidue n, and hence a different $N \equiv n^t$, the same 16 numbers N^i (mod p) will be reproduced, except that the exponents for N are different; in particular the value for μ may be different, but the solution x will be the same. More to the point is that, by this method, there is no systematic way to determine μ, except to search through the list of the numbers N^i for $i = 1, 2, 3, \ldots, 16 = 2^4$. Consequently, if s is large, such a method is better than a direct search for the solution x by only a factor of t.

Tonelli-Shanks Algorithm

The game changer is to make good use of $\mathrm{ord}_p(A)$ being a power of 2, thereby reducing the search space down to s, instead of 2^s. The idea came originally from A. Tonelli in 1891, who showed how μ can be found efficiently. Thus, letting $\bar{N}$ denote the reciprocal of N (mod p), we consider $A\bar{N}^m$ only for suitable even values of m until we reach the residue 1. The observation that $\mathrm{ord}_p(A\bar{N}^m)$ is a power of 2 means that we need only check the binary digits of m to arrive at the correct value of m in at most s steps.

In 1973 D. Shanks rediscovered and refined Tonelli's idea. The new approach is another case of saying 'Never mind the ball, just get on with the game!' This time the ball is μ, and the game is to devise a method by which the solution x can be delivered directly and efficiently.

Recall that there is a number $N \equiv n^t$ with order 2^s, and we set $A \equiv a^t$, $x_0 \equiv a^{(t+1)/2}$, so that $x_0^2 \equiv Aa$. Define an iterative scheme to deliver two sequences (A_i) and (x_i) such that $x_i^2 \equiv A_i a$. Moreover, the corresponding $\mathrm{ord}_p(A_i)$ form a strictly decreasing sequence of powers of 2, so that the sequence has to terminate with $A_k \equiv 1$ for some $k \leq s$, thereby delivering the solution $x \equiv x_k$ in at most s iterations.

Starting with $A_0 \equiv a^t$ having order $2^r < 2^s$, we define A_i inductively by setting

$$A_{i+1} \equiv A_i N^{2^{s-m}}, \qquad \text{if } \mathrm{ord}_p(A_i) = 2^m.$$

Thus, with A_i having order $2^m < 2^s$, then, since N has order 2^s, we have

$$A_{i+1}^{2^{m-1}} \equiv A_i^{2^{m-1}} N^{2^{s-1}} \equiv (-1)(-1) \equiv 1,$$

which shows that $\text{ord}_p(A_{i+1})|2^{m-1}$, so that $\text{ord}_p(A_{i+1}) = 2^h < 2^m$. The accompanying sequence (x_i) is given by

$$x_{i+1} \equiv x_i N^{2^{s-m-1}}, \qquad \text{if } \text{ord}_p(A_i) = 2^m.$$

The number k of iterations depends on the chosen value for the nonresidue n.

Returning to Example 4.9, we now have, with $n = 3$, $N = 40$,

$$A_0 \equiv 13^7 \equiv 69, \quad \text{ord}(A_0) = 2^3, \quad m_0 = 3; \quad x_0 \equiv 13^4 \equiv 85;$$

$$A_1 \equiv A_0 N^{2^{4-m_0}} \equiv -1, \quad \text{ord}(A_1) = 2, \quad m_1 = 1; \quad x_1 \equiv x_0 N^{2^{4-m_0-1}} \equiv 10;$$

$$A_2 \equiv A_1 N^{2^{4-m_1}} \equiv 1; \qquad x_2 \equiv x_1 N^{2^{4-m_1-1}} \equiv 76 \equiv -37;$$

so that the solution is delivered in only two iterations.

The following is the pseudo code for the general case.

```
Step 1. Define s,t so that p = 2^s t + 1 with t odd;
         Find a random n in 1<n<p with (n/p)=-1.
Step 2. Set N ≡ n^t; A ≡ a^t; x ≡ a^((t+1)/2);
Step 3. If A ≡ 1: return x.
Step 4. Set m = 1; v ≡ A^2;
         While v ≢ 1: v ← v^2; m ← m + 1.
Step 5. A ← AN^(2^(s-m)); x ← xN^(2^(s-m-1));
         Goto Step 3;
```

Example 4.10 Solve $x^2 \equiv 1161137 \pmod{138412033}$.

Solution Here $s = 22$, $t = 33$, $p = 2^s t + 1$ is a prime, and $a = 1161137$ is a quadratic residue (mod p). The number of iterations in the Tonelli-Shanks algorithm is at most $s = 22$, whereas $2^s = 2^{22} > 4{\cdot}10^6$.

We took two runs of the algorithm, with two randomly chosen nonresides n, to deliver the solution $x \equiv 67908183$ twice over. The results are given in Table 4.4, in which i counts the iterations, and ord is the order $\text{ord}_p(A_i)$.

$$n = 296150, \quad N = 91200064 \quad n = 17327937, \quad N = 97632270$$

The fly in the ointment is the random process used to find a nonresidue n in order to deliver $N \equiv n^t$ with order 2^s. There is no known efficient deterministic algorithm for finding a nonresidue; nevertheless, see the next subsection. However, since half of the reduced residues are nonresidues, it should not take long to find one by simply taking a random n and compute the Legendre symbol $\left(\frac{n}{p}\right)$; indeed, the expected number of attempts is only 2! The algorithm is very efficient because,

Table 4.4 Two runs of the Tonelli-Shanks algorithm

i	ord	A_i	x_i
0	17	9930419	135829031
1	13	47736456	18274908
2	12	125314188	116210564
3	10	26508016	120461516
4	8	36042692	79184189
5	5	64862387	12716174
6	2	138412032	110418377
7	1	1	67908183

i	ord	A_i	x_i
0	17	18687658	31466439
1	14	91493936	28948009
2	13	9037919	79322324
3	11	126313857	131349218
4	10	46709635	44178525
5	8	124127681	129399120
6	7	89351290	74646719
7	5	64862387	12716174
8	2	138412032	110418377
9	1	1	67908183

for $p = 2^s t + 1$ with t odd, the average number of multiplications involved is of order $\max(\log p, s^2) = O(\log^2 p)$.

Least Nonresidues (mod p)

The set of odd primes p for which a prime q is quadratic residue can be found by the law of quadratic reciprocity—they lie in various arithmetic progressions (mod m), where $m = q$ or $4q$, depending on whether $q \equiv 1$ or 3 (mod 4).

Example 4.11 Determine the set of primes of which a is a quadratic residue for (i) $a = 17$, and (ii) $a = 19$.

(i) By considering $x^2 \pmod{17}$ for $x = 1, 2, \ldots, 8$, we find that

$$\left(\frac{p}{17}\right) = \begin{cases} +1, & p \equiv \pm 1, \pm 2, \pm 4 \pm 8 \pmod{17}, \\ -1, & p \equiv \pm 3, \pm 5, \pm 6 \pm 7 \pmod{17}. \end{cases}$$

Since $17 \equiv 1 \pmod 4$, by the law of quadratic reciprocity, $\left(\frac{17}{p}\right) = \left(\frac{p}{17}\right)$, so that p lies in one of the 8 arithmetic progressions $p \equiv \pm 1, \pm 2, \pm 4 \pm 8 \pmod{17}$.

(ii) We have $\left(\frac{p}{19}\right) = +1$ for $p \in A \pmod{19}$, and $\left(\frac{p}{19}\right) = -1$ for $p \in B \pmod{19}$, where A is the union of the residue classes 1, 4, 5, 6, 7, 9, 11, 16, 17 (mod 19), and B is the union of the residue classes 2, 3, 8, 10, 12, 13, 14, 15, 18 (mod 19). If $p \equiv 1 \pmod 4$ then $\left(\frac{19}{p}\right) = \left(\frac{p}{19}\right) = 1$ for $p \in A$; and if $p \equiv -1 \pmod 4$ then $\left(\frac{19}{p}\right) = -\left(\frac{p}{19}\right) = 1$ for $p \in B$. Solving the two systems by the Chinese remainder theorem, we find that $\left(\frac{19}{p}\right) = 1$ for

$$p \equiv 1, 3, 5, 9, 15, 17, 25, 27, 31, 45, 49, 51, 59, 61, 67, 71, 73, 75 \pmod{4 \times 19}.$$

We remarked earlier that, given a large prime p, there is no known deterministic efficient method to find a nonresidue. At present, the most efficient way to find a nonresidue is to take random n repeatedly, compute the Legendre symbol $\left(\frac{n}{p}\right)$ using the law of quadratic reciprocity for the Jacobi symbol, until a nonresidue emerges—the expectation of success is only two trials! Instead of taking random n, we may take successive prime values, starting with $n = 2$. The same process then delivers $N(p)$, the least nonresidue (mod p), and it may not take too long, because it is known that $N(p) < \log^2 p$ if the generalised Riemann Hypothesis (the Riemann Hypothesis for Dirichlet L-functions) is true (see [53]).

It can be verified that $N(p) < 20$ for $p < 10000$. Instead of giving a table for $N(p)$, we consider, for a prime q,

$$\begin{aligned}\mathcal{P}_q &= \{p : N(p) = q\}\\ &= \{p : \left(\frac{q'}{p}\right) = +1 \text{ for each prime } q' < q, \text{ and } \left(\frac{q}{p}\right) = -1\},\end{aligned}$$

the set of odd primes p having q as the least nonresidue. Each $\mathcal{P}_q$ is a union of arithmetic progressions (mod m), where $m = m(q)$ depends on q, and the Chinese remainder theorem can be used to determine such progressions.

Thus, from $\left(\frac{2}{p}\right) = +1$ if and only if $p \equiv 1, 7 \pmod 8$, we find that

$$\mathcal{P}_2 = \{p : p \equiv 3, 5 (\text{mod } 8)\}.$$

Next, from $\left(\frac{3}{p}\right) = +1$ if and only if $p \equiv 1, 11 \pmod{12}$, we find that

$$\mathcal{P}_3 = \{p : p \equiv 7, 17 (\text{mod } 24)\}.$$

Next, from $\left(\frac{5}{p}\right) = +1$ if and only if $p \equiv 1, 4 \pmod 5$, we find that

$$\begin{aligned}\mathcal{P}_5 &= \{p : p \equiv 2, 3 (\text{mod } 5) \text{ and } p \equiv \pm 1 (\text{mod } 24)\}\\ &= \{p : p \equiv 23, 47, 73, 97 (\text{mod } 120)\};\end{aligned}$$

we also find that

$$\begin{aligned}\mathcal{P}_7 = \{p : p \equiv &\, 71, 191, 239, 241, 359, 409, 431, 481, 599,\\ &601, 649, 769 \,(\text{mod } 840)\}.\end{aligned}$$

Computations corresponding to primes $p < 10000$, show that

$$\begin{aligned}\mathcal{P}_{11} = \{&311, 719, 839, 1009, 1129, 1201, 1511, 1801, 2399, 2521, 3049, 3191,\\ &3359, 3361, 3889, 4079, 4201, 4561, 4679, 4729, 4871, 5039, 5209,\\ &5351, 5591, 5879, 5881, 6359, 6719, 6841, 8039, 8111, 8231, 8521,\end{aligned}$$

$$8689, 8929, 9551, 9791, \ldots\},$$

$$\mathcal{P}_{13} = \{479, 1151, 1319, 2351, 2689, 3529, 3671, 3911, 4751, 4919, 5519, 5569,$$
$$6599, 7559, 7561, 7681, 8951, 9241, 9601, 9719, 9769, \ldots\},$$

$$\mathcal{P}_{17} = \{1559, 2999, 6551, 8089, 8761, \ldots\}, \quad \text{and} \quad \mathcal{P}_{19} = \{5711, 9239, \ldots\},$$

Thus $N(p) < 20$ for each prime $p < 10000$.

4.4 Solving Quadratic Congruences (mod m)

If $m = \prod p^{\alpha}$ is a product of prime powers, then the congruence $x^2 \equiv a \pmod{m}$ can be considered as a system of simultaneous congruences $x^2 \equiv a \pmod{p^{\alpha}}$, with $p|m$. If $\left(\frac{a}{p}\right) = -1$ for some $p|m$ then $x^2 \equiv a \pmod{m}$ has no solution; otherwise each $x^2 \equiv a \pmod{p^{\alpha}}$ is soluble. An inductive scheme can be used to find the solutions of $x^2 \equiv a \pmod{p^{\alpha}}$, after which the solutions of $x^2 \equiv a \pmod{m}$ can be constructed by the application of the Chinese remainder theorem to the solutions of the system.

Solutions of $x^2 \equiv a \pmod{p^n}$

If a is a quadratic residue of an odd prime p then induction can be applied to find the solutions of $x^2 \equiv a \pmod{p^n}$ for every $n \geq 1$. First $x^2 \equiv a \pmod{p}$ has the solutions $\pm x_1 \pmod{p}$. Suppose, as induction hypothesis, that for some $n \geq 1$, the congruence $x^2 \equiv a \pmod{p^n}$ has the solutions $\pm x_n \pmod{p^n}$. Set $x_{n+1} = x_n + h_n p^n$, where h_n in $0 \leq h_n < p$ will be determined presently; meanwhile we can define b_n by $x_n^2 = a + b_n p^n$. Then

$$x_{n+1}^2 \equiv x_n^2 + 2x_n h_n p^n \equiv (a + b_n p^n) + 2x_n h_n p^n \equiv a + (b_n + 2x_n h_n) p^n \pmod{p^{n+1}};$$

since $p \nmid 2x_n$, the linear congruence $b_n + 2x_n h_n \equiv 0 \pmod{p}$ is soluble in h_n, thereby delivering the solutions $\pm x_{n+1}$ of $x^2 \equiv a \pmod{p^{n+1}}$. The induction step is complete and the solutions of $x^2 \equiv a \pmod{p^n}$ can be determined systematically.

Example 4.12 Solve $x^2 \equiv 13 \pmod{23^3}$.

Solution Here $p = 23 \equiv 3 \pmod{4}$, and $a = 13$ is a quadratic residue (mod p); the solutions of $x^2 \equiv a \pmod{p}$ are $x_1 \equiv \pm a^{(p-1)/2} \equiv \pm 6 \pmod{p}$.

Set $x_2 \equiv \pm 6 + hp \pmod{p^2}$, so that

$$x_2^2 \equiv 36 \pm 12hp \equiv (a + p) \pm 12hp \equiv a + (1 \pm 12h)p \pmod{p^2}.$$

Choosing $h = \mp 2$, we then find the solutions of $x^2 \equiv a \pmod{p^2}$ are given by

$$x_2 \equiv \pm 6 \mp 2p \equiv \mp 40 \pmod{p^2}, \qquad p^2 = 529.$$

Next, set $x_3 \equiv \mp 40 + kp^2 \pmod{p^3}$, so that

$$x_3^2 \equiv 1600 \mp 80kp^2 \equiv (a + 3p^2) \mp 80kp^2 \equiv a + (3 \mp 80k)p^2 \pmod{p^3}.$$

Choosing k to satisfy $80k \equiv \pm 3 \pmod{p}$, that is $k = \pm 17$, the required solutions are given by

$$x_3 \equiv \pm(40 - 17p^2) \equiv \pm 3214 \pmod{p^3}.$$

□

Example 4.13 Solve $x^2 \equiv 13 \pmod{m}$, where $m = 17 \cdot 23^3$.

Solution First, $x^2 \equiv 13 \pmod{17}$ is soluble, and the solutions are $\pm 8 \pmod{17}$. Together with the solutions to the previous example, the four required solutions to the $x^2 \equiv 13 \pmod{m}$ are given by the solutions to the simultaneous systems

$$X \equiv b_1 \pmod{23^3}, \qquad X \equiv b_2 \pmod{17},$$

where $b_1 = \pm 3214$ and $b_2 = \pm 8$. By the Chinese remainder theorem, the required solutions are found to be $X = \pm 27548, \pm 57621 \pmod{m}$. □

4.5 Gauss Sum and its Evaluation

There is a useful and important connection between the Legendre symbol and the complex exponential. Recall that we write

$$\mathrm{e}(x) = e^{2\pi i x} = \exp(2\pi i x), \qquad x \in \mathbb{R};$$

we now define, for a fixed odd prime p, the Gauss sum by

$$G(n) = \sum_{1 \le m < p} \left(\frac{m}{p}\right) \mathrm{e}\left(\frac{mn}{p}\right), \qquad n \in \mathbb{Z}. \tag{4.3}$$

We first show that

$$G = G(1) = S(1, p),$$

the quadratic Gauss sum introduced in Sect. 2.12. We have

$$G = \sum_{1\le m<p,\,\left(\frac{m}{p}\right)=1} \mathrm{e}\Big(\frac{m}{p}\Big) - \sum_{1\le m<p,\,\left(\frac{m}{p}\right)=-1} \mathrm{e}\Big(\frac{m}{p}\Big)$$

and

$$\sum_{1\le m<p,\,\left(\frac{m}{p}\right)=1} \mathrm{e}\Big(\frac{m}{p}\Big) + \sum_{1\le m<p,\,\left(\frac{m}{p}\right)=-1} \mathrm{e}\Big(\frac{m}{p}\Big) = -1$$

so that

$$G = 2 \sum_{1\le m<p,\,\left(\frac{m}{p}\right)=1} \mathrm{e}\Big(\frac{m}{p}\Big) + 1 = \sum_{0\le m<p} \mathrm{e}\Big(\frac{m^2}{p}\Big) = S(1, p).$$

Changing the variable of summation in (4.3) from m to m', where $m' \equiv mn \pmod p$, we find, on extending the definition of Legendre symbol with $\left(\frac{n}{p}\right) = 0$ for $p|n$, that

$$G(n) = \Big(\frac{n}{p}\Big)\, G, \qquad n \in \mathbb{Z}. \tag{4.4}$$

If $G \neq 0$ then

$$\Big(\frac{n}{p}\Big) = \frac{1}{G} \sum_{1\le m<p} \Big(\frac{m}{p}\Big)\, \mathrm{e}\Big(\frac{mn}{p}\Big), \tag{4.5}$$

so that a sum $\sum \left(\frac{n}{p}\right)$ over consecutive values of n becomes a geometric progression of complex exponentials in the inner sum on the right-hand side, albeit with $p-1$ such progressions. An observant reader will see that the sum is the value of the 'inner product of two vectors', with $\left(\frac{m}{p}\right)$ being multiplicative, and $\mathrm{e}\left(\frac{mn}{p}\right)$ additive.

For our present purpose, the connection between the Legendre symbol and the Gauss sum can be used to give another derivation of the law of quadratic reciprocity, which is essentially the sixth proof of the law given by Gauss. The idea is, for an odd prime $q \neq p$, to evaluate G^{q-1} in two ways, one involving $\left(\frac{q}{p}\right)$, and the other $\left(\frac{p}{q}\right)$. More specifically, by considering $\mathrm{e}(m/p)$, a complex pth root of unity which satisfies $\mathrm{e}(m/p)^p = 1$, to be an *algebraic integer*, and that the notion of a congruence (mod q) applies to such integers, we shall find that $G^{q-1} \equiv \left(\frac{q}{p}\right)$

(mod q); on the other hand, we also have $\left(\frac{G^2}{q}\right) \equiv G^{q-1} \pmod q$ by Euler's criterion. The first task is then the determination of G^2, which turns out to be $\pm p$. We consider

$$G^2 = \sum_{1\le m_1<p} \sum_{1\le m_2<p} \left(\frac{m_1 m_2}{p}\right) e\left(\frac{m_1+m_2}{p}\right) = \sum_{1\le m_1<p} \sum_{1\le n<p} \left(\frac{n}{p}\right) \mathrm{e}\left(\frac{m_1+m_1 n}{p}\right),$$

where we have used the substitution $m_2 \equiv m_1 n \pmod p$. Noting that

$$\sum_{1\le m_1<p} \mathrm{e}\left(\frac{m_1(n+1)}{p}\right) = \begin{cases} p-1 & \text{if} \quad p|n+1, \\ -1 & \text{otherwise,} \end{cases}$$

we then find, on interchanging summations, that

$$G^2 = p\left(\frac{-1}{p}\right) = \begin{cases} p & \text{if} \quad p \equiv 1 \pmod 4, \\ -p & \text{if} \quad p \equiv 3 \pmod 4. \end{cases}$$

Therefore $G \neq 0$; it will be convenient to write $G^2 = p^\star$, where

$$p^\star = (-1)^{(p-1)/2} p.$$

Now, let $q \neq p$ be an odd prime. For ordinary integers A_m, we find that $(\sum A_m)^q \equiv \sum A_m^q \pmod q$, because all the remaining terms in the sum are congruent to 0 (mod q) by the multinomial theorem. Accepting that such a congruence is also valid for algebraic integers such as $\mathrm{e}(m/p)$, we then find from (4.4) and (4.5) that

$$G^q = \left(\sum_{1\le m<p} \left(\frac{m}{p}\right) e\left(\frac{m}{p}\right)\right)^q \equiv \sum_{1\le m<p} \left(\frac{m}{p}\right) e\left(\frac{mq}{p}\right) = G\left(\frac{q}{p}\right) \pmod q.$$

Moreover, by Euler's criterion, together with $G^2 = p^\star$, we now find that

$$\left(\frac{p^\star}{q}\right) \equiv (p^\star)^{(q-1)/2} \equiv G^{q-1} \equiv \left(\frac{q}{p}\right) \pmod q,$$

so that

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = \left(\frac{pp^\star}{q}\right) = \left(\frac{(-1)^{p-1)/2}}{q}\right) = (-1)^{(p-1)(q-1)/4},$$

which is the statement for the law of quadratic reciprocity.

From $G^2 = p^\star$, we see that

$$G = \begin{cases} \pm\sqrt{p} & \text{if} \quad p \equiv 1 \pmod 4, \\ \pm i\sqrt{p} & \text{if} \quad p \equiv 3 \pmod 4, \end{cases}$$

but the determination of the correct signs for G is a highly non-trivial problem—it took Gauss four years, during which time he made many unsuccessful attempts.

Evaluation of G

The following evaluation of G by T. Estermann [30] is one of the simplest. Note that, for $p \equiv 1 \pmod 4$, it suffices to show that $G > -\sqrt{p}$.

Lemma 4.3 *We have, for* $p \equiv 1 \pmod 4$,

$$\frac{1+i}{2}(G-1) = \sum_{1 \le m < p} \mathrm{e}\Big(\frac{m^2}{4p}\Big);$$

moreover, writing the sum as $A + B$ *with* A, B *corresponding to the sums with* $1 \le m < \sqrt{p}$ *and* $\sqrt{p} < m < p$, *the following estimates are valid:*

$$\Re(1-i)A > \sqrt{p} - 1, \qquad |B| < \sqrt{p}.$$

Proof By considering the sum for G to run over m with m being residues and being nonresidues, we find that

$$G = 1 + 2 \sum_{1 \le m \le (p-1)/2} \mathrm{e}\Big(\frac{m^2}{p}\Big).$$

The required equation follows from the application of the identity

$$\sum_{1 \le m \le (p-1)/2} \Big(f(m) + f\big(\tfrac{p}{2} - m\big)\Big) = \sum_{1 \le m < p} f\big(\tfrac{m}{2}\big),$$

with $f(m) = \mathrm{e}(m^2/p)$, so that

$$f\big(\tfrac{p}{2} - m\big) = \mathrm{e}\Big(\frac{(p/2-m)^2}{p}\Big) = \mathrm{e}\Big(\frac{p}{4}\Big)\mathrm{e}\Big(\frac{m^2}{p}\Big) = i^p f(m) = i f(m).$$

Next, from $\cos\theta + \sin\theta \geq 1$ (triangle inequality!) for $0 \leq \theta \leq \pi/2$ we have

$$\Re(1-i)A = \sum_{1\leq m<\sqrt{p}} \Big(\cos\frac{\pi m^2}{2p} + \sin\frac{\pi m^2}{2p}\Big) \geq \sum_{1\leq m<\sqrt{p}} 1 > \sqrt{p} - 1.$$

For the estimate of B, we set

$$a_m = \mathrm{e}\Big(\frac{m(m+1)}{4p}\Big) = \cos\frac{\pi m(m+1)}{2p} + i\sin\frac{\pi m(m+1)}{2p}, \quad b_m = \operatorname{cosec}\frac{\pi m}{2\pi},$$

so that the real and imaginary parts of $a_m - a_{m-1}$ are given by

$$\cos\frac{\pi m(m+1)}{2p} - \cos\frac{\pi m(m-1)}{2p} = -2\sin\frac{\pi m^2}{2p}\sin\frac{\pi m}{2p},$$

and

$$\sin\frac{\pi m(m+1)}{2p} - \sin\frac{\pi m(m-1)}{2p} = 2\cos\frac{\pi m^2}{2p}\sin\frac{\pi m}{2p}.$$

It follows that $(a_m - a_{m-1})b_m = -2\sin\pi m^2/2p + 2i\cos\pi m^2/2p = 2i\mathrm{e}(m^2/4p)$, so that

$$2iB = \sum_{\sqrt{p}<m<p} (a_m - a_{m-1})b_m,$$

and the partial summation formula then delivers

$$2iB = \sum_{\sqrt{p}<m<p} a_m(b_m - b_{m+1}) - a_q b_{q+1} + a_{p-1}b_p, \qquad q = \lfloor\sqrt{p}\,\rfloor.$$

Since $|a_m| = 1$ and $b_1 > b_2 > \cdots > b_p > 0$, it follows from summing the telescopic series, together with the estimate $\sin\theta \geq 2\theta/\pi$ in $0 < \theta < \pi/2$ for b_{q+1} that

$$2|B| \leq \sum_{\sqrt{p}<m<p} (b_m - b_{m+1}) + b_{q+1} + b_p = 2b_{q+1} \leq \frac{2p}{q+1} < 2\sqrt{p},$$

as required. □

From the lemma we have

$$G - 1 = (1-i)(A+B) = \Re(1-i)(A+B) = \Re(1-i)A + \Re(1-i)B,$$

so that $G - 1 > \sqrt{p} - 1 - \sqrt{2}\sqrt{p}$, and hence $G > \sqrt{p} - \sqrt{2}\sqrt{p} > -\sqrt{p}$, which implies $G = \sqrt{p}$.

The same argument shows that $G = i\sqrt{p}$ when $p \equiv 3 \pmod 4$, and in fact Estermann gave the evaluation of the quadratic Gauss sum $G = G(n)$, where n need not be a prime.

Another evaluation of G using the Poisson summation formula will be given in Sect. 9.9.

4.6 Summary and Notes

Summary

- For an odd prime p not dividing a, the *Legendre symbol* $\left(\frac{a}{p}\right)$ takes the value $+1$, or -1, depending on whether a is a quadratic residue or nonresidue modulo p, that is whether the congruence $x^2 \equiv a \pmod p$ is soluble, or not.
- *Euler's criterion* states that $a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod p$, the value of which can thus be computed reasonably efficiently. The criterion can be used to strengthen Fermat's (probabilistic) primality test to *Euler's test*.
- The important *law of quadratic reciprocity*, which states that, for distinct odd primes p, q, $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{(p-1)(q-1)/4}$ is much more efficient to determine the quadratic character of a number, especially when the Legendre symbol has been extended to the Jacobi symbol.
- A number which is not a square is a quadratic nonresidue of infinitely many primes congruent to 3 (mod 4).
- Given that a is quadratic residue modulo p, there are efficient algorithms to solve $x^2 \equiv a \pmod p$, especially if $p \equiv 3, 5, 7 \pmod 8$. When $p \equiv 1 \pmod 8$, there is the *Tonelli-Shanks algorithm*, which is still very efficient, although we need to find a nonreidue modulo p first.
- The quadratic congruence $x^2 \equiv a \pmod m$ can be solved, first by solving for the case modulo a prime power p^α divisor of m, using the method of 'lifting' a solution from that for $\alpha = 1$. The Chinese remainder theorem is then deployed for the solutions when they exist.
- The quadratic Gauss sum can be converted to a Gauss sum corresponding to the Legendre symbol. We find that

$$\left(\frac{n}{p}\right) = \frac{1}{G} \sum_{1 \le m < p} \left(\frac{m}{p}\right) e^{2\pi i m n/p} \quad \text{where} \quad G = \sum_{1 \le m < p} \left(\frac{m}{p}\right) e^{2\pi i m/p} \neq 0.$$

  It is not difficult to show that $G^2 = \pm p$ corresponding to $p \equiv \pm 1 \pmod 4$, from which the law of quadratic reciprocity can be deduced.

- However, it is rather difficult to find the correct sign for the Gauss sum G, and Estermann has given an elementary evaluation of G.

Notes

1. A composite number n is SPSP(a) for at most a quarter of all bases a. Thus, given a random base a, the probability that a number is SPSP(a) is $\leq \frac{1}{4}$; this is the basis of the widely used Miller-Rabin primality test. P. Erdős and C. Pomerance showed (in 1986) that if a random n passes the Miller-Rabin primality test with respect to a randomly chosen base then it is a 'racing certainty' that n is a prime. There is also the Baillie-PSW test (named after Robert Baillie, Carl Pomerance, John Selfridge and Samuel Wagstaff) which involves Lucas sequences and the corresponding notion of probable primes which further reduces the chance that a probable prime is wrongly declared to be a prime. We leave it to the interested reader to pursuit the subject further by studying the relevant literature.
2. The method of 'lifting' solutions of $x^2 \equiv a \pmod{p^n}$ to those of $x^2 \equiv a \pmod{p^{n+1}}$ can be extended to polynomial congruences of higher degrees with integer coefficients. The general result was worked out by K. Hensel who went on to invent the p-adic numbers; see also Note 10.4.

4.7 Exercises

1. Let $p > 10$ be a prime. Show that there is always a pair of consecutive numbers which are both residues, and also a pair which are both nonresidues.
2. Prove that a prime $p > 5$ is representable as x^2+5y^2 only if $p \equiv 1, 9 \pmod{20}$.
3. Use Euler's criterion to determine the Legendre symbols $\left(\frac{17}{p}\right)$ for various primes of modest size.
4. Use Gauss' lemma instead to determine the Legendre symbols in the previous exercise.
5. Verify Eisenstein's lemma for two random odd primes p, q of modest size.
6. Take two random coprime odd numbers m, n of modest size, and evaluate the Jacobi symbol $\left(\frac{m}{n}\right)$.
7. Factorise $n = 448123436099907599$ by Fermat's method, with the enhancement of using the values of the Legendre symbol $\left(\frac{n}{p}\right)$ for primes $p < 60$. Note the number of square-tests being taken, and show that only 2 such tests are needed if 60 is replaced by 100.
8. Find all the Euler pseudoprimes base 3 (EPSP3) up to 10000.
9. Verify that 11 is a quadratic residue (mod 353) and find the solutions of the congruence $x^2 \equiv 11 \pmod{353}$ by following the solution to Example 4.9.
10. (i) Solve $x^2 \equiv 2 \pmod{2^q - 1}$, where q is an odd number.

(ii) Let $F_k = 2^{2^k} + 1$. Verify that $x \equiv X^3 - X \pmod{F_k}$, where $X = 2^{2^{k-2}}$, is the solution of $x^2 \equiv 2 \pmod{F_k}$.

11. Find a number t so that $p = 2^{50}t + 1$ is a prime. Determine the first three numbers a in the set $\{3, 33, 333, 3333, \ldots\}$ for which $x^2 \equiv a \pmod{p}$ is soluble, and find the solutions using the Tonelli-Shanks agorithm.
12. Prove that the least nonresidue $N(p)$ of a prime p is a prime less than $1 + \sqrt{p}$.
13. Find, for each $k = 1, 2, \ldots, 10$, the smallest prime p with its least nonresidue taking the value of the kth prime.
14. Find the solutions to the congruence $x^2 \equiv 5 \pmod{79^3}$.
15. Find the smallest prime q which is a quadratic non-residue (mod p) for every prime $p < 50$.

Chapter 5
Primitive Roots



Abstract Proofs of the existence of primitive roots modulo a prime, and the primitive-roots theorem are given. After the proof of Korselt's theorem there is a description of how to find Carmichael numbers. The key-exchange protocols in cryptography is explained with an application of the discrete logarithm.

Let p be a prime not dividing a. Recall that $\mathrm{ord}_p(a)$, the order of $a \pmod p$, is the least positive integer d such that $a^d \equiv 1 \pmod p$, so that $d|p-1$. A number g with order $p-1$ is called a *primitive root* of p; for such g, as n runs over $0 \le n < p-1$, the number $g^n \pmod p$ runs over the reduced residue system modulo p. The existence of a primitive root $g \pmod p$ shows that the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ of reduced residues $(\mathrm{mod}\ p)$ is *cyclic*, with g as the generator.

Take, for example, $p = 17$; then $g = 3$ is a primitive root, as shown in Table 5.1. On the other hand 2 is not a primitive root of 17 because $2^8 \equiv 1 \pmod{17}$, and only half of the reduced residue classes are represented.

The good news is that primitive roots do exist for any prime p; Euler was aware of this, although his argument for its existence was defective. Not so good news is that, given a large prime p, although there are plenty of primitive roots, there is no known efficient deterministic process by which one of them can be delivered.

We give the proof of the theorem on the existence of primitive roots, which is essentially due to Lagrange and Legendre independently. It will be relevant to recall Corollary 1.5 to Lagrange's theorem, which states that if $d|p-1$ then there are exactly d solutions to the congruence $x^d \equiv 1 \pmod p$. The proof amounts to keeping note of the number of solutions corresponding to a given order; see the example for $p = 13$ in Example 2.2.

More generally, g is a primitive root of $m \in \mathbb{N}$ if the order of g is $\phi(m)$.

In 1927, E. Artin made the conjecture that, corresponding to any g that is neither a square nor -1, there are infinitely many primes of which g is a primitive root—the conjecture also ascribes a relative asymptotic density to the corresponding set of primes. However, it is not yet established that any such g is a primitive root for infinitely many primes; see Note 1.

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Table 5.1 Index table for $g = 3$, $p = 17$

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3^n	1	3	9	10	13	5	15	11	16	14	8	7	4	12	2	6

5.1 Primitive Roots (mod p)

Theorem 5.1 *There are $\phi(p-1)$ primitive roots modulo a prime p.*

First Proof For each $d|p-1$, let $\psi(d)$ denote the number of x in $1 \le x < p$ which have order $d \pmod p$, so that $\sum_{a|d} \psi(a) = d$ by Corollary 1.5. By the Möbius inversion formula (Sect. 2.2) and Theorem 2.7 we then have

$$\psi(d) = \sum_{a|d} \mu(a)\frac{d}{a} = \phi(d).$$

In particular, $\psi(p-1) = \phi(p-1)$, as required. □

Such a proof cannot be used to construct a primitive root, so we give a second proof which appears to be constructive. We need a couple of lemmas first.

Lemma 5.1 *If $ord_p(a)$ and $ord_p(b)$ are coprime, then $ord_p(ab) = ord_p(a) \cdot ord_p(a)$.*

Proof Let $\mathrm{ord}_p(a) = h$, $\mathrm{ord}_p(b) = k$, and $\mathrm{ord}_p(ab) = d$. From $(ab)^{hk} \equiv a^{hk}b^{hk} \equiv 1 \pmod p$, we deduce that $d|hk$. Since h, k are coprime, we may write $d = d_1d_2$, where $d_1|h$, $d_2|k$ and $\gcd(d_1, d_2) = 1$. Let $h = \ell d_1$ and consider $a^{d_1d_2}b^{d_1d_2} \equiv (ab)^d \equiv 1 \pmod p$, which implies $a^{d_1d_2\ell}b^{d_1d_2\ell} \equiv 1 \pmod p$, so that $b^{d_2h} \equiv 1 \pmod p$. Thus $k|d_2h$, and hence $k|d_2$. Therefore $d_2 = k$, and hence $d_1 = h$, so that $d = hk$. □

Lemma 5.2 *Let p be a prime, and q a prime with q^t being the exact power which divides $p-1$. Then there are exactly $\phi(q^t)$ numbers which have order $q^t \pmod p$.*

Proof Numbers which have order $q^t \pmod p$ are solutions of $x^{q^t} - 1 \equiv 0 \pmod p$; those solutions which do not have order q^t may have order q^k with $0 \le k < t$, and such solutions also satisfy $x^{q^{t-1}} - 1 \equiv 0 \pmod p$. Thus, a number with order q^t is a solution of the former congruence, but not the latter one. By (1.5), the number of solutions to these congruences are q^t and q^{t-1}, respectively, so that the required number is $q^t - q^{t-1} = \phi(q^t)$. □

Second Proof of the Theorem Let $p - 1 = \prod_{q|p-1} q^t$, a product over distinct primes q. By Lemma 5.2, there are $\phi(q^t)$ numbers which have order q^t, so that, by Lemma 5.1, there are $\prod_{q|p-1} \phi(q^t) = \phi(p-1)$ numbers with order $p-1 \pmod p$. □

Although the second proof offers more information than what the first proof does, the elaborate process is still based on an existence argument; indeed, there is no

known efficient construction of a primitive root (mod p). In particular, we need the factorisation of $p-1$; however, if such a factorisation is available then a primitive root can be constructed accordingly. Indeed, looking back at Theorem 3.2 (Lucas-Lehmer), we see that the number a is a primitive root of the prime n.

Suppose that $p_1, p_2, \ldots, p_k$ are the distinct prime divisors of $p-1$; we may find the least primitive root g by setting $g = 2$ first, and check if it satisfies

$$g^{(p-1)/p_i} \not\equiv 1 \pmod{p}, \quad \text{for } i = 1, 2, \ldots, k.$$

If it does then g is a primitive root, and if not then repeat the check by taking $g \leftarrow g+1$ until it does.

5.2 Primitive Roots Theorem

Having shown that prime numbers p have primitive roots, we proceed to find all the numbers m for which there are primitive roots. First, a few trivial observations: 1 is the primitive root (mod 1) and (mod 2), and 3 is the primitive root (mod 4), but there is no primitive root (mod 8), because $x^2 \equiv 1 \pmod 8$ for every odd x, and $\phi(8) = 4$. Indeed, there is no primitive root (mod 2^a) for each $a \geq 3$; to see this we use induction to show that $x^{2^{a-2}} \equiv 1 \pmod{2^a}$ for each odd x. Thus, assuming this as induction hypothesis, we may write $x^{2^{a-2}} = 1 + K2^a$, and hence

$$x^{2^{a-1}} = \left(x^{2^{a-2}}\right)^2 = (1+K2^a)^2 \equiv 1 \pmod{2^{a+1}}.$$

Therefore, $\mathrm{ord}_{2^a}(x) < \phi(2^a) = 2^{a-1}$, so that there is no primitive root (mod 2^a).

The next step is to show that the square of an odd prime p has a primitive root. The strategy is to take a primitive root g (mod p) and consider $G \equiv g^{p-1} \pmod{p^2}$; if $G \equiv 1 \pmod{p^2}$ then $\mathrm{ord}_{p^2}(g) \leq p-1 < \phi(p^2)$, and g cannot be a primitive root (mod p^2), of course. However, the good news is that $G \not\equiv 1 \pmod{p^2}$ is a sufficient condition for g to be a primitive root (mod p^a) for every $a \geq 1$.

The case $G \equiv 1 \pmod{p^2}$ can be disposed of by replacing g with $h = g + p$, which is also a primitive root (mod p). The good news is that $H \equiv h^{p-1} \pmod{p^2}$ is distinct from $G \pmod{p^2}$, so that they cannot both be congruent to 1 (mod p^2), and we can then proceed with the argument. This will involve the identification of the residue class (mod p^a) to which a binomial expression $(x+y)^z$ belongs, and we shall apply the binomial theorem with no comment on the omission of terms which are multiples of p^a.

Lemma 5.3 *Let p be an odd prime, and $a \geq 2$. If $p \nmid k$ then $\mathrm{ord}_{p^a}(1+kp) = p^{a-1}$.*

Proof We use induction to show that, for each $a \geq 2$,

$$(1+kp)^{p^{a-2}} \equiv 1 + kp^{a-1} \pmod{p^a};$$

the case $a = 2$ being trivial. Taking the congruence as the induction hypothesis, the inductive step follows at once from

$$(1 + kp)^{p^{a-1}} \equiv (1 + kp^{a-1})^p \equiv 1 + pkp^{a-1} \pmod{p^{a+1}}.$$

Thus, for $p \nmid k$, we have

$$(1 + kp)^{p^{a-2}} \equiv 1 + kp^{a-1} \not\equiv 1 \pmod{p^a},$$

and also

$$(1 + kp)^{p^{a-1}} \equiv 1 + kp^a \equiv 1 \pmod{p^a},$$

which shows that p^{a-1} is the required order (mod p^a). □

Theorem 5.2 *Let p be an odd prime. Then, for each $a \in \mathbb{N}$, p^a and $2p^a$ have primitive roots.*

Proof We already know that there is a primitive root g (mod p). Let $G = g^{p-1}$, $H = (g + p)^{p-1}$, which are distinct (mod p^2), because

$$H \equiv (g + p)^{p-1} \equiv G + p(p - 1)g^{p-2} \equiv G - pg^{p-2} \not\equiv G \pmod{p^2},$$

so that one of them is *not* congruent to 1 (mod p^2). Since $g + p$ is also a primitive root (mod p), we may assume that $G \not\equiv 1 \pmod{p^2}$, and write $G = 1 + kp$, where $p \nmid k$. By Lemma (5.3), $\mathrm{ord}_{p^a}(G) = p^{a-1}$.

Let $n = n(a) = \mathrm{ord}_{p^a}(g)$, so that $n | \phi(p^a) = p^{a-1}(p - 1)$, and we proceed to determine n by writing $n = p^b u$, where $b \le a - 1$ and $u | p - 1$. Setting $p - 1 = uv$, so that $g^{p^b(p-1)} \equiv g^{p^b uv} \equiv g^{nv} \equiv 1 \pmod{p^a}$, we find that $G^{p^b} \equiv 1 \pmod{p^a}$. Since $\mathrm{ord}_{p^a}(G) = p^{a-1}$, it follows that $b = a - 1$.

Next, from $p - 1 | p^{a-1} - 1$, we have

$$g^u \equiv (g^{p-1})^u g^u \equiv (g^{p^{a-1}-1})^u g^u \equiv g^{p^{a-1}u} \equiv g^n \equiv 1 \pmod{p},$$

which implies $u = p - 1$, because g is a primitive root (mod p). Therefore $n = p^{a-1}(p - 1) = \phi(p^a)$, so that g is also a primitive root (mod p^a).

Lastly, we deduce from $\phi(2p^a) = \phi(p^a)$ that $2p^a$ also possesses a primitive root. Take a primitive root g (mod p^a); we may assume that g is odd, since otherwise we can replace it by $g + p^a$; let $n = \phi(p^a) = \mathrm{ord}_{p^a}(g)$. Now $g^n \equiv 1 \pmod{p^a}$ implies $g^n \equiv 1$ or $1 + p^a \pmod{2p^a}$, and the latter does not apply because g is odd. It follows that $n = \phi(2p^a)$ is also $\mathrm{ord}_{2p^a}(g)$. The theorem is proved. □

Theorem 5.3 (Primitive Roots Theorem) *The numbers $m = 1, 2, 4, p^a, 2p^a$, where p is an odd prime, and $a \in \mathbb{N}$, are the only numbers which possess primitive roots.*

Proof We have already established that $m = 1, 2, 4, p^a, 2p^a$ do possess primitive roots, and we have disposed of the case $m = 2^a$ for $a \geq 3$. The remaining values of m to be considered have the form $m = uv$, where u, v are coprime numbers with both $\phi(u)$ and $\phi(v)$ even. This is because if $u = 2^a$, where $a \geq 2$, then $\phi(u)$ is even; if $u = 2$ then v has more than one odd prime divisor, say $v = p^b Q$, where $p \nmid Q$, $Q \geq 3$, and we can replace u, v with $2p^b$, Q, so that $(u, v) = 1$, and both $\phi(u)$ and $\phi(v)$ are even. Now, for every x coprime with m, we have

$$\left(x^{\phi(u)}\right)^{\phi(v)/2} \equiv 1 \pmod{u} \quad \text{and} \quad \left(x^{\phi(v)}\right)^{\phi(u)/2} \equiv 1 \pmod{v},$$

so that, by the Chinese remainder theorem,

$$x^{\phi(m)/2} \equiv x^{\phi(u)\phi(v)/2} \equiv 1 \pmod{m}.$$

Therefore there is no primitive root for such m. □

Although there is no primitive root (mod 2^a) if $a \geq 3$, the number 5 can be used as substitute for such a root in the sense of the following theorem.

Theorem 5.4 *Let $a \geq 3$. Then, corresponding to each odd integer n, there is a unique $b = b(n)$ such that*

$$n \equiv (-1)^{(n-1)/2} 5^b \pmod{2^a}, \qquad 1 \leq b \leq \tfrac{1}{2}\phi(2^a).$$

Proof If $n \equiv 1 \pmod 4$ then $(n-1)/2$ is even, and 5^b, with $0 \leq b < 2^{a-2}$, gives 2^{a-1} distinct numbers (mod 2^a) all congruent to 1 (mod 4), so that $n \equiv 5^b \pmod{2^a}$ for a unique b. If $n \equiv 3 \pmod 4$ then $-n \equiv 1 \pmod 4$, and the required result also holds. □

Example 5.1 Identify the reduced residue classes (mod 16) in the theorem with the ordered pairs $(\pm 1, 5^b)$, $b = 1, 2, 3, 4$.

Solution For $b = 1, 2, 3, 4$, we have $5^b \equiv 5, 9, 13, 1$ and $-5^b \equiv 11, 7, 3, 15$ (mod 16), and Table 5.2 gives the required identification:

The group $(\mathbb{Z}/2^a\mathbb{Z})^\times$ is not cyclic when $a \geq 3$, but it can be considered as a direct product of a cyclic group of order 2^{a-2} with the group $\{\pm 1\}$ of order 2.

Theorem 5.5 *Suppose that m has a primitive root g. Then there are exactly $\phi(\phi(m))$ primitive roots (mod m) which are given by*

$$\mathcal{G} = \{g^n : 1 \leq n \leq \phi(m),\ \gcd(n, \phi(m)) = 1\}.$$

Table 5.2 Identification of reduced residue classes (mod 16)

1	3	5	7	9	11	13	15
$(1, 5^4)$	$(-1, 5^3)$	$(1, 5)$	$(-1, 5^2)$	$(1, 5^2)$	$(-1, 5)$	$(1, 5^3)$	$(-1, 5^4)$

Proof Let g be a fixed primitive root (mod m), so that $\mathrm{ord}_m(g) = \phi(m)$, and $\mathrm{ord}_m(g^n)$ is the least k such that $g^{kn} \equiv 1 \pmod{m}$. This means that k is the least positive integer such that $kn | \phi(m)$. Thus $\mathrm{ord}_m(g^n) = \phi(m)$ if $\gcd(n, \phi(m)) = 1$, so that each member of $\mathcal{G}$ is a primitive root (mod m).

Conversely, if h is any primitive root (mod m) then $h \equiv g^n \pmod{m}$ for some n with $1 \le n \le \phi(m)$, so that $\phi(m)$ is the order of $g^n \equiv h \pmod{m}$, and we need to have $\gcd(n, \phi(m)) = 1$. Therefore every primitive root (mod m) is a member of $\mathcal{G}$. Since $|\mathcal{G}| = \phi(\phi(m))$, the theorem is proved. □

Let $m = p^a$, where p is an odd prime, and $a \ge 1$. We already remarked that there is no known process by which a primitive root (mod m) can be delivered efficiently. The probability of a randomly chosen number being such a root (mod m) is less than $\frac{1}{2}$ but, for some m, it can be just short of $\frac{1}{2}$. It is therefore of some interest to take note of such special values of m; we ignore the case $2m = 2p^a$, because $\phi(\phi(2m)) = \phi(\phi(m))$, and the said probability is less than $\frac{1}{4}$. From $\phi(m) = p^{a-1}(p-1)$, and $p \nmid p-1$, we find that

$$\phi(\phi(m)) = \phi(p^{a-1})\phi(p-1) = \begin{cases} \phi(p-1) & \text{if } a = 1, \\ p^{a-2}(p-1)\phi(p-1) & \text{if } a > 1, \end{cases}$$

so that

$$\frac{\phi(\phi(m))}{\phi(m)} = \frac{\phi(p-1)}{p - \theta(a)}, \qquad \theta(a) = \begin{cases} 1 & \text{if } a = 1, \\ 0 & \text{if } a > 1. \end{cases}$$

Odd primes p for which $\phi(p-1)$ is near $p/2$ take the form

$$p = 2^s q + 1,$$

where q is an odd number with very few small prime divisors; in particular, if $q = 1$, then $p = 2^s + 1$ has to be a *Fermat prime*, of which only 5 are known. If q is an odd prime then $\phi(p-1) = 2^{s-1}(q-1)$, and $\phi(m) = p^{a-1}(p-1)$, so that

$$\frac{\phi(\phi(m))}{\phi(m)} = \frac{2^{s-1}(q-1)}{p - 1 + (1 - \theta(a))} = \frac{2^{s-1}(q-1)}{2^s q + (1 - \theta(a))} \approx \frac{1}{2}\Big(1 - \frac{1}{q}\Big).$$

Take, for example, $q = 97$, for which $p = 4q + 1 = 389$ is a prime. Then, for any $a > 1$ and $m = p^a$, we have

$$\frac{\phi(\phi(m))}{\phi(m)} = \frac{2 \times 96}{4 \times 97 + 1} \approx 0{\cdot}49357.$$

The following is the list of the 467 primes $p < 10000$ of which 10 is a primitive root. For such p, the period length of the decimal expansion of $1/p$ is $p - 1$.

7, 17, 19, 23, 29, 47, 59, 61, 97, 109, 113, 131, 149, 167, 179, 181, 193, 223, 229, 233, 257, 263, 269, 313, 337, 367, 379, 383, 389, 419, 433, 461, 487, 491, 499, 503, 509, 541, 571, 577, 593, 619, 647, 659, 701, 709, 727, 743, 811, 821, 823, 857, 863, 887, 937, 941, 953, 971, 977, 983, 1019, 1021, 1033, 1051, 1063, 1069, 1087, 1091, 1097, 1103, 1109, 1153, 1171, 1181, 1193, 1217, 1223, 1229, 1259, 1291, 1297, 1301, 1303, 1327, 1367, 1381, 1429, 1433, 1447, 1487, 1531, 1543, 1549, 1553, 1567, 1571, 1579, 1583, 1607, 1619, 1621, 1663, 1697, 1709, 1741, 1777, 1783, 1789, 1811, 1823, 1847, 1861, 1873, 1913, 1949, 1979, 2017, 2029, 2063, 2069, 2099, 2113, 2137, 2141, 2143, 2153, 2179, 2207, 2221, 2251, 2269, 2273, 2297, 2309, 2339, 2341, 2371, 2383, 2389, 2411, 2417, 2423, 2447, 2459, 2473, 2539, 2543, 2549, 2579, 2593, 2617, 2621, 2633, 2657, 2663, 2687, 2699, 2713, 2731, 2741, 2753, 2767, 2777, 2789, 2819, 2833, 2851, 2861, 2887, 2897, 2903, 2909, 2927, 2939, 2971, 3011, 3019, 3023, 3137, 3167, 3221, 3251, 3257, 3259, 3299, 3301, 3313, 3331, 3343, 3371, 3389, 3407, 3433, 3461, 3463, 3469, 3527, 3539, 3571, 3581, 3593, 3607, 3617, 3623, 3659, 3673, 3701, 3709, 3727, 3767, 3779, 3821, 3833, 3847, 3863, 3943, 3967, 3989, 4007, 4019, 4051, 4057, 4073, 4091, 4099, 4127, 4139, 4153, 4177, 4211, 4217, 4219, 4229, 4259, 4261, 4327, 4337, 4339, 4349, 4421, 4423, 4447, 4451, 4457, 4463, 4567, 4583, 4651, 4673, 4691, 4703, 4783, 4793, 4817, 4931, 4937, 4943, 4967, 5021, 5059, 5087, 5099, 5153, 5167, 5179, 5189, 5233, 5273, 5297, 5303, 5309, 5381, 5393, 5417, 5419, 5501, 5503, 5527, 5531, 5581, 5623, 5651, 5657, 5659, 5669, 5701, 5737, 5741, 5743, 5749, 5779, 5783, 5807, 5821, 5857, 5861, 5869, 5897, 5903, 5927, 5939, 5981, 6011, 6029, 6047, 6073, 6113, 6131, 6143, 6211, 6217, 6221, 6247, 6257, 6263, 6269, 6287, 6301, 6337, 6343, 6353, 6367, 6389, 6473, 6553, 6571, 6619, 6659, 6661, 6673, 6691, 6701, 6703, 6709, 6737, 6779, 6793, 6823, 6829, 6833, 6857, 6863, 6869, 6899, 6949, 6967, 6971, 6977, 6983, 7019, 7057, 7069, 7103, 7109, 7177, 7193, 7207, 7219, 7229, 7247, 7309, 7349, 7393, 7411, 7433, 7451, 7457, 7459, 7487, 7499, 7541, 7577, 7583, 7607, 7673, 7687, 7691, 7699, 7703, 7727, 7753, 7793, 7817, 7823, 7829, 7873, 7901, 7927, 7937, 7949, 8017, 8059, 8069, 8087, 8171, 8179, 8219, 8233, 8263, 8269, 8273, 8287, 8291, 8297, 8353, 8377, 8389, 8423, 8429, 8447, 8501, 8513, 8537, 8543, 8623, 8647, 8663, 8669, 8699, 8713, 8731, 8741, 8753, 8783, 8807, 8819, 8821, 8861, 8863, 8887, 8971, 9011, 9029, 9059, 9103, 9109, 9137, 9221, 9257, 9341, 9343, 9371, 9377, 9421, 9461, 9473, 9491, 9497, 9539, 9623, 9629, 9697, 9739, 9743, 9749, 9767, 9781, 9811, 9817, 9829, 9833, 9851, 9857, 9887, 9931, 9949, 9967.

The least primitive root (mod p) for odd primes $p < 1000$, together with list of primes p in $1000 < p < 10000$, of which the primitive root exceeds 10, are given in Table 5.3.

Table 5.3 Least primitive roots for $p < 1000$

p	g	p	g	p	g	p	g	p	g	p	g
3	2	211	2	479	13	773	2	1801	11	5591	11
5	2	223	3	487	3	787	2	1871	14	5641	14
7	3	227	2	491	2	797	2	2161	23	5689	11
11	2	229	6	499	7	809	3	2287	19	5711	19
13	2	233	3	503	5	811	3	2351	13	5879	11
17	3	239	7	509	2	821	2	2399	11	5881	31
19	2	241	7	521	3	823	3	2521	17	6079	17
23	5	251	6	523	2	827	2	2689	19	6271	11
29	2	257	3	541	2	829	2	2857	11	6359	13
31	3	263	5	547	2	839	11	2953	13	6361	19
37	2	269	2	557	2	853	2	2999	17	6551	17
41	6	271	6	563	2	857	3	3001	14	6581	14
43	3	277	5	569	3	859	2	3049	11	6599	13
47	5	281	3	571	3	863	5	3191	11	6719	11
53	2	283	3	577	5	877	2	3359	11	6841	22
59	2	293	2	587	2	881	3	3361	22	6961	13
61	2	307	5	593	3	883	2	3529	17	7559	13
67	2	311	17	599	7	887	5	3631	15	7561	13
71	7	313	10	601	7	907	2	3671	13	7681	17
73	5	317	2	607	3	911	17	3881	13	7841	12
79	3	331	3	613	2	919	7	3889	11	8011	14
83	2	337	10	617	3	929	3	3911	13	8039	11
89	3	347	2	619	2	937	5	4079	11	8089	17
97	5	349	2	631	3	941	2	4111	12	8111	11
101	2	353	3	641	3	947	2	4129	13	8191	17
103	5	359	7	643	11	953	3	4201	11	8231	11
107	2	367	6	647	5	967	5	4391	14	8521	13
109	6	373	2	653	2	971	6	4441	21	8641	17
113	3	379	2	659	2	977	3	4561	11	8681	15
127	3	383	5	661	2	983	5	4591	11	8689	13
131	2	389	2	673	5	991	6	4657	15	8761	23
137	3	397	5	677	2	997	7	4679	11	8779	11
139	2	401	3	683	5	1009	11	4729	17	8929	11
149	2	409	21	691	3	1031	14	4751	19	8951	13
151	6	419	2	701	2	1129	11	4861	11	9239	19
157	5	421	2	709	2	1151	17	4871	11	9241	13
163	2	431	7	719	11	1201	11	4919	13	9439	22
167	5	433	5	727	5	1319	13	4969	11	9551	11
173	2	439	15	733	6	1321	13	5039	11	9601	13
179	2	443	2	739	3	1399	13	5113	19	9719	17
181	2	449	3	743	5	1489	14	5209	17	9769	13

(continued)

Table 5.3 (continued)

p	g	p	g	p	g	p	g	p	g	p	g
3	2	211	2	479	13	773	2	1801	11	5591	11
181	2	449	3	743	5	1489	14	5209	17	9769	13
191	19	457	13	751	3	1511	11	5351	11	9791	11
193	5	461	2	757	2	1559	19	5519	13	9973	11
197	2	463	3	761	6	1597	11	5521	11		
199	3	467	2	769	11	1657	11	5569	13		

5.3 Carmichael Function and Carmichael Numbers

The Fermat-Euler theorem states that $a^{\phi(n)} \equiv 1 \pmod{n}$ for every a coprime with n. However, $\phi(n)$ is not always the least number m depending on n with such a property. Take, for example, $n = 8$, so that $\phi(n) = 4$, but $m = 2$ also has the same property, because $a^2 \equiv 1 \pmod{8}$ for all odd a.

The *Carmichael function*, named after R. D. Carmichael, is denoted by $\lambda(n)$; it should not be confused with the Liouville lambda function. It is defined to be the least $m \in \mathbb{N}$ such that $a^m \equiv 1 \pmod{n}$ for every a coprime with n; it is also called the *reduced totient function*, or the *least universal exponent function*. Note that, by the division process, $\lambda(n)|\phi(n)$; Table 5.4 gives the first few n at which $\lambda(n)$ is a proper divisor of $\phi(n)$.

If n has a primitive root, then $\lambda(n) = \phi(n)$, but $\lambda(n)$ is not multiplicative; for example, $\lambda(3) = 2$, $\lambda(5) = 4$, and $\lambda(15) = 4$. Also, for $a \geq 3$, we know that $x^{2^{a-2}} \equiv 1 \pmod{2^a}$ for each odd x, and that $\mathrm{ord}_{2^a}(5) = 2^{a-2}$, so that $\lambda(2^a) = 2^{a-2}$. However, it is not immediately obvious that, given $n \in \mathbb{N}$, there exists x with $\mathrm{ord}_n(x) = \lambda(n)$, the largest possible value for an order (mod n).

Theorem 5.6 *We have*

$$\lambda(n) = \begin{cases} \phi(n), & n = 1, 2, 4, \text{ and } n = p^a,\ a = 1, 2, \ldots, \\ \frac{1}{2}\phi(n), & n = 2^a,\ a = 3, 4, \ldots, \end{cases}$$

where p is an odd prime; also, for the standard factorisation $n = 2^{a_0} p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$,

$$\lambda(n) = LCM\left[\lambda(2^{a_0}), \phi(p_1^{a_1}), \phi(p_2^{a_2}), \ldots, \phi(p_k^{a_k})\right].$$

Table 5.4 Values of n with $\lambda(n)$ being a proper divisor of $\phi(n)$

n	8	12	15	16	20	21	24	28	30	32	33	35	36
$\lambda(n)$	2	2	4	4	4	6	2	6	4	8	10	12	6
$\phi(n)$	4	4	8	8	8	12	8	12	8	16	20	24	12

Corollary 5.1

(i) For each $n \in \mathbb{N}$, there exists x with $ord_n(x) = \lambda(n)$;
(ii) If $m|n$ then $\lambda(m)|\lambda(n)$, and $\lambda(n) \to \infty$ as $n \to \infty$.

Proof The first part of the theorem has already been dealt with. For the second part, write

$$N = \text{LCM}\left[\lambda(2^{a_0}), \phi(p_1^{a_1}), \phi(p_2^{a_2}), \ldots, \phi(p_k^{a_k})\right],$$

a multiple of each of the numbers $\lambda(2^{a_0}), \phi(p_1^{a_1}), \phi(p_2^{a_2}), \ldots, \phi(p_k^{a_k})$. Thus, for each x coprime with n, we have $x^N \equiv 1 \pmod{p^b}$, where p^b is a prime power factor of n, and hence $x^N \equiv 1 \pmod{n}$ by the Chinese remainder theorem. To show that $\lambda(n) = N$, it suffices to construct a number x with order $N \pmod{n}$.

Let g_i be a primitive root $\pmod{p_i^{a_i}}$, and consider the system of $k + 1$ simultaneous congruences

$$x \equiv 5 \pmod{2^{a_0}}, \qquad x \equiv g_i \pmod{p_i^{a_i}}, \quad i = 1, 2, \ldots, k.$$

By the Chinese remainder theorem, there is a simultaneous solution $x \pmod{n}$ to the system, and we proceed to show that x does have order $N \pmod{n}$.

Let M be any integer such that $x^M \equiv 1 \pmod{n}$, so that $x^M \equiv 1 \pmod{p_i^{a_i}}$, and thus M is a multiple of $\text{ord}_{p_i^{a_i}}(x)$. Since x is the solution of the simultaneous system, such an order is given by $\lambda(p_i^{a_i})$, so that $\lambda(p_i^{a_i})|M$ for each of $p_i^{a_i}$ in the factorisation of n. Therefore $N|M$ for any M satisfying $x^M \equiv 1 \pmod{n}$, so that N is the order of x, and hence $\lambda(n) = N$, as required. The corollary follows immediately. □

Example 5.2 Let $n = 180 = 2^2 3^2 5$, so that $\phi(n) = \phi(4)\phi(9)\phi(5) = 2{\cdot}6{\cdot}4 = 48$, and $\lambda(180) = \text{LCM}[2, 6, 4] = 12$. Find some x having order $12 \pmod{n}$.

Solution Each x coprime with $n = 180$ satisfies $x^{48} \equiv 1 \pmod{n}$; they are the solutions to the simultaneous systems of three congruences

$$x \equiv b_1 \pmod 4, \qquad x \equiv b_2 \pmod 9, \qquad x \equiv b_3 \pmod 5,$$

with $b_1 \in \{1, 3\}$, $b_2 \in \{1, 2, 4, 5, 7, 8\}$, $b_3 \in \{1, 2, 3, 4\}$; there are 48 such systems.

If we restrict b_1, b_2, b_3 to be primitive roots $g_1, g_2, g_3 \pmod{4,9,5}$, respectively, then the corresponding solutions will have order $\lambda(n) = 12$. There is only one primitive root (mod 4), namely $g_1 = 3$; there are $\phi(\phi(9)) = \phi(6) = 2$ primitive roots (mod 9), namely $g_2 = 2, 5$; there are $\phi(\phi(5)) = \phi(4) = 2$ primitive roots (mod 5), namely $g_3 = 2, 3$. Following the proof of the theorem, there are at least 4 values x with order 12; they are solutions to the 4 simultaneous systems of three congruences

$$x \equiv 3 \pmod 4, \qquad x \equiv 2, 5 \pmod 9, \qquad x \equiv 2, 3 \pmod 5,$$

that is $x \equiv 23, 47, 83, 167 \pmod{180}$. □

Recall that a Carmichael number is a composite number n with the property that $b^{n-1} \equiv 1 \pmod{n}$ for every b coprime with n. A direct search for Carmichael numbers is onerous; the following theorem reduces the search space considerably.

Theorem 5.7 (A. Korselt) *A necessary and sufficient condition for a number n to be a Carmichael number is that n should be odd, squarefree, and that* $p-1|n-1$ *for each prime* $p|n$.

Corollary 5.2 *Carmichael numbers have at least three distinct prime divisors.*

Proof Let n be a Carmichael number; it cannot be even since otherwise $(-1)^{n-1} \equiv 1 \pmod{n}$ implies $n = 2$. We may now write $n = \prod p_i^{a_i}$, where p_i are odd primes, and $a_i \geq 1$; we suppress the index i in the following.

Let g be a primitive root $(\text{mod } p^a)$ for each p^a. Apply the Chinese remainder theorem to the simultaneous system $x \equiv g \pmod{p^a}$, so that its solution x is coprime with n. Since n is a Carmichael number, so that $x^{n-1} \equiv 1 \pmod{n}$, and hence $x^{n-1} \equiv 1 \pmod{p^a}$, it follows that $g^{n-1} \equiv 1 \pmod{p^a}$. Since g is a primitive root $(\text{mod } p^a)$, its order, namely $\phi(p^a) = p^{a-1}(p-1)$, has to divide $n-1$. As $p \nmid n-1$, we deduce that $a = 1$ and $p-1|n-1$. The necessity part of the theorem is proved.

For the sufficiency part, we let $n = \prod p$, with $p-1|n-1$. Let $\gcd(b, n) = 1$, so that $p \nmid b$, and hence $b^{p-1} \equiv 1 \pmod{p}$ by Fermat's Little Theorem. Thus $b^{n-1} \equiv 1 \pmod{p}$, and hence $b^{n-1} \equiv 1 \pmod{n}$, that is n is a Carmichael number. □

Proof of the Corollary Let $n = pq$, a product of two distinct odd primes, be a Carmichael number. Then

$$n - 1 \equiv pq - 1 \equiv (p-1)q + q - 1 \equiv q - 1 \not\equiv 0 \pmod{p-1},$$

so that $p - 1 \nmid n - 1$. Therefore $\omega(n) \geq 3$. □

Korselt proved his theorem in 1899, but he did not find a number n satisfying his criterion. Such numbers are now named after R. D. Carmichael, who rediscovered the theorem and found that $n = 561 = 3 \cdot 11 \cdot 17$ is such a number. To search for n, we may set a squarefree P first, and try to find primes p, q such that Ppq satisfies Korselt's criterion. It has been established that, for a fixed squarefree P, there are at most finitely many such Pqr. More specifically, R. G. E. Pinch [72] has shown that there are numbers D, C satisfying $2 \leq D < P < C$ such that

$$q = \frac{(P-1)(P+D)}{CD - P^2} + 1, \quad r = \frac{(P-1)(P+C)}{CD - P^2} + 1, \quad 1 < \frac{CD}{P^2} < \frac{p+3}{p+1},$$

where p is the largest prime divisor of P, so that the problem is reduced to the search for suitable pairs C, D.

The smallest Carmichael numbers n with $\omega(n) = 4, 5, 6$ are

$$n = 41041 = 7{\cdot}11{\cdot}13{\cdot}41,$$
$$n = 825265 = 5{\cdot}7{\cdot}17{\cdot}19{\cdot}73,$$
$$n = 321197185 = 5{\cdot}19{\cdot}23{\cdot}29{\cdot}37{\cdot}137.$$

A celebrated result, established in 1994, by W. R. Alford et al. [2], states that there are infinitely many Carmichael numbers.

5.4 Discrete Logarithms

Let g be a primitive root of p, and write $m = g^n \pmod p$, so that there is a one-to-one correspondence between $n \in \{0, 1, \ldots, p-2\}$ and $m \in \{1, 2, \ldots, p-1\}$. We call n the *index* of m with respect to the primitive root g, or that n is the *discrete log* (DL) of m to the base g; its value is determined only up to (mod $p-1$). Thus

$$\mathrm{DL}(1) = 0, \quad \text{and} \quad \mathrm{DL}(-1) = \mathrm{DL}(p-1) = \frac{p-1}{2},$$

because $g^{p-1} \equiv 1 \pmod p$ and $g^{(p-1)/2} \equiv -1 \pmod p$. The other values of $\mathrm{DL}(m)$ will depend on the choice of the primitive root g.

A primitive root $g \pmod p$ is necessarily a quadratic nonresidue (mod p); in fact it is clear that, regardless of the choice of g, a number is a quadratic residue, or nonresidue, depending on whether its index is even, or odd.

Let $m_1 = g^{n_1}$ and $m_2 = g^{n_2}$. Then $m_1 m_2 \equiv g^{n_1+n_2} \pmod p$ so that

$$\mathrm{DL}(m_1 m_1) \equiv \mathrm{DL}(m_1) + \mathrm{DL}(m_2) \pmod{p-1};$$

also

$$\mathrm{DL}(m^a) \equiv a{\cdot}\mathrm{DL}(m) \pmod{p-1}.$$

Example 5.3 We find that $g = 13, 14$ are primitive roots of the prime $p = 19$. As m runs through the reduced residue system $\{1, 2, \ldots, 18\}$ (mod 19), the values n of the discrete log of m runs through the system $\{0, 1, 2, \ldots, 17\}$ (mod 18). Except for $m = 1, 18$, the corresponding values of n depend on the chosen g; in the table below, m_1, m_2 correspond to $g = 13, 14$, respectively. In the construction of such a table (for example Table 5.5) it is expedient to run through the values n of the discrete logs so that $m = g^n \pmod p$ can be delivered iteratively. It is much more onerous to find n from m.

Table 5.5 Discrete logs (mod 19), with primitive roots $g = 13, 14$

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
m_1	1	13	17	12	4	14	11	10	16	18	6	2	7	15	5	8	9	3
m_2	1	14	6	8	17	10	7	3	4	18	5	13	11	2	9	12	16	15

Binomial and Exponential Congruences

The discrete logarithm can be used to check if the binomial congruence $x^n \equiv a \pmod{p}$ is soluble, and to evaluate the solution if it exists. Suppose that p is of modest size, so that a primitive root g can be found, together with the computation of a DL table, and its inverse, or the 'anti-log' table. The congruence can be rewritten as

$$n \cdot \mathrm{DL}(x) \equiv \mathrm{DL}(a) \pmod{p-1},$$

a linear congruence in the new variable $X = \mathrm{DL}(x)$, which is soluble if $d = \gcd(n, p-1)$ divides $\mathrm{DL}(a)$. With the evaluation of $\bar{n}$, the reciprocal of $n \pmod{p-1}$ [or a reduced modulus $(p-1)/d$], that is $\bar{n}n \equiv 1 \pmod{p-1}$, we then have

$$X \equiv \bar{n} \cdot \mathrm{DL}(a) \pmod{p-1}.$$

Reading $\mathrm{DL}(a)$ from the DL table, so that X can be found, the sought after solution $x \equiv g^X \pmod{p}$ can be evaluated, or be read from the corresponding anti-log table.

A similar procedure is used to deal with the exponential congruence $a^x \equiv b \pmod{p}$; rewritten as the linear congruence

$$x \cdot \mathrm{DL}(a) \equiv \mathrm{DL}(b) \pmod{p-1},$$

it is soluble if $\gcd(\mathrm{DL}(a), p-1)$ divides $\mathrm{DL}(b)$.

Example 5.4 Examine, for the prime $p = 8887$, whether the congruences

$$33x^7 \equiv 23 \pmod{p} \quad \text{and} \quad 19^y \equiv 53 \pmod{p}$$

are soluble, and find the solutions if they exist.

Solution We have $p - 1 = 2 \cdot 3 \cdot 1481$, where 1481 is a prime. Since $7 \nmid p-1$, the first congruence is soluble. Computation delivers $\bar{7} \equiv 2539 \pmod{p-1}$. We can either evaluate the reciprocal of 33 (mod p) first, or proceed directly to find a primitive root (mod p), and move on to index calculus.

From the factorisation of $p-1$, it is not difficult to find that $g = 3, 10, 12, 13, 14$ are primitive roots (mod p). Let us take $g = 13$ and compute

$$\mathrm{DL}(33) = 3135, \quad \mathrm{DL}(23) = 4333, \quad \mathrm{DL}(19) = 7028, \quad \mathrm{DL}(53) = 2282.$$

Writing $X = \mathrm{DL}(x)$, the first congruence becomes $3135 + 7X \equiv 4333 \pmod{p-1}$, that is $7X \equiv 1198 \pmod{p-1}$. The linear congruence has the solution

$$X \equiv \bar{7} \cdot 1198 \equiv 2710 (\mathrm{mod}\ p-1), \text{ thereby delivering } x \equiv 13^X \equiv 4230 (\mathrm{mod}\ p).$$

The second congruence can be rewritten as $7028y \equiv 2282 \pmod{p-1}$ which, on division by 2, including the modulus $p-1$, becomes $3514y \equiv 1141 \pmod{q}$, where $q = \frac{1}{2}(p-1)$. The linear congruence is soluble because 3514 is coprime with $q = 4443$. The reciprocal of 3514 (mod q) is $\overline{3514} \equiv -110 \pmod{q}$, so that

$$y \equiv -110 \cdot 1141 \equiv 3337 \pmod{q}, \qquad q = \tfrac{1}{2}(p-1) = 4443.$$

Note that if 53 is replaced by b with $\mathrm{DL}(b)$ odd, then there is no solution; note also that the order of 19 (mod p) being q imposes the condition that the solutions to the congruence have to lie in a residue class (mod q). □

5.5 Key-Exchange Protocols

For a fixed prime p, a primitive root g, and an input value n, we explained in Sect. 1.9 that $m \equiv g^n \pmod{p}$ can be computed efficiently by the 'square-and-conquer' scheme. However, the inverse problem of computing the DL of an input m, that is the solution n of $g^n \equiv m \pmod{p}$, is much more difficult—the task can be likened to that of finding a needle in a hay-stack.

For example, the determination of n such that $g^n \equiv 3333 \pmod{p}$ is no easy matter. We may, of course, apply 'brute-force' by searching through $0 \le n < p-1$ systematically, or consider 'lucky-strike' by repeatedly taking random values of n, hoping that somehow $g^n = 3333$ emerges. Well, even for a modest size p, it is a racing certainty that CCH before such a scheme succeeds.

The *discrete logarithm problem* (DLP) is to find a method that can deliver the DL in reasonable time when p is no longer small. We describe Pollard's ρ-method below and, in Chap. 13, we consider the use of *smooth numbers*, to try to solve DLP when p is of rather modest size.

With no known method which can solve the DLP in reasonable time when the prime is large, say one with more than 300 decimal digits, the problem is deemed intractable enough for the function to be considered one-way by industry for business and commerce. Thus, assuming that DLP is difficult for a large prime p, the following key-exchange protocol can be used to create a common key in the form of a large number Z that is secret for two persons who need not have met previously; for example, Z can be used as 'one-time pad' for the exchange of messages between Axl and Beatrice.

Again, except for top-secret communications, large primes constructed by various methods, such as the Pocklington-Lehmer theorem will be enough. Indeed,

for ephemeral communications, even a large probable prime can be used instead. Without meeting each other, the following protocol may be adopted by Axl and Beatrice to create their secret key Z which is available only to the two of them.

- *Axl and Beatrice first agree, over an open channel, on a large prime p, and a primitive root g; such numbers can be constructed by either of them, or just taken from a public establishment.*
- *Axl then chooses his* **private key** *a, and sends Beatrice the number $A \equiv g^a \pmod p$ through the channel. Meanwhile, Beatrice chooses her* **private key** *b, and sends Axl the number $B \equiv g^b \pmod p$ through the channel.*
- *Axl uses a and the received value B to compute B^a; Beatrice uses b and the received value A to compute A^b.*
- *Since*

$$B^a \equiv (g^b)^a \equiv g^{ab} \equiv (g^a)^b \equiv A^b \pmod p,$$

Axl and Beatrice have the same $Z \equiv g^{ab} \pmod p$, which can be used as their common secret key.

The numbers $\{p, g, A, B\}$ are available to the public, but not $Z \pmod p$, which is only available to Axl and Beatrice. Because p is large and the DLP is difficult, an eavesdropper Eve cannot compute Z from the knowledge of only $\{p, g, A, B\}$. Note that Axl does not know the value of b, nor does Beatrice know the value of a, but they both can evaluate $Z \equiv g^{ab} \pmod p$, whereas Eve cannot.

Example 5.5 Find a prime $p > 10^{50}$ with a primitive root $g = 53$.

Solution Fix $a = 53$ in the Pocklington-Lehmer Theorem 3.3. By Example 3.4, $q = 123456789133 > 10^{11}$ is a prime. Let

$$F = 2q^3, \qquad R_k = 10^{20} + k, \qquad n_k = FR_k + 1, \qquad k = 0, 1, 2, \ldots,$$

and we search for a value of k so that n_k is prime according to the theorem.

We find that $n = n_{13}$ is a PRP(53), and that

$$\gcd(a^{(n-1)/2} - 1, n) = \gcd(a^{(n-1)/q} - 1, n) = 1, \qquad a = 53.$$

Therefore $g = a = 53$ is a primitive root of the prime

$$p = n = 2 \times 123456789133^3(10^{20} + 13) + 1 > 10^{50}.$$

□

Example 5.6 Suppose that Axl and Beatrice agree to use the prime p in the previous example and the primitive root $g = 53$. Choose random private keys a and b for Axl and Beatrice, respectively, and compute their common secret key Z.

Solution For random

$$a \equiv 60894101940262278462672743489624656764732494447926154,$$
$$b \equiv 37282340180337217214113144696404449092475178590708359 4,$$

modulo $p-1$, we find that

$$A \equiv g^a \equiv 22940232737093931285562591377066645021299234022987963 9,$$
$$B \equiv g^b \equiv 17983175699108977220861068199478185078411443986940501 9,$$
$$A^b \equiv B^a \equiv 33220644556582220903950642149969805464046790621181160 9,$$

modulo p. □

Pollard's ρ-Method for Discrete Logarithms

Let p be a prime of modest size, say $p < 10^{18}$, and g be one of its primitive roots. We describe Pollard's ρ-method for the computation of $m = \mathrm{DL}(n)$ for an input n.

Consider a sequence of pairs (a_i, b_i) of integers in $0 \le a_i, b_i < p-1$, defined iteratively by some specific rule. Set $x_i \equiv n^{a_i} g^{b_i} \pmod{p}$, and suppose that the terms of the sequence (x_i) appear to be random numbers in $1 \le x_i < p$. From the solution to the 'birthday problem' (see Note 3.2 in Chap. 3), we can expect that there are indices j, k with $j < k \ll \sqrt{p}$ such that $x_j = x_k$, so that $n^{a_j} g^{b_j} \equiv n^{a_k} g^{b_k} \pmod{p}$, that is $n^{a_j - a_k} \equiv g^{b_k - b_j} \pmod{p}$, and hence

$$m(a_j - a_k) \equiv b_k - b_j \pmod{p-1}.$$

Let $d = \gcd(a_j - a_k, p-1)$. If $d = 1$ then the reciprocal of $a_j - a_k \pmod{p-1}$ exists, and the required value of m can be evaluated accordingly. If $d > 1$ then we replace the congruence by

$$mA \equiv B \pmod{D}, \qquad A = \frac{a_j - a_k}{d},\ B = \frac{b_k - b_j}{d},\ D = \frac{p-1}{d},$$

and solve for $m_0 \pmod{D}$ instead. The required value of m is now given by $m = m_0 + hD$ for some $h < d$; the value of d is likely to be small, so that m can still be delivered.

As with the ρ-method for factorisation, the Floyd cycle-finding scheme in Theorem 3.4 can be used to find matching pairs with $x_j = x_k$. At the ith stage of the algorithm we compute (x_i, a_i, b_i) and (x_{2i}, a_{2i}, b_{2i}), check and halt if there is a match with $x_i = x_{2i}$. If not then move to the $(i+1)$th stage, computing

$(x_{i+1}, a_{i+1}, b_{i+1})$ from (x_i, a_i, b_i), and $(x_{2i+2}, a_{2i+2}, b_{2i+2})$ by two iterations from (x_{2i}, a_{2i}, b_{2i}). There is no problem with space requirements.

We can start with $(x_0, a_0, b_0) = (1, 0, 0)$, or any random point with $x_0 \equiv n^{a_0} g^{b_0} \pmod{p}$. There can be a host of methods to deliver the next triple. One such method is to define, for $i = 0, 1, 2, \ldots,$

$$(x_{i+1}, a_{i+1}, b_{i+1}) = \begin{cases} (nx_i,\ a_i + 1,\ b_i,), & \text{if } 0 < x_i < \frac{1}{3}p, \\ (x_i^2,\ 2a_i,\ 2b_i), & \text{if } \frac{1}{3}p < x_i < \frac{2}{3}p, \\ (gx_i,\ a_i,\ b_i + 1) & \text{if } \frac{2}{3}p < x_i < p, \end{cases}$$

reducing $x_{i+1} \pmod{p}$ and $a_{i+1}, b_{i+1} \pmod{p-1}$. Whichever rule we choose, there should be some reason to believe that x_i should appear random in the interval $0 \le x < p$. For example, for the above choice, the condition $x_{i+1} \equiv n^{a_{i+1}} g^{b_{i+1}} \pmod{p}$ is readily verified; also it seems reasonable that there is some semblance of randomness for x_i, because whichever third of the interval $0 \le x < p$ that x_i lands on has little to do with p, g, n and the initial point (x_0, a_0, b_0).

For the algorithm, define the function:

```
def xab_sequence(p,g,n,x,a,b):
  (xx,aa,bb)=((n*x)%p,(a+1)%(p-1),b%(p-1))
  if 3*x>p: (xx,aa,bb)=((x*x)%p,(a+a)%(p-1),(b+b)%(p-1))
  if 3*x>2*p: (xx,aa,bb)=((g*x)%p,a%(p-1),(b+1)%(p-1))
  return (xx,aa,bb)
```

and apply

```
(x,a,b)=xab_sequence(p,g,n,x,a,b)
(y,c,d)=xab_sequence(p,g,n,y,c,d)
(y,c,d)=xab_sequence(p,g,n,y,c,d)
```

for each step in the iteration.

It takes a few seconds to process a million iterations, but there are likely to be many millions of them before a match $x = y$ emerges.

Example 5.7 Taking $g = 5$ for the primitive root of the prime $p = 10^{16} - 63 = 9999999999999937$. It then takes about 10 minutes to deliver one DL. The two values of n, together with the chosen seeds (a_0, b_0) and the resulting (h, d), are shown in Table 5.6.

Table 5.6 Discrete logs (mod $10^{16} - 63$)

n	(a_0, b_0)	(h, d)	$m = \mathrm{DL}(n)$
1222223333344444	(111, 222)	(402, 576)	6972312130374911
5666667777788888	(555, 666)	(176, 192)	9130585130790709

5.6 Summary and Notes

Summary

- The *order* of a number a, not divisible by a prime p, is the least positive integer d such that $a^d \equiv 1 \pmod{p}$. A number g with order $p-1$ is called a *primitive root* of p.
- Primitive roots exist for every prime, and in fact there are $\phi(p-1)$ such roots.
- The *primitive roots theorem* states that the numbers $m = 1, 2, 4, p^a, 2p^a$, where p is an odd prime, and a is a positive integer, are the only numbers which possess primitive roots.
- Although 2^a $(a \geq 3)$ does not have a primitive root, the number 5 can be used essentially as a substitute in that every odd integer $n \pmod{2^a}$ may be represented in the form $\pm 5^b$, where $1 \leq b \leq \frac{1}{2}\phi(2^a)$.
- A *Carmichael number* is a composite number n with the property that $b^{n-1} \equiv 1 \pmod{n}$ for every b coprime with n. Korselt's theorem states that such a number n should be odd, squarefree, and $p-1|n-1$ for each prime $p|n$. As a consequence Carmichael numbers have at least 3 distinct prime divisors.
- A celebrated theorem of Alford, Granville and Pomerance states that there are infinitely many Carmichael numbers.
- Let g be a primitive root of p, and $m \equiv g^n \pmod{p}$. We call n the *discrete log* (DL) of m to the base g; its value is determined only up to $(\text{mod } p-1)$. Together with a table of values for DL, such a notion can be used to solve high degree, or exponential, congruences.
- The problem of the evaluation of the DL being difficult, the notion can also be used in *key exchange protocols*.
- *Pollard's ρ-method* can be use to compute the DL when the prime concerned is of very modest size.

Notes

1. *Artin's primitive root conjecture*. Let $g = bc^2$, where $b > 1$ is squarefee; define

$$A = \prod_q \Big(1 - \frac{1}{q(q-1)}\Big) = 0.3739558\cdots, \qquad q \text{ prime},$$

and

$$B_b = \begin{cases} 1, & b \not\equiv 1 \pmod 4, \\ 1 - \mu(b) \displaystyle\prod_{q|b} \frac{1}{q(q-1)-1}, & b \equiv 1 \pmod 4. \end{cases}$$

Artin's conjecture states that the proportion of primes having g as primitive root is $B_b A$. Thus, for $g = 2, 10$, the proportion is A, whereas for $g = 5$, with $B_5 = 1 + 1/19$, the proportion is $20A/19$.
The conjecture has been established by C. Hooley in 1967 under the assumption of certain cases of the generalised Riemann hypothesis.

2. D. R. Heath-Brown, improving on earlier fundamental work of R. Gupta and M. R. Murty, proved that at least one of 2, 3, or 5 is a primitive root modulo infinitely many primes; he also proved that there are at most two primes for which Artin's conjecture fails.
3. The convergence of the product for Artin's constant A is too slow for its accurate evaluation. However, the product can be rewritten as $\prod_{k\geq 2} \zeta(k)^{-e_k}$, where $\zeta(s)$ is the Riemann zeta-function, and $e_k \in \mathbb{N}$. See "Approximation of Artin type constants and automata," an article posted in April 1999 by P. Moree at the Max Planck Institute for Mathematics.
4. It has been established that the least primitive root $g = g(p) \pmod p$ satisfies

$$g \ll_\epsilon p^{\frac{1}{4}+\epsilon}, \qquad \text{for every } \epsilon > 0, \quad \text{(D. A. Burgess);}$$

$$g \ll \log^6 p, \qquad \text{under GRH,} \quad \text{(V. Shoup).}$$

5.7 Exercises

1. Find all the primitive roots (mod 101).
2. (Chebyshev) Let $p \equiv 2 \pmod 5$ be a prime, with $q = 4p + 1$ being also a prime. Show that 10 is a primitive root of q.
3. Let $p = 29$, $g = 14$ and $h = g + p = 43$.
   (i) Verify that g is a primitive root (mod p), but not one of (mod p^2);
   (ii) Verify that $G = g^{p-1} \equiv 1 \pmod{p^2}$, and $H = h^{p-1} \not\equiv 1 \pmod{p^2}$;
   (iii) Verify that h is a primitive root (mod p^2) and also (mod $2p$).
4. Identify the reduced residue classes (mod 32) using the ordered pairs $(\pm 1, 5^b)$.
5. Let g be a primitive root of an odd prime p. Show that $-g$ is also a primitive root if and only if $p \equiv 1 \pmod 4$.
6. List the primes $p < 1000$ of which 2 is a primitive root, and check that they include primes of the form $p = 4q + 1$ where q is an odd prime. Show that 2 is a primitive root of any prime p taking such a form.
7. List the primes $p < 1000$ of which 3 is a primitive root, and check that they include primes of the form $2^a + 1 > 5$. Show that 3 is a primitive root of any prime p taking such a form.
8. Let $n = 2^3 3^3 5$. Evaluate the Carmichael function $\lambda(n)$, and determine the number of $x < n$ having order $\lambda(n) \pmod n$.

9. Let $n = (6k+1)(12k+1)(18k+1)(36k+1)$. Show that if $\Omega(n) = 4$ then n is a Carmichael number. Find 4 such numbers n.
10. Find all pairs of primes q, r such that $997qr$ are Carmichael numbers.
11. Prove that each prime divisor p of a Carmichael number n is less than $\sqrt{n}$. Deduce that $\omega(n) \geq 3$.
12. Find two primitive roots g_1, g_2 of the prime $p = 23$ and construct the table of discrete logarithms with respect to g_1, g_2.
13. Examine, for the prime $p = 9941$, whether the congruences

$$33x^7 \equiv 41 \pmod{p} \quad \text{and} \quad 29^y \equiv 71 \pmod{p}$$

are soluble, and find the solutions if they exist.
14. Find a prime $p > 10^{70}$ with a primitive root $g = 37$. Use p and g to carry out the protocol for the key-exchange process for two persons.
15. Find the least prime $p > 10^{16}$ for which $g = 37$ is a primitive root, and solve the discrete log problem $37^x \equiv 123456789 \pmod{p}$.
16. Determine the lengths of the period of the decimal expansion of $1/N$ and $1/(N+1)$, when $N = 1194749$.
17. Let $(m_2, m_3, m_4, m_5, m_6, m_7) = (1, 1, 1, 2, 2, 4)$. Show that

$$\prod_{2 \leq k < 8} (1 - t^k)^{m_k} = 1 - \sum_{2 \leq n < 8} t^k + O(t^8), \qquad 0 < t < 1, \quad t \to 0,$$

and extend the result for a few more values of m_k.
Apply the result to evaluate Artin's constant to 10 decimal places.

Chapter 6
Sums of Squares



Abstract Besides the theorems of Fermat and Lagrange on the representation of numbers as sums of two and four squares, there is also a proof of the Gauss–Legendre theorem on sums of three squares based on the use of Minkowski's theorem on the existence of lattice points in a convex body. The tricky problems of finding representations of numbers as sums of squares, including Brillhart's algorithm, are considered.

The representation of numbers as sums of squares offers many challenges to mathematicians—we restrict our considerations to the classic cases of two, three and four squares. The condition for a number to be representable as a sum of two squares was first given in 1625 by the Dutch mathematician Albert Girard. It was given again by Fermat who also claimed that he had an irrefutable proof of the results by his method of infinite descent, but the first recorded proof was given by Euler in 1749.

There are obvious necessary conditions associated with such representation problems. For example, since a square is congruent to 0 or 1 (mod 4), it follows that a sum of two squares cannot be congruent to 3 (mod 4). Again, because squares are congruent to 0, 1 or 4 (mod 8), it follows that a sum of three squares cannot be congruent to 7 (mod 8). The representation problems require us to show that solutions do exist as long as such 'congruence obstructions', and their immediate consequences, are avoided. There being no such obstruction for four squares, every number is a sum of four squares, a famous result proved by Lagrange in 1770.

The three squares case turns out to be difficult. In his diary entry on 10 July 1796 Gauss wrote:

$$\text{EΥPHKA!} \quad \text{num} = \Delta + \Delta + \Delta.$$

A number of the form $\frac{1}{2}n(n+1)$ is called a triangular number, and he had succeeded in proving his 'Eureka theorem' that every number is a sum of three such numbers; since 8 times a triangular number is 1 short of a square, the result amounts to $8m+3$ being a sum of three squares. In his *Disquisitiones Arithmeticae* (1801) Gauss gave

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an extensive theory of quadratic forms, which includes a proof of the general three-squares theorem. There is no consensus view on whether Legendre also had a proof of the theorem in 1798, because his argument might have included an assumption on the existence of primes in arithmetic progressions, a much later result established by Dirichlet.

We shall prove the following three theorems, which are, together with those on the distribution of primes, landmarks in the history of number theory.

Theorem 6.1 (Fermat-Euler) *A number is representable as a sum of two squares if, and only if, it has the form PQ^2, where P is free of prime divisors $q \equiv 3 \pmod 4$.*

Theorem 6.2 (Lagrange) *Every number is representable as a sum of four squares.*

Theorem 6.3 (Gauss-Legendre) *A number is representable as a sum of three squares if, and only if, it is not of the form $4^a(8n + 7)$.*

Although the proofs of the first two theorems might be described as simple nowadays, the three-squares theorem is generally considered to be a deep result without a simple proof. Part of the difficulty with the three-squares case lies with the fact that, whereas sets corresponding to sums of two squares, and sums of four squares, are 'multiplicatively closed', the set of sums of three squares is not; thus whereas $3 = 1^2 + 1^2 + 1^2$ and $5 = 2^2 + 1^2 + 0^2$ are sums of three squares, there is no such representation for $15 = 3\times5$.

After establishing the existence of primes in arithmetic progressions, Dirichlet applied his famous result to give a greatly simplified proof of the three-squares theorem. In 1890 H. Minkowski created the subject of the geometry of numbers and established his fundamental theorem on the existence of lattice points within a convex symmetric body—the theorem can be used to give new proofs of the two-squares and four-squares theorems. In 1957 N. C. Ankeny [4] showed that the three-squares theorem can also be established in this way, and we give a rendition of his interesting proof, which also requires Dirichlet's theorem.

6.1 Sums of Two Squares

Let $\mathbb{W}$ denote the set of numbers representable as sums of two squares, so that

$$\mathbb{W} = \{1, 2, 4, 5, 8, 9, 10, \ldots\}.$$

We already remarked that numbers congruent to 3 (mod 4) are not in $\mathbb{W}$, but more can be said by recalling that -1 is a quadratic nonresidue modulo primes $q \equiv 3 \pmod 4$. If such a prime q divides $x^2 + y^2$ then it has to divide both x and y, since otherwise, by considering reciprocals, the congruence $u^2 + 1 \equiv 0 \pmod q$ would be soluble, which is not the case. It follows that if such q divides $n \in \mathbb{W}$ then q has to divide n an exact even number of times, so that a necessary condition for $n \in \mathbb{W}$ is that it has the representation $n = PQ^2$, where P is free of prime divisors $q \equiv 3$

(mod 4). Next, the identity

$$(a^2 + b^2)(c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2$$

shows that $\mathbb{W}$ is a multiplicatively closed set, and since $2 \in \mathbb{W}$, the sufficiency part of the Fermat-Euler theorem amounts to proving that every prime $p \equiv 1 \pmod 4$ belongs to $\mathbb{W}$, a result usually attributed to Fermat.

If $q \equiv 3 \pmod 4$ is a prime which divides n an odd number of times exactly, so that $n \notin \mathbb{W}$, we say that q is a *witness* to this easily verified fact. It also follows that if n is a squarefree sum of two coprime squares then each prime divisor of n is congruent to 1 (mod 4).

6.2 Fermat's Theorem for Primes $p \equiv 1 \pmod 4$

Although Fermat appreciated the importance of proofs in mathematics he rarely set them out for posterity. Instead, he was content to communicate his discoveries to his interested correspondents, and occasionally offer his results as challenges to foreign scientists. The proofs of the bulk of his discoveries became much of the life-time's work of Euler, the most prolific mathematician in the eighteenth century. Euler's legacy was taken up by Lagrange, Legendre, and followed by Gauss, who clarified and greatly expanded such works.

Fermat's theorem for primes $p \equiv 1 \pmod 4$ turns out to be one of the most interesting and important ones in elementary number theory. There are now many proofs of the theorem, using ideas from diverse areas of mathematics. Most proofs make use of the fact that $u^2 + 1 \equiv 0 \pmod p$ is soluble when $p \equiv 1 \pmod 4$ is a prime, which is part of the corollary to Wilson's theorem (Theorem 1.11). The extravagance of offering three proofs of the theorem needs no justification.

Theorem 6.4 (Fermat) *Every prime $p \equiv 1 \pmod 4$ is representable as a sum of two squares.*

First Proof Let $1 \le u < p$ satisfy $u^2 + 1 \equiv 0 \pmod p$. By Dirichlet's approximation theorem (Theorem 1.7) with $\theta = u/p$ and $Q = \sqrt{p}$, there are $a, b \in \mathbb{N}$ such that

$$\left|\frac{u}{p} - \frac{a}{b}\right| < \frac{1}{b\sqrt{p}}, \qquad 1 \le b < \sqrt{p}.$$

Set $c = ub - pa$, so that $|c| < \sqrt{p}$ and $b^2 + c^2 \equiv b^2(u^2 + 1) \equiv 0 \pmod p$. From $0 < b^2 + c^2 < 2p$, it follows that $b^2 + c^2 = p$. □

Second Proof Let $1 \le u < p$ satisfy $u^2 + 1 \equiv 0 \pmod p$. By Thue's theorem (Theorem 1.8) there are positive integers $x, y < \sqrt{p}$ such that $ux \equiv \pm y \pmod p$.

Thus $u^2x^2 \equiv y^2 \pmod{p}$, and hence $x^2 + y^2 \equiv 0 \pmod{p}$, so that $p|x^2 + y^2$. Since $x^2 + y^2 < 2p$, it follows that $p = x^2 + y^2$. □

Example 6.1 *Let* $p = 9973$. *Solve* $u^2 + 1 \equiv 0 \pmod{p}$, *and use the solution to find positive integers* x, y *such that* $p = x^2 + y^2$.

Solution Since $p \equiv 5 \pmod 8$ we find, using (4.2), that $u = 2798$.

Following the first proof of the theorem, we proceed to find a rational approximation a/b to u/p, with a denominator $b \leq \lfloor \sqrt{p} \rfloor = 99$. For each such b we take a so that $c = ub - pa$ is least in absolute value, and there should be at least one b with $|c| \leq 99$. There are in fact two solutions, namely $(a, b, c) = (16, 57, -82)$ and $(23, 82, 57)$, and we find that $p = x^2 + y^2$ for $(x, y) = (57, 82)$.

We can also use the second proof of the theorem, by finding $x, y \leq 99$ such that $ux \equiv \pm y \pmod{p}$. Thus, for each $x \leq 99$, we evaluate $y \equiv ux \pmod{p}$ with $0 \leq y < p$, and collect (x, y) if either $1 \leq y \leq 99$ or $p - y \leq 99$. The solutions are $(x, y) = (57, 9891) \equiv (57, -82)$, and $(82, 57)$. □

Finding the representation of $p = x^2 + y^2$ in these ways is no better than the inefficient method of just testing if $p - x^2$ is a square for $1 \leq x < \sqrt{p/2}$, although this 'brute force' method does not include a proof that a solution exists. In Sect. 6.5 we present an algorithm which delivers the solution efficiently.

The proofs which do not require the fact that $u^2 + 1 \equiv 0 \pmod{p}$ is soluble are often based on mappings of finite sets onto itself. A mapping $f : \mathcal{S} \to \mathcal{S}$ is called an *involution* if the composite map $f \circ f$ is the identity, so that f is its own inverse. An involution f on $\mathcal{S}$ then partitions $\mathcal{S}$ into pairs $(X, f(X))$, with some being 'self-pairs', because $f(X) = X$ is a 'fixed point'. Note that there must be a fixed point if the cardinal of $\mathcal{S}$ is odd.

Third Proof (S. Dolan [25]). For a prime $p \equiv 1 \pmod 4$, define

$$\mathcal{S} = \{(x, y, z) : f(x, y, z) = p\},$$

where $f(x, y, z) = x^2+y^2+z^2+2xy-2xz+2yz$, and x, y, z are positive integers. We shall use $f(x, y, z)$ in one of the two alternative formulations:

$$f_1(x, y, z) = (x + y + z)^2 - 4xz \quad \text{or} \quad f_2(x, y, z) = (x + y - z)^2 + 4yz;$$

for example, it is easier to see from f_2 that $\mathcal{S}$ is a finite set of lattice points in 3-dimensional space.

From $f(x, y, x) = y(y+4x)$, and p being a prime, it follows that there is exactly one point $(x, y, x) \in \mathcal{S}$, namely $y = 1$ and $x = (p - 1)/4$. Moreover, since $x + z$ and xz are symmetric functions (see more on such functions in Sect. 8.6) in f_1, the mapping $(x, y, z) \leftrightarrow (z, y, x)$ is an involution on $\mathcal{S}$, so that the points $(x, y, z) \in \mathcal{S}$, with $x \neq z$, form pairs; therefore the cardinal of $\mathcal{S}$ is *odd*, because there is exactly one fixed point. Also, because $f(x, y, y) = x^2 + (2y)^2$, the required representation of p follows if there exists $(x, y, y) \in \mathcal{S}$.

We proceed by supposing that $(x, y, y) \notin \mathcal{S}$ for any x, y, and deduce that the cardinal of $\mathcal{S}$ would then be even; the contradiction then establishes the theorem. For this purpose, we shall make use of formulation f_2.

For an ordered pair (y, z), with $y \neq z$, denote by $N(y, z)$ the number of $(x, y, z) \in \mathcal{S}$. Thus $N(y, z)$ is the number of positive integer solutions x to the quadratic equation $(x + y - z)^2 + 4yz = p$. For integer solutions, we need $p - 4yz$ to be a square, so let

$$b = \sqrt{p - 4yz} \geq 0 \quad \text{and} \quad X = |y - z| > 0;$$

actually $b > 0$, because the prime $p \neq 4yz$; also $X \neq b$, since otherwise $(y - z)^2 = p - 4yz$, that is $p = (y + z)^2$, which is impossible. If b does not exist as a positive integer then $N(y, z) + N(z, y) = 0$. Otherwise $N(y, z) + N(z, y)$ is the number of positive $x = \pm X \pm b$, and there are then two admissible values, namely $x = \max\{X, b\} \pm \min\{X, b\}$. Consequently, the cardinal of $\mathcal{S}$ is even, and we have a contradiction, so that the theorem is proved. □

Fermat also laid stress on the fact that the solution of $p = x^2 + y^2$ is unique, apart from interchanging x, y, or changing their signs. To see this, suppose that $p = X^2 + Y^2$ is another representation. There being only two solutions to the congruence $u^2 + 1 \equiv \pmod p$, say $\pm h$, it follows that $x \equiv \pm hy$ and $X \equiv \pm hY \pmod p$, and we may take $+h$ in both cases, because signs of x, y, X, Y are immaterial. Then $xX \equiv h^2 yY \equiv -yY \pmod p$ and similarly $xY \equiv yX \pmod p$. It then follows from $p^2 = (x^2 + y^2)(X^2 + Y^2) = (xX + yY)^2 + (xY - yX)^2$ that

$$1 = \Big(\frac{xX + yY}{p}\Big)^2 + \Big(\frac{xY - yX}{p}\Big)^2,$$

a representation of 1 as a sum of two squares of integers. Therefore either $xY = yX$ or $xX = -yY$, so that the representation of $p = x^2 + y^2$ is essentially unique.

The Circle Problem of Gauss

Denote by $r(n)$ the number of solutions (x, y) of $x^2 + y^2 = n$, with $x, y \in \mathbb{Z}$. Note that $r(0) = 1$, and $r(n) = 0$ if $n \notin \mathbb{W}$. The value of $r(n)$ is rather erratic: it is almost always 0, but it can be large when n is a product of many primes congruent to 1 (mod 4). Its average value over $0 \leq n \leq x$ is π; see Fig. 6.1.

Theorem 6.5 (Gauss) *We have, as $x \to \infty$,*

$$\sum_{n \leq x} r(n) = \pi x + O(\sqrt{x}).$$

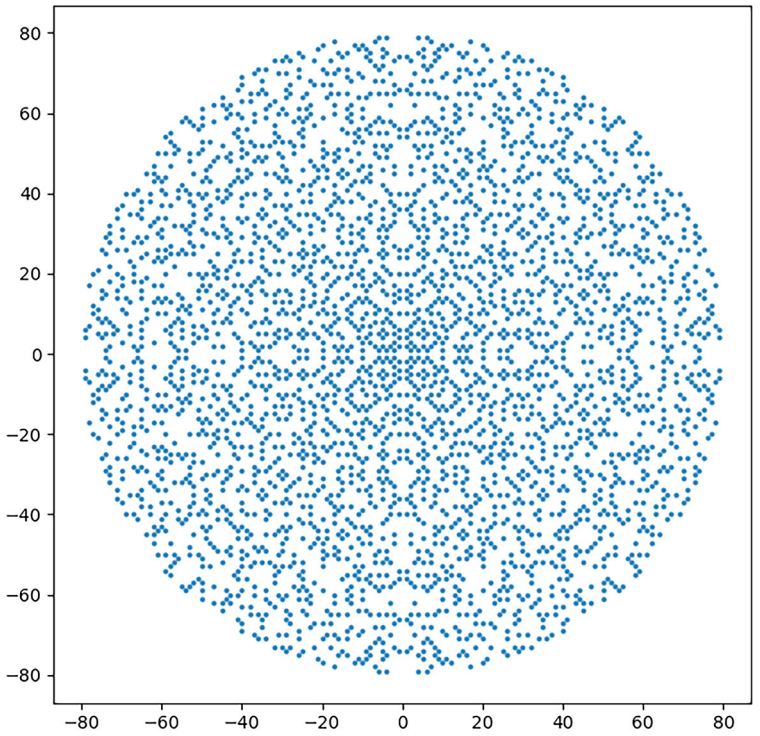


Fig. 6.1 Lattice points (x, y), with $x^2 + y^2$ being an odd prime $p < 80^2$

Proof The sum $\sum_{n\le x} r(n)$ has the interpretation of being the number of lattice points in the circular region $u^2 + v^2 \le x$, with centre 0 and radius $\sqrt{x}$, so that the area of the circle is πx. Partition the plane into unit squares with orthogonal lines passing through the lattice points. To each lattice point (u, v) in our circle we assign the square whose four corners have the coordinates (u, v), $(u+1, v)$, $(u, v+1)$, $(u+1, v+1)$. The union of these squares lie inside the circle $u^2 + v^2 = (\sqrt{x} + \sqrt{2})^2$, and it covers the circle $u^2 + v^2 = (\sqrt{x} - \sqrt{2})^2$. Therefore

$$\pi(\sqrt{x} - \sqrt{2})^2 \le \sum_{n\le x} r(n) \le \pi(\sqrt{x} + \sqrt{2})^2,$$

and the required result follows. □

Counting Sums of Two Squares

A sieve argument, similar to the sieve of Eratosthenes (see Sect. 3.3), can be used to tackle some of the problems associated with the set $\mathbb{W}$. Take, for example, the set $\mathbb{U}$ of products of distinct primes $p \equiv 1 \pmod 4$. There is then a correspondence between $\mathbb{U}$ and $\mathbb{W}$: Every $w \in \mathbb{W}$ has the form $w = x^2u$ or $w = 2x^2u$, with $u \in \mathbb{U}$. Members of $\mathbb{U}$ can be delivered by a sieve, and $\mathbb{W}$ can then be described as the unsifted elements of such a process. Consequently it is to be expected that $\mathbb{W}$ has many properties in common with the set of prime numbers. Indeed, the Meissel-Lehmer method can be applied to count the numbers in $\mathbb{W}$; see [90].

Let $W(x)$ denote the number of $n \le x$ which are representable as sums of two squares, that is, the counting function of the set $\mathbb{W}$. S. Ramanujan had anticipated the asymptotic formula

$$W(x) \sim \frac{Bx}{\sqrt{\log x}}, \quad x \to \infty, \qquad B = \frac{1}{\sqrt{2}} \prod_{p \equiv 3 \pmod 4} \left(1 - \frac{1}{p^2}\right)^{-1/2}, \tag{6.1}$$

which was established by E. Landau [54] in 1908. The analytic argument is similar to, but more complicated than, that for the prime number theorem—instead of a simple pole at $s = 1$, the corresponding analytic function has an algebraic singularity there, so that we need to take a contour round a branch point at $s = 1$, which explains the appearance of $1/\sqrt{\log x}$ in the formula for $W(x)$. Another explanation is that in the sieve argument, we are effectively sieving through only 'half' of the primes, namely primes satisfying $p \equiv 3 \pmod 4$. The number B is now called the Landau-Ramanujan constant; its defining product converges very slowly, but it can be evaluated (see [89]) without the listing of primes:

$$2B^2 = \prod_{n=1}^{\infty} \left(\left(1 - \frac{1}{2^{2^n}}\right) \frac{\zeta(2^n)}{\beta(2^n)} \right)^{1/2^n}, \qquad \beta(s) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^s};$$

it is then not difficult to show that $B = 0{\cdot}764223653589220\ldots$.

The Meissel-Lehmer method for the evaluation of $\pi(x)$ can be used to compute $W(x)$; see [90] where it is shown that

$$\begin{aligned} W(10^{10}) &= \quad 1\,637\,624\,156, \\ W(10^{11}) &= \quad 15\,570\,523\,346, \\ W(10^{12}) &= 148\,736\,628\,858. \end{aligned}$$

Thus, $Bx/\sqrt{\log x} \Big/ W(x) \approx 0{\cdot}9775$ at $x = 10^{12}$.

Gaps Between Sums of Two Squares

We already mentioned that there is no known method which can deliver arbitrarily large primes, whereas $x^2 \in \mathbb{W}$ for any $x \in \mathbb{N}$. As a consequence, many challenging problems in the distribution of primes have relatively simple solutions using elementary means when the problems are posed in $\mathbb{W}$. For example, it is a notoriously difficult problem whether 'prime gaps' $p_{n+1} - p_n$, where p_n is the nth prime, may take every even value infinitely often. However, it is relatively easy to show that the corresponding gaps for members of $\mathbb{W}$ take every positive integer value infinitely often; see [93]. Again, whether there is always a prime p satisfying $x < p \le x + \sqrt{x}$ for all large x is a problem of the same depth as that of the Riemann hypothesis, which states that all the complex zeros of $\zeta(s)$ lies on the line $\sigma = \frac{1}{2}$. However, there is a one-sentence proof of the following gap theorem for $\mathbb{W}$.

Theorem 6.6 (Bambah-Chowla [9]) *If $x \ge 1$, then the interval $x \le m < x + 2\sqrt{2}x^{1/4} + 1$ always contains a member of $\mathbb{W}$.*

Proof Set $u, v \in \mathbb{N}$ in $\sqrt{x} - 1 < u \le \sqrt{x}$ and $y < v \le y + 1$, where $y = \sqrt{x - u^2}$, so that the required result follows from

$$x = u^2 + y^2 < u^2 + v^2 = x + v^2 - y^2 \le x + 2y + 1,$$

together with the estimate $y^2 < x - (\sqrt{x} - 1)^2 = 2\sqrt{x} - 1 < 2\sqrt{x}$. □

The idea of the proof is the application of the 'greedy algorithm': Taking the largest square up to x, the error is about $2\sqrt{x}$; taking the next square nearest $2\sqrt{x}$, the error should be about $2\sqrt{2\sqrt{x}} = 2\sqrt{2}x^{1/4}$. Over the decades since, there has been no real improvement on this simple result; thus, for large x, even the reduction of the coefficient $2\sqrt{2}$ seems insurmountable, never mind the conjecture that the exponent $\frac{1}{4}$ can be reduced to any $\epsilon > 0$.

Example 6.2 Let w_n denote the nth number in $\mathbb{W}$. Show that, for any $K \in \mathbb{N}$, there exists n such that $w_{n+1} - w_n \ge K$. By appealing to Chebyshev's theorem in Sect. 9.3, show further that there are infinitely many n satisfying

$$w_{n+1} - w_n \gg \frac{\log w_n}{\log \log w_n}.$$

Solution A consecutive run of K numbers, with the kth term not in $\mathbb{W}$ by virtue of having the witness q_k, the kth prime congruent to 3 (mod 4), can be constructed using the Chinese remainder theorem. The solution X of the simultaneous system

$$x + k \equiv q_k \pmod{q_k^2}, \qquad k = 1, 2, \ldots, K,$$

satisfies $q_k | X + k$, and $q_k^2 \nmid X + k$ with $X \le (q_1 q_2 \cdots q_K)^2$. Therefore there exists n such that $w_n < X + 1 < X + K < w_{n+1}$.

Letting p_k denote the kth prime, we find that, as $K \to \infty$,

$$(q_1 q_2 \cdots q_K)^2 \ll p_1 p_2 \cdots p_{2K} = \exp \sum_{p \le p_{2K}} \log p = \exp(\theta(p_{2K})),$$

so that $\log w_n \ll \theta(p_{2K}) \ll K \log K$, and hence, for such corresponding n,

$$w_{n+1} - w_n \ge K \gg \frac{\log w_n}{\log \log w_n}.$$

□

For improvements, and the latest result on the problem, see Note 6.7.

6.3 Lagrange's Theorem

Apparently Diophantus was aware that only certain kinds of numbers can be represented as sums of two or three squares, but every number is a sum of four squares. Both Girard and Fermat stated the four-squares theorem explicitly, and Euler made many unsuccessful attempts to prove the result. Much of Euler's effort was in trying to show that every number is representable as a sum of two members of $\mathbb{W}$, which turned out to be a poor strategy, because the relationship between all four squares in the representation is being obscured by the rich structure of $\mathbb{W}$. Euler did make the crucial discovery of the identity:

$$(a^2 + b^2 + c^2 + d^2)(A^2 + B^2 + C^2 + D^2) = X^2 + Y^2 + Z^2 + W^2,$$

in which

$$X = aA + bB + cC + dD, \qquad Y = aB - bA - cD + dC,$$
$$Z = aC + bD - cA - dB, \qquad W = aD - bC + cB - dA.$$

In view of the identity and Fermat's theorem, it suffices to prove that every prime $p \equiv 3 \pmod 4$ is representable as a sum of four squares. This was proved in 1770 by Lagrange, who acknowledged his indebtedness to the work of Euler—the following is essentially Euler's simplified version of Lagrange's proof.

Proof of Lagrange's Theorem We first show that, for a prime $p \equiv 3 \pmod 4$, the congruence $x^2 + y^2 + 1 \equiv 0 \pmod p$ is soluble in x, y. This is easy, because we can take n to be the least nonresidue (mod p), so that both $n - 1$ and $-n$ are quadratic residues, and the congruence has the solution $x^2 \equiv n - 1$ and $y^2 \equiv -n$.

Therefore there exists m such that $mp = x^2 + y^2 + 1$, and we may suppose that $m < p$ because x, y can be chosen so that $0 \le x, y < p/2$. Now let $1 \le m < p$ be

such that mp is representable as a sum of four squares:

$$mp = a^2 + b^2 + c^2 + d^2.$$

We proceed to show that if $m > 1$ then there is some r satisfying $1 \leq r < m$ such that rp is also a sum of four square. The theorem then follows by applying such a 'descent' argument repeatedly, ending with $m = 1$.

Let A, B, C, D be such that $a \equiv A \pmod{m}$, with $-\frac{1}{2}m < A \leq \frac{1}{2}m$, and similarly also for B, C, D. Then $A^2 + B^2 + C^2 + D^2 \equiv 0 \pmod{m}$, so that

$$mr = A^2 + B^2 + C^2 + D^2$$

for some r, which has to be positive, since otherwise $A = B = C = D = 0$, which implies a, b, c, d being divisible by m, and thus mp would be divisible by m^2, which is impossible for the prime p unless $m = 1$.

From $mr \leq \frac{1}{4}(m^2 + m^2 + m^2 + m^2) = m^2$ we deduce that $r \leq m$. Moreover, if $r = m$ then $A = B = C = D = m/2$, which implies that m is even. It also follows that $a^2 \equiv b^2 \equiv c^2 \equiv d^2 \equiv \frac{1}{4}m^2 \pmod{m^2}$, and hence $mp \equiv 0 \pmod{m^2}$, which we already remarked to be impossible. Therefore $1 \leq r < m$, and it remains to show that rp is representable as a sum of four squares.

Since both mp and mr are sums of four squares, it follows from Euler's product identity that $m^2rp = X^2 + Y^2 + Z^2 + W^2$, with X, Y, Z, W being given by their formulae in terms of a, b, c, d, A, B, C, D. It is readily checked that each of X, Y, Z, W is divisible by m: for example,

$$X = aA + bB + cC + dD \equiv a^2 + b^2 + c^2 + d^2 \equiv 0 \pmod{m},$$
$$Y = aB - bA - cD + dC \equiv ab - ba - cd + cd \equiv 0 \pmod{m}.$$

We can now divide $m^2rp = X^2 + Y^2 + Z^2 + W^2$ by m^2 to deliver rp being a sum of four squares. The theorem is proved. □

6.4 Three-Squares Theorem

We already remarked that $8n + 7$ cannot be a sum of three squares. Moreover, if $4m$ is a sum of three squares then all three squares are necessarily even so that, on division by 2^2, m itself becomes a sum of three squares. It follows that $4^a(8n + 7)$ cannot be a sum of three squares, the necessity part of the theorem.

We mentioned earlier that Ankeny found a proof of the three-squares theorem by making use of Minkowski's theorem, which states that:

> A convex body in n-dimensional space, symmetrical about the origin, with volume exceeding 2^n, contains a lattice point distinct from the origin.

His idea is to apply the theorem to an ellipsoid in 3-dimensional space—this special case was already established by C. Hermite in 1845; the proof is given in Note 6.3. The argument then shows that a squarefree $m \not\equiv 7 \pmod 8$ has the representation $m = X^2 + Y^2 + Z^2$, where $X \in \mathbb{Z}$, and $Y, Z \in \mathbb{R}$ are such that $Y^2 + Z^2 \in \mathbb{W}$.

Consider the linear transformation $\mathbf{X} = A\mathbf{x}$, where $\mathbf{X} = (X, Y, Z)^T$ and $\mathbf{x} = (x, y, z)^T$ are column vectors in 3-dimensional vector space over $\mathbb{R}$, and A is a nonsingular matrix with elements to be chosen so that certain conditions on $\mathbf{X}$ are satisfied. Under the inverse transformation $\mathbf{x} = A^{-1}\mathbf{X}$, the sphere

$$X^2 + Y^2 + Z^2 < 2m$$

is mapped into an ellipsoid, which is symmetrical about the origin and its volume is $\frac{4}{3}\pi\,(2m)^{3/2}/|\det A|$; we shall choose A with $\det A = m^{3/2}$, so that the volume is $2^{7/2}\pi/3 > 8$. Consequently, by Minkowski's theorem, there is a nonzero vector $\mathbf{x}$ with integer coordinates x, y, z lying inside the ellipsoid. The construction of the matrix A when $m \equiv 3 \pmod 8$ requires the following lemma.

Lemma 6.1 *Let $m \equiv 3 \pmod 8$ be a squarefree number. Then there is a prime $q \equiv 1 \pmod 4$ such that the two congruences*

$$2qt^2 + 1 \equiv \pmod m \quad \textit{and} \quad b^2 + m \equiv 0 \pmod{4q},$$

are soluble in t and b.

Proof We shall require Dirichlet's theorem on the existence of primes in arithmetic progressions (see Chap. 12), the law of quadratic reciprocity, and the Chinese remainder theorem. For an odd prime p, we shall make use of

$$\left(\frac{-1}{p}\right) = \begin{cases} +1 & \text{if } p \equiv 1 \pmod 4, \\ -1 & \text{if } p \equiv 3 \pmod 4, \end{cases} \qquad \left(\frac{2}{p}\right) = \begin{cases} +1 & \text{if } p \equiv 1, 7 \pmod 8, \\ -1 & \text{if } p \equiv 3, 5 \pmod 8, \end{cases}$$

so that

$$\left(\frac{-2}{p}\right) = \begin{cases} +1 & \text{if } p \equiv 1, 3 \pmod 8, \\ -1 & \text{if } p \equiv 5, 7 \pmod 8. \end{cases}$$

Let $m = p_1 p_2 \cdots p_r$ be a product of distinct odd primes, and let q be a prime satisfying $q \equiv 1 \pmod{4H}$ and $q \equiv -2 \pmod K$, where

$$H = \prod_{p_j \equiv 1, 3 \pmod 8} p_j, \qquad K = \prod_{p_j \equiv 5, 7 \pmod 8} p_j.$$

Since $\gcd(4H, K) = 1$ and $4HK = 4m$, by the Chinese remainder theorem, the condition on q amounts to $q \equiv \ell \pmod{4m}$ for some ℓ in $1 \le \ell < 4m$, with $\gcd(\ell, 4m) = 1$. By Dirichlet's theorem there is such a prime q.

Now if $p|H$ then $\left(\frac{-2q}{p}\right) = \left(\frac{-2}{p}\right)\left(\frac{q}{p}\right) = (+1)\left(\frac{1}{p}\right) = 1$, and if $p|K$ then $\left(\frac{-2q}{p}\right) = \left(\frac{-2(-2)}{p}\right) = 1$. Therefore $\left(\frac{-2q}{p}\right) = 1$ for each $p|m$, so that the congruence $s^2 + 2q \equiv 0 \pmod{m}$ is soluble in s. Letting $st \equiv 1 \pmod{m}$, the congruence $1 + 2t^2 q \equiv 0 \pmod{m}$ is soluble in t.

Next, the condition $m \equiv 3 \pmod 8$ implies that the number of prime divisors of K is even, from which it follows that $\prod_{p|m}\left(\frac{-2}{p}\right) = 1$. The law of quadratic reciprocity then delivers

$$\left(\frac{-m}{q}\right) = \left(\frac{m}{q}\right) = \prod_{p|m}\left(\frac{p}{q}\right) = \prod_{p|m}\left(\frac{q}{p}\right) = \prod_{p|m}\left(\frac{-2q}{p}\right) = 1,$$

so that $-m$ is a quadratic residue (mod q), and because $m \equiv 3 \pmod 8$ it follows that $b^2 + m \equiv 0 \pmod{4q}$ is soluble in b. The lemma is proved. □

Proof of the Three-Squares Theorem Let $m \equiv 3 \pmod 8$ be squarefree, and choose the prime q and the numbers t, b according to the lemma. Write $b^2 + m = 4qh$, and let

$$A = \begin{pmatrix} \lambda^2 t & tb & m \\ \lambda & b/\lambda & 0 \\ 0 & \sqrt{m}/\lambda & 0 \end{pmatrix}, \qquad \lambda = \sqrt{2q},$$

so that $\det A = m^{3/2}$. For $x, y, z \in \mathbb{Z}$, let $\mathbf{X} = A\mathbf{x}$, that is

$$X = 2qtx + tby + mz, \quad Y = \lambda x + \frac{by}{\lambda}, \quad Z = \frac{\sqrt{m}y}{\lambda}.$$

Now $2q(Y^2 + Z^2) = (2qx)^2 + 4qbxy + (b^2 + m)y^2$ can be rewritten as either

$$(2qx)^2 + 4qbxy + 4qhy^2 \quad \text{or} \quad (2qx + by)^2 + my^2.$$

The first alternative allows us to write $Y^2 + Z^2 = 2v$, where

$$v = qx^2 + bxy + hy^2,$$

and the second alternative shows that

$$\begin{aligned} 2q(X^2 + Y^2 + Z^2) &\equiv 2qt^2(2qx + by)^2 + (2qx + by)^2 \pmod m \\ &\equiv (2qt^2 + 1)(2qx + by)^2 \equiv 0 \pmod m. \end{aligned}$$

Thus $X^2 + Y^2 + Z^2$ is a multiple of m, and since the only positive multiple of m which is less than $2m$ is m itself, it follows from Minkowski's theorem that there are $x, y, z \in \mathbb{Z}$ such that $X^2 + Y^2 + Z^2 = m$. It remains to show that $v \in \mathbb{W}$, so that $Y^2 + Z^2 = 2v \in \mathbb{W}$, and hence m is a sum of three squares.

Let p be any odd prime which divides v an exact odd number of times, so that we have to show that $p \equiv 1 \pmod 4$. From the definition of v, we have

$$4qv = (2qx + by)^2 + (4qh - b^2)y^2 = (2qx + by)^2 + my^2.$$

We separate the cases $p|m$, and $p \nmid m$, and proceed to show that the congruences $w^2 \pm 2q \equiv 0 \pmod p$, and $w^2 \pm m \equiv 0 \pmod p$, respectively, are soluble in w.

Suppose first that $p|m$. From $X^2 + 2v = m$, we have

$$2qX^2 + (2qx + by)^2 + my^2 = 2qm, \qquad \text{with} \quad p|X^2,\ p|(2qx + by)^2.$$

Since m is squarefree it follows that $y^2 \equiv 2q \pmod p$. On the other hand, we have $(2qt)^2 + 2q \equiv 0 \pmod p$. Thus both $w^2 \pm 2q \equiv 0 \pmod p$ are soluble, so that $\left(\frac{-1}{p}\right) = 1$, that is $p \equiv 1 \pmod 4$.

Suppose instead that $p \nmid m$. First, $X^2 + 2v = m$ implies $X^2 \equiv m \pmod p$. On the other hand, if $p = q$ then $b^2 + m \equiv 0 \pmod p$, and if $p \neq q$ then the expression for $4qv$ shows that there are $e, f \in \mathbb{Z}$ such that $e^2 + mf^2$ is divisible by an exact odd power of p, so that $w^2 + m \equiv 0 \pmod p$ is soluble. Thus both $w^2 \pm m \equiv 0 \pmod p$ are soluble, so that $\left(\frac{-1}{p}\right) = 1$, that is $p \equiv 1 \pmod 4$.

That a squarefree $m \equiv 3 \pmod 8$ is representable as a sum of three squares is established. The cases $m \equiv 1, 2, 5, 6 \pmod 8$ require only minor adjustment of the argument—we need to modify the two congruences in the lemma and also the matrix A accordingly. More specifically, $2q$ has to be replaced by q and, for $m \equiv 2, 6 \pmod 8$, we need to take $q \equiv 1, 5 \pmod 8$, respectively. We omit the similar argument.

Finally, we note that there is no loss in making the assumption that m is squarefree, because any square factor can be removed for the argument, and then multiplied back into the three squares. The theorem is proved. □

Counting Sums of Three Squares

There is an interesting dichotomy between the problems of the representation of numbers as sums of two-squares and three-squares. Whereas it is relatively easy to establish the two-squares theorem, the corresponding counting problem (Landau's theorem) is deep. In complete contrast, when it comes to the three-squares case, the representation theorem is deep but, having established it, a very satisfactory solution to the corresponding counting problem can be derived with relative ease.

The set Q of positive integers representable as a sum of three squares has asymptotic density $5/6$. To see this, we note that the number of $n \notin Q$ with $n \le x$ is the number of pairs $k, \ell \ge 0$ with $4^\ell(8k+7) \le x$, so that, writing $L = (\log x)/(\log 4)$, the number concerned is given by

$$\begin{aligned}\sum_{4^\ell(8k+7)\le x} 1 &= \sum_{0\le\ell\le L} \sum_{7\le 8k+7\le x/4^\ell} 1 = \sum_{0\le\ell\le L} \sum_{8\le 8k+8\le x/4^\ell+1} 1 \\ &= \sum_{0\le\ell\le L} \sum_{1\le k+1\le (x/4^\ell+1)/8} 1 = \sum_{0\le\ell\le L} \left\lfloor \frac{1}{8}\Big(\frac{x}{4^\ell}+1\Big)\right\rfloor \\ &= \frac{x}{8}\sum_{0\le\ell\le L}\frac{1}{4^\ell} + O(L) = \frac{x}{8}\sum_{\ell=0}^{\infty}\frac{1}{4^\ell} + O(L) \\ &= \frac{x}{6} + O(L), \qquad x\to\infty.\end{aligned}$$

Therefore the counting function $Q(x)$ of the set Q has the asymptotic formula

$$Q(x) = \frac{5x}{6} + O(\log x), \qquad x\to\infty.$$

Indeed, much more can be said about the error term. It is shown in [91] that

$$\Delta(N) = Q(N) - \frac{5N}{6}$$

is given by a 'correlated digital sum' over the base 4 representation of N. If n has the representation

$$n = \sum_{0\le r\le R} a_r(n)4^r = (a_R a_{R-1}\cdots a_1 a_0)_4,$$

where $a_r = a_r(n) = 0, 1, 2, 3$ are the digits in base 4, then

$$\Delta(n) = \frac{a_0}{6} + \sum_{1\le r\le R}\gamma(a_r, a_{r-1}), \qquad a_r = a_r(n),$$

and the values of $\gamma(a, a')$ are given in Table 6.1.

Table 6.1 Values of $\gamma(a, a')$

	$a=0$	$a=1$	$a=2$	$a=3$
$a'=0,1,2$	0	2/3	1/3	1
$a'=3$	0	−1/3	1/3	0

The formula for $\Delta(N)$ can be used to deliver the exact value of $Q(N)$ very efficiently. For example, if $N = 10^{12} = (3220311022110100000)_4$ then

$$\Delta(N) = 6\times 0 + \frac{2}{3} + 0 + \frac{2}{3} + \frac{2}{3} + \frac{1}{3} + \frac{1}{3} + 0 + \frac{2}{3} + \frac{2}{3} + 1 + 0 + \frac{1}{3} + \frac{1}{3} + 1 = \frac{20}{3},$$

giving

$$Q(10^{12}) = \frac{5}{6}\times 10^{12} + \frac{20}{3} = 833333333340.$$

The formula can also be used to establish that $\Delta(n)$ has an average order and a normal order (see Sect. 10.7), both being $3\log n/(16\log 2)$, by showing that, as $N \to \infty$,

$$\sum_{n\le N} \Delta(n) = \frac{3N\log N}{16\log 2} + O(N),$$

$$\sum_{n\le N} \Delta^2(n) = \frac{9N\log^2 N}{(16\log 2)^2} + O(N\log N).$$

6.5 Finding Representations

In finding a solution to a problem in computational number theory, we often have the paradoxical situation of being 'spoilt for choice'. By this we mean the following: If the solution is essentially unique, then there is likely to be an effective algorithm from some theory which delivers the solution; but when there are many essentially distinct solutions resulting from an existence argument, it is usually challenging to find just one!

There being an essentially unique solution to Fermat's theorem, there is a very efficient algorithm which delivers the representation of a prime $p \equiv 1 \pmod 4$ as $a^2 + b^2$. However, there is no known deterministic efficient algorithm which can deliver a solution to any of the other representation problems for sums of squares. In particular, if n is a product of two distinct large primes congruent to 1 (mod 4), there is no known efficient method which can deliver n as a sum of two squares; indeed, any such scheme may perhaps enable us to find the factors of n, which is known to be difficult.

Again, although there are plenty of representations of a given large n as a sum of four squares, it is not easy to construct a deterministic efficient algorithm to deliver just one solution, and the same can be said of finding a representation of a large suitable n as a sum of three squares.

Brillhart's Algorithm

We saw in Example 6.1 that the use of Dirichlet's approximation theorem, or Thue's theorem, does not give an efficient method for the construction of the solution to Fermat's theorem. The following algorithm by J. Brillhart [14] is the simplest and the best for the delivery of the representation of the input prime $p \equiv 1 \pmod 4$ as $a^2 + b^2$; it requires the solutions to $u^2 + 1 \equiv 0 \pmod p$, which, by Euler's criterion, are given by $u \equiv \pm n^{(p-1)/4} \pmod p$, where n is any quadratic nonresidue (mod p).

Brillhart's algorithm

- Let $p \equiv 1 \pmod 4$ and $u^2 + 1 \equiv 0 \pmod p$, with $1 < u < p/2$. If $p = u^2 + 1$: return $(a, b) = (u, 1)$.
- Apply the Euclidean algorithm to p and u, keeping three successive remainders r, r', r'' in each step. If $r > \sqrt{p} > r'$: return $(a, b) = (r', r'')$.

An explanation of why the algorithm delivers the goods is given in Sect. 7.2.

Example 6.3 Find the largest prime less than 10^8 congruent to 1 (mod 4), and express it as a sum of two squares.

Solution The prime concerned is $p = 10^8 - 11 = 99999989 \equiv 5 \pmod 8$, so that 2 is a quadratic nonresidue. We may set $u \equiv \pm 2^{(p-1)/4} \equiv \pm 33589498 \pmod p$; we take $u = 33589498 < p/2$. The output of the Euclidean algorithm for p, u is

$$\begin{aligned}
99999989 &= 2 \times 33589498 + 32820993 \\
33589498 &= 1 \times 32820993 + 768505 \\
32820993 &= 42 \times 768505 + 543783 \\
768505 &= 1 \times 543783 + 224722 \\
224722 &= 2 \times 94339 + 36044 \\
94339 &= 2 \times 36044 + 22251 \\
36044 &= 1 \times 22251 + 13793 \\
22251 &= 1 \times 13793 + 8458 \\
13793 &= 1 \times 8458 + 5335
\end{aligned}$$

and we stop here because $\lfloor \sqrt{p} \rfloor = 9999$ and $(r, r', r'') = (13793, 8458, 5335)$, so that $p = 99999989 = 8458^2 + 5335^2$. □

Representations by Sums of Two, Three and Four Squares

Consider the problem of finding representations of a large number n as a sum of two, three or four squares, assuming that the corresponding necessary conditions for n are satisfied. We first remove the largest power of 4 from n, so that we may assume that $n \not\equiv 0 \pmod 4$. Since there are product formulae for the two and four squares cases, we may also assume that n is odd for these two cases.

Two Squares Case As we already remarked, there is no known efficient algorithm which delivers the representations of a large product of two distinct primes congruent to 1 (mod 4). Since size matters, for a given large n, our only hope is to remove all the small prime factors from n; if n has a prime divisor $p \equiv 3 \pmod 4$ with an exact odd exponent, then $n \notin \mathbb{W}$. Put the remaining part m through a primality test—if m is composite, we need to remove more prime factors, or else throw the towel in; if m is a prime then apply Brillhart's algorithm for all the prime factors of n which are congruent to 1 (mod 4), and use the product formula for sums of two squares to construct the eventual representation of n. It has to be said that the chance of success is not good.

Three and Four Squares Cases There is a much better chance of finding a representation of a modest size $n \not\equiv 7 \pmod 8$ as a sum of three squares. We remove successively the largest square z^2 for which $m = n - z^2$ is suitable for an attempt to deploy the two squares case; if $m \notin \mathbb{W}$, then step down for the next z, and continue until $m \in \mathbb{W}$ is successfully decomposed as $x^2 + y^2$, so that $n = x^2 + y^2 + z^2$. A similar procedure can also be applied to find a representation of n as a sum of four squares. The chance of success should be good, and setting the procedure as an algorithm is a good exercise. Readers interested in pursuing this further should consult [75], in which much more sophisticated ideas and methods are presented.

Example 6.4 Let $a = 1+10^6+10^{12}+10^{18}+\cdots+10^{60}$, $b = 123456$, $c = 7777777$. Find $x, y, z \in \mathbb{N}$ such that $ab + c = x^2 + y^2 + z^2$.

Remark Ankeny's proof of the three-squares theorem is essentially an existence argument. There are 65 decimal digits for the number $ab + c$, and it is hopeless trying to solve the problem by starting with Lemma 6.1.

Solution Let $n = ab + c \equiv 456 + 777 \equiv 1 \pmod 8$. For $k = 0, 1, 2, \ldots$, let

$$z_k = \lfloor \sqrt{n} \rfloor - k, \quad m'_k = n - z_k^2, \quad m_k = \text{ the largest odd divisor of } m'_k.$$

We easily find witnesses for $m'_k \notin \mathbb{W}$ when $0 \le k \le 13$; for example, 3 is a witness for $m'_1, m'_3, m'_4, m'_7, m'_9, m'_{10}$. However, $m'_{14} \in \mathbb{W}$; in fact $m'_{14} = 2^8 m_{14} = 2^8 13p$, where $p = 30552189372024426799468308499 09 \equiv 1 \pmod 4$ is a prime. Brillhart's algorithm then delivers $p = 1629819715930697^2 + 631590556267290^2$, so that, together with $13 = 3^2 + 2^2$ and the product formula for sums of two squares,

we finally have $n = x^2 + y^2 + z^2$, where

$$x = 2^4 6152640260326671, \qquad y = 2^4 1364867763059524$$

and

$$z = \lfloor \sqrt{n} \rfloor - 14 = 351363235777625795878664204921991.$$

□

The following are some relevant remarks on the method. Conjecture H of Hardy and Littlewood states that all large n are representable as a sum of a square and a prime, and that the number of representations has the asymptotic value

$$\frac{\sqrt{n}}{\log n} \prod_{p>2} \left(1 - \frac{(n/p)}{p-1}\right), \qquad \left(\frac{n}{p}\right) = \text{Legendre symbol}.$$

For $x \geq 5$, the counting function for squares is $\lfloor \sqrt{x} \rfloor < \pi(x)$, so that squares are much rarer than primes, and the conjecture is considered to be deeper than the Goldbach conjecture that large even numbers are sums of two primes.

For our purpose, we want $n = x^2 + p$ or $x^2 + 2p$, where $p \equiv 1 \pmod 4$ is prime. Thus, if $n \equiv 1 \pmod 4$, and x is even then $n = x^2 + p$, where $p \equiv 1 \pmod 4$ necessarily. If $n \equiv 3 \pmod 8$, and x is odd then $n = x^2 + 2p$, where again $p \equiv 1 \pmod 4$ necessarily.

6.6 Summary and Notes

Summary

- Problems on the representation of numbers as sums of squares are very interesting, and the solutions for two, four and three squares are landmarks in the history of numbers.
- Fermat proved that every prime congruent to 1 (mod 4) is a sum of two squares.
- Lagrange proved that every number is a sum of four squares.
- The theorem of Gauss, and perhaps also Legendre, states that numbers not of the form $4^a(8n + 7)$ are sums of three squares.
- The counting function $W(x)$ for the set $\mathbb{W}$ of numbers which are sums of two squares has the asymptotic formula $W(x) \sim Bx/\sqrt{\log x}$ as $x \to \infty$, where $B \approx 0{\cdot}76422$ is the *Landau-Ramanujan* constant. As a consequence almost all numbers are not sums of two squares.

- The simple argument of Bambah and Chowla shows that the interval $[x, x + 2\sqrt{2}x^{1/4} + 1)$ always contains a sum of two squares. It is an embarrassment that, for decades, no one has been able to make a proper improvement on the result.
- It is unlikely that there is a proof of the three-squares theorem which can be described as 'simple'. Most proofs make use of Dirichlet's theorem on the existence of primes in arithmetic progressions.
- Some $5/6$ of the numbers are sums of three squares. Indeed, the counting function concerned can be given by a formula which delivers the exact count efficiently.
- The decomposition of a prime congruent to 1 (mod 4) as a sum of two squares can be delivered by the efficient Brillhart's algorithm. However, finding the representation of a general large n as a sum of a certain number of squares poses challenging problems in computational number theory.

Notes

1. There is also a proof of Fermat's theorem using lattice reduction; indeed, the corresponding algorithm is similar to Brillhart's algorithm for the delivery of the representation lattice. See [10].
2. Another proof of Fermat's theorem using involutions was given by D. R. Heath-Brown [43]. Let $p \equiv 1 \pmod{p}$ be a prime, x, y, z be positive integers, and $S = \{(x, y, z) : x^2 + 4yz = p\}$. There is a natural involution on S, namely $b : S \to S$ given by $b(x, y, z) = (x, z, y)$. The fixed points of the involution are of the form (x, y, y) which correspond to the decomposition $p = x^2 + (2y)^2$. Another involution is the mapping $h : S \to S$ where

$$h(x, y, z) = \begin{cases} (x + 2z, z, y - x - z) & y - z < x < y - 2x, \\ (2y - x, y, x - y + z) & y - 2x < x < 2y, \\ (x - 2y, x - y + z, y) & 2y < x. \end{cases}$$

   The fixed points of h now take the form (x, x, z) for which p then has the purported factorisation $p = x(x + 4z)$. That p is a prime implies $x = 1$ so that there is one, and only one, fixed point of h, namely $(1, 1, (p - 1)/4)$. Therefore the cardinal of S is odd, so that there must be a fixed point for the involution b. The argument has since been dubbed a 'one-sentence proof' of the theorem by D. Zagier [111].
3. Minkowski's theorem is needed only for 3-dimensional space $\mathbb{R}^3$ here, although the proof given below extends easily to $\mathbb{R}^n$.

   A point P in $\mathbb{R}^3$ can be represented by the vector $\mathbf{x} = (x_1, x_2, x_3)$, with $x_i \in \mathbb{R}$; we call P, or $\mathbf{x}$, a *lattice point* if $x_i \in \mathbb{Z}$. Consider the cube

$$\mathcal{U} = \{(x_1, x_2, x_3) : 0 \le x_i < 1,\ i = 1, 2, 3\},$$

which has volume 1, and its translate $\mathbf{v}+\mathcal{U} = \{\mathbf{v}+\mathbf{x},\ \mathbf{x} \in \mathcal{U}\}$ by a lattice point $\mathbf{v}$. The unit-cube $\mathcal{U}$ forms a *fundamental region* in the sense that $\mathbb{R}^3 = \bigcup(\mathbf{v} + \mathcal{U})$ is the disjoint union of all such translates over $\mathbf{v}$ of $\mathcal{U}$.

A *convex set* C in $\mathbb{R}^3$ is one with the property that if $P, Q \in C$ then the line segment PQ lies in C. A set $\mathcal{A}$ is *symmetrical* about the origin $\mathbf{O} = (0, 0, 0)$ if it has the property that $\mathbf{x} \in \mathcal{A} \iff -\mathbf{x} \in \mathcal{A}$. A *convex body* is a bounded convex open set in $\mathbb{R}^3$, and the theorem states that such a body symmetrical about the origin, and with volume exceeding 8, contains a lattice point distinct from $\mathbf{O}$.

Proof Let $\mathcal{B}$ be a symmetrical convex body with volume exceeding 8, and let

$$\mathcal{B}' = \{(\tfrac{x_1}{2}, \tfrac{x_2}{2}, \tfrac{x_3}{2}),\ (x_1, x_2, x_3) \in \mathcal{B}\},$$

the set obtained from 'shrinking' $\mathcal{B}$ by $\frac{1}{2}$ in each direction; the set $\mathcal{B}'$ is a convex body with volume exceeds 1. We proceed to show that there are two distinct points $\mathbf{x}, \mathbf{y} \in \mathcal{B}'$ such that the difference $\mathbf{x} - \mathbf{y} \neq \mathbf{O}$ is a lattice point.

Since $\mathcal{B}'$ is bounded, it intersects with only finitely many translates of the fundamental region $\mathcal{U}$; in other words, $\mathcal{B}'$ is a finite union of the sets $\mathcal{B}'_{\mathbf{v}} = \mathcal{B}' \cap (\mathbf{v} + \mathcal{U})$. On translating each $\mathcal{B}'_{\mathbf{v}}$, that is $\mathcal{B}'_{\mathbf{v}} - \mathbf{v}$, back inside $\mathcal{U}$, some of translates must intersect—this is because the volume of $\mathcal{B}'$ exceeds that of $\mathcal{U}$, and the operation of translation in $\mathbb{R}^3$ is volume invariant. Therefore there is a point $\mathbf{z} \in (\mathcal{B}'_{\mathbf{v}} - \mathbf{v}) \cap (\mathcal{B}'_{\mathbf{u}} - \mathbf{u})$, where $\mathbf{u}, v$ are distinct lattice points. Letting $\mathbf{x} = \mathbf{z} + \mathbf{u}$ and $\mathbf{y} = \mathbf{z} + \mathbf{v}$ be points in $\mathcal{B}'$, it follows that $\mathbf{x} - \mathbf{y} = \mathbf{u} - \mathbf{v} \neq \mathbf{O}$ is a lattice point.

Now $2\mathbf{x}, 2\mathbf{y} \in \mathcal{B}$, and since $\mathcal{B}$ is convex and symmetriccal about $\mathbf{O}$, it follows that $-2\mathbf{y} \in \mathcal{B}$, and the midpoint $(2\mathbf{x} - 2\mathbf{y})/2 = \mathbf{x} - \mathbf{y} \in \mathcal{B}$. The theorem is proved.

Remarks

(i) There are at least three lattice points in $\mathcal{B}$, namely $\mathbf{O}$ and $\pm(\mathbf{u} - \mathbf{v})$.

(ii) If P and Q are points in $\mathbb{R}^3$ represented by $\mathbf{x}$ and $\mathbf{y}$, then the points on the line segment PQ take the form $\lambda\mathbf{x} + \mu\mathbf{y}$, with $\lambda, \mu \geq 0$ and $\lambda + \mu = 1$. It appears that full use has not been made of the condition for convexity in the proof, in that we only apply it to the midpoint of the line segment by taking $\lambda = \mu = \frac{1}{2}$. However, it can be proved that, for a bounded open set, such a 'midpoint' property implies convexity.

4. The three-squares theorem can also be proved using Aubry's Theorem (Theorem 1.9) together with Dirichlet's theorem on the existence of primes in arithmetic progression, or the use of p-adic numbers; see [18, Postcript].
5. Theorem 6.5 is a special case of the following theorem by M. V. Jarnik: *Let ℓ be the length of a rectifiable simple closed curve, and let A be the area of the region bounded by the curve. If N is the number of lattice points inside the curve, then $|A - N| < \ell$.* The proof can be found in [47, Theorem 6.9.2].

6. The *circle problem of Gauss* is the determination of θ_1, the infimum of the set of positive real numbers θ for which the error term in Theorem 6.5 satisfies $O(x^\theta)$; see also Sect. 2.8. The proof that $\theta_1 \leq \frac{1}{3}$ can be found in [47, Theorem 6.12.1]; it is conjectured that $\theta_1 = \frac{1}{4}$.
7. On the long gaps between sums of two squares in Example 6.2, in a short and elegant paper, I. Richards [80] established that $\limsup_{n\to\infty} \frac{w_{n+1}-w_n}{\log w_n} \geq \frac{1}{4}$. The bound $\frac{1}{4}$ has been replaced by $\frac{390}{449} = 0{\cdot}868\ldots$ in the very recent paper [24]. The correct maximum order of $w_{n+1} - w_n$ is speculated to be $(\log w_n)^{3/2}$.

6.7 Exercises

1.

   (i) What is the asymptotic formula for the number of lattice points inside an ellipse centre at the origin, with major and minor axes $a > b \to \infty$?
   (ii) Show that the number of lattice points insider the sphere $u^2+v^2+w^2 \leq x$ is $\frac{4}{3}\pi x^{3/2} + O(x)$ as $x \to \infty$.
   (iii) Show that the number of lattice points inside the circle $u^2 + v^2 \leq x$ with coprime coordinates is $6x/\pi + O(\sqrt{x}\log x)$ as $x \to \infty$.

2. Use Minkowski's theorem to prove the following two results for primes p:

   (i) If $u^2 + 1 \equiv 0 \pmod{p}$ is soluble then p is a sum of two squares;
   (ii) That $u^2+v^2+1 \equiv 0 \pmod{p}$ is soluble implies p is a sum of four squares.

3. Verify that Heath-Brown's mapping given in Note 6.2 is an involution.
4. Show (without using Landau's theorem) that almost all numbers are not sums of two squares.
5. Establish the formula for $2B^2$ (which does not involve the primes) in the text.
6. Let $(w_n) = (1, 2, 4, 5, 8, 9, 10, 13, \ldots)$ denote the sequence of numbers which are sums of two squares.

   (i) Find the smallest solution of $w_{n+1} = w_n + 17$; and
   (ii) Use the Chinese remainder theorem to show that there are infinitely such solutions, producing witnesses for your claim.

7.

   (i) Count the lattice points inside, or on, the circle centre 0, radius 1000.
   (ii) Count the circles centre 0, with positive radius $r \leq 1000$, which contains at least one lattice point.

8. Determine the exact number of $n \leq 10^{200}$ which are sums of three squares.
9. Use any method to construct a prime $p \equiv 1 \pmod 4$, with $p > 10^{200}$, and find a solution of $u^2 + 1 \equiv 0 \pmod{p}$.
   Apply Brillhart's algorithm to find a, b such that $p = a^2 + b^2$.

10. (Gauss). Let $p \equiv 1 \pmod 4$ be a prime, and set

$$x \equiv \frac{((2k)!)^2}{2\,k!^2} \pmod p, \qquad y \equiv (2k)!\,x \pmod p, \qquad |x|, |y| < \frac{p}{2}.$$

Use Wilson's theorem to show that $p|x^2 + y^2$. Show further that, with adjustment of $x, y \pmod p$ if necessary, $p = x^2 + y^2$.

11. Let Q denote the set of numbers representable as sums of three squares. Show that
    (a) there are infinitely many pairs of consecutive numbers both of which do not belong to Q, and
    (b) of any 3 consecutive numbers, at least one belongs to Q.
12. Take a random number $n > 10^{150}$ with $n \equiv 1 \pmod 8$, and find one of its representations as a sum of three squares.
13. Take a random number $n > 10^{200}$, and find one of its representation as a sum of four squares.

Chapter 7
Continued Fractions

Abstract Besides the usual material on the representation of numbers as simple continued fractions, including Pell's equation, there is a description on how they can be used for factorisation of numbers. The efficient computation of continued fractions, together with some relevant metric theory, are given.

By a *(simple) finite continued fraction*, we mean an expression

$$[a_0, a_1, a_2, \ldots, a_n] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cdots + \cfrac{1}{a_n}}}}, \tag{7.1}$$

where $a_0 \in \mathbb{Z}$, $a_1, a_2, \ldots, a_n \in \mathbb{N}$; the left-hand side is just a convenient notation for the expression. Working step-by-step from the bottom line up, we find that the expression represents a fraction p_n/q_n. Take, for example, $n = 3$: the bottom line is $a_2 + 1/a_3 = (a_2a_3 + 1)/a_3$, so that the next line up is

$$a_1 + \frac{1}{a_2 + 1/a_3} = a_1 + \frac{a_3}{a_2a_3 + 1} = \frac{a_1a_2a_3 + a_1 + a_3}{a_2a_3 + 1},$$

and thus $[a_0, a_1, a_2, a_3] = a_0 + (a_2a_3 + 1)/(a_1a_2a_3 + a_1 + a_3)$, giving

$$\begin{cases} p_3 = a_0a_1a_2a_3 + a_0a_1 + a_0a_3 + a_2a_3 + 1, \\ q_3 = a_1a_2a_3 + a_1 + a_3. \end{cases} \tag{7.2}$$

Making use of $[a_0, a_1, \ldots, a_k] = a_0 + 1/[a_1, \ldots, a_k]$, mathematical induction shows that, in general, the value of $[a_0, a_1, \ldots, a_n]$ is a rational number p_n/q_n, with p_n being a sum of certain products of $a_0, a_1, \ldots, a_n$, and q_n of $a_1, a_2, \ldots, a_n$.

On the other hand, starting with a rational number p/q, the individual steps in the computation of $\gcd(p, q)$ by the Euclidean algorithm can be given their

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companion equations, which then amounts to an algorithm for the expansion of p/q as a continued fraction. Take, for example, $\gcd(179, 24) = 1$:

$$\begin{aligned} 179 &= 7\times 24 + 11, & \tfrac{179}{24} &= 7 + \tfrac{11}{24}, \\ 24 &= 2\times 11 + 2, & \tfrac{24}{11} &= 2 + \tfrac{2}{11}, \\ 11 &= 5\times 2 + 1, & \tfrac{11}{2} &= 5 + \tfrac{1}{2}, \end{aligned}$$

which enables us to write

$$\frac{179}{24} = 7 + \frac{1}{2 + \dfrac{2}{11}} = 7 + \frac{1}{2 + \dfrac{1}{5 + \dfrac{1}{2}}} = [7, 2, 5, 2].$$

In general, any rational number p/q can be represented by a simple finite continued fraction $[a_0, a_1, a_2, \ldots, a_n]$.

Suppose we allow for infinitely many a_n to form an infinite continued fraction $[a_0, a_1, a_2, \ldots]$. There is now no bottom line from which to work up, but we can define the expression by

$$[a_0, a_1, a_2, a_3, \cdots] = \lim_{n\to\infty} [a_0, a_1, a_2, \cdots, a_n], \tag{7.3}$$

although we then need to show that the limit exists for every such sequence (a_n).

We call a_n the nth *partial quotient*, and

$$\frac{p_n}{q_n} = [a_0, a_1, a_2, \ldots, a_n]$$

the nth *convergent*, of the continued fraction, finite or infinite. Every nonzero real number is representable as a continued fraction, with a_0 being the integer part of the number—sometime we write $[a_0; a_1, \ldots]$ instead of $[a_0, a_1, \ldots]$ to emphasise that a_0 is the integer part of the expression, with $a_0 \geq 0$, or $a_0 < 0$, for the positive, or negative, reals. By extending the definition of a finite continued fraction to allow the last term to be a real number, instead of restricting it to be an integer, the expression now represents a real number—indeed, continuity shows that every real number can be represented as

$$x = [a_0; a_1, a_2, \ldots, a_{n-1}, \alpha_n] = [a_0; a_1, a_2, \ldots, a_{n-1} + 1/\alpha_n] \qquad \alpha_n \geq 1;$$

we call $\alpha_n \in \mathbb{R}$ the nth *complete quotient*.

We shall establish the following:

- The continued fraction representation of a rational number is essentially unique;
- There are iterative formulae for p_n and q_n in terms of a_n;

- The limit (7.3) always exists as an irrational number; moreover every irrational number x has the unique representation $x = [a_0, a_1, a_2, \ldots]$, with a_n being given iteratively in terms of $x, a_0, a_1, \ldots, a_{n-1}$.

Continued fractions are a good thing, and their properties should be better known. They are useful because they often can be used to give explicit construction of solutions to problems when other methods will only show the existence of solutions. The first appearance of such fractions was in the work of the Indian astronomer Aryabhata in the fifth century. They were also used by R. Bombelli (1526–1573), and later by C. Huygens (1629–1695) in his selection of gears in the construction of a mechanical planetarium. Euler was the first person to make a systematic study of the subject, discovering many interesting and important facts, but many of the proofs were supplied only later by Lagrange.

7.1 Basic Properties

Recall that, for the decimal representation of a rational number $m/2^k5^\ell$, we may choose either to have a terminating expression, or an infinite one ending with 9s. Similarly there are two possible choices for the representation of a rational number by a finite continued fraction, because $[a_0, a_1, \ldots, a_n] = [a_0, a_1, \ldots, a_n - 1, 1]$ if $a_n > 1$. Thus we may choose to have the last partial quotient $a_n = 1$, or $a_n > 1$; alternatively we may choose to specify the parity of n. We show that after either such specification there is uniqueness.

Theorem 7.1 *Suppose that* $[a_0, a_1, \ldots, a_n] = [b_0, b_1, \ldots, b_m]$, *with* $a_n, b_m > 1$. *Then* $n = m$ *and* $a_i = b_i$ *for* $i = 0, 1, \ldots, n$.

Proof Suppose that $n \neq m$, say $n > m$. Equality of the two continued fractions implies equality of their integer parts, that is $a_0 = b_0$, and hence $[a_1, a_2, \ldots, a_n] = [b_1, b_2, \ldots, b_m]$. Proceeding inductively, we find that $a_i = b_i$ for $i = 0, 1, \ldots, m - 1$, and

$$[a_m, a_{m+1}, \ldots, a_n] = [b_m], \quad \text{that is} \quad a_m + \frac{1}{[a_{m+1}, \ldots, a_n]} = b_m.$$

Since a_m and b_m are integers this means that $[a_{m+1}, \ldots, a_n] = 1$, that is $m + 1 = n$ and $a_n = 1$, contradicting the hypothesis that $a_n > 1$. The theorem is proved. □

Let $a_0, a_1, \ldots$ be a finite or infinite sequence with $a_n \in \mathbb{N}$ for $n \geq 1$. Define (p_n) and (q_n) by

$$\begin{cases} p_0 = a_0, & p_1 = a_0a_1 + 1, & p_{n+2} = a_{n+2}p_{n+1} + p_n, \\ q_0 = 1, & q_1 = a_1, & q_{n+2} = a_{n+2}q_{n+1} + q_n, \end{cases} \tag{7.4}$$

for $n = 0, 1, 2, \ldots$. Much of the theory of continued fractions is based on the results in the following theorem.

Theorem 7.2 *Let $\alpha > 1$ be a real number, $(a_0, a_1, \ldots)$ be integers, with $a_n > 0$ for $n \geq 1$, and p_n, q_n be defined by (7.4). Then*

$$\text{(i)} \qquad [a_0, a_1, \ldots, a_n, \alpha] = \frac{\alpha p_n + p_{n-1}}{\alpha q_n + q_{n-1}}, \qquad n \geq 1;$$

$$\text{(ii)} \qquad [a_0, a_1, \ldots, a_n] = \frac{p_n}{q_n};$$

$$\text{(iii)} \qquad p_n q_{n+1} - p_{n+1} q_n = (-1)^{n+1}, \qquad \gcd(p_n, q_n) = 1;$$

$$\text{(iv)} \qquad \frac{p_n}{q_n} - \frac{p_{n+1}}{q_{n+1}} = \frac{(-1)^{n+1}}{q_n q_{n+1}};$$

$$\text{(v)} \qquad \frac{p_0}{q_0} < \frac{p_2}{q_2} < \cdots < \frac{p_{2n}}{q_{2n}} < \frac{p_{2n+1}}{q_{2n+1}} < \cdots < \frac{p_1}{q_1}.$$

Proof

(i) We use induction; for $n = 1$, we find that

$$[a_0, a_1, \alpha] = a_0 + \frac{1}{a_1 + 1/\alpha} = a_0 + \frac{\alpha}{\alpha a_1 + 1}$$
$$= \frac{\alpha a_0 a_1 + a_0 + \alpha}{\alpha a_1 + 1} = \frac{\alpha p_n + p_{n-1}}{\alpha q_n + q_{n-1}}.$$

Assuming as induction hypothesis that (i) holds, we have, for the inductive step,

$$[a_0, a_1, \ldots, a_{n+1}, \alpha] = [a_0, a_1, \ldots, a_n, (a_{n+1} + 1/\alpha)]$$
$$= \frac{(a_{n+1} + 1/\alpha) p_n + p_{n-1}}{(a_{n+1} + 1/\alpha) q_n + q_{n-1}}$$
$$= \frac{(\alpha a_{n+1} + 1) p_n + \alpha p_{n-1}}{(\alpha a_{n+1} + 1) q_n + \alpha q_{n-1}}$$
$$= \frac{\alpha p_{n+1} + p_n}{\alpha q_{n+1} + q_n},$$

where we have made use of (7.4).

(ii) Set $\alpha = a_{n+1}$ in (i), apply (7.4), and then replace $n + 1$ by n.

(iii) By (7.4), we have $p_0 q_1 - p_1 q_0 = -1$ and also $p_n q_{n+1} - p_{n+1} q_n = -(p_{n-1} q_n - p_n q_{n-1})$; now use induction, and $\gcd(p_n, q_n) = 1$ also follows.

(iv) Divide (iii) by $q_n q_{n+1}$.

(v) By (iv) we have

$$\frac{p_n}{q_n} - \frac{p_{n+2}}{q_{n+2}} = \left(\frac{p_n}{q_n} - \frac{p_{n+1}}{q_{n+1}}\right) + \left(\frac{p_{n+1}}{q_{n+1}} - \frac{p_{n+2}}{q_{n+2}}\right) = \frac{(-1)^n}{q_{n+1}}\left(\frac{1}{q_n} - \frac{1}{q_{n+2}}\right),$$

which is positive, or negative, depending on whether n is even, or odd. We also have $\frac{p_0}{q_0} < \frac{p_1}{q_1}$, so that the required inequalities follow. □

The results in the theorem are used so frequently that sometimes no further reference to them will be made. For example, it follows at once that the convergents $[a_0, \ldots, a_n]$ form a Cauchy sequence in $\mathbb{R}$, so that every infinite simple continued fraction exists as its limit and, as a corollary, we also have:

Theorem 7.3 *Let $\theta = [a_0, a_1, \ldots]$, and p_n/q_n and α_n be its nth convergent and complete quotient, respectively. Then, for $n = 1, 2, \ldots$,*

$$\left|\theta - \frac{p_n}{q_n}\right| = \frac{1}{q_n(\alpha_{n+1}q_n + q_{n-1})}, \tag{7.5}$$

so that

$$\left|\theta - \frac{p_n}{q_n}\right| < \frac{1}{q_n q_{n+1}} < \frac{1}{a_{n+1}q_n^2} \leq \frac{1}{q_n^2}. \tag{7.6}$$

Making use of the 'forward' iterative formulae (7.4) the convergents of a continued fraction can be evaluated much more efficiently than starting from the bottom line of a continued fraction.

Example 7.1 For $x = [2, 6, 1, 5, 6, 7]$, we use (7.4) to deliver Table 7.1.

Conversely, given a real number $x > 0$, rational or irrational, we can determine its continued fraction representation by the following procedure. Write $\{x\} = x - \lfloor x \rfloor$, the fractional part of x, so that $0 \leq \{x\} < 1$, and $\{x\} = 0$ if and only if $x \in \mathbb{N}$. Set $x_0 = x$, $a_0 = \lfloor x_0 \rfloor$ and, for $n = 0, 1, 2, \ldots$, define

$$x_{n+1} = \frac{1}{\{x_n\}}, \quad a_{n+1} = \lfloor x_{n+1} \rfloor, \quad \text{if} \quad \{x_n\} > 0. \tag{7.7}$$

If x is rational, then x_n is rational for every n and the Euclidean algorithm shows that x_n will be an integer for some n, thereby terminating the process with $x = [a_0, a_1, \ldots, a_n]$. If x is irrational then so is every complete quotient x_n, and the

Table 7.1 Convergents p_n/q_n from partial quotients a_n

n	0	1	2	3	4	5
a_n	2	6	1	5	6	7
p_n	2	13	15	88	543	3889
q_n	1	6	7	41	253	1812

process cannot terminate; we then have

$$x = [a_0, a_1, \ldots] = [a_0, a_1, \ldots, a_{n-1}, x_n], \qquad n = 1, 2, \ldots .$$

The argument in Theorem 7.1 shows that the representation is unique.

Example 7.2 Determine the first 20 partial quotients of the simple continued fractions for (i) $\sqrt{7}$, (ii) e, (iii) π.

The three numbers $x = \sqrt{7}, e, \pi$ are irrational, and we may write

$$x = [a_0, a_1, \ldots, a_{19}, x_{20}],$$

where $x_{20} > 1$ is a complete quotient, and $a_{20} = \lfloor x_{20} \rfloor$. We shall determine a_n, and the corresponding convergents, for $n \leq 20$, checking that

$$\frac{p_{20}}{q_{20}} < x < \frac{p_{19}}{q_{19}}.$$

We can apply the basic algorithm (7.7), using the truncated decimal expansion of x as input, but how accurate should we take the input approximation to be? By (7.6), for $n = 20$, the error has the bound $1/q_{20}^2$ and, by (7.4), the value of q_{20} is about the size of the product $a_1 a_2 \cdots a_{20}$, but we do not know a_n until after the evaluation!

To proceed, we make a guess of the geometric-mean of a_n. It seems likely that, for a random x, the value taken most frequently by a_n will be 1, with decreasing frequencies corresponding to 2, 3, ... Making the guess that $(a_1 a_2 \cdots a_{20})^{1/20} < 4$ does not seem unreasonable, so we assume that $q_{20}^2 < 4^{40} \approx 10^{24}$. We proceed to take 24 decimal places approximations to x, using Newton's method, for example. Anyway, we take the right-hand side of the following

$$\begin{aligned} \sqrt{7} &\approx 2{\cdot}64575\ 13110\ 64590\ 59050\ 1615, \\ e &\approx 2{\cdot}71828\ 18284\ 59045\ 23536\ 0287, \\ \pi &\approx 3{\cdot}14159\ 26535\ 89793\ 23846\ 2643, \end{aligned}$$

as the input decimal (thus rational) approximations of x in the evaluation of a_n. The iterative formulae (7.7) then delivers

$$\begin{aligned} \sqrt{7} &= [2, 1, 1, 1, 4, 1, 1, 1, 4, 1, 1, 1, 4, 1, 1, 1, 4, 1, 1, 1, 4, \ldots], \\ e &= [2, 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, 1, 10, 1, 1, 12, 1, 1, \ldots], \\ \pi &= [3, 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, 1, \ldots]. \end{aligned}$$

(i) For $\sqrt{7}$, we have

$$\frac{p_{20}}{q_{20}} = \frac{2388325}{902702} < \sqrt{7} < \frac{514088}{194307} = \frac{p_{19}}{q_{19}};$$

here $q_{19}q_{20} = 194307 \times 902702 = 17\,54013\,17514 < 2 \times 10^{11}$, so that the error bound 10^{-24} was extravagant for the purpose; but we did not know that the geometric-mean of $1, 1, 1, 4$, being $\sqrt[4]{4} = \sqrt{2} \approx 1{\cdot}4$, is so much smaller than our guess value 4. The expansion appears to be periodic with the period $\overline{1, 1, 1, 4}$.

(ii) For e, we have

$$\frac{p_{20}}{q_{20}} = \frac{410105312}{150869313} < e < \frac{28245729}{10391023} = \frac{p_{19}}{q_{19}}.$$

Again, with $q_{19}q_{20} = 10391023 \times 150869313 < 2 \times 10^{14}$, so that 10^{-24} was still extravagant; the geometric mean of $a_1, \ldots, a_{20}$ is about $1{\cdot}95$, which is still much less than 4. The expansion appears to have the pattern of successive even numbers $2, 4, 6, \ldots$ being separated by '1, 1'; as we shall see in Sect. 8.6 and Example 7.9, this is so, and that the geometric mean tends to infinity as $n \to \infty$, so that the extravagance comes from $n = 20$ being rather small.

(iii) For π, we have

$$\frac{p_{20}}{q_{20}} = \frac{21053343141}{6701487259} < \pi < \frac{14885392687}{4738167652} = \frac{p_{19}}{q_{19}}.$$

Now $q_{19}q_{20} = 4738167652 \times 6701487259 \approx 4 \times 10^{19}$, so that 10^{-24} was perhaps generous but not unreasonable—the geometric-mean of $a_1, a_2, \ldots, a_{20}$ is about $2{\cdot}6$. The sequence of partial quotients does not appear to have any pattern. □

Example 7.3 Show that $\sqrt{7} = [2, \overline{1, 1, 1, 4}]$, where the bar indicates the period, which is repeated indefinitely.

We need to use exact arithmetic, so keep $\sqrt{7}$ as a number in (7.7). Let $x_0 = \sqrt{7}$, so that $a_0 = \lfloor x_0 \rfloor = 2$, and

$$x_1 = \frac{1}{x_0 - a_0} = \frac{1}{\sqrt{7} - 2} = \frac{\sqrt{7} + 2}{3}, \quad a_1 = \lfloor x_1 \rfloor = 1,$$

$$x_2 = \frac{1}{x_1 - a_1} = \frac{1}{(\sqrt{7} + 2)/3 - 1} = \frac{3}{\sqrt{7} - 1} = \frac{\sqrt{7} + 1}{2}, \quad a_2 = \lfloor x_2 \rfloor = 1,$$

$$x_3 = \frac{1}{x_2 - a_2} = \frac{1}{(\sqrt{7} + 1)/2 - 1} = \frac{2}{\sqrt{7} - 1} = \frac{\sqrt{7} + 1}{3}, \quad a_3 = \lfloor x_3 \rfloor = 1,$$

$$x_4 = \frac{1}{x_3 - a_3} = \frac{1}{(\sqrt{7}+1)/3 - 1} = \frac{3}{\sqrt{7}-2} = \frac{\sqrt{7}+2}{1}, \quad a_4 = \lfloor x_4 \rfloor = 4,$$

$$x_5 = \frac{1}{x_4 - a_4} = \frac{1}{\sqrt{7}-2} = x_1.$$

It follows that $x_{n+4} = x_n$ and $a_{n+4} = a_n$ for $n = 1, 2, \ldots$. □

We show later that the continued fraction of x is eventually periodic if and only if x is a quadratic irrational number, and that, for $\sqrt{N}$, there is a special pattern for the period. Euler found that the continued fraction for e does have the stated pattern. However, nothing concrete is known concerning the continued fraction for π; for example, we do not even know whether its sequence of partial quotients (a_n) is bounded or not.

We shall say more on the value of the geometric-mean $(a_1 \cdots a_n)^{1/n}$, and describe in Sect. 7.5 a more efficient method for the evaluation of a large number of partial quotients.

7.2 Continuants

Denote the denominator q_n of the nth convergent $[a_0, a_1, a_2, \ldots, a_n]$ by

$$K(a_1, a_2, \ldots, a_n).$$

Then, by the iterative formula for q_n in (7.7),

$$K(a_1) = a_1, \qquad K(a_1, a_2) = a_1 a_2 + 1,$$
$$K(a_1, \ldots, a_n) = a_n K(a_1, \ldots, a_{n-1}) + K(a_1, \ldots, a_{n-2}), \quad n = 3, 4, \ldots.$$

We have slightly abused the notation by using the same letter K when the function depends also on the number of parameters involved. Anyway, we then find that $p_n = K(a_0, a_1, a_2, \ldots, a_n)$ and, on setting $K() = 1$,

$$[a_0, a_1, a_2, \ldots, a_n] = \frac{K(a_0, a_1, a_2, \ldots, a_n)}{K(a_1, a_2, \ldots, a_n)}, \qquad n = 0, 1, 2, \ldots.$$

The functions $K(a_1, a_2, \ldots, a_n)$ are called *continuants*, or Euler polynomials because Euler gave the following rule for their formation: Take the sum of all possible products of $a_1, a_2, \ldots, a_n$ in which any number of disjoint pairs of consecutive terms are deleted; for example, $K(a_1, a_2, a_3, a_4, a_5)$ is given by

$$a_1a_2a_3a_4a_5 + a_3a_4a_5 + a_1a_4a_5 + a_1a_2a_5 + a_1a_2a_3 + a_1 + a_3 + a_5.$$

It follows from Euler's rule that the value of $K(a_0, a_1, \ldots, a_n)$ is unaltered if the terms are written in the reversed order:

$$K(a_0, a_1, \ldots, a_n) = K(a_n, a_{n-1}, \ldots, a_0).$$

The continuant can be given by the determinant of a tridiagonal matrix:

$$K(a_1, a_2, \ldots, a_n) = \begin{vmatrix} a_1 & 1 & 0 & \ldots & 0 & 0 \\ -1 & a_2 & 1 & \ldots & 0 & 0 \\ 0 & -1 & a_3 & \ldots & 1 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \ldots & a_{n-1} & 1 \\ 0 & 0 & 0 & \ldots & -1 & a_n \end{vmatrix},$$

from which various properties associated with the functions become transparent. For example, it can be shown that

$$K(a_0, \ldots, a_n) = K(a_0, \ldots, a_\ell)K(a_{\ell+1}, \ldots, a_n) + K(a_0, \ldots, a_{\ell-1})K(a_{\ell+2}, \ldots, a_n). \tag{7.8}$$

Although rarely used nowadays, continuants sometimes provide new insight into problems involving continued fractions. For example, H. J. S. Smith [95] used them to give the first proof of Fermat's theorem on the representation of a prime $p \equiv 1 \pmod 4$ as a sum of two squares *without* making use of the fact that -1 is a quadratic residue (mod p). The following is a sketch of his argument.

Let $p \equiv 1 \pmod 4$, $\mathcal{Q} = \{2, 3, \ldots, (p-1)/2\}$, and consider the continued fractions

$$\frac{p}{q} = [a_0, a_1, \ldots, a_n], \qquad q \in \mathcal{Q}, \quad a_0, a_n \geq 2.$$

Corresponding to each $q \in \mathcal{Q}$ there is a unique $r \in \mathcal{Q}$ such that the ordering of the partial quotient for p/r is reversed:

$$\frac{p}{r} = [a_n, a_{n-1}, \ldots, a_0];$$

the mapping $q \leftrightarrow r$ is an *involution* on the set $\mathcal{Q}$; this means that the inverse of the mapping is itself. Members of $\mathcal{Q}$ now form pairs (q, r) and, since $|\mathcal{Q}| = (p-3)/2$ is odd, there is at least one 'self-paired' (u, u). For such a pair the continued fraction for p/u is palindromic:

$$\frac{p}{u} = [a_0, a_1, \ldots, a_1, a_0] = \frac{K(a_0, a_1, \ldots, a_1, a_0)}{K(a_1, a_2, \ldots, a_1, a_0)}.$$

Smith then showed that there is no central term in the continuant K for p, because the existence of such a term a_ℓ, together with (7.8), led to p having the proper divisor $K(a_0, \ldots, a_{\ell-1})$ which is impossible for a prime. Therefore $p = K(a_0, \ldots, a_\ell, a_\ell, \ldots, a_0)$, and a further application of (7.8) delivers

$$p = K(a_0, \ldots, a_{\ell-1})^2 + K(a_0, \ldots, a_\ell)^2.$$

That $p|u^2 + 1$ is not involved in the proof, although Smith mentioned it as a consequence, whereas other proofs made use of it as an initial stage. More interesting is that Smith's argument led to the very efficient algorithm (see Sect. 6.5) for the evaluation of the representation of p as a sum of two squares.

7.3 Periodic Continued Fractions

The method of evaluating $\sqrt{7} = [2; \overline{1, 1, 1, 4}]$ in Example 7.3 extends easily to show that, for $k = \pm 1, \pm 2$, the period lengths of $\sqrt{a^2 + k}$ do not depend on a. We shall see later that this is true also for $k = 4$, but not for $k = 3$.

Example 7.4 For $a = 1, 2, 3, \ldots$, we have

$$\begin{aligned}
&\text{(i)} && \sqrt{a^2+1} = [a; \overline{2a}] = [a; 2a, 2a, \ldots],\\
&\text{(ii)} && \sqrt{a^2+2} = [a; \overline{a, 2a}] = [a; a, 2a, a, 2a, \ldots],\\
&\text{(iii)} && \sqrt{(a+1)^2-2} = [a; \overline{1, a-1, 1, 2a}]\\
&&& \qquad = [a; 1, a-1, 1, 2a, 1, a-1, 1, 2a, \ldots],\\
&\text{(iv)} && \sqrt{(a+1)^2-1} = [a; \overline{1, 2a}] = [a; 1, 2a, 1, 2a, \ldots],
\end{aligned}$$

so that the period lengths are 1, 2, 4, 2, respectively,

The derivations of (i), (ii), (iv) are similar to, and easier than, that of (iii), which is given here. Let $N = (a+1)^2 - 2$ and $\sqrt{N} = [a_0; a_1, a_2 \ldots]$. From $a^2 < N < (a+1)^2$, we have $a_0 = a$, and

$$\alpha_1 = \frac{1}{\sqrt{N} - a} = \frac{\sqrt{N} + a}{N - a^2} = \frac{\sqrt{N} + a}{2a - 1} = 1 + \frac{\sqrt{N} - (a-1)}{2a - 1},$$

$$\alpha_2 = \frac{2a - 1}{\sqrt{N} - (a-1)} = \frac{\sqrt{N} + (a-1)}{2} = (a-1) + \frac{\sqrt{N} - (a-1)}{2},$$

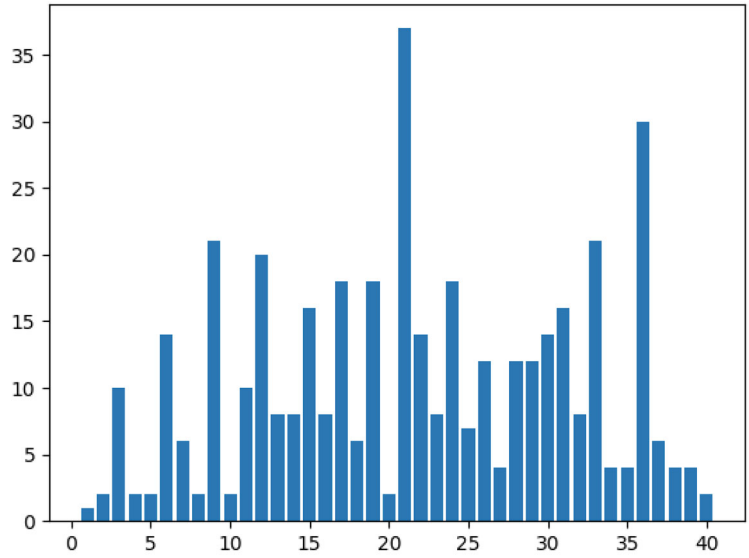


Fig. 7.1 Period lengths for $\sqrt{a^2+k}$, $a = 20$, $1 \le k \le 40$

$$\alpha_3 = \frac{2}{\sqrt{N}-(a-1)} = \frac{\sqrt{N}+(a-1)}{2a-1} = 1 + \frac{\sqrt{N}-a}{2a-1},$$
$$\alpha_4 = \frac{2a-1}{\sqrt{N}-a} = \sqrt{N}+a = 2a+(\sqrt{N}-a) = 2a+\frac{1}{\alpha_1},$$

so that $(a_1, a_2, a_3, a_4) = (1, a-1, 1, 2a)$, and $a_{n+4} = a_n$ for $n = 1, 2, \ldots$.

Let $\ell(N)$ denote the period lengths for $\sqrt{N}$, with $N = a^2 + k$, $1 \le k \le 2a$. There are also other explicit formulae for $\ell(N)$: For example, $\ell(a^2+4) = 5$ or 2, depending on whether $a \ge 3$ is odd or even. The values for $\ell(N)$ for $3 \le a \le 18$ with $3 \le k \le 2a-2$ are given in Sect. 7.5, where there is also a graphical display of the lengths in Fig. 7.1 for $a = 20$.

Meanwhile we shall prove that, in general,

$$\sqrt{N} = [a_0; \overline{a_1, a_2, \ldots, a_2, a_1, 2a_0}], \qquad a_0 = \lfloor \sqrt{N} \rfloor; \tag{7.9}$$

the period begins immediately after the integer part a_0, and it consists of a symmetrical, or palindromic, part $a_1, a_2, \ldots, a_2, a_1$, followed by $2a_0$. The symmetrical part may, or may not, have a central term, and it may be empty, reducing the period to the single number $\overline{2a_0}$; for example, $\sqrt{a^2+1} = [a; \overline{2a}]$. Note that, from (7.9),

$$\sqrt{N} + a_0 = [2a_0, \overline{, a_1, a_2, \ldots, a_2, a_1, 2a_0}] = [\overline{2a_0, a_1, a_2, \ldots, a_2, a_1}],$$

where the period starts right from the beginning.

Besides the rationals, the only class of numbers about which much is known concerning their continued fractions are the quadratic irrationals—real number solutions of quadratic equations $ax^2 + bx + c = 0$, with integer coefficients a, b, c, and $D = b^2 - 4ac > 0$ not a square. Quadratic irrationals then take the form

$$\theta = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{P \pm \sqrt{D}}{Q}, \tag{7.10}$$

where $P = -b$, $Q = 2a \neq 0$, so that $Q | P^2 - D$. That D is not a square means that there are two (algebraic) conjugates roots θ and $\bar{\theta}$; note that $\theta + \bar{\theta}$ and $\theta\bar{\theta}$ are rational, and that $\overline{1/\theta} = 1/\bar{\theta}$.

If $p, q \in \mathbb{Z}$ then $p\theta + q$ is also a quadratic irrational; indeed, from

$$\frac{1}{\theta} = \frac{Q}{P \pm \sqrt{D}} = \frac{Q(P \mp \sqrt{D})}{P^2 - D}$$

it follows that $(p\theta + q)/(u\theta + v)$ is also a quadratic irrational. The coefficient of $\sqrt{D}$ can be normalised to be 1 by changing the value of D; also, after such transformations, the resulting quadratic irrational may no longer satisfy the condition that '$Q | P^2 - D$', but we may rewrite it as '$(PQ \pm \sqrt{DQ^2})/Q^2$', which then satisfies the condition.

The algorithm (7.7) for the computation of the partial quotients becomes rather simple for a quadratic irrational, involving only integers.

Lemma 7.1 *Let $\theta = (P_0 + \sqrt{N})/Q_0$ be a quadratic irrational, with $Q_0 | N - P_0^2$. Then, for $k = 0, 1, \ldots$, the complete quotients α_k, and partial quotients a_k, are given by*

$$\alpha_k = \frac{P_k + \sqrt{N}}{Q_k}, \qquad Q_k \neq 0, \qquad a_k = \lfloor \alpha_k \rfloor, \tag{7.11}$$

where P_k, Q_k are integers given by

$$P_{k+1} = a_k Q_k - P_k, \qquad Q_{k+1} = \frac{N - P_{k+1}^2}{Q_k}; \tag{7.12}$$

in particular, $Q_k | N - P_{k+1}^2$.

Proof For $k \geq 0$, we have

$$\begin{aligned} \alpha_{k+1} &= \frac{1}{\alpha_k - a_k} = \frac{1}{(P_k + \sqrt{N})/Q_k - a_k} = \frac{Q_k}{P_k - a_k Q_k + \sqrt{N}} \\ &= \frac{Q_k(P_k - a_k Q_k - \sqrt{N})}{(P_k - a_k Q_k)^2 - N} = \frac{a_k Q_k - P_k + \sqrt{N}}{(N - (P_k - a_k Q_k)^2)/Q_k}. \end{aligned}$$

For $k = 0$, we take $P_1 = a_0 Q_0 - P_0$ and $Q_1 = (N - P_1^2)/Q_0$, as in (7.12), so that (7.11) holds for $k = 1$. Mathematical induction then shows that (7.11) holds for all k.

It remains to verify that $Q_k | N - P_{k+1}^2$ to ensure that P_k, Q_n are integers. Assume, as induction hypothesis, that $P_k \in \mathbb{Z}$ and $Q_{k-1} Q_k = N - P_k^2$. Then $P_{k+1} = a_k Q_k - P_k$ is an integer. Now from

$$N - P_{k+1}^2 = N - (a_k Q_k - P_k)^2 = (N - P_k^2) + Q_k(2a_k P_k - Q_k a_k^2),$$

and the induction hypothesis, Q_k divides the right-hand side of the equation, so that it divides $N - P_{k+1}^2$, completing the induction step. The lemma is proved. □

Note that only integer arithmetic is involved in the computation of the partial quotients a_k, because it is easy to check that

$$\left\lfloor \frac{P + \sqrt{N}}{Q} \right\rfloor = \left\lfloor \frac{P + \lfloor \sqrt{N} \rfloor}{Q} \right\rfloor.$$

Let $\theta = [a_0; a_1, a_2, \ldots]$ be a quadratic irrational with $a_0 \geq 1$, so that $\theta > 1$. We show that the continued fraction is eventually periodic: there exist i, n such that

$$\theta = [a_0, a_1, a_2, \ldots, a_i, \overline{a_{i+1}, a_{i+2}, \ldots, a_{i+n}}],$$

where the terms under the bar is the *period*, which repeats indefinitely; the smallest such n is the length of the period, but when the context is clear this length is also called the period. The $(i + 1)$th complete quotient of θ is given by

$$\phi = [\overline{a_{i+1}, a_{i+2}, \ldots, a_{i+n}}],$$

a *purely periodic continued fraction*, in which the period starts from the beginning. Being periodic means being eventually periodic; since every periodic continued fraction ends repeatedly with a purely periodic complete quotient, we investigate the properties of purely periodic continued fractions first.

Purely Periodic Continued Fractions

A purely periodic continued fraction $\theta = [\overline{a_0, a_1, \ldots, a_n}] > a_0 \geq 1$ can be written as $\theta = [a_0, a_1, \ldots, a_n, \theta]$, so that, by (i) of Theorem 7.2,

$$\theta = \frac{\theta p_n + p_{n-1}}{\theta q_n + q_{n-1}}, \quad \text{that is} \quad q_n \theta^2 - (p_n - q_{n-1})\theta - p_{n-1} = 0,$$

where p_k/q_k is the kth convergent. Now let $\phi = [\overline{a_n, a_{n-1}, \ldots, a_0}]$, the purely periodic continued fraction with the terms of the period in the reversed order. Letting p'_n/q'_n denote the nth convergent we then find, using continuants, that $(p'_n, p'_{n-1}, q'_n, q'_{n-1}) = (p_n, q_n, p_{n-1}, q_{n-1})$; for example,

$$p_n = K(a_0, a_1, \ldots, a_n) = K(a_n, a_{n-1}, \ldots, a_0) = p'_n.$$

It follows that

$$\phi = \frac{\phi p_n + q_n}{\phi p_{n-1} + q_{n-1}}, \quad \text{that is} \quad p_{n-1}\phi^2 - (p_n - q_{n-1})\phi - q_n = 0.$$

Setting $\phi = -1/\theta$, the transformed quadratic equation in θ is the same as that given earlier; this means that θ and $-1/\theta$ are the conjugates roots. Writing $\bar{\theta} = -1/\theta$, it follows from $\theta > 1$ that $-1 < \bar{\theta} < 0$.

Definition 7.1 A quadratic irrational $\theta > 1$ is said to be *reduced* if its conjugate $\bar{\theta}$ satisfies $-1 < \bar{\theta} < 0$.

Lemma 7.2 *If θ is a reduced quadratic irrational then so is $1/(\theta - \lfloor\theta\rfloor)$, so that the complete quotients of reduced numbers are reduced.*

Proof Let $\phi = 1/(\theta - a)$, where $a = \lfloor\theta\rfloor$, so that $\phi > 1$. Taking the conjugate of $\theta = a + 1/\phi$ we have $\bar{\theta} = a + 1/\bar{\phi}$, and deduce from $\bar{\theta} < 0$ that $1/\bar{\phi} < -a \leq -1$. This implies $-1 < \bar{\phi} < 0$, so that ϕ is a reduced quadratic irrational. □

Theorem 7.4 (Galois) *A necessary and sufficient condition for a quadratic irrational θ to have a purely periodic continued fraction is that θ is reduced. Furthermore, if θ is reduced and $\theta = [\overline{a_0, a_1, \ldots, a_n}]$ then $-1/\bar{\theta} = [\overline{a_n, a_{n-1}, \ldots, a_0}]$.*

Proof The necessity part, and the last part, of the theorem have been established. For the sufficiency part, let $\theta > 1$ be a reduced irrational satisfying (7.10); we can choose the $+$ sign, for we can change the signs of P, Q if necessary, so that

$$\theta = \frac{P + \sqrt{D}}{Q}, \qquad \bar{\theta} = \frac{P - \sqrt{D}}{Q}, \qquad Q | P^2 - D.$$

Since θ is reduced, we have $-1 < \bar{\theta} < 0$; together with $\theta > 1$, it follows that

$$0 < P < \sqrt{D}, \quad 0 < Q < 2\sqrt{D}, \tag{7.13}$$

which shows that there are only finitely many reduced irrationals with a fixed D. Now let $\theta = [a_0, a_1, \ldots a_{n-1}, \alpha_n]$, where the complete quotient α_n is also a reduced irrational. From $\theta = \alpha_0 = a_0 + 1/\alpha_1$, we have $\alpha_1 = 1/(\alpha_0 - a_0)$ and, in general, $\alpha_{n+1} = 1/(\alpha_n - a_n)$. Setting $(P_0, Q_0) = (P, Q)$ we can define (P_n, Q_n) by means of Lemma 7.1. Now, because of the bounds for P_n, Q_n, there can only be finitely

many pairs (P_n, Q_n), so that eventually some pair must repeat the same values that have been taken before; this means that $\alpha_n = \alpha_m$ for some $n < m$.

We now show that $\alpha_n = \alpha_m$ implies $\alpha_{n-1} = \alpha_{m-1}$; for this we need to bring the conjugates $\bar{\alpha}_n$ into the argument. We let $\phi_n = -1/\bar{\alpha}_n$, so that $\phi_n > 1$ because $-1 < \bar{\alpha}_n < 0$. From

$$\alpha_n = a_n + \frac{1}{\alpha_{n+1}}, \quad \text{we deduce} \quad \bar{\alpha}_n = a_n + \frac{1}{\overline{\alpha_{n+1}}} = a_n - \phi_{n+1};$$

that is $\phi_{n+1} = a_n + 1/\phi_n$, which shows that a_n is not only the integer part of α_n, but also the integer part of ϕ_{n+1}.

Suppose now that $\alpha_n = \alpha_m$ for some $n < m$. Then $\phi_n = \phi_m$; and a_{n-1} and a_{m-1} are the integer parts of ϕ_n and ϕ_m, so that $a_{n-1} = a_{m-1}$. Next, from $\alpha_{n-1} = a_{n-1} + 1/\alpha_n$ and $\alpha_{m-1} = a_{m-1} + 1/\alpha_m$, we deduce that $\alpha_{n-1} = \alpha_{m-1}$. Repeating the argument we arrive at $\alpha_0 = \alpha_{m-n}$, which then delivers

$$\theta = \alpha_0 = [a_0, a_1, \ldots, a_k, 1/\alpha_0] = [\,\overline{a_0, a_1, \ldots, a_k}\,], \qquad k = m - n.$$

That is, the continued fraction for θ is purely periodic. The theorem is proved.

Corollary 7.1 *Let $a \geq 1$, and $N = a^2 + k$ for $k = 1, 2, \ldots, 2a$. Then the continued fraction for $\sqrt{N}$ is given by* (7.9).

Proof Let $\theta = \sqrt{N} + a > 1$. Then $\bar{\theta} = -\sqrt{N} + a$ lies between -1 and 0, so that θ is a reduced irrational. By the theorem, we may write

$$\theta = \sqrt{N} + a = [\overline{2a, a_1, a_2, \ldots, a_n}], \quad \text{for some } n,$$

from which we deduce that $\sqrt{N} - a = [0; \overline{a_1, a_2, \ldots, a_n, 2a}] < 1$, and hence

$$-\frac{1}{\bar{\theta}} = \frac{1}{\sqrt{N} - a} = [\overline{a_1, a_2, \ldots, a_n, 2a}].$$

Since we also have $-1/\bar{\theta} = [\overline{a_n, a_{n-1}, \ldots, a_2, a_1, 2a}]$, it follows that $a_i = a_j$ for $i + j = n + 1$, so that the corollary is established.

Example 7.5 Evaluate $\theta = [7; 1, \overline{2, 3}]$ to verify that is a quadratic irrational.

We first determine $\phi = [\overline{2, 3}] = [2, 3, 2, 3, \ldots]$ by considering

$$\phi = 2 + \frac{1}{3 + \frac{1}{\phi}} = 2 + \frac{\phi}{3\phi + 1} = \frac{7\phi + 2}{3\phi + 1},$$

so that $3\phi^2 - 6\phi - 2 = 0$. Ignoring the negative root, we find that $\phi = (3+\sqrt{15})/3$. From $\theta = [7; 1, \overline{2, 3}] = [7, 1, \phi]$, we arrive at

$$\theta = 7 + \frac{1}{1 + \frac{1}{\phi}} = 7 + \frac{\phi}{\phi + 1} = \frac{8\phi + 7}{\phi + 1} = \frac{50 + \sqrt{15}}{7}.$$

□

Theorem 7.5 (Lagrange) *The continued fraction representation of θ is periodic if and only if θ is a quadratic irrational.*

Proof Suppose first that $\theta = [a_0, a_1, \ldots, a_{k-1}, \overline{a_k, a_{k+1}, \ldots, a_{k+n-1}}]$, a periodic continued fraction with period n. Writing $\phi = [\overline{a_k, a_{k+1}, \ldots, a_{k+n-1}}]$, so that $\phi = [a_k, a_{k+1}, \ldots, a_{k+n-1}, \phi]$, and hence

$$\phi = \frac{\phi p'_{k-1} + p'_{k-2}}{\phi q'_{k-1} + q'_{k-2}},$$

where p'_i/q'_i is the ith convergent of the quadratic irrational ϕ. We also have

$$\theta = \frac{\phi p_{k-1} + p_{k-2}}{\phi q_{k-1} + q_{k-2}},$$

where p_i/q_i is now the ith convergent of θ. Therefore θ is a quadratic irrational.

Conversely, suppose that $\theta = \alpha_0 = (P_0 + \sqrt{D})/Q_0 = [a_0, a_1, \ldots] > 0$ is a quadratic irrational, with convergents p_k/q_k. The argument used in the proof of Galois' theorem for (P_n, Q_n) applies. Set $\bar{\phi}_n = (P_n - \sqrt{D})/Q_n$, so that

$$\bar{\phi}_n = -\frac{q_{n-2}}{q_{n-1}} \cdot \frac{\bar{\phi}_0 - p_{n-2}/q_{n-2}}{\bar{\phi}_0 - p_{n-1}/q_{n-1}}.$$

Since the second factor of the right-hand side of the equation tends to 1 as $n \to \infty$, it follows that $\bar{\phi}_n < 0$, so that $2\sqrt{D}/Q_n = \phi_n - \bar{\phi}_n > 0$, and hence $Q_n > 0$ for all large n. We also have

$$Q_n \le Q_n Q_{n+1} = D - P_{n+1} < D \quad \text{and} \quad P_{n+1}^2 < D.$$

Consequently there are only finitely many distinct pairs (P_n, Q_n), which implies that there exists j such that $a_{n+i} = a_{n+j+i}$ for all $i \ge 0$. Therefore $\theta = [a_0, a_1, \ldots]$ is periodic. The theorem is proved. □

Pell's Equation

There is an interesting history connected to what is now known as Pell's equation,

$$x^2 = Ny^2 + 1, \tag{7.14}$$

where $N \in \mathbb{N}$ is not a square, and with solutions $x, y \in \mathbb{N}$. Reference to isolated cases of N can be traced back to antiquity—see, for example, Note 7.2. Results of significance were first given by the Indian mathematician Brahmagupta (c. 600) who is credited with the useful identity

$$(a^2 - Nb^2)(x^2 - Ny^2) = (ax \pm Nby)^2 - N(ay \pm bx)^2,$$

which can be used to compute large solutions, when one exists, by setting $(a, b) = (x, y)$. Brahmagupta opined that *whoever can solve* $x^2 - 92y^2 = 1$ *within a year is a mathematician*, and a fellow countryman Bhaskara (c. 1100) had also contributed to the equation.

In Europe, Frénicle de Bessy tabulated solutions[1] of (7.14) for $N \leq 150$, and he challenged the English mathematicians to find a solution for $N = 313$. Lord Brouncker[2] duly gave a solution, adding that he had found it in an hour or two—his method of solution was rendered by J. Wallis, who claimed to have proved that the equation is always soluble. Commenting on the work, Fermat stated that a solution always exists, and indeed that there are infinitely many solutions. Although the English mathematician J. Pell had nothing to do with it, his name is forever stuck to the equation because of Euler's mistaken attribution.

Solutions of Pell's equation are closely linked to the continued fraction expansion of $\sqrt{N}$, particularly the convergents and the period—the difficulty of finding a solution depends much on the length of the period. Anyway, it is no longer difficult to show that Pell's equation is always soluble, and indeed with infinitely many solutions.

Theorem 7.6 *Suppose that the period of the continued fraction for* $\sqrt{N}$ *has length* $n + 1$. *Then the convergent* p_n/q_n *satisfies*

$$p_n^2 - Nq_n^2 = (-1)^{n+1}.$$

Corollary 7.2 *If* n *is odd then* $(x, y) = (p_n, q_n)$ *is a solution of Pell's equation; if* n *is even then* $(x, y) = (p_n^2 + Nq_n^2, 2p_nq_n)$ *is a solution. Moreover, there are always infinitely many solutions to Pell's equation.*

[1] The work has not survived.

[2] William Brouncker, 2nd Viscount Brouncker, was the first president of the Royal Society.

Proof Let $\sqrt{N} = [a_0; \overline{a_1, a_2, \ldots, a_n, 2a_0}]$, which has period length $n+1$. We have

$$\sqrt{N} = \frac{\alpha_{n+1} p_n + p_{n-1}}{\alpha_{n+1} q_n + q_{n-1}},$$

where α_{n+1} is the $(n+1)$th complete quotient, which is given by

$$\alpha_{n+1} = [\overline{2a_0, a_1, a_2, \ldots, a_n}] = \sqrt{N} + a_0,$$

so that

$$\sqrt{N}(\sqrt{N} + a_0)q_n + \sqrt{N}q_{n-1} = (\sqrt{N} + a_0)p_n + p_{n-1}.$$

Since the numbers here are integers, apart from the irrational $\sqrt{N}$, we deduce that

$$p_{n-1} = Nq_n - a_0 p_n \quad \text{and} \quad q_{n-1} = p_n - a_0 q_n,$$

and hence

$$p_n(p_n - a_0 q_n) - q_n(Nq_n - a_0 p_n) = p_n q_{n-1} - p_{n-1} q_n = (-1)^{n+1}.$$

This means $p_n^2 - Nq_n^2 = (-1)^{n+1}$, as required. □

Proof of the Corollary If n is odd, then $(-1)^{n+1} = 1$, so that (p_n, q_n) is a solution of Pell's equation. If n is even, then we square the equation and apply Brahmagupta's identity to give the solution $(x, y) = (p_n^2 + Nq_n^2, 2p_n q_n)$. Indeed, repeated application of the identity shows that the number of solutions of Pell's equation is infinite. □

In fact $(p_n^2 + Nq_n^2, 2p_n q_n) = (p_{2n+1}, q_{2n+1})$. This is because the period of the continued fraction for $\sqrt{N}$ has length $n+1$, so that by periodicity the same argument used in the proof of the theorem applied to the two convergents p_k/q_k and p_{k-1}/q_{k-1} now delivers $p_k^2 - Nq_k^2 = (-1)^{k+1}$ for each $k \equiv n \pmod{n+1}$. In particular, for $k = 2n+1$ we find that $p_{2n+1}^2 - Nq_{2n+1}^2 = 1$, so that $(x, y) = (p_{2n+1}, q_{2n+1})$ is a solution of Pell's equation regardless of the parity of n.

The use of Brahmagupta's identity to compute convergents with a large index is much more efficient than having to evaluate each intermediate convergent. It can be shown that if (x, y) is a solution of Pell's equation and we write

$$(x + y\sqrt{N})^k = x_k + y_k\sqrt{N}, \quad k = 1, 2, \ldots,$$

then (x_k, y_k) is also a solution. Indeed, by taking (x, y) to be the solution with $x^2 + y^2$ least, we may expect all the solutions to be expressed in this way. We omit the proof, which belongs properly to the study of quadratic fields and Dirichlet's unit theorem.

Example 7.6 Solve Pell's equation for $N = 92$ and $N = 313$.

For $N = 92$, we have $\sqrt{N} = [9; \overline{1, 1, 2, 4, 2, 1, 1, 18}]$, with period length $7 + 1$, and

$$\left(\frac{p_k}{q_k}\right) = \left(\frac{9}{1}, \frac{10}{1}, \frac{19}{2}, \frac{48}{5}, \frac{211}{22}, \frac{470}{49}, \frac{681}{71}, \frac{1151}{120}\right), \quad k = 0, 1, 2, \ldots, 7,$$

so that $(x, y) = (p_7, q_7) = (1151, 120)$ is a solution. If this appears easy we should remind ourselves that Brahmagupta found it some 1400 years earlier, and that Bhaskara must have known how to generate larger solutions.

For $N = 313$, we have $\sqrt{N} = [17; \overline{1, 2, 4, 11, 1, 1, 3, 2, 2, 3, 1, 1, 11, 4, 2, 1, 34}]$, with period length $16 + 1 = 17$, and

$$\left(\frac{p_k}{q_k}\right) = \left(\frac{17}{1}, \frac{18}{1}, \frac{53}{3}, \cdots, \frac{126862368}{7170685}\right), \quad k = 0, 1, 2, \ldots, 16.$$

If $(p, q) = (p_{16}, q_{16})$ then $p^2 - 313q^2 = -1$ and, on squaring, we have the solution

$$(x, y) = (p^2 + 313q^2, 2pq) = (32188120829134849, 1819380158564160).$$

It can be checked that $(x, y) = (p_{33}, q_{33})$; in any case, Lord Brouncker did well finding the solution with only quill pen and paper in under two hours.

There is the interesting problem of the condition on N under which the 'negative' equation $x^2 - Ny^2 = -1$ is soluble. It depends, of course, on the parity of the period length $n + 1$ of the continued fraction of $\sqrt{N}$; if the length is odd, so that n is even, then the equation is soluble, otherwise there is no solution. However, there is no known general criterion for the parity of the period, and the problem seems to be deep.

There are some obvious partial answers: for example, since $x^2 + y^2 \not\equiv 3 \pmod 4$, it follows that there is no solution if $N \equiv 3 \pmod 4$. Indeed, if the equation is soluble then $N = (x/y)^2 + 1/y^2$ is a sum of two rational squares, so that it is a sum of two integer squares by a simplified version of Aubry's theorem (Theorem 1.9).

The cases $N \equiv 1, 2 \pmod 4$ remain open. We saw that, for the de Bessy challenge $N = 313 \equiv 1 \pmod 4$, the period length is 17, and $x^2 - 313y^2 = -1$ has the solution (p_{16}, q_{16}). On the other hand, for $N = 34, 221$, we have $\sqrt{34} = [5; \overline{1, 4, 1, 10}]$ with period length 4, and $\sqrt{221} = [14; \overline{1, 6, 2, 6, 1, 28}]$ with period length 6, so that $x^2 - Ny^2 \equiv -1$ has no solution for these N. Note that $34 = 2 \times 17$, $221 = 13 \times 17$, whereas 313 is a prime.

Theorem 7.7 (Lagrange) *If $p \equiv 1 \pmod 4$ is a prime, then $x^2 - py^2 = -1$ is soluble, so that the period length of the continued fraction of $\sqrt{p}$ is odd.*

Proof Pell's equation with $N = p$ being soluble, we can apply a descent process to construct a solution to $x^2 - py^2 = -1$.

Let (a, b) be a solution of $x^2 - py^2 = 1$. Modulo 4 consideration shows that a is odd and b is even, and from $(a + 1)(a - 1) = pb^2$, it follows that p divides either

$a-1$ or $a+1$. We show that if $p|a-1$ then a 'smaller' solution (a', b') of Pell's equation can be constructed, and the descent argument applies until we arrived at a solution with $p|a'+1$. Essentially the same construction then delivers a solution of $x^2 - py^2 = -1$ instead.

If $p|a-1$ then $n = (p-1)/2p$ and $m = (a+1)/2$ are coprime integers with $mn = (2b)^2$, so that we may write $m = u^2$, $n = v^2$, whereupon $u^2 - pv^2 = ((a+1) - (a-1))/2 = 1$, delivering a smaller solution (u, v) of Pell's equation. Replace (a, b) with (u, v) and repeat the process until we reached the 'least' solution in which the argument does not apply. Therefore we must have $p|a+1$ instead.

For $p|a+1$, the same argument now shows that we need to have $a+1 = 2pv^2$ and $a-1 = 2u^2$, giving instead $u^2 - pv^2 = ((a-1) - (a+1))/2 = -1$. Therefore the equation $x^2 - py^2 = -1$ does have a solution. □

Example 7.7 From Brouncker's solution (a, b) of $x^2 - 313y^2 = 1$, we can apply descent to compute the solution (u, v) of $x^2 - 313y^2 = -1$:

$$(u, v) = \left(\sqrt{\frac{a-1}{2}}, \sqrt{\frac{a+1}{2\times 313}}\right) = (126862368, 7170685),$$

which corresponds to the convergent p_{16}/q_{16} of $\sqrt{N}$, of course.

We show in Example 8.2 that if p, q satisfy $p^2 - Nq^2 = \pm 1$ then p/q is a convergent to the continued fraction of $\sqrt{N}$.

7.4 Factorisation by Continued Fractions

The use of continued fractions for factorisation was envisaged even before the advent of powerful computing machines. It was first proposed by D. H. Lehmer and R. E. Power in 1931, and the method was systematically developed and implemented into computer programs during the years 1970–1985. In 1974 M. A. Morrison and J. Brillhart applied the method successfully to factorise the seventh Fermat number

$$F_7 = 2^{128} + 1 = 59649589127497217\times 5704689200685129054721.$$

Recall that Pollard's ρ-method of factorisation of N is based on the idea of Legendre and Kraitchik that we should consider the congruence $x^2 \equiv y^2 \pmod N$, so that $N|(x-y)(x+y)$, and a proper divisor of N could be delivered from a suitable pair (x, y) by the application of the Euclidean algorithm for $\gcd(N, x \pm y)$. Instead of applying the algorithm many times to random input pairs (x, y), the new method offers much more promising pairs systematically from the continued fraction expansion of $\sqrt{N}$, so that the computing time is spent mostly on the production of such pairs in lieu of the Euclidean algorithm.

Let N be a composite number, not a square, so that we may write

$$\sqrt{N} = [a_0; \overline{a_1, a_2, \ldots, a_n, 2a_0}];$$

the symmetry of the partial quotients $a_1, a_2, \ldots, a_n$ will not be used, but n has to be reasonably large. The complete and partial quotients α_k and a_k, together with Q_k can be given by (7.11), and by (7.12). If p_k/q_k is the corresponding convergent then the argument in the proof of Theorem 7.6 shows that

$$p_k^2 - Nq_k^2 = (-1)^{k+1} Q_{k+1}, \qquad k = 0, 1, 2, \ldots,$$

which implies that $p_k^2 \equiv (-1)^{k+1} Q_{k+1} \pmod{N}$. Now if k is odd and Q_{k+1} is a square then we have the desired congruence $x^2 \equiv y^2 \pmod{N}$, with $(x, y) = (p_k, \sqrt{Q_{k+1}})$, but this may be too much to ask for. However, we can generate several pairs (p_k, Q_{k+1}), and try to find a suitable combination of them which has the property that the product $\prod Q_{k+1}$ is a square, together with the matching product $\prod(-1)^{k+1}$ taking the value $+1$, so that

$$x^2 \equiv y^2 \pmod{N} \quad \text{is satisfied by} \quad x = \prod p_k, \ y = \sqrt{\prod Q_{k+1}}.$$

The method of finding a suitable combination of Q to be a square involves the selection of a *factor base* $\mathcal{B}$ of small primes, and using trial division to find and collect those Q which can be factorised completely with the prime divisors in $\mathcal{B}$. The idea is that when a fair number of Q_i have been collected, a suitable combination of them will have all the exponents of their primes even, so that we have a square. If the factor base $\mathcal{B}$ is very small then such squares are just 'low-hanging fruits' to be picked; otherwise we may need to use linear algebra to construct the squares.

There is then an interesting dichotomy concerning the choice of the factor base. First, the smaller the base, the easier it is to find the squares. However, if N is large, so that Q_i are also large, then it becomes difficult to find suitable Q_i that can be factorised completely with primes in a small base. If we just want to pick low-hanging fruits, then a small base would be necessary, but if N is large then cows may come home before any fruit appears. For a large base we need linear algebra software to construct the squares. With experience from practice, given an input N of a certain size, an individual should know how to choose $B = B(N)$ so that on setting $\mathcal{B} = \{p : p \le B\}$ there will be enough Q_i either for picking, or to be put through the linear algebra machinery.

It has to be said that some squares deliver only trivial factorisations, so we may need several squares before the sought after factors emerge. Here are some relevant comments before we give an example.

• To save embarrassment, Fermat's test should be applied first to ensure that N is composite. If the period length of $\sqrt{N}$ is small then we may not have enough Q_i for the task. In this case we can replace N by hN with some small h so that the new period length is reasonably long. For example, if $N = a^2 + 1$, then $\sqrt{N} = [a; \overline{2a}]$,

but the period length of $\sqrt{3N}$ is usually more than long enough, although two new trivial factors, namely 3, $N/3$, may emerge from the method as a spoiler.

• If $p|Q_{k+1}$ then $p_k^2 \equiv Nq_k^2 \pmod{p}$, so that N is a quadratic residue (mod p). Consequently we need only take primes $p \in \mathcal{B}$ for which $\left(\frac{N}{p}\right) = 1$, which should reduce $|\mathcal{B}|$ by about half. We use $p \in \mathcal{B}$ without a suffix so that it will not be confused with p_k, the numerator of the kth convergent.

• We only require the evaluations of P_k, Q_k, a_k, p_k, and not q_k, but it will be prudent to have the initial few q_k at least so that the check $p_k q_{k+1} - p_{k+1}q_k = (-1)^{k+1}$ can be carried out. More important is that we should stick with arithmetic modulo N. Although the sequences (P_k), (Q_k) are bounded, and a_k is usually small, the terms p_k increase rapidly with k, so that it becomes unmanageable unless it is reduced (mod N). On the other hand the square roots $\sqrt{\prod Q_{k+1}}$ should be evaluated before reduction (mod N), of course.

The factorisation of the fifth Fermat number $F_5 = 2^{32} + 1 = 641 \times 6700417$ was given by Euler in 1732, and the factors for $F_6 = 2^{64} + 1$ were found by Landry and Le Lasseur in 1880. We recover these known factors using continued fractions as an illustration.

Example 7.8 Factorise $F_6 = 2^{64} + 1 = 18446744073709551617$.

We let $N = 3F_6$ to increase the period length of $\sqrt{N}$. Since $N < 10^{20}$ is small for the method, we consider only the primes $p \le 17$ for the factor base. From $\left(\frac{N}{p}\right) = -1$ for $p = 3, 11, 17$, we settle with

$$\mathcal{B} = \{-1, 2, 5, 7, 13\},$$

where -1 is included to cater for the parity of k, so that $|\mathcal{B}| = 5$. We proceed to find 6 values of k for which Q_{k+1} can be completely factorised in $\mathcal{B}$; there should then be enough low-hanging fruits to deliver the factorisation.

Since we are looking for squares in the form of a product $\prod Q_{k+1}$, we need only take note of the parity of the exponents of a prime divisor $p \in \mathcal{B}$ for each Q being collected. We consider the coding

$$(-1)^{k+1} Q_{k+1} \mapsto (d_1, d_2, d_3, d_4, d_5),$$

where $d_1 = 1$ or 0, depending on whether k is even or odd and, for $i = 2, 3, 4, 5$, the value of d_i is the exponent, reduced (mod 2), of the primes $p = 2, 5, 7, 13 \in \mathcal{B}$. For example, we found that $Q_{k+1} = 1338257375 = 5^3 7^7 11$ when $k = 1416789$, so that the image vector is $(0, 0, 1, 1, 1)$, where $d_1 = 0$ corresponds to k being odd. The result of the search for 6 values of k for which Q_{k+1} can be factorised completely in $\mathcal{B}$ is set out in Table 7.2.

There are now some very easy pickings. For example, the two vectors in the second and third rows are identical—a gift begging to be picked! Thus $Q_{i+1}Q_{j+1}$ is a square and i, j are both even, so that the congruence $x^2 \equiv y^2 \pmod{N}$ has

Table 7.2 Factorisation of Q_{k+1} in factor base $\mathcal{B}$

	k	$p_k \pmod N$	$(-1)^{k-1}Q_{k+1}$	$\{-1, 2, 5, 7, 13\}$
1	1416789	30187367987260976369	+1338257375	(0, 0, 1, 1, 1)
2	5841804	43499416313812236494	−66015625	(1, 0, 0, 0, 0)
3	6979456	19462812996344101564	−1714374025	(1, 0, 0, 0, 0)
4	8133778	8178077520898704994	−7535710	(1, 1, 1, 1, 1)
5	8395031	17800904125503738203	+2354909375	(0, 0, 1, 1, 1)
6	9549777	54761463763881316111	+3428748050	(0, 1, 0, 0, 0)

the solution $x = p_i p_j$ and $y = \sqrt{Q_{i+1}Q_{j+1}}$. Unfortunately, the computation of $\gcd(N, x \pm y)$ delivers the spoiler factorisation $N = (N/3)\times(3)$.

There are other easy pickings. For example, the vectors in the first and the fifth rows are also identical, and this time they deliver the goods. Thus for $i = 1416789$ and $j = 8395031$, we find that

$$x = p_i p_j \equiv 72544441456319417 29 \pmod N$$

and $Q_{i+1}Q_{j+1}$ is a square, with

$$y = \sqrt{Q_{i+1}Q_{j+1}} \equiv 1775239375 \pmod N.$$

From $\gcd(N, x \pm y) = 67280421310721, 822531$, where $822531 = 3\times 274177$, we deduce that

$$F_6 = 2^{64} + 1 = 274177 \times 67280421310721.$$

Factorisation can also be delivered by combining the vectors in rows 1, 2, 4, 6, and 1, 3, 4, 6. With no linear algebra work to speak of, and only a few gcd computations, the whole minute of computing time is spent on the elimination of those Q with a prime divisor not in $\mathcal{B}$ so that 6 values of Q_{k+1} for $k < 10^7$ are collected. If we include 19 in the factor base then the computing time is halved, and if we take 8 primes then the time drops down to only 2 seconds, but then we may need linear algebra to find the squares. □

In general, for the factorisation of N, we construct an $m\times n$ matrix

$$A = A_{m,n} = \begin{pmatrix} d_{1,1} & d_{1,2} & \cdots & d_{1,n} \\ d_{2,1} & d_{2,2} & \cdots & d_{2,n} \\ \vdots & \vdots & \cdots & \vdots \\ d_{m,1} & d_{m,2} & \cdots & d_{m,n} \end{pmatrix}, \qquad d_{i,j} \in \{1, 0\},$$

in which a row vector $\mathbf{v}_i = (d_{i,1}, d_{i,2}, \ldots d_{i,n})$ is the image of $(-1)^{k_i+1}Q_{k_i+1}$. The linear algebra problem of finding a square boils down to finding a nonzero row

vector $\mathbf{u} = (u_1, u_2, \ldots u_m)$, with $u_j \in \{1, 0\}$, such that $\mathbf{u}A = \mathbf{0}$, the zero column vector with dimension n. If m exceeds the rank of A, then Gaussian elimination can be applied to find the solution space for $\mathbf{u}$. In the example, we have

$$A = \begin{pmatrix} 0\,0\,1\,1\,1 \\ 1\,0\,0\,0\,0 \\ 1\,0\,0\,0\,0 \\ 1\,1\,1\,1\,1 \\ 0\,0\,1\,1\,1 \\ 0\,1\,0\,0\,0 \end{pmatrix},$$

which has rank 3, so that there are $6-3 = 3$ linearly independent solution vectors $\mathbf{u}$. The four solutions k in the example correspond to the vectors

$$\mathbf{u} = (0, 1, 1, 0, 0, 0),\ (1, 0, 0, 0, 1, 0),\ (1, 1, 0, 1, 0, 1),\ (1, 0, 1, 1, 0, 1);$$

they are not independent, because $\mathbf{u}_3 + \mathbf{u}_4 = \mathbf{u}_1$.

It has been established that the number of operations involved in the method of factorising N is of order $\exp(2\sqrt{\log N \log\log N})$; also, practitioners found that, for $N < 10^{50}$, the number of operations is much smaller than the bound.

7.5 Computing Continued Fractions

Implementing the basic algorithm (7.7) on a small machine it takes only a few seconds to evaluate $N = 10^4$ partial quotients of the continued fraction of a number from its input decimal approximation, and perhaps still only minutes for $N = 10^5$. For many numbers, there is another more efficient way to compute their partial quotients; for example, for an algebraic number (see Sect. 10.5), we may use function evaluations from the defining polynomial instead of relying on an input approximation. Before considering such computations and their results, it will be useful to mention some accompanying theory.

Suppose that we wish to compute the first N partial quotients $a_1, a_2, \ldots, a_N$ of the continued fractions of a real number α. As we surmised earlier there is no a priori condition that will ensure any precomputed approximation to α will be sufficiently accurate to deliver all the required a_n. This is because the denominator q_N of the last convergent exceeds the product $a_1 a_2 \cdots a_N$, which has no fixed upper bound, so that $1/q_N^2$ has no fixed positive lower bound for the accuracy of the input approximation of α. Nevertheless, A. Ya. Khinchin [48] has proved the remarkable theorem that, for almost all α in the sense of Lebesgue measure, the sequence of geometric means $(a_1 a_2 \cdots a_n)^{1/n}$ has the limit

$$K = \prod_{k=1}^{\infty} \left(1 + \frac{1}{k(k+2)}\right)^{\log k/\log 2} = 2{\cdot}68545\,20010\,65306\,\ldots\,.$$

and P. Lévy [59] has proved that, for almost all α, the sequence $(q_n^{1/n})$ has the limit

$$L = \exp\left(\frac{\pi^2}{12\log 2}\right) = 3{\cdot}27582\,29187\,21811\ldots .$$

Since the error of the approximation is about $1/q_n^2 \approx 1/L^{2n}$, for almost all α, the number n of partial quotients that can be obtained from an input approximation having m decimal digits satisfies

$$\frac{n}{m} \approx \frac{\log 10}{2\log L} = \frac{6\log 2\log 10}{\pi^2} \approx 0{\cdot}97.$$

In other words, *we would normally require about an extra 3% in the decimal digits in the input in order to obtain the desired number of partial quotients*. For example, if we want 10,000 partial quotients then it will be wise to take 10,500 decimal places for an input approximation. Nevertheless we cannot be sure until after the computation that a certain preassigned number of digits will be sufficient to deliver all N partial quotients.

Gauss-Kuzmin Distribution

The *Gauss-Kuzmin distribution* is a discrete probability distribution arising from the limit probability distribution of the partial quotients of the continued fraction of a random variable uniformly distributed in the interval $0 \le \alpha < 1$. In 1800 Gauss, from a "very simple argument", formulated that, for almost all $\alpha = [a_0; a_1, a_2, \ldots]$, the probability that $a_m = j$ is given by

$$p(j) = \frac{1}{\log 2}\log\frac{(j+1)^2}{j(j+2)} = \log_2\left(1 + \frac{1}{j(j+2)}\right), \qquad j = 1, 2, \ldots .$$

We have no idea what his argument was, and we can only speculate on some of the ideas involved. As α varies in the interval $[0, 1)$, we may hope that their continued fractions may, in some way, be a 'reflection' of such fractions for k/N, where N is large and $0 \le k < N$. For a uniform distribution $X = X_0$, define the distribution function

$$F_n(\alpha) = \text{probability that } X_n \le \alpha \text{ for } 0 \le \alpha < 1.$$

Then, starting with $F_0(\alpha) = \alpha$, we have

$$\begin{aligned} F_{n+1}(\alpha) &= \sum_{k\geq 1} (\text{probability that } k \leq 1/X_N \leq k+\alpha) \\ &= \sum_{k\geq 1} (\text{probability that } 1/(k+\alpha) \leq X_N \leq 1/k) \\ &= \sum_{k\geq 1} \Big(F_n\Big(\frac{1}{k}\Big) - F_n\Big(\frac{1}{k+\alpha}\Big)\Big). \end{aligned}$$

Assuming that the sequence of distributions $F_n(\alpha)$ converges, the limiting distribution $F(\alpha)$ should then satisfy

$$F(\alpha) = \sum_{k\geq 1} \Big(F_n\Big(\frac{1}{k}\Big) - F_n\Big(\frac{1}{k+\alpha}\Big)\Big).$$

It is readily checked that the equation has the solution $F(\alpha) = \log_b(1+\alpha)$, where the base of the logarithm can be any real $b > 1$: we need only note that

$$\begin{aligned} \sum_{k\geq 1} \log_b \frac{1+1/k}{1+1/(k+\alpha)} &= \log_b \prod_{k\geq 1} \Big(\frac{k+1}{k} \cdot \frac{k+1+\alpha}{k+\alpha} \Big) \\ &= \log_b \prod_{k\geq 1} \Big(\frac{2}{1} \cdot \frac{3}{2} \cdots \Big) + \log_b \Big(\frac{1+\alpha}{2+\alpha} \cdot \frac{2+\alpha}{3+\alpha} \cdots \Big) \\ &= \log_b 1 + \log_b(1+\alpha) = \log_b(1+\alpha). \end{aligned}$$

The normalisation $F(1) = 1$ shows that $b = 2$ is the suitable base. Thus, the probability that a given partial quotient a_m should take the value j is given by

$$\log_2\Big(1+\frac{1}{j}\Big) - \log_2\Big(1+\frac{1}{j+1}\Big) = \log_2\Big(\frac{(j+1)^2}{j(j+2)}\Big), \qquad j = 1, 2, \ldots .$$

This is, of course, speculation using notions which had still to be put forward more than a century later. In a letter dated 30 January 1812, Gauss wrote to Laplace that he did not succeed in solving the "curious problem", and that his efforts in estimating the error term "were unfruitful". In 1928, R. Kuzmin established the theorem to the effect that

$$F_n(\alpha) = \log_2(1+\alpha) + O(e^{-A\sqrt{n}}), \qquad n \to \infty,$$

for some positive constant A. With contributions from Khinchin and Lévy, the metric theory of continued fractions was born, but the subject is beyond our scope.

Verification of Laws

Although almost all numbers satisfy the laws in the metric theory, in practice it is hopelessly difficult to verify, or to prove, that a given α does satisfy them, apart from specifically constructing one to do so. Quadratic irrationals, with their periodic continued fractions, do not satisfy any of the laws, and are therefore exceptional numbers with respect to such laws. Perhaps to emphasise our inadequacy on this, it is an open question whether all, or indeed any, algebraic numbers of degree at least 3, have unbounded partial quotients. On the other hand, if we know the continued fraction of α, it seems inevitable that it is an exceptional number. We let

$$K(\alpha, n) = (a_1 a_2 \cdots a_n)^{1/n},$$

so that $K(\alpha, n) \to K$, as $n \to \infty$, for almost all α. An amusing but quite intractable problem is whether Khinchin's constant K satisfies Khinchin's law, that is whether $K(K, n) \to K$ as $n \to \infty$?!

Example 7.9 The number e is an exceptional number to Khinchin's law.

We shall prove in Sect. 8.6 that $e = [2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, \ldots]$, so that

$$K^{3k}(e, 3k) = \prod_{m \le 3k} a_m = \prod_{m \le k} (2m) = 2^k k!.$$

Stirling's formula implies $(k!)^{1/k} \sim k/e$ as $k \to \infty$ so that, for $3|n$,

$$K(e, n) \sim \left(\frac{2n}{3e}\right)^{1/3} \to \infty, \quad \text{as} \quad n \to \infty,$$

and Khinchin's law is not satisfied. □

Computation of Continued Fractions by Newton's Method

Let α be a simple zero of a differentiable function $f(t)$, of which the logarithmic derivative $f'(t)/f(t)$ can be determined with arbitrary accuracy at rational points t. The following are four examples of such $(\alpha, f(t))$ pairings:

$$(\pi, \sin t), \quad (\log 2, e^t - 2), \quad (e^\pi, \sin(\log t)), \quad (2^{\sqrt{2}}, \log^2 t - 2\log^2 2).$$

If $t \neq \alpha$ then, by the mean-value theorem of calculus, $f(t) = (t-\alpha)f'(\beta)$ for some β between α and t, so that

$$|t - \alpha| = \frac{|f(t)|}{|f'(\beta)|}.$$

This is the basis of Newton's method for the computation of α, and it applies also to the computation of the continued fractions for α directly, using function evaluations at appropriate places, instead of having to rely on a precomputed input approximation. From the well known quadratic convergence of Newton's method, an iterative scheme can be devised in which each iteration involving function evaluation delivers nearly double the previous number partial quotients.

Let $t_n = p_n/q_n$ be the nth convergent of

$$\alpha = [a_0; a_1, a_2, \ldots], \quad \text{and} \quad \alpha_n = [a_n, a_{n+1}, \ldots],$$

its nth complete quotient. The idea is to use (i) and (iii) of Theorem 7.2 to write

$$\alpha = [a_0; a_1, \ldots, a_n, \alpha'] = \frac{\alpha' p_n + p_{n-1}}{\alpha' q_n + p_{q-1}}, \qquad \alpha' = \alpha_{n+1},$$

and to solve for

$$\alpha' = \frac{(-1)^{n+1}}{q_n(p_n - \alpha q_n)} - \frac{q_{n-1}}{q_n},$$

so that α' can be obtained from functional evaluation near α. More specifically, from $|t_n - \alpha| = |f(t_n)/f'(\beta)|$, we find that

$$\alpha' = \alpha_{n+1} = \frac{|f'(\beta)|}{q_n^2 |f(t_n)|} - \frac{q_{n-1}}{q_n},$$

where β lies between α and t_n. Recall that t_{n-1} and t_n lie on the opposite sides of α, so that we have bounds for α' by replacing β with t_{n-1} and t_n, and the process involves only rational arithmetic. When the two bounds are developed into continued fractions by the basic algorithm, all the initial partial quotients that are in agreement must also be the leading partial quotients $a_{n+1}, a_{n+2}, \ldots$ for α'. In practice we simply replace β by t_n and use the error bound $|f''(t_n)/q_n^2 f'(t_n)|$.

In this way only a small number of function evaluations are required, and many new partial quotients can be obtained in one single iterative step. The need for an input value from an external source to calculate the complete quotient has been eliminated by function evaluation at the stage when more partial quotients are needed, so that there is no need to guess the required accuracy of an input value. The method is particularly useful if a seriously large number N of partial quotients are required.

In the following procedure it is assumed that there is a library of functions to call on for the values of $f(t)$, $f'(t)$, to be evaluated to a chosen accuracy at rational points t. Two initial convergents t_0, t_1 are needed as inputs for the iteration—in practice we compute a few more convergents and use t_m, t_{m+1} so that they are already near α. Also, for robustness, we may set $\min(2n + 100, 1{\cdot}05N)$ for the number of decimal places in the evaluation of α' at t_n. The results for $\alpha = \pi, \log 2, e^{\pi}, 2^{\sqrt{2}}$ with $N = 10^5$ are given in the next subsection.

- While $n \le N$: Do
  Check that $p_n q_{n+1} - p_{n+1} q_n = (-1)^{n+1}$;
  Set an appropriate accuracy and compute

$$\alpha' = \frac{|f'(t_n)|}{q_n^2 |f(t_n)|} - \frac{q_{n-1}}{q_n}.$$

  Set $B = q_n^2/200$ (say, or $q_n^{1{\cdot}9}$ for robustness).
  While $q_n < B$: Do
  Replace $n \leftarrow n + 1$;
  Set $a_n = \lfloor \alpha' \rfloor$, the next partial quotient;
  Process the relevant information on a_n (or store it);
  Use a_n and the last two convergents to compute $t_n = p_n/q_n$;
  Update the previous convergent $t_{n-1} = p_{n-1}/q_{n-1}$;
  Replace $\alpha' \leftarrow 1/(\alpha' - a_n)$, the next complete quotient.

Computation from Input Values

We do not know of an appropriate pair $(\alpha, f(t))$ when $\alpha = e + \pi, e\pi, \pi^e$ for the purpose of the computation of the continued fraction for α without input values. Let $N = 10^5$, and

$$\alpha = [a_0; a_1, a_2, \ldots, a_N, \ldots], \qquad \alpha = e + \pi,\ e\pi,\ \pi^e;$$

we compute a_i for $1 \le i \le N$ for each of α, ignoring the value of a_0. Following our remark on Lévy's theorem, we take 105×10^3 decimal places as rational inputs.

Instead of simply applying the basic algorithm rigidly, we use the following indirect procedure first proposed by D. H. Lehmer [56]. The computation is divided up into sections inside which the appropriate accuracy for the input is set, with the initial sections not requiring the full accuracy, so that the arithmetic therein is simpler. Thus, for $N = 10^5$, we may divide the computation into 6 sections: $1 \le n \le 50000$ and $10000k + 1 \le n \le 10000(k + 1)$ for $k = 5, 6, 7, 8, 9$; only the last section require the full decimal input value.

Table 7.3 Data for 10,000 partial quotients for the continued fractions of α

α	$a = 1$	$a = 2$	$a = 3$	$a = 4$	$a = 5$	$a = 6$	[7, 100]	> 100
π	41495	17002	9236	6034	4118	2930	17796	1389
$\log 2$	41430	16927	9435	5883	3988	2954	17963	1420
e^{π}	41436	17072	9398	5886	4047	2955	17769	1437
$2^{\sqrt{2}}$	41767	16942	9375	5795	4026	2908	17768	1419
$e + \pi$	41779	16972	9271	5864	4121	3089	17520	1384
$e\pi$	41765	17136	9220	5761	4030	2941	17742	1405
π^e	41504	17056	9229	5994	4049	2984	17721	1463
Predict	41503	16992	9310	5889	4064	2974	17828	1435

Table 7.4 Data for 10000 partial quotients for the continued fractions of α

α	$\max a_n; n \le N$	Least $b \ne a_n$	$K(\alpha, N)$	$\lfloor 2 \log_{10} q_N \rfloor$
π	$a_{28421} = 78629$	$b = 192$	2·6795	102850
$\log 2$	$a_{9168} = 963664$	$b = 187$	2·6945	103338
e^{π}	$a_{82893} = 219932$	$b = 241$	2·6807	102891
$2^{\sqrt{2}}$	$a_{57958} = 177830$	$b = 197$	2·6731	102770
$e + \pi$	$a_{93853} = 131059$	$b = 176$	2·6589	102347
$e\pi$	$a_{80269} = 174895$	$b = 214$	2·6689	102621
π^e	$a_{24734} = 104639$	$b = 209$	2·6841	103048

The results of the calculations of the partial quotients a_n of the continued fractions for $\alpha = [a_0; a_1, a_2, \ldots, a_N]$, with $N = 10^5$, excluding a_0, are given in the following two tables.

In Table 7.3 we list the number of partial quotients taking the values $a = 1, \ldots, 6$, and also those satisfying $7 \le a_n \le 100$, and $a_n > 100$, in the last two columns. The last row gives the corresponding 'expected' numbers for an α which is not an exceptional number; that is $\lfloor Np(a) \rfloor$, where $p(a)$ is the Gauss-Kuzmin discrete probability. In Table 7.4 the largest values for a_n, and the smallest values b not taken by a_n, are set out in columns 2 and 3.

The data for $\alpha = \pi, \log 2, e^{\pi}, 2^{\sqrt{2}}$ are obtained from the direct (Newton) method, and those for $\alpha = e + \pi, e\pi, \pi^e$ from input values, with the interval $1 \le n \le N$ divided into 5 sections. The final number of decimal places for function evaluation and input value is set at 105,000, and the entries $\lfloor 2 \log_{10} q_N \rfloor$ indicate that the choice is more than enough for the computations.

The computing times by the direct method is about 650 seconds for each of the four numbers, and about 1800 seconds by input values for each of the three numbers.

Computation for Quadratic Irrationals

Since the continued fraction for a quadratic irrational is periodic, only exact arithmetic is involved in the evaluation of all the partial quotients. In particular the algorithm in Lemma 7.1 should be used to evaluate such quotients for $\sqrt{N}$.

For $a = 1, 2, \ldots$, let

$$N = a^2 + k, \quad 1 \le k \le 2a; \qquad \sqrt{N} = [a_0; \overline{a_1, a_2, \ldots, a_n, 2a_0}].$$

We already know from Example 7.4 that the period lengths of $\sqrt{N}$ are $1, 2, 4, 2$ corresponding to $k = 1, 2, 2a - 2, 2a - 1$; it can also be shown that, for $k = 4$, the length is 2 or 5 depending on whether a is even or odd, but there seems to be no general results for other values of k.

The graphic display of period lengths for $\sqrt{N}$ with $a = 20$ and $1 \le k \le 40$ is given in Fig. 7.1.

The two longest periods shown in the figure are at $k = 21, 36$ with period lengths $n + 1 = 37, 30$, respectively. The two longest periods are:

$N = 421, n + 1 = 37$:

$$(1, 1, 13, 5, 1, 3, 1, 2, 1, 1, 1, 2, 9, 1, 7, 3, 3, 2, 2, 3, 3,$$
$$7, 1, 9, 2, 1, 1, 1, 2, 1, 3, 1, 5, 13, 1, 1, 40);$$

$N = 436, n + 1 = 30$;

$$(1, 7, 2, 1, 1, 1, 13, 3, 2, 2, 5, 1, 1, 4, 10, 4, 1, 1, 5, 2, 2, 3, 13, 1, 1, 1, 2, 7, 1, 40).$$

For $N = a^2 + k$, with $a = 1, 2, \ldots, 18$ and $3 \le k \le 2a - 2$, the period lengths of $\sqrt{N}$ are tabulated in Table 7.5. Take, for example, $N = 190$, with $a = 13$, $k = 21$; the period length of $\sqrt{190}$ is 14. There is a notable surplus of periods with even length, which seems typical.

Algebraic Numbers with Degree $d \ge 3$

We know next to nothing of the continued fraction of a number which is a root of a polynomial equation of degree $d \ge 3$ with integer coefficients. However, there is an integer arithmetic algorithm for the computation of its partial quotients which is easier to implement than the use of Newton's method. See, for example, [12].

Let $\mathcal{F}_d$ be the family of polynomials f of degree $d \ge 3$, with the following properties:

(i) $f(x)$ has integer coefficients, with the leading one being positive;

Table 7.5 Period lengths of $\sqrt{a^2+k}$, with $a = 1, 2, \ldots, 18$ and $3 \le k \le 2a-2$

$a\backslash k$	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
3	2	5	4	2	*	*	*	*	*	*	*	*	*	*	*	*
4	6	2	6	6	4	2	*	*	*	*	*	*	*	*	*	*
5	4	5	2	8	4	4	4	2	*	*	*	*	*	*	*	*
6	2	2	3	2	10	8	6	12	4	2	*	*	*	*	*	*
7	6	5	6	4	2	6	7	6	4	11	4	2	*	*	*	*
8	10	2	8	6	8	2	7	5	4	12	6	4	4	2	*	*
9	2	5	10	2	6	5	2	8	8	10	16	4	4	11	4	2
10	12	2	2	9	6	8	15	2	6	6	9	6	10	10	6	10
11	16	5	4	12	4	10	3	6	2	16	14	8	4	9	4	18
12	2	2	9	2	20	2	8	10	4	2	17	8	10	8	10	10
13	16	5	4	6	4	8	6	14	4	21	2	6	12	5	10	6
14	20	2	14	7	2	8	8	8	8	6	8	2	26	14	12	26
15	2	5	2	2	6	11	8	2	12	10	8	12	2	17	10	10
16	10	2	16	14	12	2	9	4	6	20	3	6	24	2	6	7
17	10	5	6	12	6	10	11	4	4	26	16	6	12	4	2	14
18	2	2	12	2	34	8	2	28	4	2	19	5	10	14	14	2

$a\backslash k$	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
11	4	4	4	2	*	*	*	*	*	*	*	*	*	*	*	*
12	18	6	6	22	4	2	*	*	*	*	*	*	*	*	*	*
13	8	4	14	16	4	13	4	2	*	*	*	*	*	*	*	*
14	4	6	16	5	4	4	6	4	4	2	*	*	*	*	*	*
15	26	10	10	12	4	16	7	14	4	22	4	2	*	*	*	*
16	6	12	21	6	8	6	13	8	18	16	6	10	4	2	*	*
17	6	26	16	16	4	17	7	4	8	11	4	14	4	4	4	2
18	16	10	10	5	14	8	7	6	12	8	15	10	12	14	6	26

(ii) $f(x)$ is irreducible over $\mathbb{Z}$, meaning that $f(x)$ is not a product of two polynomials with integer coefficients and smaller degrees;
(iii) $f(x)$ has exactly one positive root, which is a simple and irrational.

The properties are set out so that the proof of Lemma 7.3 below can be carried with ease. Some simple transformation may be required to manipulate a given polynomial so that it belongs to $\mathcal{F}_d$.

From the *intermediate value theorem* for continuous functions in analysis, we see that if $f \in \mathcal{F}_d$, with the root $\alpha > 0$, then

$$f(x) < 0 < f(y), \quad \text{for } 0 \le x < \alpha < y.$$

From $0 \le \lfloor\alpha\rfloor < \alpha < \lfloor\alpha\rfloor + 1$, it follows that $f(\lfloor\alpha\rfloor) < 0$; also, if $n \in \mathbb{N}$ and $n > \lfloor\alpha\rfloor$ then $n \ge \lfloor\alpha\rfloor + 1 > \alpha$, so that $f(n) > 0$. Consequently, $\lfloor\alpha\rfloor$ is the largest integer m such that $f(m) < 0$.

Lemma 7.3 *Let $f \in \mathcal{F}_d$. Define g by*

$$g(x) = -x^d f(1/x + \lfloor \alpha \rfloor), \quad x \neq 0, \qquad g(0) = -A,$$

where $f(x)$ has the leading coefficient A, and positive root α. Then $g \in \mathcal{F}_d$, with $g(x)$ having the leading coefficient $-f(\lfloor \alpha \rfloor)$, and the positive root $1/(\alpha - \lfloor \alpha \rfloor)$.

Proof From the defining equations for $g(x)$, it is readily verified that

$$\lim_{x \to 0} g(x) = g(0) \quad \text{and} \quad \lim_{x \to \infty} \frac{g(x)}{x^d} = -f(\lfloor \alpha \rfloor),$$

so that g is continuous in $\mathbb{R}$, and that, indeed, g is a polynomial of degree d, with integer coefficients, and a positive leading one; irreducibility also follows so that conditions (i) and (ii) are satisfied. Moreover, for $x > 0$, we find that $g(x) = 0$ if and only if $f(1/x + \lfloor \alpha \rfloor) = 0$, which happens when $1/x + \lfloor \alpha \rfloor = \alpha$, that is $x = 1/(\alpha - \lfloor \alpha \rfloor)$. The lemma is proved.

Example 7.10 Take $f(x) = x^3 - 8x - 10$, so that $f \in \mathcal{F}_3$, with the positive root

$$x = \sqrt[3]{5 + \sqrt{\frac{163}{27}}} + \sqrt[3]{5 - \sqrt{\frac{163}{27}}} \approx 3{\cdot}3186,$$

and $f(\lfloor \alpha \rfloor) = f(3) = -7$, $f(4) = 24$. Also

$$\begin{aligned} f(1/x + 3) &= \Big(\frac{1}{x} + 3\Big)^3 - 8\Big(\frac{1}{x} + 3\Big) - 10 \\ &= \Big(\frac{1}{x^3} + \frac{3^2}{x^2} + \frac{3^3}{x} + 3^3\Big) - 8\Big(\frac{1}{x} + 3\Big) - 10 \\ &= \frac{1}{x^3} + \frac{9}{x^2} + \frac{19}{x} - 7 \end{aligned}$$

so that $g(x) = 7x^3 - 19x^2 - 9x - 1$, a cubic with leading coefficient 7. Thus $g(3) = -10$, $g(4) = 107$, and the positive root is $\alpha' = 1/(\alpha - \lfloor \alpha \rfloor) \approx 3{\cdot}1387$.

We can now apply the lemma to give an efficient algorithm for the delivery of the continued fraction of α corresponding to the positive root a polynomial $f \in \mathcal{F}_d$. Let $f_0 = f$, $\alpha_0 = \alpha$ and

$$\alpha = [a_0; a_1, a_2, \ldots], \qquad a_0 = \lfloor \alpha_0 \rfloor.$$

Define

$$f_{k+1}(x) = -x^d f_k(1/x + a_k), \qquad k = 0, 1, 2, \ldots,$$

where $a_k = \lfloor \alpha_k \rfloor$, and α_k is the unique positive root of the polynomial f_k, which belongs to $\mathcal{F}_d$ according to Lemma 7.3. Then each partial quotient a_k of α is the largest $m \in \mathbb{N}$ such that $f_k(m) < 0$, so that the value of a_k can be evaluated using the method of repeated bisection of the relevant interval.

A Not Particularly Special Cubic Number

In the 1960s, when computing machines were in their infancy, attempts were made to find any 'unusual phenomenon' associated with the continued fractions of various cubic irrationals. Experiments were carried out systematically on solutions of $x^3 + ax + b = 0$ with increasing bounds for $|a|$, $|b|$. It was reported by J. Brillhart that the continued fraction for the solution of the cubic in Example 7.10 is rather unusual in that exceptionally large partial quotients a_n had been found with $n \leq 200$. Questions were then asked if such large values persisted to the extent that the Gauss-Kuzmin law would be violated, thereby establishing that x was an exceptional number. From modular functions, quadratic fields and other sophisticated areas of mathematics, reasons were found for the existence of these large partial quotients with small indices n, and that the phenomenon should fizzle out as n increases. We need only say that the phenomenon is related to the fact that the discriminant of the cubic has the value -4×163, and that the class-number of the quadratic field $Q(\sqrt{-163})$ is 1. Computation shows that

$$x = [3; 3, 7, 4, 2, 30, 1, 8, 3, 1, 1, 1, 9, 2, 2, 1, 3, 22986, 2, 1, 32, \ldots],$$

and that, for $n \leq 10000$, there are only 8 partial quotients $a_n > 20000$:

$$a_{17} = 22986, \quad a_{33} = 1501790, \quad a_{59} = 35657, \quad a_{81} = 49405,$$
$$a_{103} = 53460, \quad a_{121} = 16467250, \quad a_{139} = 48120, \quad a_{161} = 325927,$$

and $K(x, 10000) \approx 2.7337$. The number of a_n taking the values $b = 1, 2, \ldots, 10$ are

$$4096, 1683, 928, 619, 392, 314, 238, 196, 148, 129,$$

respectively, so that they are close to the predictions from the Gauss-Kuzmin theorem. The numerical evidence suggests that x is not an exceptional number.

7.6 Summary and Notes

Summary

- Every real number has a continued fraction expansion, finite if and only if the number is rational. There is an iterative scheme which delivers the *convergents* from the *partial quotients*. The basic results concerning the properties of continued fractions are set out in Theorem 7.2.
- The continued fraction expansion of a number is periodic if and only if the number is a *quadratic irrational*. There is then an integer arithmetic algorithm which delivers all the partial quotients. The continued fraction for $\sqrt{N}$ has the special form:

$$\sqrt{N} = [a_0; \overline{a_1 a_2, \ldots, a_2, a_1, 2a_0}].$$

- There are infinitely many solutions to *Pell's equation* $x^2 = Ny^2 + 1$, where N is a positive integer which is not a square. The solutions can be obtained from the convergents of $\sqrt{N}$. The 'negative' Pell equation $x^2 = Ny^2 - 1$ is soluble if and only if the period length of $\sqrt{N}$ is odd. There is no known criterion on the parity of this period length. Partial solutions include the result of Lagrange, which states that the parity concerned is odd when N is a prime congruent to 1 (mod 4).
- The factorisation of N can be obtained from the continued fraction expansion of $\sqrt{N}$. The method involves the selection of a suitable *factor base* for the complete factorisation of the denominators of the convergents in order to construct a square.
- For almost all real numbers the sequence of geometric means $(a_1 a_2 \ldots a_n)^{1/n}$ of the partial quotients has the limit $K \approx 2{\cdot}68545$, known as Khinchin's constant.

  If a truncated approximation is used as input for the continued fraction, then the number of correct partial quotients being delivered is some 3% short of the number of decimal places of the input.
- The values of the partial quotients of almost all real numbers satisfy the Gauss-Kuzmin distribution law.
- For real algebraic numbers there is an integer arithmetic algorithm which delivers the partial quotients without an input value.
- Base on Newton's method the continued fraction expansion of some suitable transcendental numbers can also be delivered without an input value.
- We know next to nothing of the continued fraction expansion of any algebraic number of degree at least 3, or of the number π. We do not even know if the sequence of partial quotients for any, or all, of such numbers is bounded, or not.

Notes

1. A. J. van der Poorten [101] has put forward a 2×2 matrix scheme in the study of continued fractions which gives some new insight, simplifications and clarification in the presentation of the theory.
2. In 1773 G. E. Lessing found a poem in an ancient manuscript in the library of Wolfenbuttel, which he then translated and published. The poem describes a problem involving 8 kinds of cattle satisfying certain linear equations, together with quadratic conditions. It is generally agreed that the problem, now known as the *Cattle Problem of Archimedes*, was constructed by the great mathematician, probably for Eratosthenes. Solving the linear equations, the problem reduces to the equation (7.14) with $N = 4729494$. Although it is unlikely that even Archimedes could have found a solution, the fact that the problem was propounded suggests that he might have some knowledge about the equation. The least solution of the problem consists of numbers with some two hundred thousand decimal digits. Several solutions have now been posted on `www`.
3. The proof of the Gauss-Kuzmin law can be found in the delightful book [49] by A. Ya. Khinchin.

7.7 Exercises

1. Develop $123456789/234567892$ into a continued fraction.
2. Evaluate the continued fractions $[1, 2, 3, 4, 5, 6, 7, 8, 9]$ and $[\overline{1, 2, 3, 4, 5, 6, 7, 8, 9}]$.
3. Perform the calculations in Example 8.2 for the numbers (i) $\sqrt{7}+e$, (ii) $\sqrt{7}+\pi$, (iii) $\pi + e$.
4. Evaluate the continued fractions for $\sqrt{a^2 + 4}$ for $a = 2, 3, 4, \ldots$.
5. Let s_n denote the nth squarefull number (See Exercises 6.5, 6.6). Show that there are infinitely many n for which $s_{n+1} = s_n + 1$.
6. Find the 'least' solution of $x^2 = 7788y^2 + 1$, and that of $x^2 = 7789y^2 - 1$.
7. Show that if p/q is a convergent of $\sqrt{n}$ then $|p^2 - nq^2| < 1 + 2\sqrt{n}$.
8. Use the continued fractions method to factorise $N = 9643\times 3733^5$.
9. (i) Construct α_1 which satisfies Khinchin's law, but not Gauss-Kuzmin's law.
   (ii) Construct α_2 which satisfies Gauss-Kuzmin's law.
10. Use function evaluations, instead of an input value, to compute 10000 partial quotients of the continued fraction for e^e.
11. Compute enough decimal digits for π^π in order to deliver 10000 partial quotients of its continued fraction.
12. Construct the histogram for the period lengths for the continued fraction of $\sqrt{1600 + k}$ with $1 \le k \le 80$.
13. When projected onto the star sphere, the great circle path of the moon intersects the ecliptic (the projected path of the sun) at the ascending and the descending

nodes. A draconic month is the period of time for the moon to return to the same node, and it is 27 days, 5 hours, 5 minutes and 35.8 seconds (averaged over many years). A synodic month is the period of time between two new moons, and it is 29 days, 12 hours, 44 minutes and 2.9 seconds (again, averaged). Show that solar and lunar eclipses occur in cycles with period of about 18 years 10 days.

Part II
Analytic Number Theory

Chapter 8
Diophantine Approximations



Abstract Using π as an example, it is explained that even bad approximations can be very accurate, and useful for their conversion into good approximations. The condition for an approximation of a number to be a convergent of the continued fraction is established. Liouville's theorem on the approximation of algebraic numbers is proved, thereby showing the existence of real transcendental numbers. The theorems of Hermite and Lindemann on the transcendence of e and π are given. The continued fraction for e, and the proof of the irrationality of $\zeta(3)$ are also included.

Since the set $\mathbb{Q}$ of rational numbers is dense in the set $\mathbb{R}$ of reals, given any $\theta \in \mathbb{R}$, there are rationals $p/q \in \mathbb{Q}$ arbitrarily close to θ; we call such p/q rational approximations to θ. In considering how efficient such an approximation is, besides the size of the error $|\theta - p/q|$, we also take into account the size of its denominator q. For example, the fraction $3{\cdot}1416 = 31416/10000 = 3927/1250$ is a reasonably close upper approximation to π, with the denominator 1250, but, by taking 4 partial quotients in the continued fraction of π, we find that the convergent $[3, 7, 15, 1] = 355/113 = 3{\cdot}1415929\ldots$ is not only a closer upper approximation, but also with the much smaller denominator 113, so that it is more efficient.

The branch of mathematics concerned with approximations to irrational numbers by rational numbers is called *Diophantine approximations*, in honour of Diophantus (c.210–c.294), the first person to use fractions as numbers in the study of solutions of algebraic equations. The subject is rich with deep results and many unsolved problems, and continued fractions often play a central role in the investigations. We establish some of the basic facts in this chapter.

Let $\theta \in \mathbb{R}$ be fixed. For *any* positive integer q, we may set $p = \lfloor q\theta \rfloor$ so that $p \le q\theta < p + 1 = \lceil q\theta \rceil$, and hence $0 \le \theta - p/q < 1/q$; thus p/q and $(p + 1)/q$ are approximations to θ from below and from above, respectively, with an error bound $1/q$. We may say that p and $p + 1$ are the correctly chosen integers with respect to q for the approximations. For a certain set of denominators q corresponding to an irrational θ, the error associated with one of the correctly

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chosen numerators p can be very much smaller than $1/q$, leading to the definitions in Sect. 8.2.

8.1 Accurate But Bad Approximations to π

Let $\theta > 0$ be an irrational number. A fraction p/q, with $p \neq \lfloor q\theta \rfloor, \lceil q\theta \rceil$, has necessarily an error $|\theta - p/q| > 1/q$; with p being inappropriately set (with respect to θ and q), the fraction p/q can hardly be described as an approximation to θ. Nevertheless, the corresponding error may still be small, because it is possible that $|\theta - p/q| < 1/\sqrt{q}$, or even $|\theta - p/q| < (\log q)/q$; if the error is small when q is large, such fractions are accurate, but bad, approximations to θ. Why bother with bad approximations? The answer is that there is no simple procedure to find good approximations to θ directly, whereas there may be a simple arithmetic algorithm which delivers accurate, though bad, approximations to θ. Such accurate approximations can then be used as inputs into the continued fraction algorithm to deliver the convergents which are good approximations. We consider the following example for π.

Example 8.1 Let

$$f_k(x) = (x(1-x))^{4k}, \qquad k = 1, 2, \ldots,$$

so that $0 \leq f_k(x) \leq f_k(\frac{1}{2}) = (\frac{1}{4})^{4k}$, for $0 \leq x \leq 1$, and $f_k(i) = (-4)^k$, $i = \sqrt{-1}$. The division algorithm, applied to the polynomial $f_k(x)$, with the divisor $x^2 + 1$, delivers

$$f_k(x) = (x^2 + 1)g_k(x) + (-4)^k, \tag{8.1}$$

where the quotient $g_k(x)$ is a polynomial with degree $8k - 2$, and with integer coefficients. In fact, letting

$$g_k(x) = \sum_{0 \leq h \leq 8k-2} b_h x^h,$$

the coefficients b_h are given by the following downward iterative scheme:

$$b_{8k-2} = +1, \qquad b_{8k-3} = -4k,$$

$$b_{8k-j} = (-1)^j \binom{4k}{4k-j+2} - b_{8k-j+2}, \qquad 4 \leq j \leq 4k+2,$$

$$b_h = \begin{cases} 0, & h = 1, 3, 5, \ldots, 4k-3, \\ (-1)^{h/2} 4^k, & h = 0, 2, 4, \ldots, 4k-4. \end{cases}$$

The proof follows at once from the definition of $g_k(x)$ in (8.1),

$$(x^2+1)g_k(x) = \sum_{2\le h\le 8k-2} (b_{h-2}+b_h)x^h + b_0 + b_1 x + b_{8k-3}x^{8k-1} + b_{8k-2}x^{8k}.$$

and the expansion of $f_k(x)$ by the binomial theorem. We may now write

$$\int_0^1 g_k(x)dx = \sum_{0\le h\le 8k-2} \int_0^1 b_h x^h dx = \sum_{0\le h\le 8k-2} \frac{b_h}{h+1} = \frac{u_k}{v_k}, \tag{8.2}$$

where the integers u_k and v_k can be determined, and we also have

$$\frac{u_k}{v_k} = \int_0^1 \frac{f_k(x)}{x^2+1}dx - (-4)^k \int_0^1 \frac{dx}{x^2+1}.$$

Noting that

$$\int_0^1 \frac{dx}{x^2+1} = \frac{\pi}{4}, \qquad 0 < \int_0^1 \frac{f_k(x)}{x^2+1}dx < \frac{1}{4^{4k}}\frac{\pi}{4} < \frac{1}{4^{4k}},$$

we then find that

$$0 < \frac{u_k}{v_k} - (-4)^{k-1}\pi < \frac{1}{4^{4k}}, \qquad k = 1, 2, \ldots .$$

In particular, we have

$$0 < \frac{u_k}{V_k} - \pi < \frac{1}{4^{5k-1}}, \quad V_k = 4^{k-1}v_k, \qquad k = 1, 3, 5, \ldots ,$$

delivering rational approximations to π from above, with an error bound $1/4^{5k-1}$.

However, such approximations, though accurate, are likely to be bad ones.[1] To see this, we note from (8.2) that the size of q_k is essentially that of the product of primes not exceeding $8k$, so that, by Chebyshev's theorem, $v_k > 4^{4k}$ for large k, and therefore $V_k > 4^{5k-1}$ is too large for the fraction to be a suitable approximation.

Nevertheless, a stand alone integer arithmetic program can now be written to deliver rational approximations to π with arbitrarily small errors. Suppose, for example, that we want a rational approximation to π which is correct to more than

[1] As a rule, fractions with denominators having some prescribed form, such as having a large power of a number as a factor, are almost certain to be bad approximations of the number concerned.

40 decimal digits. We may take $k = 15$, so that $4^{5k-1} \approx 10^{44\cdot5}$, and the program delivers, for $4k = 60 \le h \le 118 = 8k - 2$,

$$(b_h) = (4^{15} + 1, -60, -1073740055, -34160, \ldots, 485866, -34160, 1769, -60, 1),$$

and

$$\frac{u_{15}}{v_{15}} = \frac{3737271881299351053666441876167156192604573934552162661}{443164477934791771527293523410856183608702310 0},$$

$$\frac{u_{15}}{V_{15}} = 3{\cdot}14159265358979323846264338327950288419716939962995 12\ldots,$$

$$\pi = 3.14159265358979323846264338327950288419716939937510 58\ldots;$$

the last equation for π is inserted here for comparison to show that the underlined digits of the approximation are in disagreement. Anyway, the simple program delivers a rational approximation to π which is correct up to 45 decimal digits. It can be readily checked that u_{15} exceeds $\lceil V_{15}\pi \rceil$ by 303166777, so that the approximation is bad; indeed, $V_{15} \approx 10^{54}$, whereas the error is about $1/10^{45}$.

In practice, the scheme can be used for input values for the continued fractions of π. With $k = 15$, the denominator of the corresponding convergent nearest π should then be of order 10^{23} only.

The same scheme can be applied to $\log 2$ instead of π, by making use of

$$\int_0^1 \frac{2x}{x^2+1}dx = \log 2 \quad \text{and} \quad (x(1-x))^{4k+2} \equiv 2(-4)^k x \pmod{x^2+1}.$$

8.2 Good and Best Approximations

Definition 8.1 Let θ be an irrational number, and $\kappa \ge 2$. A fraction p/q satisfying

$$\left|\theta - \frac{p}{q}\right| < \frac{1}{q^\kappa} \tag{8.3}$$

is called a *good* approximation to θ, and θ is said to be *well-approximated* to level κ if there are infinitely many such approximations.

Let us first consider the case $\kappa = 2$. It follows from Dirichlet's approximation theorem (Theorem 1.7) that there are always infinitely many good approximations to an irrational θ and, indeed, by Theorem 7.3, the convergents of the continued fraction for θ are good approximations; moreover, such good approximations can be from above or from below θ by setting the index of the convergent to be odd or even.

Definition 8.2 Let $\theta > 0$ be an irrational number, and $n \in \mathbb{N}$. An approximation p/q to θ is *best* relative to n if $q \leq n$ and, among all the errors associated with the fractions u/v where $v \leq n$, the one associated with p/q is the least; in other words every rational approximation u/v to θ with $|\theta - u/v| < |\theta - p/q|$ has $v > n$. If best is used with no reference to n, it means best relative to its own denominator.

Theorem 8.1 *Let $\theta = [a_0, a_1, \ldots,]$ be a finite or infinite continued fraction, and $[a_0, a_1, \ldots, a_n] = p_n/q_n$. Suppose that $n \geq 1$, $0 < q \leq q_n$, and $p/q \neq p_n/q_n$. Then $|p_n/q_n - \theta| < |p/q - \theta|$.*

Corollary 8.1 *The convergents of an irrational θ are best approximations.* □

Proof Since $q \leq q_n$, it suffices to prove that $|p_n - q_n\theta| < |p - q\theta|$. If $\theta = \lfloor\theta\rfloor + \frac{1}{2}$ then $\theta = p_1/q_1$, and we are done.

If $\theta < \lfloor\theta\rfloor + \frac{1}{2}$ then the result holds for $n = 0$, and if $\theta > \lfloor\theta\rfloor + \frac{1}{2}$ then the result holds for $n = 1$. We now assume as induction hypothesis that the result holds for $n - 1$, and proceed to prove the result by induction.

If $q \leq q_{n-1}$ then, from the induction hypothesis $|p_{n-1} - q_{n-1}\theta| < |p - q\theta|$, so that we may assume that $q_{n-1} < q \leq q_n$. If $q = q_n$ then

$$\left|\frac{p_n}{q_n} - \frac{p}{q}\right| \geq \frac{1}{q_n}, \quad p \neq p_n, \quad \text{and also} \quad \left|\frac{p_n}{q_n} - \theta\right| \leq \frac{1}{q_n q_{n+1}} \leq \frac{1}{2q_n}.$$

If $q_{n+1} = 2$ then $n = 1$ and $a_1 = a_2 = 1$, giving $\theta = [a_0, 1, 1, a_3, \ldots]$ which shows that $a_0 + \frac{1}{2} < \theta < a_0 + 1$, and the require result holds. We may therefore assume that $q_{n+1} > 2$, so that the bound holds with strict inequality, and hence

$$\left|\frac{p}{q} - \theta\right| \geq \left|\frac{p}{q} - \frac{p_n}{q_n}\right| - \left|\frac{p_n}{q_n} - \theta\right| \geq \frac{1}{q_n} - \left|\frac{p_n}{q_n} - \theta\right| > \left|\frac{p_n}{q_n} - \theta\right|.$$

We may now assume that $q_{n-1} < q < q_n$. Let us write $up_n + vp_{n-1} = p$, $uq_n + vq_{n-1} = q$, so that $u(p_n q_{n-1} - p_{n-1}q_n) = pq_{n-1} - qp_{n-1}$. We then have $u = \pm(pq_{n-1} - qp_{n-1})$ and $v = \pm(pq_n - qp_n)$. The numbers u, v cannot be zero, and in fact from $q_n > q = uq_n + vq_{n-1}$ we see that they are of opposite signs. Now $p_n - q_n\theta$ and $p_{n-1} - q_{n-1}\theta$ are also of opposite signs, so that $u(p_n - q_n\theta)$ and $v(p_{n-1} - q_{n-1}\theta)$ are of the same sign. Finally, from $p - q_\theta = u(p_n - q_n\theta) + v(p_{n-1} - q_{n-1}\theta)$, it follows that $|p - q\theta| > |p_{n-1} - q_{n-1}\theta| > |p_n - q_n\theta|$, completing the inductive argument. □

Thus the convergents of an irrational θ are both good and best approximations. However, there are approximations which are good but not best, and also some which are best but not good. There is, of course, a linguistic inconsistency here, but mathematicians do occasionally take such liberty and abuse ordinary language in order to accommodate obtuse definitions. For example (see Exercise 8.6), there are approximations which are good and best, but are not convergents; and also approximations which are best, but are not good.

8.3 Conditions for Being Convergents

It is not difficult to show that if θ is irrational then there are infinitely many fractions p/q satisfying the slightly more stringent condition

$$\left|\theta - \frac{p}{q}\right| < \frac{1}{2q^2}. \tag{8.4}$$

Indeed, of any two consecutive convergents of θ, at least one satisfies the inequality. This is because such convergents lie on opposite sides of θ, so that

$$\left|\frac{p_{n+1}}{q_{n+1}} - \frac{p_n}{q_n}\right| = \left|\frac{p_{n+1}}{q_{n+1}} - \theta\right| + \left|\frac{p_n}{q_n} - \theta\right|,$$

and if neither fraction satisfies (8.4), then

$$\frac{1}{q_n q_{n+1}} \geq \frac{1}{2q_{n+1}^2} + \frac{1}{2q_n^2},$$

implying $q_{n+1} = q_n$, which is plainly false.

Being a good approximation is a necessary condition to be a convergent, but not a sufficient one. Legendre found a necessary and sufficient condition for p/q to be a convergent of θ, which shows that (8.4) is a sufficient condition.

Theorem 8.2 (Legendre) *A necessary and sufficient condition for p/q to be a convergent of θ is that*

$$|q_n\theta - p_n| \leq \frac{1}{q_n + q_{n-1}},$$

where $p/q = p_n/q_n$, and q_{n-1} is the denominator of the previous convergent.

Corollary 8.2 *If p/q satisfy (8.4) then it is a convergent of θ.*

Proof We may assume that $p/q = [a_0, a_1, \ldots, a_n]$ is a good approximation, and write

$$\theta - \frac{p}{q} = \frac{\epsilon\delta}{q^2}, \qquad \epsilon = \pm 1, \quad 0 < \delta < 1;$$

by choosing the parity of n, we may also assume that $(-1)^n = \epsilon$. Define β by

$$\theta = \frac{\beta p_n + p_{n-1}}{\beta q_n + q_{n-1}},$$

so that

$$\frac{\epsilon\delta}{q_n^2} = \frac{\beta p_n + p_{n-1}}{\beta q_n + q_{n-1}} - \frac{p_n}{q_n} = \frac{p_{n-1}q_n - p_n q_{n-1}}{q_n(\beta q_n + q_{n-1})} = \frac{(-1)^n}{q_n(\beta q_n + q_{n-1})}.$$

Then

$$\delta = \frac{q_n}{\beta q_n + q_{n-1}}, \qquad 0 < \delta < 1, \tag{8.5}$$

giving $\beta = (q_n - \delta q_{n-1})/\delta q_n > 0$, so that $\theta = [a_0, a_1, \ldots, a_n, \beta]$.

If $\beta \geq 1$ then $\beta = [a_{n+1}, a_{n+2}, \ldots]$, and p/q is a convergent to θ. On the other hand, if $0 < \beta < 1$ then $[a_n + 1/\beta] = a_n + c$, where $c \in \mathbb{N}$, so that

$$\theta = [a_0, a_1, \ldots, a_{n-1}, a_n + c, \ldots]$$

which means that $p/q = [a_0, a_1, \ldots, a_{n-1}, a_n]$ cannot be a convergent. Therefore $\beta \geq 1$ is a necessary and sufficient condition for p/q to be a convergent of θ. Returning to (8.5), the condition becomes

$$\frac{1}{q_n + q_{n-1}} \geq \frac{1}{\beta q_n + q_{n-1}} = \frac{\delta}{q_n} = |q_n\theta - p|,$$

so that the theorem is proved. For the corollary, we find that if (8.4) holds then

$$|q\theta - p| = q\left|\theta - \frac{p}{q}\right| < \frac{1}{2q} < \frac{1}{q_n + q_{n-1}}.$$

□

Example 8.2 If $p^2 - Nq^2 = \pm 1$ then p/q is a convergent to $\sqrt{N}$.

From $|(p - \sqrt{N}q)(p + \sqrt{N}q)| = 1$ we deduce that

$$|p - \sqrt{N}q| = \frac{1}{p + \sqrt{N}q} < \frac{1}{2q}, \quad \text{so that} \quad \left|\frac{p}{q} - \sqrt{N}\right| < \frac{1}{2q^2}.$$

□

We say that the rationals are the worst numbers to be approximated to; by this we mean the following. If $\theta = m/n$ is rational and $q > n$ then $|\theta - p/q| = |mq - np|/qn > 1/q^2$, so that there are at most n good approximations to θ. We may also say that quadratic irrationals are the 'next worst case' for the purpose of approximations for the following reasoning.

The error bound (7.6) shows that, as a general rule, the larger the partial quotient a_n is, the smaller the error is likely to be. For a fixed quadratic irrational, because of periodicity, the partial quotients a_n are bounded. Indeed, if $a_n = 1$ for all large n, then the corresponding number may not have infinitely many approximations p/q with an error less than $1/Aq^2$, if we fix $A > 10$, say.

Theorem 8.3 (Hurwitz) *Of any three consecutive convergents p_n/q_n to θ, at least one of them satisfies*

$$\left|\theta - \frac{p_n}{q_n}\right| < \frac{1}{\sqrt{5}q_n^2},$$

so that, if θ is irrational then there are infinitely such convergents. Moreover, if $A > \sqrt{5}$ then there exists θ such that there can only be finitely many fractions p/q satisfying $|\theta - p/q| < 1/Aq^2$.

Proof Let $\phi_m = q_{m-2}/q_{m-1}$, and α_m be the mth complete quotient, so that

$$\left|\theta - \frac{p_n}{q_n}\right| = \frac{1}{q_n(\alpha_{n+1}q_n + q_{n-1})} = \frac{1}{q_n^2(\alpha_{n+1} + \phi_{n+1})}.$$

We proceed to show that

$$\alpha_m + \phi_m \le \sqrt{5}$$

cannot hold for three consecutive values of m. Suppose that it holds for $m = n-1, n$. Then, from $\alpha_{n-1} = a_{n-1} + 1/\alpha_n$ and

$$\frac{1}{\phi_n} = \frac{q_{n-1}}{q_{n-2}} = \frac{a_{n-1}q_{n-2} + q_{n-3}}{q_{n-2}} = a_{n-1} + \phi_{n-1},$$

we deduce that $1/\alpha_n + 1/\phi_n = \alpha_{n-1} + \phi_{n-1} \le \sqrt{5}$, giving

$$1 = \frac{1}{\alpha_n}\alpha_n \le \Big(\sqrt{5} - \frac{1}{\phi_n}\Big)(\sqrt{5} - \phi_n).$$

Thus $\phi_n + 1/\phi_n \le \sqrt{5}$ and hence $\phi_n > (\sqrt{5} - 1)/2$, because $\phi_n < 1$ is rational. Similarly, if $\alpha_m + \phi_m \le \sqrt{5}$ holds for $m = n, n+1$, then $\phi_{n+1} > (\sqrt{-}1)$, and we arrive at

$$a_n = \frac{q_n - q_{n-2}}{q_{n-1}} = \frac{1}{\phi_{n+1}} - \phi_n < \sqrt{5} - \phi_{n+1} - \phi_n < 1,$$

which is impossible. The first part of the theorem is established.

For the remaining part of the theorem, we set $\theta = (\sqrt{5} - 1)/2$, fix any $A > \sqrt{5}$, and let p/q be any fraction. Write $\theta - p/q = \delta/q^2$, and suppose that, for infinitely many p/q, we have $|\delta| < 1/A$. Then

$$\frac{\delta}{q} - \frac{\sqrt{5}q}{2} = -\Big(\frac{q}{2} + p\Big),$$

and, on squaring and rearranging,

$$\frac{\delta^2}{q^2} - \sqrt{5}\delta = \left(\frac{q}{2} + p\right)^2 - \frac{5q^2}{4} = pq - q^2 + p^2.$$

Since $\sqrt{5}|\delta| < \sqrt{5}/A < 1$, it follows that, for all large q, the left-hand side of the equation is less than 1 in magnitude, so that $pq - q^2 + p^2 = 0$, or $(2p+q)^2 = 5q^2$, which is impossible because $\sqrt{5}$ is irrational. The theorem is proved. □

The reader may care to check that $\theta = (\sqrt{5} - 1)/2 = [\bar{1}] = [1; 1, 1, \ldots]$, and that any quadratic irrational with a continued fraction having $\bar{1}$ as the period can be used in the last part of the proof of the theorem.

8.4 Fractional Parts of $n\theta$

Let θ be an irrational number. It follows from Dirichlet's approximation theorem(Theorem 1.7) that the fractional parts of $n\theta$, that is $n\theta - \lfloor n\theta \rfloor$, can be arbitrarily small, for suitable choices of n; in other words, 0 is a *limit point* of the set of such fractional parts, with $n = 1, 2, \ldots$. Actually the set concerned is *dense* in the interval [0, 1] in the sense that every point in the interval is a limit point of the set.

Theorem 8.4 (Chebyshev) *Let $\theta, \beta \in \mathbb{R}$, with θ being irrational. Then there are infinitely many integers x, y satisfying*

$$|x\theta - y - \beta| < \frac{3}{x}.$$

Proof Let p/q be a convergent of θ, so that we may write

$$\theta = \frac{p}{q} + \frac{\delta}{q^2}, \qquad |\delta| < 1.$$

Let t be (one of) the integer nearest to $q\beta$, so that $|q\beta - t| \le \frac{1}{2}$, and thus

$$\beta = \frac{t}{q} + \frac{\delta'}{2q}, \qquad |\delta'| \le 1.$$

Next, choose x, y satisfying $px - qy = t$, with $\frac{q}{2} \le x < \frac{3q}{2}$; this is possible according to Theorem 1.1. We then have

$$|x\theta - y - \beta| = \left|\frac{xp}{q} + \frac{x\delta}{q^2} - y - \frac{t}{q} - \frac{\delta'}{2q}\right| = \left|\frac{x\delta}{q^2} - \frac{\delta'}{2q}\right| < \frac{x}{q^2} + \frac{1}{2q} < \frac{9}{4x} + \frac{3}{4x} = \frac{3}{x}.$$

There being infinitely many convergents of the irrational θ, it follows that there are infinitely many corresponding integers x, y. The theorem is proved.

Remarks

(i) By a more careful argument, the constant 3 may be reduced to any real number exceeding $1/\sqrt{5}$.

(ii) A deeper result is that the fractional parts $\{n\theta\}$ are *uniformly distributed* in the interval $(0, 1)$. Roughly speaking, being uniformly distributed means that if (a, b) is any subinterval of $(0, 1)$ then, as n takes the values $1, 2, \ldots, N$, the subinterval received the 'correct proportion' of fractional parts, that is about $(b - a)/N$ fractional parts $\{n\theta\}$, when N is large.

8.5 Liouville's Theorem and Diophantine Equations

A complex number α is said to be *algebraic* if it is a zero of a polynomial with integer coefficients; there is then a polynomial with the least degree. Such a polynomial is *irreducible* in the sense that it is not a product of two polynomials with integer coefficients and smaller degrees—it is said to be *minimal*, and the degree of α is the degree of its minimal polynomial. A *transcendental number* is a complex number which is *not* algebraic. We shall be concerned with real algebraic and transcendental numbers in this section. For example, rationals are of degree 1, quadratic irrational are of degree 2, and $\sqrt[3]{2}$, being the real root of $x^3 - 2 = 0$, is of degree 3. We prove in Sect. 8.6 that e and π are transcendental.

Corresponding to any fixed $\kappa > 2$, there are infinitely many θ which are well-approximated to level κ. For example, we may take any $\theta = [a_0; a_1, a_2, \ldots,]$, with $a_{n+1} > q_n^n$, where p_n/q_n is the nth convergent, so that, for $n > \kappa$,

$$\left|\theta - \frac{p_n}{q_n}\right| < \frac{1}{a_{n+1}q_n^2} < \frac{1}{a_{n+1}} < \frac{1}{q_n^n} < \frac{1}{q_n^\kappa}.$$

Liouville was the first person to consider such inequalities, and he established that an algebraic number cannot be well-approximated to a level beyond its degree.

Theorem 8.5 (Liouville) *Let θ be an algebraic number of degree n, and $\kappa > n$. Then there are only finitely many rational numbers p/q satisfying (8.3).*

Proof The theorem is true for rational numbers, so we assume that $n > 1$. Let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be the defining polynomial for θ, so that $f(\theta) = 0$, and $f(p/q) \neq 0$ for any rational p/q. Indeed, since $a_i, p, q \in \mathbb{Z}$, we have the following lower bound:

$$\left|f\left(\frac{p}{q}\right)\right| = \frac{|a_n p^n + \cdots + a_1 pq^{n-1} + a_0 q^n|}{q^n} \geq \frac{1}{q^n}.$$

We are interested in $\theta - p/q$, which can be related to $f(p/q)$ via the mean-value theorem of differential calculus, which states that there is a number η in the interval between θ and p/q such that

$$-f\Big(\frac{p}{q}\Big) = f(\theta) - f\Big(\frac{p}{q}\Big) = \Big(\theta - \frac{p}{q}\Big) f'(\eta),$$

where f' is the derivative of f. Thus $0 < |f'(\eta)| \le \mu$ where μ depends only on f. It then follows that

$$\left|\theta - \frac{p}{q}\right| = \left|\frac{f(p/q)}{f'(\eta)}\right| \ge \frac{1}{\mu q^n}.$$

Consequently, if $\kappa > n$, there can only be finitely many p/q satisfying (8.3). □

Corollary 8.3 *There are real transcendental numbers.*

Proof If any of the numbers $\theta = [a_0; a_1, a_2, \ldots]$, with $a_{n+1} > q_n^n$, were algebraic with degree K, say, then, for $n > K$, the nth convergent is an approximation to level beyond K. Therefore any such θ cannot be algebraic. □

The proof of the corollary amounts to showing that there are numbers close to rationals with exceedingly small errors. Such an argument need not involve continued fractions. Take, for example, $\theta = 0{\cdot}x_1x_2\ldots$ in the decimal representation, with the digits $x_i = 0$ except for a sequence of indices i which increases rapidly enough to ensure that any finite truncation of the representation at a value $a_i = 1$ has to be a very good approximation to θ. Thus, using binary representation, we find, for example, that $\theta = \sum_{k\ge1} 1/2^{k!}$ will do. If θ were algebraic of degree n, say, then we may take any $m > n$, and consider

$$\frac{p}{q} = \sum_{1\le k\le m} \frac{1}{2^{k!}},$$

which satisfies

$$\left|\theta - \frac{p}{q}\right| = \sum_{k>m} \frac{1}{2^{k!}} < \frac{1}{2^{(m+1)!}}\Big(1 + \frac{1}{2^{(m+2)!}} + \cdots\Big) < \frac{2}{2^{(m+1)!}} < \frac{1}{q^m} \le \frac{1}{q^{n+1}}.$$

Therefore θ is well-approximated to level $n+1$, which contradicts the theorem.

Liouville first demonstrated the existence of such numbers in 1844; half a century later, G. Cantor used his famous 'diagonal argument' to show that the algebraic numbers form a countable set, whereas $\mathbb{R}$ is uncountable, so that the set of transcendental numbers is uncountable. This can also be deduced from Liouville's argument. Thus, modifying the definition of θ to form the set $\mathcal{L}$ of numbers $\sum_{k\ge1} \epsilon_k/2^{k!}$, where $\epsilon_k \in \{\pm1\}$, the same argument still applies, showing that each member of $\mathcal{L}$ is transcendental. The sets $\mathcal{L}$ and $\mathbb{R}$ have the same cardinal.

Solutions of polynomial equations with integer coefficients may be complex numbers, and the notion of being algebraic or transcendental applies also to complex numbers. It can be shown that the set of algebraic numbers form a field. Numbers which cannot be well approximated to any finite order are called *Liouville numbers*, which are necessarily transcendental. However, as we shall see, there are transcendental numbers which are not Liouville numbers.

Diophantine equations are polynomial equations, with at least two unknowns, of which solutions in integers are sought. The method of rational approximations to algebraic numbers is at the kernel of the problem of finding solutions. Modern development of the subject is based on a method originating from A. Thue in 1908 who, by a long and difficult train of reasoning, gave the first substantial improvement of Liouville's theorem, namely that the lower bound for κ can be reduced from n to $\frac{1}{2}n + 1$. This was further reduced by C. L. Siegel in 1921 to $2\sqrt{n}$, and then by F. Dyson in 1947 to $\sqrt{2n}$. Finally, in 1955, K. F. Roth achieved the bound 2, which is best possible because of Hurwitz's theorem for quadratic irrationals.

Roth's Theorem *Let θ be an irrational algebraic number, and $\epsilon > 0$. Then there are at most a finite number of fractions p/q such that*

$$\left|\theta - \frac{p}{q}\right| < \frac{1}{q^{2+\epsilon}}.$$

As Siegel remarked to Davenport, Roth's work will be remembered as long as mankind is interested in mathematics; and, on the occasion when the Fields Medal was awarded to Roth for this piece of work, Davenport said:

The achievement is one that speaks for itself: it closes a chapter, and a new chapter is opened. Roth's theorem settles a question which is both of a fundamental nature and of extreme difficulty. It will stand as a landmark in mathematics for as long as mathematics is cultivated.

The motivation of Thue's work is that results in Diophantine approximations lead to bounds for the number of solutions to Diophantine equations. For example, after Roth's theorem, it can be shown that if $f(x, y)$ is an irreducible polynomial with integer coefficient without multiple roots, and of degree $n \geq 3$, then there are only finitely many solutions to $f(x, y) = 0$. Note that this is not true for $n = 2$, because there are infinitely many solutions to Pell's equation $x^2 = 2y^2 + 1$, for example.

The Thue-Siegel-Roth method has one peculiar feature ('defect' seems inappropriate following what has been said), namely that the method is non-effective. Although the argument shows that various types of equations in two variables x and y have only a finite number of solutions, the argument does not provide a bound for x, y beyond which there is no solution. Indeed in some interesting particular cases, there is a known bound for all but possibly one other solution, but there is no known bound beyond which there is no solution.

8.6 Transcendence of e and π

The irrationality of the number e follows almost at once from its definition, and the criterion established in Corollary 1.4. From

$$e^{-1} = \sum_{0\le n\le k} \frac{(-1)^n}{n!} + \sum_{n>k} \frac{(-1)^n}{n!} = A_k + B_k \text{ (say)}, \qquad k = 1, 2, \dots,$$

we find that $k!\,A_k \in \mathbb{N}$ and $0 < k!\,|B_k| < 1/(k+1)$. Thus $|k!/e - k!A_k| \to 0$ as $k \to \infty$, so that $1/e$, and hence also e, are irrational according to the criterion.

That π is irrational was first established by J. H. Lambert in 1761, using the continued fraction expansion of $\tan x$. In the nineteenth century, C. Hermite introduced radical ideas from which irrationality of various numbers related to the exponential and logarithm functions, including π, can be tackled. His method has been much simplified to give slick proofs of some of them.

Theorem 8.6 *The number π is irrational.*

Proof (Niven) Suppose the contrary, so that $\pi = a/b$, where a, b are coprime positive integers. For $n \in \mathbb{N}$, define

$$f(x) = \frac{x^n(a - bx)^n}{n!}, \qquad x \in \mathbb{R},$$

and

$$F(x) = f(x) - f^{(2)}(x) + f^{(4)}(x) - \cdots + (-1)^n f^{(2n)}(x).$$

It is easy to see that $f(x)$, together with its derivatives, are integers at $x = 0$ and π, so that $F(0)$ and $F(\pi)$ are integers. Now

$$\frac{d}{dx}\big(F'(x)\sin x - F(x)\cos x\big) = \Big(F''(x) + F(x)\Big)\sin x = f(x)\sin x,$$

so that

$$\int_0^\pi f(x)\sin x\,dx = F(\pi) - F(0) \in \mathbb{Z}. \tag{8.6}$$

However, for $0 < x < \pi$ and n sufficiently large, we have

$$0 < f(x)\sin x < \frac{\pi^n a^n}{n!} < \frac{1}{\pi},$$

so that

$$0 < \int_0^\pi f(x) \sin x \, dx < 1,$$

contradicting (8.6). The theorem is proved. □

Hermite's method can also be used to establish the transcendence of various numbers related to the exponential function. Naturally, the method is much more complicated, and it is not immediately obvious how to construct the auxiliary function. The following lemma is used to establish the transcendence of e and π.

Lemma 8.1 *Let*

$$f(x) = \sum_{0 \le m \le n} a_m x^m, \quad F(x) = \sum_{0 \le k \le m} f^{(k)}(x), \quad \text{and} \quad Q(x) = F(0)e^x - F(x).$$

Then

$$|Q(x)| \le e^{|x|} \sum_{0 \le m \le n} |a_m| |x|^m.$$

Proof Using the formula for $f^{(k)}(x)$ we have

$$\begin{aligned} F(x) &= \sum_{0 \le k \le m} \sum_{k \le m \le n} a_m \frac{m!}{(m-k)!} x^{m-k} \\ &= \sum_{0 \le m \le n} a_m m! \sum_{0 \le k \le m} \frac{x^{m-k}}{(m-k)!} = \sum_{0 \le m \le n} a_m m! \sum_{0 \le k \le m} \frac{x^k}{k!}, \end{aligned}$$

and, in particular,

$$F(0) = \sum_{0 \le m \le n} a_m m!.$$

Therefore

$$\begin{aligned} |Q(x)| &= \left| \sum_{0 \le m \le n} a_m m! \sum_{k=0}^{\infty} \frac{x^k}{k!} - \sum_{0 \le m \le n} a_m m! \sum_{0 \le k \le m} \frac{x^k}{k!} \right| \\ &= \left| \sum_{0 \le m \le n} a_m m! \sum_{k=m+1}^{\infty} \frac{x^k}{k!} \right| \\ &\le \sum_{0 \le m \le n} |a_m| \sum_{k=m+1}^{\infty} \frac{|x|^k}{(k-m)!} \end{aligned}$$

$$= \sum_{0\le m\le n} |a_m||x|^m \sum_{\ell=1}^{\infty} \frac{|x|^\ell}{\ell!} = e^{|x|} \sum_{0\le m\le n} |a_m||x|^m.$$

□

Theorem 8.7 (Hermite) *The number e is transcendental.*

Proof Suppose, if possible, that e is algebraic, so that it is a zero of a polynomial

$$P(x) = \sum_{0\le h\le m} g_h x^h, \qquad g_h \in \mathbb{Z},\ g_0 \neq 0, \quad m \ge 1.$$

Let $p > \max(m, |g_0|)$ be a prime, and let

$$f(x) = \frac{x^{p-1}\prod_{1\le h\le m}(h-x)^p}{(p-1)!} = \sum_{0\le k\le n} a_k x^k,$$

where the coefficients a_k of the polynomial so obtained depend on p. Since h is a zero of order p in the product, the numerator of $f(x)$ can be written as

$$(m!)^p x^{p-1} + A_p x^p + \cdots = B_{p,h}(x-h)^p + B_{p+1,h}(x-h)^{p+1} + \cdots$$

where A_u and $B_{u,v}$ are integers. We proceed to construct $F(x)$ and $Q(x)$ according to the hypothesis of the lemma. First, because the derivative of e^x is e^x, we have

$$0 = F(0)P(e) = F(0)\sum_{0\le h\le m} g_h e^h = \sum_{0\le h\le m} g_h F(h) + \sum_{0\le h\le m} g_h Q(h). \tag{8.7}$$

Now

$$\begin{aligned}\sum_{0\le h\le m} g_h F(h) &= g_0 \sum_{0\le k\le n} f^{(k)}(0) + \sum_{1\le h\le m} g_h \sum_{0\le k\le n} f^{(k)}(h)\\ &= g_0\big((m!)^p + pA_p + \cdots\big)\\ &\quad + \sum_{1\le h\le m} g_h\big(pB_{p,h} + p(p+1)B_{p+1,h} + \cdots\big)\end{aligned}$$

in an integer. Moreover, since each term on the right-hand side of the equation is a multiple of p, except $g_0(m!)^p$, which is not divisible by the prime $p > \max(m, |g_0|)$, we deduce that $\left|\sum_{0\le h\le m} g_h F(h)\right| \ge 1$. On the other hand, from

$$\sum_{0\le k\le n} |a_k||x|^k \le \frac{|x|^{p-1}\prod_{1\le h\le m}(h+|x|)^p}{(p-1)!} \to 0 \quad \text{as} \quad p\to\infty,$$

and the lemma, it follows that $\left|\sum_{0\le h\le m} g_h Q(h)\right| < 1$ when p is large. The two inequalities together contradict (8.7). The theorem is proved.

The proof of the transcendence of π is similar, but more complicated—the argument requires the symmetric polynomial theorem (see the next subsection) applied to the functions being constructed. It suffices to prove that $i\pi$ is not algebraic.

Theorem 8.8 (Lindemann) *The number π is transcendental.*

Proof Suppose, if possible, that $i\pi$ is algebraic, so that it is a zero of a polynomial

$$g(x) = ax^m + b_1 x^{m-1} + \cdots + b_m, \qquad a \in \mathbb{N}, \quad b_h \in \mathbb{Z};$$

then $ai\pi$ is also algebraic because it satisfies $a^{m-1}g(x/a) = 0$. For the required contradiction, we now suppose that $ai\pi$ is a root of a certain polynomial equation with integer coefficients, and with degree m. Let the roots of the equation be $a\alpha_1, a\alpha_2, \ldots, a\alpha_m$, and consider the product

$$R = \prod_{1\le h\le m} (1 + e^{\alpha_h}).$$

Since $ai\pi$ is one of the roots, so that $\alpha_h = i\pi$ for some h, it follows from $1+e^{i\pi} = 0$ that $R = 0$. We proceed to derive a contradiction by showing that $R \neq 0$.

Expanding out the product for R into 2^m terms, we have

$$R = c + \Sigma e^{\alpha} + \Sigma e^{\alpha+\alpha'} + \cdots = c + e^{\beta_1} + e^{\beta_2} + \cdots + e^{\beta_r},$$

where c counts the terms with the value $e^0 = 1$, and $\beta_1, \beta_2, \ldots, \beta_r \neq 0$. Next, let $p > \max(c, a, \prod_{1\le h\le r} a|\beta_h|)$ be a prime, and define $f(x)$ by

$$f(x) = \frac{(ax)^{p-1}\prod_{1\le h\le r}(ax - a\beta_h)^p}{(p-1)!} = \sum_{0\le k\le n} a_k x^k.$$

Similar to the proof of the transcendence of e, we now find that

$$f(x) = \frac{A_{p-1}x^{p-1} + A_p x^p + \cdots}{(p-1)!} = \frac{\gamma_{p,h}(x-\beta_h)^p + \gamma_{p+1,h}(x-\beta_h)^{p+1} + \cdots}{(p-1)!},$$

where A_ℓ are symmetric functions (see next section) of $a\beta_1, \ldots, a\beta_r$, and hence symmetric functions of $a\alpha_1, \ldots, a\alpha_r$, so that they are integers, and with $p \nmid A_{p-1}$.

For the construction of $F(x)$ and $Q(x)$ in the lemma, we have

$$F(0)R = F(0)\left(c + \sum_{1\le h\le r} e^{\beta_h}\right) = cF(0) + \sum_{1\le h\le r} F(\beta_h) + \sum_{1\le h\le r} Q(\beta_h), \qquad (8.8)$$

so that $cF(0) = c(A_{p-1} + pA_p + \cdots)$ is an integer not divisible by the prime $p > c$. Also, on expanding $(x - \beta_h)^{p+j}$ by the binomial theorem, we find that

$$\begin{aligned}\sum_{1\le h\le r} F(\beta_h) &= \sum_{1\le h\le r} \left(p\gamma_{p,h} + p(p+1)\gamma_{p+1,h} + \cdots\right)\\ &= p\sum_{1\le h\le r} \gamma_{p,h} + p(p+1)\sum_{1\le h\le r} \gamma_{p+1,h} + \cdots\\ &= pc_p + p(p+1)c_{p+1} + \cdots,\end{aligned}$$

where $c_p, c_{p+1}, \ldots$, being symmetric functions of $a\beta_1, \ldots, a\beta_r$, are integers. It follows that $\sum_{1\le h\le r} F(\beta_h)$ is a multiple of p and whence $\left|cF(0) + \sum_{1\le h\le r} F(\beta_h)\right| \ge 1$.

On the other hand, from

$$\sum_{0\le k\le m} |a_k||x|^k \le \frac{a|x|^{p-1}\prod_{1\le h\le r}(a|x| + a|\beta_h|)^p}{(p-1)!} \to 0 \quad \text{as} \quad p \to \infty,$$

and the lemma, it follows that $\left|\sum_{0\le h\le r} Q(\beta_h)\right| < 1$ if p is large. Returning to (8.8) we now find, by the triangle inequality, that

$$|F(0)R| \ge \left|cF(0) + \sum_{1\le h\le r} F(\beta_h)\right| - \left|\sum_{0\le h\le r} Q(\beta_h)\right| > 1 - 1 = 0,$$

so that $R \ne 0$, the sought after contradiction. The theorem is proved.

Hermite's method eventually led to the famous Gelfond-Scheinder theorem which states that $\alpha^\beta = \exp(\beta \log \alpha)$ is transcendental if α, β are algebraic, with $\alpha \ne 0, 1$ and β is not a real rational number. This includes numbers such as $2^{\sqrt{2}}$ and $e^\pi = (-1)^{-i}$, where $-i$ is considered to be irrational. The generalisation of the theorem by A. Baker is concerned with linear forms in logarithms of algebraic numbers in transcendental number theory.

Symmetric Functions and Polynomials

A function in n variables $x_1, x_2, \ldots, x_n$ is said to be *symmetric* if its value is unchanged under all permutations of the subscripts i. Take $n = 3$ for example; the following three polynomials

$$\sigma_1 = x_1 + x_2 + x_3, \quad \sigma_2 = x_1x_2 + x_2x_3 + x_3x_1, \quad \sigma_3 = x_1x_2x_3,$$

are symmetric—they are the coefficients of the cubic expression in t:

$$(t - x_1)(t - x_2)(t - x_3) = t^3 - \sigma_1 t^2 + \sigma_2 t - \sigma_3,$$

which takes the value 0 at $t = x_1, x_2, x_3$. The polynomials $\sigma_1, \sigma_2, \sigma_3$ are called the *elementary symmetric polynomials* in 3 variables. In general, such polynomials in n-variables are

$$\sigma_1 = \sum_i x_i, \quad \sigma_2 = \sum_{i<j} x_i x_j, \quad \sigma_3 = \sum_{i<j<k} x_i x_j x_k, \quad \ldots, \quad \sigma_n = x_1 x_2 \cdots x_n;$$

they are the coefficients of the polynomial $\prod_k (t - x_k)$, which has zeros at $t = x_r$.

For each $k > 1$, we find that the symmetric polynomial $x^k + y^k$ in two variables can be expressed as a polynomial in σ_1, σ_2. Thus, for $k = 2, 3$ we have

$$x^2 + y^2 = (x + y)^2 - 2xy = \sigma_1^2 - 2\sigma_2,$$
$$x^3 + y^3 = (x + y)^3 - 3xy(x + y) = \sigma_1(\sigma_1^2 - 3\sigma_2),$$

In general, we have the following theorem due to Isaac Newton:

Theorem 8.9 (Fundamental Theorem on Symmetric Polynomials) *Every symmetric polynomial in variables $x_1, x_2, \ldots, x_n$ can be expressed as a polynomial in the elementary symmetric polynomials $\sigma_1, \sigma_2, \ldots, \sigma_n$ of the variables x_i.* □

We omit the proof, which can be based on the double induction on the number of variables as well as the degree of the polynomial.

Continued Fraction for e

The continued fraction for e was first discovered by Euler when he analysed the Ricatti differential equation. Hermite gave a proof which is a byproduct of his argument that e is transcendental. The following elegant proof is by H. Cohen [17]; the paper includes also some of the background of Hermite's argument.

Theorem 8.10 *The number e has the continued fraction expansion*

$$e = [a_0; a_1, a_2, \dots] = [2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, 1, 10, \dots],$$

where the partial quotients are given by $a_0 = 2$, $a_1 = 1$, *and*

$$a_{3k-1} = 2k, \quad a_{3k} = a_{3k+1} = 1, \qquad k = 1, 2, \dots .$$

Proof (Cohen) Let p_k/q_k denote the kth convergent, so that $p_0 = 2$, $p_1 = 3$, $q_0 = 1$, $q_1 = 1$; and, for $k = 1, 2, \dots$,

$$\begin{aligned} p_{3k-1} &= 2kp_{3k-2} + p_{3k-3}, & q_{3k-1} &= 2kq_{3k-2} + q_{3k-3}, \\ p_{3k} &= p_{3k-1} + p_{3k-2}, & q_{3k} &= q_{3k-1} + q_{3k-2}, \\ p_{3k+1} &= p_{3k} + p_{3k-1}, & q_{3k+1} &= q_{3k} + q_{3k-1}. \end{aligned}$$

It suffices to show that there is a subsequence of (p_k/q_k) which converges to e, for which purpose Cohen introduced, for $k = 0, 1, 2, \dots$, the integrals

$$\begin{aligned} A_k &= \int_0^1 \frac{x^k(x-1)^k}{k!} e^x dx, \\ B_k &= \int_0^1 \frac{x^{k+1}(x-1)^k}{k!} e^x dx, \\ C_k &= \int_0^1 \frac{x^k(x-1)^{k+1}}{k!} e^x dx. \end{aligned}$$

Note that $A_k, B_k, C_k \to 0$ as $k \to \infty$, that $A_0 = e - 1$, $B_0 = 1$, $C_0 = 2 - e$, and that, for $k = 1, 2, \dots$,

$$A_k + C_k = \int_0^1 \frac{x^k(x-1)^k(1 + (x-1))}{k!} e^x dx = B_k.$$

Indeed, we also have $A_k + B_{k-1} + C_{k-1} = 0$ and $B_k + 2kA_k = C_{k-1}$, which can be established by integration by parts, or just by applying the product rule for differentiation to the two products $(x^k(x-1)^k/k!)e^x$ and $(x^k(x-1)^{k+1}/k!)e^x$, and then integrate the result by the fundamental theorem of calculus.

Anyway, we then find that B_k and C_k satisfy the reduction formulae given by

$$\begin{pmatrix} B_k \\ C_k \end{pmatrix} = M_k \begin{pmatrix} B_{k-1} \\ C_{k-1} \end{pmatrix}, \qquad M_k = \begin{pmatrix} 2k & 2k+1 \\ 2k+1 & 2k+2 \end{pmatrix}, \qquad k = 1, 2, \dots ,$$

so that, on writing $N_k = M_k M_{k-1} \cdots M_1$,

$$\begin{pmatrix} B_k \\ C_k \end{pmatrix} = N_k \begin{pmatrix} B_0 \\ C_0 \end{pmatrix} = N_k \begin{pmatrix} 1 \\ 2-e \end{pmatrix} = N_k \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ e \end{pmatrix}.$$

We claim that

$$N_k \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} p_{3k-1} & -q_{3k-1} \\ p_{3k} & -q_{3k} \end{pmatrix}, \qquad k = 1, 2, \ldots,$$

which can be established by induction on k running over different arithmetic progressions (mod 3). First, it is readily checked that the claim is valid for $k = 1, 2, 3$. Suppose, as induction hypothesis, that the claim holds for $k = 3n-2$ for some $n \geq 1$. Then, for $k = 3n+1$, we have

$$\begin{aligned} N_{3n+1} \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} &= M_{3n+1} N_{3n} \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} = M_{3n+1} \begin{pmatrix} p_{9n-1} & -q_{9n-1} \\ p_{9n} & -q_{9n} \end{pmatrix} \\ &= \begin{pmatrix} 6n+2 & 6n+3 \\ 6n+3 & 6n+4 \end{pmatrix} \begin{pmatrix} p_{9n-1} & -q_{9n-1} \\ p_{9n} & -q_{9n} \end{pmatrix} = \begin{pmatrix} p_{9n+2} & -q_{9n+2} \\ p_{9n+3} & -q_{9n+3} \end{pmatrix}, \end{aligned}$$

in which the last equality is derived from the recurrence relations for p_k and q_k; for example, from the recurrence we have

$$p_{9n+2} = 2(3n+1)p_{9n+1} + p_{9n} = (6n+2)p_{9n-1} + (6n+3)p_{9n},$$

and similar equations for p_{9n+3}, q_{9n+2} and q_{9n+3}. Thus the claim holds for $k = 3n+1$, completing the inductive step for $k = 3n-2$. The same argument applies for the inductive steps when $k = 3n-1$ and $k = 3n$, so that we have proved that

$$\begin{pmatrix} B_k \\ C_k \end{pmatrix} = \begin{pmatrix} p_{3k-1} & -q_{3k-1} \\ p_{3k} & -q_{3k} \end{pmatrix} \begin{pmatrix} 1 \\ e \end{pmatrix}, \qquad k = 1, 2, \ldots.$$

In particular, $C_k = p_{3k} - q_{3k}e$, so that

$$\frac{p_{3k}}{q_{3k}} = e + \frac{C_k}{q_{3k}} \to e \qquad \text{as} \quad k \to \infty.$$

The theorem is proved.

Corollary 8.4 *The number e is not well approximated to any order $\kappa > 2$.*

Proof We show, more generally, that if the partial quotients a_k of an irrational θ satisfies $a_k \ll k$ then, for all p/q with a large q,

$$\left| \theta - \frac{p}{q} \right| \gg \frac{1}{q^2 \log q}.$$

We may assume that p/q is a convergent p_k/q_k to θ, since otherwise the required result follows at once from Theorem 8.4. Also, it is trivial that $\log q_k \gg k$, because q_k is at least the kth Fibonacci number.

Now let α_k denote the kth complete quotient of θ so that, $\alpha_k \le a_k + 1 \ll k$. It then follows from (7.5) that, for all large k,

$$\left|\theta - \frac{p_k}{q_k}\right| = \frac{1}{(\alpha_{k+1}q_k + q_{k-1})q_k} \gg \frac{1}{kq_k^2} \gg \frac{1}{q_k^2 \log q_k} \gg \frac{1}{q_k^\kappa}.$$

The corollary is proved.

The corollary shows that a transcendental number need not be a Liouville number, but actually there are plenty of such numbers. Take, for example, the set of θ for which the partial quotients a_k taking only the values 1 or 2. Such θ cannot be Liouville numbers because a_k is bounded, so that θ cannot be well approximated to an order exceeding 2. Moreover, the set of such numbers has the cardinal of the continuum, whereas the subset algebraic numbers is countable, so that there are uncountably many transcendental numbers which are not Liouville numbers.

8.7 Irrationality of $\zeta(3)$

Before 1978, the problem of the irrationality of $\zeta(3)$ was considered to be one that was as difficult as that for Euler's constant γ, which currently still seems to be hopelessly difficult. At the *Journées Arithmétiques de Marseille-Luminy* in June 1978, R. Apéry confronted his audience with a miraculous proof for the irrationality of $\zeta(3)$. His proof, although elementary, was complicated and lengthy, with various astonishing formulae, and many participants were more than sceptical that there was a proof. However, the claimed formulae were soon verified numerically by computational experiments carried out by H. Cohen. The proof itself was then independently established by Cohen and D. Zagier during the *International Congress of Mathematicians* (ICM 1978) in Helsinki, and the theorem was reported by Cohen to a packed audience of delegates of the Congress at 5pm on Friday, 18 August 1978.

Not long after the Congress, F. Beukers [11] gave a much shorter proof which is being reproduced here. The elegant new proof contains transparent formulae in terms of triple integrals, the shape of which being motivated by Apéry's original formulae. The useful bound (see Sect. 11.3)

$$d_n < 3^n, \quad \text{for all large } n, \tag{8.9}$$

where

$$d_n = \mathrm{LCM}(1, 2, \ldots, n), \qquad n = 1, 2, \ldots,$$

plays a crucial role in the argument.

In the following, integrands of triple integrals over the unit cube $0 \le x, y, z \le 1$ may not be well-defined on the faces of the cube. However, the integrals concerned do exist as improper integrals, and their manipulation can be justified by considering instead the smaller cube $\epsilon \le x, y, z \le 1-\epsilon$, and letting $\epsilon \to 0$. As Beukers showed, the argument applies equally well for the irrationality of $\zeta(2) = \pi^2/6$, which is, of course, known to be transcendental by Lindemann's theorem (Theorem 8.8). Following Beukers we deal with this easier case first. Let, for non-negative real numbers r, s,

$$J_{r,s} = \int_0^1 \int_0^1 \frac{x^r y^s}{1-xy}\, dxdy. \tag{8.10}$$

Lemma 8.2 *For integers $r > s \ge 0$, the number $J_{r,s}$ is a fraction with denominator being a divisor of d_r^2. Also,*

$$J_{0,0} = \zeta(2) \quad \textit{and} \quad J_{r,r} = \zeta(2) - \frac{1}{1^2} - \frac{1}{2^2} - \cdots - \frac{1}{r^2}, \quad r > 0. \tag{8.11}$$

Proof From

$$\frac{1}{1-xy} = 1 + xy + x^2y^2 + \cdots, \qquad 0 \le xy < 1,$$

term-by-term integration applied to $J_{r,s}$, with integer values for r, s, yields

$$\begin{aligned} J_{r,s} &= \sum_{k=0}^{\infty} \frac{1}{(k+r+1)(k+s+1)} \\ &= \sum_{k=0}^{\infty} \frac{1}{r-s}\Big(\frac{1}{k+s+1} - \frac{1}{k+r+1}\Big) \\ &= \frac{1}{r-s}\Big(\frac{1}{s+1} + \frac{1}{s+2} + \cdots + \frac{1}{r}\Big). \end{aligned} \tag{8.12}$$

Thus, $J_{r,s}$ is a product of two fractions, each of which with a denominator being a divisor of $d_r = \mathrm{LCM}(1, 2, \ldots, r)$, so that the first part of the lemma is established, and (8.11) follows from

$$J_{r,r} = \sum_{k=0}^{\infty} \frac{1}{(k+r+1)^2}, \qquad r \ge 0.$$

□

Now let

$$I(P_n) = \int_0^1 \int_0^1 \frac{(1-y)^n P_n(x)}{1-xy}\, dxdy, \tag{8.13}$$

where $P_n(x)$ is a polynomial to be chosen with degree n, and with integer coefficients. It then follows from Lemma 8.2 that there are integers A_n, B_n such that

$$I(P_n) = \frac{A_n + B_n\zeta(2)}{d_n^2}; \tag{8.14}$$

it will be relevant to remark that $I(P_n) \neq 0$, so that $A_n + B_n\zeta(2) \neq 0$. The idea then is to choose P_n with the corresponding integers A_n, B_n satisfying

$$A_n + B_n\zeta(2) \to 0 \quad \text{as} \quad n \to \infty, \tag{8.15}$$

which then delivers the irrationality of $\zeta(2)$ according to the criterion given in Corollary 1.4 in Chap. 1. A suitable choice for P_n is the Legendre polynomial

$$P_n(x) = \frac{1}{n!}\left\{\frac{d}{dx}\right\}^n x^n(1-x)^n, \qquad n = 1, 2, \ldots, \quad 0 \le x \le 1, \tag{8.16}$$

which has integer coefficients, and we proceed to find a good upper bound for the double integral $I(P_n)$ defined in (8.13). The nth order derivative in P_n appearing in the integrand can be removed by applying an n-fold integration by parts to the inner integral with respect to x; after such a manoeuvre, we arrive at

$$I(P_n) = (-1)^n \int_0^1 \int_0^1 \frac{(x(1-x)y(1-y))^n}{(1-xy)^{n+1}}\, dxdy. \tag{8.17}$$

Lemma 8.3 *For* $0 \le x \le 1$, $0 \le y \le 1$ *and* $(x, y) \neq (1, 1)$, *we have*

$$\frac{x(1-x)y(1-y)}{1-xy} \le \phi^5, \qquad \phi = \frac{\sqrt{5}-1}{2}.$$

Proof Partial differentiation shows that $f(x, y)$, the left-hand side of the inequality, attains a unique maximum at (x, y), when $x = y$ and $1 - 2x + x^2y = 0$. From $1-2x+x^3 = (x-1)(x^2+x-1)$, we see that $f(x, y)$ is maximum when $x = y = \phi$, so that

$$\frac{x(1-x)y(1-y)}{1-xy} \le \frac{\phi^2(1-\phi)^2}{1-\phi^2} = \phi^5.$$

□

It now follows from (8.17) and (8.11) that

$$|I(P_n)| \le \phi^{5n} \int_0^1 \frac{dx\,dy}{1-xy} = \phi^{5n} J_{0,0} = \phi^{5n}\zeta(2),$$

and hence, by (8.14) and (8.9), that, for all large n,

$$0 < |A_n + B_n\zeta(2)| \le d_n^2 \phi^{5n}\zeta(2) < (9\phi^5)^n \zeta(2) < \Big(\frac{5}{6}\Big)^n.$$

The condition (8.15) is satisfied, so that $\zeta(2)$ is irrational. □

Theorem 8.11 (Apéry) *The number $\zeta(3)$ is irrational.*

Proof We extend the argument for $\zeta(3)$. Instead of $J_{r,s}$ in (8.10), we consider

$$K_{r,s} = \int_0^1 \int_0^1 \frac{-\log xy}{1-xy}\, x^r y^s dx dy.$$

Making use of

$$\frac{d}{dt}(xy)^t\Big|_{t=0} = \log xy,$$

we then find, from term-by-term differentiation of (8.12), that

$$K_{r,s} = -\frac{d}{dt} J_{r+t,s+t}\Big|_{t=0} = \frac{1}{r-s}\Big(\frac{1}{(s+1)^2} + \frac{1}{(s+2)^2} + \cdots + \frac{1}{r^2}\Big),$$

a fraction with denominator being a divisor of d_r^3. Similarly, it also follows that

$$K_{0,0} = 2\zeta(3) \qquad \text{and} \qquad K_{r,r} = 2\zeta(3) - \frac{2}{1^3} - \frac{2}{2^3} - \cdots - \frac{2}{r^3}.$$

Instead of $I(P_n)$ in (8.13), consider now the integral

$$H(P_n) = \int_0^1 \int_0^1 \frac{-\log xy}{1-xy}\, P_n(x) P_n(y)\, dx dy,$$

where P_n is the same Legendre polynomial defined in (8.16). The argument then shows that there are integers A_n, B_n (different from the previous ones for $\zeta(2)$, of course) depending on P_n such that

$$H(P_n) = \frac{A_n + B_n\zeta(3)}{d_n^3} \ne 0,$$

and it remains to find a suitable bound for $|H(P_n)|$.

Observe that the first factor of the integrand has the integral representation

$$\frac{-\log xy}{1-xy} = \int_0^1 \frac{1}{1-(1-xy)z}\,dz,$$

turning $H(P_n)$ into the triple integral

$$H(P_n) = \int_0^1\int_0^1\int_0^1 \frac{P_n(x)P_n(y)}{1-(1-xy)z}\,dxdydz.$$

After an n-fold integration by parts for the inner most integral with respect to x, we find that

$$\begin{aligned} H(P_n) &= \int_0^1\int_0^1\int_0^1 \frac{(xyz)^n(1-x)^n P_n(y)}{(1-(1-xy)z)^{n+1}}\,dxdydz \\ &= \int_0^1\int_0^1\int_0^1 \frac{((1-x)(1-w)^n P_n(y)}{1-(1-xy)w}\,dxdydw, \qquad w = \frac{1-z}{1-(1-xy)z}. \end{aligned}$$

Another n-fold integration by parts, this time with respect to y for the second inner integral, yields

$$H(P_n) = \int_0^1\int_0^1\int_0^1 \frac{(x(1-x)y(1-y)w(1-w))^n}{(1-(1-xy)w)^{n+1}}dxdydw,$$

and we proceed to bound the integrand.

Lemma 8.4 *For $0 < x, y, w < 1$, we have*

$$\frac{x(1-x)y(1-y)w(1-w)}{1-(1-xy)w} \le \theta^4, \qquad \theta = \sqrt{2}-1.$$

Proof Let $g(x, y, w)$ denote the left-hand side of the inequality, so that $g(x, y, w) = g(y, x, w)$ and, for fixed w, the maximum of $g(x, y, w)$ occurs when $x = y$. Partial differentiation with respect to w applied to

$$g(x, x, w) = \frac{x^2(1-x)^2w(1-w)}{1-(1-x^2)w}$$

shows that the maximum is attained when $w = 1/(1+x)$, at which $g(x, x, w) = \big(x(1-x)/(1+x)\big)^2$. Differentiation with respect to x then shows that the maximum is attained at $x = \theta$, with $\theta^2 + 2\theta - 1 = 0$, that is $\theta = \sqrt{2}-1$, so that

$$g(x, y, w) \le g\Big(\theta, \theta, \frac{1}{1+\theta}\Big) = \Big(\frac{\theta(1-\theta)}{1+\theta}\Big)^2 = (\theta^2)^2 = \theta^4.$$

As from before for the $\zeta(2)$ case, it now follows that

$$|H(P_n)| \le \theta^{4n} \int_0^1 \int_0^1 \int_0^1 \frac{dxdydw}{1-(1-xy)w} = \theta^{4n} K_{0,0} = 2\zeta(3)\theta^{4n},$$

and hence, for all large n,

$$0 < |A_n + B_n\zeta(3)| < 2\zeta(3)d_n^3\theta^{4n} < 2\zeta(3)(3^3\theta^4)^n < \left(\frac{4}{5}\right)^n.$$

Therefore $\zeta(3)$ is irrational.

8.8 Summary and Notes

Summary

- An irrational number θ is said to be *well-approximated* to level $\kappa \ge 2$ if there are infinitely fractions p/q satisfying $|\theta - p/q| < 1/q^\kappa$. Such a fraction p/q is called a *good* approximation to θ.
- By Dirichlet's approximation theorem (Theorem 1.7) there are infinitely many good approximations to any irrational number. Moreover, the convergents are such approximations, from below and from above alternately.
- The condition $|\theta - p/q| < 1/2q^2$ is a sufficient condition for p/q to be a convergent of the irrational θ. The necessary and sufficient condition is set out in Legendre's theorem.
- Of any three consecutive convergents p/q of an irrational θ, at least one of them satisfies $|\theta - p/q| < 1/(\sqrt{5}q^2)$; the number $\sqrt{5}$ is best possible.
- The fractional parts of $n\theta$, where θ is irrational and $n = 1, 2, \ldots$, is dense in the interval $(0, 1)$; in fact the sequence is *uniformly distributed* in the interval.
- Liouville's theorem states that an algebraic number is not well-approximated to a level beyond its degree. Consequently there must be real transcendental numbers. The result has been improved to the best possible one, namely Roth's theorem, which states that an algebraic number is not well-approximated to a level $\kappa > 2$.

  Results on Diophantine approximations lead to bounds for the number of solutions to Diophantine equations; for example, after Roth's theorem, an irreducible polynomial with integer coefficients of degree at least 3 has only finitely many zeros.
- Hermite introduced radical ideas from which irrationality of various numbers can be proved. With the construction of complicated auxiliary functions the transcendence of e and π (by Lindemann) have been established.
- Hermite's method also led to a nice simple proof, by Cohen, of the continued fraction expansion of e, which was first discovered by Euler.

- That $\zeta(3)$ is irrational was sensational claimed and established by R. Apéry in 1978. Beukers has given a simplified proof which requires a bound for the LCM of the first n numbers, namely that $\mathrm{LCM}(1, 2, \ldots, n) < 3^n$ will do.

Notes

1. The proof of Roth's theorem can be found in [15].
2. Unlike e, we know very little about the continued fraction of π, and how well can π be approximated by rationals is a much more difficult problem. In 1953 K. Mahler [61], using a much more sophisticated method, proved the remarkable result that, for all p/q with $q > 1$,

$$\left|\pi - \frac{p}{q}\right| > \frac{1}{q^{42}},$$

   so that π too is not a Liouville number. More recently [112], the exponent 42 has been reduced to about 7·1, when q is sufficiently large.
3. Although the proof of irrationality of $\zeta(3)$ has not yet been extended to the irrationality of $\zeta(s)$ for any specific odd $s > 3$, it has been established that at least one of the four numbers $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational. Indeed, for each odd $s > 3$, at least one of $\zeta(s + 2), \zeta(s + 4), \ldots, \zeta(8s - 1)$ is irrational, which implies that there are infinitely many irrationals among the numbers $\zeta(5), \zeta(7), \zeta(9), \ldots$. The references for these results are in the article by W. Zudilin [113], who gave a new proof of the irrationality of $\zeta(3)$ which dispenses of Beukers' integrals, and is therefore technically more elementary than the one here.
4. More on the topics in this chapter can be found in the recent publication [3].

8.9 Exercises

1. Prove that $\cos\frac{\pi}{9}$ and $e^{\sqrt{2}}$ are irrational numbers.
2. Obtain a reduction formula for $I_n = \int_{-1}^{1}(1-x^2)^n \cos \pi x \, dx$, and use it to show that π^2 is irrational.
3. Apply Example 12.1 to deliver an approximation to π as input value for the evaluation of 100 partial quotients of its continued fraction.
4. Let

$$J_h = \int_0^1 \frac{\big(x(1-x)\big)^{4h}}{x^2+1}\, dx, \qquad h = 0, 1, 2, \ldots,$$

and P, Q be coprime positive integers satisfying $\pi < P/Q < 22/7$. Show that there exist an odd h, and coprime positive integers r, s, such that

$$\frac{r J_h + s J_{h+2}}{4^{h-1}(r + 16s)} = \frac{P}{Q} - \pi.$$

Determine h, r, s corresponding to $P = 355$, $Q = 113$.

5. Determine the value of the largest positive integer relative to which $\frac{22}{7}$ is the best approximation to π.
6. Find approximations to π which are: (i) Good, and best, but is not a convergent; (ii) Best, but is not good.
7. Verify that the convergent $[3, 7, 15, 1, 292, 1, 1, 1]$ of π satisfies Legendre's criterion. Show also that $[3, 7, 15, 1, 292, 1, 1, 2]$ cannot be a convergent of π by applying the criterion.
8. Ramanujan discovered that π^4 has a large partial quotient with a small suffix. Find the partial quotient concerned, and the fraction p/q such that $\pi \approx \sqrt[4]{p/q}$.
9. Let $a_n = \{n\pi\}$, the fractional part of $n\pi$, for $1 \le n \le 10000$, and

$$A_k = \left|\{a_n : \frac{k-1}{K} < a_n \le \frac{k}{K},\ 1 \le n \le 1000\}\right|, \qquad k = 1, 2, \ldots, K.$$

Build the histogram for A_k with various choices of K.

10. Let u_n/v_n denote the nth convergent of the continued fraction for

$$\alpha = [a_0, a_1, a_2, \ldots], \qquad a_n = 4n + 2, \quad n = 0, 1, 2, \ldots .$$

Show that $(u_n + v_n)/(u_n - v_n)$ is a convergent of the continued fraction for e, and deduce that $\alpha = (e + 1)/(e - 1)$.

11. Show that if θ is well approximated to order κ, then θ^2 is well approximated to order $\kappa/2$.
12. Let $a, b, c \in \mathbb{N}$. Show, by considering the fact that $\sqrt[3]{a/b}$ cannot be well approximated to a certain level, that there are only finitely many integer solutions to the equation $ax^3 - by^3 = c$.
13. Write $x_1^2x_2+x_2^2x_3+x_3^2x_1+x_1^2x_3+x_2^2x_1+x_3^2x_2$ as a polynomial in the elementary symmetric polynomials σ_i of the variables x_j.

Chapter 9
Distribution of Prime Numbers



Abstract After establishing Chebyshev's theorem, Euler summation formula and Mertens' theorem, there is description of the use of Dirichlet characters to isolate members of an arithmetic progression from other integers. The chapter includes material on Dirichlet series, the Riemann zeta-function, Poisson summation, the Mellin transform and Perron's formula.

The prime number theorem states that

$$\pi(x) \sim \frac{x}{\log x}, \quad \text{as} \quad x \to \infty.$$

The main difficulty of the proof is to show that the ratio $\pi(x) : x/(\log x)$ has a limit as $x \to \infty$. Some of the proofs are considered to be 'elementary', meaning that the arguments used are not too far removed from the properties of whole numbers. Being elementary is a rather vague notion with no general agreement, but such an argument should not involve the theory of analytic functions, for example. A deep result established by an elementary argument is necessarily complicated, and it often seems opaque and lack motivation; see Note 9.1. On the other hand, an argument based on analytic function theory, once the necessary machinery has been developed, is often transparent and convincing, although there will be technical details to overcome still, of course.

None of the proofs of the prime number theorem can be described as being straightforward, let alone simple or easy. In Chap. 11, we give one of the simplest known proofs, which still involves much intricacy. The argument involved is mainly analytic, and a few words here may be helpful.

Suppose that we wish to investigate the properties of a sequence whose members possess certain arithmetic properties. We use the sequence to form an appropriate *generating function*, hoping that the analytic properties of the function may allow us to deduce the distributional characteristics of the sequence. For the simplest case of the constant sequence $(1, 1, 1, \ldots)$, one appropriate generating function is the Riemann zeta-function $\zeta(s) = \sum 1/n^s$; indeed Euler's argument that the series $\sum 1/p$ over primes p is divergent is based on the fact that $\zeta(s) \to \infty$ as $s \to 1$

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from above. Thus Euler may be considered the forerunner of the subject of analytic number theory but, in his time, he could only deal with such functions when s is real. For the property of being multiplicative, the appropriate generating function is the *Dirichlet series* (see Sect. 9.6) because $1/(mn)^s = (1/m^s)(1/n^s)$.

By mid-nineteenth century function theory was being developed earnestly, and analytic number theory was transformed into a rich theory by two seminal papers by Dirichlet in 1837, and by Riemann in 1859. Thus Dirichlet and Riemann are considered the pioneers of the subject, with Euler being a worthy forerunner. In can be said that the impetus for the development of function theory itself was the search for the proof of the prime number theorem.

9.1 Chebyshev's Theorem

Meanwhile P. L. Chebyshev found the correct order of $\pi(x)$, and he also showed that *if* the ratio $\pi(x) : x/(\log x)$ has a limit, then its value is 1. Indeed, Chebyshev's theorem can be considered the 'base camp' for the ascent of the prime number summit. Recall that $\Lambda(n) = \log p$ if $n = p^a$ is a power of a prime p, and zero otherwise; we now formally define

$$\pi(x) = \sum_{p \le x} 1, \quad \theta(x) = \sum_{p \le x} \log p, \quad \psi(x) = \sum_{n \le x} \Lambda(n).$$

Theorem 9.1 (Chebyshev) *For $x > 4$, we have*

$$\frac{1}{2}\frac{x}{\log x} < \pi(x) < \frac{5x}{\log x}.$$

Proof (Erdős) Much of what we want to know about the distribution of primes can be found from a close scrutiny of the factorisation of

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2},$$

the middle coefficient in the binomial expansion of $(1+1)^{2n}$. Being the largest of such coefficients it lies between the average and the sum of all the terms of the coefficients, so that

$$\frac{4^n}{2n+1} < \binom{2n}{n} < 4^n.$$

By Theorem 1.6, the exponent of the prime p in the prime factorisation of $m!$ is given by $\nu_p(m) = \sum_{a\geq 1} \lfloor m/p^a \rfloor$, so that

$$\frac{(2n)!}{(n!)^2} = \prod_{p\leq 2n} p^{\nu_p(2n)-2\nu_p(n)},$$

and since $\lfloor 2\lambda \rfloor - 2\lfloor \lambda \rfloor = 1$ or 0, it follows that

$$\nu_p(2n) - 2\nu_p(n) = \sum_{p^a \leq 2n} 1 = \sum_{a\leq \log(2n)/\log p} 1 \leq \frac{\log 2n}{\log p},$$

delivering the upper bound

$$\frac{(2n)!}{(n!)^2} \leq \prod_{p\leq 2n} p^{\log(2n)/\log p} = \prod_{p\leq 2n} (2n) = (2n)^{\pi(2n)}.$$

On the other hand, for $n < p \leq 2n$, we have

$$\nu_p(2n) - 2\nu_p(n) \geq \left\lfloor \frac{2n}{p} \right\rfloor - 2\left\lfloor \frac{n}{p} \right\rfloor \geq 1 - 0 = 1,$$

delivering the lower bound

$$\frac{(2n)!}{(n!)^2} \geq \prod_{n<p\leq 2n} p \geq \prod_{n<p\leq 2n} n = n^{\pi(2n)-\pi(n)}.$$

Recalling the earlier bounds for $\binom{2n}{n}$, we now have

$$\frac{4^n}{2n+1} < (2n)^{\pi(2n)} \qquad \text{and} \qquad n^{\pi(2n)-\pi(n)} < 4^n,$$

from which bounds for $\pi(x)$ can be obtained by taking logarithms.

First, for $x > 4$, we choose n so that $2n \leq x < 2n+2$, and find that

$$\pi(x) \geq \pi(2n) > \frac{2n\log 2}{\log 2n} - \frac{\log(2n+1)}{\log 2n} > \frac{(x-2)\log 2}{\log x} - 2 > \frac{1}{2}\frac{x}{\log x}.$$

Next, setting $n = 2^k$ in $(\pi(2n) - \pi(n))\log n < n \log 4$, we have

$$\pi(2^{k+1}) - \pi(2^k) < \frac{2^{k+1}\log 2}{k\log 2} = \frac{2^{k+1}}{k}, \qquad k = 1, 2, \ldots,$$

that is $k\Big(\pi(2^{k+1}) - \pi(2^k)\Big) < 2^{k+1}$, so that

$$(k+1)\pi(2^{k+1}) - k\pi(2^k) < \pi(2^{k+1}) + 2^{k+1} < 3{\cdot}2^k,$$

because $\pi(2z) < z$ when $z \geq 2$. Summing over $0 \leq k < K$, the left-hand side telescopes and we find that

$$K\pi(2^K) < 3(1 + 2 + 2^2 + \cdots + 2^{K-1}) < 3{\cdot}2^K.$$

For $x > 4$, we choose K so that $2^{K-1} < x \leq 2^K$. Then

$$\pi(x) \leq \pi(2^K) < \frac{3{\cdot}2^K}{K} < \frac{6(\log 2)x}{\log x} < \frac{5x}{\log x}.$$

The theorem is proved. □

Corollary 9.1 *Let p_k denote the kth prime. Then, for $k > 5$, we have*

$$\frac{1}{5}k\log k < p_k < 4k\log k.$$

Proof It follows from $\pi(p_k) = k$ that

$$\frac{1}{2}\frac{p_k}{\log p_k} < k < \frac{5p_k}{\log p_k},$$

and the required result follows from $\log k < \log p_k < 2\log k$.

9.2 Euler Summation Formula

Euler was the first person to make a serious study of the partial sums of the harmonic series. The logarithm function can be defined by

$$\log x = \int_1^x \frac{dt}{t}, \qquad x > 0;$$

note that, since $1/t$ is decreasing in $t > 0$, we have

$$0 \leq \int_m^{m+1} \frac{dt}{t} - \frac{1}{m+1} \leq \frac{1}{m} - \frac{1}{m+1}, \qquad m = 1, 2, \ldots .$$

Setting

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}, \quad \gamma_n = H_n - \log n, \qquad n = 1, 2, \ldots,$$

we then find that

$$0 \le \gamma_m - \gamma_{m+1} \le \frac{1}{m} - \frac{1}{m+1} < 1, \qquad m = 1, 2, \ldots.$$

Thus (γ_n) is a bounded decreasing sequence, which has to converge to a limit

$$\gamma = \lim_{n\to\infty} \gamma_n = \lim_{n\to\infty} (H_n - \log n)$$

called *Euler's constant*. Moreover, summing over $n \le m < N$, the two telescopic expressions collapse down to $0 \le \gamma_n - \gamma_N \le \frac{1}{n} - \frac{1}{N}$, and, letting $N \to \infty$, we have $0 \le \gamma_n - \gamma \le \frac{1}{n}$, for $n \ge 1$. Therefore

$$H_n = \log n + \gamma + O\Big(\frac{1}{n}\Big), \qquad n \to \infty.$$

In theory, the asymptotic formula can be used to evaluate γ, but the convergence is too slow, and it is not easy to determine H_n with high accuracy.

The method works equally well by replacing $1/t$ with $f(t)$ which decreases to 0. The argument then shows that there is a constant $\delta = \delta(f)$ such that

$$\sum_{1\le m\le n} f(m) = \int_1^n f(t)\,dt + \delta(f) + O(f(n)), \qquad n \to \infty.$$

The *Euler summation formula* gives an exact expression for the difference between the sum $\sum_{x<n\le y} f(n)$ and the integral $\int_x^y f(t)\,dt$; the expression becomes rather neat if $f(x)$ has a continuous derivative. We write $\{\lambda\} = \lambda - \lfloor\lambda\rfloor$, the fractional part of $\lambda \in \mathbb{R}$.

Theorem 9.2 (Euler Summation Formula) *Let f have a continuous derivative f' on the interval $[x, y]$. Then*

$$\sum_{x<n\le y} f(n) = \int_x^y f(t)\,dt + \int_x^y \{t\} f'(t)\,dt + f(x)\{x\} - f(y)\{y\}.$$

Proof First, on integration by parts, we have

$$\int_x^y f(t)\,dt = \Big[tf(t)\Big]_x^y - \int_x^y tf'(t)\,dt = yf(y) - xf(x) - \int_x^y tf'(t)\,dt.$$

Next, for $x \le n-1 < n \le y$, we have

$$\int_{n-1}^{n} \lfloor t \rfloor f'(t)\,dt = (n-1)\big(f(n)-f(n-1)\big) = \Big(nf(n)-(n-1)f(n-1)\Big) - f(n).$$

Summing over $\lfloor x \rfloor = m < n \le k = \lfloor y \rfloor$, we find that

$$\int_{m}^{k} \lfloor t \rfloor f'(t)\,dt = kf(k) - mf(m) - \sum_{x<n\le y} f(n),$$

so that

$$\begin{aligned}\sum_{x<n\le y} f(n) &= -\int_{m}^{k} \lfloor t \rfloor f'(t)\,dt + kf(k) - mf(m)\\ &= -\int_{x}^{y} \lfloor t \rfloor f'(t)\,dt + kf(y) - mf(x).\end{aligned}$$

The required result follows from the notation for the fractional parts $\{t\}$, $\{x\}$, $\{y\}$. □

Corollary 9.2 *We have, for $s > 1$, and $x \to \infty$,*

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \sum_{n\le x} \frac{1}{n^s} + \frac{1}{(s-1)x^{s-1}} + O\Big(\frac{1}{x^s}\Big).$$

Proof Let $f(t) = 1/t^s$. Then, taking $x = 1$, $y = X \in \mathbb{N}$ in the theorem, we have

$$\sum_{1<m\le X} \frac{1}{m^s} = \int_{1}^{X} \frac{dt}{t^s} - s\int_{1}^{X} \frac{\lfloor t \rfloor}{t^{s+1}}dt,$$

so that

$$\sum_{m\le X} \frac{1}{m^s} = 1 + \int_{1}^{X} \frac{dt}{t^s} - s\int_{1}^{X} \frac{\lfloor t \rfloor}{t^{s+1}}dt = 1 + \frac{1-X^{1-s}}{s-1} - s\int_{1}^{X} \frac{\lfloor t \rfloor}{t^{s+1}}dt.$$

As $X \to \infty$, the last integral converges, so that we may write

$$\sum_{m\le X} \frac{1}{m^s} = -\frac{1}{(s-1)X^{s-1}} + C(s) + O\Big(\frac{1}{X^s}\Big), \qquad X \to \infty,$$

where necessarily $C(s) = \sum_{m\ge 1} 1/m^s$. The result is established.

Corollary 9.3 (Stirling's Formula) *As $n \to \infty$, we have*

$$\log n! = n \log n - n + \frac{1}{2} \log n + \log\sqrt{2\pi} + o(1),$$

so that

$$n! \sim \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n}.$$

Proof Relating the sum $\log n! = \sum_{1<m\le n} \log m$ to the integral $\int_1^n \log t\, dt$, it follows at once that the first two dominating terms are $n \log n$ and $-n$. The summation formula then delivers the next dominating term $\frac{1}{2} \log n$, and also the existence of a constant term. That the constant involved has the value $\log\sqrt{2\pi}$ requires a result by J. Wallis. The following are the details.

Let $f(t) = \log t$, for $t \ge 1$, and $x = 1$, $y = n$, in the summation formula. Then

$$\sum_{1<m\le n} \log m = \int_1^n \log t\, dt + \int_1^n \frac{\{t\}}{t}\, dt = n \log n - n + 1 + \int_1^n \frac{\{t\}}{t}\, dt,$$

where the last integral is of order $\log n$, because the fractional part $\{t\}$ has the average value $\frac{1}{2}$ over any interval of length 1. We therefore expect the value of the integral to be about $\frac{1}{2} \log n$ and, indeed, it follows from the second mean-value theorem of integration that

$$\int_1^n \frac{\{t\}}{t}\, dt - \frac{1}{2} \log n = \int_1^n \frac{\{t\} - \frac{1}{2}}{t}\, dt \to C \quad \text{as} \quad n \to \infty,$$

for some constant C. Therefore, as $n \to \infty$,

$$\log n! = n \log n - n + \frac{1}{2} \log n + 1 + C + o(1),$$

and we may now write

$$n! = \exp(\log n!) = n^{n+\frac{1}{2}} e^{-n} K_n$$

where $K_n \to K = \exp(1 + C)$ as $n \to \infty$. Wallis' formula is

$$\lim_{n\to\infty} \frac{2{\cdot}2{\cdot}4{\cdot}4{\cdot}\cdots(2n){\cdot}(2n)}{1{\cdot}3{\cdot}3{\cdot}5{\cdot}5{\cdot}\cdots(2n-1){\cdot}(2n+1)} = \frac{\pi}{2}, \quad \text{that is} \quad \lim_{n\to\infty} \frac{(2^n n!)^4}{(2n)!^2(2n+1)} = \frac{\pi}{2}.$$

Substituting our newly found formula for $n!$, it now follows that

$$\lim_{n\to\infty} \frac{K_n^4 (2^n n^{n+\frac{1}{2}} e^{-n})^4}{K_{2n}^2 ((2n)^{2n+\frac{1}{2}} e^{-2n})^2 (2n+1)} = \frac{\pi}{2},$$

which delivers $K^2 = 2\pi$, that is $1 + C = \log\sqrt{2\pi}$, as required. □

Example 9.1 Find the asymptotic for the binomial coefficient $\binom{2n}{n}$.

As $n \to \infty$, we have

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2} = \frac{(2n)^{2n+\frac{1}{2}} e^{-2n} K_{2n}}{n^{2n+1} e^{-2n} K_n^2} \sim \frac{4^n \sqrt{2}}{\sqrt{n}\, K} \sim \frac{4^n}{\sqrt{\pi n}}.$$

□

For most applications of Stirling's formula, the estimate

$$\log n! = n \log n - n + O(\log n) \tag{9.1}$$

is often more than sufficient.

Evaluation of $\zeta(2)$

In 1735, Euler succeeded in showing that

$$\sum_{m=1}^{\infty} \frac{1}{m^2} = \frac{\pi^2}{6},$$

a result for which he was acknowledged as the leading mathematician in Europe. He used his summation formula, and its refinements, to give extensive computation of the partial sum $\sum_{m\le x} 1/m^2$, which he then suspected, or realised, was converging to $\pi^2/6$ before he embarked on a proof. The result appears in many formulae in number theory, and there are now dozens of proofs using a variety of themes. We offer the following one based on the integral calculus.

With the substitution $t = \tanh x$, we find that

$$\frac{2}{2k+1} = \int_{-1}^{1} t^{2k}\, dt = \int_{-\infty}^{\infty} \Big(\frac{\tanh x}{\cosh x}\Big)^{2k} dx, \qquad k = 0, 1, 2, \ldots,$$

so that

$$4\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2}=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\sum_{k=0}^{\infty}\Big(\frac{\tanh x\ \tanh y}{\cosh x\ \cosh y}\Big)^{2k}dx\,dy.$$

Summing the geometric progression under the integral sign the integrand becomes $1/\cosh(x+y)\cosh(x-y)$, and another substitution $u=x+y$, $v=x-y$ then delivers

$$8\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2}=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\frac{du\,dv}{\cosh u\ \cosh v}=\Big(\int_{-\infty}^{\infty}\frac{du}{\cosh u}\Big)^2=\pi^2,$$

where the last integral is evaluated by the substitution $w=e^u$. The required result follows from

$$\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2}=\sum_{n=1}^{\infty}\frac{1}{n^2}-\sum_{m=1}^{\infty}\frac{1}{(2m)^2}=\frac{3}{4}\sum_{n=1}^{\infty}\frac{1}{n^2}.$$

9.3 Sums Related to the Distribution of Primes

We have the following corollary to Chebyshev's theorem:

Corollary 9.4 *For $x>4$, we have $\frac{1}{6}x<\theta(x)<5x$. Also, as $x\to\infty$,*

$$\sum_{p\le x}\frac{\log p}{p}=\log x+O(1). \tag{9.2}$$

Proof First, we have $\theta(x)\le(\log x)\pi(x)<5x$, and

$$\theta(x)>\sum_{\sqrt{x}<p\le x}\log p>\frac{1}{2}\log x(\pi(x)-\pi(\sqrt{x}))>\frac{1}{3}(\log x)\pi(x)>\frac{x}{6}.$$

The exponent $\nu_p(N)$ has the bounds

$$\frac{N}{p}-1<\Big\lfloor\frac{N}{p}\Big\rfloor<\nu_p(N)<\frac{N}{p}+\frac{N}{p^2}+\cdots<\frac{N}{p}+\frac{N}{p(p-1)}.$$

On multiplication by $\log p$ and summing over $p\le N$, we have, as $N\to\infty$,

$$N\sum_{p\le N}\frac{\log p}{p}-\sum_{p\le N}\log p<\sum_{p\le N}\nu_p(N)\log p<N\sum_{p\le N}\frac{\log p}{p}+O(N).$$

Now $\sum_{p\le N}\log p=\theta(N)<5N$, and $\sum_{p\le N}\nu_p(N)\log p=\log N!=N\log N+O(N)$, by Stirling's formula (see (9.1)), so that

$$N\sum_{p\le N}\frac{\log p}{p}=N\log N+O(N).$$

The error involved in replacing x by N in the formula in the corollary is $O(1)$, so that the corollary is established. □

From

$$0<\log\Big(\frac{1}{1-1/p}\Big)-\frac{1}{p}=\frac{1}{2p^2}+\frac{1}{3p^3}+\cdots<\frac{1}{2p(p-1)},$$

and that the series $\sum 1/p(p-1)$ is convergent, we see that the series

$$\sum_p\left\{\log\Big(\frac{1}{1-1/p}\Big)-\frac{1}{p}\right\}$$

is also convergent. It follows that the series $\sum 1/p$ and the product $\prod(1-1/p)^{-1}$ are either both convergent, or both divergent. We now derive an accurate estimate for the partial sum $\sum_{p\le x}1/p$, from which an asymptotic formula for the partial product $\prod_{p\le x}(1-1/p)$ is derived.

Theorem 9.3 (Mertens [65]) *There are constants B_0 and B_1 such that, as $x\to\infty$,*

$$\text{(i)}\qquad \sum_{p\le x}\frac{1}{p}=\log\log x+B_0+O\Big(\frac{1}{\log x}\Big),$$

$$\text{(ii)}\qquad \prod_{p\le x}\Big(1-\frac{1}{p}\Big)=\frac{B_1}{\log x}+O\Big(\frac{1}{\log^2 x}\Big).$$

Proof It suffices to establish the results for $x=N\in\mathbb{N}$; in the following the numbers c_1,c_2,c_3,c_4,c_5 are well defined real constants.

(i) The factor $\log p$ in the sum $\sum_{p\le x}(\log p)/p$ in the Corollary to Chebyshev's theorem can be removed by applying partial summation (the discrete version of integration by parts), but care is required in the manipulation so that an appropriate error term is retained. Write

$$S(n)=\sum_{p\le n}\frac{\log p}{p}=\log n+r_n,\qquad r_n=O(1).$$

Then

$$\sum_{p\le N}\frac{1}{p}=\sum_{n\le N}\frac{\log p}{p}\cdot\frac{1}{\log p}=\sum_{2\le n\le N}\frac{S(n)-S(n-1)}{\log n}=\Sigma_1+\Sigma_2,$$

where

$$\Sigma_1=\sum_{2\le n\le N}\frac{\log n-\log(n-1)}{\log n},\qquad \Sigma_2=\sum_{2\le n\le N}\frac{r_n-r_{n-1}}{\log n}.$$

Define

$$f(t)=-\frac{\log(1-1/t)}{\log t},\qquad t\ge 2,\quad t\in\mathbb{R},$$

so that $\Sigma_1=\sum_{2\le n\le N}f(n)$; note that $f(t)$ decreases to 0 as $t\to\infty$, and that

$$f(t)=\frac{1}{t\log t}+O\Big(\frac{1}{t^2\log t}\Big).$$

Then

$$\int_N^\infty\Big\{f(t)-\frac{1}{t\log t}\Big\}\,dt=O\Big(\int_N^\infty\frac{dt}{t^2\log t}\Big)=O\Big(\frac{1}{N\log N}\Big),$$

so that we may set

$$c_1=\int_2^\infty\Big\{f(t)-\frac{1}{t\log t}\Big\}\,dt.$$

Replacing the sum Σ_1 by the corresponding integral in the summation formula, we then find that there is a constant c_2 such that

$$\begin{aligned}\Sigma_1&=\int_2^N f(t)\,dt+c_2+O(f(N))\\&=\int_2^N\frac{dt}{t\log t}+\int_2^N\Big\{f(t)-\frac{1}{t\log t}\Big\}\,dt+c_2+O\Big(\frac{1}{N\log N}\Big)\\&=\log\log N-\log\log 2+c_1+c_2+O\Big(\frac{1}{N\log N}\Big)\\&=\log\log N+c_3+O\Big(\frac{1}{N\log N}\Big),\qquad c_3=c_1+c_2-\log\log 2.\end{aligned}$$

Next, because r_n is bounded, we can set

$$c_4 = \sum_{n=2}^{\infty} r_n\Big(\frac{1}{\log n} - \frac{1}{\log(n+1)}\Big);$$

the tail end of the series satisfies

$$\sum_{n>N} r_n\Big(\frac{1}{\log n} - \frac{1}{\log(n+1)}\Big) \ll \sum_{n>N}\Big(\frac{1}{\log n} - \frac{1}{\log(n+1)}\Big) \ll \frac{1}{\log N}.$$

Then

$$\begin{aligned}\Sigma_2 &= \sum_{2\le n\le N} \frac{r_n - r_{n-1}}{\log n} = \sum_{2\le n\le N} \frac{r_n}{\log n} - \sum_{1\le n<N} \frac{r_n}{\log(n+1)} \\ &= \sum_{2\le n\le N} r_n\Big(\frac{1}{\log n} - \frac{1}{\log(n+1)}\Big) + O\Big(\frac{1}{\log N}\Big) \\ &= c_4 + O\Big(\frac{1}{\log N}\Big).\end{aligned}$$

Collecting the results, and setting $B_0 = c_3 + c_4$, we have

$$\begin{aligned}\sum_{p\le N} \frac{1}{p} &= \Sigma_1 + \Sigma_2 = \log\log N + c_3 + O\Big(\frac{1}{N\log N}\Big) + c_4 + O\Big(\frac{1}{\log N}\Big) \\ &= \log\log N + B_0 + O\Big(\frac{1}{\log N}\Big).\end{aligned}$$

(ii) We start with

$$\sum_{p>N}\left(\log\Big(1-\frac{1}{p}\Big) + \frac{1}{p}\right) \ll \sum_{p>N}\frac{1}{p^2} \ll \sum_{n>N}\frac{1}{n^2} \ll \frac{1}{N},$$

so that we may set

$$c_5 = \sum_{p\ge 2}\left(\log\Big(1-\frac{1}{p}\Big) + \frac{1}{p}\right)$$

for the sum of the convergent series. It follows from (i) that

$$\begin{aligned}\log \prod_{p\le N}\Big(1-\frac{1}{p}\Big) &= -\sum_{p\le N}\frac{1}{p}+\sum_{p\le N}\left(\log\Big(1-\frac{1}{p}\Big)+\frac{1}{p}\right)\\ &= -\log\log N - B_0 + O\Big(\frac{1}{\log N}\Big)\\ &\quad + c_5 - \sum_{p>N}\left(\log\Big(1-\frac{1}{p}\Big)+\frac{1}{p}\right)\\ &= -\log\log N - B_0 + c_5 + O\Big(\frac{1}{\log N}\Big).\end{aligned}$$

On exponentiation and noting that $\exp(O(1/\log N)) = 1 + O(1/\log N)$, we have

$$\prod_{p\le N}\Big(1-\frac{1}{p}\Big) = \frac{\exp(-B_0+c_5)}{\log N}\Big(1+O\Big(\frac{1}{\log N}\Big)\Big) = \frac{B_1}{\log N}+O\Big(\frac{1}{\log^2 N}\Big),$$

which is the required formula with $B_1 = \exp(-B_0 + c_5)$. □

Corollary 9.5 *Let $\phi(n)$ be Euler's totient function. As $n \to \infty$, we have*

$$\frac{n}{\phi(n)} \le \Big(\frac{1}{B_1}+o(1)\Big)\log\log n.$$

Proof Write $n = PQ$, where P and Q are products of primes $p|n$, with $p < \log n$ and $p > \log n$, respectively, so that $\phi(n)/n = \Pi(P)\cdot\Pi(Q)$, where

$$\Pi(P) = \prod_{p|P}\Big(1-\frac{1}{p}\Big) \ge \prod_{p<\log n}\Big(1-\frac{1}{p}\Big) \ge \frac{B_1+o(1)}{\log\log n},$$

$$\Pi(Q) = \prod_{p|Q}\Big(1-\frac{1}{p}\Big) \ge \Big(1-\frac{1}{\log n}\Big)^{\omega(Q)} \ge 1+o(1),$$

because $(\log n)^{\omega(Q)} < Q \le n$, so that $\omega(Q) < \log n/\log\log n$. The required result follows.

Making use of the fact that an absolutely convergent series can be summed in any way, and that $\sum_{d|k}\mu(d) = 1$, or 0, depending on whether $k = 1$, or $k > 1$, we have

$$\sum_{m=1}^{\infty}\frac{1}{m^s}\sum_{n=1}^{\infty}\frac{\mu(n)}{n^s} = \sum_{m,n=1}^{\infty}\frac{\mu(n)}{(mn)^s} = \sum_{k=1}^{\infty}\Big\{\frac{1}{k^s}\sum_{d|k}\mu(d)\Big\} = 1, \qquad s > 1.$$

In particular, at $s = 2$, we find that

$$\sum_{n=1}^{\infty} \frac{\mu(n)}{n^2} = \frac{1}{\zeta(2)} = \frac{6}{\pi^2},$$

a result we shall use in Sects. 10.1 and in 10.3. Also, because $\zeta(s) \to \infty$ as $s \to 1$,

$$\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} \to 0 \quad \text{as} \quad s \to 1.$$

Indeed, we may expect that the infinite series $\sum \mu(n)/n$ to be convergent to the sum 0; this is true, but the proof is as deep as that of the prime number theorem. Perhaps to emphasise this, the two results can be shown to be 'equivalent'. Roughly speaking being 'equivalent' here means that either result can be used to deduce the other 'without the heavy machinery used in establishing either result'. For example, in [5, Theorem 4.16], a relatively simple argument is used to deduce the prime number theorem from the assumption that $\sum \mu(n)/n$ converges. It has been established that $\sum_{n \le x} \mu(n) = o(x)$ as $x \to \infty$ is also equivalent to the prime number theorem.

9.4 Arithmetic Progressions and Dirichlet Characters

In 1837 Dirichlet proved his famous theorem that, for coprime numbers a, m, there are infinitely many primes in the arithmetic progression $n \equiv a \pmod{m}$. His starting point was Euler's proof that the series $\sum 1/p$, summed over primes p, is divergent, and his aim was to show that the series, when summed over primes $p \equiv a \pmod{m}$, still diverges. The proof involves many auxiliary results in analysis and algebra which first appeared in the memoir.

The first step is the construction of a suitable expression so that members of an arithmetic progression can be isolated from the other integers. For the required analytic argument, being suitable here means being a linear combination of periodic functions which are completely multiplicative. The laborious process then involves the construction of such functions called *Dirichlet characters*, of which an appropriate linear combination then delivers the required isolation.

Dirichlet's work preceded by decades the proper development of group theory; it was also in the middle of the period when function theory of a complex variable was being developed, mainly by Cauchy and Riemann. We follow essentially his construction of the characters, sprinkling with modern nomenclatures where appropriate, but we shall deviate significantly when we come to the analytic part of the proof of his theorem; see Chap. 12.

We start with an example, which appears to be a trick pulled out of a hat. The substance behind the magic will be revealed.

Table 9.1 Characters χ (mod 8)

n	1	3	5	7
$\chi_0(n)$	1	1	1	1
$\chi_1(n)$	1	−1	−1	1
$\chi_2(n)$	1	−1	1	−1
$\chi_3(n)$	1	1	−1	−1

Example 9.2 Find four distinct completely multiplicative functions $\chi_0, \chi_1, \chi_2, \chi_3$ with the properties that, for each odd a,

$$\frac{1}{4}\sum_{0\le r<4}\chi_r(a)\chi_r(n) = \begin{cases} 1 & \text{if} \quad n \equiv a \pmod 8, \\ 0 & \text{if} \quad n \not\equiv a \pmod 8. \end{cases}$$

Solution Define $\chi_r(n)$ at $n = 1, 3, 5, 7$ by Table 9.1: and $\chi_r(n) = 0$ for even n; and extend $\chi_r(n)$ periodically with period 8. It is readily checked that $\chi_r(n)$ are completely multiplicative functions with the required properties. □

In algebra, a *homomorphism* is a map from one group G to another group G' which preserves the action of group operations; this implies that the image of the identity of G is the identity of G', and the inverse of an element of G is mapped to the inverse of the image element in G'. Let $m > 1$, and consider the multiplicative groups $G = (\mathbb{Z}/m\mathbb{Z})^\times$ of coprime residues (mod m), and $G' = \mathbb{C}^\star$ of nonzero complex numbers. A homomorphism

$$\chi : (\mathbb{Z}/m\mathbb{Z})^\times \to \mathbb{C}^\star$$

is called a *character* (mod m). Since the order of $(\mathbb{Z}/m\mathbb{Z})^\times$ is $\phi(m)$, it follows from the Fermat-Euler theorem that

$$(\chi(n))^{\phi(m)} = 1, \qquad n \in (\mathbb{Z}/m\mathbb{Z})^\times;$$

this means that the range of $\chi(n)$ is the set of mth roots of unity. As usual, we have considered n here to be an integer representing the class to which it belongs, and treat $\chi(n)$ as a function with domain the set of numbers coprime with m; the function then satisfies the conditions that, for $\gcd(ab, m) = 1$,

(i) $\chi(a) \neq 0$, (ii) if $a \equiv b \pmod m$ then $\chi(a) = \chi(b)$, (iii) $\chi(ab) = \chi(a)\chi(b)$.

Indeed, by a *Dirichlet character* χ (mod m) we mean such a function when its domain is extended to $\mathbb{N}$ by further setting

$$\chi(n) = 0, \qquad \text{if} \quad \gcd(m, n) > 1.$$

As an arithmetic function, $\chi(n)$ is completely multiplicative, but perhaps we need some clarification. The same letter χ is used here for the two related, but different, objects: a group character, and a Dirichlet character. As a group character, the domain of χ is $(\mathbb{Z}/m\mathbb{Z})^\times$, and the number 0 is excluded from the range, whereas, as a Dirichlet character, it is an arithmetic function with domain $\mathbb{Z}$, and the range includes 0; we may say that the group character induces the Dirichlet character. With the clarification and historical justification, it does not matter if a Dirichlet character is called a character in the following.

The function $\chi(n) = 1$, for all n coprime with m, is clearly a character—we call it the *principal character*, and we denote it by χ_0. It follows from the definition of a character that $\chi(1) = 1$; that $\bar{\chi}(n) = \overline{\chi(n)}$ is also a character; and that the 'product' of two characters is a character. For conveniency, we shall write

$$\mathrm{e}(x) = \exp(2\pi i x), \quad x \in \mathbb{R}.$$

Example 9.3 Let $m = p$ be an odd prime, g a primitive root (mod p), and denote by $\nu(n)$ the index (or discrete logarithm) of n with respect to g. For $1 \le a < p$, the function defined by

$$\chi_a(n) = \mathrm{e}\Big(\frac{a\,\nu(n)}{p-1}\Big) \quad \text{is a character.}$$

Thus we find that (i) $\chi_a(1) = 1$; (ii) if $n \equiv n'$ (mod p) then $\nu(n) \equiv \nu(n')$ (mod $p-1$), so that $\chi_a(n) = \chi_a(n')$; and, for (iii),

$$\chi_a(nn') = \mathrm{e}\Big(\frac{a\nu(nn')}{p-1}\Big) = \mathrm{e}\Big(\frac{a(\nu(n)+\nu(n'))}{p-1}\Big) = \chi_a(n)\chi_a(n').$$

If $a = (p-1)/2$, then $\chi_a(n) = \mathrm{e}(\nu(n)/2) = 1$, or -1, depending on whether $\nu(n)$ is even, or odd; this means that $\chi_a(n) = \big(\frac{n}{p}\big)$, the Legendre symbol, is a character.

Construction of Characters

First, let $m = p^\ell$, a power of an odd prime p. By Theorem 5.2, there is a primitive root (mod m), and the theory of indices (or discrete logarithm) applies; let $\nu(n)$ denote the index of n to a fixed primitive root. Next, letting ω be any complex number satisfying $\omega^{\phi(m)} = 1$, we may then define a character by

$$\chi(n) = \omega^{\nu(n)}, \qquad p \nmid n.$$

The number of characters is then $\phi(m) = p^{\ell-1}(p-1)$.

The case $m = 2^\ell$ is slightly more complicated. If $\ell = 1$ then $m = 2$ and there is only the principal character χ_0; if $\ell = 2$ then $m = 4$; besides the principal character χ_0, there is also the character given by $\chi_1(1) = 1$ and $\chi_1(3) = -1$. For $\ell \geq 3$, there is no primitive root (mod p^ℓ), but we can appeal to Theorem 5.4 and use the fact that reduced residue classes are representable uniquely as $(-1)^\nu 5^{\nu'}$, where ν is defined (mod 2), and ν' is defined (mod $2^{\ell-2}$), so that we may set

$$\chi(n) = \omega^\nu (\omega')^{\nu'}, \qquad \text{where} \quad \omega^2 = 1, \quad (\omega')^{2^{\ell-2}} = 1.$$

Thus, for $\ell = 3$, we have $\omega = \pm 1$ and $\omega' = \pm 1$, from which the 4 characters in Example 9.2 are delivered. The number of such characters is then $2^{\ell-1} = \phi(2^\ell)$.

In general, when $m = 2^\ell p_1^{\ell_1} p_2^{\ell_2} \cdots p_k^{\ell_k}$, we define the characters (mod m) as products of arbitrary characters to the various prime power moduli:

$$\chi(n) = \chi(n; 2^\ell)\chi(n; p_1^{\ell_1})\chi(n; p_2^{\ell_2}) \cdots \chi(n; p_k^{\ell_k}), \qquad \gcd(n, m) = 1,$$

where $\chi(n; q)$ denotes any character (mod q) with q being a prime power. Such characters are distinct (mod m), and one of them is the principal character χ_0, which corresponds to taking all the ω values to be 1. The number of characters so constructed is then $\phi(2^\ell)\phi(p_1^{\ell_1})\phi(p_2^{\ell_2}) \cdots \phi(p_k^{\ell_k}) = \phi(m)$; we shall see that they are all the possible characters.

Example 9.4 Find the characters (mod 9), and use them to derive an expression which isolates members of individual arithmetic progressions (mod 9).

For $m = 9$, there are $\phi(9) = 6$ characters, and only two primitive roots, namely $g = 2$ and 5 (mod 9). Taking $g = 2$, and $\nu = 0, 1, 2, 3, 4, 5$, we have $n = g^\nu = 1, 2, 4, -1, -2, -4$, respectively.

Now let $\omega = e^{\pi i/3}$, a 6th root of unity, with $\omega^3 = -1$, and set $\omega_a = \omega^a$, for $a = 0, 1, 2, 3, 4, 5$. The 6 characters (mod 9) are then given by

$$\chi_a(n) = \omega_a^{\nu(n)} = \omega^{a\nu(n)}, \qquad n \not\equiv 0 \,(\text{mod } 3), \quad a = 0, 1, 2, 3, 4, 5;$$

their values are the elements of the matrix in Table 9.2. Setting $\chi_r(n) = 0$ for $3|n$, and $r = 0, 1, 2, 3, 4, 5$, we have, for $3 \nmid a$,

$$\frac{1}{6} \sum_{0 \leq r \leq 5} \overline{\chi_r(a)} \chi_r(n) = \begin{cases} 1 & \text{if} \quad n \equiv a \pmod 9, \\ 0 & \text{if} \quad n \not\equiv a \pmod 9, \end{cases}$$

the expression isolating the terms of the arithmetic progression $n \equiv a$ (mod 9). Taking the primitive root $g = 5$, instead of 2, only reorders the same characters. □

Table 9.2 Characters χ (mod 9)

n	1	2	4	-1	-2	-4
$\chi_0(n)$	1	1	1	1	1	1
$\chi_1(n)$	1	ω	ω^2	-1	$-\omega$	$-\omega^2$
$\chi_2(n)$	1	ω^2	$-\omega$	1	ω^2	$-\omega$
$\chi_3(n)$	1	-1	1	-1	1	-1
$\chi_4(n)$	1	$-\omega$	ω^2	1	$-\omega$	ω^2
$\chi_5(n)$	1	$-\omega^2$	$-\omega$	-1	ω^2	ω

Orthogonality Properties and Isolation Expression

In algebraic ring theory, the Chinese remainder theorem is used to show that $\mathbb{Z}/m\mathbb{Z} \simeq \oplus_j \mathbb{Z}/p_j^{\ell_j}\mathbb{Z}$ is an isomorphism of rings, so that

$$(\mathbb{Z}/m\mathbb{Z})^\times \simeq \oplus_j (\mathbb{Z}/p_j^{\ell_j}\mathbb{Z})^\star.$$

The characters $\chi : (\mathbb{Z}/m\mathbb{Z})^\times \to \mathbb{C}^\star$, as such, form themselves a group, called the dual, if the product characters is taken as the group operation, and the principal character the identity. Indeed, the dual group can be shown to be isomorphic to $(\mathbb{Z}/m\mathbb{Z})^\times$, so that there are $\phi(m)$ characters; the proof (see, for example, Theorem 6.8 [5]) will involve the basis theorem for Abelian groups.

Instead of such an algebraic approach, we have followed the historical development and constructed the $\phi(m)$ Dirichlet characters (mod m), but we now need to show that they are the only possible characters. We do this by first establishing the orthogonality properties associated with them. In the following $\sum_n$ denotes summation over a reduced set of residues (mod m), and $\sum_\chi$ summation over all the Dirichlet characters (mod m).

Theorem 9.4 *We have*

$$\sum_n \chi(n) = \begin{cases} \phi(m) & \text{if} \quad \chi = \chi_0, \\ 0 & \text{if} \quad \chi \neq \chi_0, \end{cases}$$

and

$$\sum_\chi \chi(n) = \begin{cases} \phi(m) & \text{if} \quad n \equiv 1 \pmod m, \\ 0 & \text{if} \quad n \not\equiv 1 \pmod m. \end{cases}$$

There are no other characters besides the constructed Dirichlet characters.

Proof If $\chi = \chi_0$ then $\chi(n) = 1$ for every n coprime with m, and 0 otherwise, so that $\sum_n \chi(n) = \phi(m)$. If $\chi \neq \chi_0$ then there exists a coprime with m such that $\chi(a) \neq 1$. As n runs over a residue system, so does an; thus

$$\chi(a)\sum_n \chi(n) = \sum_n \chi(an) = \sum_n \chi(n), \quad \text{so that} \quad (\chi(a)-1)\sum_n \chi(n) = 0,$$

and hence $\sum_n \chi(n) = 0$.

Let h denote the number of characters constructed. From $n^{\phi(m)} \equiv 1 \pmod m$ we have $(\chi(n))^{\phi(m)} = 1$, so that h is finite. If $n \equiv 1 \pmod m$ then $\chi(n) = 1$ for every χ, so that $\sum_\chi \chi(n) = h$, and

$$h = \sum_n \sum_\chi \chi(n) = \sum_\chi \sum_n \chi(n) = \phi(m).$$

If $n \not\equiv 1 \pmod m$ then there is a character χ with $\chi(n) \neq 1$, and we deploy

$$\chi(n)\sum_\chi \chi(n) = \sum_\chi \chi(an) = \sum_\chi \chi(n), \quad \text{so that} \quad (\chi(n)-1)\sum_\chi \chi(n) = 0,$$

and hence $\sum_\chi \chi(n) = 0$.

For the last part, let $X(n)$ be a function which is multiplicative, periodic with period m, and is 0 for $\gcd(n, m) > 1$. Then, unless $X(n)$ is identically zero, it is one of the constructed characters. To see this we let $\gcd(b, m) = 1$ and consider

$$\sum_n X(n)\bar\chi(n) = \sum_n X(bn)\bar\chi(bn) = X(b)\bar\chi(b)\sum_n X(n)\bar\chi(n),$$

which shows that either $X(b) = \chi(b)$ for some χ and all such b, or else the sum is 0. If the latter holds for every χ then, for arbitrary q,

$$0 = \sum_\chi \chi(q)\sum_n X(n)\bar\chi(n) = \sum_n X(n)\sum_\chi \chi(q)\bar\chi(n) = \phi(m)X(q).$$

Therefore if $X(q) \neq 0$ for some q, then $X(n)$ has to be some constructed $\chi(n)$. □

The promised expression of a linear combination of multiplicative functions which isolates the terms $n \equiv a \pmod m$ can now be delivered.

Theorem 9.5 *Let* $\gcd(a, m) = 1$. *Then, for all* $n > 0$,

$$\frac{1}{\phi(m)}\sum_\chi \frac{\chi(n)}{\chi(a)} = \begin{cases} 1 & \text{when} \quad n \equiv a \pmod m, \\ 0 & \text{when} \quad n \not\equiv a \pmod m. \end{cases}$$

Also, for any $u < v$,

$$\left| \sum_{u \le n < v} \chi(n) \right| \le \phi(m).$$

Proof Let $\bar{a}$ be the reciprocal of $a \pmod m$, so that $\bar{a}a \equiv 1 \pmod m$, and thus $\chi(\bar{a})\chi(a) = 1$. Then, by the second part of the last theorem,

$$\sum_{\chi} \frac{\chi(n)}{\chi(a)} = \sum_{\chi} \chi(\bar{a}n) = \begin{cases} \phi(m) & \text{if} \quad \bar{a}n \equiv 1, \text{ i.e., } n \equiv a \pmod m, \\ 0, & \text{if} \quad \bar{a}n \not\equiv 1, \text{ i.e., } n \not\equiv a \pmod m. \end{cases}$$

The last part of the theorem is a consequence of the observation that $\chi(n)$ is periodic with period m, and $\sum_n \chi(n) = 0$ if the sum is over all the reduced residues. The theorem is proved. □

9.5 Euler Products

Euler's argument that $\sum 1/p$ diverges is based on his study of the function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \qquad s > 1,$$

which, by the integral test, is convergent. He realised, of course, that $\zeta(s) \to \infty$ as $s \to 1$ and, indeed, he was the first person to make extensive study of the partial sums of the divergent harmonic series; see Sect. 9.2. Euler's more important discovery is that

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots\right) = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1}, \qquad s > 1; \tag{9.3}$$

here is an identity in s for a sum over all the natural numbers n, and a product over all the primes p. To see this, we note that if the product is to be multiplied out formally, we obtain the series instead because, by the fundamental theorem of arithmetic, each n can be expressed as a product of prime powers in just one way; indeed, the identity can be considered as the analytic equivalent of the fundamental theorem.

To make the argument rigorous, we first take a finite number of factors, with $p \le x$, say. Since absolutely convergent series can be summed in anyway, the value

of the finite product is just the series $\sum' 1/n^s$, where $'$ indicates that the sum is over those n having no prime divisors exceeding x. It then follows that, for $s > 1$,

$$0 \le \sum_{n=1}^{\infty} \frac{1}{n^s} - \prod_{p \le x} \Big(1 - \frac{1}{p^s}\Big)^{-1} \le \sum_{k=1}^{\infty} \frac{1}{(x+k)^s} \to 0 \quad \text{as} \quad x \to \infty,$$

which then delivers the identity (9.3).

More generally, if $f(n)$ is a multiplicative arithmetic function such that $\sum f(n)$ is absolutely convergent, then the sum of the series can be expressed as an absolutely convergent infinite product over the primes:

$$\sum_{n=1}^{\infty} f(n) = \prod_{p} \big(1 + f(p) + f(p^2) + \cdots\big);$$

moreover, if $f(n)$ is completely multiplicative, so that $f(p^m) = (f(p))^m$, and the series becomes a geometric progression with the sum $\Big(1 - f(p)\Big)^{-1}$, we then have

$$\sum_{n=1}^{\infty} f(n) = \prod_{p} \frac{1}{1 - f(p)}.$$

In any such cases, the product is called the *Euler product* of the series.

9.6 Dirichlet Series and Riemann Zeta-Function

Following Riemann it is traditional in analytic number theory to denote a complex variable by

$$s = \sigma + it \in \mathbb{C}, \qquad \sigma, t \in \mathbb{R},$$

so that σ and t are the real and imaginary parts of $s \in \mathbb{C}$. Thus, for $n \in \mathbb{N}$,

$$n^s = e^{s \log n} = e^{(\sigma + it) \log n} = n^{\sigma} e^{it \log n}; \qquad |n^s| = n^{\sigma}, \quad |n^{it}| = 1.$$

For an arithmetic function $f(n)$ we call a series of the form

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s}, \qquad s = \sigma + it, \tag{9.4}$$

a *Dirichlet series*; the simplest case is when $f(n) = 1$, and we then have the *Riemann zeta-function* $\zeta(s)$—the same function introduced by Euler, but Riemann

considered it with the complex variable s. The standard reference for the theory of the Riemann zeta-function is the classical text [97].

Riemann made many important discoveries related to $\zeta(s)$ in an epoch making paper in 1859, his only paper written on number theory. He showed that $\zeta(s)$ has an analytic continuation into the whole complex plane $\mathbb{C}$, apart from a simple pole at $s = 1$. He then showed that many of the secrets of the distribution of primes are tied up with his zeta-function, particularly with the complex zeros of $\zeta(s)$. His ideas led to further development of function theory in order to carry out his program. As we already remarked earlier, it culminated in 1896 when Hadamard and de la Vallée Poussin independently proved the prime number theorem.

For our purpose we first note that

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \qquad \sigma > 1,$$

is analytic and not zero in $\sigma > 1$. For if $\sigma \geq a > 1$, we have

$$\left| \sum_{n=N}^{\infty} \frac{1}{n^s} \right| \leq \sum_{n=N}^{\infty} \frac{1}{n^\sigma} \leq \sum_{n=N}^{\infty} \frac{1}{n^a} \to 0, \qquad N \to \infty,$$

so that the series converges uniformly in $\sigma \geq a$. Since a can be any real number greater than 1, it follows that $\zeta(s)$ is an analytic function in $\sigma > 1$. The product representation is valid in $\sigma > 1$, wherein $\zeta(s) \neq 0$. Indeed, for $\sigma > 1$, we have

$$\frac{1}{|\zeta(s)|} \leq \prod_p \left(1 + \frac{1}{p^\sigma}\right) < \sum_{n=1}^{\infty} \frac{1}{n^\sigma} \leq 1 + \int_1^{\infty} \frac{du}{u} = \frac{\sigma}{\sigma - 1},$$

so that $|\zeta(s)| > 1 - 1/\sigma > 0$.

Next, let $m > 1$, and $\chi(n)$ be a Dirichlet character (mod m). The *Dirichlet L-function* is defined by

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}, \qquad \sigma > 1. \tag{9.5}$$

Since $|\chi(n)| \leq 1$, it also follows from the above that $L(s, \chi)$ is analytic in $\sigma > 1$, and that it has the Euler product representation

$$L(s, \chi) = \prod_p \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}, \qquad \sigma > 1.$$

Note that, for the principal character χ_0, we have

$$L(s, \chi_0) = \prod_{p \nmid m} \left(1 - \frac{1}{p^s}\right)^{-1} = \zeta(s) \cdot \prod_{p \mid m} \left(1 - \frac{1}{p^s}\right), \qquad \sigma > 1. \tag{9.6}$$

The coefficients of the Dirichlet series for the convolution is given by the coefficient of the product of the two Dirichlet series, that is, if $h = f * g$, so that $h(k) = \sum_{mn} f(n)g(m)$, then

$$\sum_{k=1}^{\infty} \frac{h(k)}{k^s} = \left(\sum_{n=1}^{\infty} \frac{f(n)}{n^s}\right)\left(\sum_{m=1}^{\infty} \frac{g(m)}{m^s}\right),$$

which is valid in the half-plane where the two series on the right-hand side converge absolutely. To see this we simply multiply the two series concerned and rearrange the terms appropriately—this can be justified by absolute convergence. By appropriately we mean collecting those terms for which $mn = k$ for each $k = 1, 2, \ldots$. Thus

$$\left(\sum_{n=1}^{\infty} \frac{f(n)}{n^s}\right)\left(\sum_{m=1}^{\infty} \frac{g(m)}{m^s}\right) = \sum_{k=1}^{\infty}\left(\sum_{mn=k} \frac{f(n)g(m)}{k^s}\right) = \sum_{k=1}^{\infty} \frac{h(k)}{k^s}.$$

The converse also holds, although we need to use the fact that the Dirichlet series of a given sum function is unique.

Example 9.5 Show that

$$\text{(i)} \quad \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)}, \qquad \sigma > 1;$$

$$\text{(ii)} \quad \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s} = -\frac{\zeta'(s)}{\zeta(s)}, \qquad \sigma > 1;$$

$$\text{(iii)} \quad \sum_{n=1}^{\infty} \frac{d(n)}{n^s} = \zeta^2(s), \qquad \sigma > 1;$$

$$\text{(iv)} \quad \sum_{n=1}^{\infty} \frac{\mu^2(n)}{n^s} = \frac{\zeta(s)}{\zeta(2s)}, \qquad \sigma > 1;$$

$$\text{(v)} \quad \sum_{n=1}^{\infty} \frac{d^2(n)}{n^s} = \frac{\zeta^4(s)}{\zeta(2s)}, \qquad \sigma > 1;$$

$$\text{(vi)} \quad \sum_{n=1}^{\infty} \frac{2^{\omega(n)}}{n^s} = \frac{\zeta^2(s)}{\zeta(2s)}, \qquad \sigma > 1;$$

$$\text{(vii)} \quad \sum_{n=1}^{\infty} \frac{d(n^2)}{n^s} = \frac{\zeta^3(s)}{\zeta(2s)}, \qquad \sigma > 1;$$

$$\text{(viii)} \quad \sum_{n=1}^{\infty} \frac{\lambda(n)}{n^s} = \frac{\zeta(2s)}{\zeta(s)}, \qquad \sigma > 1;$$

$$\text{(ix)} \quad \sum_{n=1}^{\infty} \frac{\phi(n)}{n^s} = \frac{\zeta(s-1)}{\zeta(s)}, \qquad \sigma > 2,$$

and write down the corresponding Euler products.

Solution We need only employ Dirichlet convolution, and a few selected examples will suffice. For $\sigma > 1$, we have

(i)

$$\zeta(s) \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} = 1 \quad \text{and} \quad \frac{1}{\zeta(s)} = \prod_p (1 - p^{-s});$$

(viii)

$$\zeta(s) \sum_{n=1}^{\infty} \frac{\lambda(n)}{n^s} = \sum_{\substack{n=1 \\ n=\text{square}}}^{\infty} \frac{1}{n^s} = \sum_{m=1}^{\infty} \frac{1}{m^{2s}} = \zeta(2s) \quad \text{and}$$

$$\frac{\zeta(2s)}{\zeta(s)} = \prod_p \frac{1}{1 + p^{-s}};$$

(ix) for $\sigma > 2$,

$$\zeta(s) \sum_{n=1}^{\infty} \frac{\phi(n)}{n^s} = \sum_{n=1}^{\infty} \frac{n}{n^s} = \zeta(s-1) \quad \text{and} \quad \frac{\zeta(s-1)}{\zeta(s)} = \prod_p \frac{1 - p^{-s}}{1 - p^{1-s}}.$$

Analytic Continuation into $\sigma > 0$

Riemann found that his zeta-function $\zeta(s)$ satisfies a functional equation, from which it follows that $\zeta(s)$ has an analytic continuation into the whole complex plane $\mathbb{C}$, with a simple pole at $s = 1$; see [97]. For our purpose, we need only

continuation into the half-plane $\sigma > 0$, and this can be delivered by partial summation with relative ease. It is convenient to use integral rather than sums—we have, for $s = \sigma + it$, and $\sigma > 1$,

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \sum_{n=1}^{\infty} n\Big(\frac{1}{n^s} - \frac{1}{(n+1)^s}\Big) = s\sum_{n=1}^{\infty} n \int_n^{n+1} \frac{dx}{x^{s+1}} = s\int_1^{\infty} \frac{\lfloor x \rfloor}{x^{s+1}}\, dx.$$

Letting $\{x\} = x - \lfloor x \rfloor$, the 'fractional part' of $x \in \mathbb{R}$, we then find that

$$\zeta(s) = \frac{s}{s-1} - s\int_1^{\infty} \frac{\{x\}}{x^{s+1}}\, dx, \qquad \sigma > 1. \tag{9.7}$$

Since $0 \le \{x\} < 1$, this last integral is convergent in $\sigma > 0$, and uniformly so in $\sigma \ge \delta$ for any $\delta > 0$, so that it represents an analytic function of s in $\sigma > 0$. The equation therefore delivers $\zeta(s)$ as a meromorphic function in $\sigma > 0$, its only singularity being a simple pole at $s = 1$ with residue 1.

From (9.6), we see that $L(s, \chi_0)$ also has a continuation into $\sigma > 0$, and that there is a simple pole at $s = 1$, with residue $\prod_{p|m}(1 - 1/p) = \phi(m)/m$.

If $\chi \ne \chi_0$ is not the principal character, then the defining series (9.5) for $L(s, \chi)$ converges locally uniformly in $\sigma > 0$; 'locally uniformly in a set $\mathcal{A}$' means 'each point belonging to $\mathcal{A}$ is an interior point of a compact set in which there is uniformity.' Readers will be familiar with the alternating-sign test for series: If (b_n) is a real sequence which deceases to 0, then the series $\sum(-1)^n b_n$ converges. This can be generalised to Dirichlet's test for $\sum a_n b_n$, in which the partial sums for $\sum a_n$ are bounded—the proof amounts to making use of partial summation, the special case of which is given in the proof of the following theorem.

Theorem 9.6 *Let $\sum a_n$ be a series with bounded partial sums. Then the series $\sum_{n\ge 1} a_n/n^s$ converges locally uniformly in $\sigma > 0$.*

Proof Write $A_n = a_1 + a_2 + \cdots + a_n$, and let $|A_n| \le K$ for all $n \ge 1$. Then, for $M \le n \le N$, we have, by partial summation,

$$\sum_{n=M}^{N} \frac{a_n}{n^s} = \sum_{n=M}^{N} \frac{A_n - A_{n-1}}{n^s} = \frac{A_N}{N^s} - \frac{A_{M-1}}{(M-1)^s} + \sum_{n=M}^{N} A_{n-1}\Big(\frac{1}{(n-1)^s} - \frac{1}{n^s}\Big).$$

Letting $s = \sigma + it$, so that

$$\left|\frac{1}{(n-1)^s} - \frac{1}{n^s}\right| = \left|s\int_{n-1}^{n} \frac{du}{u^{s+1}}\right| \le |s| \int_{n-1}^{n} \frac{du}{u^{\sigma+1}} \le \frac{|s|}{(n-1)^{\sigma+1}},$$

and thus

$$\sum_{n\ge M} \left|\frac{1}{(n-1)^s} - \frac{1}{n^s}\right| \ll \frac{|s|}{M^{\sigma}}, \qquad s = \sigma + it,$$

we then find that, for $\sigma \geq r > 0$ and $|t| \leq T$,

$$\left| \sum_{n=M}^{N} \frac{a_n}{n^s} \right| \ll \frac{K|s|}{M^\sigma} \ll \frac{K(T+r)}{M^r} \to 0 \quad \text{as} \quad M \to \infty.$$

By Cauchy's criterion, the series $\sum_{n\geq 1} a_n/n^s$ converges uniformly in the half-strip $\sigma \geq r$, $|t| \leq T$. Since r can be arbitrarily small, and T can be arbitrarily large, the proposition is proved. □

Corollary 9.6 *Let χ be a nonprincipal character* (mod m). *Then the series in (9.5) converges locally uniformly in $\sigma > 0$, so that $L(s, \chi)$ is an analytic function in $\sigma > 0$. In particular, $L(s, \chi)$ is well defined at $s = 1$.*

Proof The partial sums of $\sum \chi(n)$ are bounded by $\phi(m)$.

Example 9.6 Use the Dirichlet series for $(1 - 2^{1-s})\zeta(s)$ in $\sigma > 1$ to deliver an analytic continuation for $\zeta(s)$ into $\sigma > 0$, and verify that it has a simple pole with residue 1 at $s = 1$.

From

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \qquad \frac{2}{2^s}\zeta(s) = \sum_{n=1}^{\infty} \frac{2}{(2n)^s}, \qquad \sigma > 1,$$

we find, on subtraction, that

$$(1 - 2^{1-s})\zeta(s) = \sum_{n \text{ odd}} \frac{1}{n^s} - \sum_{n \text{ even}} \frac{1}{n^s} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s}, \qquad \sigma > 1. \tag{9.8}$$

The alternating-sign series is convergent in $\sigma > 0$, and $1 - 2^{1-s}$ has a simple zero at $s = 1$, so that

$$\zeta(s) = \frac{1}{1 - 2^{1-s}} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s}, \qquad \sigma > 0, \quad s \neq 1.$$

The residue of the simple pole of $\zeta(s)$ at $s = 1$ is given by

$$\lim_{s\to 1} \frac{s-1}{1 - 2^{1-s}} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} = \lim_{s\to 1} \frac{s-1}{1 - 2^{1-s}} \log 2 = 1,$$

because $2^{1-s} = \exp((1-s)\log 2) = 1 - (s-1)\log 2 + O(|s-1|^2)$.

It turns out that $\zeta(s) \neq 0$ on the vertical line $\sigma = 1$, $s \neq 1$, and also that $L(1, \chi) \neq 0$ for any nonprincipal character χ. Indeed, these two facts are indispensable in our proofs of the prime number theorem, and Dirchlet's theorem, respectively. We shall cross these bridges when we get there.

Example 9.7 Evaluate $L(1, \chi)$ for each nonprincipal character $\chi \pmod 8$.

The three nonprincipal characters $\chi_r \pmod 8$ $(r = 1, 2, 3)$ are given in (9.2), so that $L(1, \chi_r)$ are given by the sums of the conditionally convergent series:

$$L(1, \chi_1) = 1 - \frac{1}{3} - \frac{1}{5} + \frac{1}{7} + \frac{1}{9} - \frac{1}{11} - \frac{1}{13} + \frac{1}{15} + \cdots,$$
$$L(1, \chi_2) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \frac{1}{13} - \frac{1}{15} + \cdots,$$
$$L(1, \chi_3) = 1 + \frac{1}{3} - \frac{1}{5} - \frac{1}{7} + \frac{1}{9} + \frac{1}{11} + \frac{1}{13} - \frac{1}{15} - \cdots.$$

Partial sums at multiples of 4 terms reveal that the sum of each series is positive. In fact, for $L(1, \chi_2)$, we find that

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \sum_{n=0}^{\infty} \int_0^1 (-1)^n x^{2n} dx = \int_0^1 \sum_{n=0}^{\infty} (-1)^n x^{2n} dx = \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4},$$

assuming that it is legitimate to change the order of summation and integration. For justification, let $0 < a < 1$, and consider

$$1 - x^2 + x^4 - x^6 + \cdots = \frac{1}{1+x^2}, \qquad 0 \le x \le a;$$

term-by-term integration is then legitimate because the power series converges uniformly in $0 \le x \le a$. Moreover, since the series for $L(1, \chi_2)$ is known to be convergent, by Abel's continuity theorem, the result still holds at $a = 1$.

For $L(1, \chi_1)$ and $L(1, \chi_3)$, we consider, for $0 < x < 1$,

$$\frac{1}{1+x^4} = 1 - x^4 + x^8 - x^{12} + x^{16} - x^{20} + \cdots,$$
$$\frac{x^2}{1+x^4} = x^2 - x^6 + x^{10} - x^{14} + x^{18} - x^{22} + \cdots,$$
$$\frac{1 \pm x^2}{1+x^4} = 1 \pm x^2 - x^4 \mp x^6 + x^8 \pm x^{10} - x^{12} \mp x^{14} + \cdots,$$

so that

$$L(1,\chi_3)=\int_0^1\frac{1+x^2}{1+x^4}\,dx,\qquad L(1,\chi_1)=\int_0^1\frac{1-x^2}{1+x^4}\,dx,$$

which can be evaluated using the factorisation $1+x^4=(1+\sqrt{2}x+x^2)(1-\sqrt{2}x+x^2)$.

9.7 Poisson, Mellin and Perron

In analytic number theory, there are three very useful tools for the practitioner: *Poisson summation formula, Mellin transform* and *Perron's formula*. We give reasonable detail for the first item, using it for another evaluation of the quadratic Gauss sum; the remaining two items are given only rough sketches.

Poisson Summation Formula

The formula states that, under certain conditions on a function $f(x)$, the sum $\sum f(n)$ over $A\le n\le B$ is given by

$$\sum_{A\le n\le B}{}'\ f(n)=\sum_{v\in\mathbb{Z}}\int_A^B f(x)e^{2\pi ivx}dx, \tag{9.9}$$

where $\sum'$ means that if A and/or B are integers, then the terms concerned are to be replaced by $\frac{1}{2}f(A)$ and $\frac{1}{2}f(B)$.

Readers familiar with the representation of a function $f(x)$ by its Fourier series may expect that formula (9.9) is related to such a theory. Indeed we shall derive the formula from the simplest convergence theorem for Fourier series. The formula is even more closely related to the Fourier transform of a continuous function $F(x)$ such that $\int_{\mathbb{R}}|F(x)|dx<\infty$, which is given by the formula

$$\hat{F}(u)=\int_{-\infty}^{\infty}F(x)e^{-2\pi ixu}dx.$$

The transform $\hat{F}(u)$ is continuous, and if $\int_{\mathbb{R}}|\hat{F}(u)|du<\infty$ then there is the Fourier inversion formula

$$F(x)=\int_{-\infty}^{\infty}\hat{F}(u)e^{2\pi ixu}du,$$

but we shall not consider them here.

We suppose that $f(x)$ is real, and that it is monotonic in stretches, a condition[1] that can be verified for most problems in number theory. Let $f_1(x) = f(x)$ for $0 \le x < 1$, and extend the domain of f_1 to $\mathbb{R}$ by periodicity with period 1. The Fourier series of $f_1(x)$ is

$$\tfrac{1}{2}a_0 + \sum_{v=1}^{\infty}(a_v \cos 2\pi vx + b_v \sin 2\pi vx),$$

where the coefficients are given by Fourier's formulae:

$$\tfrac{1}{2}a_v = \int_0^1 f(x)\cos 2\pi vx\,dx, \qquad \tfrac{1}{2}b_v = \int_0^1 f(x)\sin 2\pi vx\,dx.$$

A theorem of Dirichlet's states that the Fourier series for such a function converges for all $x \in \mathbb{R}$, and that its sum is $f_1(x)$ at a point of continuity, and is otherwise the average of the left-right limits of f_1 at x, in other words the value of the sum is

$$\frac{1}{2}\Big(\lim_{t\to x_+} f(t) + \lim_{t\to x_-} f(t)\Big);$$

the limits exist because f is monotonic in stretches. In particular

$$\frac{f(0)+f(1)}{2} = \frac{a_0}{2} + \sum_{v=1}^{\infty} a_v = \sum_{v=-\infty}^{\infty} f(x)\cos 2\pi vx\,dx.$$

This is the case of the Poisson formula for $A = 0$ and $B = 1$, and the general case follows on replacing $f(x)$ by $f(x+n)$ for $A \le n < B$ and adding the result.

Another Evaluation of the Gauss Sum

Recall that the quadratic Gauss sum is given by

$$G = G(p) = \sum_{0\le n<p} e^{2\pi i n^2/p};$$

[1] In mathematical analysis, however, being monotonic in stretches is a severe condition, because it implies that there are only finitely many maxima and minima in a finite interval. Readers may be aware that a monotonic function is differentiable almost everywhere, whereas some continuous functions may be differentiable nowhere.

we do not restrict p to be a prime in the following. On application of the Poisson summation formula with $f(x) = \cos 2\pi x^2/p$ and $f(x) = \sin 2\pi x^2/p$, we find that

$$\sum_{n=0}^{p}{}' \cos\frac{2\pi n^2}{p} = \sum_{v\in\mathbb{Z}} \int_0^p \cos\frac{2\pi x^2}{p} e^{2\pi i v x}\,dx,$$

$$\sum_{n=0}^{p}{}' \sin\frac{2\pi n^2}{p} = \sum_{v\in\mathbb{Z}} \int_0^p \sin\frac{2\pi x^2}{p} e^{2\pi i v x}\,dx,$$

so that, halving the values of the end terms of the sums, we find that

$$\begin{aligned} G = \sum_{0\le n<p} e^{2\pi i n^2/p} &= \sum_{v\in\mathbb{Z}} \int_0^p e^{2\pi i v x + 2\pi i x^2/p}\,dx \qquad (x \mapsto px) \\ &= p\sum_{v\in\mathbb{Z}} \int_0^1 e^{2\pi i p(x^2+vx)}\,dx \qquad (x+\tfrac{1}{2}v = y) \\ &= p\sum_{v\in\mathbb{Z}} e^{-\frac{1}{2}\pi i p v^2} \int_{\frac{1}{2}v}^{1+\frac{1}{2}v} e^{2\pi i p y^2}\,dy. \end{aligned}$$

The value of the coefficient of the integral depends on the parity of v; more precisely we find that

$$e^{-\frac{1}{2}\pi i p v^2} = \begin{cases} 1, & \text{if } n = 2u \text{ is even,} \\ e^{-\pi i p/2} = i^{-p}, & \text{if } n = 2u-1 \text{ is odd.} \end{cases}$$

Thus

$$G = p\sum_{u\in\mathbb{Z}} \int_u^{u+1} e^{2\pi i p y^2}\,dy + p i^{-p} \sum_{u\in\mathbb{Z}} \int_{u-\frac{1}{2}}^{u+\frac{1}{2}} e^{2\pi i p y^2}\,dy,$$

and

$$\sum_{u\in\mathbb{Z}} \int_u^{u+1} e^{2\pi i p y^2}\,dy = \sum_{u\in\mathbb{Z}} \int_{u-\frac{1}{2}}^{u+\frac{1}{2}} e^{2\pi i p y^2}\,dy = \int_{-\infty}^{\infty} e^{2\pi i p y^2}\,dy.$$

With the substitution $\sqrt{p}y = u$, the value of the last integral is $C/\sqrt{p}$, where C is a constant, so that $G = \sqrt{p}(1 + i^{-p})C$. The value of C can now be determined by some specific choice of p, and since the above analysis applies equally well to

any positive integer p which need not be a prime, we may set $p = 1$, upon which $G = 1$, delivering $C = (1 + i^{-1})^{-1}$, and thus

$$G(p) = \frac{1 + i^{-p}}{1 + i^{-1}} \sqrt{p}.$$

The value of i^n is periodic in n with period length 4, so that the coefficient of $\sqrt{p}$ depends only on the residue class (mod 4) to which p belongs; in particular,

$$G(p) = \begin{cases} \sqrt{p}, & \text{for } p \equiv 1 \pmod 4, \\ i\sqrt{p}, & \text{for } p \equiv 3 \pmod 4; \end{cases}$$

there is also $G(n) = 0$ if $n \equiv 2 \pmod 4$, and $G(n) = (1 + i)\sqrt{n}$ if $4|n$.

Mellin Transform

Suppose that $f(x)$ is continuous, has a finite number of maxima and minima in every finite interval, and that $f(x)x^{\sigma-1}$ is integrable over $0 < x < \infty$. Its Mellin transform is defined by

$$F(s) = \int_0^\infty f(x)\, x^{s-1} dx, \qquad s = \sigma + it, \quad \sigma, t \in \mathbb{R}.$$

The function $f(x)$ can be recovered in terms of $F(s)$ by Mellin's inversion formula

$$f(x) = \frac{1}{2\pi i} \lim_{T\to\infty} \int_{\sigma - iT}^{\sigma + iT} F(s) x^{-s} ds.$$

The conditions on $f(x)$ can be much relaxed; for example, at points of discontinuity of $f(x)$, the formula still holds by making the usual adjustments.

Perron's Formula

A useful method in analytic number theory enables us to represent the sum function of an arithmetic function as a contour integral involving its generating function. For generating functions with Dirichlet series, we have *Perron's formula*: If $A(s) = \sum_{n\geq 1} a(n)/n^s$ is absolutely convergent in $\sigma \geq c > 0$, then

$$\sum_{n\leq x} a(n) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{A(s)x^s}{s} ds;$$

here, if x is an integer then the last term in the sum should be taken as $\frac{1}{2}a(n)$ instead of $a(n)$. The formula is an example of an inverse Mellin transform.

By itself, Perron's formula is of not much use. In practice, we only want an estimate for the sum function when x is large, so we set $T = T(x) \to \infty$ as $x \to \infty$, and replace the infinite line of integration by the finite segment $\Gamma = \{c-iT, c+iT\}$, and then choose T large enough to ensure that the incurred error is smaller than the expected dominating terms from the sum.

To see that the formula is valid, we note that, since the series for $A(s)$ converges absolutely, we may perform term-by-term integration and apply the formula:

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{X^s}{s}\,ds = \begin{cases} 0, & 0 \le X < 1, \\ \frac{1}{2}, & X = 1, \\ 1, & X > 1, \end{cases}$$

which can be established by the residue theorem. We then have

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{A(s)x^s}{s}ds = \sum_{n\ge 1}\frac{a(n)}{2\pi i}\int_{c-i\infty}^{c+i\infty}\left(\frac{x}{n}\right)^s\frac{ds}{s} = \sum_{n\le x}a(n),$$

although the interchange of limits require justification.

9.8 Summary and Notes

Summary

- Chebyshev established that $x/\log x \ll \pi(x) \ll x/\log x$. The simple proof, based on the idea from Erdős of examining the factorisation of the binomial coefficient $\binom{2n}{n}$, is set out here. Chebyshev also showed that if the ratio $\pi(x)\Big/(x/\log x)$ has a limit then the value of the limit is 1.
- Asymptotic formulae for various prime number sums and products can be derived from Chebyshev's theorem, as shown by Mertens.
- The very useful *Euler summation formula* gives an exact relationship between the sum $\sum_{x\le n\le y} f(n)$ and the integral $\int_x^y f(t)\,dt$. As a by-product we find that the partial sums of the harmonic series satisfy: $1+\frac{1}{2}+\cdots+\frac{1}{n} = \log n+\gamma+O(1/n)$, as $n \to \infty$, where $\gamma \approx 0{\cdot}5772$ is *Euler's constant*. There is also the asymptotic formula $n! \sim \sqrt{2\pi}n^{n+\frac{1}{2}}e^{-n}$, and Euler found that $\sum_{n\ge 1} 1/n^2 = \pi^2/6$.
- Dirichlet gave a construction of his completely multiplicative periodic functions with period q called *Dirichlet characters* $\chi \pmod q$, a linear combination of which then isolates the terms of an arithmetic progression $n \equiv a \pmod q$.

- Euler discovered the identity (9.3) relating a sum over the natural numbers to a product over all the primes. The method extends to sums of multiplicative functions, and completely multiplicative functions, so that such sums can also be represented by *Euler products*.
- A series $\sum_{n\geq 1} f(n)/n^s$, where $f(n)$ is an arithmetic function, is called a *Dirichlet series*. Following a tradition in analytic number theory, set by Riemann, $s = \sigma + it$ is a complex number with real and imaginary parts σ and t. It can be shown that the series converges in $\sigma > \sigma_0$ for some σ_0, so that its sum function is analytic in the region concerned. The simplest case is when $f(n) = 1$ for all n, and the resulting function $\zeta(s)$ is called the *Riemann zeta-function*. The series converges in $\sigma > 1$, and $\zeta(s)$ then also has an Euler product representation.
- Let $\chi(n)$ be a Dirichlet character (mod q). The Dirichlet L-function is defined by

$$L(s,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \Big(1 - \frac{\chi(p)}{p^s}\Big)^{-1}, \quad \sigma > 1;$$

  if χ_0 is the principal character then $L(s,\chi_0) = \zeta(s)\prod_{p|q}(1-p^{-s})^{-1}$ in $\sigma > 1$.
- The Riemann zeta-function $\zeta(s)$ has an analytic continuation into the whole complex plane $\mathbb{C}$, with a simple pole having a residue 1 at $s = 1$, via a *functional equation* which is not considered here. For our purpose there is a simple continuation into the region $\sigma > 0$ using partial summation, or integration by parts. There is then also a similar continuation for the Dirichlet L-function $L(s,\chi)$.
- There are three very useful tools in analytic number theory: *Poisson summation formula, Mellin transform*, and *Perron's formula*. For example, the Poisson summation formula can be used to give a rather natural derivation of the value of the quadratic Gauss sum. Mellin transform can be used in the proof of the prime number theorem, and Perron's formula can be used to give flexible approximations to the partial sums of the sum function of a Dirichlet series.

Notes

1. The very recent elementary proof of the prime number theorem, which involves the Liouville function $\lambda(n)$ and one of Landau's theorem, by F. K. Richter [81] can be considered relatively transparent.
2. The use of the prime factorisation of the binomial coefficient $\binom{2n}{n}$ was first introduced by the nineteen year old P. Erdős in his proof of Bertrand's postulate.

3. It can be shown that the values of the constants B_0 and B_1 in Theorem 9.3 are given by

$$B_0 = \gamma + \sum_p \left(\log\left(1 - \frac{1}{p}\right) + \frac{1}{p} \right) \quad \text{and} \quad B_1 = e^{-\gamma},$$

where γ is Euler's constant, and p denotes a prime. See [42, Theorems 428 and 429].

3. The solution to Exercise 9.1 is closer to Chebyshev's own proof of his theorem.
5. Although the series on the right-hand side of (9.8) converges in $\sigma > 0$, the convergence is delicate when s is near 0, and the series diverges at $s = 0$. See Exercise 15 below.
6. The two evaluations of the Gauss sum G given in Sects. 4.5 and in Sect. 9.9 illustrate the interesting situation in number theory concerned with a reasonably deep result: A proof by elementary means often lacks motivation and requires some ingenuity, whereas, once the required machinery for some advanced theory is in place, the delivery of the result becomes transparent from the beginning.

9.9 Exercises

1. Show that

$$\sum_{k=1}^{\infty} \psi\left(\frac{x}{k}\right) = T(x), \quad \text{and} \quad \psi(x) - \psi\left(\frac{x}{6}\right) \le T^\star(x) \le \psi(x),$$

where

$$T(x) = \sum_{n \le x} \log n, \quad \text{and} \quad T^\star(x) = T(x) - T\left(\frac{x}{2}\right) - T\left(\frac{x}{3}\right) - T\left(\frac{x}{5}\right) + T\left(\frac{x}{30}\right).$$

Deduce that

$$\kappa x + O(\log x) \le \psi(x) \le \frac{6}{5}\kappa x + O(\log^2 x), \qquad x \to \infty,$$

where

$$\kappa = \frac{\log 2}{2} + \frac{\log 3}{3} + \frac{\log 5}{5} - \frac{\log 30}{30} = 0{\cdot}921292\ldots .$$

2. Let $p(n)$, and $P(n)$, denote the least, and the greatest, prime divisor of $n > 1$, respectively. Prove that

$$\sum_{n>1} \frac{1}{n\,p(n)} \quad \text{is divergent, and} \quad \sum_{n>1} \frac{1}{n\,P(n)} \quad \text{is convergent.}$$

3. Show that the set of fractions $\phi(n)/n$ is dense in the interval $(0, 1)$.
4. Find all positive integers n for which $\phi(n)|n$.
5. Show that, corresponding to each k, there exists n such that $\phi(n + k) = \phi(n)$.
6. Let $g = p^a q^b r^c$, where p, q, r are distinct primes, and $a, b, c > 0$. Construct n such that $\gcd(n, \phi(n)) = g$.
7. Use integration by parts to derive a reduction formula for $\int_0^{\pi/2} \cos^n t\, dt$, and apply the result to prove Wallis' formula

$$\lim_{n\to\infty} \frac{2{\cdot}2{\cdot}4{\cdot}4\cdots(2n){\cdot}(2n)}{1{\cdot}3{\cdot}3{\cdot}5{\cdot}5\cdots(2n-1){\cdot}(2n+1)} = \frac{\pi}{2}.$$

8. Determine the factorisation of $\binom{100}{50}$, and compare its logarithm with that from Stirling's formula.
9. Build the table of characters (mod 15).
10. Give the remaining derivations of the functions with the given Dirichlet series in Example 6.4.
11. Show that

$$\zeta\Big(\frac{1}{2}\Big) = \frac{1}{1-\sqrt{2}} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}},$$

and deduce that $\zeta(s)$ has no zero in $0 < s < 1$.

12. Show that

$$\sum_{n=2}^{\infty} \frac{\zeta(n) - 1}{n} = 1 - \gamma, \qquad \gamma = \text{Euler's constant.}$$

13. Use the factorisation $1 + x^4 = (1 + \sqrt{2}x + x^2)(1 - \sqrt{2}x + x^2)$ to evaluate

$$\int_0^1 \frac{1 + x^2}{1 + x^4}\, dx \qquad \text{and} \qquad \int_0^1 \frac{1 - x^2}{1 + x^4}\, dx.$$

14. Use Corollary 9.2 to evaluate $\zeta(s)$ for $1 < s < 4$, and plot its graph.
15. You are given the (correct) information that $\zeta(0) = -\frac{1}{2}$. Explain why it is hopeless trying to show that $\zeta(s) \to -\frac{1}{2}$ as $s \to 0_+$ using the series in (9.8). How then should $\zeta(s)$ be evaluated when s is small?

Chapter 10
Arithmetic Functions II



Abstract The deeper problems associated with some arithmetic functions, such as the gap problem of squarefree numbers, are consider. Such items include also moment sums, and the notion of normal order. The Hardy–Ramanujan theorem on the normal order of the prime divisor function is given, and there is a proof of Segal's that Euler's totient function does not have a normal order. There are descriptions of highly composite numbers and sparsely totient numbers.

We revisit the arithmetic functions introduced in Chap. 2, and derive some deeper results for such functions. Generally speaking, because factorisation is difficult, there is no known efficient method that can be used to evaluate the values of a multiplicative function $f(n)$; this applies even to the Möbius function $\mu(n)$, or the Liouville function, the range of which is only $\{0, \pm 1\}$, or $\{\pm 1\}$, respectively. Indeed, almost as a rule, it is extremely difficult to determine the parity of the number of prime divisors of a given number, even when averaged over an interval.

10.1 Moment Sums

There are situations in which the average value of the numbers in a set $\mathcal{S}$ is not a good description of the size of its members. Take an artificial example: Suppose that 50 members of $\mathcal{S}$ lie in the interval $70 \le n \le 80$, and the 4 remaining members in $3000 \le n \le 3050$. The average value is then some number between 287 and 300, a not particularly useful piece of information concerning $\mathcal{S}$. That most of the numbers in $\mathcal{S}$ are less than 100, but a few of them exceed 1000, is a much more helpful remark.

From Theorem 2.11, we know that the average value of $d(n)$ over an interval of $x \le n < x + h$ is $\log x$, provided that $h > \sqrt{x}$. It turns out that, for most numbers n near a large x, the value of $d(n)$ is substantially smaller than the average value $\log x$. As we shall see in Sect. 10.7, $d(n) < (\log n)^{0.7}$ for most n, a bound significantly smaller than $\log n$. How then does $d(n)$ manage to have the larger $\log n$ as its average

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value? The answer is similar to that for the artificial example in the above: the average value has been jacked-up by the presence of a few n with exceptionally large $d(n)$. More specifically, for any K however large, by Theorem 2.2 there exists n with $d(n) > (\log n)^K$.

Another way of detecting the presence of large $d(n)$ is to consider higher moment sums; this is because, for $k > 1$, the value of $d^k(n) = (d(n))^k$ will be accentuated when $d(n)$ is unusually large. Thus, if $d(n)$ were near the average value $\log n$ for most n then the mean value of $d^k(n)$ would have the order $\log^k n$, whereas it turns out that, for $k > 1$,

$$\sum_{n\le x} d^k(n) \sim A_k x \log^K x, \qquad \text{as} \quad x \to \infty; \qquad K = 2^k - 1, \tag{10.1}$$

where A_k are positive constants depending only on k; see Note 10.1. Since $K > k$, the result is a reflection of the presence of some unusually large $d(n)$. The following is a sketch of the proof of the asymptotic formula for $k = 2$, so that $K = 3$.

Let $d_4(n)$ be the multiplicative function defined to be the number of ways of writing n as a product of 4 positive integers. Then, essentially the same method as that used in the Dirichlet divisor problem shows that

$$D_4(x) = \sum_{n\le x} d_4(n) = \frac{1}{6} x \log^3 x + O(x \log^2 x), \qquad \text{as} \quad x \to \infty.$$

Next, we have

$$d^2(n) = \sum_{ab^2=n} d_4(a)\mu(b),$$

which can be established by noting that both sides of the equation are multiplicative functions, and equality holds when n is a prime power. It then follows that

$$\sum_{n\le x} d^2(n) = \sum_{ab^2\le x} d_4(a)\mu(b) = \sum_{b\le\sqrt{x}} \mu(b) \sum_{a\le x/b^2} d_4(b) = \sum_{b\le\sqrt{x}} \mu(b) D_4\Big(\frac{x}{b^2}\Big);$$

the absolutely convergent series $\sum_{b\ge1} \mu(b)/b^2$ has the sum $6/\pi^2$, so that the asymptotic formula (10.1) holds with $A_2 = 1/\pi^2$.

10.2 Highly Composite Numbers

In 1915, S. Ramanujan [78] wrote an interesting paper on *highly composite numbers*, which he defined to be numbers in the set

$$\mathcal{H} = \{n : d(m) < d(n) \text{ for every } m < n\}.$$

From listing the values of $d(m)$, we find that

$$(h_n) = (1, 2, 4, 6, 12, 24, 36, \ldots), \qquad h_n \in \mathcal{H}.$$

The highly composite numbers can be given a pictorial interpretation: Consider flagpoles with height $d(m)$ placed at unit distance apart along a straight road, leading from east to west, with the rising sun shining from the east. The numbers h_n then correspond to the flagpoles which receive sunlight; see Fig. 6.1.

The sequence (h_n) increases rather rapidly, so that a very long listing for $d(m)$ will be required to deliver h_n even for a modest size n. In [78], Ramanujan offered the much better method of evaluating each h_n from the value of its predecessor. He wrote that he did "not know of any method for determining consecutive highly composite numbers except by trial," and went on to explain how to restrict the number of 'trials' required to determine the next highly composite number.

In Table 10.1 we give the first 28 highly composite numbers h_n obtained from listing $d(m)$ in $1 \le m \le 55440$; each h_n has the factorisation $2^{a_1}3^{a_2}5^{a_3}7^{a_4}11^{a_5}$. Note that $11|h_{25}, h_{28}$, but $11 \nmid h_{26}, h_{27}$.

Let p_k be the kth prime. We make the following observations from the table:

(i) Each $h \in \mathcal{H}$, with $\omega(h) = k$, takes the form

$$2^{a_1}3^{a_2}5^{a_3}\cdots p_k^{a_k}, \qquad a_1 \ge a_2 \ge a_3 \ge \cdots \ge a_k.$$

For if $p^a|h$ and $q^b|h$, with $p < q$, $a < b$ and a, b are the exact exponents for the primes, then

$$\frac{p^b q^a}{p^a q^b} = \left(\frac{p}{q}\right)^{b-a} < 1,$$

so that $h' = (p/q)^{b-1}h < h$ and $d(h') = d(h)$, which is not possible for $h \in \mathcal{H}$.

(ii) Since $d(2h_n) > d(h_n)$, it follows that $h_{n+1} \le 2h_n$. Ramanujan proved that $h_{n+1} \sim h_n$ as $n \to \infty$, so that for any fixed $\epsilon > 0$, there exists $N = N(\epsilon)$ such that $h_{n+1} < (1+\epsilon)h_n$ for $n \ge N$. However, for the purpose of computation, it is necessary to determine the value of $N(\epsilon)$ for a given ϵ, such as $\epsilon = \frac{1}{100}$.

(iii) It appears from the table that if $h_n > 48$ then $p_k \ge 5$ and $a_k = 1$. The proof makes use of only a very modest result on prime gaps established by Chebyshev.

Proof Let $h = 2^a 3^b \in \mathcal{H}$, so that $a \ge b \ge 1$, and suppose that $h > 48$. If $b = 1$ then $a \ge 5$, so that on taking $h' = 2^{a-3}3{\cdot}5 = 5h/8 < h$, we find that $d(h') = 4(a-2) \ge 2(a+1) = d(h)$, which is impossible. If $b \ge 2$ then set

Table 10.1 Highly composite numbers

n	h_n	a_1	a_2	a_3	a_4	a_5	$d(h_n)$
1	1	0	0	0	0	0	1
2	2	1	0	0	0	0	2
3	4	2	0	0	0	0	3
4	6	1	1	0	0	0	4
5	12	2	1	0	0	0	6
6	24	3	1	0	0	0	8
7	36	2	2	0	0	0	9
8	48	4	1	0	0	0	10
9	60	2	1	1	0	0	12
10	120	3	1	1	0	0	16
11	180	2	2	1	0	0	18
12	240	4	1	1	0	0	20
13	360	3	2	1	0	0	24
14	720	4	2	1	0	0	30
15	840	3	1	1	1	0	32
16	1260	2	2	1	1	0	36
17	1680	4	1	1	1	0	40
18	2520	3	2	1	1	0	48
19	5040	4	2	1	1	0	60
20	7560	3	3	1	1	0	64
21	10080	5	2	1	1	0	72
22	15120	4	3	1	1	0	80
23	20160	6	2	1	1	0	84
24	25200	4	2	2	1	0	90
25	27720	3	2	1	1	1	96
26	45360	4	4	1	1	0	100
27	50400	5	2	2	1	0	108
28	55440	4	2	1	1	1	120

$h' = 2^{a-1}3^{b-1}5 = 5h/6 < h$, with $d(h') = 2ab \geq (a+1)(b+1) = d(h)$, which is again impossible. Therefore, for $h > 48$ we need to have

$$h = 2^{a_1}3^{a_2}5^{a_3}\cdots p_i^{a_i}\cdots p_k^{a_k}, \qquad p_k \geq 5, \quad a_1 \geq a_2 \geq \cdots \geq a_i \geq \cdots \geq a_k.$$

Finally, suppose that $a_{p_k} \geq 2$. Since it is not possible to have $a_1 = a_2 = \cdots = a_k$, we may take a prime $p_i < p_k$ with $a_i > a_{i+1}$ and set

$$h' = \frac{hp_{k+1}}{p_i p_k} \leq \frac{hp_{k+1}}{2p_k} < h, \quad \text{making use of} \quad p_{k+1} < 2p_k.$$

We then find that $d(h') \geq d(h)$, which is impossible. Therefore $a_{p_k} = 1$ and, in particular, $h \in \mathcal{H}$ cannot be a square if $h > 36$.

(iv) For computing purpose, the most useful result in Ramanujan's paper is the following explicit bounds for the exponent a_i in $2^{a_1}3^{a_2}5^{a_3}\cdots p_i^{a_i}\cdots p_k^{a_k} \in \mathcal{H}$:

$$\left\lfloor \frac{\log p_k}{\log p_i} \right\rfloor \leq a_i \leq 2\left\lfloor \frac{\log p_{k+1}}{\log p_i} \right\rfloor.$$

We omit the proof, which is an elaboration of the argument set out in (iii).

Anyway we now have, in particular, $\left\lfloor \frac{\log p_k}{\log 2} \right\rfloor \leq a_1 \leq 2\left\lfloor \frac{\log p_{k+1}}{\log 2} \right\rfloor$, which reduces the search for the next $h \in \mathcal{H}$; for example, if $p_k = 7$ then $2 \leq a_1 \leq 6$, and if $p_k = 11$ then $3 \leq a_1 \leq 6$. □

Sets of Numbers Related to the Divisor Function

There are four interesting sets which are related to the divisor function:

$$\mathcal{H} = \{n : d(m) < d(n) \text{ for every } m < n\},$$

$$\mathcal{A} = \{n = p_1^{a_1}p_2^{a_2}\cdots p_k^{a_k} : a_1 \geq a_2 \geq \cdots \geq a_k,\ k \geq 0\},$$

$$\mathcal{B} = \{n = p_1^{q_1-1}p_2^{q_2-1}\cdots p_k^{q_k-1} : q_1 \geq q_2 \geq \cdots \geq q_k,\ q_i \text{ prime},\ k \geq 0\},$$

$$\mathcal{D} = \{n : d(m) \neq d(n) \text{ for every } m < n\}.$$

From Sect. 10.2, we have $\mathcal{H} \subset \mathcal{A}$, and we remark that we also have $\mathcal{H} \subset \mathcal{D}$ and $\mathcal{B} \subset \mathcal{A}$. Let $H(x)$, $A(x)$, $B(x)$, $D(x)$ denote the counting functions corresponding to these sets so that $H(x) \leq A(x)$, $H(x) \leq D(x)$ and $B(x) \leq A(x)$.

There are positive constants a, b such that, for all large x,

$$(\log x)^a < H(x) < (\log x)^b;$$

the bounds are due to Erdős (lower, 1944) and J.-L. Nicolas (upper, 1988). The sets $\mathcal{B}$ and $\mathcal{D}$ were introduced by P. Erdős and L. Mirsky [27] in order to study $D(x)$, which counts the number of distinct values taken by $d(n)$ in $1 \leq n \leq x$. Since $n = p_1^{q_1-1}p_2^{q_2-1}\cdots p_k^{q_k-1} \in \mathcal{B}$ is the unique number in $\mathcal{B}$ such that $d(n) = q_1q_2\cdots q_k$, there is a bijection between $\mathcal{D}$ and $\mathcal{B}$; from the minimal property of the members of $\mathcal{B}$, we see that $D(x) - B(x)$ counts the numbers $n \leq x$ with $n \in \mathcal{D} \setminus \mathcal{B}$.

It is difficult to determine the asymptotic formulae for these counting functions; however, Erdős and Mirsky managed to prove that

$$\log B(x) \sim \frac{2\pi\sqrt{2\log x}}{\sqrt{3}\log\log x} \qquad \text{as} \quad x \to \infty,$$

and deduce that

$$\log D(x) \sim \frac{2\pi\sqrt{2\log x}}{\sqrt{3}\log\log x} \qquad \text{as} \quad x \to \infty;$$

they also proved that $D(x) - B(x) \gg \log\log\log x$, and that $D(x) \sim B(X)$ as $x \to \infty$. These results are clearly set out in [27], in which Erdős and Mirsky also made the conjecture that there are infinitely n such that $d(n+1) = d(n)$; the conjecture was established by D. R. Heath-Brown [44].

10.3 Euler Totient Function II

Compared to the divisor function $d(n)$, Euler's totient function $\phi(n)$ behaves rather regularly; however, it does not mean that the corresponding problems are any easier. Anyway, the order of magnitude of $\phi(n)$ is 'nearly n'; more precisely, if $0 < \delta < 1$ then $\phi(n)/n^{1-\delta} \to \infty$ as $n \to \infty$. Take the case $\delta = \frac{1}{2}$, and consider the multiplicative function

$$f(n) = \frac{\phi(n)}{\sqrt{n}}.$$

At each prime power $n = p^m$, we find that

$$f(p^m) = \frac{\phi(p^m)}{p^{m/2}} = p^{m/2}\Big(1 - \frac{1}{p}\Big) \geq \frac{1}{2}p^{m/2} \to \infty \quad \text{as} \quad p^m \to \infty,$$

so that $f(n) \to \infty$ as $n \to \infty$.

On the other hand, taking $n_k = \prod_{p \leq p_k} p$, the product of the first k primes, we find, by Theorem 9.3, that

$$\frac{\phi(n_k)}{n_k} = \prod_{p \leq p_k}\Big(1 - \frac{1}{p}\Big) \to 0, \qquad k \to \infty.$$

The range of $\phi(n)$ are called the *totient numbers*. Since $\phi(n)$ is even for $n > 2$, it follows that, apart from 1, totient numbers are even, but there are infinitely many even numbers which are not in the range, the smallest one being 14.

Theorem 10.1 *We have, as $x \to \infty$,*

$$\sum_{n \leq x} \phi(n) = \frac{3x^2}{\pi^2} + O(x\log x).$$

Proof From the representation of $\phi(n)$ as a certain sum over $d|n$, we find that

$$\begin{aligned}\sum_{n\le x}\phi(n) &= \sum_{n\le x}\sum_{d|n}\mu(d)\frac{n}{d} = \sum_{ad\le x}\mu(d)a = \sum_{d\le x}\mu(d)\sum_{a\le x/d}a\\ &= \frac{1}{2}\sum_{d\le x}\mu(d)\left(\left(\frac{x}{d}\right)^2+O\left(\frac{x}{d}\right)\right) = \frac{x^2}{2}\sum_{d\le x}\frac{\mu(d)}{d^2}+O\left(x\sum_{a\le x}\frac{1}{a}\right)\\ &= \frac{x^2}{2}\sum_{d=1}^{\infty}\frac{\mu(d)}{d^2}+O\left(x^2\sum_{d>x}\frac{1}{d^2}\right)+O(x\log x)\\ &= \frac{x^2}{2}\frac{6}{\pi^2}+O(x\log x), \qquad \text{as} \quad x\to\infty,\end{aligned}$$

as required. □

Corollary 10.1 *The probability that two positive integers are coprime is* $6/\pi^2$.

Proof The sum $\sum_{n\le N} n$ can be considered to be the number of lattice points (m, n) in the triangular region $1 \le n \le m \le N$, and the sum $\sum_{n\le N}\phi(n)$ is then the number of such points with coprime coordinates. Now

$$\lim_{N\to\infty}\frac{\sum_{n\le N}\phi(n)}{\sum_{n\le N} n} = \frac{6}{\pi^2},$$

the required probability. □

We also say that the average value of $\phi(n)$ is $6/\pi^2$. In any case, we may say that $\phi(n)$ is not too often being too far way from n. Taking the special case $a = 1$ in Lemma 12.5, we have

$$\sum_{n\le x}\frac{1}{\phi(n)} = \frac{\zeta(2)\zeta(3)}{\zeta(6)}\log x + O(1), \qquad x\to\infty.$$

which is a weaker version of a theorem due to E. Landau [55]:

$$\sum_{n\le x}\frac{1}{\phi(n)} = \frac{\zeta(2)\zeta(3)}{\zeta(6)}\left(\log x+\gamma-\sum_{p}\frac{\log p}{p^2-p+1}\right)+O\left(\frac{\log x}{x}\right), \tag{10.2}$$

where γ is Euler's constant.

10.4 Sparsely Totient Numbers

In 1986, D. W. Masser and I [64] considered the problem for Euler's totient function $\phi(n)$ which is the analogue of the problem on highly composite numbers for the divisor function $d(n)$. We call members of the set

$$\mathcal{F} = \{n : n > 1, \phi(n) < \phi(m) \text{ for all } m > n\}$$

the *sparsely totient numbers*; note that $\mathcal{F}$ is an infinite set because $\phi(n) \to \infty$.

Sparsely totient numbers can also be given a pictorial interpretation: Consider each $\phi(m)$ to be the bottom of an 'upside-down' pole dangling from the sky above, set at unit distance apart along the road leading from east to west, and the setting sun is now shining from the west instead. Members of $\mathcal{F}$ then correspond to the inverted poles which receive sunlight; see Fig. 2.2 in Sect. 2.3.

Since $\phi(2m) = \phi(m)$ for m odd, it follows that every element of $\mathcal{F}$ is even. It fact, the product of the first k primes divides all large members of $\mathcal{F}$, but the factorisations of sparsely totient numbers are very different from those of the highly composite numbers. Indeed, unlike the highly composite numbers, we can construct easily arbitrarily large members of $\mathcal{F}$.

Theorem 10.2 *Let p_k denote the kth prime. Suppose that $k \geq 2$, $d \geq 1$, $\ell \geq 0$ and (i) $d < p_{k+1} - 1$, (ii) $d(p_{k+\ell} - 1) < (d+1)(p_k + 1)$. Then $dp_1 p_2 \cdots p_{k-1} p_{k+\ell} \in \mathcal{F}$.* □

Taking $d = p_k$, $\ell = 0$, conditions (i) and (ii) are satisfied, so that

$$p_1 p_2 \cdots p_{k-1} p_k^2 \in \mathcal{F} \qquad \text{for} \quad k = 2, 3, \ldots,$$

and taking $d = 1$, we find that

$$p_1 p_2 \cdots p_{k-1} p_{k+\ell} \in \mathcal{F} \qquad \text{for} \quad \ell \geq 0, \quad p_{k+\ell} < 2(p_k - 1).$$

Indeed, a more intricate argument in [64] based on the theorem shows that the ratio of large consecutive elements of $\mathcal{F}$ tends to 1, and that the counting function $F(x)$ has the lower bound $\log^2 x / \log\log x$ as $x \to \infty$. It is also proved that if P is the largest prime divisor of a large $n \in \mathcal{F}$ then $P^3 \nmid n$, and that $\log F(x) \ll (\log x)^{1/2}$ as $x \to \infty$. The arguments involve Diophantine analysis and results on prime gaps.

In the study of sparsely totient numbers there is much mileage to be gained from adopting the following strategy. If necessary replace a number n, with a divisor k, by a number $m > n$ not divisible by k, but with a chosen new divisor h. Thus, if $n \notin \mathcal{F}$ then, by definition, there exists $m > n$ such that $\phi(m) \leq \phi(n)$, and a promising idea is to construct such a suitable m by replacing a 'wasteful' factor k of n by a more amenable factor $h < k$; by this we means that h is made up of small prime divisors not dividing n. Every interval with length h contains precisely one multiple of h, and the multiple in interval $k < t \leq k + h$ is not divisible by k. So a good candidate

is the number

$$m = \frac{nh}{k}\left(\left\lfloor \frac{k}{h} \right\rfloor + 1\right) > n,$$

which is a promising choice because it is not too far away from n, and it has replaced the divisor k by a certain multiple of h. For example, we suspect that $n = 2{\cdot}3{\cdot}5{\cdot}7{\cdot}13{\cdot}31 \notin \mathcal{F}$, because the prime 11 is absent, and 31 appears relatively large to be considered wasteful. So we take $k = 31$, and $h = 11$, so that $\lfloor k/h \rfloor + 1 = 3$. Now set $m = n{\cdot}11{\cdot}3/31$, that is $m = 2{\cdot}3^2{\cdot}5{\cdot}7{\cdot}11{\cdot}13$. Then $m > n$ because $3{\cdot}11 > 31$; moreover, because $\phi(3^2 11) = 60 = \phi(3{\cdot}31)$, we deduce that $\phi(m) = \phi(n)$, so that $n \notin \mathcal{F}$. Incidentally, the idea fails when applied to $n' = 2{\cdot}3{\cdot}5{\cdot}7{\cdot}13{\cdot}29$, but then for the very good reason that $n' \in \mathcal{F}$.

For the listing of sparsely totient numbers, we may apply the lower bound:

$$\phi(n) > \frac{n}{e^{\gamma} \log\log n + 3/\log\log n}, \qquad n > 2,$$

where γ is Euler's constant. Thus, setting $n = N + h$ in the bound, with h being relatively small compare to N, we can find all the sparsely totient numbers up to N from the listing of $\phi(n)$ in $1 \le n \le N + h$. For example, taking $N = 10^6$ and $h = 6000$, so that $\phi(n) > 173533$ for $n > N + h$, and the listing of $\phi(n)$ in $1 \le n \le N + h$, all $n \in \mathcal{F}$ with $n \le 10^6$ are found; thus the largest one is $n = 930930 = 2{\cdot}3{\cdot}5{\cdot}7{\cdot}11{\cdot}13{\cdot}31$, with $\phi(n) = 172800$. The theorem can be used to check the calculations.

Carmichael's Conjecture and Menon's Identity

There are many challenging problems concerning $\phi(n)$ which can be stated easily, and we mention only two of them. First, from listing $\phi(n)$ in $1 \le n \le 1000$, we find that $\phi(n) = \phi(n + 1)$ for $n = 1, 3, 15, 104, \ldots, 975$, but it is not known if there are infinitely many such n. Next, the notorious[1] Carmichael conjecture states that, given any $n \in \mathbb{N}$, there always exists $m \ne n$ such that $\phi(m) = \phi(n)$. It has been established by K. Ford [34] that if there is a counter example then there are infinitely many of them, the least of which exceeds $10^{10^{10}}$; Ford also proved Sierpiński's conjecture that if $k \ge 2$ then there are infinitely many n such that $\phi(m) = \phi(n)$ has exactly k solutions in m.

We end the section with an interesting identity discovered by P. K. Menon relating $\phi(n)$ and $d(n)$, which has application in some arithmetical sums of functions; see, for example, [98]. We first establish a lemma.

[1] R. D. Carmichael converted what he thought was a theorem to a conjecture, after which there had been many false claims of the proof.

Lemma 10.1 *Let*

$$F(n) = \sum_{(k,n)=1} f(k,n),$$

where k runs over any reduced residue system (mod n). *Suppose that $f(k,mn) = f(k,m)f(k,n)$ for all k, and $(m,n) = 1$. Then $F(n)$ is multiplicative.*

Proof The argument used in Sect. 2.3 to show that $\phi(n)$ is multiplicative applies equally well here: Let m, n be coprime. Then

$$\begin{aligned} F(mn) &= \sum_{(k,mn)=1} f(k,mn) \\ &= \sum_{(a,m)=1} \sum_{(bm+a,n)=1} f(bm+a,mn) \\ &= \sum_{(a,m)=1} f(a,m) \sum_{(b,,n)=1} f(bm+a,n) \\ &= \sum_{(a,m)=1} f(a,m) \sum_{(b,,n)=1} f(b,n) \\ &= F(m)F(n). \end{aligned}$$

The lemma is proved. □

Theorem 10.3 (Menon's Identity) *For all $n \in \mathbb{N}$,*

$$\sum_{(k,n)=1} \gcd(k-1,n) = \phi(n)d(n).$$

Proof The function $f(k,n) = \gcd(k-1,n)$ satisfies the conditions in the lemma, so that the left-hand side $F(n)$ of the identity is a multiplicative function of n. Since $\phi(n)d(n)$ is multiplicative, it remains to verify the identity for a prime power $n = p^h$; however, the verification is not trivial. We have

$$\begin{aligned} F(p^h) &= \sum_{1 \le k < p^h,\, k \not\equiv 0 \pmod{p}} (k-1,\, p^h) = \sum_{0 \le \ell < p^h - 1,\, \ell \not\equiv -1 \pmod{p}} (\ell,\, p^h) \\ &= p^h + \sum_{1 \le \ell < p^h - 1,\, \ell \not\equiv -1 \pmod{p}} (\ell,\, p^h) = p^h + \Sigma_1 - \Sigma_2, \end{aligned}$$

where ℓ runs over $1 \le \ell < p^h - 1$ for both sums Σ_1, Σ_2, with Σ_2 having the extra condition that $\ell \equiv -1 \pmod{p}$. For Σ_2, we put $\ell = mp - 1$, so that $1 \le \ell < p^h - 1$ becomes $1 \le m < p^{h-1}$, and $(\ell, p^h) = 1$. Thus

$$p^h - \Sigma_2 = p^h - \sum_{1 \le m < p^{h-1}} 1 = p^h - p^{h-1} + 1 = \phi(p^h) + 1.$$

In Σ_1, we separate the cases $p|\ell$ and $p \nmid \ell$. For $p|\ell$, let $\ell = mp^r$, with $(m, p) = 1$, so that p^r is the exact prime power divisor of ℓ. Then, $(\ell, p^h) = p^r$, so that

$$\begin{aligned}
\Sigma_1 &= \sum_{1 \le r < h} \sum_{1 \le \ell < p^h - 1,\ p^r | \ell,\ p^{r+1} \nmid \ell} + \sum_{1 \le \ell < p^h - 1,\ (\ell, p) = 1} 1 \\
&= \sum_{1 \le r < h} p^r \sum_{1 \le m < p^{h-r},\ (m,p) = 1} 1 + \Big(\phi(p^h) - 1\Big) \\
&= \sum_{1 \le r < h} p^r \phi(p^{h-r}) + \Big(\phi(p^h) - 1\Big) \\
&= \sum_{1 \le r < h} \phi(p^h) + \Big(\phi(p^h) - 1\Big) \\
&= h\phi(p^h) - 1,
\end{aligned}$$

giving $F(p^h) = (h + 1)\phi(p^h) = d(p^h)\phi(p^h)$, as required. □

10.5 Squarefree Numbers

Although $\mu(n)$ may take only the values $0, \pm 1$, there is no known efficient method to determine which value it does take when n is large; indeed the only method seems to rely on the factorisation of n.

It is not difficult to show that a positive proportion of numbers are squarefree. Indeed, it turns out that nearly 61% of the numbers are squarefree; more precisely, we have the following theorem, of which two proofs are offered. The first one makes use of the identity $\mu^2(n) = \sum_{a^2|n} \mu(a)$, which follows easily from the fact that if m, n are coprime and $a^2|mn$ then $a = bc$ with $b^2|m$, $c^2|n$ and $(b, c) = 1$. The second one makes use of Möbius inversion.

Theorem 10.4 *We have, as $x \to \infty$,*

$$\sum_{n \le x} \mu^2(n) = \frac{6x}{\pi^2} + O(\sqrt{x}).$$

Proof We have, as $x \to \infty$,

$$\begin{aligned}\sum_{n\le x}\mu^2(n) &= \sum_{n\le x}\sum_{a^2|n}\mu(a) = \sum_{a\le\sqrt{x}}\mu(a)\sum_{n\le x/a^2}1\\ &= \sum_{a\le\sqrt{x}}\mu(a)\Big\lfloor\frac{x}{a^2}\Big\rfloor = \sum_{a\le\sqrt{x}}\mu(a)\frac{x}{a^2}+O(\sqrt{x})\\ &= x\sum_{a=1}^{\infty}\frac{\mu(a)}{a^2}+O\Big(\sum_{a>\sqrt{x}}\frac{x}{a^2}\Big)+O(\sqrt{x})\\ &= \frac{6x}{\pi^2}+O(\sqrt{x}).\end{aligned}$$

□

Second Proof Let $Q(x)$ count the squarefree numbers up to x. The number of $n \le x$ with the largest square divisor q^2 is then $Q(x/q^2)$, so that

$$\lfloor x\rfloor = \sum_{q\le\sqrt{x}} Q\Big(\frac{x}{q^2}\Big).$$

Setting $x = y^2$ and applying Möbius inversion (2.5) to $\lfloor y^2\rfloor$, the argument used in the first proof then delivers

$$Q(x) = \sum_{k\le y}\mu(k)\Big\lfloor\frac{y^2}{k^2}\Big\rfloor = \frac{6y^2}{\pi^2}+O(y) = \frac{6x}{\pi^2}+O(\sqrt{x}).$$

□

That a positive proportion of numbers are squarefree was used by P. Erdős in his simple argument giving a lower bound for the prime counting function $\pi(x)$. Let p_k denote the kth prime. Corresponding to any $x > 5$, define k by

$$p_1p_2\cdots p_k \le x < p_1p_2\cdots p_{k+1},$$

so that $\pi(x) \ge k$. Now each of the numbers $p_1^{\epsilon_1}p_2^{\epsilon_2}\cdots p_k^{\epsilon_k}$, where $\epsilon_i \in \{0,1\}$, is squarefree and is at most x, so that $2^k \ge Q(x) > x/2 > \sqrt{x}$, and hence $\pi(x) \ge k > c\log x$, with $c = 1/2\log 2$ being admissible.

Erdős' bound is much better than $\pi(x) > \log\log x$ obtained by Euclid's argument, but not a match for Chebyshev's bound $\pi(x) \gg x/\log x$. It is a challenge to find an argument which is substantially less complicated than a Chebyshev's type argument and still manages to deliver $\pi(x) > \sqrt{x}$, say.

One may expect that some half of the squarefree numbers will have an odd number of prime divisors, but the proof is as difficult as that of the prime number

theorem. On the other hand it is an easy exercise to show that, for the set of squarefree numbers not exceeding a large x, the odd and even members are in the ratio of about 2 : 1.

Example 10.1 Evaluate $Q(10^k)$ for $k = 2, 3, \ldots, 8$.

Solution We apply the formula $Q(x) = \sum_{n \le \sqrt{x}} \mu(n) \lfloor x/n^2 \rfloor$, for which a table for $\mu(n)$ with $n \le \sqrt{x}$ is required. Let $y = \lfloor \sqrt{x} \rfloor$, $z = \lfloor \sqrt{y} \rfloor$, and use the sieve of Eratosthenes to collect the primes $p \le z$, and sieve again to extend it for the primes $p \le y$. For the $\mu(n)$ table, first set the stores $\mathtt{mu} = [1] * \mathtt{y}$. Then

(i) Sieve out multiples of p^2 for $p \le z$;
(ii) Run over $p \le y$, and change the sign of each unsifted number which is a multiple of p.
(iii) Set $\mu(n) = -1$ for those prime value $n = p$ with $z < p \le y$. The values for $\mu(n)$ are now in the stores for the use in the sum for $Q(x)$.

The required computing time for $x = 10^8$ is about $\frac{1}{3}$ second. In Table 10.2, 'Main' is $\lfloor 6x/\pi^2 \rfloor$, and 'Error' is the integer part of the error term:

Gap Problems

There are now interesting problems associated with the gaps between squarefree numbers. Let q_n denote the nth squarefree number, $q_0 = 0$, and define

$$g_n = q_{n+1} - q_n, \qquad n = 0, 1, 2, \ldots,$$

and

$$G = \{g_n : g_n > g_m \text{ for each } m \text{ in } 1 \le m < n\}.$$

The first remark is that, since the asymptotic density of squarefree numbers is $6/\pi^2$, the average gaps value in the interval should be $\pi^2/6$.

Table 10.2 Values of $Q(x)$

x	y	z	$Q(x)$	Main	Error
10^2	10	3	61	60	1
10^3	31	5	608	607	1
10^4	100	10	6083	6079	4
10^5	316	17	60794	60792	2
10^6	1000	31	607926	607927	−1
10^7	3162	56	6079291	6079271	20
10^8	10000	100	60792694	60792710	−16

Example 10.2 Show that the sequence (g_n) is unbounded, and compute the first 10 members of G.

Solution Let p_k denote the kth prime. We show that, for any $K > 1$, there is a run of K consecutive numbers with the kth term being divisible by p_k^2. By the Chinese remainder theorem, the simultaneous system

$$x + k \equiv 0 \pmod{p_k^2}, \qquad k = 1, 2, \ldots, K,$$

has a unique solution $X \leq (p_1 p_2 \cdots p_K)^2 = Y_K$, so that $X + k$ is a run of K numbers which are not squarefree. Therefore $g_n > K$ for some n with $q_n < Y_k$, so that g_n is unbounded. In particular, there exists $g_n \geq 10$ for some n with

$$q_n < Y_9 = (2{\cdot}3 \cdots 23)^2 = 49770428644836900 < 5{\times}10^{16},$$

an unnecessarily large bound. However, there is little one can do to lower the general bound Y_k in any significant way, apart from the listing of q_n to evaluate (g_n).

Gaps in the set G increase rapidly, and we may need $q_n \leq 10^8$ to deliver the first 10 such gaps. This can be done by sieving the interval $1 \leq m \leq 10^8$ with p^2 for primes $p \leq 10^4$. To reduce storage space, the sieving can be done in blocks of intervals of the form $kh \leq m < kh + h$; for example, with $h = 10^5$ and $k = 0, 1, 2, \ldots, 999$.

The task is more onerous than just counting in the previous example, and the computing time is about 50-seconds. The results are given in Table 10.3.

For example, it is readily checked that 8870033 and 8870023 are squarefree, and that, for $h = 1, 2, \ldots, 9$, we find that $8870023 + h$ are divisible by p^2, where $p = 2, 5, 11, 29, 2, 7, 31, 3, 2$, respectively. Thus, $8870027 = 29^2 53{\cdot}199$ and $8870030 = 2{\cdot}5{\cdot}13{\cdot}31^2 71$, and $g_n = 10$ for a value of $q_n < 9{\times}10^6$. □

Table 10.3 Gaps between squarefree numbers

g_n	n	q_{n+1}	q_n
1	0	1	0
2	3	5	3
3	6	10	7
4	31	51	47
5	150	246	241
6	515	849	843
7	13391	22026	22019
8	131964	217077	217069
9	664313	1092755	1092746
10	5392318	8870033	8870023

Example 10.3 Use the prime number theorem to show that if $\epsilon > 0$, then there exists n such that

$$g_n > \frac{(\frac{1}{2} - \epsilon)\log n}{\log\log n}.$$

Solution For each K there exists n such that $g_n \geq K$ and $q_n \leq (\prod p)^2$, where the product is over primes $p \leq p_K$, the Kth prime. Thus

$$\log q_n \leq 2\log \prod_{p \leq p_K} p = 2\sum_{p \leq p_K} \log p = 2\theta(p_K);$$

here $\theta(z)$ is the Chebyshev prime counting function with the weight $\log p$. By the prime number theorem, we have $\theta(p_K) \sim p_K \sim K\log K$ as $K \to \infty$. Also, $\log K \sim \log\log n$ and $\log q_n \sim \log n$, so that

$$g_n \geq K \sim \frac{\theta(p_K)}{\log K} \geq \frac{\frac{1}{2}\log q_n}{\log K} \sim \frac{\frac{1}{2}\log n}{\log\log n},$$

and the required result follows. □

There is also the problem of finding reasonably good upper bounds for g_n, which amounts to finding the smallest $h = h(x)$ such that the interval $x < q \leq x + h$ contains a squarefree q, that is $Q(x + h) > Q(x)$. A trivial deduction from the subtraction of the corresponding asymptotic formulae in Theorem 10.4 shows that we may take $h = a\sqrt{x}$ for some suitable a. For, by the theorem, $Q(x) = 6x/\pi^2 + E(x)$, where $|E(x)| \leq C\sqrt{x}$, for some $C > 0$, so that

$$Q(x + a\sqrt{x}) - Q(x) \geq \frac{6a\sqrt{x}}{\pi^2} - 2C\sqrt{x} > 0, \qquad \text{if} \quad 6a > 2\pi^2 C.$$

Therefore there is a squarefree q in the interval $x < q \leq x + a\sqrt{x}$, so that $g_n \ll \sqrt{n}$.

Using still only an elementary argument, T. Estermann and K. F. Roth [84] improved the bound to $g_n \ll n^{1/3}$.

Theorem 10.5 (Estermann-Roth) *Let $1 < a < \log x$. Then, as $x \to \infty$,*

$$Q(x + ax^{1/3}) - Q(x) = \frac{6ax^{1/3}}{\pi^2} + O(x^{1/3}),$$

where the implied constant does not depend on a.

Proof Letting $h = ax^{1/3}$ and making use of $\mu^2(n) = \sum_{m\ell^2} \mu(\ell)$, we have

$$Q(x + h) - Q(x) = \sum_{x < n \leq x+h} \mu^2(n) = \sum_{x < m\ell^2 \leq x+h} \mu(\ell) = \Sigma_1 + \Sigma_2,$$

where Σ_1 and Σ denote the double sums over m, ℓ, with $\ell \le x^{1/3}$ and $\ell > x^{1/3}$, respectively. Thus

$$\begin{aligned}
\Sigma_1 &= \sum_{\ell \le x^{1/3}} \mu(\ell) \sum_{x/\ell^2 < m \le (x+h)/\ell^2} 1 \\
&= \sum_{\ell \le x^{1/3}} \mu(\ell) \left\{ \left\lfloor \frac{x+h}{\ell^2} \right\rfloor - \left\lfloor \frac{x}{\ell^2} \right\rfloor \right\} \\
&= \sum_{\ell \le x^{1/3}} \mu(\ell) \left\{ \frac{h}{\ell^2} + O(1) \right\} \\
&= \sum_{\ell=1}^{\infty} \frac{h\mu(\ell)}{\ell^2} + O\Big(\sum_{\ell > x^{1/3}} \frac{h}{\ell^2} \Big) + O(x^{1/3}) \\
&= \frac{6h}{\pi^2} + O\Big(\frac{h}{x^{1/3}} \Big) + O(x^{1/3}) \\
&= \frac{6h}{\pi^2} + O(\log x) + O(x^{1/3}),
\end{aligned}$$

and

$$|\Sigma_2| \le \sum_{\substack{x < m\ell^2 \le x+h \\ \ell > x^{1/3}}} 1.$$

The condition on m in this last sum implies that $m \le (x+h)/\ell^2 < 2x^{1/3}$; moreover, corresponding to each such m, there is at most one ℓ, because the length of the interval $\sqrt{x/m} < \ell \le \sqrt{(x+h)/m}$ is less than 1. Therefore $|\Sigma_2| \le 2x^{1/3}$, and

$$Q(x+h) - Q(x) = \frac{6h}{\pi^2} + O(x^{1/3}), \qquad h = ax^{1/3}.$$

□

Again, if a is bigger than the constant implied in the error term, then $Q(x+h) > Q(x)$, so that there must be a squarefree number in the interval $x < q \le x+h$, and hence $g_n \ll n^{1/3}$. See Note 10.3 for the current best result for this gap problem.

There cannot be a run of four consecutive squarefree numbers, because one of them is divisible by 4. However, there is no local obstruction to having 'squarefree twins and triplets'. Denote by $Q_2(x)$ and $Q_3(x)$ the counting functions for such twins and triplets; with our list of squarefree numbers, we can determine their values for $x \le 10^8$. The results are given in Table 10.4. Note that a triplet $(q-1, q, q+1)$ includes the two twins $(q, q \pm 1)$. The 14 triplets up to 100 correspond to $q \in \{2, 6, 14, 22, 30, 34, 38, 42, 58, 66, 70, 78, 86, 94\}$, and the 5 twins $(q, q+1)$ not belonging to a triplet correspond to $q \in \{10, 46, 61, 73, 82\}$. It is clear from the

Table 10.4 Squarefree twins and triplets

x	$Q(x)$	$Q_2(x)$	$Q_3(x)$
10^2	61	33	14
10^3	608	323	127
10^4	6083	3230	1260
10^5	60794	32269	12553
10^6	607926	322619	125476
10^7	6079291	3226343	1254869
10^8	60792694	32263377	12548680

counts that the set of twins and triplets possess asymptotic densities. The interested reader may find out more about such numbers in [66].

Mertens Function

The Mertens function is defined by

$$M(x) = \sum_{n \le x} \mu(n).$$

Again, that $M(x)/x \to 0$ as $x \to \infty$ is 'equivalent' to the prime number theorem; see [5, Theorem 4.14–15]. Indeed the order of magnitude of $M(x)$ is related to that of the error term in the stronger form of the prime number theorem. In 1897, Mertens conjectured that $|M(x)| < \sqrt{x}$ for $x > 1$. The conjecture was verified for $x \le 10^{10}$ before it was disproved by A. Odlyzko and H. te Reile [71] in 1985. Their indirect treatment involves a large amount of computation, and it shows that there is a counter example $x_0 < 3.21 \times 10^{64}$, and that there are x_1, x_2 such that

$$M(x_1) > 1.06\sqrt{x_1} \qquad \text{and} \quad M(x_2) < -1.009\sqrt{x_2}.$$

Liouville Function

For the Liouville function $\lambda(n) = (-1)^{\Omega(n)}$, where $\Omega(n)$ counts the prime divisors of n, including repeated ones, the behaviour of the sum function $\sum_{n \le N} \lambda(n)$ is also tricky. One would expect that, for about half of the numbers $n \le N$, the value of $\Omega(n)$ should be even, and odd for the remaining half; again we need to quantify this, perhaps to mean that

$$\frac{1}{N} \sum_{1 \le n \le N} \lambda(n) \to 0 \qquad \text{as} \quad N \to \infty. \tag{10.3}$$

This turn out to be true, but it has been established by E. Landau that the statement is elementarily equivalent to the prime number theorem. See also Note 9.1 in Chap. 9.

10.6 Prime Divisor Functions

An arithmetic function $f(n)$ is said to be *additive* if

$$f(mn) = f(m) + f(n), \qquad \text{for coprime numbers } m, n.$$

The prime divisor functions $\omega(n)$ and $\Omega(n)$ introduced in Sect. 1.2 are additive; their values are also difficult to determine without factorisation, but their behaviour is reasonably regular. First, we have

$$\omega(p) = \Omega(p) = 1 \quad \text{and} \quad \Omega(n) = \frac{\log n}{\log 2}, \quad \text{for a prime } p, \text{ and } n = 2^k.$$

The value for $\omega(n)$ can also be large; thus if n is the product of the first k primes, then $\omega(n) = k$, which, by Chebyshev's theorem, is about $\log n / \log\log n$. We show that the values of $\omega(n)$ and $\Omega(n)$ are, on average, $\log\log n$. In the next section, we show that they are also 'usually about' their common average value.

Theorem 10.6 *There are positive constants B_0 and B_2 such that, as $x \to \infty$,*

$$\sum_{n\le x} \omega(n) = x \log\log x + B_0 x + o(x),$$

$$\sum_{n\le x} \Omega(n) = x \log\log x + B_2 x + o(x).$$

Proof Using p to denote a prime, we have, as $x \to \infty$,

$$\sum_{n\le x} \omega(n) = \sum_{n\le x} \sum_{p\mid n} 1 = \sum_{p\le x} \left\lfloor \frac{x}{p} \right\rfloor = \sum_{n\le x} \frac{x}{p} + o(\pi(x)) = x \sum_{p\le x} \frac{1}{p} + o(x),$$

and the required result follows from the asymptotic formula for $\sum_{p\le x} 1/p$.

Also,

$$\sum_{n\le x} \Omega(n) - \sum_{n\le x} \omega(n) = \sum_{n\le x} \sum_{p^a\mid n, a\ge 2} 1 = \sum_{p^a\le x, a\ge 2} \left\lfloor \frac{x}{p^a} \right\rfloor,$$

the last double sum being over $p \le x^{1/a}$ and $2 \le a \le \log x/\log 2$. Now

$$\sum_{2\le a\le \log x/\log 2}\ \sum_{p\le x^{1/a}} 1 \le \sum_{2\le a\le \log x/\log 2}\ \sum_{p\le x^{1/2}} 1 \le \frac{\log x}{\log 2}\cdot \pi(\sqrt{x}) \ll \sqrt{x},$$

and the series

$$\sum_{a=2}^{\infty}\sum_{p}\frac{1}{p^a} = \sum_{p}\sum_{a=2}^{\infty}\frac{1}{p^a} = \sum_{p}\frac{1}{p(p-1)}$$

converges to the sum c, say, so that

$$\sum_{p^a\le x, a\ge 2}\left\lfloor \frac{x}{p^a}\right\rfloor = \sum_{p^a\le x, a\ge 2}\frac{x}{p^a} + O(\sqrt{x}) = cx + o(x) + O(\sqrt{x}).$$

The required result follows with $B_2 = B_0 + c$. □

10.7 Normal Order

The *normal order* of an arithmetic function measures the 'usual size' of the function: A function $\psi(n) \ge 0$ is said to have a normal order $f(n)$ if, to every $\epsilon > 0$, the number of $n \le x$ for which $|\psi(n) - f(n)| > \epsilon f(n)$ is $o(x)$, as $x \to \infty$. It is assumed that $f(n)$ is increasing—otherwise, every such $\psi(n)$ has itself as normal order.

The notion was first introduced by G. H. Hardy and S. Ramanujan [40], who proved that $\omega(n)$ and $\Omega(n)$ have the normal order $\log\log n$. Their proof was much simplified by P. Turán [100], who applied the asymptotic formulae for the second moment sum of $\omega(n)$ to establish the theorem; it is sometimes said that probabilistic number theory stems from such an argument.

Theorem 10.7 (Hardy-Ramanujan) *Let $g(n)$ be $\omega(n)$ or $\Omega(n)$. Then, for every $\epsilon > 0$, the number of $n \le x$ for which*

$$|g(n) - \log\log n| > (\log\log n)^{\frac{1}{2}+\epsilon},$$

is $o(x)$ as $x \to \infty$, so that the normal order of $\omega(n)$ and $\Omega(n)$ is $\log\log n$.

Proof (Turán) Since $\log\log x - 1 < \log\log n \le \log\log x$ for $x^{1/e} < n \le x$, and the number of $n \le x^{1/e}$ is cerainly $o(x)$, it suffices to prove that the number of $n \le x$ satisfying

$$|g(n) - \log\log x| > (\log\log x)^{\frac{1}{2}+\epsilon}$$

is $o(x)$. Also, $\Omega(n) \geq \omega(n)$ and $\sum_{n\leq x}\big(\Omega(n)-\omega(n)\big) \ll x$, so that the number of $n \leq x$ satisfying $\Omega(n)-\omega(n) > \log\log x)^{\frac{1}{2}}$ is $\ll (x/(\log\log x)^{\frac{1}{2}} = o(x)$. Therefore we need only deal with the case $g(n)=\omega(n)$.

Consider a pair p, q of distinct prime divisors of n (p, q and q, p are treated as two distinct pairs). Each p may take $\omega(n)$ values and, for each fixed p, the prime q may take $\omega(n)-1$ values. Therefore we have

$$\omega(n)\big(\omega(n)-1\big) = \sum_{pq|n,\, p\neq q} 1 = \sum_{pq|n} 1 - \sum_{p^2|n} 1.$$

Summing over $1 \leq n \leq x$, we have

$$\sum_{n\leq x}\omega^2(n) - \sum_{n\leq x}\omega(n) = \sum_{n\leq x}\Big(\sum_{pq|n} 1 - \sum_{p^2|n} 1\Big) = \sum_{pq\leq x}\Big\lfloor \frac{x}{pq}\Big\rfloor - \sum_{p^2\leq x}\Big\lfloor\frac{x}{p^2}\Big\rfloor.$$

Now

$$\sum_{p^2\leq x}\Big\lfloor\frac{x}{p^2}\Big\rfloor \leq \sum_{p^2\leq x}\frac{x}{p^2} \leq x\sum_{p}\frac{1}{p^2} \ll x$$

and

$$\sum_{pq\leq x}\Big\lfloor\frac{x}{pq}\Big\rfloor = x\sum_{pq\leq x}\frac{1}{pq} + O\Big(\sum_{pq\leq x} 1\Big),$$

where the last error term is $O(x)$ because, by Chebyshev's theorem,

$$\sum_{pq\leq x} 1 = \sum_{p\leq x}\pi\Big(\frac{x}{p}\Big) = \sum_{p\leq x/2}\pi\Big(\frac{x}{p}\Big) \leq 5x\sum_{p\leq x/2}\frac{1}{p\log x/p}$$

and

$$\sum_{p\leq\sqrt{x}}\frac{1}{p\log x/p} \leq \frac{2}{\log x}\sum_{p\leq\sqrt{x}}\frac{1}{p} = o(1),$$

$$\sum_{\sqrt{x}<p\leq x/2}\frac{1}{p\log x/p} \ll \sum_{\sqrt{x}<p\leq x/2}\frac{1}{p} \ll 1.$$

It follows that

$$\sum_{n\leq x}\omega^2(n) = x\sum_{pq\leq x}\frac{1}{pq} + O(x\log\log x).$$

Next, if $p, q \le \sqrt{x}$ then $pq \le x$, and if $pq \le x$ then $p, q < x$, so that

$$\Big(\sum_{p\le\sqrt{x}} \frac{1}{p}\Big)^2 \le \sum_{pq\le x} \frac{1}{pq} \le \Big(\sum_{p\le x} \frac{1}{p}\Big)^2,$$

and since the two outside terms are each $(\log\log x + O(1))^2$, it follows that

$$\sum_{n\le x} \omega^2(n) = x(\log\log x)^2 + O(x\log\log x).$$

Consequently

$$\begin{aligned}\sum_{n\le x}(\omega(n) - \log\log x)^2 &= \sum_{n\le x}\omega^2(n) - 2\log\log x \sum_{n\le x}\omega(n) + \lfloor x\rfloor(\log\log x)^2\\ &= x(\log\log x)^2 + O(x\log\log x)\\ &\qquad - 2(\log\log x)\Big(x\log\log x + O(x)\Big)\\ &\qquad + x(\log\log x)^2 + O((\log\log x)^2)\\ &= O(x\log\log x).\end{aligned}$$

Now if, for any $\delta > 0$, there are more than δx integers $n \le x$ satisfying the displayed inequality in the theorem then

$$\sum_{n\le x}(\omega(n) - \log\log x)^2 \ge \delta x(\log\log x)^{1+2\epsilon},$$

which is at variance with the bound for the sum. Therefore the number of $n \le x$ satisfying the displayed inequality in the theorem is $o(x)$, as $x \to \infty$ □

Corollary 10.2 *The function* $\log d(n)$ *has the normal order* $\log 2 \log\log n$.

Proof From $2^{\omega(n)} \le d(n) \le 2^{\Omega(n)}$ we havet $\omega(n)\log 2 \le \log d(n) \le \Omega(n)\log 2$, and hence the required result. □

This means that $d(n)$ is 'usually' about $2^{\log\log n} = (\log n)^{\log 2} = (\log n)^{0.691\cdots}$. However, this is **not** the normal order of $d(n)$, because there is too much fluctuation in $d(n)$ for it to have such an order.

No Normal Orders for $\phi(n)$ and $d(n)$

By applying Turán's method 'in reverse', so to speak, S. L. Segal [86] showed that Euler's totient function $\phi(n)$ does not have a normal order. We distil his argument, thereby extending his result to a certain class of arithmetic functions.

Let $\mathcal{M}$ denote the class of arithmetic functions ψ for which there are positive constants A, B, C such that $0 \le \psi(n) < Cn$ and, as $x \to \infty$,

$$\sum_{n\le x} \psi(n) \sim \frac{Ax^2}{2} \qquad \text{and} \qquad \sum_{n\le x} \psi^2(n) \sim \frac{Bx^3}{3}. \tag{10.4}$$

Theorem 10.8 *Let $\psi \in \mathcal{M}$. If $A^2 < B$ then ψ does not have a normal order.*

Proof Let A, B, C be constants associated with $\psi \in \mathcal{M}$, and set

$$R(x) = \sum_{n\le x} \Big(\psi(n) - An\Big) = o(x^2), \qquad \text{as} \quad x \to \infty. \tag{10.5}$$

Suppose that $\psi(n)$ has the normal order $f(n)$; we may assume without loss that $f(n) < 2Cn$, so that $|\psi(n) - f(n)| \le \max\{\psi(n), f(n)\} < 2Cn$. Making use of $\psi \in \mathcal{M}$, and $f(n)$ being increasing, we find, by partial summation, that

$$\Big|\sum_{n\le x} \Big(\psi(n) - An\Big) f(n)\Big| \le \max_{n\le x} |R(n)| \Big\{ \sum_{n\le x-1} \Big(f(n+1) - f(n)\Big) + f(x)\Big\}$$
$$= o(x^3) \quad \text{as} \quad x \to \infty.$$

Let $\epsilon > 0$. Appealing to the definition of normal order and separating terms depending on whether $|\psi(n) - f(n)| < \epsilon f(n)$, or not, we then have, as $x \to \infty$,

$$\sum_{n\le x} (\psi(n) - f(n))^2 \le 4\epsilon^2 C^2 \sum_{n\le x} n^2 + 4C^2x^2 o(x) = \frac{4\epsilon^2 C^2 x^3}{3} + o(x^3). \tag{10.6}$$

Making use of

$$\psi^2(n) = A^2n^2 + (\psi(n) - f(n))^2 + 2(\psi(n) - An) f(n) - (f(n) - An)^2$$
$$\le A^2n^2 + (\psi(n) - f(n))^2 + 2(\psi(n) - An) f(n),$$

we now have, on summing over $n \le x$,

$$\frac{Bx^3}{3} + o(x^3) \le \frac{A^2x^3}{3} + \frac{4\epsilon^2 C^2 x^3}{3} + o(x^3).$$

If $\epsilon = \epsilon(A, B, C)$ is sufficiently small, and x is large, then the inequality here is untenable for $A^2 < B$. The theorem is proved. □

Theorem 10.9 (Segal) *Euler's function $\phi(n)$ does not have a normal order.*

We need the asymptotic formulae for the two sums $\sum_{n\le x}\phi(n)$ and $\sum_{n\le x}\phi^2(n)$ to deduce the theorem. We already have that for the former sum, and Segal found that for the latter.

Lemma 10.2 *We have, as $x\to\infty$,*

$$\sum_{n\le x}\phi^2(n)=\frac{Bx^3}{3}+O(x^2\log^2 x),\quad \textit{where}\quad B=\prod_p\Big(1-\frac{2}{p^2}+\frac{1}{p^3}\Big). \tag{10.7}$$

Proof From the functions involved being multiplicative, it can be checked that

$$\Big(\sum_{d|n}\frac{\mu(d)}{d}\Big)^2=\sum_{a|n}\frac{\mu^2(a)}{a^2}g(a),\qquad \text{where}\quad g(a)=\prod_{p|a}(1-2p). \tag{10.8}$$

Thus, as $x\to\infty$,

$$\begin{aligned}\sum_{n\le x}\phi^2(n)&=\sum_{ab\le x}a^2\mu^2(b)g(b)=\sum_{b\le x}\mu^2(b)g(b)\Big\{\frac{x^3}{3b^3}+O\Big(\frac{x^2}{b^2}\Big)\Big\}\\&=\frac{Bx^3}{3}+E_3(x)+E_4(x)\end{aligned}$$

where

$$B=\sum_{b=1}^{\infty}\frac{\mu^2(b)g(b)}{b^3}=\prod_p\Big(1-\frac{2}{p^2}+\frac{1}{p^3}\Big),$$

$$E_3(x)=O\Big(x^3\sum_{b>x}\frac{|g(b)|}{b^3}\Big),\qquad E_4(x)=O\Big(x^2\sum_{b\le x}\frac{|g(b)|}{b^2}\Big).$$

Apply the bound $|g(b)|\le\prod_{p|b}(2p)\le 2^{\omega(b)}b\le d(b)b$, where $d(n)$ is the divisor function, and consider

$$\begin{aligned}\sum_{b>x}\frac{d(b)}{b^2}&=\sum_{uv>x}\frac{1}{u^2v^2}=\sum_{u\le x}\frac{1}{u^2}\sum_{v>x/u}\frac{1}{v^2}+\sum_{u>x}\frac{1}{u^2}\sum_{v=1}^{\infty}\frac{1}{v^2}\\&=O\Big(\frac{1}{x}\sum_{u\le x}\frac{1}{u}\Big)+O\Big(\sum_{u>x}\frac{1}{u^2}\Big)=O\Big(\frac{\log x}{x}\Big),\\\sum_{b\le x}\frac{d(b)}{b}&=\sum_{uv\le x}\frac{1}{uv}=O(\log^2 x).\end{aligned}$$

Thus $E_3(x)=O(x^2\log x)$ and $E_4(x)=O(x^2\log^2 x)$, and the lemma is proved. □

Proof of Segal's Theorem We have, as $x \to \infty$,

$$\sum_{n\le x} \phi(n) = \frac{Ax^2}{2} + O(x \log x) \qquad A = \prod_p \Big(1 - \frac{1}{p^2}\Big) = \frac{6}{\pi^2}. \tag{10.9}$$

Thus $\phi \in \mathcal{M}$, and $A^2 < B$, so that $\phi(n)$ does not have a normal order. □

Turán's method is more flexible than what is used to establish the theorem for the class $\mathcal{M}$. The argument applies to any $\psi(n)$ for which the second moment sum $\sum_{n\le x} \psi^2(n)$ is substantially larger than what 'might be expected' from the bound for the first moment sum $\sum_{n\le x} \psi(n)$. For example, the divisor function $d(n)$ is not a member of $\mathcal{M}$, but

$$\sum_{n\le x} d(n) \sim x \log x \qquad \text{and} \qquad \sum_{n\le x} d^2(n) \sim \frac{x \log^3 x}{\pi^2}, \qquad \text{as} \quad x \to \infty, \tag{10.10}$$

so that the average value for $d(n)$ is $\log n$, whereas the average value for $d^2(n)$ is $\log^3 n/\pi^2$, which is significantly larger than $\log^2 n$.

Theorem 10.10 *The divisor function $d(n)$ does not have a normal order.*

Proof Suppose, if possible, that $d(n)$ has a normal order $f(n)$. Then, for large x and for at least half of the numbers $n \le 2x$, we have $d(n) > \frac{1}{2} f(n)$, so that

$$\sum_{n\le x} f(n) < 2 \sum_{n\le 2x} d(n) < 5x \log x.$$

Since $f(n)$ is increasing, it follows that $f(n) < 10 \log n$ for all large n, and hence

$$\sum_{n\le x} (d(n) - f(n)) f(n) = O(x \log^2 x), \qquad x \to \infty.$$

Appealing to the notion of normal order again, and separating the following sum into two parts, depending on whether $|d(n) - f(n)| \le f(n)$, or not, we find that

$$\begin{aligned} \sum_{n\le x} (d(n) - f(n))^2 &\le \sum_{n\le x} f^2(n) + o\Big(\sum_{n\le x} \big(d^2(n) + f^2(n)\big)\Big) \\ &\le o\Big(\sum_{n\le x} d^2(n)\Big) + O(x \log^2 x). \end{aligned}$$

Summing the identity

$$d^2(n) = (d(n) - f(n))^2 + f^2(n) + 2(d(n) - f(n)) f(n)$$

over $1 \le n \le x$, we now find that

$$\sum_{n\le x} d^2(n) = o\Big(\sum_{n\le x} d^2(n)\Big) + O(x\log^2 x),$$

which is at variance with (10.10). Therefore $d(n)$ does not have a normal order. □

10.8 Summary and Notes

Summary

- The average value of a function may not always be a good description of its 'typical' value.
- It is not easy to find what Ramanujan called *highly composite numbers*, but he has given a method of finding, from such a number, the next one.
- The probability that two positive integers are coprime is $6/\pi^2$.
- It is easy to find a large n with the property that $\phi(m) > \phi(n)$ for all $m > n$.
- Nearly 60% of the numbers are *squarefree*.
- The problem of the gaps between successive squarefree numbers involves the study of *exponential sums*.
- The problems of the average values of the Möbius function $\mu(n)$ and the Liouville function $\lambda = (-1)^{\Omega(n)}$ can be related to the prime number theorem.
- The average values of the prime divisor functions $\omega(n)$ and $\Omega(n)$ are both $\log\log n$. Moreover, they have *normal order* $\log\log n$, which means roughly that their usual size is near its average value, so that, for almost all n, their values are near $\log\log n$.
- The functions $\phi(n)$ and $d(n)$ do not have a normal order. Such, and similar, results can be established by examining their first and second moment sums, and apply Turán's method.

Notes

1. For $k \ge 2$, the constant A_k in the asymptotic formula (10.1) is given by

$$A_k = \frac{1}{(2^k-1)!}\prod_p \left(\Big(1-\frac{1}{p}\Big)^{2^k}\Big(\sum_{a\ge 0}\frac{(a+1)^k}{p^a}\Big)\right), \quad p \text{ prime}.$$

For $k = 2$, Ramanujan stated that there are constants B, C, D, such that

$$\sum_{n\le x} d^2(n) = \frac{1}{\pi^2}x\log^3 x + Bx\log^2 x + Cx\log x + Dx + \Delta(x),$$

where $\Delta(x) \ll x^\theta$ for any $\theta > \frac{3}{5}$; this was established by B. M. Wilson in 1922, with $\theta > \frac{1}{2}$. The latest (2014) estimate is $\Delta(x) \ll x^{\frac{1}{2}} \log^5 x$ by Chaohua Jia and Ayyadurai Sankaranarayanan.

2. The reader may ask: What about the set

$$\mathcal{F}^\star = \{n : n > 1, \phi(n) > \phi(m) \text{ for all } m < n\}$$

of highly totient numbers? Clearly $\mathcal{F}^\star$ contains all the primes, and it is very probable that there is no other elements of $\mathcal{F}^\star$; however, we can establish this only with the help of a suitable hypothesis on prime gaps.

3. On the gap problem for squarefree numbers, in [84] Roth used exponential sums to bound the contribution from fractional parts associated with the floor function $\lfloor x/\ell^2 \rfloor$, by which he was able to show that the bound for the gaps can be reduced n^θ, where any $\theta > 3/13 = 0{\cdot}23\ldots$ is admissible. After successive improvements by other mathematicians, the current best result [36] is that $\theta = 0{\cdot}2209$ is admissible; readers interested in the problem should read [33, 84].

10.9 Exercises

1. Determine the sum of the series

$$\sum_{n\geq 1} \frac{\mu^2(n)}{n^2}.$$

2. Show that, as $x \to \infty$,

$$\sum_{\substack{n\leq x \\ n\equiv a \pmod{Q}}} \mu^2(n) \sim \frac{x}{Q} \prod_{p\nmid Q} \Big(1 - \frac{1}{p^2}\Big), \qquad \gcd(a, Q) = 1.$$

3. Verify that, for $n \leq x$, with x of modest size, the odd and even squarefree ones are in the ratio of about 2:1. Establish the corresponding asymptotic formulae, with $x \to \infty$, for the counting functions.
4. Find, for each $a = 0, 1, 2, \ldots, 11$, the value of $C(a)$ such that

$$\sum_{\substack{n\leq x \\ n\equiv a \pmod{12}}} \mu^2(n) \sim C(a)x, \qquad x \to \infty.$$

Evaluate the 12 sums at $x = 10^8$.

5. A number n with the property that $p|n$ for a prime p implies $p^2|n$ is said to be squarefull. Show that squarefull numbers are products of squares and cubes.

Show further that the counting function for such numbers is asymptotic to $C\sqrt{x}$, where C can be expressed in terms of the Riemann-zeta function $\zeta(s)$.

6. Find the smallest n such that the interval $(n-1)^2 < s < n^2$ contains exactly 4 squarefull numbers s.
7. Given that 240940299600 is a highly composite number, find the next such number.
8. Let p_i denote the i th prime, and let n be a sparsely totient number satisfying $p_1 p_2 \cdots p_k \le n < p_1 p_2 \cdots p_{k+1}$. Show that $\omega(n)$ is either $k-1$ or k.
9. Let $P = P(n)$ and $Q = Q(n)$, where $n > 1$, denote the largest prime divisor P of n, and the smallest prime Q not dividing n. Also, write $n = R(n)\prod_{p|n} p$, so that $R(n)$ is the 'repeated prime factors' of n. Denote by $\mathcal{F}$ the set of sparsely totient numbers:

   (i) Show that 5610 is the largest $n \in \mathcal{F}$ with $Q = 7$, and find the largest $n \in \mathcal{F}$ with $Q = 11$;
   (ii) Show that $2{\cdot}3{\cdot}5{\cdot}7{\cdot}11^2{\cdot}19 \notin \mathcal{F}$.
   (iii) Show that, for $n \in \mathcal{F}$, we have $\max(P, R) < Q^2$. Deduce that $P^4 \nmid n$.

10. Verify Menon's identity for various choices of n of your's.
11. (a) Evaluate Mertens function $M(x) = \sum_{n \le x} \mu(n)$ for $1 \le x \le 10^6$.

    (i) Count the zeros, and list any run of 6 consecutive zeros;
    (ii) Find $(X_1, M(X_1))$ and $(X_2, M(X_2))$, where

$$M(X_1) = \max_{1 \le x \le 10^6} M(x) \qquad \text{and} \qquad M(X_2) = \min_{1 \le x \le 10^6} M(x).$$

    (b) Let $\lambda(n) = (-1)^{\Omega(n)}$ be Liouville's function, and $L(x) = \sum_{n \le x} \lambda(n)$.

    For the range $2 \le x \le 10^6$, verify that $L(x) \le 0$, and find its minimum value; find also the largest $x \le 10^6$ with $L(x) = 0$.

12. Verify the inequality

$$\prod_p \Big(1 - \frac{2}{p} + \frac{1}{p^3}\Big) > \frac{36}{\pi^4}$$

used in the proof of Segal's theorem that Euler's function does not have a normal order.

13. Let $f(n)$ be an arithmetic function which has a normal order, and let $F(x)$ count the number of pairs (m, n), with $1 \le m \le n \le x$, such that $100 f(n) \le 99 f(m)$.

    Show that $F(x) = o(x^2)$ as $x \to \infty$.

Chapter 11
Prime Number Theorem



Abstract Following a brief sketch of the prime number theorem, the proof is given along the lines of Newman's simplified proof of the Ikehara–Wiener tauberian theorem. There are discussions on the error-term of the prime number theorem, the zero-free region of the zeta-function, the Riemann hypothesis, and the problem of prime gaps.

For comprehension of the analytic proof of the prime number theorem, the reader will need to have some basic knowledge of analytic function theory, including analytic continuation and contour integration.

The classical proofs of the prime number theorem make use of the properties of $\zeta(s)$ in some region inside the half-plane $\sigma < 1$; see (9.7) in which $\zeta(s)$ is defined. In more recent times the requirement of such information has been reduced to its bare minimum, which is that $\zeta(s)$ does not have a zero on the vertical line $\sigma = 1$; we know that there is a simple pole at $s = 1$, of course. The following is a sketch of the some of the arguments involved.

11.1 Brief Sketch

Since $\zeta(s)$ has an Euler product over the primes, it is natural to consider the function $\log \zeta(s)$, which is well defined in $\sigma > 1$ if we take its value to be real and positive for $s > 1$. It can then be shown by reasonably elementary means that $\zeta(s)$ is connected to $\pi(x)$ by the formula

$$\log \zeta(s) = s \int_2^\infty \frac{\pi(x)}{x(x^s - 1)} dx, \qquad \sigma > 1. \tag{11.1}$$

On 'inverting' this relation, $\pi(x)$ can be recovered as an integral of $(\log \zeta(s))/s$, which is technically rather difficult to handle. The function $(\log \zeta(s))/s$ may be replaced by the logarithmic derivative of $\zeta(s)$ instead; indeed we then find the

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more amenable function $\zeta'(s)/\zeta(s)$ can be represented by a Dirichlet seires with coefficient closely related to the primes.

Recall the definition of the Mangoldt function in (2.9)

$$\Lambda(n) = \begin{cases} \log p, & \text{if } n = p^a, \ p \text{ prime}, \\ 0, & \text{otherwise}; \end{cases}$$

and Theorem 2.12 which states that $\sum_{m|n} \Lambda(m) = \log n$. Let

$$\psi(x) = \sum_{n \le x} \Lambda(n) = \sum_{p \le x} \left\lfloor \frac{\log x}{\log p} \right\rfloor \log p,$$

the prime power counting function, with each prime power p^a being given the weight $\log p$. Since prime powers are literally few and far between among the integers, it is not difficult to show that the prime number theorem is equivalent to $\psi(x) \sim x$ as $x \to \infty$, and analytic proofs of the prime number theorem can be based on the fact that $\Lambda(n)$ are the coefficients of the Dirichlet series for the negative of the logarithmic derivative of $\zeta(s)$. Thus, from $\log \zeta(s) = -\sum_p \log(1 - p^{-s})$, and the derivative of p^{-s} being $-(\log p)p^{-s}$, we find that

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_p \frac{\log p}{p^s(1 - p^{-s})} = \sum_p \sum_{m=1}^{\infty} \frac{\log p}{p^{ms}} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}, \qquad \sigma > 1. \tag{11.2}$$

This is the 'interesting analytic property' mentioned in Sect. 2.10 and, as we shall see in Theorem 11.3, it enables us to write

$$-\frac{\zeta'(s)}{\zeta(s)} = s \int_1^{\infty} \psi(x) x^{-s-1} dx, \qquad \sigma > 1,$$

the Mellin transform (see Sect. 9.9) of $\psi(x)$. Several proofs of the prime number theorem are based on the inversion of the formula to arrive at

$$\psi(x) = -\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\zeta'(s)}{\zeta(s)} \frac{x^s}{s} ds, \tag{11.3}$$

where the integral is along the vertical line $s = c > 1$, and with an appropriate adjustment if x is an integer, where $\psi(x)$ may have a jump discontinuity.

The method then involves the replacement of the line $s = c$ by a finite segment L through c, symmetrical about the real axis, with the length depending on x. Cauchy's theorem is then applied over a contour C, part of which is L, and with the point $s = 1$ inside C, which then has to include either part of the line $\sigma = 1$, or inside the region $\sigma < 1$. The pole of the integrand at $s = 1$ then delivers the main term x for $\psi(x)$, and the remaining part of the argument is to show that the contribution

of the integral from $C \setminus L$ can be absorbed into the error term, so that $\psi(x) \sim x$ as $x \to \infty$. The work involved in this last part is considerable; in particular it will require estimates for $|\zeta'(s)|$ and $1/|\zeta(s)|$ for s on C.

An analytic proof of the prime number theorem will then require the fact that $\zeta(s)$ does not have a zero on the line $\sigma = 1$. By foraging into a part of the region $\sigma < 1$ in which $\zeta(s)$ has no zeros, an error term for the prime number theorem can be obtained; see, for example, the result due to de la Vallée Poussin mentioned in Sect. 3.1.

Contributions from Landau to Ingham

In 1908 E. Landau gave a relatively simple proof of the prime number theorem based on function theory which requires the condition that $\zeta'(s)/\zeta(s) \ll |s|^k$ for some fixed k, as $|s| \to \infty$ in $\sigma \geq 1$; the polynomial bound was relaxed to an exponential bound in 1918 by G. H. Hardy and J. E. Littlewood in connection to their work on Tauberian theory (see Note 1). In 1928 N. Wiener made a useful contribution to this theory for Fourier integrals in which his student S. Ikehara later (1931) showed that the prime number theorem can be deduced, and with the said bound condition being dispensed completely. The method was simplified by A. E. Ingham (see [46]), but it still involves much of the machinery associated with Fourier integrals.

11.2 Newman's Simplification

In 1980, D. J. Newman proved a useful Tauberian theorem for Dirichlet series with bounded coefficients by simple contour integration. As a consequence, the Ikehara-Wiener theorem can be established without involving Fourier integrals, thereby giving a much simplified proof of the prime number theorem.

Theorem 11.1 (Ikehara-Wiener) *Let $F(x)$ be bounded and piecewise continuous in $x \geq 0$, so that its Laplace transform*

$$G(s) = \int_0^\infty F(x)e^{-sx}dx$$

exists and is analytic in $\sigma > 0$. Suppose that $G(s)$ has an analytic extension to a region which includes $\sigma \geq 0$. Then the improper integral $\int_0^\infty F(x)\,dx$ converges to $G(0)$.

Corollary 11.1 *Let $0 \leq f(x) \ll x$ be increasing, and piecewise continuous in $x \geq 1$, so that its Mellin transform*

$$g(s) = s \int_1^\infty f(x) x^{-s-1} dx, \qquad \sigma > 1,$$

exists, and is analytic in $\sigma > 1$. Suppose that there is a constant $c > 0$ such that

$$g(s) - \frac{c}{s-1}$$

has an analytic extension to a region which includes $\sigma \geq 0$. Then

$$f(x) \sim cx \quad as \quad x \to \infty.$$

***Proof** (**Newman**)* Each point on the vertical line $\sigma = 0$ is an interior point of the region in which $G(s)$ is analytic. By a compactness argument using, for example, the Heini-Borel theorem, corresponding to each $R > 0$, there exists $\delta = \delta(R) > 0$ such that $G(s)$ is analytic in an open region containing

$$S = \{s : s = \sigma + it,\ \sigma \geq -\delta,\ |s| \leq R\},$$

the union of a semicircle in the half-plane $\sigma \geq 0$, and a thin strip in the half-plane $\sigma \leq 0$. Let C be the positively orientated contour which forms the boundary of S, so that $G(s)$ is analytic inside and on C.

Next, let $B = \sup |F(x)|$ and, for $X > 0$, define

$$G_X(s) = \int_0^X F(x) e^{-sx} dx,$$

an entire function of s, that is a function analytic on the whole of $\mathbb{C}$. We then need to show that $G_X(0) \to G(0)$ as $X \to \infty$; it will be relevant to remark that the parameters R and X are independent of each other.

Newman's idea is to represent $G(0) - G_X(0)$ by Cauchy's integral formula over C at $s = 0$:

$$G(0) - G_X(0) = \frac{1}{2\pi i} \int_C \frac{G(s) - G_X(s)}{s} ds,$$

and to find suitable estimates for the integral along C. First, we note that

$$|G(s) - G_X(s)| = \left| \int_X^\infty F(x) e^{-sx} dx \right| \leq B \int_X^\infty e^{-\sigma x} dx = \frac{B e^{-\sigma X}}{\sigma}, \qquad \sigma > 0.$$

For the purpose of bounding the integral representation suitably, Newman introduced two mollifying factors to the integrand: the first factor is e^{sX}, which corresponds to e^{-sx} in the Laplace transform of $G(s)$, and the second factor is $(1+s^2/R^2)$ to cater for the effect of $1/s$ when σ is small. Since $e^{sX}(1+s^2/R^2)$ is an entire function taking the value 1 at $s=0$, its introduction has no material effect on the integral formula, so that

$$G(0)-G_X(0)=\frac{1}{2\pi i}\int_C \frac{G(s)-G_X(s)}{s}e^{sX}\Big(1+\frac{s^2}{R^2}\Big)\,ds.$$

Now let $C=C_+\cup C_-$, where C_+ is semicircle part of C in the half-plane $\sigma\geq 0$, and C_- is the part of C corresponding to the thin strip lying in the half-plane $\sigma\leq 0$. On C_+, we have the estimate

$$\left|\frac{e^{sX}}{s}\Big(1+\frac{s^2}{R^2}\Big)\right|=\frac{e^{\sigma T}}{R^2}\left|\frac{R^2}{s}+s\right|=\frac{e^{\sigma X}(2\sigma)}{R^2},$$

so that the contribution to $G(0)-G_X(0)$ from the integral over C_+ is bounded in absolute value by $\frac{\pi R}{2\pi}\{Be^{-\sigma X}/\sigma\}\{e^{\sigma X}(2\sigma)/R^2\}=B/R$.

For the integral over C_-, we deal with $G(s)$ and $G_X(s)$ separately. With $G_X(s)$ being an entire function, we may replace C_- by the semicircle C' in $\sigma\leq 0$, which is the reflection of C_+ along $\sigma=0$, so that, on C',

$$|G_X(s)|=\left|\int_0^X F(x)e^{-sx}dx\right|\leq B\int_{-\infty}^X e^{-\sigma x}dx\leq\frac{Be^{-\sigma X}}{-\sigma},\qquad \sigma<0,$$

and hence, again,

$$\left|\frac{1}{2\pi i}\int_{C'}\frac{G_X(s)}{s}e^{sX}\Big(1+\frac{s^2}{R^2}\Big)\,ds\right|\leq\frac{\pi R}{2\pi}\cdot\frac{Be^{-\sigma X}}{-\sigma}\times\frac{e^{\sigma X}|2\sigma|}{R^2}=\frac{B}{R}.$$

Collecting the results with the two bounds we now have

$$|G(0)-G_X(0)|\leq\frac{2B}{R}+\frac{1}{2\pi}\left|\int_{C_-}e^{sX}\cdot\frac{G(s)}{s}\Big(1+\frac{s^2}{R^2}\Big)\,ds\right|.$$

The only involvement with X in the integral is the factor e^{sX}, which tends to 0 locally uniformly in s on C_-, as $X\to\infty$; indeed, for any fixed $\sigma_0<0$, and $s=\sigma+it$ with $\sigma\leq\sigma_0$, we have

$$|e^{sX}|=e^{\sigma X}\leq e^{\sigma_0 X}\to 0,\quad \text{uniformly in} s \text{ as } X\to\infty.$$

Recalling that X and R are independent, we now have, for each $R > 0$,

$$\limsup_{X\to\infty} |G(0) - G_X(0)| \le \frac{2B}{R},$$

so that $\lim_{X\to\infty} G_X(0) = G(0)$. The theorem is proved. □

Proof of the Corollary Define, for the given $f(x)$ and c in the corollary,

$$F(x) = e^{-x} f(e^x) - c,$$

so that $F(x)$ satisfies the hypothesis of the theorem. Its Laplace transform is

$$G(s) = \int_0^\infty F(x)e^{-sx}dx = \int_0^\infty \Big(e^{-x} f(e^x) - c\Big)e^{-sx}dx,\ s = \sigma + it,\ \sigma > 0,$$

and, with the substitution $u = e^x$,

$$\begin{aligned} G(s) &= \int_1^\infty \Big(\frac{f(u)}{u} - c\Big)u^{-s}\frac{du}{u} \\ &= \int_1^\infty f(u)u^{-s-2}du - c\int_1^\infty u^{-s-1}du \\ &= \frac{g(s+1)}{s+1} - \frac{c}{s} = \frac{g(s+1) - c/s - c}{s+1}, \qquad \sigma > 0, \end{aligned}$$

where $g(s)$ is the Mellin transform of $f(x)$. By hypothesis, $g(s+1) - c/s$ has an analytic extension to a region containing the line $\sigma = 0$, so that $G(s)$ also has such an extension. Thus $F(x)$ and $G(s)$ satisfy the hypothesis of the theorem, so that the improper integral

$$\int_1^\infty \Big(\frac{f(u)}{u} - c\Big)\frac{du}{u} \quad \text{is convergent.}$$

It remains to show that the integrand tends to 0 as $u \to \infty$, for which the condition $f(x)$ being increasing is required.

Suppose that there exists $\lambda > 1$ such that $f(x) \ge \lambda cx$ for an unbounded set of numbers x. Since $f(u)$ is increasing, we have, for each such x,

$$\int_x^{\lambda x} \Big(\frac{f(u)}{u} - c\Big)\frac{du}{u} \ge \int_x^{\lambda x} \Big(\frac{\lambda cx}{u} - c\Big)\frac{du}{u} = c\int_1^\lambda \Big(\frac{\lambda}{y} - 1\Big)\frac{dy}{y},$$

a positive lower bound depending only on λ; this is impossible for the convergence of the improper integral.

Similarly, if there exists $\lambda < 1$ such that $f(x) \le \lambda cx$ for an unbounded set of x, then, for each such x,

$$\int_{\lambda x}^{x} \Big(\frac{f(u)}{u} - c\Big)\frac{du}{u} \le \int_{\lambda x}^{x} \Big(\frac{\lambda cx}{u} - c\Big)\frac{du}{u} = c\int_{\lambda}^{1} \Big(\frac{\lambda}{y} - 1\Big)\frac{dy}{y},$$

which is again impossible. Therefore

$$1 \le \liminf_{x\to\infty} \frac{f(x)}{cx} \le \limsup_{x\to\infty} \frac{f(x)}{cx} \le 1,$$

so that $f(x) \sim cx$ as $x \to \infty$. □

11.3 Proof of the Prime Number Theorem

We already remarked that all analytic proofs of the prime number theorem require the vital step that $\zeta(s) \neq 0$ on $\sigma = 1$, and indeed both Hadamard and de la Vallée Poussin gave rather elaborate proofs of the result. Later, Mertens gave a much simplified argument based on the use of the inequality

$$3 + 4\cos\theta + \cos 2\theta \ge 0,$$

which holds for all θ because the left-hand side is $2(1 + \cos\theta)^2$.

Lemma 11.1 *For $\sigma > 1$ and $t \neq 0$, we have $\zeta^3(\sigma)|\zeta(\sigma + it)|^4|\zeta(\sigma + 2it)| \ge 1$.*

Proof From

$$\log\zeta(s) = \sum_p \sum_{m=1}^{\infty} \frac{1}{mp^{ms}} = \sum_p \sum_{m=1}^{\infty} \frac{\exp(-imt\log p)}{mp^{m\sigma}}$$

we deduce that

$$|\zeta(s)| = \exp\left\{\sum_p \sum_{m=1}^{\infty} \frac{\cos(mt\log p)}{mp^{m\sigma}}\right\}.$$

In view of Mertens' cosine inequality, the application of this formula at $s = \sigma$, $s = \sigma + it$, $s = \sigma + 2it$ delivers $\log\big(\zeta^3(\sigma)|\zeta(\sigma + it)|^4|\zeta(\sigma + 2it)|\big) \ge 0$, so that the required result follows. □

Theorem 11.2 *The Riemann zeta-function $\zeta(s)$ does not have a zero on the vertical line $\sigma = 1, t \neq 0$.*

Proof Recall that $\zeta(s)$ has a simple pole at $s = 1$, so that $\zeta^3(s)$ has a triple pole there. If $\zeta(s)$ has a zero at $1 + it$, then $\zeta^4(s)$ has a zero of order at least 4 there. Since $\zeta(s)$ is bounded near $1 + 2it$, the triple pole at $s = 1$ is not enough to cancel the effect of the zero of order 4 in order to satisfy the inequality in the lemma. To put matter more clearly, we rewrite the inequality in the lemma as

$$((\sigma - 1)\zeta(\sigma))^3 \left|\frac{\zeta(\sigma + it)}{\sigma - 1}\right|^4 |\zeta(\sigma + 2it)| \geq \frac{1}{\sigma - 1}, \qquad \sigma > 1,$$

which is not tenable when σ is near 1 if there were a zero at $1 + it$. □

Corollary 11.2 *The function $\zeta'(s)/\zeta(s)$ has a meromorphic extension to $\sigma > 0$; the only pole on the line $\sigma = 1$ is at $s = 1$, with residue -1.* □

Theorem 11.3 *We have*

$$-\frac{\zeta'(s)}{\zeta(s)} = s\int_1^\infty \psi(x)x^{-s-1}dx, \qquad \sigma > 1.$$

Proof We start with

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^\infty \frac{\Lambda(n)}{n^s} = \sum_{n=1}^\infty \frac{\psi(n) - \psi(n-1)}{n^s},$$

and, by partial summation,

$$\sum_{1\leq n\leq N} \frac{\psi(n) - \psi(n-1)}{n^s} = \frac{\psi(N)}{(N+1)^s} + \sum_{1\leq n\leq N} \psi(n)\Big(\frac{1}{n^s} - \frac{1}{(n+1)^s}\Big).$$

For $\sigma > 1$, we have, by Chebyshev's theorem, $\psi(N)/(N+1)^s = o(1)$, as $N \to \infty$. Also, writing $1/n^s - 1/(n+1)^s = s\int_n^{n+1} x^{-s-1}dx$, and noting that $\psi(x)$ is constant for $n \leq x < n + 1$, the last sum becomes

$$\sum_{1\leq n\leq N} s\int_n^{n+1} \psi(x)x^{-s-1}dx = s\int_1^{N+1} \psi(x)x^{-s-1}dx,$$

and the required result follows by letting $N \to \infty$. □

Corollary 11.3 (Prime Number Theorem) *As $x \to \infty$, we have*

$$\psi(x) \sim x, \qquad \textit{and} \qquad \pi(x) \sim \frac{x}{\log x}.$$

Proof We can now apply Corollary 11.1 to the Ikehara-Wiener theorem to $\psi(x)$, which is an increasing step function, satisfying $\psi(x) \ll x$ by Chebyshev's theorem;

its Mellin transform is the meromorphic function $-\zeta'(s)/\zeta(s)$, and $-\zeta'(s)/\zeta(s) - 1/(s-1)$ does have an analytic extension to a region containing the vertical line $\sigma = 1$. Therefore $\psi(x) \sim x$ as $x \to \infty$.

Elementary arguments, such as partial summation, can be used to show that the three asymptotic formulae

$$\pi(x) \sim \frac{x}{\log x}, \qquad \theta(x) \sim x, \qquad \psi(x) \sim x, \qquad \text{as} \quad x \to \infty,$$

are equivalent to each other. In particular we can deduce the asymptotic formula for $\pi(x)$ from that for $\psi(x)$. First, we have

$$\psi(x) = \sum_{p \le x} \left\lfloor \frac{\log x}{\log p} \right\rfloor \log p \le \sum_{p \le x} \log x = (\log x)\pi(x).$$

Next, for $1 < y < x$, we consider

$$\pi(x) \le y + \sum_{y < p \le x} 1 \le y + \frac{1}{\log y} \sum_{y < p \le x} \log p \le y + \frac{\psi(x)}{\log y},$$

so that

$$\psi(x) \le (\log x)\pi(x) \le y(\log x) + \frac{\log x}{\log y}\psi(x).$$

Setting $y = x/\log^2 x$, so that $\log y \sim \log x$ as $x \to \infty$, we deduce that $(\log x)\pi(x) \sim \psi(x) \sim x$, as required. □

Corollary 11.4 *Let p_k denote the kth prime. Then $p_k \sim k \log k$ as $k \to \infty$.*

Proof As $k \to \infty$, we have $p_k \sim \pi(p_k) \log p_k = k \log p_k \sim k \log k$. □

Estimates for $\prod_{p \le n} p$ and LCM(1, 2, . . . , n)

Since the exact prime exponent k of a prime p which divides $\mathrm{LCM}(1, 2, \ldots, n)$ satisfies $p^k \le n < p^{k+1}$, we have

$$\mathrm{LCM}(1, 2, \ldots, n) = \Big(\prod_{p \le n} p\Big)\Big(\prod_{p^2 \le n} p\Big)\Big(\prod_{p^3 \le n} p\Big) \cdots,$$

and hence

$$\log(\mathrm{LCM}(1, 2, \ldots, n) = \sum_{p \le n} \log p + \sum_{p^2 \le n} \log p + \cdots = \psi(n).$$

Writing

$$P_n = \prod_{p \le n} p, \quad \text{and} \quad d_n = \mathrm{LCM}(1, 2, \ldots, n),$$

we then find that $\log P_n = \theta(n) \le \psi(n) = \log d_n$. Now, by the prime number theorem, as $n \to \infty$,

$$\theta(n) \sim \psi(n) \sim n, \qquad \text{that is} \qquad \log P_n \sim \log d_n \sim n.$$

It follows that if $A < e < B$, then there exists $N = N(A, B)$, depending on A, B, such that $A^n < P_n \le d_n < B^n$ for $n \ge N$. Once A and B are specified, we should be able to compute an admissible value for N; indeed, we do not require the full force of the prime number theorem, just an enhanced Chebyshev-type argument should do, although the task will be tedious if A, B are near e. In particular, there are integers N_1, N_2 with the properties that $P_n > 2^n$ for $n \ge N_1$, and $d_n < 3^n$ for $n \ge N_2$, so that we have the rather useful bounds

$$2^n < P_n \le d_n < 3^n, \quad \text{for } n \ge \max(N_1, N_2).$$

Corollary 9.4 states that $\exp(\theta(n)) > \exp(n/6)$, which value is less than 2^n, so we need to sharpen the lower estimate in Chebyshev's theorem. It can be shown that $N_1 = 31$ is an admissible value. Readers may wish to study the nice ad hoc solution presented as Exercise 1.28 in [19]. There is also a charming proof of the related bound $d_n \ge 2^n$, for $n \ge 7$, by M. Nair [70]. Incidentally, we find that $P_n < 2^n$ only for $n = 1, 2, \ldots, 10, 12, 15, 16, 28$, and

$$2^{27} < P_{23} = P_{24} = P_{25} = P_{26} = P_{27} = P_{28} < 2^{28}$$

whereas

$$P_{29} = P_{30} = 6\,469\,693\,230 > 1\,073\,741\,824 = 2^{30} > 2^{29},$$

so that $N_1 = 29$ is the smallest admissible value.

In applications of $d_n < 3^n$, such as that in Beukers' proof of Apéry's theorem on the irrationality of $\zeta(3)$ in Sect. 8.7, we only require that n is large. Computation shows that the bound holds for $1 \le n \le 100$, so that it is possible that $N_2 = 1$ is an admissible value.

11.4 Error Term of the Prime Number Theorem

Calculations of $\pi(x)$ made by Legendre led him to suggest that there should be a constant A such that, for large x,

$$\pi(x) \approx \frac{x}{\log x - A},$$

and he even tentatively offered that perhaps $A = 1{\cdot}08\ldots$. Since

$$\frac{x}{\log x - A} = \frac{x}{\log x} + \frac{Ax}{\log x(\log x - A)},$$

his suggestion amounts to the conjecture that

$$\pi(x) = \frac{x}{\log x} + \frac{(A + o(1))x}{\log^2 x}, \qquad x \to \infty,$$

which implies the prime number theorem, and indeed with the claimed error term.

Gauss wrote in a letter of 1849 that, as a boy, he had investigated on the likelihood of a number near x being a prime, by examining numbers in blocks of varying length. He came to the conclusion that the 'density' of primes near x is $1/\log x$, and had suggested that $\pi(x) \approx \mathrm{li}(x)$, where

$$\mathrm{li}(x) = \int_2^x \frac{dt}{\log t}.$$

On integration by parts, we find that, for any fixed q and $x \to \infty$,

$$\int_2^x \frac{dt}{\log t} = \frac{x}{\log x} + \frac{1!x}{\log^2 x} + \cdots + \frac{q!x}{\log^{q+1} x}(1 + o(1)),$$

so that Gauss too had anticipated the prime number theorem. Indeed, on writing Gauss' conjecture is

$$\pi(x) = \int_2^x \frac{dt}{\log x} + E(x) = \mathrm{li}(x) + E(x), \tag{11.4}$$

and treating the integral $\mathrm{li}(x)$ as the dominating term so that $E(x) = o(\mathrm{li}(x))$ is an error term, we then find that $E(x) = O(x/\log^q x)$, for any q however large, with the O-constant depending on q. It also implies that the number A in Legendre's guess should take the value 1. Very roughly speaking, $\mathrm{li}(x)$ counts powers of primes up to x, and that Gauss' reasoning is correct. However, more work is required to establish the prime number theorem with an error term, and the problem of how accurate the approximation $\mathrm{li}(x)$ is to $\pi(x)$ is delicate; see more in Note 2.

11.5 Zero-Free Region and the Riemann Hypothesis

As we already remarked, a proof of the prime number theorem based on the use of $\zeta(s)$ will require Theorem 11.2 which states that $\zeta(s)$ does not have a zero on the vertical line $\sigma = 1$. For the deeper result of finding an appropriate error term of the prime number theorem, knowledge of $\zeta(s)$ inside the *critical strip*, that is the region $\{s : s = \sigma + it, 0 < \sigma < 1\}$, is required. In particular, we need to show that $\zeta(s)$ does not have zeros in a certain region inside the strip, so that $\zeta'(s)/\zeta(s)$ is regular in the extended domain. This may then enable us to deal with, for example, the Mellin transform of $\psi(x)$ in (11.3), in a more flexible way.

Thus, in 1899, de la Vallée Poussin established that *there exists a constant* $c > 0$ *such that* $\zeta(s)$ *has no zero in the region*

$$\sigma \geq 1 - \frac{c}{\log t}, \qquad t \geq 2;$$

from which he succeeded in proving (11.4) with $E(x) \ll x \exp(-c'\sqrt{\log x})$, where $c' = 1/200$, say, is admissible.

Much of the work on $\pi(x)$ since 1900 is concerned with the improvement of the error term in the prime number theorem, with the aim of establishing the *Riemann Hypothesis*, which states that

$$\zeta(s) \neq 0 \quad \text{for} \quad \sigma > \frac{1}{2}. \tag{11.5}$$

The hypothesis is equivalent to the statement that, for every $\epsilon > 0$,

$$\pi(x) = \int_2^x \frac{dt}{\log t} + O(x^{\frac{1}{2}+\epsilon}), \qquad x \to \infty. \tag{11.6}$$

For computation purposes we often need effective bounds for $\pi(x)$, and the following useful results are due to J. B. Rosser and L. Scoenfeld [83]:

$$\frac{x}{\log x}\Big(1 + \frac{1}{2\log x}\Big) < \pi(x) < \frac{x}{\log x}\Big(1 + \frac{3}{2\log x}\Big), \qquad x \geq 59;$$

if p_k is the kth prime then

$$k\big(\log(k\log k) - 1.0073\big) < p_k < k\big(\log(k\log k) - 0.9385\big), \qquad k \geq 7022.$$

11.6 Prime Gaps

After the prime number theorem, we have the more challenging gap problems associated with the sequence of primes $(p_n) = (2, 3, 5, \dots)$. Let $g_n = p_{n+1} - p_n$, so that each gap g_n is even for $n > 1$. It is easy to see that g_n can be arbitrarily large for suitably chosen n. Thus, for any even K, we can construct a run of $K - 1$ consecutive composite numbers by taking

$$K! + k, \qquad k = 2, 3, \dots, K,$$

so that k is a proper divisor of $K! + k$. Letting n be specified by $p_n \le K! + 1 < K! + K < p_{n+1}$, we then have $g_n = p_{n+1} - p_n \ge K$.

Example 11.1 Take $K = 6$, so that $K! = 6! = 720$. Then each of

$$720 \pm k = \begin{cases} 722, 723, 724, 725, 726, \\ 718, 717, 716, 715, 714, \end{cases} \qquad k = 2, 3, 4, 5, 6,$$

is necessarily composite; indeed, we have the 'bonus' composites 713, 712, 711, 710, and 721, 720, so that we have the prime gaps $g = 727 - 719 = 8$, and $g' = 719 - 709 = 10$. □

Given K, the corresponding p_n in the argument here satisfies

$$\log p_n \sim \log K! \sim K \log K \qquad \text{as} \quad K \to \infty.$$

We now show that there is a better solution, better in the sense that there is already a solution with a smaller value for the prime p_n.

First, let $p_{m-1} < K \le p_m$, and $P(m) = p_1 p_2 \cdots p_m$, so that $P(m)$ is a multiple of each $k \le p_m$, and hence $P(m) + k$ is composite for $1 < k \le K$. Now let n be specified by $p_n \le P(m) + 1 < P(m) + K < p_{n+1}$, so that $g_n \ge K$, and that

$$\log p_n \sim \log P(m) \sim \sum_{p \le K} \log p = \theta(K) \sim K \qquad \text{as} \quad K \to \infty,$$

by the prime number theorem. Therefore the new solution is substantially smaller.

Example 11.2 For $K = 6$, we may take $P = p_1 p_2 p_3 = 2{\cdot}3{\cdot}5 = 30$. Again, each of

$$30 \pm k = \begin{cases} 32, 33, 34, 35, 36, \\ 28, 27, 26, 25, 24, \end{cases} \qquad k = 2, 3, 4, 5, 6,$$

is composite, so that $g = 37 - 31 = 6$ and $g' = 29 - 23 = 6$; there is no bonus!

For $K = 10$, we may take $P = p_1p_2p_3p_5 = 2{\cdot}3{\cdot}5{\cdot}7 = 210$. Then, since $k|P$ for $k \le 10$, it follows that each of

$$210 \pm k = \begin{cases} 212, 213, 214, 215, 216, 217, 218, 219, 220, \\ 208, 207, 206, 205, 204, 203, 202, 201, 200, \end{cases} \qquad k = 2, 3, \dots, 10,$$

is necessarily composite, and we have the bonus composites, $209 = 11{\times}19$, 210; $221 = 210{+}11 = 13{\times}17$, 222. Therefore the three successive primes $199 < 211 < 223$ deliver two successive gaps $g = 12$. □

It is a very difficulty problem to find good lower bounds for g_n which is valid for infinitely many n. After successive improvements since the 1930s, the current best result is that, for infinitely many n,

$$g_n \gg \frac{\log n \ \log\log n \ \log\log\log\log n}{\log\log\log n},$$

by the combined efforts of Ford-Green-Konyagin-Maynard-Tao; and T. Tao has offered 10000USD reward for an improvement.

Good upper bounds for g_n are just as difficult. Bertrand's postulate, which was established by Chebyshev with his theorem, states that, for $x \ge 1$, there is a prime p satisfying[1] $x < p \le 2x$, so that $\pi(2x) - \pi(x) \ge 1$, and hence $g_n < p_n$.

Example 11.3 Use the prime number theorem to show that, if $\epsilon > 0$, then there exists $X = X(\epsilon)$ with the property that there is a prime p satisfying $x < p \le x{+}\epsilon x$ when $x \ge X$.

Solution For $0 \le h \le x$ and $x \to \infty$, we have, by the prime number theorem,

$$\pi(x+h) - \pi(x) = \frac{x+h}{\log(x+h)} - \frac{x}{\log x} + o\Big(\frac{x}{\log x}\Big) = \frac{h + o(x)}{\log x}.$$

Therefore, for $0 < \epsilon < 1$, $h = \epsilon x$, and x large, we have $\pi(x+h) > \pi(x)$. □

The result here implies that $g_n < \epsilon p_n$ for all large n. In 1930, G. Hoheisel showed that $g_n \ll p_n^{\theta}$, where $\theta = 1 - 1/33000$ is admissible; again, after successive improvement, the current best result (by R. C. Baker, G. Harman and J. Pintz) is that any $\theta > 0{\cdot}525$ is admissible. Since $\theta = \frac{2}{3}$ is admissible, it follows that there is a prime between large successive cubes. The conjecture that there is already a prime between large successive squares amounts to $\theta = \frac{1}{2}$ being admissible, but the conjecture is of similar depth to the Riemann hypothesis which, by (11.6), implies that $g_n \ll p_n^{\frac{1}{2}+\epsilon}$. Meanwhile we can only contemplate Cramér's conjecture that $g_n \ll \log^2 p_n$, which is based on a model suggesting how primes may behave.

[1] In 1845, Bertrand verified his 'postulate' based on primes $p < 3000000$.

There is therefore a huge lacuna when it comes to the problem on how large can prime gaps be.

It is a famous conjecture that g_n takes every even value infinitely often but, by the close of the twentieth century, no one had been able to disprove that $g_n \to \infty$ as $n \to \infty$. It was a spectacular feat in the early part of the twenty-first century when Yitang Zhang succeeded in establishing that $g_n \leq 7\times 10^7$ holds for infinitely many n. A *Polymath Project* collaborative effort managed to optimise Zhang's bound down to 4680. Later J. Maynard introduced a new refinement of the sieve used, enabling him to reduce further the bound to 600; using his idea, the Polymath Project improved this to 246. The final goal is, of course, the notorious prime-twins conjecture which states that $g_n = 2$ for infinitely many n.

11.7 Summary Notes

Summary

- Classical analytic proofs of the *prime number theorem* make use of the properties of $\zeta(s)$ inside the *critical strip* $0 < \sigma \leq 1$. Over the years the requirement of information on $\zeta(s)$ has been reduced to its bare minimum, which is that $\zeta(s)$ does not have a zero on the vertical line $\sigma = 1$. The much simplified proof is based on the work of many mathematicians, culminating in the establishment of a Tauberian theorem for Dirichlet series using contour integration. Specifically, the prime number theorem can now be deduced from the Ikehara-Wiener theorem, with a proof by Newman using contour integration.
- The vital step that $\zeta(s) \neq 0$, on $\sigma = 1$, can now be established using the elementary inequality $3 + 4\cos\theta + \cos 2\theta \geq 0$, which implies that

$$\zeta^3(\sigma)|\zeta(\sigma + it)|^4|\zeta(\sigma + 2it)| \geq 1;$$

  the desired result follows from the fact that the triple pole of $\zeta^3(s)$ at $s = 1$ is not enough to cope with a possible zero of order 4 for $\zeta^4(s)$ at $s = 1 + it$.
- The prime number theorem was conjectured by Legendre based on his computed values of $\pi(x)$, and also by Gauss from his empirical observation of the likelihood of a number being a prime in its neighbourhood. More specifically their suggestions were, respectively:

$$\pi(x) \approx \frac{x}{\log x - \text{constant}}, \quad \text{and} \quad \pi(x) \approx \text{li}(x) = \int_2^x \frac{dt}{\log t}.$$

- The prime number theorem may be stated as any one of the equivalent formulae:

$$\pi(x) \sim \frac{x}{\log x}, \quad \psi(x) = \sum_{n\leq x} \Lambda(n) \sim x, \quad \theta(x) = \sum_{p\leq x} \log p \sim x, \quad x \to \infty.$$

Some immediate deductions are that the nth prime $p_n \sim n \log n$, and that

$$\text{if} \quad P_n = \prod_{p \le n} p \quad \text{and} \quad d_n = \text{LCM}(1, 2, \ldots, n), \quad \text{then} \quad \log P_n \sim \log d_n \sim n,$$

so that $2^n \le P_n \le d_n < 3^n$ for all large n.

- Writing $\pi(x) = \int_2^x \frac{dt}{\log t} + E(x)$, and considering $E(x)$ to be the 'error-term', estimates can be obtained by establishing a 'zero-free' region for $\zeta(s)$ inside the critical strip. For example, de le Vallée Poussin proved that $\zeta(s) \neq 0$ in $\sigma \ge 1 - c/\log t$ for a positive constant c and $t \ge 2$, from which it follows that $E(x) \ll x \exp(-\frac{1}{200}\sqrt{\log x})$.
- The *Riemann hypothesis* states that $\zeta(s) \neq 0$ for $\sigma > \frac{1}{2}$, which is equivalent to $E(x) \ll x^{\frac{1}{2}+\epsilon}$ for every $\epsilon > 0$.
- The problem of finding good estimates for the prime gap $g_n = p_{n+1} - p_n$ is notoriously difficult. It is conjectured that g_n takes every even number infinitely often, in particular, that $g_n = 2$, the twin-primes conjecture. The prime number theorem implies that $g_n < \epsilon p_n$ for all large n, and it is known that there is always a prime between successive large cubes, but the problem of whether there is a prime between successive large squares is as deep as the Riemann hypothesis.

Based on a model of how primes might behave, it is suggested that $g_n \ll \log^2 p_n$. More established results on g_n are given in Sect. 11.6.

Notes

1. *Tauberian theory*. Readers may recall Abel's continuity theorem (1826), which states that if the convergent series $\sum_{n\ge 0} a_n$ has the sum A (meaning that its sequence of partial sums $\sum_{k\le n} a_k \to A$ as $n \to \infty$) then, for $0 < x < 1$ and $x \to 1$,

$$\sum_{n=0}^{\infty} a_n x^n \to A;$$

the converse of the theorem is not true unless some condition is imposed on the sequence (a_n). In 1897, A. Tauber proved that the condition $na_n \to 0$ is sufficient for the converse. In 1911, Littlewood famously established that an optimal order condition, namely that na_n is bounded, for the converse of Abel's theorem. Subsequently, Hardy and Littlewood proved numerous other converse theorems of a similar type, and they coined the name *Tauberian theorems*. Broadly speaking nowadays, Tauberian theory is concerned with the search for suitable conditions under which various 'weak' notions of convergence can deliver 'ordinary' convergence.

2. *Littlewood's theorem and Skewes' number*. Computations show that, up to $x = 10^{22}$, the approximation $\mathrm{Li}(x)$ to $\pi(x)$ is an overestimate by about $\sqrt{\pi(x)}$. Indeed, the difference seems to be monotonically increasing, which suggests that there might be a smooth secondary term, such as $-\frac{1}{2}\mathrm{Li}(\sqrt{x})$, which is included in Riemann's memoir. However, such a term does not always dominate because, back in 1914, J. E. Littlewood had proved a famous theorem which states that, for certain arbitrarily large x,

$$\pi(x) > \mathrm{li}(x) + \frac{\sqrt{x}\log\log\log x}{3\log x},$$

and for other large x,

$$\pi(x) < \mathrm{li}(x) - \frac{\sqrt{x}\log\log\log x}{3\log x}.$$

In particular, the set $S = \{x : \pi(x) > \mathrm{li}(x), x \in \mathbb{N}\}$ is infinite, and there is a least member $x_1 \in S$. In 1933, Littlewood's student S. Skewes used the argument in the proof of the theorem to show that, under the assumption of the Riemann Hypothesis, $x_1 < \exp\exp\exp(7{\cdot}8)$, and later that, unconditionally, $x_1 < \exp\exp\exp\exp(7{\cdot}8)$. In more recent times, using large-scale computer calculations of zeros of $\zeta(s)$, the unimaginably large Skewes bound has been reduced to something more tangible. Thus, in 2000, C. Bays and R. H. Hudson showed that there are at least 10^{153} consecutive values $x \in S$ near $1{\cdot}3982\times10^{316}$. However, there is no known explicit value $x \in S$, mainly because of the apparent need for the value of $\pi(x)$. See also the recent book [73].

The set S does not have an asymptotic density but, in 1941, A. Wintner showed that it has a positive logarithmic density; see the next Note. More recently Rubinstein and Sarnak [85] found that, under a very plausible hypothesis (see Sect. 12.6), the value of this latter density is the very small number

$$\lim_{X\to\infty}\frac{1}{\log X}\sum_{x\le X,\, x\in S}\frac{1}{x} = 0{\cdot}00000026\ldots,$$

which may explain why it is not easy to find an explicit value $x \in S$.

3. *Asymptotic and logarithmic densities*. Let $A(n)$ denote the counting function of $A \subset \mathbb{N}$. By the *upper and lower asymptotic densities* of A, we mean the limits

$$\bar{d}(A) = \limsup_{n\to\infty}\frac{A(n)}{n} \quad \text{and} \quad \underline{d}(A) = \liminf_{n\to\infty}\frac{A(n)}{n};$$

if $\bar{d}(A) = \underline{d}(A)$, so that $\lim_{n\to\infty} A(n)/n$ exists, then we denote the common value by $d(A)$ and call it the *asymptotic density* of A.

Next, consider the attachment of the weight of $1/n$ to each $n \in A$, and normalise the corresponding weighted sum by dividing it by the partial sum of the harmonic series $\sum_{k\le n} 1/k$, which has the asymptotic value $\log n$, as $n \to \infty$. After such a manoeuvre we define the *upper and lower logarithmic densities* of A by

$$\bar{\delta}(A) = \limsup_{n\to\infty} \frac{1}{\log n} \sum_{k\le n,\, k\in A} \frac{1}{k} \quad \text{and} \quad \underline{\delta}(A) = \liminf_{n\to\infty} \frac{1}{\log n} \sum_{k\le n,\, k\in A} \frac{1}{k};$$

if $\bar{\delta}(A) = \underline{\delta}(A)$, then we denote the common value by $\delta(A)$ and call it the *logarithmic density* of A.

Logarithmic density is a finer notion than asymptotic density in the sense that $\underline{d}(A) \le \underline{\delta}(A) \le \bar{\delta}(A) \le \bar{d}(A)$ (see Exercise 11.10), so that if a set fails to have an asymptotic density, we may consider its logarithmic density, if it exists, instead.

11.8 Exercises

1. Establish the formula

$$\log \zeta(s) = s \int_2^\infty \frac{\pi(x)}{x(x^s - 1)}\, dx, \qquad \sigma > 1.$$

2. Establish the formula

$$\psi(x) = -\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\zeta'(s)}{\zeta(s)} \frac{x^s}{s}\, ds, \qquad c > 1.$$

3. Show that, for all θ,

$$5 + 8\cos\theta + 4\cos 2\theta + \cos 3\theta \ge 0,$$

and use the inequality to give a similar proof to that in Sect. 11 that $\zeta(s)$ has no zero on the vertical line $\sigma = 1$, $t \ne 0$.
4. Sketch Newman's contour C in the proof of the Ikehara-Wiener theorem.
5. Show that

$$\sum_{p\le x} p \sim \frac{x^2}{2\log x}, \qquad x \to \infty.$$

6. Find two pairs of primes with gap values 100.
7. Use the prime number theorem to show that the set of fractions p/p', where p, p' are primes, is dense in the set of positive real numbers.

8.

(i) Verify that

$$\prod_{p\le n} p < 2^n \quad \text{if} \quad n = 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 16, 28.$$

(ii) Let

$$M_n = \text{LCM}[1, 2, \ldots, n], \qquad n = 1, 2, \ldots .$$

Compute M_n, for $1 \le n \le N$, where $n = 1000$, say, by an iterative scheme without making use of a precomputed list of primes.

(iii) Verify that

$$2^n < \prod_{p\le n} p \le M_n < 3^n, \quad \text{for} \quad \text{for} \quad 29 \le n \le N.$$

(iv) Plot the points $f(n) = M_n^{1/n}$ for $1 \le n \le 400$, and find n_0 and n_1 such that

$$f(n_0) = \max_{1\le n\le N} f(n) \quad \text{and} \quad f(n_1) = \min_{n_0<n\le N} f(n).$$

9. Show that the following two asymptotic formulae are equivalent to each other:

$$\pi(x) = \frac{x}{\log x} + O\Big(\frac{x}{\log^2 x}\Big), \quad \psi(x) = x + O\Big(\frac{x}{\log x}\Big), \qquad x \to \infty.$$

10. Let $A(n)$ be the counting function of a set $A \subset \mathbb{N}$. Use partial summation to show that

$$\sum_{k\le n,\, k\in A} \frac{1}{k} = \frac{A(n)}{n} + \sum_{k<n} \frac{A(k)}{k}\cdot\frac{1}{(k+1)},$$

and deduce that $\underline{d}(A) \le \underline{\delta}(A) \le \bar{\delta}(A) \le \bar{d}(A)$.

(i) Construct a set A with $\underline{d}(A) < \underline{\delta}(A) = \bar{\delta}(A) < \bar{d}(A)$;

(ii) Construct a set A with $\underline{\delta}(A) < \bar{\delta}(A)$.

11. Show that

$$e^{-x} = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \Gamma(s)x^{-s}dx, \qquad x > 0, \quad c > 0.$$

12. Use the Mellin transform to show that

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx, \qquad \sigma > 1.$$

13. The Bernoulli numbers B_k are defined by

$$\frac{z}{e^z - 1} = \sum_{k \geq 0} \frac{B_k z^k}{k!}, \qquad |z| < 2\pi,$$

so that $B_0 = 1$, $B_1 = -1$ and $B_{2k+1} = 0$ for $k \geq 1$. By substituting $x = -iu/2$ into the formula

$$\sin x = x \prod_{k \geq 1} \Big(1 - \frac{x^2}{\pi^2 k^2}\Big),$$

and then taking logarithmic differentiation, show that

$$\frac{1}{e^u - 1} = \frac{1}{u} - \frac{1}{2} + \sum_{k \geq 1} \frac{2u}{4k^2\pi^2 + u^2}, \qquad u \neq \pm 2m\pi i.$$

Show that

$$\zeta(2k) = \frac{(-1)^{k+1}(2\pi)^{2k} B_{2k}}{2(2k)!}, \qquad k = 1, 2, \ldots,$$

by manipulating the power series $\sum B_{2k} u^{2k}/(2k)!$ into one involving $\zeta(2k)$.

14. Evaluate $f(x) = 2\pi(x) - \pi(2x)$ for $x = 10, 10^2, 10^3, 10^4, 10^5, 10^6$.

Explain why "$f(x) > 0$ for all large x" does not follow from the prime number theorem. Show, however, that if there are positive constants $A < B$ such that

$$\frac{x}{\log x - A} < \pi(x) < \frac{x}{\log x - B}, \quad \text{for all large } x, \text{ then } \quad f(x) \gg \frac{x}{\log^2 x}.$$

Use de la Vallée Poussin's form of the prime number theorem to show that

$$f(x) \sim \frac{(2 \log 2)x}{\log^2 x}, \qquad x \to \infty.$$

15. Let

$$f(x) = \frac{ex}{\log x}\pi\left(\frac{x}{e}\right) - \pi^2(x) \quad \text{and} \quad x_0 = 38\,358\,837\,682.$$

(i) Use de la Vallée Poussin's form of the prime number theorem to confirm Ramanujan's observation that $f(x) > 0$ for all large x.
(ii) Use Meissel's method to evaluate $f(x_0)$ and $f(x_0 + 1)$.

16. Determine the maximum of $\psi(n)/n$ over $1 \le n \le 1000$.

Chapter 12
Primes in Arithmetic Progressions

Abstract The plan for the proof of Dirichlet's theorem is given before embarking on the proof itself. There follow descriptions of the prime number theorem in arithmetic progressions, the generalised Riemann hypothesis, the Siegel–Walfisz theorem, Linnik's theorem, the Brun–Titchmarsh theorem, Bombieri's theorem, and the Titchmarsh–Linnik theorem.

The very difficult problems concerning Chebyshev's bias, or 'prime number races', are considered, including the Rubinstein–Sarnark hypothetical theorem. The chapter ends with the notion of primitive characters and the Pólya–Vinogradov inequality.

Our proof of Dirichlet's theorem on the existence of primes in an arithmetic progression $a \pmod m$ deviates significantly from his original proof in two places. First, instead of considering the series $\sum 1/p$, summed over primes p in the progression, we find it expedient to deal with the series $\sum(\log p)/p$ over the progression instead—it enables us to dispense with complex logarithms. The argument is heavily involved with the Dirichlet characters $\chi \pmod m$ and the corresponding Dirichlet functions $L(s, \chi)$. In particular it hinges on the fact that $L(1, \chi) \neq 0$ when $\chi \neq \chi_0$, the principal character, and the case when χ is real requires particular attentions. Dirichlet showed that, for such χ, the value of $L(1, \chi)$ is connected with the number of quadratic forms with a fundamental discriminant related to m, and that $L(1, \chi) > 0$ follows from his famous *class-number formula* for such forms. There are now elegant and relatively short proofs of $L(1, \chi) \neq 0$ using function theory (see Note 12.1), but we shall apply a more lengthy basic analytic argument instead.

Theorem 12.1 *Let m and a be coprime positive integers. Then, as $x \to \infty$,*

$$\sum_{\substack{p \leq x \\ p \equiv a \pmod m}} \frac{\log p}{p} = \frac{\log x}{\phi(m)} + O(1).$$

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Since $\log x \to \infty$ as $x \to \infty$, Dirichlet's theorem is a corollary to the theorem. The constant for $O(1)$ depends on m, but it need not depend on a because there are only $\phi(m)$ relevant values for a.

12.1 Plan for the Proof

The proof of Theorem 12.1 is so intricate and lengthy that it may appear convoluted on a first reading. We give the outline of the argument, and the underlying ideas in the various stages. The obvious first step is to use the characters $\chi \pmod m$ to isolate the primes $p \equiv a \pmod m$ in the sum in the theorem. Thus Theorem 9.5 enables us to write

$$\phi(m) \sum_{\substack{p\le x\\ p\equiv a \pmod m}} \frac{\log p}{p} = \sum_{p\le x}\sum_{\chi} \frac{\chi(p)\log p}{\chi(a)p} = \sum_{\chi}\frac{1}{\chi(a)}\sum_{p\le x}\frac{\chi(p)\log p}{p},$$

where $\sum_\chi$ denotes summation over all the $\phi(m)$ characters $\chi \pmod m$. The sum corresponding to the principal character χ_0, for which $\chi_0(p) = \chi_0(a) = 1$, is $\sum_{p\le x}(\log p)/p$ if we also throw in the finitely many terms with $p|m$. It is the dominating sum and its value, according to Corollary 9.4 to Chebyshev's theorem, is $\log x + O(1)$, so that

$$\phi(m) \sum_{\substack{p\le x\\ p\equiv a \pmod m}} \frac{\log p}{p} = \log x + \sum_{\chi\ne\chi_0}\frac{1}{\chi(a)}\sum_{p\le x}\frac{\chi(p)\log p}{p} + O(1). \tag{12.1}$$

The analytic part of the argument takes over, and the task is to show that each of the inner sums of the double sum is bounded. Now, for a nonprincipal character $\chi \ne \chi_0$, the factor $\chi(p)$ is a complex number on the unit circle, delivering a directional 'twist' to the terms of the sum. It is to be hoped, or even expected, that the anticipated cancellation effect arising from such a twist will deliver the desired

$$\sum_{p\le x}\frac{\chi(p)\log p}{p} = O(1), \quad \text{for each } \chi \ne \chi_0. \tag{12.2}$$

This is true, but the argument is far from being simple.

The next idea is to replace the sum over primes $p \le x$ by integers $n \le x$, and $\log p$ by the Mangoldt function $\Lambda(n)$; prime powers being few and far between among the integers means that the difference can be absorbed by the error term. We shall see that such a manoeuvre delivers

$$\sum_{p\le x}\frac{\chi(p)\log p}{p} = -L'(1,\chi)\sum_{n\le x}\frac{\mu(n)\chi(n)}{n} + O(1), \quad \text{for each } \chi \ne \chi_0, \tag{12.3}$$

where

$$-L'(1,\chi)=\sum_{n=1}^{\infty}\frac{\chi(n)\log n}{n} \tag{12.4}$$

is obtained from term-by-term differentiation of $L(s,\chi)=\sum\chi(n)/n^s$ at $s=1$. The convergent infinite series is related to the sum over primes p; indeed the effect of the manoeuvre is to have the factor $\log p$ taken out of the sum, replacing it by the familiar $\mu(n)$ from the application of the Möbius inversion formula on $\Lambda(n)$. The 'double-twist' $\mu(n)\chi(n)$ is still just a twist, now on $1/n$ only, and the anticipated cancellation should deliver the much wanted result (12.2). This is true, but now $L(1,\chi)$ emerges from the wash as the coefficient of the new sum:

Lemma 12.1 *For each $\chi\neq\chi_0$, and $x\to\infty$, we have*

$$L(1,\chi)\sum_{n\le x}\frac{\mu(n)\chi(n)}{n}=O(1). \tag{12.5}$$

Now if $L(1,\chi)\neq 0$ then we may delete it from (12.5) to deliver the desired (12.2) via (12.3). Theorem 12.1 then follows from (12.1), so that everything hinges on the following:

Theorem 12.2 *If $\chi\neq\chi_0$ then $L(1,\chi)\neq 0$.*

The main tool involved in the analysis is the generalised Möbius inversion formula (Theorem 2.5). Apart from this and the given ideas in the above, care has to be taken in keeping the accompanying error terms under control. We need estimates from the tail ends of various convergent series; for example,

$$\sum_{n\le x}\log\frac{x}{n}=\sum_{n\le x}(\log x-\log n)=\lfloor x\rfloor\log x-\log\lfloor x\rfloor!=O(x),\quad x\to\infty, \tag{12.6}$$

which follows from Stirling's formula (see (9.1)).

12.2 Replacing Sums Over Primes with Sums Over Integers

Assume, for the moment, that in replacing $\log p$ by $\Lambda(n)$ in (12.2), the result becomes a sum over $n\le x$, with only a bounded error term. Then, by Lemma 2.2,

$$\begin{aligned}\sum_{p\le x}\frac{\chi(p)\log p}{p}&=\sum_{n\le x}\frac{\chi(n)\Lambda(n)}{n}+O(1)\\&=\sum_{n\le x}\frac{\chi(n)}{n}\sum_{d|n}\mu(d)\log\frac{n}{d}+O(1)\end{aligned}$$

$$= \sum_{dd' \le x} \frac{\chi(d)\chi(d')}{dd'} \mu(d) \log d' + O(1)$$

$$= \sum_{d \le x} \frac{\mu(d)\chi(d)}{d} \sum_{d' \le x/d} \frac{\chi(d') \log d'}{d'} + O(1),$$

and, as $x \to \infty$, the inner sum converges to $-L'(1, \chi)$ given by (12.4), with the tail end of the series being bounded by $O\big((\log x/d)/(x/d)\big)$. We then need to have

$$\sum_{d \le x} \frac{1}{d} \frac{\log x/d}{x/d} = O(1),$$

which follows from (12.6). Finally, the difference arising from the replacement of $\log p$ by $\Lambda(n)$ has the bound

$$\left| \sum_{\ell \ge 2} \sum_{p \le x^{1/\ell}} \frac{\chi(p^\ell) \log p}{p^\ell} \right| \le \sum_p \sum_{\ell \ge 2} \frac{\log p}{p^\ell} = \sum_p \frac{\log p}{p(p-1)} = O(1).$$

Collecting our results we have

$$\sum_{p \le x} \frac{\chi(p) \log p}{p} = -L'(1, \chi) \sum_{d \le x} \frac{\mu(d)\chi(d)}{d} + O(1)$$

which is (12.3).

Emergence of $L(1, \chi)$

Since $\chi(n)$ is completely multiplicative we can apply the genralised Möbius inversion Theorem 2.5, with $F(x) = x$. Forming

$$G(x) = \sum_{n \le x} F\Big(\frac{x}{n}\Big) \chi(n) = x \sum_{n \le x} \frac{\chi(n)}{n} = x\, L(1, \chi) + O(1),$$

we then find that

$$x = F(x) = \sum_{n \le x} \mu(n)\, G\Big(\frac{x}{n}\Big) \chi(n) = x\, L(1, \chi) \sum_{n \le x} \frac{\mu(n)\chi(n)}{n} + O(x),$$

and Lemma 12.1 follows from dividing the equation by x. □

12.3 Nonvanishing of $L(1, \chi)$ for Characters $\chi \neq \chi_0$

Recall that $L(s, \chi_0)$ has a pole at $s = 1$, but $L(1, \chi)$ is well defined for $\chi \neq \chi_0$. We single out the case when $\chi \neq \chi_0$ is a real character.

Lemma 12.2 *Let $\chi \neq \chi_0$ be a real character* (mod m)*, and $A(n) = \sum_{d|n} \chi(d)$. Then $A(n) \geq 0$, and $A(n) \geq 1$ if $n = k^2$.*

Proof The sum function of a multiplicative function being multiplicative, we need only consider prime powers $n = p^m$ with $A(p^m) = 1 + \chi(p) + \chi(p)^2 + \cdots \chi(p)^m$. That χ is real means that $\chi(p) = 0, \pm 1$ for each prime p, so that

$$A(p^m) = \begin{cases} m+1, & \text{if} \quad \chi(p) = 1, \\ 1, & \text{if} \quad \chi(p) = 0, \text{ or } \chi(p) = -1, \ m \text{ even}, \\ 0, & \text{if} \quad \chi(p) = -1, \ m \text{ odd}. \end{cases}$$

The required result follows. □

Lemma 12.3 *Let $\chi \neq \chi_0$ be a real character* (mod m)*. Then, $L(1, \chi) > 0$.*

Proof Let $A(n)$ be defined in the previous lemma, so that

$$\sum_{n \leq x} \frac{A(n)}{\sqrt{n}} \geq \sum_{1 \leq k \leq \sqrt{x}} \frac{1}{k} \to \infty, \quad \text{as} \quad x \to \infty. \tag{12.7}$$

On the other hand, we have

$$\begin{aligned} \sum_{n \leq x} \frac{A(n)}{\sqrt{n}} &= \sum_{n \leq x} \frac{1}{\sqrt{n}} \sum_{d|n} \chi(d) = \sum_{dd' \leq x} \frac{\chi(d)}{\sqrt{dd'}} \\ &= \sum_{d' \leq \sqrt{x}} \frac{1}{\sqrt{d'}} \sum_{\sqrt{x} < d \leq x/d'} \frac{\chi(d)}{\sqrt{d}} + \sum_{d \leq \sqrt{x}} \frac{\chi(d)}{\sqrt{d}} \sum_{d' \leq x/d} \frac{1}{\sqrt{d'}}, \end{aligned}$$

because $dd' \leq x$ implies either $d' \leq \sqrt{x}$ or $d \leq \sqrt{x}$. Next, by the Euler summation formula (Theorem 9.2),

$$\sum_{\sqrt{x} < d \leq x/d'} \frac{\chi(d)}{\sqrt{d}} \ll \frac{1}{\sqrt{x}} \quad \text{and} \quad \sum_{d' \leq x/d} \frac{1}{\sqrt{d'}} = 2\sqrt{\frac{x}{d}} + c + O\Big(\sqrt{\frac{d}{x}}\Big),$$

where c is a constant. It follows that

$$\sum_{n\le x}\frac{A(n)}{\sqrt{n}} = 2\sqrt{x}\sum_{d\le\sqrt{x}}\frac{\chi(d)}{\sqrt{d}} + O(1) = 2\sqrt{x}\,L(1,\chi) + O(1),$$

and that $L(1,\chi) > 0$ is a consequence of (12.7). □

Nonvanishing of $L(1,\chi)$ for Complex Characters χ

We next show that *if* $L(1,\chi) = 0$ then, as $x\to\infty$,

$$L'(1,\chi)\sum_{n\le x}\frac{\mu(n)\chi(n)}{n} = \log x + O(1); \tag{12.8}$$

we choose not to set this out as a lemma because the hypothesis $L(1,\chi) = 0$ is false. However, we do not yet know this at the present stage of the proof and, indeed, (12.8) will be used to establish that $L(1,\chi)\neq 0$.

We again apply the generalised Möbius inversion theorem; this time with $F(x) = x\log x$, so that on writing

$$x\log x = \sum_{n\le x}\mu(n)\chi(n)G\Big(\frac{x}{n}\Big), \tag{12.9}$$

we have

$$\begin{aligned} G(x) &= \sum_{n\le x}\chi(n)\frac{x}{n}\log\frac{x}{n} = x\log x\sum_{n\le x}\frac{\chi(n)}{n} - x\sum_{n\le x}\frac{\chi(n)\log n}{n}\\ &= x\log x\left\{L(1,\chi) + O\Big(\frac{1}{x}\Big)\right\} + x\left\{L'(1,\chi) + O\Big(\frac{\log x}{x}\Big)\right\}\\ &= x\,L'(1,\chi) + O(\log x), \end{aligned}$$

in which the last line follows from the (unfounded) hypothesis that $L(1,\chi) = 0$. Returning to (12.9) we now have

$$\begin{aligned} x\log x &= \sum_{n\le x}\mu(n)\chi(n)\left\{\frac{x\,L'(1,\chi)}{n} + O\Big(\log\frac{x}{n}\Big)\right\}\\ &= x\,L'(1,\chi)\sum_{n\le x}\frac{\mu(n)\chi(n)}{n} + O(x), \end{aligned}$$

and (12.8) follows on division by x.

Proof of Theorem 12.2

Let χ be a complex character (mod m), so that its complex conjugate $\bar{\chi}$ is also a character. From

$$L(\sigma, \bar{\chi}) = \sum_{n=1}^{\infty} \frac{\bar{\chi}(n)}{n^{\sigma}} = \overline{\sum_{n=1}^{\infty} \frac{\chi(n)}{n^{\sigma}}} = \overline{L(\sigma, \chi)}, \qquad \sigma > 0,$$

it follows that $L(1, \chi) = 0$ if and only if $L(1, \bar{\chi}) = 0$.

Let $N(m)$ denote the total number of characters $\chi \neq \chi_0$ with $L(1, \chi) = 0$, so that, after Lemma 12.3, $N(m)$ is an even number.

Lemma 12.4 *As $x \to \infty$, we have*

$$\phi(m) \sum_{\substack{p \leq x \\ p \equiv 1 \pmod m}} \frac{\log p}{p} = (1 - N(m)) \log x + O(1). \tag{12.10}$$

Proof Setting $a = 1$ in (12.1) yields

$$\phi(m) \sum_{\substack{p \leq x \\ p \equiv 1 \pmod m}} \frac{\log p}{p} = \log x + \sum_{\chi \neq \chi_0} \sum_{p \leq x} \frac{\chi(p) \log p}{p} + O(1),$$

where the inner sums have been transformed into (12.3), and for which Lemma 12.1 and (12.8) are relevant. Thus, for each χ with $L(1, \chi) \neq 0$, the said inner sum is bounded according to Lemma 12.1, whereas, for each χ with $L(1, \chi) = 0$, the sum has the value $-\log x + O(1)$ according to (12.8). The required result follows from the definition of $N(m)$. □

The left-hand side of (12.10) is non-negative whereas, because $N(m)$ is even, the right-hand side is negative for large x, unless $N(m) = 0$. Therefore we need to have $N(m) = 0$, which is Theorem 12.2, and hence also Theorem 12.1.

Perhaps we should recapitulate the intricate argument for Theorem 12.1 in the forward manner. From the fact that $L(1, \chi) \neq 0$ for each nonprincipal character χ, we deduce from Lemma 12.1 that the sum $\sum_{n \leq x} \mu(n)\chi(n)/n$ is bounded. It then follows from (12.3) that the sum $\sum_{p \leq x} \chi(p) \log p/p$ over primes is bounded, and the required result follows from (12.1). The whole proof depends, of course, on the use of Dirichlet characters χ to isolate the primes in an arithmetic progression in the first place, and to use $L(1, \chi) \neq 0$ for $\chi \neq \chi_0$ in the second place.

12.4 Prime Number Theorem in Arithmetic Progressions

There is the weight $(\log p)/p$ attached to each prime p in the sum in Theorem 12.1. Let us define, for $\gcd(m, a) = 1$,

$$\pi(x; m, a) = \sum_{\substack{p \le x \\ p \equiv a \pmod m}} 1, \qquad \psi(x; m, a) = \sum_{\substack{n \le x \\ n \equiv a \pmod m}} \Lambda(n),$$

which are sums with unit weight, and with weight $\log p$ at prime powers $n = p^k$. The expected asymptotic formulae for these sums are that, as $x \to \infty$,

$$\pi(x; m, a) \sim \frac{1}{\phi(m)} \frac{x}{\log x}, \qquad \psi(x; m.a) \sim \frac{x}{\phi(m)};$$

the two results are elementarily equivalent, and either of them is called the *prime number theorem in arithmetic progressions*. The proof depends on $L(s, \chi) \neq 0$ on $\sigma = 1$, for all nonprincipal character $\chi \pmod m$. There is an important provision, namely that the modulus m, although arbitrary, is fixed.

The problem becomes seriously difficult if m is allowed to increase with x; this is to be expected because the arithmetic progression is 'thin' when m is large, so that there are fewer terms for the average argument to arrive at the conclusion.

The deep results that have been established are beyond our scope—we can only mention some of them, and give an example to illustrate their usefulness and importance. See, for example, [20]. The preferred symbol for the modulus of the arithmetic progression is q in such topics, so we comply by changing m to q in the following.

Generalised Riemann Hypothesis

The Generalised Riemann Hypothesis (GRH) states that all the zeros of $L(s, \chi)$ in $0 < \sigma < 1$ lie on the vertical line $\sigma = \frac{1}{2}$, for every character $\chi \pmod q$. Under the hypothesis we have

$$\psi(x; q, a) = \frac{x}{\phi(q)} + O\big(\sqrt{x}\, \log^2 x\big); \tag{12.11}$$

note that although the hypothetical result is strong if $q < \sqrt{x}$, the main term no longer dominates if $q > \sqrt{x}$.

Siegel Zero and Siegel-Walfisz Theorem

The zero-free regions of $L(s, \chi)$ can be obtain in a way similar to that for $\zeta(s)$ when χ is a complex character (mod q). However, when χ is real, there is a potential counterexample to GRH in that it is possible to have such a character with a real zero very near $s = 1$. In 1935 C. L. Siegel [94] established an important result on the value of $L(1, \chi)$ for a real primitive character χ (mod q) (see Sect. 12.7), from which A. Walfisz [106] deduced the following theorem:

Theorem 12.3 (Siegel-Walfisz) *Corresponding to any $A > 0$, there is a positive constant c with the property that for $q \leq \log^A x$, $\gcd(a, q) = 1$ and $x \to \infty$, we have*

$$\psi(x; q, a) = \frac{x}{\phi(q)} + O\left(x \exp(-c\sqrt{\log x})\right).$$

The dependence of c on A is not effective in the sense that its value cannot be computed in terms of A; in particular, the implied constant is not effective.

Linnik's Theorem

After Dirichlet's theorem on the existence of primes p in the arithmetic progression a (mod q), it is natural to ask for the size of $p(q, a)$, the least prime in the progression. The Siegel-Walfisz theorem shows that $\psi(x; q, a) > 0$ if q is small relative to x; for example, it implies $p(q, a) < \exp(q^\epsilon)$ for $\epsilon > 0$ and large q.

In 1944 Yu. V. Linnik proved that there is an absolute constant L such that $p(q, a) \ll q^L$. The proof can be used to derive a bound for L, and C. T. Pan (1957) showed that $L \leq 10000$; after a succession of improvements, the current best bound is $L \leq 5$ by T. Xylouris (2011). It is known that $L \leq 2$ for almost all q and that, under GRH, $p(q, a) \leq (1 + o(1))(\phi(q) \log q)^2$.

Brun-Titchmarsh Theorem

For applications of results in the distribution of primes in arithmetic progressions, we may need the modulus q to vary in a long range, usually well beyond a power of $\log x$, say up to a small power of x. Often we also need to have the range even beyond $\sqrt{x}$, although only a correct order of magnitude for the prime count will be needed for such large q. For such applications we have the useful

Brun-Titchmarsh Theorem

$$\pi(x;q,a) \le \frac{2x}{\phi(q)\log(x/q)}, \qquad \textit{for all} \quad q < x.$$

In practice an unspecified constant in place of 2 will suffice, but it is nice to have the clean bound obtained by Montgomery and Vaughan [67]; their argument shows that it will be difficult to replace 2 by a smaller number.

Bombieri's Theorem and Elliott-Halberstam Conjecture

In many problems which involve the distribution of primes in arithmetic progressions, conditional results can be derived under the assumption of GRH. Sometimes, instead of the full strength of GRH, it suffices to have the expected result of such distributions *on average* as q ranges over an interval. In 1965 E. Bombieri proved his celebrated theorem which can be used to establish various unconditional estimates by the removal of GRH from the argument.

Bombieri's Theorem *Corresponding to any $A > 0$, there exists $B = B(A) > 0$ such that,*

$$\sum_{q \le \sqrt{x}/\log^B x} \max_{(a,q)=1} \left|\psi(x;q,a) - \frac{x}{\phi(q)}\right| \ll \frac{x}{\log^A x}, \qquad x \to \infty.$$

The difference $\psi(x;q,a) - x/\phi(q)$ can be replaced by $\pi(x;q,a) - \mathrm{Li}(x)/\phi(q)$.

The left-hand side can be considered a sum of the error terms of the prime number theorem in arithmetic progressions—it has the trivial bound $\ll x\log^2 x$. The theorem says that an arbitrary power of $\log x$ can be saved on the estimate, provided the range of q is correspondingly short of $\sqrt{x}$. As such, the theorem is as strong as GRH because if we apply (12.11) to the sum the resulting estimates is essentially the same. The implied constant is not effectively computable in terms of A, because the Siegel-Walfisz theorem is used in the proof.

It would be wonderful if a Bombieri's type theorem could be extended with q beyond the $\sqrt{x}$ barrier, thereby even beyond what can be expected from GRH. For this P. D. T. A. Elliott and H. Halberstam proposed the following:

Conjecture (Elliott-Halberstam) For any $\epsilon > 0$ and $A > 0$, we have

$$\sum_{q \le x^{1-\epsilon}} \max_{(a,q)=1} \left|\psi(x;q,a) - \frac{x}{\phi(q)}\right| \ll \frac{x}{\log^A x}, \qquad \text{as} \quad x \to \infty;$$

the implied constant may depend on ϵ and A.

12.5 Titchmarsh-Linnik Theorem

The occurrence of primes among the integers being unpredictable, what can be said of the values taken by the primes after they have been 'shifted' by a fixed number? For example, it will be interesting to investigate the sum

$$D_a(x) = \sum_{a<p\le x} d(p-a),$$

where $d(n)$ is the divisor function. Considering $p-a$ to be just an integer near p, the average value of $d(p-a)$ over $a < p \le x$ is expected to be about $\log x$, according to Theorem 2.11. The number of terms in $D_a(x)$ is $\pi(x) - \pi(a) \sim x/\log x$, so that the asymptotic value of the sum should be linear in x, with a coefficient depending on a. The problem is closely related to our knowledge of the distribution of primes in arithmetic progressions and, in 1931, E. C. Titchmarsh established the asymptotic formula for $D_a(x)$ under GRH:

Theorem 12.4 (Titchmarsh Divisor Problem) *Under the assumption of GRH, we have, as $x \to \infty$,*

$$D_a(x) = E_a x + O\Big(\frac{x \log\log x}{\log x}\Big),$$

where

$$E_a = \prod_{p\nmid a}\Big(1+\frac{1}{p(p-1)}\Big)\prod_{p\mid a}\Big(1-\frac{1}{p}\Big).$$

In 1961 Linnik applied his difficult *dispersion method* to give an unconditional proof of Titchmarsh's asymptotic formula, which we now call the **Titchmarsh-Linnik Theorem**. With Bombieri's theorem and the Brun-Titchmarsh theorem at our disposal, the proof is no longer formidable. We need first to deal with a rather awkward sum, wrapped up in which is the number E_a.

An Awkward Sum and the Number E_a

Lemma 12.5 *For $a > 0$, we have, as $x \to \infty$,*

$$\sum_{n\le x,(n,a)=1}\frac{1}{\phi(n)} = E_a \log x + O(1).$$

Proof *(**T. Estermann**)* Let $f(n) = 1/\phi(n)$ if $(n, a) = 1$, and 0 otherwise, so that

$$nf(n) = \prod_{p|n,\, p\nmid a} \frac{1}{1-1/p} \quad \text{if} \quad (n,a) = 1,$$

and the sum concerned is $\sum_{n\le x} f(n)$. Estermann introduced the multiplicative function

$$g(m) = \mu^2(m) \prod_{p|m,\, p\nmid a} \frac{1}{p-1} \prod_{p|(m,a)} (-1),$$

which satisfies $g(p) = 1/(p-1)$ if $p \nmid a$, so that

$$nf(n) = \prod_{p|n} (1 + g(p)) = \sum_{m|n} g(m).$$

Thus

$$f(n) = \sum_{mr=n} \frac{g(m)}{mr},$$

and hence

$$\sum_{n\le x} f(n) = \sum_{mr\le x} \frac{g(m)}{mr} = \sum_{r\le x} \frac{1}{r} \sum_{m\le x/r} \frac{g(m)}{m}. \tag{12.12}$$

The inner sum converges as $x \to \infty$; indeed, by the definitions for $f(n)$, $g(m)$ and E_a,

$$\sum_{m=1}^{\infty} \frac{g(m)}{m} = \prod_p \Big(1 + \frac{g(p)}{p}\Big) = E_a.$$

We proceed to bound the tail-end of the series by considering

$$m|g(m)| \le a \prod_{p|m} \frac{1}{1-1/p} = a \prod_{p|m} \frac{1}{1-1/p^2} \prod_{p|m} \Big(1 + \frac{1}{p}\Big);$$

the first product on the right-hand side is at most $\zeta(2)$, and the second product is at most $\sum_{h|m} 1/h$, so that

$$\frac{|g(m)|}{m} < \frac{\zeta(2)a}{m^2} \sum_{hk=m} \frac{1}{h} < 2a \sum_{hk=m} \frac{1}{h^3 k^2}.$$

Therefore, for $r \geq 1$,

$$\left| E_a - \sum_{m \leq x/r} \frac{g(m)}{m} \right| < 2a \sum_{hk > x/r} \frac{1}{h^3 k^2} = 2a \Bigg(\sum_{k \leq x/r} \frac{1}{k^2} \sum_{h > x/rk} \frac{1}{h^3} + \sum_{k > x/r} \frac{1}{k^2} \sum_{h=1}^{\infty} \frac{1}{h^3} \Bigg).$$

Making use of

$$\sum_{h > X} \frac{1}{h^3} < \frac{1}{X^2} \quad \text{and} \quad \sum_{k > Y} \frac{1}{k^2} < \frac{1}{Y},$$

with $X = x/rk$ and $Y = x/r$, and $\zeta(3) < 3/2$, we arrive at the bound

$$\left| E_a - \sum_{m \leq x/r} \frac{g(m)}{m} \right| < 2a \sum_{k \leq x/r} \frac{r^2}{x^2} + 3a \sum_{k > x/r} \frac{1}{k^2} < \frac{10ar}{x}.$$

Returning to (12.12) we now have

$$\sum_{n \leq x, (n,a)=1} \frac{1}{\phi(n)} = \sum_{r \leq x} \frac{1}{r} \left(E_a + O\left(\frac{r}{x}\right) \right) = E_a \sum_{r \leq x} \frac{1}{r} + O(1) = E_a \log x + O(1),$$

as required. □

We need only $E = E_1$ for the computation of E_a, because we may write

$$E_a = E \times B_a, \quad \text{where} \quad B_a = \prod_{p \mid a} \frac{p^2 - 2p + 1}{p^2 - p + 1}$$

is a fraction. From the proof of the lemma, it will not be surprising to find that $\zeta(2)\zeta(3)$ is a 'factor' of E, and in fact

$$E = \frac{\zeta(2)\zeta(3)}{\zeta(6)} = \frac{315\zeta(3)}{2\pi^4} = 1{\cdot}9435964\ldots,$$

which follows from the identity

$$\left(1 - \frac{1}{p^2}\right)^{-1} \left(1 - \frac{1}{p^3}\right)^{-1} \left(1 - \frac{1}{p^6}\right) = \left(1 + \frac{1}{p(p-1)}\right).$$

Also, if a, b are coprime then $B_{ab} = B_a B_b$, and if $a = p^\ell$ then $B_a = B_p$. Therefore we need only B_p, and we find, for example, that

$$B_p = \frac{1}{3}, \frac{4}{7}, \frac{16}{21}, \frac{36}{43}, \frac{100}{111}, \frac{144}{157}, \frac{256}{273}, \frac{324}{343}, \qquad p = 2, 3, 5, 7, 11, 13, 17, 19.$$

Proof of the Titchmarsh-Linnik Theorem

Recalling that the divisors of $n \neq m^2$ form pairs q_1, q_2 with $q_1 < \sqrt{n} < q_2$, we let $d(n) = 2\sum_{q|n,\, q<\sqrt{n}} 1 + \delta_n$, where $\delta_n = 1$ or 0, depending on whether n is a square or not. Then

$$\begin{aligned} D_a(x) &= 2 \sum_{a<p\le x} \sum_{q|(p-a),\, q<\sqrt{p-a}} 1 + O(\sqrt{x}) \\ &= 2 \sum_{q<\sqrt{x-a}} \sum_{\substack{q^2+a<p\le x \\ p\equiv a \pmod q}} 1 + O(\sqrt{x}) \\ &= 2 \sum_{q<\sqrt{x},\, (q,a)=1} \Big(\pi(x;q,a) - \pi(q^2+a;q,a)\Big) + O(\sqrt{x}). \end{aligned}$$

First, by the Brun-Titchmarsh theorem, the sum $\sum \pi(q^2+a;q,a)$ over $q < \sqrt{x}$ is

$$\ll \sum_{q<\sqrt{x}} \frac{q^2}{\phi(q)\log x/q} \ll \frac{1}{\log x} \sum_{q<\sqrt{x}} \frac{q^2}{\phi(q)} \ll \frac{x \log\log x}{\log x},$$

because $q/\phi(q) \ll \log\log q$ by Corollary 9.5.

For the sum $\sum \pi(x;q,a)$ over $q < \sqrt{x}$, we apply Bombieri's theorem, with $A = 1$, so that there is a corresponding value of B for us to break the sum up into the sums Σ_1 and Σ_2, for which the ranges are $q \le \sqrt{x}/\log^B x$ and $\sqrt{x}/\log^B x < q < \sqrt{x}$, respectively, together with the coprime condition $(q,a) = 1$. Then, on application of Bombieri's theorem to Σ_1, together with the lemma, we have

$$\Sigma_1 = \mathrm{Li}(x) \sum_{q\le\sqrt{x}/\log^B x,\, (q,a)=1} \frac{1}{\phi(q)} + O\Big(\frac{x}{\log x}\Big) = \tfrac{1}{2}(E_a \log x)\,\mathrm{Li}(x) + O\Big(\frac{x}{\log x}\Big).$$

On application of the Brun-Titchmarsh estimate to the terms in Σ_2, together with the lemma applied to a short interval by taking the difference, we find that

$$\Sigma_2 \ll \frac{x}{\log x} \sum_{\sqrt{x}/\log^B x<q<\sqrt{x},\, (q,a)=1} \frac{1}{\phi(q)} \ll \frac{x\log\log x}{\log x}.$$

Collecting all the estimates, together with

$$\mathrm{Li}(x) = \frac{x}{\log x} + O\Big(\frac{x}{\log^2 x}\Big),$$

the required result follows. □

12.6 Prime Number Races

Suppose that $\gcd(a, m) = \gcd(b, m) = 1$, with a fixed modulus $m > 2$, and x is an integer. By the prime number theorem in arithmetic progressions,

$$\pi(x; m, a) \sim \pi(x; m, b) \sim \frac{x}{\phi(m) \log x} \quad \text{as} \quad x \to \infty,$$

so that, for large x,

$$\Delta_{a,b}(x, m) = \pi(x; m, a) - \pi(x; m.b)$$

is relatively small—we shall write $\Delta_{a,b}(m)$, or just $\Delta_{a,b}$, when there is no need to mention x or m. There are some very interesting and difficult problems on $\Delta_{a,b}$.

Back in 1853, Chebyshev had already observed the bias that $\Delta_{3,1}(x, 4) > 0$ for many more values of x. In the 1914 paper mentioned in Note 11.2, Littlewood also proved that both $\Delta_{3,1}(x, 4)$ and $\Delta_{2,1}(x, 3)$ change signs infinitely often as $x \to \infty$. In a series of many papers starting in 1962, S. Knapowski and P. Turán established various similar results and stated some relevant conjectures—in particular, that $\Delta_{a,1}$ changes signs infinitely often for any $m > 2$ and $a \neq 1$, and they conjectured that the set $\{x : \Delta_{3,1}(x, 4) < 0\}$ should have zero asymptotic density.

The topic of *Chebyshev's bias* is now an actively researched subject called *comparative prime number theory*. We shall adopt the more colourful name of *prime number races*, which is the title of the beautifully written expository article [37] on the subject by A. Granville and G. Martin.

By a (mod m) *race at distance* x, between Team-a and Team-b, we mean the examination of the sign of $\Delta_{a,b}$. Since both prime counts are unbounded, the race never finishes, unless a distance is specified. We are interested in

- whether the sign of $\Delta_{a,b}$ changes infinitely often, and if so,
- whether the 'frequency' of $\Delta_{a,b} > 0$ can be quantified, so that we may predict which team usually dominates.

Insight from computer experiments is particularly important for this type of investigations. Let us take the simplest cases for m with $\phi(m) = 2$, so that there are only two teams in each of the races. This means that $m = 3, 4, 6$, and we need only deal with $m = 3$ and 4, because an odd prime $p \equiv \pm 1 \pmod 3$ implies $p \equiv \pm 1 \pmod 6$. From the list of primes $p < 100$, it is easy to check that, for $m = 4$, Team-3 is always ahead, and for $m = 3$, Team-2 is always ahead, at each distance $x < 100$.

Example 12.1 Let $m = 4$, and $m = 3$, and $x = 10^k$, $k = 2, 3, 4, 5, 6$. Then

Although the same two teams again emerge as leaders at each such distances x in Table 12.1, we may have missed some drama at some distance not covered in the table—it is possible at some $x < 10^6$ when Team-1 may have stolen the lead, if only for a few steps. Indeed there must be such dramas throughout the races as x increases, according to Littlewood's theorem. In particular, there is a least distance

Table 12.1 Prime number races (mod 3) and (mod 4)

	(mod 4)		(mod 3)	
x	Team-1	Team-3	Team-1	Team-2
10^2	11	13	11	13
10^3	80	87	80	87
10^4	609	619	611	617
10^5	4783	4808	4784	4807
10^6	39175	39322	39231	39266

x_1 in the (mod 4) race at which Team-1 surges ahead for the first time. The drama occurs in the interval containing the five successive primes

$$p = 26821, 26833, 26839, 26849, 26861,$$

with only $26839 \equiv 3 \pmod 4$. Thus, at $x = 26820$, the prime counts for Team-1 and Team-3 are 1469 and 1471, respectively. Then Team-1 pulls level after the next two primes, only to lose back again at $x = 26839$. Then the next two primes congruent to 1 (mod 4) enable Team-1 to be ahead at x_1 for the first time with

$$\pi(x_1; 4, 1) = 1473 > 1472 = \pi(x_1; 4, 3), \qquad x_1 = 26861.$$

Team-3 restores the lead after the next two primes 26863 and 26879, which are congruent to 3 (mod 4), and stays ahead for quite a while. As promised by Littlewood, there are more such dramas, and the next two occur in the intervals $616841 \le x \le 633798$ and $12306137 \le x \le 12382326$.

For the (mod 3) race, it seems that Team-2 dominates much more—so much so that it is not easy to find the least x_2 at which Team-1 first takes over the lead. The corresponding drama was unfolded on Christmas Day of 1976 when C. Bays and R. H. Hudson found that

$$\pi(x_2; 3, 1) > \pi(x_2; 3, 2), \qquad x_2 = 608\,981\,813\,029.$$

It would be extremely tedious to verify that $\pi(x; 3, 1) \le \pi(x; 3, 2)$ for *each* $x < x_2$, and the value of x_2 is unlikely to emerge before CCH by such exhaustive search. The sophisticated method used involves computed values of the complex zeros of the $L(s, \chi)$, for the nonprincipal character $\chi \pmod 3$, from which regions of x with $\pi(x; 3, 1) > \pi(x; 3, 2)$ can be located. A value $x^\star$ is then selected in the first region, and $\pi(x^\star; 3, 1)$ is evaluated, using the Meissel-Lehmer method, for example. The sieve of Eratosthenes is then applied to isolate the primes near $x^\star$, so that the exact value of x_2 is identified.

We still need to explain why Team-3 and Team-2 dominate in the (mod-4) and (mod 3) races respectively. Let us gather more data by taking $m = 8$ and 10, so that there are now $\phi(m) = 4$ teams in these races.

Table 12.2 Prime number race (mod 8)

x	Team-1	Team-3	Team-5	Team-7
10^2	5	7	6	6
10^3	37	44	43	43
10^4	295	311	314	308
10^5	2384	2409	2399	2399
10^6	19552	19653	19623	19669

Table 12.3 Prime number race (mod 10)

x	Team-1	Team-3	Team-7	Team-9
10^2	5	7	6	5
10^3	40	42	46	38
10^4	306	310	308	303
10^5	2387	2402	2411	2309
10^6	19617	19665	19621	19593

Example 12.2 Let $m = 8$, and $x = 10^k$, $k = 2, 3, 4, 5, 6$. Then, see Table 12.2.

Example 12.3 Let $m = 10$, and $x = 10^k$, $k = 2, 3, 4, 5, 6$. Then, see Table 12.3.

Example 12.4 Consider the 'Grand National' with $m = 100$, so that there are $\phi(m) = 40$ teams at distance $x = 10^k$, $k = 3, 4, 5, 6$. The prime counts of leaders at distance x are set out in brackets: At $x = 10^3$, Team-57 and Team-83 are joint-leaders; at $x = 10^4$, Team-19 is the leader; at $x = 10^5$, Team-97 is the leader; at $x = 10^6$, Team-71 is the leader. The results are set out in Table 12.4.

Rubinstein-Sarnak Theorem

The data for $m = 3, 4, 8$, over the chosen distances, indicate that Team-1 is not a leader, and a possible explanation for this is that *an odd square is congruent to* 1 (mod 8). This may be relevant because prime squares are likely to be of secondary importance in prime counts, so it would appear that Team-1 should lose to the other teams, for most of the distances, in the (mod 8) race. The data for $m = 10$ seem to indicate that Team-1 and Team-9 tend to trail behind Team-3 and Team-7, and this is consistent with the speculation: $p^2 \equiv 1 \pmod 4$, $p^2 \equiv 1 \pmod 3$, $p^2 \equiv \pm 1 \pmod{10}$ if $p > 10$. Indeed, by the same token, we may speculate that, in a (mod m) race, Team-a is more likely to lose out to Team b at most distances x if a is a square (mod m) and b is not. If such reasoning is not just speculation, we may ask if there can be any quantitative results accompanying the predictions? These are seriously difficult problems because the assumption that prime squares are likely to be of secondary importance in prime counts depends on the truth of the Riemann hypothesis. Nevertheless some very encouraging hypothetical results have been obtained in recent times.

Table 12.4 Grand national for prime numbers

Team	10^3	10^4	10^5	10^6	Team	10^3	10^4	10^5	10^6
1	4	30	243	1964	51	3	33	243	1970
3	3	31	237	1969	53	5	30	242	1977
7	5	32	237	1932	57	(6)	32	239	1961
9	5	29	237	1957	59	4	29	242	1967
11	5	32	244	1958	61	4	30	229	1960
13	4	28	243	1965	63	5	31	237	1962
17	3	29	244	1970	67	5	32	240	1943
19	5	(35)	238	1973	69	3	28	235	1955
21	3	34	243	1937	71	4	27	238	(1986)
23	4	35	246	1976	73	5	32	235	1956
27	4	28	244	1976	77	5	30	233	1960
29	4	34	245	1926	79	4	30	242	1960
31	5	31	245	1964	81	3	30	238	1942
33	3	32	236	1967	83	(6)	30	244	1969
37	4	33	243	1973	87	4	31	244	1984
39	5	31	248	1970	89	2	29	240	1958
41	5	28	228	1955	91	4	31	236	1981
43	4	31	236	1959	93	3	30	246	1965
47	5	31	237	1956	97	5	30	(250)	1966
49	3	30	233	1960	99	3	28	230	1967

First, it has been shown that the Knapowski-Turán conjecture is false because, similar to the set S in Littlewood's theorem (see Note 11.2), the counting function of the set $\{x : \pi(x; 4, 3) < \pi(x; 4, 1)\}$ fluctuates to such an extend that it does not have an asymptotic density. In particular, there is a fixed $\epsilon_0 > 0$ with the property that there are arbitrarily large X such that the number of $x \le X$ satisfying $\pi(x; 4, 3) < \pi(x; 4, 1)$ is at least $\epsilon_0 X$. Again, instead of asymptotic density, it turns out that the correct measure for sets involved in prime number races is the logarithmic measure.

In the following theorem of Rubinstein and Sarnak [85] we shall assume the Generalised Riemann Hypothesis so that the zeros of $L(s, \chi)$ are of the form $\frac{1}{2} \pm i\gamma_n$, where $(\gamma_n) = (\gamma_1, \gamma_2, \dots)$ is an increasing sequence of positive terms. We further assume that these zeros are linearly independent over the rationals, so that, in particular, they are simple zeros of $L(s, \chi)$. It has to be said that, although these are highly plausible assumptions, their proofs are unlikely to be forthcoming, certainly not in the near future.

Theorem 12.5 (Rubinstein and Sarnak [85]) *Let* $S = \{x : \pi(x; 4, 3) > \pi(x; 4, 1)\}$ *and* $T = \{x : \pi(x; 3, 2) > \pi(x; 3, 1)\}$. *Then, as* $X \to \infty$,

$$\frac{1}{\log X} \sum_{x \le X,\, x \in S} \frac{1}{x} \to 0{\cdot}9959\dots, \quad \textit{and} \quad \frac{1}{\log X} \sum_{x \le X,\, x \in T} \frac{1}{x} \to 0{\cdot}9990\dots.$$

The hypothetical theorem shows that, with respect to logarithmic measure, Chebyshev was correct about 99·6% of the time! It also explains why x_2 in the (mod 3) race is more difficult to locate than x_1 is in the (mod 4) race. Returning to our speculation concerning races between Team-a and Team-b, taking into account the quadratic characters of a and b (mod m), Rubinstein and Sarnak also demonstrated, under the same assumptions, that the set $\{x : \pi(x; m, a) > \pi(x; m, b)\}$ has logarithmic density $\frac{1}{2}$ if a, b have the same quadratic character (mod m)—that is either both residues, or both nonresidues (mod m).

What about predictions for the Grand National in Example 12.4? According to the Shanks-Rényi conjecture, given any ordering of the 40 teams, there are infinitely many distances x at which the ordering holds, if only fleetingly. Indeed, the method of Rubinstein and Sarnak may even deliver the associated positive odds for this to happen in logarithmic measure, albeit hypothetically.

12.7 Primitive Characters

The non-principal characters χ (mod q) can be divided into two types, called *primitive* and *imprimitive* characters. Many results about χ, and the corresponding function $L(s, \chi)$, take a simple form only for primitive characters. Some such results can be extended to imprimitive characters, but usually either with a fair amount of complications, or a loss of some sharpness of the result.

Let ψ be a character (mod d), with $d|q$, and let χ (mod q) be defined by

$$\chi(n) = \begin{cases} \psi(n), & \text{if} \quad \gcd(n, q) = 1; \\ 0, & \text{otherwise.} \end{cases}$$

Then $\chi(n)$ should be a character (mod q), but we need to check that it is *completely* multiplicative, that is $\chi(mn) = \chi(m)\chi(n)$, for any pair $m, n \in \mathbb{Z}$. The equation is trivial if $\gcd(mn, q) > 1$, so we suppose that $\gcd(m, q) = \gcd(n, q) = 1$. Now, for any such pair m, n there exists k such that $n + kq \equiv 1$ (mod m); this is because, as k runs over the residues (mod m), so does $n + kq$, and particularly 1 (mod m). Thus, for the specific k, we have $\gcd(m, n + kq) = 1$, and hence that

$$\chi(mn) = \chi(mn + mkq) = \chi(m(n + kq)) = \chi(m)\chi(n + kq) = \chi(m)\chi(n).$$

With χ so defined, we say that ψ (mod d) *induces* χ (mod q), which is then said to be imprimitive; we also call d the induced modulus. A character χ (mod q) is primitive if it is not induced by any character other than χ itself; every non-principal character χ modulo an odd prime q is then primitive.

If $\psi \pmod d$ induces $\chi \pmod q$ then the two characters are essentially the same in the sense that $\chi(p) = \psi(p)$ for all primes p apart for a finite number of exceptions, and in fact

$$L(s, \chi) = L(s, \psi) \prod_{p|q} \Big(1 - \frac{\psi(p)}{p^s}\Big).$$

The product of a character $\pmod{q_1}$ and a character $\pmod{q_2}$ is a character with modulus $\mathrm{LCM}(q_1, q_2)$. If a character with a minimum modulus can be factorised into two characters, one of a smaller modulus and the other being principal, then the character is imprimiitve; otherwise it is primitive. If $\chi \pmod q$ is imprimitive then there is a unique primitive character $\psi \pmod d$ that induces $\chi \pmod q$. It can be established that if χ_1 and χ_2 are characters modulo coprime integers q_1 and q_2, and $\chi(n) = \chi_1(n)\chi_2(n)$, then χ is primitive modulo q_1q_2 if and only if both χ_1 and χ_2 are primitive.

We need perhaps a more suitable condition to see whether χ is a primitive character or not.

Theorem 12.6 *The following is a necessary and sufficient condition for a non-principal $\chi \pmod q$ to be primitive: If $d|q$ and $d < q$, then there exists c such that*

$$c \equiv 1 \pmod d, \qquad \gcd(c, q) = 1, \qquad \chi(c) \neq 1. \tag{12.13}$$

Proof Necessity. Suppose that $\chi \pmod q$ is primitive, and let $d|q$ and $d < q$. Being primitive, χ is not induced by a character $\pmod d$, so that there are m, n such that

$$m \equiv n \pmod d, \qquad \chi(m) \neq \chi(n), \qquad \chi(mn) \neq 0.$$

We may now choose c so that $\gcd(c, d) = 1$, $cm \equiv n \pmod q$. Then $cm \equiv n \pmod d$ and $\chi(c)\chi(m) = \chi(n)$, and they imply $c \equiv 1 \pmod d$, and $\chi(c) \neq 1$.

Sufficiency. Write $q = dq'$, let c satisfy (12.13), and let also $a \in \mathbb{Z}$. Then, as k runs through a complete residue system $\pmod{q'}$, the numbers $n = ac + kcd$ run through the complete residue system $\pmod q$ with $n \equiv ac \equiv a \pmod d$. It follows that

$$\begin{aligned}
\sum_{\substack{0\le n<q \\ n\equiv a \pmod d}} \chi(n) &= \sum_{0\le k<q'} \chi(ac + kcd) \\
&= \chi(c) \sum_{0\le k<q'} \chi(a + kd) \\
&= \chi(c) \sum_{\substack{0\le n<q \\ n\equiv a \pmod d}} \chi(n),
\end{aligned}$$

so that the value of the sum concerned is 0, because $\chi(c) \neq 1$.

Now set $a = 1$, so that $\chi(1) = 1$ is a term in the sum, and we deduce that there must be another term $\chi(n)$ which is neither 1, nor 0, and $n \equiv 1 \pmod{d}$. This means that d cannot be an induced modulus of q, so that χ is primitive. □

Gauss Sums Associated with Dirichlet Characters

We now extend the notion of a quadratic Gauss sum (see Sects. 2.12 and 4.5) by replacing the Legendre symbol with a Dirichlet character $\chi \pmod{q}$, that is

$$\tau(\chi) = \sum_{0 \le m < q} \chi(m) \mathrm{e}\Big(\frac{m}{q}\Big).$$

We may generalise the sum by setting

$$c_\chi(n) = \sum_{0 \le m < q} \chi(m) \mathrm{e}\Big(\frac{mn}{q}\Big);$$

note that if $\chi = \chi_0$ is the principal character, then we have the Ramanujan sum (see Sect. 2.12)

$$c_q(n) = \sum_{\substack{0 \le m < q \\ \gcd(m,q)=1}} \mathrm{e}\Big(\frac{mn}{q}\Big).$$

For a character χ a very useful relation is the equation

$$\chi(n)\tau(\bar{\chi}) = \sum_{0 \le m < q} \bar{\chi}(m) \mathrm{e}\Big(\frac{mn}{q}\Big), \qquad \gcd(n, q) = 1. \tag{12.14}$$

To see this, we note that the condition $\gcd(n, q) = 1$ implies that as m runs over the residues $\pmod{q}$, so does mn, so that

$$\chi(n)c_\chi(n) = \sum_{0 \le m < q} \chi(mn) \mathrm{e}\Big(\frac{mn}{q}\Big) = \tau(\chi),$$

and replacing χ by $\bar{\chi}$ delivers (12.14).

What if $\gcd(n, q) > 1$? Then $\chi(n) = 0$, of course, and (12.14) still holds provided that χ is primitive. For the proof, we appeal to Theorem 12.6, and choose a and d so that $\gcd(a, d) = 1$ and $a/d = n/q$. Then

$$\sum_{0 \le m < q} \bar{\chi}(m) \mathrm{e}\Big(\frac{mn}{q}\Big) = \sum_{0 \le h < d} \mathrm{e}\Big(\frac{ah}{d}\Big) \sum_{\substack{0 \le m < q \\ m \equiv h \pmod{d}}} \chi(m).$$

Since $d|q$ and $d < q$, we see from the proof of Theorem 12.6 that the value of each inner sum is 0, so that (12.14) holds also for primitive characters.

As a corollary, we can deduce that $|\tau(\chi)| = \sqrt{q}$ for primitive characters $\chi \pmod q$. Thus we have

$$\begin{aligned}\phi(q)|\tau(\chi)|^2 &= \sum_{0\le n<q}\Big|\sum_{0\le m<q}\chi(m)\mathrm{e}\Big(\frac{mn}{q}\Big)\Big|^2\\ &= \sum_{0\le a,b<q}\chi(a)\bar\chi(b)\sum_{0\le n<q}\mathrm{e}\Big(\frac{(a-b)n}{q}\Big).\end{aligned}$$

The innermost sum is 0, except when $a \equiv b \pmod q$, in which case its value is q. Therefore $\phi(q)|\tau(\chi)|^2 = \phi(q)q$, and hence $|\tau(\chi)| = \sqrt{q}$.

With $\tau(\chi) \neq 0$ for a primitive character χ, we may rewrite (12.14) as

$$\chi(n) = \frac{1}{\tau(\bar\chi)}\sum_{0\le m<q}\bar\chi(m)\mathrm{e}\Big(\frac{mn}{q}\Big),$$

which may be considered as the finite Fourier expansion of the Dirichlet character; see Sect. 2.11.

Pólya-Vinogradov Inequality

We saw in Theorem 9.5 that for a Dirichlet character $\chi \pmod q$,

$$\Big|\sum_{n\le x}\chi(n)\Big| \le \phi(q).$$

There can be no improvement if $\chi = \chi_0$ is the principal character and $x = q$, but there can be considerable saving when χ is a primitive character.

Theorem 12.7 (Pólya-Vinogradov) *Let χ be a primitive character* $(\mathrm{mod}\, q)$. *Then*

$$\Big|\sum_{n\le x}\chi(n)\Big| < \sqrt{q}\log q.$$

Proof Since χ is primitive, we may apply (12.14), and replace the summation range there to $-\frac{q}{2} < m < \frac{q}{2}$. Thus

$$\tau(\bar\chi)\sum_{n\le x}\chi(n) = \sum_{0<|m|\le\frac{q}{2}}\overline{\chi(m)}\sum_{n\le x}\mathrm{e}\Big(\frac{mn}{q}\Big).$$

The inner sum is a geometric progression, with $q > |m|$, so that $q \nmid m$, and

$$\sum_{n\le x} \mathrm{e}\Big(\frac{mn}{q}\Big) = 1 + r + r^2 + \cdots + r^{\lfloor x\rfloor} = \frac{r^{\lfloor x\rfloor+1} - 1}{r-1}, \qquad r = e^{2\pi i m/q} \neq 1.$$

Making use of $|r^{\lfloor x\rfloor+1} - 1| \le 2$, and

$$e^{2i\theta} - 1 = e^{i\theta}(e^{i\theta} - e^{-i\theta}) = 2e^{i\theta}\sin\theta, \qquad \frac{\sin\theta}{\theta} > \frac{2}{\pi}, \qquad \text{for} \quad 0 < \theta < \frac{\pi}{2},$$

we find (with $\theta = \frac{\pi m}{q}$), that

$$\left|\sum_{n\le x} \mathrm{e}\Big(\frac{mn}{q}\Big)\right| \le \frac{1}{|\sin \pi m/q|} < \frac{q}{2m}.$$

It now follows that

$$|\tau(\bar\chi)| \left|\sum_{n\le x} \chi(n)\right| \le \frac{q}{2} \sum_{0<|m|\le \frac{q}{2}} \frac{1}{m} < q\log q,$$

and the required result follows from $|\tau(\bar\chi)| = \sqrt{q}$. □

Corollary 12.1 *Let $\epsilon > 0$. Then for any sufficiently large odd q, there is some $a < q^{\frac{1}{2}+\epsilon}$ which is a quadratic nonresidue (mod q).*

Proof Taking $\chi(n) = \left(\frac{n}{q}\right)$, the theorem implies that we cannot have $\left(\frac{a}{q}\right) = 1$ for all $a < q^{\frac{1}{2}+\epsilon}$ coprime with q, so that there must be some such a with $\left(\frac{a}{q}\right) = -1$. □

It can be shown, by an averaging argument, that for χ primitive,

$$\max_{u,v} \left|\sum_{u\le n<u+v} \chi(n)\right| \ge \frac{\sqrt{q}}{\pi},$$

and R. E. A. C. Paley has established in 1932 that there are infinitely many quadratic characters χ with

$$\max_{u,v} \left|\sum_{u\le n<u+v} \chi(n)\right| \gg \sqrt{q}\log\log q.$$

Anyway, the Pólya-Vinogradov inequality is near best possible apart from a factor of $\log q$. There are various other applications of the Pólya-Vingradov inequality, including the proof of Bombieri's theorem mentioned in the above. The case when

χ is imprimitive is more complicated, but it can be established that

$$\sum_{u \le n < v} \chi(n) \ll \sqrt{q} \log q.$$

12.8 Summary and Notes

Summary

- For coprime numbers a, m, there are infinitely many primes p in the arithmetic progression $a \pmod m$; indeed, as $x \to \infty$,

$$\sum_{\substack{p \le x \\ p \equiv a \pmod m}} \frac{\log p}{p} = \frac{\log x}{\phi(m)} + O(1).$$

  The proof makes use of Dirichlet character $\chi \pmod m$ to isolate the terms in the arithmetic progression concerned. The dominating term in the sum then manifests itself, but the argument that the remaining terms are bounded is lengthy. The vital auxiliary result is that $L(1, \chi) \neq 0$ when χ is a non-principal character, and the case when χ is a real character requires special attention.
- Let $\pi(x; m, a)$ count the primes in the arithmetic progression $p \equiv a \pmod m$, and let $\psi(x; m, a)$ and $\theta(x; m, a)$ be the corresponding counts with the usual weights attached. The method used in the proof of the prime number theorem can be extended to show that, for any fixed m,

$$\pi(x; m, a) \sim \frac{x}{\phi(m) \log x}, \quad x \to \infty, \quad \gcd(a, m) = 1,$$

  and the corresponding similar formulae for $\psi(x; m, a)$ and $\theta(x; m, a)$. The result is called the *prime number theorem in arithmetic progressions*.
- The *generalised Riemann hypothesis* states that all the zeros of $L(s, \chi)$ in $0 < \sigma < 1$ lie on the vertical line $\sigma = \frac{1}{2}$, for every character $\chi \pmod q$. Under the hypothesis we have the estimate $\psi(x; q, a) = x/\phi(q) + O(\sqrt{x} \log^2 x)$, uniformly in q, a, with $q < \sqrt{x}$.
- The Siegel-Walfisz theorem gives $\psi(x; q, a) = x/\phi(q) + O(x \exp(-c\sqrt{\log x}))$, with $c > 0$, but the implied constant is non-effective. The theorem gives an estimate of prime counts in an arithmetic progression with $q < \log^B x$, for any B.
- Linnik's theorem states that the least prime $p(q, a)$ in the arithmetic progression $a \pmod q$ satisfies $p(q, a) \ll q^L$, with $L = 5$ being admissible from recent results.
- The Brun-Titchmarsh theorem states that $\pi(x; q, a) \le 2x \big/ \big(\phi(q) \log(x/q)\big)$; the important point being that it holds for all $q < x$.

- Bombieri's theorem (see the text) shows that an arbitrary power of $\log^A x$ can be saved when the error terms in prime counts over arithmetic progressions are averaged over $\sqrt{x}/\log^B x$.
- The Titchmarsh-Linnik theorem states that the sum $\sum_{a<p\le x} d(p-a)$, where $d(n)$ is the divisor function, and p denotes a prime, has the asymptotic value $E_a x$, where the constant E_a is given in the text.
- There are some very interesting but difficult problem when considering the existence of bias in the distribution of primes in different arithmetic progressions with the same modulus. See the text for some results, actual, or hypothetical.
- Results about Dirichlet characters χ, and the corresponding L-functions, take a simpler form when χ is a *primitive* character.
- The Pólya-Vinogradov inequality is a bound for an incomplete character sum: For a primitive character χ,

$$\left|\sum_{n\le x}\chi(n)\right| < \sqrt{q}\log q;$$

an immediate application is that the least non-residue (mod q) satisfies the bound $\ll q^{\frac{1}{2}+\epsilon}$ for any $\epsilon > 0$.

Notes

1. The following argument showing that $L(1,\chi) \neq 0$ for a complex character χ (mod q) appeals more directly to analytic function theory.

   Let χ be any one of the $\phi(q)$ character (mod q); we shall make use of $\sum_\chi \chi(n) = \phi(q)$ if $n \equiv 1 \pmod q$, and 0 otherwise. Meanwhile, consider

$$L(s,\chi) = \sum_{n=1}^{\infty}\frac{\chi(n)}{n^s} = \prod_p\left(1-\frac{\chi(p)}{p^s}\right)^{-1}, \qquad \sigma > 1.$$

   Making the substitution $z = \chi(p)/p^s$, so that $|z| < 1$, in the series expansion for $\log(1-z)$, and noting that $\chi(p)^k = \chi(p^k)$, because $\chi(n)$ is completely multiplicative, we find that

$$\log L(s,\chi) = \sum_p\sum_{k=1}^{\infty}\frac{\chi(p^k)}{kp^{ks}}, \qquad \sigma > 1.$$

   The double sum is essentially a single sum over the prime powers $n = p^k$. Indeed, with $\log n = k\log p$ and recalling that, for such n, the Mangoldt function

is given by $\Lambda(n) = \log p$, and 0 otherwise, we now have

$$\log L(s, \chi) = \sum_{n=2}^{\infty} \frac{\Lambda(n)\,\chi(n)}{(\log n)\, n^s}, \qquad \sigma > 1.$$

On exponentiating, taking the product over all characters $\chi \pmod q$, we arrive at

$$\prod_{\chi} L(s, \chi) = \exp\left(\sum_{\chi} \sum_{n=2}^{\infty} \frac{\Lambda(n)\,\chi(n)}{(\log n)\, n^s} \right)$$
$$= \exp\left(\phi(q) \sum_{\substack{n \geq 2 \\ n \equiv 1 \pmod q}} \frac{\Lambda(n)}{(\log n)\, n^s} \right), \quad \sigma > 1.$$

Now set $s = \sigma > 1$, so that the sum concerned is a non-negative real number, which implies $\prod_{\chi} L(\sigma, \chi) \geq 1$. Since $L(s, \chi)$ is analytic in $\sigma > 0$, except for a simple pole at $s = 1$ when $\chi = \chi_0$ is the principal character $\pmod q$, it follows from the bound for the product that there can be at most one χ for which $L(1, \chi) = 0$. As we already saw, a complex character χ with $L(1, \chi) = 0$ implies $L(1, \bar{\chi}) = 0$, and $\bar{\chi} \neq \chi$. Therefore $L(1, \chi) \neq 0$ for complex χ.

2. The proof of Bombieri's theorem and the Siegel-Walfisz theorem can be found in [22]; the book also contains an explanation of why GRH and the linear independence of zeros are relevant in some problems concerned with primes in arithmetic progressions.
3. A second dominating term of the Titchmarsh-Linnik theorem has been obtained; see (10.2), the proof of which can be found in [13].

   Let $d_3(n) = \sum_{abc=n} 1$. It can been shown, under the assumption of the Elliott-Halberstam conjecture, there is a constant E'_a such that

$$\sum_{a < p \leq x} d_3(p - a) = E'_a x \log x + O(x), \qquad x \to \infty.$$

12.9 Exercises

1. Evaluate $\pi(10^6; 7, a)$ for each $a = 1, 2, 3, 4, 5, 6$.
2. Verify that $L(1, \chi) > 0$ for each of the two real nonprincipal character $\chi \pmod{15}$.
3. Prove that, for the principal character $\chi_0 \pmod q$, as $s \to 1$ from $s > 1$,

$$(s - 1)L(s, \chi_0) \to \frac{\phi(q)}{q}.$$

4. Prove that, for each $\chi \neq \chi_0 \pmod q$,

$$\sum_{n>x} \frac{\chi(n)}{n} \ll \frac{1}{x}, \quad x \to \infty.$$

5. Let $\gcd(m, a) = 1$. Prove that

$$\sum_{\substack{p \leq x \\ p \equiv a \pmod m}} \frac{1}{p} \sim \frac{\log\log x}{\phi(m)} \quad \text{as} \quad x \to \infty.$$

6. Show that there is a prime p such that both $p \pm 1$ have more than 1000 divisors.
7. Let $K \in \mathbb{N}$. Show that there exists a prime p such that $p \pm k$ is composite for each k in $1 \leq k \leq K$.
8. Let $D_a(x) = \sum_{a<p\leq x} d(p-a)$. Compute $D_a(10^6)$ for $a = 1, 2, 3, 4, 5, 6, 7$. What can be said of the error term in the Titchmarsh-Linnik theorem?
9. Run the prime number races for the moduli $m = 3, 4, 8, 10$ at distances $x = 10^k$ for $k = 7, 8$.
10. Tabulate the results of the 'Regional National' for the modulus $m = 11$ at distances $x = 10^k$ for $k = 6, 7, 8$.

Chapter 13
Smooth Numbers



Abstract It is explained why smooth numbers are important in practice and also in theory. Ramanujan's claim on numbers that are 3-smooth is given a thorough investigation. The Dickman function $\rho(\mathrm{u})$ is introduced together with discussions on its asymptotic formula, starting with the work of de Bruijn, and the more recent contributions from Hildebrand. It is explained how the values of $\rho(\mathrm{u})$ can be computed accurately with the use of power series. Rankin's trick is explained with the application of the estimate of the count of smooth numbers. There is an example of the use of smooth numbers to help to solve the discrete logarithm problem.

Driven mainly by the need to solve concrete problems in computational number theory, there is now a small cottage industry in the investigation of the very useful subject of *smooth numbers*, which are, roughly speaking, numbers without large prime divisors. For example, we saw in Sect. 7.4 that in some factorisation schemes a significant step involves the search for numbers whose product is a square, and knowledge of the distribution of smooth numbers enables us to know how likely we are able to find such a product. In Sect. 13.6 below, we show that smooth numbers can help to solve the discrete logarithm problem. Beside such practical applications smooth numbers are also involved in the study of some deep problems in analytic number theory, such as the gaps between primes, and the fact that there are infinitely many Carmichael numbers.

Denote by $P(n)$ the largest prime divisor of $n > 1$, and set $P(1) = 1$. A number n is said to be y-*smooth* if $P(n) \le y$, and we let

$$\Psi(x, y) = |\{n : n \le x,\ P(n) \le y\}| = \sum_{n \le x,\ P(n) \le y} 1$$

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count such numbers up to x. Obviously if $y \geq x$ then $\Psi(x, y) = \lfloor x \rfloor$; also because a number $n \leq x$ can have at most one prime divisor $p > \sqrt{x}$, it follows that if $\sqrt{x} \leq y \leq x$ then

$$\begin{aligned}\Psi(x, y) &= \lfloor x \rfloor - \sum_{y < p \leq x} \sum_{\substack{n \leq x \\ p \mid n}} 1 = \lfloor x \rfloor - \sum_{y < p \leq x} \left\lfloor \frac{x}{p} \right\rfloor \\ &= x - x \sum_{y < p \leq x} \frac{1}{p} + O(\pi(x)) = x\Big(1 - \log \frac{\log x}{\log y}\Big) + O\Big(\frac{x}{\log x}\Big),\end{aligned} \tag{13.1}$$

by Mertens' theorem (Theorem 9.3 (i)).

13.1 Introduction

In his famous letter to G. H. Hardy on 16 January 1913, S. Ramanujan wrote that

$$\Psi(n, 3) = \frac{1}{2} \frac{\log(2n) \log(3n)}{\log 2 \log 3}. \tag{13.2}$$

The formula is, of course, meant to be an approximation—Ramanujan was not concerned with the proof, and he cared less about the quality of the error involved. We shall see that if p_k denotes the kth prime then, for a fixed k,

$$\Psi(x, p_k) \sim \frac{\log^k x}{k! \log p_1 \log p_2 \cdots \log p_k}, \qquad x \to \infty. \tag{13.3}$$

It is important to have estimates of $\Psi(x, y)$ with y being allowed to vary with x, and good explicit bounds are particularly useful.

In 1930 K. Dickman showed that, corresponding to each fixed $u > 0$, there is a positive $\rho(u)$ such that

$$\Psi(x, x^{1/u}) \sim \rho(u)x, \qquad x \to \infty;$$

his argument also indicates that $\rho(u)$ satisfies the differential-difference equation

$$u\, \rho'(u) = -\rho(u - 1), \qquad u > 1, \tag{13.4}$$

with the obvious initial condition that $\rho(u) = 1$ for $0 < u \le 1$; we define $\rho(0) = 1$, so that ρ is continuous in $[0, \infty)$. Setting $u = (\log x)/\log y$, it follows from (13.1) that

$$\rho(u) = \begin{cases} 1, & 0 \le u < 1, \\ 1 - \log u, & 1 \le u < 2, \end{cases} \tag{13.5}$$

and, from (13.4), that

$$\rho(u) = 1 - \log u + \int_2^u \frac{\log(v-1)}{v}\, dv, \qquad 2 \le v \le 3.$$

It becomes tedious to derive a useful explicit formula for $\rho(u)$ when $u > 3$, but its value can be computed numerically. We call $\rho(u)$ the *Dickman function* and, in the following, unless otherwise stated, we shall assume that

$$y = x^{1/u}, \quad \text{so that} \quad x = y^u \quad \text{and} \quad u = \frac{\log x}{\log y}, \qquad x \ge y \ge 2.$$

Rigorous proofs for Dickman's heuristic argument were given subsequently. In particular, in 1951, N. G. de Bruijn made significant progress by establishing an asymptotic formula for $\Psi(x, y)$ as $x \to \infty$, (see (13.8) below) which is valid in a certain interval for u. His basic tool is the *Buchstab iterative formula*

$$\Psi(x, y) = \Psi(x, z) - \sum_{y < p \le z} \Psi\Big(\frac{x}{p}, p\Big), \quad p \text{ prime}, \quad 1 \le y \le x; \tag{13.6}$$

this is a consequence of the simpler case that $\Psi(x, y) = 1 + \sum_{p \le y} \Psi(x/p, p)$, which follows from the observation that $\Psi(x/p, p)$ counts the numbers $n \le x$ with $P(n) = p$. The identity allows one to approximate $\Psi(x, y)$ by induction on $\lfloor u \rfloor$. As a corollary to his asymptotic formula for $\Psi(x, y)$, de Bruijn also derived an asymptotic formula for $\rho(u)$ as $u \to \infty$ (see (13.9) below). The function $\rho(u)$ decreases very rapidly to 0 as $u > 1$ increases; it can be shown that (see also (13.10))

$$\frac{1}{2\Gamma(2u+1)} \le \rho(u) \le \frac{1}{\Gamma(u+1)},$$

where Γ is the gamma function; see [68].

We also require estimates for $\Psi(x, y)$ when y is much smaller than a power of x; for example,

$$y = L_c(x) = \exp(c\sqrt{\log x \log\log x}\,), \qquad c \text{ being a positive constant},$$

or just $y = \log x \log\log x$, or even for a fixed but large y. When it comes to practical problems, we need good bounds for $\Psi(x, y)$, with all the constants involved being

made explicit. A beautiful clean result by A. Hildebrand states that

$$\Psi(x+z,y) \le \Psi(x,y) + \Psi(z,y), \qquad x, z > y,$$

provided that y is large, but we need to wait and see if this is true for all $y \ge 2$.

13.2 Asymptotic Formula for $\Psi(x, p_k)$

Let us first consider Ramanujan's assertion for $\Psi(x,3)$ in (13.2). A 3-smooth number has the form $2^a 3^b$, with $a, b \ge 0$, so that, on writing

$$A = \log 2, \quad B = \log 3, \quad X = \log x,$$

we have

$$\Psi(x,3) = \sum_{1 \le 2^a 3^b \le x} 1 = \sum_{0 \le a \log 2 + b \log 3 \le \log x} 1 = \sum_{0 \le aA + bB \le X} 1.$$

The value of $\Psi(x,3)$ then has the geometric interpretation of being the number of lattice points (a,b) in the triangular region below the diagonal

$$Ax + By = X \text{ of the rectangle } \Big\{(x,y) : 0 \le x \le \frac{X}{A},\ 0 \le y \le \frac{X}{B}\Big\}.$$

The area of the triangle is $X^2/2AB$, and its perimeter is $\ll X$, so that by Jarnik's theorem (see Note 7.5),

$$\Psi(x,3) = \frac{X^2}{2AB} + O(X) = \frac{\log^2 x}{2 \log 2 \log 3} + O(\log x), \qquad x \to \infty.$$

The asymptotic formula can also be derived by the evaluation of the defining double sum: we have

$$\begin{aligned}
\Psi(x,3) &= \sum_{0 \le a \le X/A} \sum_{0 \le b \le (X - aA)/B} 1 = \sum_{0 \le a \le X/A} \Big(1 + \Big\lfloor \frac{X - aA}{B} \Big\rfloor\Big) \\
&= \frac{X}{B}\frac{X}{A} - \frac{A}{B} \sum_{a \le X/A} a + O(X) = \frac{X^2}{AB} - \frac{A}{B}\frac{1}{2}\frac{X^2}{A^2} + O(X) \\
&= \frac{X^2}{2AB} + O(X) = \frac{\log^2 x}{2 \log 2 \log 3} + O(\log x).
\end{aligned}$$

Note that the result is less accurate than Ramanujan's claim (13.2), which is

$$\frac{\log(2x)\log(3x)}{2\log 2\log 3} = \frac{\log^2 x}{2\log 2\log 3} + \frac{1}{2}\Big(\frac{1}{\log 2} + \frac{1}{\log 3}\Big)\log x,$$

and it will be relevant to remark that the lengths of the horizontal and vertical sides of the said triangle are $X/A = \log x/(\log 2)$ and $X/B = \log x/(\log 3)$. Indeed, if we apply Diophantine analysis to deal with the integer part $\lfloor (X - aA)/B \rfloor$ in the sum, it is possible to reveal the second dominating term, thereby delivering the smaller error term $o(\log x)$. The interested reader should consult Lecture V of the beautiful book *Ramanujan: Twelve lectures on subjects suggested by his life and work*, by Hardy, who gave the analysis which results in the smaller error term $O(\log x/\log\log x)$, and indeed the even better $o(\log x/\log\log x)$.

The generalisation (13.2) is essentially the number of lattice points in the corresponding k-dimensional tetrahedron, and S. S. Pillai had established that, for a fixed k, and $x \to \infty$,

$$\Psi(x, p_k) = \frac{\log^k x}{k!\log p_1 \cdots \log p_k}\Big(1 + \frac{k(\log p_1 + \cdots + \log p_k)}{\log x}\Big) + o(\log^{k-1} x).$$

In fact, for primes p, q,

$$\frac{1}{k!}\prod_{p\le y}\frac{\log x}{\log p} \le \Psi(x, y) \le \frac{1}{k!}\prod_{p\le y}\frac{\log x + \sum_{q\le y}\log q}{\log p}.$$

13.3 Table of Values for $\Psi(x, y)$

The function $\Psi(x, y)$ is increasing with respect to both x and y. For a feel of the subject, readers should compute $\Psi(x, y)$ for various x of modest size, and y being very small, and try also to employ Buchstab's iterative formula (13.6). A small set of values at selected points are given in Table 13.1.

For example, the numbers $19a$, $23b$, for $a = 1, 2, 3, 4, 5$, $b = 1, 2, 3, 4$, are counted by $\Psi(100, 23)$, but not by $\Psi(100, 17)$, so that $\Psi(100, 23) = \Psi(100, 17) +$

Table 13.1 Values of $\Psi(x, y)$

x/y	7	17	23	31	47	59
100	46	67	76	82	90	92
1000	141	287	369	434	529	563
10000	338	945	1385	1778	2463	2756
100000	694	2579	4298	6070	9639	11331
1000000	1273	6179	11654	18083	32876	40636

9, because $19\times23 > 100$. However, it is not easy to keep track of $\Psi(x, y)$ as x, y vary independently.

The analysis of $\Psi(x, 3)$ in Sect. 13.2 shows that it is not difficult to determine its exact value even for rather large values of x, and that, by Hardy's result, Ramanujan's count should be very accurate. Let $E(x)$ denote the excess of $\Psi(x, 3)$ over the integer nearest to $\log(2x)\log(3x)/2\log 2\log 3$. We find that

$$\Psi(X, 3) = 3481222694429, \qquad \text{for} \quad X = 10^{1000000},$$

with the required computing time being only a single second, and that there are plenty of numbers x near X with $|E(x)| = 4, 5$, but none is found with $|E(x)| \ge 6$. The following is the result of an experiment for 10000 random numbers x near X:

$$1139,\ (1091, 1136),\ (1151, 1130),\ (1118, 1132),\ (846, 941),\ (126, 190);$$

the first number is the count of x with $E(x) = 0$, and the five pairs of numbers are the counts with $E(x) = \pm a$ for $a = 1, 2, 3, 4, 5$. From such experiments, it seems that there is room for improvement in Hardy's result and, indeed, one may also speculate that $E(x) \gg \log\log x$ for infinitely many integers x.

13.4 Dickman Function

From probabilistic consideration it is tempting to argue that, for large x, y, we should have, by Mertens' theorem (Theorem 9.3 (ii)),

$$\Psi(x, y) \approx x \prod_{y<p\le x}\Big(1-\frac{1}{p}\Big) \sim x\,\frac{\log y}{\log x} = \frac{x}{u}, \qquad y = x^{1/u}, \quad u > 1,$$

but this is **wrong**, because it disagrees with (13.5), but why? The explanation is that a probabilistic argument involving a product $\prod(1 - 1/p)$ is valid only when divisibility conditions for two different primes are essentially independent; in the present case $P(n)$ is the only prime divisor in the interval $x^{1/u} < m \le x$, so that such divisibility conditions cannot be statistically independent. Anyway, we have established (13.5), but the case $u > 2$ is complicated.

De Bruijn's derivation of (13.4) involves double induction, based on the Buchstab iterative formula (13.6), and on k for the intervals $k \le u < k + 1$. Later, A. Hildebrand simplified and enhanced the argument by replacing the Buchstab formula with the more useful functional equation

$$\Psi(x, y)\log x = \int_1^x \frac{\Psi(t, y)}{t}dt + \sum_{p^m\le x,\ p\le y} \Psi\Big(\frac{x}{p^m}, y\Big)\log p, \quad x \ge 1,\ y \ge 2. \tag{13.7}$$

The equation can be established by evaluating the sum $S = \sum_{n\le x,\, P(n)\le y} \log n$ in two ways. The first way is to use partial summation, or integration by parts after rewriting the sum as an integral:

$$S = \int_1^x (\log t)\, d\,\Psi(t, y) = \Psi(x, y)\log x - \int_1^x \frac{\Psi(t, y)}{t} dt;$$

the second way is to write $\log n = \sum_{p^m|n} \log p$, for $n = kp^m \le x$, so that

$$\begin{aligned} S &= \sum_{n\le x,\, P(n)\le y} \sum_{p^m|n} \log p = \sum_{p^m\le x,\, p\le y} \log p \sum_{k\le x/p^m,\, P(k)\le y} 1 \\ &= \sum_{p^m\le x,\, p\le y} (\log p)\Psi\Big(\frac{x}{p^m}, y\Big), \end{aligned}$$

and the functional equation (13.7) is proved.

The following is a rough sketch of Hildebrand's derivation of (13.4): The value of the integral in (13.7) is at most x, and, as for the sum, the contribution of the terms from $m \ge 2$ is at most $\sum_{p\le y}(x \log p)/p(p-1) \ll x$, so that

$$\Psi(x, y)\log x = \sum_{p\le y} \Psi\Big(\frac{x}{p}, y\Big)\log p + O(x).$$

Now write the sum as an integral, using Chebyshev's prime counting function $\theta(t)$ with weight. On replacing $\Psi(z, y)$ by their asymptotic values in terms of the ρ function, we then find that

$$x(\log y)\, u\, \rho(u) \approx \int_1^y \Psi\Big(\frac{x}{t}, y\Big)\, d\theta(t) \approx \int_1^y \frac{x}{t}\rho\Big(u - \frac{\log t}{\log y}\Big)\, dt,$$

and, with the substitution $t = y^v$, the last integral becomes

$$x\int_0^1 \rho(u - v)(\log y)\, dv = x\log y \int_{u-1}^u \rho(w)\, dw, \qquad w = u - v.$$

Assuming that all the error terms involved in the approximations tend to zero as $x, y \to \infty$, we may then conclude that

$$u\,\rho(u) = \int_{u-1}^u \rho(w)\, dw,$$

and hence the differential-difference equation (13.4).

De Bruijn's asymptotic formula for $\Psi(x, y)$ is

$$\Psi(x, y) = x\rho(u)\Big\{1 + O\Big(\frac{\log(u+1)}{\log y}\Big)\Big\}, \qquad y = x^{1/u}, \tag{13.8}$$

provided that $1 \le u \le \log^A y$, $0 < A < \frac{3}{5}$. Besides simplifying the derivation of the differential-difference equation by using the functional equation (13.7) instead of the Buchstab identity (13.6), Hildebrand's superior method also shows that the asymptotic formula (13.8) is valid in the much longer interval

$$\exp\{(\log\log x)^{5/3+\epsilon}\} \le y \le x, \quad \text{for any fixed } \epsilon > 0.$$

He also showed that, under the assumption of the Riemann hypothesis, the lower bound for y can be further reduced to $(\log x)^{2+\epsilon}$, and he conjectured that (13.8) is no longer valid for $y < (\log x)^{2-\epsilon}$; the conjecture has very recently been established unconditionally by Gorodetsky [35].

The accompanying asymptotic formula for $\rho(u)$, as $u \to \infty$, is

$$\rho(u) = \exp\Big(-u\Big(\log u + \log\log u - 1 + \frac{\log\log u - 1}{\log y} + O\Big(\frac{(\log\log u)^2}{\log^2 u}\Big)\Big)\Big); \tag{13.9}$$

the explicit bounds

$$\exp\Big(-u\Big(\log u + \log\log u + \frac{6\log\log u}{\log u}\Big)\Big) < \rho(u) \le \frac{1}{\lfloor u \rfloor!} \tag{13.10}$$

may be more useful in some situations.

Readers interested in the proofs of the asymptotic formulae (13.8) and (13.9) should consult [96].

The Values for $\rho(u)$

The values of $\rho(u)$ for $0 < u \le 2$ can be determined from its explicit formula, and by numerical methods for $u > 2$.

The differential-difference equation (13.4) implies that ρ is piecewise analytic, in the sense that, for each $k = 1, 2, \ldots$, there is an analytic function ρ_k agreeing with $\rho(x)$ when $k - 1 \le x \le k$. Thus we have $\rho_1(x) = 1$, and $\rho_2(x) = 1 - \log x$; we call the points $x = k$ the *knots* for the function ρ, which has a continuous kth derivative in $x > k$, but with a jump discontinuity at $x = k$.

Up to about 1990, authors applied various methods of numerical integration, such as Romberg's method together with Richardson extrapolation, to compute the values

of $\rho(x)$. Such methods require careful error analysis involving the magnitude of high-order derivatives, and the results become quite unsatisfactory even for x of very modest size.

In 1989, G. Marsaglia et al. [63] introduced a much simpler and superior method which enables them to evaluate $\rho(x)$ to hundreds of decimal digits, and even up to $x = 500$. The underlying idea is that, between knots, the function $\rho(x)$ can be expressed as a power series. The coefficients of successive power series over different intervals can be obtained by converting (13.4) into its equivalent integral difference-equation, such as:

$$\rho(k+1+t) = \rho(k+1) - \int_0^t \frac{\rho(k+y)}{k+1+y} dy, \tag{13.11}$$

so that the coefficients can be expressed easily in terms of those for the previous interval. The method has the enormous advantage that only rational arithmetic is involved in such an operation, so that there is no propagation of errors in approximations to speak of—indeed, only the initial value of $\log 2$ has to be evaluated to a high accuracy as an input value. The following simplified version will be enough to illustrate the wonderful method for the evaluation of $\rho(x)$.

Suppose that we wish to evaluate $\rho(x)$ to $N-2$ decimal places, in the interval $0 \le x < K$, with N, K being of rather modest size. For $k \le K$, and $k-1 \le x \le k$, we may assume that there are coefficients $c_{k,i}$ such that

$$\rho(x) = \rho_k(x) = \sum_{i=0}^{\infty} c_{k,i} t^i, \qquad x = k - t, \quad 0 \le t \le 1.$$

In practice, we may scale up calculations by the factor $M = 10^N$, taking the integer parts of any real number in $0 \le z \le M$; we shall not bother with the floor function notation by assuming integer arithmetic in the following. First, we evaluate $\log 2$ to N decimal places, and set

$$L = (1 - \log 2)M.$$

Next, write $I = 7N/2$, so that $2^I > M$, and proceed to construct a $K \times I$ matrix $\{c_{k,i}\}$ with integer elements as follows. Define the first two rows of the matrix by setting

$$c_{1,0} = M, \quad c_{2,0} = L; \qquad c_{1,i} = 0, \quad c_{2,i} = \frac{M}{i2^i}, \quad i = 1, 2, \ldots, I.$$

The rows for $3 \le k \le K$ are defined by

$$c_{k,i} = \sum_{0 \le j < i} \frac{c_{k-1,j}}{ik^{i-j}}, \qquad \text{for} \quad i = 1, 2, \ldots, I,$$

and then set

$$c_{k,0} = \frac{1}{k-1} \sum_{1 \le j \le I} \frac{c_{k,j}}{j+1};$$

for example,

$$c_{3,1} = \frac{c_{2,0}}{3}, \cdots, c_{3,I} = \frac{1}{I}\Big(\frac{c_{2,0}}{3^I} + \frac{c_{2,1}}{3^{I-1}} + \cdots + \frac{c_{2,I-1}}{3}\Big), \quad c_{3,0} = \frac{1}{2} \sum_{1 \le j \le I} \frac{c_{3,j}}{j+1}.$$

It is readily verified by induction that the rule for setting the values $c_{k,i}$ follows from the integral-difference equation.

For the evaluation of $\rho(x)$ at rational values of x, we can choose and fix $b = 10$, or 100, say. Set $t = a/b$, with a running over $0 \le a \le b$, (so that $0 \le t \le 1$) backwards for increasing values of $x = k - t$. Then

$$\rho(x) = \rho_k(k-t) \approx \frac{1}{M} \sum_{0 \le i \le I} c_{k,t} t^i = \frac{1}{M} \sum_{0 \le i \le I} \frac{c_{k,t} a^i}{b^i}, \qquad 0 \le a \le b.$$

It can be shown that, without the scaling factor M, each element of the matrix satisfies the bound $0 \le c_{k,i} \le 2^{-i}$, so that the series concerned approximates ρ_k within an absolute error bound of $2^{-I} < 10^{-N}$. Note that the 'convergence' is faster when t is near 0, and slower when t is near 1; so that, without scaling, the approximation is better when x is just below k. Indeed, Marsaglia, Zaman, and Marsaglia used series expansion centred around the midpoint $x = k + \frac{1}{2}$, so that convergence is excellent throughout the interval $k - 1 \le x \le k$.

We give, in Table 13.2, the values of $\rho(x)$, *truncated* after 10 decimal places, for $1 < x \le 5$, with step size $\frac{1}{10}$. The calculations were done with the scaling factor 10^{16}, and we set $I = 56$, and $L = \lfloor 10^{16}(1 - \log 2) \rfloor = 3068528194400547$. The function $\rho(x)$ has a rather simple looking structure—its graph, as a scattered plot of 500 points over the interval $0 < u \le 5$, is displayed in Fig. 13.1.

13.5 Rankin's Trick Applied to $\Psi(x, y)$

In number theory, we often need an upper estimate of the sum $\sum_{n \le x} f(n)$, where $f(n)$ is a non-negative function. Knowledge of the structure of $f(n)$ may enable us to derive a suitable estimate, but if $f(n)$ is complicated, or there is no known structure to speak of, then Rankin's trick (see Note 1) may be considered. We impose some structure by attaching the weight $(x/n)^\delta$ to the term $f(n)$, where $\delta \ge 0$ is to

Table 13.2 Values for $\rho(x)$ in $1 < x \leq 5$, with spacing $\frac{1}{10}$

x	$\rho(x)$	x	$\rho(x)$
1·1	0·9046898201	3·1	0·0393229695
1·2	0·8176784432	3·2	0·.0317034445
1·3	0·7376357355	3·3	0·0254647238
1·4	0·6635277633	3·4	0·0203717790
1·5	0·5945348918	3·5	0·0162295932
1·6	0·5299963707	3·6	0·0128754341
1·7	0·4693717489	3·7	0·0101728378
1·8	0·4122133350	3·8	0·0080068721
1·9	0·3581461138	3·9	0·0062803730
2·0	0·3068528194	4·0	0·0049109256
2·1	0·2604057801	4·1	0·0038285861
2·2	0·2203571379	4·2	0·0029754747
2·3	0·1857994615	4·3	0·0023050505
2·4	0·1559912638	4·4	0·0017799424
2·5	0·1303195618	4·5	0·0013701177
2·6	0·1082724429	4·6	0·0010514448
2·7	0·0894185657	4·7	0·0008045586
2·8	0·0733915807	4·8	0·0006139573
2·9	0·0598781159	4·9	0·0004672798
3·0	0·0486083882	5·0	0·0003547247

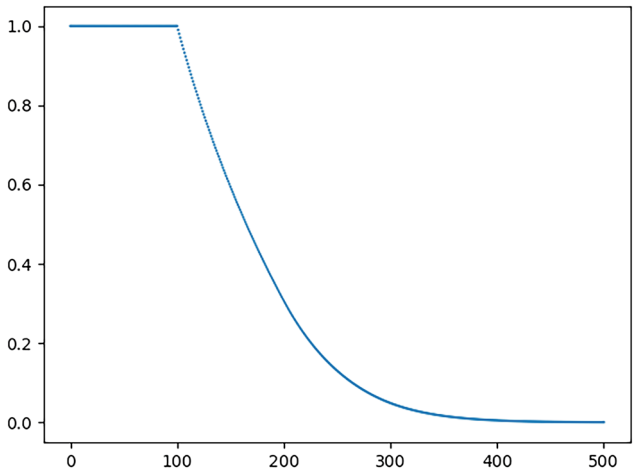


Fig. 13.1 500 points for $\rho(u)$ over $0 < u \leq 5$

be chosen in due course. The trick hinges on the entirely trivial inequality

$$\sum_{n\le x} f(n) \le \sum_{n\le x} \Big(\frac{x}{n}\Big)^{\delta} f(n) = x^{\delta} \sum_{n\le x} \frac{f(n)}{n^{\delta}}.$$

There may be some structure to the new series—for example, if $f(n)$ is the characteristic function of y-smooth numbers, then we may consider it to be the partial sum of a Dirichlet series, so that

$$\Psi(x, y) \le x^{\delta} \sum_{P(n)\le y} \frac{1}{n^{\delta}} = x^{\delta} \prod_{p\le y} \Big(1 - \frac{1}{p^{\delta}}\Big)^{-1}.$$

Anyway, the idea is that the factor $n^{-\delta}$ may enable us to derive a suitable estimate for the modified sum, thereby delivering a non-trivial estimate for the original sum if δ can be chosen appropriately—the choice may, but need not, depend on x.

A good example to illustrate the trick is to apply the idea to bound $\Psi(x, y)$, where y is much smaller than a power of x.

Theorem 13.1 *For all large x, we have*

$$\Psi(x, \log x \log\log x) \le \exp\Big(\frac{5\log x}{\sqrt{\log\log x}}\Big).$$

Corollary 13.1 *Let $K, \epsilon > 0$. Then $\Psi(x, K\log x) < x^{\epsilon}$ for all large x.*

Proof Let $\delta > 0$ be specified later and consider, for large x and y,

$$\begin{aligned}
\Psi(x, y) &= \sum_{n\le x,\, P(n)\le y} 1 \le \sum_{n\le x,\, P(n)\le y} \Big(\frac{x}{n}\Big)^{\delta} = x^{\delta} \sum_{n\le x,\, P(n)\le y} \frac{1}{n^{\delta}} \\
&\le x^{\delta} \prod_{p\le y} \Big(1 + \frac{1}{p^{\delta}} + \frac{1}{p^{2\delta}} + \cdots\Big) = x^{\delta} \prod_{p\le y} \Big(1 + \frac{1}{p^{\delta} - 1}\Big) \\
&= \exp\Big(\delta\log x + \sum_{p\le y} \log\Big(1 + \frac{1}{p^{\delta} - 1}\Big)\Big) \\
&\le \exp\Big(\delta\log x + \sum_{p\le y} \frac{1}{p^{\delta} - 1}\Big) \le \exp\Big(\delta\log x + \frac{1}{\delta} \sum_{p\le y} \frac{1}{\log p}\Big) \\
&\le \exp\Big(\delta\log x + \frac{1}{\delta} \frac{4y}{\log^2 y}\Big).
\end{aligned}$$

Here we have made use of the fundamental theorem of arithmetic: as n runs over the set of y-smooth numbers, the sum $\sum 1/n^{\delta}$ is given by the product shown with primes p running over $p \le y$; also, exp and log are inverses to each other, and

$\log(1+t) \le t$ for $t \ge 0$. The last inequality follows from Corollary 9.4 of Chebyshev's theorem.

Letting $y = \log x \log\log x$, so that $\log y \sim \log\log x$ as $x \to \infty$, and hence

$$\frac{y}{\log^2 y} \sim \frac{\log x \log\log x}{(\log\log x)^2} \sim \frac{\log x}{\log\log x},$$

we see that the required result follows by setting $\delta = 1/\sqrt{\log\log x}$.

For the Corollary, we note that $\log\log x > K$ when x is large, so that

$$\Psi(x, K\log x) \le \exp\left(\frac{5\log x}{\sqrt{\log\log x}}\right) < \exp\left(\epsilon \log x\right) = x^{\epsilon}.$$

□

13.6 Application to the Discrete Logarithm Problem

Let g be a primitive root of a prime p, and $1 < b < p$. Recall in Sect. 5.5 that the discrete logarithm problem is to find the index, or the DL, m such that

$$g^m \equiv b \pmod{p},$$

and that the direct search through $1 \le m < p$ is liken to finding a needle in a hay-stack. We now show that the use of smooth numbers may help the situation, at least to the extend that the problem can be solved if p is of modest size, say $p \approx 10^{20}$.

For an analogy, we may say that smooth numbers can be used to construct a suitable 'metal detector' by which the 'needle in the hay-stack' can be located. A more colourful analogy is to say that smooth numbers are used to extract statements (the DL) from the usual suspects (the small primes in a base), from which the actual villain can be tracked down.

The Detector

We first choose a suitable y, depending on the size of p, and let $p_1, p_2, \ldots, p_k$ denote the primes up to y, so that $k = \pi(y)$. Every y-smooth number up to p can be factorised into a product of p_i, by trial division, say.

The proposed method requires the DL d_i of each prime $p_i \le y$, that is

$$p_i \equiv g^{d_i} \pmod{p}, \qquad i = 1, 2, \ldots, k; \tag{13.12}$$

in other words, we need the answers to k similar problems in the first place! The reason for this convoluted way is because, although it is very difficult to find the DL of a general input b, smooth numbers can be used to find the DLs of small primes. Moreover, once such d_i are found, they can then be used to deliver the DL of any input value; we deal with this easier second part first.

Let us then (pre)suppose that the components of the column vector

$$\mathbf{d} = (d_1, d_2, \ldots, d_k)^T \tag{13.13}$$

have been found, and proceed to show how the DL of an input b can be delivered. Compute $g^r \pmod p$ for *random* choices of r and, rather than just hoping for a miracle that $r = m$ is chosen to deliver $g^m \equiv b \pmod p$, we only require that $u \equiv bg^r \pmod p$ be y-smooth, so that we may write

$$u \equiv p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}, \qquad \text{and set} \quad \mathbf{b} = (b_1, b_2, \ldots, b_k). \tag{13.14}$$

It then follows from (13.12) and (13.13) that

$$u \equiv (g^{d_1})^{b_1} (g^{d_2})^{b_2} \cdots (g^{d_k})^{b_k} \equiv g^{b_1 d_1 + b_2 d_2 + \cdots + b_k d_k} \equiv g^{\mathbf{bd}} \pmod p,$$

where **bd** is the scalar product of the two vectors, that is the product of a $1 \times k$ row vector, and a $k \times 1$ column vector. From $b \equiv ug^{-r} \equiv g^{\mathbf{bd}-r} \pmod p$, the required DL of b is given by

$$m \equiv \mathbf{bd} - r \pmod{p-1}. \tag{13.15}$$

The method therefore hinges on the knowledge of the vector **d** in (13.13), which will be called the *detector*. Indeed, most of the required work is involved in the construction of such a detector **d**.

Construction of the Detector

The detector **d** in (13.13) is the solution of a certain matrix congruence

$$A\mathbf{d} \equiv \mathbf{n} \pmod{p-1}, \tag{13.16}$$

where A is a non-singular square matrix, the construction of which is as follows. Choose random numbers $n_i < p - 1$, compute $u_i \equiv g^{n_i} \pmod p$, retaining those u_i that are y-smooth, write

$$u_i \equiv p_1^{a_{i,1}} p_2^{a_{i,2,}} \cdots p_k^{a_{i,k}} \pmod p,$$

and let the exponents form the vector:

$$\mathbf{a}_i = (a_{i,1}, a_{i,2}, \ldots, a_{i,k}). \tag{13.17}$$

We need to discard u_{i+1} if $\mathbf{a}_{i+1}$ is a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_i$, and we may stop when k linearly independent vectors $\mathbf{a}_i$ have been collected, so that we can form the non-singular matrix

$$A = \begin{pmatrix} \mathbf{a_1} \\ \mathbf{a_2} \\ \vdots \\ \mathbf{a_k} \end{pmatrix} = \begin{pmatrix} a_{i,1} & a_{i,2} & \cdots & a_{i,k} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k,2} & \cdots & a_{k,k} \end{pmatrix}. \tag{13.18}$$

By (13.12) and (13.17), we may also write

$$u_i \equiv (g^{d_1})^{a_{i,1}} (g^{d_2})^{a_{i,2}} \cdots (g^{d_k})^{a_{i,k}} \equiv g^{\mathbf{a}_i \mathbf{d}} \pmod{p},$$

where $\mathbf{a}_i \mathbf{d}$ is the scalar product of the two vectors. Setting

$$\mathbf{n} = (n_1, n_2, \ldots, n_k)^T \tag{13.19}$$

and recalling $u_i \equiv g^{n_i} \pmod{p}$, so that $n_i \equiv \mathbf{a}_i \mathbf{d} \pmod{p-1}$, the congruence (13.16) follows from (13.17), (13.18) and (13.19).

Remarks on the Proposed Method

(i) The choice of y presents a dichotomy: On one hand, if y is small then the probability of finding random numbers (up to p) being y-smooth is also small. On the other hand, y should not be too large because the linear algebra involved becomes more difficult with increasing y; the order of the matrix A is k^2 and, by the prime number theorem, $k^2 \approx y^2 / \log^2 y$. The method is a good example of one in which there is a tradeoff between the computing time and storage requirement.

(ii) The components b_i of $\mathbf{b}$ satisfy $2^{b_i} \leq p_i^{b_i} < p$, so that $b_i \leq \lfloor \log_2 p \rfloor$, and the bound applies also to the elements $a_{i,j}$ of A. Moreover, most such exponents take the value 0 because, by the theorem of Hardy and Ramanujan (Theorem 10.7), almost all numbers less than p have about $\log \log p$ distinct prime divisors, whereas k should be much bigger. Thus the congruence (13.16) is not too onerous to solve because A is a sparse matrix. Familiarity with linear systems in general and sparse matrices in particular will therefore be useful.

(iii) The congruence (13.16) can be solved as an equation by Gaussian elimination—the components of the solution $\mathbf{d}$ will be fractions which are

to be identified with the corresponding residue classes (mod $p-1$). If the denominators of the fractions are coprime with $p-1$ then, on multiplication of the appropriate reciprocals, the solution $\mathbf{d}$ to the congruence is obtained. We can either choose A so that this prevails, or we can solve the congruence modulo each prime power divisor of $p-1$, and then construct the solution by the Chinese remainder theorem.

(iv) The case (mod 2) for the system (13.16) can be avoided. The number $n \equiv g^m \pmod{p}$ is a quadratic residue, or nonresidue, depending on whether m is even, or odd. In particular, we have

$$d_i \equiv \frac{1}{2}\Big(3 + \Big(\frac{p_i}{p}\Big)\Big) \pmod{2}, \tag{13.20}$$

where the Legendre symbol $\big(\frac{p_i}{p}\big)$ can be obtained from the law of quadratic reciprocity.

(v) Roughly speaking, the detector $\mathbf{d}$ in (13.13) enables us to increase the chance of success in finding the required DL by the factor of $1/\Psi(p, y)$.

We illustrate the theory with two small examples.

Example 13.1 Let $p = 10^7 - 9 = 9999991$, so that $p - 1 = 2.3^3 5.7.11.13.37$. There are plenty of primitive roots and we choose $g = 59$, say.

Taking $y = 7$, there are 2154 y-smooth numbers up to p, so that the chance of a random $u < p$ being y-smooth is about 1 in 4600. We can improve this by taking $y = 19$ instead; there are then 20197 y-smooth numbers up to p, and the chance of finding such a number is about 1 in 500. The order of the matrix A for the two choices is $k = \pi(7) = 4$, or $k = \pi(19) = 8$.

Settling for $y = 19$, so that $k = 8$, and running some 4000 random $n_i < p$, we manage to collect 8 suitable y-smooth numbers $u_i \equiv 59^{n_i} \pmod{p}$, with their factorisations shown:

$$(n_1, u_1) = (3098973, 38896) = (3098973, 2^4 11.13.17)$$

$$(n_2, u_2) = (1743953, 8001504) = (1743953, 2^5 3^6 7^3)$$

$$(n_3, u_3) = (9746927, 88920) = (9746927, 2^3 3^2 5.13.19)$$

$$(n_4, u_4) = (1418516, 7116564) = (1418516, 2^2 3.7^4 13.19)$$

$$(n_5, u_5) = (8490766, 1834560) = (8490766, 2^6 3^2 5.7^2 13)$$

$$(n_6, u_6) = (3305020, 14400) = (3305020, 2^6 3^2 5^2)$$

$$(n_7, u_7) = (7290964, 609444) = (7290964, 2^2 3^6 11.19)$$

$$(n_8, u_8) = (8479967, 1880064) = (8479967, 2^{12} 3^3 17).$$

Each factorisation of u_i delivers a vector $\mathbf{a}$ with entries being the exponents of the first 8 primes for the factorisation. Thus $u_1 = 38896 = 2^4 11.13.17$ delivers $\mathbf{a}_1 = (4, 0, 0, 0, 1, 1, 1, 0)$, and we now have the congruence (13.16):

$$\begin{pmatrix} 4 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 5 & 6 & 0 & 3 & 0 & 0 & 0 & 0 \\ 3 & 2 & 1 & 0 & 0 & 1 & 0 & 1 \\ 2 & 1 & 0 & 4 & 0 & 1 & 0 & 1 \\ 6 & 2 & 1 & 2 & 0 & 1 & 0 & 0 \\ 6 & 2 & 2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 6 & 0 & 0 & 1 & 0 & 0 & 1 \\ 12 & 3 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \\ d_6 \\ d_7 \\ d_8 \end{pmatrix} \equiv \begin{pmatrix} 3098973 \\ 1743953 \\ 9746927 \\ 1418516 \\ 8490766 \\ 3305020 \\ 7290964 \\ 8479967 \end{pmatrix} \pmod{p-1}.$$

Treating the congruence as an equation, the solutions d_i do not belong to residue classes modulo $p - 1$, because they are fractions with denominators 2, 7 or 14, which are divisors of $p-1$. Instead we consider the congruence modulo each prime power divisor of $p - 1$; from Gaussian eliminations, we obtain:

$$\begin{aligned} \mathbf{d} &\equiv (1, 3, 23, 9, 19, 23, 2, 7)^T \pmod{3^3} \\ \mathbf{d} &\equiv (1, 4, 3, 3, 3, 3, 3, 0)^T \pmod{5} \\ \mathbf{d} &\equiv (2, 0, 0, 4, 1, 5, 3, 4)^T \pmod{7} \\ \mathbf{d} &\equiv (6, 2, 4, 5, 8, 10, 0, 0)^T \pmod{11} \\ \mathbf{d} &\equiv (11, 9, 12, 8, 0, 3, 12, 7)^T \pmod{13} \\ \mathbf{d} &\equiv (9, 19, 7, 7, 4, 4, 31, 15)^T \pmod{37} \end{aligned}$$

and $\mathbf{d} \equiv (0, 1, 0, 1, 1, 0, 0, 1)^T \pmod 2$ is obtained from (13.20). The Chinese remainder theorem then delivers

$$\mathbf{d} = (5807566, 8362389, 867398, 4177233, 8468083, 7698668, 3701918, 7033345)^T$$

for the solution (mod $p-1$). With the detector in place we can now evaluate the DL of any b, say $b = 3333333$; we found, by random choice, the suitable

$$r = 7094492, \quad \text{with} \quad u \equiv bg^r \equiv 1892800 \equiv 2^6 5^2 7.13^2 \pmod{p-1},$$

being smooth, so that the vector in (13.14) is $\mathbf{b} = (6, 0, 2, 1, 0, 2, 0, 0)$. By (13.15) the sought after DL is

$$m \equiv \mathbf{bd} - r \equiv 9060309 \pmod{p-1}.$$

There was no need for the testing of smooth numbers because such numbers can be listed; indeed, p is small enough for all the DLs to be listed anyway.

Example 13.2 Let $p = 10^{16} - 63 = 9\,99999\,99999\,99937$, with

$$p - 1 = 2^6 3^4 154321 \times 12499999,$$

and take $g = 5$ to be the primitive root.

It will be too laborious to list the relevant smooth numbers, whatever meaningful value of y we care to set. Instead we shall apply an algorithm for the testing of smooth numbers.

For $y = 1000$, so that $k = \pi(10^3) = 168$, the computing time for 172 y-smooth numbers $u \equiv g^n \pmod p$ from some 800000 random $n < p$ is about 75 seconds. For $y = 200$, so that $k = 46$, the computing time for 48 y-smooth numbers from some 14 million random n is about 20 minute; we settle for this so that the size of the matrix A is more manageable. Gaussian elimination then delivers the following components of the detector $\mathbf{d}$ satisfying (13.16):

$$\begin{aligned} d_1 &= 1\,07566\,61436\,34676, \\ d_2 &= 8\,59280\,93811\,67746, \\ d_3 &= \qquad\qquad 1, \\ d_4 &= \quad 2503\,95358\,19340, \\ &\vdots \\ d_{46} &= 1\,98530\,18570\,55517; \end{aligned}$$

note that $d_3 = 1$, because $p_3 = 5 = g$. With the detector $\mathbf{d}$ in place we can now find the DL of any number $m < p$. For example, the DLs of

$$a = 1\,22222\,33333\,44444, \qquad c = 5\,66666\,77777\,88888,$$

are

$$m_a = 6\,97231\,21303\,74911, \qquad m_c = 9\,13058\,51307\,90709.$$

We then find that

$$\begin{aligned} m_a + m_c &\equiv 6\,10289\,72611\,65684 \pmod{p-1}, & ac &\equiv 1\,41115\,79757\,84071 \pmod p, \\ c \times m_a &\equiv 7\,32741\,95594\,36008 \pmod{p-1}, & a^c &\equiv 5\,29875\,70226\,64905 \pmod p, \\ a \times m_c &\equiv 7\,96573\,40131\,88716 \pmod{p-1}, & c^a &\equiv 4\,07746\,61159\,26364 \pmod p, \end{aligned}$$

where the values on the right-hand side can be computed either directly, or by making use of their corresponding DLs on the left-hand side.

13.7 Summary and Notes

Summary

- There are applications, both theoretical and practical, from knowledge of the distribution of *smooth numbers*, which are, roughly speaking, numbers without a large prime divisor.
- Let $\Psi(x, y)$ count the numbers $n \leq x$ with the largest prime divisor $P(n) \leq y$. As predicted by Ramanujan, $\Psi(x, 3)$ is very close to $\log(2x)\log(3x)/(2\log 2\log 3)$. It is easy to compute $\Psi(x, 3)$ at $x = 10^{10^6}$, and experiments with counts near x indicate that the error in Romanian's prediction is indeed relatively small.
- Dickman showed that, for each fixed $u > 0$, there is a positive $\rho(u)$ such that $\Psi(x, x^{1/u}) \sim \rho(u)x$ as $x \to \infty$. The Dickman function $\rho(u)$ satisfies the differential-difference equation $u\rho'(u) + \rho(u-1) = 0$ for $u > 1$, but there is no simple explicit formula for $\rho(u)$ when $u > 3$.

  The function $\rho(u)$ decreases to 0 rapidly, and its value can be computed accurately and reasonably efficiently.
- An asymptotic formula for $\Psi(x, y)$, with (x, y) varying uniformly in a region, is available, but explicit bounds for $\Psi(x, y)$ are more useful in practice.
- An upper estimate of the sum $\sum_{n \leq x} f(n)$, where $f(n)$ is non-negative, can be delivered by the application of what is known as *Rankin's trick* in the trade. It shows, for example, that $\Psi(x, K\log x) < x^{\epsilon}$ for any fixed $K, \epsilon > 0$ and all large x.
- The use of smooth numbers to solve the *DL problem* is described in the text.

Notes

1. *Rankin's trick*. A trick is a sneaky idea that works, but a mathematical trick often also works in other similar problems. For example, as a party trick, the value of the square of a number congruent to 5 (mod 10) can be blurted out easily from

$$(10n + 5)^2 = 100n(n + 1) + 25;$$

   the formula may be generalised to the product $(10n + a)(10n + b)$ in which $a + b = 10$, but there is not much else to the rather cheap trick. However, the

related trick of the 'completion of a square' is worth much more, because it can be applied to any quadratic expression.

A trick worth its salt is a treat for the practitioner; indeed if it works in a variety of problems, then it acquires the status of being a *method*. Perhaps the mother of any method is a good trick, and it may be argued that many a recondite method, such as even the esoteric Hardy-Littlewood circle method, is based on a trick, albeit a very sophisticated one.

Besides theorems, mathematicians have all sorts of items named after them—such items may involve *axioms, criteria, equations, hypotheses*, ... The Scottish mathematician Robert Rankin (1915–2001) had the enviable distinction of having a *trick* named after him. It is one of those really sneaky ideas that, from the initial introduction of something which is apparently worth nothing, and thus with no prospect of success, the trick somehow manages to deliver a good dividend. Rankin first applied the trick in his famous paper [79] on the difficult problem concerning the gaps between prime numbers. It soon became a treat for others, who were so impressed by the sneaky idea that they christened it *Rankin's trick* in their papers. Apparently Rankin was initially somewhat unhappy to have the trick named after him, not because he objected to have a beautiful idea considered a 'trick'. The reason for his unease was that the idea was a gift from Albert Ingham, one of his research supervisors (his other supervisor was G. H. Hardy). Having clarified the situation he reluctantly accepted that the method is now called 'Rankin's trick'.

2. Rankin's trick can also be applied to the case $y = \log^K x$, where $K > 1$. We may set $\delta = 1 - 1/K$ to deliver

$$\Psi(x, \log^K x) = x^{1-1/K+O(1/\log\log x)}.$$

In fact it is known that $\Psi(x, \log^K x) \geq x^{1-1/K}$ (Konyagin and Pomerance); in particular, $\Psi(x, \log^2 x) \geq \sqrt{x}$ for $x \geq 4$.

3. *Finding smooth numbers*. Besides the obvious method of trial division, there are other more efficient ways of testing whether a number is smooth. Here is an algorithm which delivers the largest y-smooth divisor of $m < p$. It is a simplified, and much less efficient, version of Daniel Bernstein's smoothness test given as Algorithm 3.3.1 in [19] which is suitable for batches of numbers.

Let p and y be specified, and precompute

$$P = p_1 p_2 \cdots p_k, \quad p_k \leq y < p_{k+1}, \qquad b = \lfloor \log_2 p \rfloor;$$

it may be necessary to rewrite P as a product of several smaller sub-products in order to accommodate the capacity of the machine. From the input n, compute

$$r \equiv P \pmod{n}, \quad 0 \leq r < n, \qquad s \equiv r^b \pmod{n}, \quad 0 \leq s < n,$$

and return $t = \gcd(s, n)$. Then (see Exercise 11 below) the input n is y-smooth if, and only if, $t = n$.

13.8 Exercises

1. Build the table for $\Psi(x, y)$ with $x = 10^k$, $k = 2, 3, 4, 5, 6$ and $y = 61, 67, 71, 73, 79, 83, 89, 97$.
2. Evaluate $\Psi(X, 3)$, for $X = 10^{10^k}$, $k = 1, 2, 3, 4, 5$. For each chosen x near X, denote by $E_k(x)$ the excess of $\Psi(x, 3)$ over the integer nearest to

$$\frac{\log(2x)\log(3x)}{2\log 2\ \log 3}.$$

   Take 10000 random x near X, and build the tables for $E_k(x) = 0, 1, 2, 3, 4, 5$.
3. Let $K = \log 2 \log 3 \log 5$, and $x = 10^{10k}$ for $k = 1, 2, 3, 4, 5$. Evaluate $\Psi(x, 5)$ and verify that

$$\log^3 x \le 6K\ \Psi(x, 5) \le \Big(\log x + K\Big)^3.$$

4. Use the formula

$$\rho(u) = \begin{cases} 1 - \log u, & 1 \le u < 2, \\ 1 - \log u + \int_2^u \frac{\log(v-1)}{v}\, dv, & 2 \le u \le 3, \end{cases}$$

   to evaluate $\rho(u)$ for in $1 < x \le 3$, with spacing 0.1.
5. Use the Marsaglia-Zaman-Marsaglia method to compute

   (i) $\rho(5)$ to 10 significant figures, and
   (ii) $\rho(10)$ to 6 significant figures.

   Check that the value satisfies the bound (13.10).
6. Show that $\rho(u) \le 1/\Gamma(u + 1)$, where $\Gamma(u)$ is the gamma function.
7. Show that $\sum_{n \le x,\ P(n) \le y} n^\delta \ll x$, as $x \to \infty$, when $\delta = \delta(y) > 0$ is suitably chosen to be small. Use Rankin's trick to prove that

$$\Psi(x, y) \ll x e^{-u/2}, \quad u = \frac{\log x}{\log y}, \quad 2 \le y \le x \to \infty.$$

8. Let $k(n)$ denote the largest squarefree divisor of $n > 1$, that is $k(n) = \prod_{p|n} p$, and let $N(x, y)$ count the numbers $n \le x$ with $k(n) \le y$. Show that $N(x, y) \ll yx^\delta$, for any $\delta > 0$, and $x \to \infty$. Show further, by Rankin's trick, that

$$N(x, y) \ll y(\log y)e^{\sqrt{8v}}, \quad v = \log(x/y), \quad 2 \le y \le x \to \infty.$$

9. Take a prime $p \approx 10^{14}$, and find a primitive root $g \pmod{p}$. Choose a suitable y and construct the corresponding detector for the evaluation of discrete logarithms. Use the detector to solve the congruence $g^m = 888888888 \pmod{p}$.
10. Find 200 numbers m in $10^{10} < m < 10^{18}$ which are 50-smooth.
11. Establish the claim in Remark 3 that the input n is y-smooth if, and only if, $t = n$.

Chapter 14
Circle Method



Abstract The background to the invention of the circle method is given, with a description of the Hardy–Ramanujan asymptotic series for the partition function, and the eventual solution in the form of the Rademacher series. Waring's problem on sums of powers, particularly of cubes, and the problem of the determination, or estimation, of the numbers g(k) and G(k) are considered. There is a proof of the Hardy-Littlewood asymptotic formula of the number of representation of a large numbers as a sum of $s \geq 2^k + 1$ lots of k-th powers, with the coefficient of the main term involving the singular series. The third example of the circle method is Vinogradov's theorem which solves the ternary Goldbach conjecture on the representation of odd numbers as sums of three primes.

The most significant contribution to analytic number theory in the twentieth century is the invention and development of the *circle method* by G. H. Hardy and J. E. Littlewood. The method has its genesis in the celebrated paper [41] of Hardy and S. Ramanujan in their search for an accurate formula for the value of the partition function. Following the untimely death of Ramanujan back in India in 1920, Hardy and Littlewood produced a series of stunning papers, under the general title "*Some problems of partitio numerorum*", heralding the very powerful new method which is particularly effective in the study of additive problems.

There is also a brief discussion of the problem of the representation of numbers as the sum of a fixed number s of squares in [41]; the solution was given by Hardy, and also by L. J. Mordell using the modular group. A complete and self-contained proof of the theorem for $s = 5, 6, 7, 8$ can be found in [28] by T. Estermann.

In this final chapter we give a description of the circle method, with guided tours for the solutions of the following three additive problems:

(•) the evaluation of the partition function,
(•) the representation of numbers as sums of powers,
(•) the representation of odd numbers as sums of three primes.

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The solutions to these famous difficult problems are necessarily lengthy and intricate, and the tours are meant to be introductions for students who want to study the proofs. A proper account of the circle method, suitable for graduate students, is set out in the definitive text [104] by R. C. Vaughan. There is also an excellent survey paper by Vaughan and T. D. Wooley [105] on Waring's problem.

14.1 Partition Function: Rademacher Series

A fundamental additive problem is the study of $p(n)$, the number of unrestricted partitions of n. By a partition of n we mean a representation of n as a sum of any number of positive integral parts, and the partition is unrestricted in the sense that repetitions of the same parts are allowed, and the ordering of the parts is not relevant. For example, $p(5) = 7$, and the partitions are

$$5,\ \ 4+1,\ \ 3+2,\ \ 3+1+1,\ \ 2+2+1,\ \ 2+1+1+1,\ \ 1+1+1+1+1.$$

Euler saw that the key to the study of a given partition function $f(n)$ is to consider the corresponding *generating function* represented by power series

$$F(x) = \sum_{n=0}^{\infty} f(n)x^n,$$

because $x^n x^m = x^{n+m}$, the idea being that $F(x)$ may have another representation from which properties of $f(n)$ may be derived. Thus, setting $p(0) = 1$, Euler found that

$$\sum_{n=0}^{\infty} p(n)x^n = \prod_{m=1}^{\infty} \frac{1}{1-x^m}, \qquad |x| < 1. \tag{14.1}$$

To see this we consider the product by letting $1/(1-x^m) = 1 + x^m + x^{2m} + \cdots$, and multiplying such series together as if they were polynomials, and collect the like powers of x to deliver a power series $1 + \sum_{k\geq 1} f(k)x^k$. It is not difficult to see that $f(k) = p(k)$ and, although such an explanation does not constitute a proof, the idea can be made into one.

Euler also discovered his *pentagonal-number formula*:

$$\prod_{m=1}^{\infty}(1-x^m) = 1 - x - x^2 + x^5 + x^7 - x^{12} + \cdots = \sum_{n\in\mathbb{Z}} (-1)^n x^{\omega(n)}, \quad |x| < 1,$$

where $\omega(\pm n) = (3n^2 \mp n)/2$ are called the pentagonal-numbers; his proof is by induction. There is an elegant proof, by D. Shanks [88], of

$$\left| \prod_{m=1}^{n} (1 - x^m) - \sum_{|m| \le n} (-1)^m x^{\omega(m)} \right| \le n x^{n+1}, \qquad 0 \le x < 1,$$

from which the formula follows, because $nx^{n+1} \to 0$ as $n \to \infty$. The pentagonal-number formula can be used to give the useful recurrence formula for $p(n)$:

$$p(n) = \sum_{m=1}^{\infty} (-1)^{m+1} \Big(p\big(n - \omega(m)\big) + p\big(n - \omega(-m)\big) \Big),$$

in which $p(k) = 0$ for $k < 0$, so that the sum is over m such that $\omega(-m) \le n$; in particular the terms become zero when $m > \sqrt{2n/3}$. The proof of the formula amounts to equating the coefficient of x^n in the identity

$$\left(1 + \sum_{m=1}^{\infty} (-1)^m \Big(x^{\omega(m)} + x^{\omega(-m)} \Big) \right) \left(\sum_{m=0}^{\infty} p(m) x^m \right) = 1;$$

the recurrence formula can be used to compute $p(n)$ when n is of modest size. There is no particular problem in the implementation of the formula for $p(n)$; the number of additions required to find $p(m)$ with $0 \le m \le n$ is of order $n^{3/2}$. Although it takes only one second to evaluate $p(10^4)$, the required time increases to 40 seconds for $p(10^5)$, and 26 minutes for $p(10^6)$. There is also the problem of storage requirement for the values of $p(m)$, for $m \le n$, when n is much larger than a million. Anyway, the following is value for $p(10^6)$:

$$147168498635822339863100476 \ldots 430312131563000346710467 3818,$$

a number with 1108 decimal digits.

There was no interest in finding reasonable bounds for $p(n)$ before Hardy and Ramanujan, although Euler could have shown that $\log p(n) \approx \sqrt{n}$. Using a Tauberian argument, Hardy and Ramanujan found the asymptotic formula

$$p(n) \sim \frac{e^{K\sqrt{n}}}{4n\sqrt{3}} \quad \text{as} \quad n \to \infty; \qquad K = \pi \sqrt{\frac{2}{3}}.$$

Circle Method for $p(n)$

The starting point of the circle method for $p(n)$ is to consider, for $n = 0, 1, 2, \ldots,$

$$\frac{F(x)}{x^{n+1}} = \sum_{k=0}^{\infty} \frac{p(k)x^k}{x^{n+1}}, \qquad 0 < |x| < 1,$$

where $F(x)$ is given by the infinite product in (14.1). Treating x as a complex variable, the function $F(x)/x^{n+1}$ has a pole of order $n+1$ at $x = 0$, and the equation serves as its Laurent expansion in the punctured disc $0 < |x| < 1$. Cauchy's integral formula then delivers

$$p(n) = \frac{1}{2\pi i} \int_C \frac{F(x)}{x^{n+1}} dx, \qquad n = 0, 1, 2, \ldots,$$

where C is any positively oriented contour inside the disc enclosing the origin. The problem then becomes one of estimating, or evaluating, the integral.

If $1 - x^m = 0$, then x is an mth root of unity, so that each such root is a singularity of $F(x)$, near which the function exhibits prominent 'peaks'. The unit circle $|x| = 1$ forms a 'natural boundary' for $F(x)$, across which there can be no analytic continuation. However, $F(x)$ is essentially a *modular form* (see Note 1), so that it can be replaced by an elementary function which shares the features of $F(x)$ when x is near a singularity. More specifically making the transformation $x = e^{2\pi i\tau}$, where $\tau \in \mathbb{H}$, the upper half of the complex plane $\mathbb{C}$, so that τ has positive imaginary part, and thus $|x| < 1$, we may then write $F(e^{2\pi i\tau}) = e^{2\pi i\tau/12}/\eta(\tau)$, where

$$\eta(\tau) = e^{2\pi i\tau/12} \prod_{n=1}^{\infty} (1 - e^{2\pi in\tau})$$

is the Dedekind eta-function. It then becomes necessary to evaluate the contribution to $p(n)$ from the elementary function, and to keep the resulting error arising from the replacement under control.

When arranged in an increasing order, the set of fractions h/k, with $0 \le h \le k \le n$, and $\gcd(h, k) = 1$, is called the *Farey sequence* of order n. The circle method amounts to the following: Take C to be the circular contour centre 0, radius $e^{-2\pi/n^2} < 1$, and divide C into Farey arcs $C_{h,k}$ given by

$$x = \exp\Big(\frac{2\pi ih}{k} - \frac{2\pi}{n^2} + 2\pi i\phi\Big), \qquad \theta'_{h,k} \le \phi \le \theta''_{h,k},$$

so that

$$p(n) = \frac{1}{2\pi i} \sum_{1 \le k \le n} \sum_{\substack{0 \le h < k \\ \gcd(h,k)=1}} \int_{C_{h,k}} \frac{F(x)}{x^{n+1}} dx.$$

We shall not continue with the intricate argument for the solution of the problem. It suffices to say that the Dedekind eta-function $\eta(\tau)$ is essentially the 24th root of a certain discriminant with a functional equation from which the said elementary function can be extracted.

From the method, Hardy and Ramanujan obtained the asymptotic expansion:

$$p(n) \sim \frac{1}{2\pi\sqrt{2}} \sum_{1 \le k < \alpha\sqrt{n}} A_k(n)\sqrt{k}\,\frac{d}{dn}\left(\frac{\exp\left\{\frac{\pi}{k}\sqrt{\frac{2}{3}\left(n-\frac{1}{24}\right)}\right\}}{\sqrt{n-\frac{1}{24}}}\right),$$

where α is a constant, and

$$A_k(n) = \sum_{\substack{0 \le h < k \\ \gcd(h,k)=1}} e^{\pi i s(h,k) - 2\pi i n h/k}, \quad s(h,k) = \sum_{1 \le r < k} \frac{r}{k}\left(\frac{hr}{k} - \left\lfloor \frac{hr}{k} \right\rfloor - \frac{1}{2}\right). \tag{14.2}$$

The expression $s(h, k)$ is called a *Dedekind sum*, and $A_k(n)$ has its root in the theory of the transformation of modular functions. The error involved in the expansion is of order $1/n^{\frac{1}{4}}$. Since $p(n)$ is an integer, the expansion can be used to determine its exact value when k is large enough for the error term to be less than $\frac{1}{2}$. The infinite expansion is shown later to be divergent.

Rademacher's Series for $p(n)$

Apparently Ramanujan had always believed that there is an 'exact' formula for $p(n)$. In 1937 H. Rademacher, while preparing lecture notes on the method, made a change in the analysis of the solution to the problem; roughly speaking, he replaced the contour C by a contour in the τ-plane, where $x = e^{2\pi i\tau}$, and the exponential function by the hyperbolic sine function. It turns out that the apparently small change has a profound effect on the result, namely that the revised solution now delivers an absolutely convergent series for $p(n)$. The following spectacular theorem [76] is considered the jewel in the crown of the circle method.

Theorem 14.1 (Rademacher) *For $n \ge 1$, the value of $p(n)$ is given by*

$$p(n) = \frac{1}{\pi\sqrt{2}} \sum_{k=1}^{\infty} A_k(n)\sqrt{k}\,\frac{d}{dn}\left(\frac{\sinh\left\{\frac{\pi}{k}\sqrt{\frac{2}{3}\left(n-\frac{1}{24}\right)}\right\}}{\sqrt{n-\frac{1}{24}}}\right),$$

where $A_k(n)$ is given by (14.2).

Implementation of Rademacher's Formula

Unlike the pentagonal-number formula, we can evaluate $p(n)$ directly without needing any value of $p(m)$ for $m < n$. The advantage of such a method can be explained by the following analogy. Suppose that we wish to have access information at a particular point in a long manuscript. If the manuscript is preserved in the form of a scroll, then we need to unroll and scan from the beginning of the scroll until we reach the relevant point. It is a much easier task if the manuscript is put in the form of a book with direct access to individual pages. Similarly, it is easier to retrieve information stored in disc form than in tape form.

The implementation of the method as an algorithm is not difficult, provided there is a suitable library of functions which can be evaluated with arbitrary accuracy; indeed, storage space is only required for function evaluation to such accuracy. The formula can be reset in the more convenient form

$$p(n) = C \sum_{k=1}^{\infty} A_k(n) \frac{g(v)}{k^{5/2}}, \tag{14.3}$$

where

$$C = \frac{\pi^2}{3\sqrt{3}}, \qquad g(v) = \frac{v \cosh v - \sinh v}{v^3}, \qquad v = v(n,k) = \frac{\pi\sqrt{24n-1}}{6k}.$$

The value of $g(v)$ can be computed by calls to a library of functions; similarly the evaluation of $A_k(n)$, which is real, will require many calls on the cosine function. We also need to evaluate the Dedekind sums $s(h,k)$ first. In fact, in the implementation of the formula for $p(n)$, the time consuming part of the calculations lies in the evaluation of $A_k(n)$, which depends on k and the residue class (mod k) to which n belongs; in other words $A_k(n)$ is periodic in n with period k. Rademacher himself has given an expository account of the beautiful formula in his book [77] in which there is a more sophisticated approach to the efficient evaluation of $A_k(n)$ using the esoteric Kloosterman sum.

For a given value of n, we still have to know how far we need to calculate the individual terms of the series for $p(n)$ before we can be sure that the tail-end has contribution less than $\frac{1}{2}$. Using the crude bound $|A_k(n)| \le k$ we find that

$$\left| C \sum_{k>H} A_k(n) \frac{g(v)}{k^{5/2}} \right| \le C g(v^\star) \sum_{k>H} \frac{1}{k^{3/2}} \le \frac{C g(v^\star)}{\sqrt{H}}, \qquad v^\star = v(n,H).$$

14.2 Sums of Powers: Waring's Problem

Even Diophantus (c. 250) was aware that every number is a sum of four squares, and he knew, of course, that the 'awkward number' 7 does require four squares. Nobody asked a similar question concerning cubes or higher powers until 1770 when E. Waring made an interesting statement (in Latin) to the effect that "*every number is a sum of 4 squares, 9 cubes, 19 biquadrates*,[1] *and so on*." This became the famous Waring's problem which he could have stated more lengthily as "I believe that, corresponding to any $k \geq 2$, there exists $s = s(k)$ such that every n is representable as

$$n = x_1^k + x_2^k + \cdots + x_s^k, \qquad x_i \geq 0. \tag{14.4}$$

Moreover, if $g(k)$ denotes the smallest s then I conjecture that $g(2) = 4$, $g(3) = 9$ and $g(4) = 19$, and I have a procedure to determine $g(k)$ for every k."

In the same year when the statement was made, Lagrange proved that $g(2) = 4$. However, by the end of the nineteenth century, the existence of $g(k)$ was known only for $k \leq 8$ and $k = 10$. In 1909, D. Hilbert [45] succeeded in proving that $g(k)$ does indeed exist for every k, that is $g(k) < \infty$, and much progress was made in the determination of $g(k)$ during the twentieth century. Thus, from collective results of many mathematicians, there is now a simple criterion set out in Theorem 14.2 below from which the value of $g(k)$ can be determined for any $k \geq 2$. Nevertheless, we still cannot say that the value of each $g(k)$ has been found, not even if we add the qualification 'apart from a finite amount of computation.'

Value of g(k)

It is not difficult to find a good lower bound for $g(k)$ by considering the 'awkward number' corresponding to k. Take $k = 3$, and note that $23 < 3^3$, so that any representation of 23 as a sum of cubes can only involve 2^3 and 1^3, and the representation

$$23 = 2^3 + 2^3 + 1^3 + 1^3 + 1^3 + 1^3 + 1^3 + 1^3 + 1^3$$

shows that 9 cubes are necessary for the awkward number 23. Similarly, we find from $79 < 3^4$, and $79 = 4\times 2^4 + 15\times 1^4$, that 79 is the awkward number which requires 19 biquadrates for its representation. Therefore $g(3) \geq 9$ and $g(4) \geq 19$.

The following is possibly what Waring had in mind in relation to the awkward number phenomenon. With k fixed, we consider the number $n = q2^k - 1$, where q

[1] Numbers which are fourth powers of an integer.

is set as large as possible subject to $n < 3^k$. This then specifies $q = \lfloor (3/2)^k \rfloor$, and the most economical way to represent n as a sum of kth powers is

$$n = (q-1)2^k + (2^k - 1)1^k,$$

a sum with $2^k + q - 2$ terms. The argument then delivers a lower bound for $g(k)$:

$$g(k) \geq 2^k + \left\lfloor \left(\frac{3}{2}\right)^k \right\rfloor - 2,$$

and it is conjectured that equality holds for every $k \geq 2$. The value of $g(k)$ is governed by the effect of the first few kth powers, and there is still the possibility of the existence of a very large k with the property that there is a corresponding 'really awkward' number n which requires more than the conjectured number of kth powers. To appreciate this, we note that there are numbers which require the expected number of kth powers, but with the representations making use of 3^k, 4^k or 5^k. For example, 23 is not the only number that requires 9 cubes, because so does 239, with two different representations:

$$\begin{aligned} 239 &= 5^3 + 3^3 + 3^3 + 3^3 + 2^3 + 2^3 + 2^3 + 2^3 + 1^3 \\ &= 4^3 + 4^3 + 3^3 + 3^3 + 3^3 + 3^3 + 1^3 + 1^3 + 1^3. \end{aligned}$$

It is highly non-trivial to eliminate such similar possibilities; indeed the tasks of establishing $g(5) = 37$ and $g(4) = 19$ came as late as 1964 and 1986, respectively; see [16] and [8]. Significant progress has been make to resolve this unsatisfactory situation and we now have the following:

Theorem 14.2 *Let $k \geq 2$. Then*

$$g(k) = 2^k + \left\lfloor \left(\frac{3}{2}\right)^k \right\rfloor - 2, \qquad \textit{provided that} \qquad 2^k \left\{ \left(\frac{3}{2}\right)^k \right\} + \left\lfloor \left(\frac{3}{2}\right)^k \right\rfloor \leq 2^k,$$

where $\{\lambda\} = \lambda - \lfloor \lambda \rfloor$ is the fractional part of λ. Furthermore,

$$g(k) = 2^k + \left\lfloor \left(\frac{3}{2}\right)^k \right\rfloor + \left\lfloor \left(\frac{4}{3}\right)^k \right\rfloor - \theta, \qquad \textit{if} \qquad 2^k \left\{ \left(\frac{3}{2}\right)^k \right\} + \left\lfloor \left(\frac{3}{2}\right)^k \right\rfloor > 2^k;$$

where

$$\theta = \begin{cases} 2 & \textit{when} \quad \left\lfloor \left(\frac{4}{3}\right)^k \right\rfloor \left\lfloor \left(\frac{3}{2}\right)^k \right\rfloor + \left\lfloor \left(\frac{4}{3}\right)^k \right\rfloor + \left\lfloor \left(\frac{3}{2}\right)^k \right\rfloor = 2^k, \\ 3 & \textit{when} \quad \left\lfloor \left(\frac{4}{3}\right)^k \right\rfloor \left\lfloor \left(\frac{3}{2}\right)^k \right\rfloor + \left\lfloor \left(\frac{4}{3}\right)^k \right\rfloor + \left\lfloor \left(\frac{3}{2}\right)^k \right\rfloor > 2^k. \end{cases}$$

To complete the proof of the conjectural formula for all k it suffices to verify that $\{(3/2)^k\} \leq 1 - (3/4)^{k-1}$. On this, K. Mahler [62] has shown that there are at

most a finite number of exceptions to this inequality. However, his proof makes use of Roth's theorem (see Sect. 10.5) and is therefore not effective in the production of a bound for k beyond which the inequality is valid. The right-hand side of the inequality to be verified increases to 1 exponentially, so it is not surprising that no one has found k for which it fails. It is known to hold (see [50]) for $k \leq 471600000$.

Modern Problem of G(k)

Hilbert's argument is combinatoric in nature, involving algebraic identities, and it is not easy to derive an upper bound for $g(k)$ from his proof of the theorem. Hardy and Littlewood applied their circle method to give a new proof of the theorem which delivers an explicit bound for $g(k)$.

Since the value of $g(k)$ is dictated by the values of the first few kth powers, the modern Waring's problem is the much more challenging one of the determination of $G(k)$, the smallest number s such that every *sufficiently large* number is the sum of at most s kth powers. Thus $G(k) \leq g(k)$, and $G(2) = g(2) = 4$, the first equality follows from $8m + 7$ not being a sum of three squares for any m.

Hardy and Littlewood established the first upper bound for $G(k)$, namely

$$G(k) \leq (k-2)2^{k-1} + 5,$$

and later that $G(k) \leq k2^{k-2}$ for large k. They also proved that $G(4) \leq 19$, so that $G(4) \leq 19 \leq g(4)$, which implies that there can only be finitely many n that require more than 19 biquadrates for their representations; nevertheless, as we noted earlier, we had to wait until 1986 to establish that $g(4) = 19$.

In 1934 I. M. Vinogradov made enormous progress on the problem by proving that $G(k) \ll (k \log k)^2$, which implies that $G(k)$ is much smaller than the conjectured value for $g(k)$; it became the basis for the work on the determination of $g(k)$. There is a useful description in the Notes section of Chapter XXI in [42] on how bounds for $G(k)$, coupled with numerical work, can be used to determine $g(k)$.

In more recent times, the upper bound for $G(k)$ has been reduced by T. D. Wooley [110] to $G(k) \leq k(\log k + \log\log k + O(1))$, and the latest (2022) bound, by J. Brüdern and Wooley, is the tidier

$$G(k) \leq \lceil k(\log k + 4{\cdot}20032) \rceil, \qquad \text{for all } k \geq 2.$$

Incidentally, smooth numbers are brought into the investigation; see [103]. It was conjectured by Hardy and Littlewood that $G(k) \leq 2k + 1$, except for $k = 2^m \geq 4$, when $G(k)$ has to take a larger value because of 'local obstructions', that is congruence considerations. Anyway, $G(k) < g(k)$ for $k > 2$, and the growth of $G(k)$ is 'nearly' linear in k, whereas that of $g(k)$ is exponential in k. We remark that $G(2) = 4 \leq G(3) \leq 7$, and that $G(4) = 16$; more bounds with other small k, such as $6 \leq G(5) \leq 17$ and $9 \leq G(6) \leq 24$, can be found in [105].

Lower Bounds for G(k)

From $x^4 \equiv 0, 1 \pmod{16}$ it follows that at least 15 biquadrates are required to represent numbers congruent to 15 (mod 16), so that $G(4) \geq 15$. Indeed, more can be said because if $16n = x_1^4 + x_2^4 + \cdots + x_{15}^4$, then $\gcd(x_1, x_2, \ldots, x_{15})$ is even, so that (on replacing x_j by $x_j/2$) $n = x_1^4 + x_2^4 + \cdots + x_{15}^4$. Moreover, 31 cannot be a sum of fewer than 16 biquadrates, so that, for $a = 1, 2, \ldots$, the numbers $16^a 15$ cannot be a sum of 15 biquadrates, delivering $G(4) \geq 16$. More generally, we find from $x^k \equiv 0, 1 \pmod{4k}$ that

$$G(k) \geq 4k \qquad \text{if} \quad k = 2^m \geq 4.$$

For suppose that n is odd and that $4kn$ is a sum of $4k - 1$ or fewer kth powers. Then each of these powers must be even, and hence a multiple of $2^k > 4k$, so that the supposition is impossible. Hardy and Littlewood conjectured that $G(k) = 4k$ for each such k and, in 1939, H. Davenport proved that $G(4) = 16$; together with $G(2) = 4$, these are the only two known values for $G(k)$.

Congruence consideration also shows that $G(3) \geq 4$, because $x^3 \equiv 0, \pm 1 \pmod{9}$ implies at least 4 cubes are required to represent numbers congruent to $\pm 4 \pmod{9}$. Actually a simple counting argument shows that

$$G(k) \geq k + 1 \qquad \text{for all} \quad k \geq 2.$$

To see this, fix $s = k \geq 2$ in (14.4), take any N, and set $M = N^k + N < (N + 1)^k$, so that $x \geq N + 1$ implies $x^k > M$. Thus, if $n \leq M$ has the representation (14.4) then $x_1, x_2, \ldots, x_k \leq N$, so that the number of such n is at most $N^k = M - N$. This means that there are at least N values of n with no such representation; since N is arbitrary, it follows that $G(k) \geq k + 1$.

Sums of Cubes

Waring's problem for cubes already offers challenges. Some such challenges can be met with methods that relate cubes to quadratic forms, especially sums of squares, via the use of identities. We already mentioned that $g(3) = 9$, that is, every number is a sum of at most 9 cubes. This was established in 1909 by A. Wieferich and, in the same year, E. Landau proved that $G(3) < 9$, so that there is a largest number which requires 9 cubes for its representation. That $G(3) < 8$ was proved in 1941 by Yu. V. Linnik, so that there is a corresponding largest number which requires 8 cubes. Later, G. L. Watson [107] gave a much simplified proof of Linnik's theorem using results in the distribution of primes in arithmetic progressions, and he also gave a rather simple proof of Landau's theorem in [108].

Define, for $5 \le s \le 9$, the number n_s, if finite, to be the largest number that requires s positive cubes for its representation, so that $G(3) < s$ means that $n_s < \infty$, and that $n_9 \le n_8 < \infty$ by the theorems of Landau and Linnik. It has been established that

$$n_9 = 239, \quad \text{and} \quad n_8 = 454,$$

the former by Dickson [23] and the latter, very recently, by Siksek [87]. Assuming that $G(3) = 4$, so that $454 \le n_7 \le n_6 \le n_5 < \infty$, it is suspected that

$$n_7 = 8042, \quad n_6 = 1290740, \quad n_5 = 7373170279850.$$

For our rather limited scope, we carried out verifications up to 10^7:

(i) The only numbers that require 9 cubes are 23, 239;
(ii) There are 15 numbers which require 8 cubes, namely

$$15, 22, 50, 114, 167, 175, 186, 212, 231, 238, 303, 364, 420, 428, 454;$$

(iii) There are 121 numbers which require 7 cubes: they are

$$7, 14, 21, 42, 47, 49, 61, \ldots, 5306, 5818, 8042;$$

(iv) There are 3922 numbers which require 6 cubes: they are

$$6, 13, 20, 34, \ldots, 1026545, 1083983, n_6.$$

There are 288 ways to represent n_6 as a sum of 6 cubes; letting the representation

$$n_6 = 1290740 = x_1^3 + x_2^3 + x_3^3 + x_4^3 + x_5^3 + x_6^3,$$

be ordered lexicographically, that is by $(x_1, x_2, x_3, x_4, x_5, x_6)$ with $1 \le x_i \le x_{i+1}$, we find that the first and the last representations are

$$(1, 1, 25, 49, 54, 100) \quad \text{and} \quad (50, 50, 56, 57, 59, 78).$$

Next, let a_n denote the least required number of cubes for the representation of n. The following are three distributions of 10^5 values of $a_n \in \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$:

$(3, 295, 10150, 66414, 23136, 2, 0, 0, 0)$ for $10^6 < n \le 10^6 + 10^5$;

$(3, 275, 10227, 67162, 22332, 1, 0, 0, 0)$ for $10^6 + 2\times 10^5 < n \le 10^6 + 3\times 10^5$.

$(1, 143, 10150, 76794, 12912, 0, 0, 0, 0)$ for $10^7 - 10^5 < n \le 10^7$.

Thus

$$a_n = 1 \quad \text{for} \quad n = 101^3, 102^3, 103^3;\ 107^3, 108^3, 109^3; \quad \text{and} \quad 215^3;$$

$$a_n = 6 \quad \text{for} \quad n = 1026545, 1083983; \quad \text{and} \quad n_6.$$

It will be a good exercise for the reader to build histograms for a_n with n lying in various intervals. The algorithm for the computation of the values of a_n for $A < n \le B$ is only a small variation of that for the sieve of Eratosthenes. Each store for numbers in $A < n \le B$ is given the initial value 10. For $B \le 10^4$, take 9 loops to run over $1 \le x_1 \le x_2 \le \ldots \le x_9$, reducing the value of each relevant store whenever there is a smaller required number of cubes. For $10^4 < A < B \le 2\times 10^6$, we need only take 6 loops over $1 \le x_1 \le x_2 \le \ldots \le x_6$, making a check that the number of $a_n \le 6$ do add up to the length of the interval to confirm that there is no $a_n = 7, 8, 9$ in the interval concerned. For $A > 2\times 10^6$, we need only take 5 loops over $1 \le x_1 \le x_2 \le \ldots \le x_5$, again checking that no $a_n \ge 6$ has been missed.

From the circle method to be described in the next subsection, the number $R_4(n)$ of representations of n as a sum of 4 cubes has the conjectured asymptotic formula:

$$R_4(n) \sim \Gamma\Big(\frac{4}{3}\Big)^3 \mathfrak{S}(n)\, n^{1/3}, \quad n \to \infty,$$

where $\Gamma(s)$ is the gamma function, and $\mathfrak{S}(n)$ is an arithmetic function which satisfies $1/(\log\log n)^c < \mathfrak{S}(n) < (\log\log n)^c$, with c being a positive constant; see Theorem 14.3 below which, for $k = 3$, applies only for $s \ge 9$. Indeed, as a by-product of his proof of the asymptotic formula for $R_8(n)$, R. C. Vaughan also showed that almost all numbers n satisfy the conjectural asymptotic formula for $R_4(n)$. The smart money is therefore on $G(3) = 4$, but do not expect payment for quite a while, never mind the proof of the suspected value of n_5, after which all the entries for $a_n = 5$ will also be 0.

14.3 Circle Method for $G(k)$

Let $k \ge 2$, and s be sufficiently large, depending on k; with k and s fixed, we denote by $R(n) = R_s(n)$ the number of representations of n in (14.4). The task is to show that $R_s(n) > 0$ for all large n, and it is already clear that the smaller the chosen value of s is, the more difficult the task will be.

Hardy and Littlewood applied the circle method to determine the asymptotic formula for $R_s(n)$. After much simplification of the method, mainly by I. M. Vinogradov, and also by H. Davenport, and L.-K. Hua, the still intricate proof of the formula for the lower bound $s \ge 2^k + 1$ (which implies $G(k) \le 2^k + 1$) can be given an outline which contains the salient features of the method; indeed not too much is lost by taking $k = 3$ and $s = 9$ on a first reading of the following.

Theorem 14.3 (Hardy-Littlewood) *Let $k \geq 2$, and $s \geq 2^k + 1$. Then there exists $\delta = \delta(k, s) > 0$ such that, as $n \to \infty$,*

$$R(n) = R_s(n) = \frac{\Gamma(1 + 1/k)^s}{\Gamma(s/k)} \mathfrak{S}(n)\, n^{s/k-1} + O(n^{s/k-1-\delta}),$$

where $\mathfrak{S}(n)$ is an arithmetic function satisfying $0 < c_1 < \mathfrak{S}(n) < c_2$, with c_1, c_2 and the implied constant depending on k and s.

Let us first take an informal approach by writing, for real α,

$$f(\alpha) = \sum_{m=1}^{\infty} e^{2\pi i \alpha m^k};$$

this is, of course, mathematically reckless because the series is divergent. Pushing the bottomless boat further out, we consider

$$f(\alpha)^s = \sum_{m_1=1}^{\infty} \cdots \sum_{m_s=1}^{\infty} e^{2\pi i \alpha (m_1^k + \cdots + m_s^k)} = \sum_{\ell=1}^{\infty} R_s(\ell) e^{2\pi i \alpha \ell},$$

and, by appealing to the orthogonal property of the exponential function

$$\int_0^1 e^{2\pi i m \alpha} e^{-2\pi i \ell \alpha} d\alpha = \begin{cases} 1 & \text{if} \quad m = \ell, \\ 0 & \text{if} \quad m \neq \ell, \end{cases}$$

then write

$$R_s(n) = \int_0^1 f(\alpha)^s e^{-2\pi i \alpha n} d\alpha.$$

Well, this is a pie in the sky, because $f(\alpha)^s$ makes no mathematical sense.

Vinogradov's idea is that the approach becomes legitimate if we take only a finite number of terms in the series for $f(\alpha)$, so that there is now a solid bottom for the new boat! From such a modification of the argument, a digestible formula for $R_s(n)$ can be recovered.

To restart let, for real α, and a large fixed n,

$$f(\alpha) = \sum_{1 \leq m \leq N} e^{2\pi i \alpha m^k}, \qquad N = \lfloor n^{1/k} \rfloor, \tag{14.5}$$

a trigonometric polynomial. Then

$$f(\alpha)^s = \sum_{1 \leq m_1 \leq N} \cdots \sum_{1 \leq m_s \leq N} e^{2\pi i \alpha (m_1^k + \cdots + m_s^k)} = \sum_{s \leq m \leq sn} R_s(m, n) e^{2\pi i \alpha m},$$

where $R_s(m,n)$ is the number of representations of m as the sum of s kth powers, none of which exceeds n. Thus $R_s(m,n) = R_s(m)$ when $m \le n$ and, in particular, $R_s(n,n) = R_s(n)$. Again, by orthogonality, we now have a legitimate formula

$$R(n) = R_s(n) = \int_0^1 f(\alpha)^s e^{-2\pi i n\alpha}\, d\alpha. \tag{14.6}$$

The use of a trigonometric polynomial and Fourier analysis replaces that of power series and Cauchy's theorem. Following such a manoeuvre, the unit circle $e^{2\pi i\alpha}$ in $\mathbb{C}$ is replaced by the unit interval $0 < \alpha \le 1$ in $\mathbb{R}$; for historic reason, in such a setting of the problem, a subinterval of the unit interval is sometimes called an 'arc'. It is also traditional in this and related topics to write

$$\mathrm{e}(\alpha) = e^{2\pi i\alpha}, \qquad \alpha \in \mathbb{R},$$

and denote by $\|\alpha\|$ the distance between α and the nearest integer.

Diophantine approximation is at the heart of the argument. It is applied, in a delicate manner, to α in the estimation of the integrand in (14.6). More specifically, the integrand is found to 'peak' when α is near rational numbers with relatively small denominators. Estimates of the sum $\sum \mathrm{e}(n\alpha)$ over intervals will be needed; in the following, an expression $\min(A, 1/0)$ is taken to be A.

Lemma 14.1 *For fixed α, and large N_1, N_2, N, we have*

$$\sum_{N_1 \le m \le N_2} \mathrm{e}(m\alpha) \ll \min\Big(N_2 - N_1, \frac{1}{\|\alpha\|}\Big),$$

and

$$\sum_{t \le N} \max_w \Big| \sum_{w \le r \le N/t} \mathrm{e}(rt\alpha)\Big| \ll \sum_{t \le N} \min\Big(\frac{N}{t}, \frac{1}{\|\alpha\|}\Big).$$

Proof If α is a multiple of 2π then $\mathrm{e}(m\alpha) = 1$. In any case, the sum $\sum \mathrm{e}(m\alpha)$ is a geometric progression, so that

$$\sum_{N_1 \le m \le N_2} \mathrm{e}(m\alpha) = \frac{\mathrm{e}((N_2+1)\alpha) - \mathrm{e}(N_1\alpha)}{\mathrm{e}(\alpha) - 1}, \qquad \mathrm{e}(\alpha) \ne 1,$$

in which $|\mathrm{e}((N_2+1)\alpha) - \mathrm{e}(N_1\alpha)| \le 2$ and $|\mathrm{e}(\alpha) - 1| = |2\sin(x/2)| \gg \|\alpha\|$. The first part of the lemma is proved, and the second part follows at once. □

Lemma 14.2 *Suppose that* $X, Y \geq 1$ *and* $|\alpha - a/q| \leq 1/q^2$, *with* $\gcd(a, q) = 1$. *Then*

$$\sum_{x \leq X} \min\Big(\frac{XY}{x}, \frac{1}{\|\alpha x\|}\Big) \ll XY\Big(\frac{1}{q} + \frac{1}{Y} + \frac{q}{XY}\Big)\log(2Xq)$$

Proof Dividing the interval $1 \leq x \leq X$ into $\lfloor X/q \rfloor$ sections, with $x = qj + r$ for $1 \leq r \leq q$ in an interval, the sum S concerned has the bound

$$S \leq \sum_{0 \leq j \leq X/q} \sum_{1 \leq r \leq q} \min\Big(\frac{XY}{qj + r}, \frac{1}{\|\alpha(qj + r)\|}\Big).$$

For each j, let $y_j = \lfloor \alpha j q^2 \rfloor$, and write $\theta = q^2\alpha - qa$. Then

$$\alpha(qj + r) = \frac{y_j + ar}{q} + \frac{\alpha j q^2}{q} + \frac{\theta r}{q^2}.$$

When $j = 0$ and $r \leq \frac{1}{2}q$,

$$\|\alpha(qj + r)\| = \Big\|\frac{ar}{q}\Big\| - \frac{1}{2q} \geq \frac{1}{2}\Big\|\frac{ar}{q}\Big\|;$$

otherwise, for each j there are a bounded number of r for which $\|\alpha(qj + r)\| \geq \frac{1}{2}\|y_j + ar\|/q$ fails to hold, and moreover $qj + r \gg q(j + 1)$. Therefore

$$\begin{aligned} S &\ll \sum_{1 \leq r \leq q/2} \frac{1}{\|ar/q\|} + \sum_{0 \leq j \leq X/q} \Big(\frac{XY}{q(j+1)} + \sum_{\substack{1 \leq r \leq q \\ q \mid y_j + ar}} \frac{1}{\|y_j + ar/q\|}\Big) \\ &\ll \frac{XY}{q} \sum_{0 \leq j \leq X} \frac{1}{j+1} + \Big(\frac{X}{q} + 1\Big) \sum_{1 \leq h \leq q/2} \frac{q}{h}, \end{aligned}$$

and the required result follows. □

Major and Minor Arcs

In general, a major arc is an interval centred at a/q with a relatively small denominator q. In Waring's problem, the bound for q can be rather generous and flexible, and we choose to set

$$1 \leq q \leq P, \qquad P = N^{\frac{1}{10}}, \qquad N = \lfloor n^{1/k} \rfloor.$$

For such q, and $1 \le a \le q$ with $\gcd(a, q) = 1$, the interval

$$\mathfrak{M}(q, a) = \Big\{\beta : \frac{a}{q} - \frac{1}{N^{k-\frac{1}{10}}} \le \beta \le \frac{a}{q} + \frac{1}{N^{k-\frac{1}{10}}}\Big\},$$

is called a *major arc*. Such arcs are pairwise disjoint because the distance between any two centres is at least $1/P^2$, which is much larger than their combine lengths $2/N^{k-1/10}$; indeed, even the sum of the lengths of all the arcs is still small because

$$\frac{2}{N^{k-\frac{1}{10}}} \sum_{1\le q\le P} \phi(q) \le \frac{2}{N^{k-\frac{1}{10}}} \sum_{1\le q\le P} q \ll \frac{P^2}{N^{k-\frac{1}{10}}} \ll \frac{1}{N^{k-\frac{3}{10}}} \to 0 \quad \text{as} \quad n \to \infty.$$

Let $\mathfrak{M}$ denote the union of all the major arcs $\mathfrak{M}(q, a)$, considered as a subset of the unit interval $0 < \alpha \le 1$, shifting by the amount $N^{-1/10}$ if necessary to cover an arc at the end of the interval. We proceed to define

$$R_{\mathfrak{M}}(n) = \int_{\mathfrak{M}} f(\alpha)^s \mathrm{e}(-n\alpha)\, d\alpha,$$

which turns out to be the dominating term for $R(n)$, that is

$$R_{\mathfrak{M}}(n) \sim \frac{\Gamma(1 + 1/k)^s}{\Gamma(s/k)} \mathfrak{S}(n) n^{s/k-1} \quad \text{as} \quad n \to \infty. \tag{14.7}$$

The complementary set $\mathfrak{m} = (0, 1] \setminus \mathfrak{M}$ of intervals form the *minor arcs*. Although such arcs cover nearly the whole of the unit interval, we show that its contribution is small by comparison, that is

$$R_{\mathfrak{m}}(n) = \int_{\mathfrak{m}} f(\alpha)^s \mathrm{e}(-n\alpha)\, d\alpha = o(n^{s/k-1}) \quad \text{as} \quad n \to \infty. \tag{14.8}$$

By (14.6), and the decomposition $(0, 1] = \mathfrak{M} \cup \mathfrak{m}$, the two very distinct tasks for the method are to establish (14.7) and (14.8), with appropriate error terms if possible.

Singular Series and Singular Integral

For the major arcs, let $\alpha \in \mathfrak{M}(q, a)$, so that $\alpha = a/q + \beta$, with $|\beta| \le 1/N^{k-\frac{1}{10}}$. From $\mathrm{e}(\alpha m^k) = \mathrm{e}(am^k/q)\mathrm{e}(\beta m^k)$, we find that $f(\alpha)$ is essentially the product of two functions:

$$f(\alpha) = \sum_{1\le m\le N} \mathrm{e}(\alpha m^k) = \frac{S(q, a)}{q}\, v(\beta) + O(N^{1/5}), \qquad N = \lfloor n^{1/k} \rfloor,$$

where $q \le P = N^{\frac{1}{10}}$, $\gcd(a,q) = 1$,

$$S(q,a) = \sum_{1 \le m \le q} \mathrm{e}\Big(\frac{am^k}{q}\Big), \tag{14.9}$$

and

$$v(\beta) = \sum_{1 \le m \le n} \frac{m^{1/k-1}\mathrm{e}(\beta m)}{k}. \tag{14.10}$$

The factor $S(q,a)$ caters for the fact that the kth powers are not uniformly distributed modulo q, and the factor $v(\beta)$ is that obtained from $f(\alpha)$ by replacing the characteristic function of the kth power by the probability of m being a kth power: thus

$$\sum_{1 \le m \le N^k} \frac{m^{1/k-1}}{k} = \int_1^{N^k} \frac{\alpha^{1/k-1}}{k}\, d\alpha + O(1) = N + O(1).$$

Our first task is to study the structure of $S(q,a)$, which forms the *singular series* $\mathfrak{S}(n)$ in (14.7), with the aim of showing that $\mathfrak{S}(n)$ is bounded below by a positive constant c_1 depending only on k and s, and not on n. We first show how $\mathfrak{S}(n)$ becomes embedded in $f(\alpha)$ in (14.5).

Take $\alpha = a/q$ and consider

$$\begin{aligned} f\Big(\frac{a}{q}\Big) &= \sum_{1 \le m \le N} \mathrm{e}\Big(\frac{am^k}{q}\Big) \\ &= \sum_{1 \le m \le N} \sum_{\substack{1 \le \ell \le q \\ \ell \equiv m \pmod q}} \mathrm{e}\Big(\frac{a\ell^k}{q}\Big) \\ &= \sum_{1 \le \ell \le q} \mathrm{e}\Big(\frac{a\ell^k}{q}\Big) \sum_{\substack{1 \le m \le N \\ m \equiv \ell \pmod q}} 1 \\ &= \sum_{1 \le \ell \le q} \mathrm{e}\Big(\frac{a\ell^k}{q}\Big)\Big(\frac{N}{q} + O(1)\Big) \\ &= \frac{NS(q,a)}{q} + O(q). \end{aligned}$$

The expression $S(q,a)$ has a 'multiplicative' property:

$$S(qr, ar + bq) = S(q,a)S(r,b), \quad \text{if } \gcd(a,q) = \gcd(b,r) = \gcd(q,r) = 1.$$

This follows from the observation that each $m \pmod{qr}$ can be represented uniquely in the form $tr + uq$, with t, u running over $1 \le t \le q$ and $1 \le u \le r$, so that

$$\begin{aligned} S(qr, ar + bq) &= \sum_{1 \le m \le qr} \mathrm{e}\Big(\frac{(ar + bq)m^k}{qr}\Big) \\ &= \sum_{1 \le t \le q} \sum_{1 \le u \le r} \mathrm{e}\Big(\frac{at^k}{q}\Big)\mathrm{e}\Big(\frac{bu^k}{r}\Big) \\ &= S(q, a)S(r, b). \end{aligned}$$

Next, there is the trivial estimate $|S(q, a)| \le q$, but we expect, and need, some cancellation in the sum. If k is large then m^k are spaced far apart, and cancellation becomes relatively difficult to obtain. It can be shown that $S(q, a) \ll q^{1-1/k}$, but the weaker bound

$$S(q, a) \ll q^{1-1/K+\epsilon}, \qquad K = 2^{k-1}, \quad \epsilon > 0,$$

is enough for our purpose; such a bound follows from Weyl's inequality, in the later consideration of the contribution from the minor arcs. We next introduce

$$S(q) = \sum_{\substack{1 \le a \le q \\ \gcd(a,q)=1}} \Big(\frac{S(q, a)}{q}\Big)^s \mathrm{e}\Big(\frac{-an}{q}\Big), \tag{14.11}$$

which has the bound $S(q) \ll q^z$, where, for $s \ge 2^k + 1$, the exponent satisfies

$$z = (\epsilon - 1/K)s + 1 = 1 - s/K + \epsilon s < -1 - 1/2^k, \qquad \text{if} \quad \epsilon < 1/k.$$

The expression $S(q)$, which depends also on n, is real, because on replacing a by $-a$ in the defining sum, the result becomes $\bar{S}(q)$.

As we shall see, in virtue of the bound for $S(q)$, the singular series in (14.7) is then given by the absolutely convergent series

$$\mathfrak{S}(n) = \sum_{q=1}^{\infty} S(q);$$

the convergence is also uniform with respect to n. The next task is to convert the series into an Euler product. For this, we need the fact that the function $S(q)$ is multiplicative, which follows from the said multiplicative property of $S(q, a)$. Thus, if q, r are coprime then

$$S(qr) = \sum_{\substack{1 \le a \le q \\ \gcd(a,q)=1}} \sum_{\substack{1 \le b \le r \\ \gcd(b,r)=1}} \frac{S(qr, ar + bq)^s \mathrm{e}(-(ar + bq)n/qr)}{q^s r^s} = S(q)S(r).$$

Now define, for primes p,

$$T(p) = \sum_{h=0}^{\infty} S(p^h) = 1 + S(p) + S(p^2) + \cdots, \tag{14.12}$$

an absolutely convergent series, so that there exists a constant c_2, depending only on k and s, such that $|T(p)| < c_2$ for every p, and we may write

$$\mathfrak{S}(n) = \prod_{p \text{ prime}} T(p),$$

an absolutely convergent product. In as much as the tail end of a convergent series has to tend to 0, the tail end of a convergent product has to tend to 1, so that there exists $C > 0$ such that

$$\frac{1}{2} < \prod_{p \geq C} T(p) < \frac{3}{2}.$$

As a consequence, it suffices to show that, for each p, the value for $T(p)$ is real and is bounded away from 0, but the required argument is highly nontrivial.

Lest we forget, the problem is on the representations of n in (14.4), and there is a close connection between $T(p)$ and the number $M_n(q)$ of the solutions of the congruence

$$m_1^k + m_2^k + \cdots + m_s^k \equiv n \pmod{q}, \qquad 1 \leq m_i \leq q,$$

namely that

$$T(p) = \sum_{h=0}^{\infty} S(p^h) = \lim_{\ell \to \infty} \frac{M_n(p^\ell)}{p^{\ell(s-1)}}. \tag{14.13}$$

For the proof, we first show that

$$\sum_{d|q} S(d) = \frac{M_n(q)}{q^{s-1}}.$$

To see this, we start with $M_n(q)$ and relate it to $S(q)$ by making use of the fact that value of $\sum_{1 \leq a \leq q} \mathrm{e}(ah/q)$ is q or 0, depending on whether $q|h$ or not. Thus

$$M_n(q) = \frac{1}{q} \sum_{1 \leq m_1, m_2, \ldots, m_s \leq q} \sum_{a=1}^{q} \mathrm{e}\Big(\frac{a(m_1^k + m_2^k + \cdots + m_s^k - n)}{q}\Big)$$

$$= \frac{1}{q}\sum_{a=1}^{q} S(q,a)^s \mathrm{e}\Big(-\frac{an}{q}\Big).$$

Now rearranging the sum over a into subsums according to value of $d = \gcd(a,q)$, so that

$$\begin{aligned} M_n(q) &= \frac{1}{q}\sum_{d|q}\sum_{\substack{1\le a\le d\\ \gcd(a,q)=d}} d^s S\Big(\frac{q}{d},\frac{a}{d}\Big)^2 \mathrm{e}\Big(-\frac{a/d}{q/d}n\Big) \\ &= q^{s-1}\sum_{d|q}\sum_{\substack{1\le a\le d\\ \gcd(a,q)=d}} d^s \frac{S(q/d,a/d)^s}{(q/d)^s}\mathrm{e}\Big(-\frac{a/d}{q/d}n\Big) \\ &= q^{s-1}\sum_{d|q}\sum_{\substack{1\le b\le q/d,\\ \gcd(b,q/d)=1}} d^s \frac{S(q/d,b)^s}{(q/d)^s}\mathrm{e}\Big(-\frac{bn}{q/d}\Big) \\ &= q^{s-1}\sum_{d|q} S\Big(\frac{q}{d}\Big), \end{aligned}$$

as required, because the summand here can be replaced by $S(d)$. Formula (14.13) for $T(p)$ now follows from $\sum_{d|p^\ell} S(d) = S(1) + S(p) + \cdots + S(p^\ell)$.

Let k be given the standard prime factorisation, with τ for the exponent of p, so that $p^\tau | k$, and $p^{\tau+1} \nmid k$, and we may write

$$k = p^\tau k_0, \qquad \tau \ge 0, \quad p \nmid k_0.$$

If τ is large then the residues x^k modulo p^τ are comparatively scarce, and $M_n(p^\tau)$ is relatively difficult to estimate. We now define, for each prime p,

$$\gamma = \gamma(p) = \begin{cases} \tau+1 & \text{if} \quad p>2, \quad \text{or if} \quad p=2 \ \text{ and } \ \tau = 0, \\ \tau+2 & \text{if} \quad p=2 \ \text{ and } \ \tau > 0. \end{cases}$$

Lemma 14.3 *Suppose that $p \nmid m$, and that the congruence $x^k \equiv m \pmod{p^\gamma}$ is soluble. Then the congruence $y^k \equiv m \pmod{p^\ell}$ is soluble for every $\ell \ge \gamma$.*

Proof Suppose that $p > 2$, so that $\ell \ge \tau + 1$. A solution of $y^k \equiv m \pmod{p^\ell}$ can be constructed from the solution of $x^k \equiv m \pmod{p^\gamma}$ by making use of a primitive root g modulo p^ℓ, which is then also a primitive root g modulo p^γ. Set

$$x \equiv g^u \pmod{p^\gamma} \quad \text{and} \quad m \equiv g^v \pmod{p^\gamma},$$

so that $ku \equiv v \pmod{\phi(p^\gamma)}$, which implies $\gcd(k, \phi(p^\gamma)) | v$. Since

$$\gcd(k,\phi(p^\ell)) = \gcd(p^\tau k_0, (p-1)p^{\ell-1}) = \gcd(k_0, p-1)\times p^\tau = \gcd(k,\phi(p^\tau),$$

we deduce that $\gcd(k, \phi(p^\ell)|v$, so that there exists w such that $kw \equiv v \pmod{\phi(p^\ell)}$. Letting $y \equiv g^w \pmod{\phi(p^\ell)}$, we then find that $y^k \equiv g^{kw} \equiv g^v \equiv m \pmod{p^\ell}$.

For $p = 2$, there is no primitive root for 2^ℓ, but the usual argument of using 5 as a generator of a cyclic subgroup of order $2^{\ell-2}$ can be applied for the result. □

Lemma 14.4 *Let p be a prime for which the congruence $x_1^k + x_2^k + \cdots + x_s^k \equiv n \pmod{p^\gamma}$ has a solution with some x_i not divisible by p. Then*

$$T(p) \geq \frac{1}{p^{\gamma(s-1)}}. \tag{14.14}$$

Proof Assume without loss that $p \nmid x_1$, and let $\ell > \gamma$. Then, for each $i = 2, 3, \ldots, s$, there are $p^{\ell-\gamma}$ distinct residues $y_i \pmod{p^\ell}$ such that $y_i \equiv x_i \pmod{p^\gamma}$, so that the congruence

$$y_1^k \equiv n - x_2^k - x_3^k - \cdots x_s^k \pmod{p^\gamma}$$

has a solution $y_1 = x_1$ with $p \nmid y_1$. By Lemma 14.3, the congruence is soluble with γ being replaced by any $\ell > \gamma$, so that $M_n(p^\ell) \geq p^{(\ell-\gamma)(s-1)}$. The required result follows from (14.13). □

We still need to show that the hypothesis in Lemma 14.4 is satisfied.

Lemma 14.5 *Let*

$$s \geq \begin{cases} 2k, & k \text{ odd}, \\ 4k, & k \text{ even}. \end{cases}$$

Then, the congruence

$$x_1^k + x_2^k + \cdots + x_s^k \equiv n \pmod{p^\gamma} \tag{14.15}$$

has a solution with at least one x_i not divisible by p.

Proof Suppose first that $p > 2$. If $p \nmid n$, and the congruence (14.15) is soluble, then $p \nmid x_i$ for some x_i. If $p|n$, then we may set $x_s \equiv 1$ in (14.15), which then becomes

$$x_1^k + x_2^k + \cdots + x_{s-1}^k \equiv n - 1 \pmod{p^\gamma}.$$

Since $p \nmid n - 1$ the congruence in the lemma has, in effect, become one in which $p \nmid n$, and s is reduced by 1; that is, we now need to show that there is a solution for $s \geq 2k - 1$ for odd k, or $s \geq 4k - 1$ for even k, and $p \nmid n$.

Take a primitive root g modulo p^γ, and let $a \equiv g^\alpha$ and $b \equiv g^\beta \pmod{p^\gamma}$. Then $a^k \equiv b \pmod{p^\gamma}$ becomes $k\alpha \equiv \beta \pmod{\phi(p^\gamma)}$, which is soluble only if $\gcd(k, \phi(p^\gamma))|\beta$. As β runs over $\phi(p^\gamma) = (p-1)p^{\gamma-1} = (p-1)p^\tau$ distinct

values, the number of $b \pmod{p^\gamma}$ for which the congruence is soluble is

$$J = \frac{\phi(p^\gamma)}{\gcd(k_0, p-1)p^\gamma} = \frac{p-1}{\gcd(k_0, p-1)}.$$

Denote by s_n the least s for which the congruence is soluble, and divide the set of values of n into sets B_i, according to the values of s_n, so that B_i is the set of n with $s_n = i$. A set B_j is either empty, or else it has at least J members, and we proceed to show that no two consecutive sets are empty. Note that B_j is closed with respect to multiplication by x^k, because if the congruence is soluble for some n, then it is soluble for $x^k n$.

Let n' be the least number coprime with p and not belonging to B_i for $i \le j-1$. Then, because $p > 2$, one of $n'-1, n'-2$ is not divisible by p, and since $n' = (n'-1)+1^k = (n'-2)+1^k+1^k$, whichever one not divisible by p is a sum of s kth powers for some $s \le j-1$, so that $s_{n'} \le j+1$, whence $s_{n'} = j$ or $j+1$.

Let B_s be the last nonempty set. Then at least half of the first $s-1$ sets are nonempty, so that there are at least $(s+1)/2$ nonempty sets. Since each such sets has at least J members, if they are disjoint then $\frac{1}{2}(s+1)J \le (p-1)p^\tau$, so that

$$s_n \le s \le \frac{2(p-1)p^\tau \gcd(k_0, p-1)}{(p-1)} - 1 \le 2p^\tau k_0 - 1 = 2k-1.$$

This means that the congruence is soluble when $s \ge 2k-1$.

For $p = 2$, if $\tau = 0$, then k is odd and $x_1 \equiv 1 \pmod 2$ will do as a solution. If $\tau \ge 1$ then k is even and $\gamma = \tau + 2$. By setting $x_i = 1$ or 0 we can assume that $n \le 2^\gamma - 1$. If $s = 2^\gamma - 1 = 4\times 2^\tau - 1 \le 4k-1$, then the congruence (14.15) can be solved by setting $x_i = 1$ or 0. Thus $s_n \le 4k-1$ as required, and the lemma is proved. □

The condition for s in Lemma 14.5 is satisfied when $s \ge 2^k + 1$, so that, for every prime p, the bound (14.14) is valid. Therefore there exist c_1, c_2, depending on k, s and, not on n, such that

$$0 < c_1 < \mathfrak{S}(n) < c_2.$$

The remaining factor in the dominating term of the asymptotic formula for $R_s(n)$ is called the *singular integral* $J(n)$ for Waring's problem; it is given by

$$J(n) = J_s(N) = \int_{-1/2}^{1/2} v(\beta)^s \mathrm{e}(-\beta n)\, d\beta \sim \frac{\Gamma(1+1/k)^s}{\Gamma(s/k)} n^{s/k-1}, \tag{14.16}$$

with an error of order $n^{s/k-1-1/k}$. To see this, we consider in (14.10),

$$v(\beta)^s = \frac{1}{k^s} \sum_{m_1=1}^{n} \cdots \sum_{m_1=1}^{n} (m_1 \cdots m_s)^{1/k-1} \mathrm{e}(\beta(m_1 + \cdots + m_s)),$$

from which it follows that

$$J_s(n) = \frac{1}{k^s} \sum_{m_1+\cdots+m_s=n} (m_1 \cdots m_s)^{1/k-1}.$$

Now, for $A \geq B > 0$ and $B < 1$,

$$\sum_{1 \leq < n} m^{A-1}(n-m)^{B-1} = \int_0^n t^{B-1}(n-t)^{A-1}\, dt + O(n^{A-1}),$$

and the value of the integral here is

$$n^{A+B-1} \frac{\Gamma(A)\Gamma(B)}{\Gamma(A+B)}.$$

Taking $A = B = 1/k$, and incorporating the factor $1/k^2$ into $\Gamma(A)\Gamma(B)$, we have

$$J_2(n) = \frac{\Gamma(1+1/k)^2}{\Gamma(2/k)} N^{2/k-1} + O(n^{1/k-1}),$$

and (14.16) can be established by induction on s.

Weyl's Inequality and Hua's Lemma

The validity of the asymptotic formula for $R_s(n)$ depends on the admissible value of s, which depends ultimately on the quality of the eventual error term that can be obtained. The contribution from the minor arcs forms part of the error term, and this is perhaps the more delicate part of the argument, especially if s is to be set relatively small.

Since $\mathrm{e}(-n\alpha)$ no longer offers discernible savings from cancellations when $\alpha \in \mathfrak{m}$, we ignore the factor by taking its trivial bound, and consider

$$\left| \int_{\mathfrak{m}} f(\alpha)^s \mathrm{e}(-n\alpha)\, d\alpha \right| \leq \int_{\mathfrak{m}} |f(\alpha)|^s\, d\alpha \leq \sup_{\alpha \in \mathfrak{m}} |f(\alpha)| \int_0^1 |f(\alpha)|^{s-1}\, d\alpha. \tag{14.17}$$

Non-trivial estimates for $|f(\alpha)|$, when $\alpha \in \mathfrak{m}$, can be obtained from the work of H. Weyl [109] in 1916; in particular his proof that if α is irrational then the sequence (αm^k), $m = 1, 2, \ldots$, is uniformly distributed modulo 1.

For $s = 2t + 1$, the integral factor is the $2t$ th moment of $|f(\alpha)|$ over $0 \leq \alpha \leq 1$, and it has the interpretation of being the number of solutions of

$$x_1^k + x_2^k + \cdots + x_t^k = y_1^k + y_2^k + \cdots + y_t^k \tag{14.18}$$

with $1 \le x_j, y_j \le N$, and bounds can be obtained by first restricting the variables in $X_j \le x_j, y_j \le 2X_j$, with increasing X_j. There is an important lemma due to L.-K. Hua [60] which delivers significant savings, especially when t is a power of 2. Thus, for $k = 3$, $s = 9$, so that $s/k - 1 = 2$, Hua's lemma can be used to deliver $\int_0^1 |f(\alpha)|^8 \, d\alpha \ll n^{2-\frac{1}{3}}$; see Theorem 14.5 below.

The following is a description of Weyl's method of dealing with exponential sums corresponding to polynomials. The linear case, being a geometric progression, can be dealt with by Lemma 14.1, and the quadratic case can then be dealt with by a differencing procedure, reducing it to the linear case with hardly any loss. In general, for a polynomial with degree k, we introduce a forward difference operator, to be applied in conjunction with Cauchy's inequality, to reduce the problem to one with degree $k - 1$. However, the induction process may appear complicated, so we only describe the first step.

Let α be an irrational number, and $H(m)$ be a polynomial with real coefficients and degree $k > 2$. There may then be savings in an exponential sum $\sum \mathrm{e}(\alpha H(m))$ over $1 \le m \le N$, by taking approximations to α with fractions having denominators lying in a certain range. By Dirichlet's approximation theorem there is a fraction a/q such that

$$\left|\alpha - \frac{a}{q}\right| < \frac{1}{qQ}, \quad \text{with} \quad 1 \le q \le Q, \quad \gcd(a, q) = 1.$$

Thus, for $\alpha \in \mathfrak{m}$, we need to have $N^{\frac{1}{10}} < q \le Q$, and we may set $Q = N^{k-\frac{1}{10}}$.

Starting from (14.5) with the trivial estimate $|f(\alpha)| \le N$, we consider

$$|f(\alpha)|^2 = f(\alpha)\overline{f(\alpha)} = \sum_{1 \le \ell \le N} \sum_{1 \le m \le N} \mathrm{e}(\alpha(m^k - \ell^k)),$$

and replace the variable m in the inner sum by letting $h = m - \ell$. From $m^k - \ell^k = (h + \ell)^k - \ell^k = h^k + H(h, \ell)$, where $H(h, \ell) = kh^{k-1}\ell + \cdots + kh\ell^{k-1}$ is a polynomial in h, or ℓ, with degree $k - 1$, the expression for $|f(\alpha)|^2$ can be given a temporary bound. Thus, reversing the order of summations of the double sum, the manoeuvre then delivers

$$|f(\alpha)|^2 \le \sum_{|h| < N} |S_h|, \quad \text{where} \quad S_h = \sum_{\ell} \mathrm{e}(\alpha H(h, \ell)),$$

with ℓ lying in the intersection of the two intervals $1 \le \ell \le N$ and $1 \le \ell + h \le N$. An application of Cauchy's inequality then delivers $|f(\alpha)|^4 \le 2N \sum_{|h| < N} |S_h|^2$, which is just the first step in the differencing process. The bound now involves a polynomial with degree $k - 1$ and, by a forward difference operator on polynomials and repeating the process successively, the exponential sum is brought down to the linear case.

Theorem 14.4 (Weyl's Inequality) *Let $g(x) = \alpha x^k + \cdots$ be a polynomial of degree $k \geq 2$ with real coefficients, and suppose that $|\alpha - a/q| < 1/q^2$, where $\gcd(a, q) = 1$. Then, for $K = 2^{k-1}$, $\epsilon > 0$ and $N \to \infty$,*

$$\sum_{1 \leq \ell \leq N} \mathrm{e}(g(\ell)) \ll N^{1+\epsilon}\Big(\frac{1}{N} + \frac{1}{q} + \frac{q}{N^k}\Big)^{1/K},$$

where the implied constant depends on k and ϵ.

For our purpose, we take $g(x) = x^k$, and the earlier required estimate for $S(q, a)$ follows by setting $N = q$. Here, for $N^{\frac{1}{10}} < q < N^{k-\frac{1}{10}}$, the bound becomes

$$N^{1+\epsilon}\Big(\frac{1}{N} + \frac{2}{N^{\frac{1}{10}}}\Big)^{1/K} \ll N^{1+\epsilon-\frac{1}{10K}} = N^{1-\frac{1}{20K}}, \qquad K = 2^{k-1}, \quad \epsilon = \frac{1}{20K}.$$

We also have the following useful lemma to bound the integral factor in (14.17).

Theorem 14.5 (Hua's Lemma) *Let $k \geq 2$, $T(\alpha) = \sum_{1 \leq \ell \leq N} \mathrm{e}(\alpha \ell^k)$, and $\epsilon > 0$. Then, as $N \to \infty$,*

$$\int_0^1 |T(\alpha)|^{2^k} d\alpha \ll N^{2^k - k + \epsilon}.$$

The lemma can be established by induction on k; for $k = 1$, we have

$$\int_0^1 |T(\alpha)|^2 d\alpha = \sum_{1 \leq m, \ell \leq N} \int_0^1 \mathrm{e}(\alpha(m^k - \ell^k))\, d\alpha = \sum_{\substack{1 \leq m, \ell \leq N \\ m = \ell}} 1 = N.$$

The inductive step involves the forward difference operator applied to $|T(\alpha)|^{2^k}$, the divisor function, and also the number of solutions to (14.18).

From Weyl's inequality and Hua's lemma, we then find that the bound (14.17) for the contribution from the minor arcs is $\ll N^{s-k-\delta} \ll n^{s/k-1-\delta/k}$, for some $\delta > 0$, so that the asymptotic formula in Theorem 14.3 follows.

It is conjectured that, for every $\epsilon > 0$ and $N \to \infty$,

$$\int_0^1 |f(\alpha)|^{2s} d\alpha \ll N^{\epsilon} \max(N^s, N^{2s-k});$$

this would imply that the asymptotic formula for $R_s(n)$ holds for $s \geq 2k + 1$.

Let $\tilde{G}(k)$ denote the value of the smallest s for which the asymptotic value holds, so that $G(k) \leq \tilde{G}(k)$; there is likely to be strict inequality here because, for some s, we may have $R(n) > 0$ for large n already without the asymptotic formula being valid. Indeed, the current upper bound for $\tilde{G}(k)$ is larger than that for $G(k)$ by about a factor of k. Much more details are given in [105].

14.4 Sums of Three Primes: Vinogradov's Theorem

In 1742 C. Goldbach communicated to Euler his famous conjecture that every even number greater than 4 is a sum of two odd primes. Goldbach also realised that if his 'binary conjecture' is true then every odd number exceeding 7 is a sum of three odd primes, his 'ternary conjecture'. Let $R_2(n)$ denote the number of ordered representations of an even number n as a sum of two odd primes, and $R_3(n)$ the number of ordered representations of an odd number n as a sum of three odd primes. Heuristic reasoning suggests that $R_2(n) \gg n/\log^2 n$ and $R_3(n) \gg n^2/\log^3 n$, and Table 14.1 gives the results of some experiments.

For example, the 25 representations of $101 = p_1 + p_2 + p_3$ as a sum of three odd primes (p_1, p_2, p_3) with $p_1 \le p_2 \le p_3$ are:
(3, 19, 79), (3, 31, 67), (5, 7, 89), (5, 13, 83), (5, 17, 79), (5, 23, 73), (5, 29, 67), (7, 11, 83), (7, 23, 71), (11, 11, 79), (11, 17, 73), (11, 19, 71), (11, 23, 67), (11, 29, 61), (11, 31, 59), (13, 17, 71), (13, 29, 59), (17, 17, 67), (17, 23, 61), (17, 31, 53), (19, 23, 59), (19, 29, 53), (23, 31, 47), (29, 29, 43), (29, 31, 41).
The time taken for the computation is not significant until that for $R_3(10^4 + 1) = 28722$, which took over 100 minutes, and $(\star)$ means that the task is onerous on a small machine. We should add that $R_2(n)$ and $R_3(n)$ are **not** increasing functions, because their values are subject to local perturbations, depending on the number and size of the odd prime divisors of n. Such perturbation may be relatively large; for example, we find that

$$R_2(n+8) = 2310, \quad R_2(n) = 9493, \qquad \text{when} \quad n = 510510,$$

with the factorisations

$$510510 = 2\times3\times5\times7\times11\times13\times17, \qquad 510518 = 2p, \quad p = 255259 \text{ prime}.$$

The phenomenon will be quantified in the form of the *singular series* which emerges from the analysis of the problems.

An additive problem concerned with prime numbers is perhaps somewhat intractable, because prime numbers are defined, ostensibly, in terms of multiplication only. Indeed there was no method to attack the conjectures until Hardy and Littlewood came along with their circle method. Their wonderful idea not only lends

Table 14.1 Counts for Goldbach's binary and ternary conjectures

n	$R_2(n)$	$R_3(n+1)$
10	2	1
10^2	6	25
10^3	28	745
10^4	127	28722
10^5	810	$(\star)$
10^6	5402	$(\star)$

credence to the conjectures but also shows that, if the argument can be successfully completed, then asymptotic formulae for the numbers of representations concerned are valid. Besides Cauchy's theorem, or Fourier analysis, there are three essential ingredients for the solution of the ternary conjecture:

- Diophantine approximations;
- Distribution of primes in arithmetic progressions;
- Estimates of exponential sums over primes.

Again, although Diophantine approximation is at the heart of the problem, the required result from it is straightforward. However, there are serious obstacles corresponding to the remaining two ingredients that have to be overcome. Thus, although there is the theorem on primes in arithmetic progressions, we now require results with the modulus being allowed to vary. A standard hypothesis may imply such a result and, indeed, Hardy and Littlewood had shown, assuming (a weakened version of) the generalised Riemann hypothesis, that large odd numbers are sums of three primes.

In 1935 C. L. Siegel [94] established an important result on the value of $L(1, \chi)$ for a real primitive character $\chi \pmod q$, from which A. Walfisz [106] deduced the following theorem in which

$$\theta(x; q, a) = \sum_{\substack{p \le x \\ p \equiv a \pmod q}} \log p, \qquad p \text{ prime},$$

is the weighted prime counting function over an arithmetic progression, and $\phi(q)$ is the Euler totient function:

Theorem 14.6 (Siegel-Walfisz) *Let $C > 0$. Then, as $x \to \infty$,*

$$\theta(x; q, a) = \frac{x}{\phi(q)} + O\Big(\frac{x}{\log^C x}\Big),$$

uniformly in q, a with $\gcd(q, a) = 1$.

Note that the main term $x/\phi(q)$ manifests itself only for $q < \log^C x$. The proof of the theorem is 'ineffective' in the sense that it does not provide a procedure to determine the value of the implied constant from the value of C.

There still remains the finding of savings from estimates of exponential sums over primes in order to exercise control over the error terms involved in the method. In 1937 I. M. Vinogradov succeeded in overcoming this last obstacle, thereby delivering the unconditional proof of the ternary Goldbach conjecture for large odd numbers. Even after the simplification of the circle method by Vinogradov, and more recently by Vaughan, it is still a formidable task, but we hope the reader can comprehend the wonderful ideas involved in the proof of the theorem.

Counting Weighted Representations

As in the solution to Waring's problem, it is more appropriate to consider the apparently more difficult task of finding the number of representations of a large odd number as a sum of three primes. Although this might appear too tall an order, such a bold approach may deliver at least some partial success still to be celebrated. For example, we may be able to show that, for almost all odd numbers, there is an abundance of representations.

Since the primes are involved, it is expedient (though not necessary) to count the representations with an appropriate weight by setting

$$R(n) = \sum_{p_1+p_2+p_3=n} (\log p_1)(\log p_2)(\log(p_3), \qquad p_1, p_2, p_3 \text{ primes}; \tag{14.19}$$

the weights can be removed later by partial summation (see Sect. 11.3). The generating function for the problem is now (the cube of)

$$f(\alpha) = \sum_{p\le n} (\log p)\mathrm{e}(\alpha p), \qquad p \text{ prime}, \ \alpha \in \mathbb{R}; \tag{14.20}$$

so that $f(\alpha)$ depends on n, which is considered fixed in the following. We then find, as in Waring's problem, that

$$R(n) = \int_0^1 f(\alpha)^3 \mathrm{e}(-n\alpha)\, d\alpha. \tag{14.21}$$

By applying Diophantine approximations to α in the integrand, and dividing the unit interval into major and minor arcs, the dominating expression for $R(n)$ may be extracted from the major arcs.

Major and Minor Arcs

For Vinogradov's theorem, we need to have a rather small bound[2] for the denominators of the major arcs. Let

$$P = \log^B n, \qquad Q = \frac{n}{P} = \frac{n}{\log^B n};$$

[2] This is because our current knowledge of the distribution of primes in arithmetic progressions, although adequate for the purpose here, is rather poor compared to the probable truth.

the parameter $B > 0$ will be set to deliver an appropriate error term in the eventual asymptotic formula for $R(n)$. For each fraction a/q with

$$1 \le a \le q \le P, \quad \gcd(a, q) = 1,$$

denote the interval $|\alpha - a/q| \le 1/Q$ by $\mathfrak{N}(q, a)$; such major arcs $\mathfrak{N}(q, a)$ are pairwise disjoint, and their union is denoted by $\mathfrak{N}$. The complementary set $\mathfrak{n} = (0, 1] \setminus \mathfrak{N}$ of intervals form the minor arcs.

Dirichlet's approximation theorem states that for any $\alpha \in \mathbb{R}$, and $Q \ge 1$, there is a fraction a/q such that

$$\left|\alpha - \frac{a}{q}\right| < \frac{1}{qQ}, \quad \text{with} \quad 1 \le q \le Q, \quad \gcd(a, q) = 1.$$

Thus, for $\alpha \in \mathfrak{n}$ and the chosen values for P and Q, we have $P < q \le Q$.

The integral (14.21) is now decomposed into the sum

$$R(n) = \int_{\mathfrak{N}} f(\alpha)^3 \mathrm{e}(-n\alpha)\, d\alpha + \int_{\mathfrak{n}} f(\alpha)^3 \mathrm{e}(-n\alpha)\, d\alpha, \tag{14.22}$$

and the aim is to show that the first integral over the major arcs $\mathfrak{N}$ is $\gg n^2$ when n is odd, and the second integral over the minor arcs $\mathfrak{n}$ is $o(n^2)$.

Contribution from the Major Arcs

For the purpose of analysing the contribution from the major arcs, we introduce the function

$$v(\beta) = \sum_{1 \le m \le n} \mathrm{e}(\beta m).$$

We shall make use of the fact that

$$\int_{-1/2}^{1/2} v(\beta)^3 \mathrm{e}(-n\beta)\, d\beta = \sum_{1 \le m_1, m_2, m_3 \le n} \int_{-1/2}^{1/2} \mathrm{e}((m_1 + m_2 + m_3 - n)\beta)\, d\beta,$$

which is the number of solutions to $m_1 + m_2 + m_3 = n$, that is $\frac{1}{2}(n-1)(n-2) = \frac{1}{2}n^2 + O(n)$. Anyway, we then find that, for $\alpha = a/q + \beta$, as $n \to \infty$,

$$f(\alpha) = \frac{\mu(q)}{\phi(q)} v(\beta) + O\Big(\frac{qn}{\log^C n}\Big).$$

The derivation is based on

$$\begin{aligned} f\Big(\frac{a}{q}\Big) &= \sum_{\substack{1\le r\le q\\ \gcd(r,q)=1}} \mathrm{e}\Big(\frac{ar}{q}\Big)\theta(n;q,r) + O(\log(nq)) \\ &= \frac{n}{\phi(q)} \sum_{\substack{1\le r\le q\\ \gcd(r,q)=1}} \mathrm{e}\Big(\frac{ar}{q}\Big) + O\Big(\frac{qn}{\log^C n}\Big) \\ &= \frac{\mu(q)n}{\phi(q)} + O\Big(\frac{qn}{\log^C n}\Big), \end{aligned}$$

where we have invoked the Sielgel-Walfisz theorem, and that the Ramanujan sum

$$c_q(a) = \sum_{\substack{1\le r\le q\\ \gcd(r,q)=1}} \mathrm{e}\Big(\frac{ra}{q}\Big)$$

takes the value of the Möbius function $\mu(q)$ when $\gcd(a,q) = 1$; see Sect. 2.12. Suppose now that $\alpha \in \mathfrak{N}(q,a)$, so that we may write

$$\alpha = \frac{a}{q} + \beta, \qquad |\beta| < \frac{1}{Q}, \quad q \le P.$$

We then find that

$$f(\alpha)^3 = \frac{\mu(q)}{\phi(q)^3} v(\beta)^3 + O\Big(\frac{n^3}{\log^C n}\Big),$$

and thus that

$$\int_{\mathfrak{M}} f(\alpha)^3 \mathrm{e}(-n\alpha)\,d\alpha = \mathfrak{S}(n,P) \int_{-P/n}^{P/n} v(\beta)^3 \mathrm{e}(-n\beta)\,d\beta + O\Big(\frac{n^3}{\log^C n}\Big),$$

where

$$\mathfrak{S}(n,P) = \sum_{q\le P} \sum_{\substack{1\le a\le q\\ \gcd(a,q)=1}} \frac{\mu(q)}{\phi(q)^3} e\Big(\frac{-an}{q}\Big), \qquad P = (\log n)^B.$$

The Ramanujan sum $c_q(n)$ is a multiplicative function of q and satisfies (see (2.14))

$$c_q(n) = \frac{\mu(q/d)\phi(q)}{\phi(q/d)}, \qquad d = \gcd(q,n),$$

so that, as $P \to \infty$, the expression for $\mathfrak{S}(n,P)$ converges absolutely to the singular series $\mathfrak{S}(n)$ for the ternary Goldbach problem. The series can also be given by an

absolutely convergent product over primes p:

$$\mathfrak{S}(n) = \sum_{q=1}^{\infty} \frac{\mu(q)}{\phi(q)^3} c_q(n) = \prod_p \left(1 - \frac{c_p(n)}{(p-1)^3}\right) \tag{14.23}$$
$$= \prod_{p|n} \left(1 - \frac{1}{(p-1)^2}\right) \prod_{p\nmid n} \left(1 + \frac{1}{(p-1)^3}\right).$$

Note that if n is even then $2|n$ and $\mathfrak{S}(n) = 0$, but if n is odd then

$$\prod_{p|n} \left(1 - \frac{1}{(p-1)^2}\right) > \prod_p \left(1 - \frac{1}{p^2}\right) = \frac{1}{\zeta(2)}, \quad \prod_{p\nmid n} \left(1 + \frac{1}{(p-1)^3}\right) > 2,$$

so that $\mathfrak{S}(n) > 1$.

Next, because $v(\beta) \ll 1/\|\beta\|$, the integral factor over $-P/n \le \beta \le P/n$ can be replaced by that over the interval $-1/2 \le \beta \le 1/2$. Therefore, by our initial remark on the value of the integral, we now have, for any $A > 0$, as $n \to \infty$,

$$\int_{\mathfrak{M}} f(\alpha)^3 \mathrm{e}(-n\alpha)\, d\alpha = \frac{1}{2}\mathfrak{S}(n)\, n^2 + O\Big(\frac{n^2}{\log^A n}\Big). \tag{14.24}$$

Contribution from the Minor Arcs

As in Waring's problem, we apply

$$\left| \int_{\mathfrak{m}} f(\alpha)^3 \mathrm{e}(-n\alpha)\, d\alpha \right| \le \Big(\max_{\alpha\in\mathfrak{m}} |f(\alpha)|\Big) \int_0^1 |f(\alpha)|^2\, d\alpha; \tag{14.25}$$

there is then no corresponding Hua's lemma to speak of because the last integral is

$$\sum_{p_1,p_2\le n} (\log p_1)(\log p_2) \int_0^1 \mathrm{e}((p_1 - p_2)\alpha)\, d\alpha = \sum_{p\le n} (\log p)^2 \ll n \log n,$$

and the bound is already sharp. Instead of Weyl's inequality, the required saving from $|f(\alpha)|$ when $\alpha \in \mathfrak{m}$ now comes from a result of Vinogradov in 1937.

Vaughan's Identity

Many theorems in analytic number theory depend on estimates for finite sums $\sum_p f(p)$ over primes p, which amounts to $\sum_n \Lambda(n) f(n)$, where $f(x) = e(\theta_x)$ is a complex number on the unit circle for $x \geq 1$, so that some cancellation is expected.[3] Such estimates depend in turns on estimates of bilinear forms of the types

$$\text{(I)} \quad \sum_m \sum_n x_m f(mn), \qquad \text{(II)} \quad \sum_m \sum_n x_m y_n f(mn).$$

Thus, Vinogradov related sums over primes to such bilinear forms via the sieve of Eratosthenes:

$$f(1) + \sum_{\sqrt{X} < p \leq X} f(p) = \sum_{m \mid P(\sqrt{X})} \sum_{n \leq X/m} \mu(m) f(mn), \qquad P(Y) = \prod_{p \leq Y} p,$$

which is a sum of type I. However, when m is near X the length of the inner sum is rather short for any effective cancellation to take place. Vinogradov overcame the difficulty by dividing the interval $1 < n \leq X$ into $\log X$ subintervals of the form $M < n \leq M'$, with $M' \leq 2M$. In effect, his argument shows that those m close to X with all their prime factors small occur relatively rarely, and so can be neglected. The remaining terms can be rearranged into the form

$$\sum_{V < p \leq \sqrt{X}} \sum_{k \leq X/p} x_p y_k f(pk),$$

which is a manageable bilinear form of type (II), but the argument involved is extremely complicated.

In 1977 R. C. Vaughan [102] introduced an identity which greatly simplifies Vinogradov's method; indeed a more flexible scheme also applies to other problems; for example, it has led to a new short proof of Bombieri's theorem, and also the proof that the values of cubic Gauss sums are uniform distributed.

The main idea is the introduction of the truncated sum function

$$M(x) = M(x, X) = \sum_{d \mid x,\, d \leq X} \mu(d), \qquad x \in \mathbb{N};$$

thus $M(1) = 1$ and $M(x) = 0$ for $1 < x \leq X$. For an arithmetic function $f(k, \ell)$ of two variables, Vaughan's identity is

$$\sum_{X < y \leq n} f(1, y) + \sum_{X < x \leq n} \sum_{X < y \leq n/x} M(x) f(x, y) = \sum_{d \leq X} \sum_{X < y \leq n/d} \sum_{z \leq n/\ell d} \mu(d) f(dz, y).$$

[3] The generic function $f(x)$ here is not to be confused with $f(\alpha)$ defined in (14.20).

The left-hand side of the identity is just the double sum

$$\sum_{1\le x\le n}\ \sum_{X\le y\le n/x} M(x)f(x,y),$$

and the right-hand side is the evaluation of the double sum by setting $x = dz$ and interchanging the order of summation.

Lemma 14.6 (Vinogradov) *Let $f(\alpha)$ be given by (14.20). Suppose that $|\alpha - a/q| \le 1/q^2$, where $\gcd(a,q) = 1$ and $q \le n$. Then, as $n \to \infty$,*

$$f(\alpha) \ll (\log n)^4\Big(\frac{n}{\sqrt{q}} + n^{4/5} + n^{1/2}q^{1/2}\Big).$$

Proof (Vaughan) Let $\Lambda(m)$ be the Mangoldt function. Taking $f(x,y) = \Lambda(y)\mathrm{e}(\alpha xy)$, and $X = n^{2/5}$, Vaughan's identity delivers

$$f(\alpha) = S_1 - S_2 - S_3 + O(n^{1/2}),$$

where

$$S_1 = \sum_{x\le n^{2/5}}\ \sum_{y\le n/x} \mu(x)(\log y)\mathrm{e}(\alpha xy),$$

$$S_2 = \sum_{x\le n^{4/5}}\ \sum_{y\le n/x} c_x\,\mathrm{e}(\alpha xy), \quad \text{with} \quad c_x = \sum_{\substack{d,y\le n^{2/5}\\ dy=x}} \mu(d)\Lambda(y),$$

$$S_3 = \sum_{\substack{x,y>n^{2/5}\\ xy\le n}} M(x,n^{2/5})\Lambda(y)\mathrm{e}(\alpha xy).$$

The inner sum in S_1 is

$$\mu(x)\int_1^{n/x} \mathrm{e}(\alpha xy)\frac{d\gamma}{\gamma},$$

and $c_x \ll \log x$. Hence, by Lemma 14.1,

$$S_1, S_2 \ll (\log n)\sum_{x\le n^{4/5}} \min\Big(\frac{n}{x}, \frac{1}{\|\alpha x\|}\Big),$$

and so, by Lemma 14.2,

$$S_1, S_2 \ll (\log n)^2\Big(\frac{n}{q} + n^{4/5} + q\Big).$$

For S_3, we let

$$\mathcal{A} = \{X, 2X, 4X, \dots, 2^k X : 2^k X^2 < n \le 2^{k+1} X^2\},$$

so that

$$S_3 = \sum_{Y \in \mathcal{A}} S(Y), \qquad S(Y) = \sum_{Y < x \le 2Y} \sum_{X < y \le n/x} M(x)\Lambda(y)\mathrm{e}(\alpha xy).$$

By Cauchy's inequality,

$$|S(y)|^2 \ll \Big(\sum_{x \le 2Y} d(x)^2 \Big) \sum_{Y < x \le 2Y} \Big| \sum_{X < y \le n/x} \Lambda(y)\mathrm{e}(\alpha xy) \Big|^2,$$

where $d(x)$ is the divisor function, so that, by (10.1), the first factor of the bound is $\ll Y(\log Y)^3$. The inner sum in the second factor is of the form S_2, so that

$$\begin{aligned} |S(Y)|^2 &\ll Y(\log n)^5 \sum_{y \le n/Y} \sum_{z \le n/Y} \min\Big(Y, \frac{1}{\|\alpha(y-z)\|}\Big) \\ &\ll n(\log n)^6 \Big(\frac{n}{q} + Y + \frac{n}{Y} + q \Big), \end{aligned}$$

by another application of Lemma 14.2. Consequently

$$\begin{aligned} S_3 &\ll \sum_{Y \in \mathcal{A}} (\log n)^3 \Big(\frac{n}{\sqrt{q}} + \sqrt{nY} + \frac{n}{\sqrt{Y}} + \sqrt{nq} \Big) \\ &\ll (\log n)^4 \Big(\frac{n}{\sqrt{q}} + n^{4/5} + \sqrt{nq} \Big), \end{aligned}$$

as required. □

Returning to the contribution from the minor arcs, let $\alpha \in \mathfrak{m}$. By Dirichlet's approximation theorem there is a fraction a/q satisfying

$$\Big| \alpha - \frac{a}{q} \Big| \le \frac{1}{q^2}, \qquad \text{with} \quad 1 \le q \le Q, \quad \gcd(a, q) = 1,$$

Thus $P < q \le Q = n/\log^B n$, so that $f(\alpha) \ll n/\log^{B/2-4} n$ by Vinogradov's lemma, and hence, by (14.25),

$$\int_{\mathfrak{m}} f(\alpha)^3 \mathrm{e}(-n\alpha)\, d\alpha \ll \frac{n^2}{\log^{B/2-5} n}, \tag{14.26}$$

where B can be arbitrarily large.

Vinogradov's Theorem

From (14.22) together with the estimates (14.24) and (14.26), we arrive at the following theorem.

Theorem 14.7 (I. M. Vinogradov) *For any fixed* $A > 0$*, and* $\mathfrak{S}(n)$ *given by* (14.23)*, we have*

$$R(n) = \frac{1}{2}\mathfrak{S}(n)n^2 + O\Big(\frac{n^2}{\log^A n}\Big), \qquad n \to \infty.$$

If n is even then $\mathfrak{S}(n) = 0$, and the theorem is not relevant; indeed there is then the better estimate $R(n) \ll n\log^2 n$, because one of the primes concerned is 2. However, if n is odd then $\mathfrak{S}(n) > 1$, so that $R(n) > \frac{1}{2}n^2$ for large n, and hence all large odd numbers n are representable as sums of three primes in $\gg n^2/\log^3 n$ ways. The ternary conjecture for large odd numbers follows as a corollary.

If, instead of (14.19), we take $r(n)$ to be the number of unordered representations of n as a sum of three odd primes then

$$r(n) = \frac{\mathfrak{S}(n)n^2}{2\log^3 n} + O\Big(\frac{n^2}{\log^A n}\Big), \qquad n \to \infty.$$

For $n = 10^4 + 1 = 73\times137$, the dominating term of the formula gives $\frac{1}{6}r(n) \approx 24500$, which is not unreasonable when compared to the experimental count of 28722.

According to Vinogradov's theorem, there can only be a finite number of exceptions to the ternary Goldbach conjecture. However, we still need to have an 'effective' bound for n for the application of theorem in order to reduce the conjecture to a matter of verification by computation. In other words, we need to set a value for A and then make explicit the value of the implied constant associated with the error term $O(n/\log^A n)$; in particular, we need somehow to 'side-step' the non-effective aspect of the Siegel-Walfisz theorem. This was first done in 1956 by Vinogradov's student K. Borozdin who showed that all exceptional values are less than $10^{7000000}$. The huge bound was reduced to the smaller, but still enormous, value of $e^{3100} \approx 2\times10^{1346}$ by M. C. Liu and T. Z. Wang in 2002.

Recently, H. Helfgott has completed the solution of the ternary Goldbach problem. More specifically, by making improvements on the estimate for the minor arcs and the error term in the major arcs, he has lowered the bound to 10^{29}, but computer verification can reach only up to about 10^{20}. Then, by making use of extensive numerical verifications of the Generalised Riemann Hypothesis, he has completed the solution, posted on arXiv in 2013; his claim is broadly accepted.

Binary Conjecture

We end the guided tour of the circle method for the Goldbach conjectures with a few words on why there is no corresponding result for the binary case. The asymptotic formula for $R(n) = \sum_{p_1+p_2=n}(\log p_1)(\log p_2)$, which is expected to be $R(n) \sim \mathfrak{S}(n)n$, as $n \to \infty$, where the singular series is given by

$$\mathfrak{S}(n) = \sum_{q=1}^{\infty} \frac{\mu^2(q)c_q(n)}{\phi^2(q)} = \prod_{p\nmid n}\Big(1 - \frac{1}{(p-1)^2}\Big)\prod_{p|n}\Big(1 + \frac{1}{p-1}\Big),$$

can be derived in a similar way from the corresponding integral over the major arcs. However, we now encounter an apparently insurmountable difficulty of finding an adequate estimate of the integral

$$\int_{\mathfrak{m}} f(\alpha)^2 \mathrm{e}(-n\alpha)\, d\alpha$$

over the minor arcs, because the required bound is now $o(n)$. Finding such savings directly from the integrand seems hopeless, so the only hope left is to enlarge the major arcs $\mathfrak{M}$ to such an extent that the minor arcs $\mathfrak{m}$ becomes a thin set instead, but that would require assumptions perhaps beyond what can be expected from the uniform distribution of primes in arithmetic progressions. There being little doubt that the conjectural asymptotic formula is valid, we have to wait for someone to come up with a radical new idea.

14.5 Summary and Notes

Summary

- The *circle method* is a powerful analytic tool which is particularly useful in tackling additive problems. The method has its genesis in a paper by Hardy and Ramanujan in their search for an accurate formula for the partition function, and it was followed by a series of paper under the general title "*Some problems of partitio numerorum*", by Hardy and Littlewood.
- Meanwhile, following from the work of Hardy and Ramanujan, Rademacher succeeded in giving the value of the partition function as the sum of a convergent series.
- *Waring's problem* is concerned with the representation of n as a sum of s lots of kth powers. Denoting by $g(k)$ the least s such that $n = x_1^k + \cdots + x_s^k$ is soluble for every $n \in \mathbb{N}$, Waring predicted that $g(2) = 4$, $g(3) = 9$, $g(4) = 19$, 'and so on'. That $g(k)$ is finite was established by Hilbert, and later Hardy and

Littlewood applied the circle method giving bounds for $g(k)$, which together with much further work by others, the value of $g(k)$ is now given in Theorem 14.2.

- The modern Waring's problem is the much more challenging one of the determination of $G(k)$, the least s for which the representation equation is soluble for all sufficiently large n. The circle method delivers an asymptotic formula (see Theorem 14.3) for the number of solutions of the representation equation, which is valid provided that s is not too small corresponding to a given value of k. The method requires Diophantine approximations and delicate estimates of exponential sums. Even the task of extracting the dominating term in the asymptotic formula is formidable.
- The circle method can be applied to tackle the problem of the representation of large odd numbers as sums of three primes, the ternary Goldbach conjecture. The required ingredients are Diophantine approximations, distribution of primes in arithmetic progressions, and the estimation of exponential sums over primes. The result was deduced by Hardy and Littlewood, under the assumption that $L(s, \chi) \neq 0$ in $\sigma > \frac{3}{4}$, in order to overcome the problem of finding savings from exponential sums over primes.
- Vinogradov succeeded in finding savings from exponential sums over primes, thereby giving an unconditional proof of the ternary Goldbach conjecture.
- There remains the problem of the conjecture being valid for *all* odd numbers. The first obstacle to overcome is to bypass the problem of the non-effectiveness of the Siegel-Walfisz theorem for the distribution of primes in arithmetic progression, and the finding explicit constants so that machine verification can complete the task. This has been done, but only recently.

Notes

1. *Modular group, modular functions and modular forms*. The set of 2×2 matrices

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad a, b, c, d \in \mathbb{Z}, \quad ad - bc = 1,$$

form a group. On writing $A\tau = (a\tau + b)/(c\tau + b)$ for a bilinear transformation of $\tau \in \mathbb{C}$, the set of such transformations is called the *modular group* Γ, if we take A to be its representative element. The group Γ is then generated by the two matrices $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

A function f is *modular* if it is meromorphic in the upper half-plane $\mathbb{H}$, with $f(A\tau) = f(\tau)$ for every $A \in \Gamma$, and its Fourier expansion has the form $f(\tau) = \sum_{n\in\mathbb{Z}} a(n)e^{2\pi in\tau}$. A *modular form* f of weight $k \in \mathbb{Z}$ is an analytic function

in $\mathbb{H}$ satisfying $f(A\tau) = (c\tau + d)^k f(\tau)$ for every $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$, and its Fourier expansion has the form $f(\tau) = \sum_{n \geq 0} c(n) e^{2\pi i n \tau}$.

The book [6] is devoted to modular functions and forms, including the proof of Rademacher's formula for the partition function $p(n)$.

2. For Waring's problem, there can be much local fluctuations among the values of $R_k(n)$ when n lies in different arithmetic progressions with the same modulus. Thus, in his inaugural lecture at Oxford University (1920), G. H. Hardy wrote that "We can show, for example, that a large number of the form $16n + 10$ can be expressed by 21 biquadrates in about 200 times more ways than one of the form $16n + 2$." Hardy chose '21 biquadrates' because, for $k = 4$, the then bound for s obtained by him and Littlewood for the validity of the asymptotic formula for $R_k(n)$ was $s \geq 21$.
3. The circle method has been applied to show that almost all even numbers are sums of two primes (see, for example, [29]); indeed letting $E(x)$ count the even numbers n in $4 < n \leq x$ which are **not** sums of two odd primes, it has been established that there exists a positive $\delta < 1$ such that $E(x) \ll x^{\delta}$. The binary conjecture states, of course, that $E(x) = 0$.
4. There is a proof of Vinogradov's theorem in [32].
5. The very recent publication [69] on the circle method is pitched at a similar level to ours here.

14.6 Exercises

1. Use Euler's recurrence formula to evaluate $p(m)$ for $0 \leq m \leq n$ for a reasonably large n, and apply the results to

   (i) Verify that if $\delta = 5^a, 11^a$ and $24m \equiv 1 \pmod{\delta}$ then $p(m) \equiv 0 \pmod{\delta}$;
   (ii) Find the three least values m such that $24m \equiv 1 \pmod{7^3}$, but $7^3 \nmid p(m)$.

2. Denote by $H_+(n)$, and $H_-(n)$, the numbers of $m \leq n$ with $p(m)$ odd, and even, respectively, and let $H(n) = H_+(n) - H_-(n)$. It seems likely that

$$\limsup_{n \to \infty} H(n) = \infty \quad \text{and} \quad \liminf_{n \to \infty} H(n) = -\infty,$$

   so that $H(n) = z$ is soluble for every integer $z \in \mathbb{Z}$.

   (i) Verify that $H(21016) = -241$ and $H(78822) = 809$;
   (ii) Find the three least solutions of $H(m) = 0$.

3. Show that $\sqrt{n} \ll \log p(n) \ll \sqrt{n}$ as $n \to \infty$.
4. Derive the implementation from (14.3) of Rademacher's formula for $p(n)$, and use to verify the value of $p(10^6)$ given in the text.

5.

(i) Show that if n a sum of two positive cubes then

(a) n has a divisor d satisfying $\sqrt[3]{n} \le d \le \sqrt[3]{4n}$;
(b) $n \not\equiv 3, 4 \pmod 7$ and $n \not\equiv 3, 4, 5, 6 \pmod 9$.

(ii) The famous taxicab number of Hardy-Ramanaujan, namely 1729, is the smallest number representable as a sum of two cubes in two (essentially) different ways; thus $1729 = 10^3 + 9^3 = 12^3 + 1^3$. Find the next number with this property.

(iii) It is known that 87539319 is a sum of two cubes in three different ways. Find the representations concerned.

6. Verify that $g(k) = 2^k + \lfloor (3/2)^k \rfloor - 2$ for $2 \le k \le 10^6$.
7. Write out the argument given in the text for the specific case $G(7) \ge 8$.
8. Verify the claims (i), (ii), (iii) and (iv) in the text concerning numbers representable as sums of three cubes.
9.

(i) Show that there is a unique cube which is the sum of three cubes of consecutive numbers.

(ii) Verify that 239 is not a sum of 8 nonnegative cubes.

10. Perform, as described in the text, experiments on a_n, the least number of cubes for the representation of n.
11. Let $R_9(n)$ denote the number of representations of n as a sum of 9 cubes, and $\mathfrak{S}(n)$ the corresponding singular series. Evaluate $R_9(n)$ for various values of n and compare the result to the Hardy-Littlewood formula.
12. Write out the proof of Hua's lemma for the case $k = 3$.
13. Verify Hardy's statement in Note 2.
14. Show that $G(6) \ge 9$ by making use of $a^6 \equiv 0, 1 \pmod 9$ to deduce that numbers congruent to 9 (mod 27) cannot be a sum of 8 sixth powers.

Answers/Hints/Solutions to Selected Exercises

Chapter 1

1.1 *Hints:* Well ordering $\Longrightarrow$ induction. The set of n for which $P(n)$ is false is either empty, or else it has a least member. Induction $\Longrightarrow$ well ordering. Suppose that $S \subset \mathbb{N}$ has no least member. Let $P(n)$ be the proposition that $n \notin S$. Then $P(1)$ is true, and if $P(k)$ is true then ....

1.2 *Hint:* Consider, for example, $(k+1)^3 - k^3 = 3k^2 + 3k + 1$ for $0 \leq k \leq n$.

1.3 *Hint:* Write the sum as A/B, and consider the parities of A and B.

1.4 *Answer:* Either $m = n \in \mathbb{N}$ or $\{m, n\} = \{2, 4\}$.

1.5 *Hint:* Suppose (if possible) that there are coprime integers m, n such that $\sqrt[5]{3} = m/n$, a fraction in 'lowest form', so that $3^{4/5}n^4 = m^4$ and hence $\sqrt[5]{3}m^4 = 3n^4$. Apply the division process to m^4 with the divisor n.

1.6 *Hint:* Numbers congruent to 2 (mod 3) must have a prime divisor $p \equiv 2$ (mod 3).

1.7 *Hint:* Apply Dirichlet's approximation theorem to the irrational $\phi = \theta + \sqrt{1+\theta^2}$.

1.9 *Answer:* 7.

1.10 *Solution:* If $a \geq 1$ then $2 \leq a + 1 \leq 2^a$, so that $2 \leq d(p^a) \leq 2^a$. Thus if $n = \prod_{p|n} p^a$ then $2^{\sum_{p|n} 1} \leq d(n) \leq 2^{\sum_{p|n} a}$, which is the required result.

1.12 *Answer:* $(x, y) = (6, 11), (13, 8), (20, 5), (27, 2)$ for $n = 95$;
$(x, y) = (4, 12), (11, 9), (18, 6), (25, 3), (32, 0)$ for $n = 96$.

1.13 *Hint:* Apply Thue's theorem.

1.14 *Hint:* Apply Exercise 1.13.

1.15 *Hint:* Use the method in Example 1.6.

1.16 *Solution:* When Team-i plays Team-j, we let the home team be $\min(i, j)$ or $\max(i, j)$ depending on whether $i + j$ is odd or even. Teams with numbers in

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brackets are home teams. Observe that the home teams in a round are the five teams immediately 'in front of' the bye in that round.

Round	Team										
	1	2	3	4	5	6	7	8	9	10	11
1	(11)	(10)	(9)	(8)	(7)	bye	5	4	3	2	1
2	bye	11	10	9	8	7	(6)	(5)	(4)	(3)	(2)
3	2	(1)	(11)	(10)	(9)	(8)	bye	6	5	4	3
4	(3)	bye	1	11	10	9	8	(7)	(6)	(5)	(4)
5	4	3	(2)	(1)	(11)	(10)	(9)	bye	7	6	5
6	(5)	(4)	bye	2	1	11	10	9	(8)	(7)	(6)
7	6	5	4	(3)	(2)	(1)	(11)	(10)	bye	8	7
8	(7)	(6)	(5)	bye	3	2	1	11	10	(9)	(8)
9	8	7	6	5	(4)	(3)	(2)	(1)	(11)	bye	9
10	(9)	(8)	(7)	(6)	bye	4	3	2	1	11	(10)
11	10	9	8	7	6	(5)	(4)	(3)	(2)	(1)	bye

1.17 *Answer:* 235309 (mod 1616615).

1.18 *Hint:* Apply the Chinese remainder theorem.

1.19 *Hint:* Consider the 'conjugate' number $\sqrt{2}+1$.

1.20 *Answer:* The number concerned is $N = 160878900622575730954212 5$. The outputs from the implementation of Aubry's theorem are:

d'	m_1	m_2	m_3
2297	419250897899726	1120209687330243	2656626803819300
573	308811603025386	656959325126223	35382546427380
102	126153896868276	17723384027118	22559184035520
54	43622793691212	46446269573310	25120204128384
24	942896841072	16247236455240	25725487764096
1	39287368378	676968185635	1071895323504

Chapter 2

2.1 *Answer:* The 16 odd numbers in $3, 5, 7, \ldots, 37$, together with $14, 26, 34, 38$.

2.2 *Answer:* Solutions of $\phi(n) = 24$ are $n = 35, 39, 45, 52, 56, 70, 72, 78, 84, 90$. *Hint:* Take $m_k = (p_1 - 1)(p_2 - 1)\cdots(p_k - 1)$, where p_k is the k th prime, so that $\phi(p_1 p_2 \cdots p_k) = m_k$, and use the fact that if $m|n$ then $\phi(mn) = m\phi(n)$.

2.3 *Hint:* The formula $\phi(n) = n\prod_{p|n}(1 - 1/p)$ shows that the value of $\phi(n)$ is relatively small if n has many small prime divisors p. It is thus more likely

that $\phi(6n+6) \le \phi(6n+2)$, for randomly chosen values of n, because $6n+6$ is divisible by the two smallest primes 2 and 3, whereas $6n+2$ is not divisible by 3. To find a counter example, choose n so that $6n+2$ has more small prime divisors (such as 2, 5, 7, 11, 13, 17) than those of $6n+6$. Take $n = 128$, for example.

More generally, let $Q_k = 2p_3p_4\cdots p_k$, where p_k denotes the kth prime. Take any $k \ge 5$ for which there are an odd number of prime divisors in the arithmetic progression $p \equiv 2 \pmod 3$, so that $Q_k \equiv 2 \pmod 6$, and set $n_k = (Q_k - 2)/6$.

2.4 *Hint:* If n is a prime power then $\phi(n) = \ldots$. If n is a product of two coprime numbers, then $\phi(n) = \ldots$, and recall the means inequality.

2.5 *Hint:* Consider the (length of the) interval $7/x \le t < 8/x$, where $\frac{1}{10} \le x < 1$, and wait for a wet weekend before contemplating the last part.

2.6 *Answer:* The order of 10 (mod p) is $p-1$ when $p = 7, 17, 19, 23, 29, 47, 59, 61, 97$, and

$$\frac{1}{37} = 0{\cdot}\overline{027}, \quad \frac{1}{41} = 0{\cdot}\overline{02439}, \quad \frac{1}{43} = 0{\cdot}\overline{023255813953488372093},$$

$$\frac{1}{47} = 0{\cdot}\overline{021276595744680851063829787234042553191489361 7}.$$

2.7 *Answer:* The digital representations of $n = 10^{20}$ in base b are:

$$\begin{aligned}
n =&(1,0,1,0,1,1,0,1,0,1,1,1,1,0,0,0,1,1,1,0,1,0,1,1,1,1,0,\\
&0,0,1,0,1,1,0,0,1,1,0,1,0,1,0,0,0,1,1,0,0,0,1,0,0,0,0,\\
&0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0)_2,\\
n =&(2,2,0,2,0,0,0,2,0,1,2,2,1,2,0,1,1,2,0,1,0,2,2,2,0,2,2,\\
&1,2,2,0,0,2,0,0,0,1,0,0,2,0,1)_3,\\
n =&(2,3,2,0,2,3,3,0,1,0,0,0,0,0,0,0,0,0,0,0,0,\\
&0,0,0,0,0,0,0)_5,\\
n =&(3,3,0,3,4,3,0,4,0,1,3,0,2,3,4,4,0,0,4,2,0,5,4,5,4,4)_6,\\
n =&(3,4,4,0,1,5,3,1,3,5,6,1,6,2,1,0,0,1,4,5,2,5,6,2)_7.
\end{aligned}$$

2.8 *Solution:* From $13 \equiv 3 \pmod{10}$ and $\phi(10) = 4$ together with $2345 \equiv 1 \pmod 4$, we have $13^{2345} \equiv 3^{2345} \equiv 3 \pmod{10}$.

2.9 *Hint:* Consider (mod m) and (mod n) separately.

2.10 *Solution:* For the construction of m, we note that a product of 8 coprime numbers, each with at least 10 divisors, has at least 10^8 divisors, but we need to ensure that the value of product is at most 10^{100}. There is much leeway, because we may also take 6 suitable coprime numbers M in $10^{12} <$

$M < 10^{20}$, with $15 < d(M) < 70$, say, to deliver the goods. We take $m = M_1M_2M_3M_4$, where

$$\begin{aligned} M_1 &= 2^{60} = (2^{10})^6 = 1024^6 \approx (10^3)^6 = 10^{18}, \\ M_2 &= 3^{40} = (3^2)^{20} < 10^{20}, \\ M_3 &= 5^{20} = 10^{20}/2^{20} < 10^{20}/10^6 = 10^{14}, \\ M_4 &= 1001^{15} \approx (10^3)^{15} = 10^{45}, \end{aligned}$$

so that $m = 10^{18+20+14+45+\Delta} = 10^{97+\Delta} < 10^{100}$, if $\Delta < 3$; in fact $\Delta < 0$. Also, from $1001 = 7\times11\times13$, we have $d(m) = 61\times41\times21\times16^3 = 215126016 > 10^8$.

For the construction of n, we take the primes in $2 = p_1 < p_3 < \cdots < p_{20} = 71$. Let $i = 1, 2, \ldots, 20$, $j = 21 - i$, and $n_i = p_i^{p_j - 1}$, so that $d(n_i) = p_j$. Now take $n = \prod n_i$, so that $d(n) = d(\prod p_j) = 2^{20} > 10^6$, and that $n < 10^{1000}$.

2.12 *Solution:* The number $n + 1$, being coprime with n, is a product of h (not necessarily distinct) primes $q_i > p_k$, so that $h < k$. By Exercise 1.10, we now have $d(n + 1) \le 2^{\Omega(n+1)} = 2^h \le 2^{k-1}$. The case for $n - 1$ is similar.

2.13 *Answer:* The first 10 values n with $d(n) = d(n + 1)$ are

$$2, 14, 21, 26, 33, 34, 38, 44, 57, 75, \text{ and the number of such } n < 10000 \text{ is } 1119.$$

2.14 *Answer:* The first 10 values n with $\phi(n) = \phi(n + 1)$ are

$$1, 3, 15, 104, 164, 194, 255, 495, 584, 975.$$

2.15 *Hint:* Make use of the sum $\sum_{d|\gcd(n,m)} \mu(d)$.

2.16 *Hint:* The simple formula for the sum is $\lfloor\sqrt{N}\,\rfloor$.

2.17 *Hint:* The functions concerned are multiplicative.

2.18 *Hint:* Consider $k \le \dfrac{\log m/n}{\log p} < k + 1 \iff \dfrac{m}{p^{k+1}} < n \le \dfrac{m}{p^k}$, for $k \in \mathbb{N}$ and $p \in \mathbb{P}$.

Chapter 3

3.2 *Answers:* $U(10^k) = 2, 8, 34, 205, 1223, 8169$ for $k = 1, 2, 3, 4, 5, 6$. $V(10^6) = 166$.

3.3 *Answer:* The difference $492227 - 492113 = 114$ is the largest prime gap up to 10^6.

3.4 *Answer:* $\pi(10^9) = 50847534$, the corrected value of Meissel's heroic effort (+56) without a machine in 1885.
3.5 *Answer:* $\pi(10^{10}) = 455052511$, the corrected value of Lehmer's result, the first count (−1) with a machine in 1959.
3.7 *Hints and answers:* Make use of the following facts: (i) almost all numbers are composite; (ii) if $f(n) = k\pi(n) - n$ then $f(m-1) \le f(m) + 1$; (iii) all the solutions of $10\pi(n) = n$ are in the interval $64540 \le n \le 64720$; (iv) $11\pi(n) = n$ if and only if $n = 175197$. More on the problem may be found in [92].
3.10 *Answer:* $490191327 = 1791 \times 273697$. The number of steps involved is 115603, and $(1791 + 273697)/2 - \sqrt{490191327} \approx 115603.7$.
3.12 *Answer:* $(n_1, n_2) = (17, 64)$.
3.14 *Hint:* After passage of time, some secret messages may no longer be secret.

Chapter 4

4.1 *Solution:* At least one of the pairs $(1, 2), (4, 5), (9, 10)$ has the property of being both residues; this is because $1, 4, 9$ are residues, and if 2 and 5 are both nonresidues then 10 is a residue. The remaining part follows, because the set of residues and the set of nonresidues have the same cardinal, namely $(p-1)/2$.
4.2 *Hint:* The necessary condition follows by considering the equation $p = x^2 + 5y^2$ as a congruence (mod 5), and (mod 4). *Remark*: The condition is also sufficient, but the proof requires the theory of quadratic forms.
4.7 *Answer:* $448123436099907599 = 789101123 \times 567891013$. The number of steps in the iterative scheme is 9075844, which is reduced to 93 if the process is enhanced with the use of the Legendre symbol for primes $p < 60$, and further reduced to only 2 if applied with $p < 100$.
4.8 *Answer:* The 9 Euler pseudoprimes base 3 less that 1000 are: [121, 703, 1729, 1891, 2821, 3281, 7381, 8401, 8911] 9
4.9 *Answer:* $x \equiv \pm 94 \pmod{353}$.
4.11 *Answer:* $p = 2^{50}7 + 1 = 7881299347898369$ is a prime, and $a = 33, 333, 3333$ are quadratic residues (mod p). The corresponding solutions of $x^2 \equiv a \pmod{p}$ are

$$x = \pm 3846243273383196, \quad \pm 7672682142152315, \quad \pm 3519171874720845.$$

4.12 *Hint:* First, N has to be a prime, because .... Next, show that the number k satisfying $0 < kN - p < N$ has to be a nonresidue, because ....
4.13 *Answer:* $p = 3, 7, 23, 71, 311, 479, 1559, 5711, 10559, 18191$. Thus $p = 10559$ and 18191 are the two smallest primes with least nonresidues 23 and 31, the 9th and 10th primes.

4.14 *Answer:* $x \equiv \pm 200482 \pmod{493039}$.
4.15 *Answer:* $q = 1427911$.

Chapter 5

5.1 *Answer:* The $\phi(\phi(101)) = \phi(100) = 40$ primitive roots (mod 101) are

$$2, 3, 7, 8, 11, 12, 15, 18, 26, 27, 28, 29, 34, 35, 38, 40, 42, 46, 48, 50, 51,$$
$$53, 55, 59, 61, 63, 66, 67, 72, 73, 74, 75, 83, 86, 89, 90, 93, 94, 98, 99.$$

5.2 *Hints:* Consider (and justify):

$$10^{2p} \equiv \left(\frac{10}{q}\right) \equiv \left(\frac{2}{q}\right)\left(\frac{5}{q}\right) \equiv \left(\frac{2}{q}\right)\left(\frac{q}{5}\right) \equiv (-1)(+1) \equiv -1 \pmod{q}.$$

5.4 *Answer:* For $b = 0, 1, 2, 3, 4, 5, 6, 7$, we have

$$\pm 5^b \equiv 1, 5, 25, 29, 17, 21, 9, 13 \pmod{32}.$$

5.5 *Solution:* Note that $(-g)^a \equiv \pm g^a \pmod{p}$, depending on the parity of a, and also $g^{(p-1)/2} \equiv -1 \pmod{p}$. If $p \equiv 3 \pmod 4$ then $(p-1)/2$ is odd, so that $(-g)^{(p-1)/2} \equiv (-1)(-1) \equiv 1 \pmod{p}$, which implies that $-g$ is not a primitive root.

If $p \equiv 1 \pmod 4$ and $a = 1, 3, 5, \ldots, (p+1)/2$ then $(-g)^a \equiv -g^a \pmod{p}$, from which it follows that $-g$ is also a primitive root.

5.6 *Answer/Hint:* The 14 primes of the form $p = 4q + 1 < 1000$, with q being prime, are

$$13, 29, 53, 149, 173, 269, 293, 317, 389, 509, 557, 653, 773, 797.$$

They form a subset of the 67 primes $p = 3, 5, 11, 13, \ldots 941, 947 < 1000$, of which 2 is a primitive root. *Hint:* The order of a number modulo a $p = 4q + 1$ is restricted to ....

5.7 *Answer/Hint:* There are only three primes $p < 1000$ of the form $p = 2^a + 1 > 3$, namely, $p = 5, 17, 257$. The number 3 is a primitive root of the 68 primes $p = 5, 7, 17, 19, \ldots, 257, \ldots, 953, 977 < 1000$.

Hint: The order of a number modulo a $p = 2^a + 1$ is restricted to ....

5.8 *Answer:* There are 96 numbers $\{7, 13, 23, \ldots, 1057, 1067, 1073\}$ having order $\lambda(1080) = 36$.

5.9 *Hint/Answer:* $\Omega(n) = 4$ when $k = 1, 45, 56, 121$.

5.10 *Answer:* The 6 Carmichael numbers are $997qr$, with

$$(q, r) = (8467, 367027),\ (12451, 4137883),\ (17929, 130477),$$
$$(29383, 450691),\ (167329, 15166093),\ (1002973, 99996409).$$

5.11 *Hint:* Write $n = pQ$ and consider $n - 1 = (p - 1)Q + Q - 1$.

5.13 *Answer:* For $p = 9941$ there is no solution to $33x^7 \equiv 41 \pmod{p}$; there are two solutions to $29^y \equiv 71 \pmod{p}$, given by $y = 3639, 8609$. *Remark*: Note that $8609 - 3639 = 4970 = (p - 1)/2$.

5.15 *Answer:* The least prime $p > 10^{16}$ is $p = 10^{16}+61$, of which $g = 37$ happens to be a primitive root. The discrete log of 123456789 is 1611008384357885.

5.16 *Answers:* $N = 97\times109\times113$ and $N + 1 = 2\times5^3 3^4 59$. The number 10 is a primitive root of each of the primes 59, 97, 109, 113, so that the length of the period of the decimal expansion of $1/N$ is $\phi(N) = 96\times108\times112 = 1161216$. From $1/3^4 = 0.\overline{012345679}$, it follows that the length of the period of $1/(N + 1)$ is $9\times58 = 522$.

5.17 *Hint:* The equation follows from expansion of the product concerned, and it is not onerous to show that $m_8 = 5$, $m_9 = 8$ are valid. Setting $t = 1/q$, so that $t^2 + t^3 + \cdots = 1/p(p - 1)$, we then find that Artin's constant can be rewritten as $A = \prod_{k\geq 2} 1/\zeta(k)^{m_k}$. The value of A can now be evaluated from the accurate values of $\zeta(k)$, giving $A = 0.37395\,59136\ldots$.
Remark: P. Moree showed, in the quoted article, that $m_k = \frac{1}{k}\sum_{d|k} a_d \mu(k/d)$, where a_k satisfies $a_1 = 0$, $a_2 = 2$, and $a_k = a_{k-1} + a_{k-2} + 1$ for $k \geq 3$, and μ is the Möbius function.

Chapter 6

6.1 *Hints:* (i) What is the area of the ellipse? (ii) What is the volume of a sphere with radius $\sqrt{x}$? (iii) Apply the argument used in the proof of Theorem 6.5.

6.2 *Hints:* (i) Use u to define a suitable lattice, and consider a circle centre the origin with area exceeding $4p$; (ii) Use u, v to define a 4-dimensional lattice with fundamental domain having volume p^2, and consider a 4-dimensional sphere centre at the origin with volume exceeding $16p^2$.

6.4 *Hint:* About half of the primes p are congruent to 3 (mod 4), and the product $\prod(1 + 1/p)$ over such p is divergent. Also, for each such p, a number $n \equiv kp \pmod{p^2}$ with $1 \leq k < p$ has a witness for $n \notin \mathbb{W}$.

6.5 *Hint:* Start from $4B^4 = \prod_r (1 - 1/r^2)^{-2}$, where r runs over primes congruent to 3 (mod 4), and consider, for a certain character χ,

$$\prod_r \Big(1 - \frac{1}{r^2}\Big)^{-1}\Big(1 + \frac{1}{r^2}\Big) = \prod_{p>2} \Big(1 - \frac{1}{p^2}\Big)^{-1}\Big(1 - \frac{\chi(p)}{p^2}\Big).$$

6.6 (i) *Answer:* $w_{2677} = 9721$, $w_{2676} = 9704$;
(ii) *Hint:* Let m be an even number satisfying $m \equiv 1 \pmod 9$, and consider the 16 numbers $z = m^2 + u$, where $8^2 < u < 9^2$. If $u \equiv 3 \pmod 4$, or $u \equiv 2, 5 \pmod 9$, then z cannot be a sum of two squares. Use the Chinese remainder theorem to find m so that each of the remaining z has a witness. Then $(w_n, w_{n+1}) = (m^2 + 8^2, m^2 + 9^2)$ is a solution.

6.7 *Answers:* (i) 3141549; (ii) 216341.

6.8 *Solution:* Let $n = 10^{200} = (a_R a_{R-1} \cdots a_1 a_0)_4$ in base 4. The number $\Delta(n) = a_0/6 + \sum_{1 \le r \le R} \gamma(a_r, a_{r-1})$ can be obtained from Table 6.1, giving $\Delta(n) = 676/6$, and the required number is given by $5n/6 + \Delta(n) = 8333 \cdots 333446$.

6.11 *Hint:* (a) Observe that $111, 112 \notin Q$; (b) If two of them are odd then ...; and if two of them are even, and the odd one is congruent to 7 (mod 8), then the smaller of the even one ....

Chapter 7

7.1 *Answer:* [0, 1, 1, 8, 1, 173881, 1, 9, 7].

7.2 *Answer:* Let $a_n = n + 1$, for $n = 0, 1, 2, \ldots, 8$, and $x = [a_0, a_1, \ldots, a_8]$. Then the last two convergents for x are

$$\frac{p_7}{q_7} = \frac{81201}{56660}, \quad \frac{p_8}{q_8} = \frac{740785}{516901} = x.$$

The purely periodic continued fraction $[\overline{a_0, a_1, \ldots, a_8}]$ is the positive root of the quadratic $q_8\theta^2 - (p_8 - q_7)\theta - p_7 = 0$.

7.3 *Answers:* The decimal approximations for $\sqrt{7}, e, \pi$ in the text should be enough for the purpose of the determination of the 20 partial quotients a_n for the three numbers $x = \sqrt{7} + e, \sqrt{7} + \pi, e + \pi$. We then find that

$$\sqrt{7} + e = [5, 2, 1, 2, 1, 20, 9, 1, 4, 1, 2, 1, 7, 1, 1, 3, 1, 1, 3, 1, 2, \ldots],$$
$$\sqrt{7} + \pi = [5, 1, 3, 1, 2, 2, 1, 3, 2, 2, 2, 13, 1, 70, 1, 2, 1, 3, 1, 2, 4, \ldots],$$
$$e + \pi = [5, 1, 6, 7, 3, 21, 2, 1, 2, 2, 1, 1, 2, 3, 3, 2, 5, 2, 1, 1, 1, \ldots].$$

and that

$$\frac{p_{20}}{q_{20}} = \frac{468018457}{87251224} < x < \frac{168303572}{31376311} = \frac{p_{19}}{q_{19}};$$
$$\frac{p_{20}}{q_{20}} = \frac{3208271381}{554359893} < x < \frac{736096058}{127190653} = \frac{p_{19}}{q_{19}};$$
$$\frac{p_{20}}{q_{20}} = \frac{2836559813}{484064944} < x < \frac{1780479469}{303842595} = \frac{p_{19}}{q_{19}}.$$

7.4 *Answer:*

$$\sqrt{a^2+4} = \begin{cases} [a, \overline{b, 2a}], & a = 2b \geq 2 \\ [a, \overline{b, 1, 1, b, 2a}], & a = 2b+1 \geq 3. \end{cases}$$

7.5 *Hint:* Think Pell's equation.

7.6 *Solutions:* For $N = 7788$, we have $\lfloor \sqrt{N} \rfloor = 88$, and

$$\frac{1}{\sqrt{N}-88} = \frac{\sqrt{N}+88}{44} = 4 + \frac{\sqrt{N}-88}{44}, \quad \frac{44}{\sqrt{N}-88} = 176 + (\sqrt{N}-88),$$

giving $\sqrt{7788} = [88, \overline{4, 176}]$. The first convergent is $88 + \frac{1}{4} = \frac{353}{4}$, so that the least solution of $x^2 = 7788y^2 + 1$ is $(x, y) = (353, 4)$.

The number $N = 7789$ is a prime congruent to 1 (mod 4), so that the negative Pell equation $x^2 = 7789y^2 - 1$ is soluble by Lagrange's theorem (Theorem 7.7). Computation shows that

$$\sqrt{7789} = [88, \overline{3, 1, 11, 58, 1, \ldots, 2, 2, 1, 1, 2, 2, \ldots, 1, 58, 11, 1, 3, 176}]$$

with period length 111. The least solution of $x^2 = 7789y^2 - 1$ is given by $(x, y) = (p, q)$, where p/q is the 110th convergent, and computation delivers

$$p = 4975862822894008560408239687255191680998469880 1$$
$$2076378629818,$$
$$q = 5638032137518668487771917797885014649796028953714788160325.$$

7.7 *Hint:* If α is near θ then $\theta \pm \alpha$ is either near zero, or near 2θ.

7.9 *Hints:* (i) Let $\alpha = [a_0 : a_1, a_2, \ldots]$, with $a_n = 2$ or 3 according to some rule; (ii) Let $\alpha = [a_0 : a_1, a_2, \ldots]$, with $a_n = j$ for the least j with $\ldots$.

7.13 *Solution:* A solar eclipse occurs when there is a new moon, but only when the projected image of the moon onto the star sphere is at, or very near, one of the nodes, in order for the three heavenly bodies *Sun, Moon, Earth* to be in near alignment in space. Similarly, a lunar eclipse occurs when there is a full moon, with *Sun, Earth, Moon* in near alignment.

The time for a draconic month is 2351135·8 seconds, and the time for a synodic month is 2551442·9 seconds. From

$$\frac{25514429}{23511358} = [1, 11, 1, 2, 1, 4, 3, 5, 6, 7, 1, 8, 1, 1, 6],$$

with the convergents

$$\frac{12}{11}, \quad \frac{13}{12}, \quad \frac{38}{35}, \quad \frac{51}{47}, \quad \frac{242}{223}, \quad \frac{777}{716}, \quad \cdots, \quad \frac{25514429}{23511358},$$

it follows from taking the fifth convergent $242/223$ that 242 draconic months is pretty near 223 synodic months. Thus, eclipses occur in cycles with approximately such a length of time, and the required conclusion follows from

$$242 \times 2351135{\cdot}8 \text{ seconds} \approx 6585 \text{ days} = 18 \times 365 + 15 \text{ days},$$

and noting that there are 4 or 5 leap years in 18 years.
Remark. There is, of course, no need for such accurate results for the draconic and synodic months in order to arrive at the conclusion, which explains why the phenomenon (Saros cycle) was already known even in antiquity.

Chapter 8

8.1 *Hint:* Consider $\cos\frac{\pi}{3}$ and $\cosh\sqrt{2}$.
8.2 *Hint:* $\pi^2 I_{n+2} = 2(2n+3)(n+2)I_{n+1} - 4(n+2)(n+1)I_n$ for $n = 0, 1, 2, \ldots$, and if $\pi^2 = a/b$ then $a^n I_n/n!$ is an integer.
8.4 *Hint/Answer:* The expression $\big((x(1-x))^{4h} - (-4)^h\big)/(x^2+1)$ is a polynomial with degree $8h-2$. If h is odd, then there are positive integers u_h, v_h, with v_h being a divisor of $\mathrm{LCM}(1, 2, \ldots, 8h-1)$, such that

$$J_h = \frac{u_h}{v_h} - 4^{h-1}\pi, \qquad \text{so that} \quad \mu_h = \frac{u_h}{4^{h-1}v_h}, \qquad h = 1, 3, 5 \ldots,$$

are rational approximations of π. The intervals $\mathcal{I}_h = \{\mu_{h+2} \le \mu < \mu_h\}$ form a partition of the interval $\pi < \mu < 22/7 = \mu_1$.
For $(P, Q) = 355/113)$ we find that $(h, r, s) = (1, 4127, 1225785)$.
8.5 *Answer:* 56.
8.6 *Answers:* (i) 19/6, (ii) 179/57.
8.7 *Answers:* The values of the two convergents concerned are 312689/99532 and 521030/165849; it is readily verified that the former one satisfies the criterion, whereas the latter one does not.

8.8 *Answer:* $\pi^4 = [97, 2, 2, 3, 1, 16539, \ldots]$, and the convergents are

$$\frac{195}{2}, \quad \frac{487}{5}, \quad \frac{195}{2}, \quad \frac{1656}{17}, \quad \frac{2143}{22}, \quad \frac{35444733}{363875},$$

so that $\pi \approx \sqrt[4]{35444733/363875} = 3{\cdot}141592653589\ldots$.

8.10 *Hint:* Let p_n/q_n be the nth convergent of the continued fraction for e as given in the proof of Theorem 8.10. Use induction to show that $(u_n+v_n)/(u_n-v_n) = p_{3n}/q_{3n}$, and hence that $(\alpha+1)/(\alpha-1) = e$.

8.11 *Hint:* Consider a difference of two squares.

8.12 *Hint:* Obtain a bound for $\sqrt[3]{a/b} - y/x$ by considering that if ϕ is near θ then $(\theta^3 - \phi^3) = (\theta-\phi)(\theta^2+\theta\phi+\phi^2) \approx 3\theta^2(\theta-\phi)$.

8.13 *Answer:* $\sigma_1\sigma_2 - 3\sigma_3$.

Chapter 9

9.1 *Hint:* Use $\log n = \sum_{d|n} \Lambda(d)$ to show that the sum $T(x)$ becomes the sum $\sum \psi(x/n)$, and that $T^\star(x)$ can be written as a sum $\sum a_n \psi(x/n)$. By considering the value of a_n for $n \pmod{30}$, show that the non-zero terms of the sum is an alternating-sign series, so that $\cdots$.

9.2 *Hint:* If n is even then $n\,p(n) = 4m$, and if $P(n) = q$ then $n\,P(n) = q^2 m$, where $m = 1$ or $P(m) \le q$.

9.3 *Solution:* Let $0 < a < b < 1$, $p_0 = 1$, p_i denote the i th prime, and write $F_0 = 1$, $F_k = \prod_{1\le i\le k}(1 - 1/p_i)$. Define k by $F_{k+1} \le b < F_k$, and set $n = p_1 p_2 \cdots p_k p$, with a sufficiently large p, depending on a.

9.4 *Answer:* $n = 2^a$ for $a \ge 0$, or $n = 2^a 3^b$ with $ab > 0$.

9.5 *Hint:* Suppose that there exists a prime p not dividing k, but $p-1$ does divide k. Then, $n = (p-1)k$ is a solution. Now show that p always exists.

9.6 *Hint:* An obvious first try is to set $n = gpqr$, but it works only if $\phi(pqr)$ is coprime with g.

9.8 *Answer:* Let $N = \binom{2n}{n}$. Stirling's formula then gives $\log N = L(n) + O(\log n)$, where $L(n) = \log n + (2\log 2 - 1)n$. For $n = 50$, we find that

$$N = 2^{50}3^{26}5^{12}7^8 11^5 13^4 17^3 19^3 23^2 29^2 31^2 \prod_{31<p<100} p, \quad p \text{ prime},$$

so that $\log N = 215{\cdot}26\ldots$. We also find that $L(50) = 214{\cdot}91\ldots$.

9.9 *Answer:*

n	1	2	4	7	8	11	13	14
$\chi_0(n)$	1	1	1	1	1	1	1	1
$\chi_1(n)$	1	$-i$	-1	i	i	-1	$-i$	1
$\chi_2(n)$	1	-1	1	-1	-1	1	-1	1
$\chi_3(n)$	1	i	-1	i	$-i$	1	$-i$	-1
$\chi_4(n)$	1	i	-1	$-i$	$-i$	-1	i	1
$\chi_5(n)$	1	-1	1	1	-1	-1	1	-1
$\chi_6(n)$	1	$-i$	-1	$-i$	i	1	i	-1
$\chi_6(n)$	1	1	1	-1	1	-1	-1	-1

9.11 *Solution:* Make use of Example 9.5.

9.12 *Hint:* Rewrite the series as $\sum_{m,k\geq 2} 1/(km^k)$, and recall the definition of γ.

9.13 *Answer:*

$$\int_0^1 \frac{1+x^2}{1+x^4}\,dx = \frac{\tan^{-1}(\sqrt{2}+1)+\tan^{-1}(\sqrt{2}-1)}{\sqrt{2}},$$
$$\int_0^1 \frac{1-x^2}{1+x^4}\,dx = \frac{\log(3+2\sqrt{2})}{2\sqrt{2}}.$$

9.14 *Solution*: The `Python` code and the accompanying plot are given below (Fig. 1):

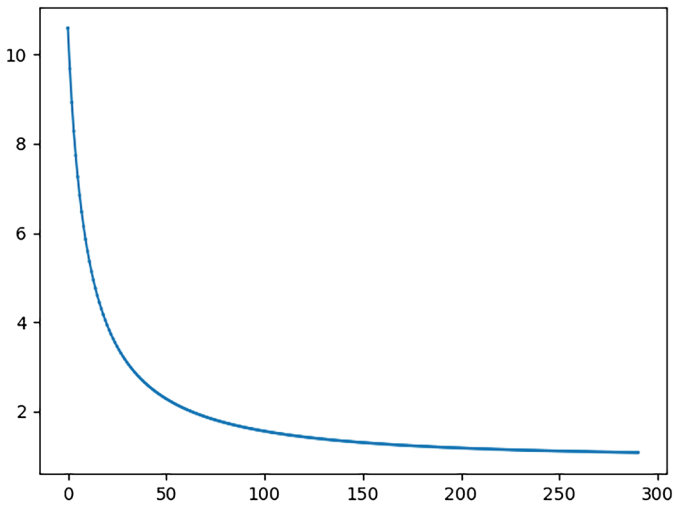


Fig. 1 $\zeta(s)$: $s = 1 + \frac{h}{300}$, $10 < h < 290$

```
import math
import matplotlib.pyplot as plt
import numpy as np
# Computing and plotting ζ(s) in 1 < s < 4
# using its series expansion
xx=100000
zeta_data=[]
def zeta(s):
    zz=0
    for n in range(1,xx+1): zz+=1/n**s
    s1=s-1
    xs=xx**s1
    adjust=1/(s1*xs)
    return zz+adjust
hh=100
start=10
finish=3*hh+1
for h in range(start,finish):
    s=1+h/hh
    zeta_data.append(zeta(s))
    if s==2: print ("zeta(2)",h,zeta(s),"for checking")
    if s==3: print ("zeta(3)",h,zeta(3),"for checking")
    if s==4: print ("zeta(3)",h,zeta(4),"for checking")
howmany=finish-start
s=[]
for n in range(howmany): s.append(0.1)
mylist=[zeta_data[x] for x in range(howmany)]
x=range(0,howmany)
plt.scatter(x,mylist,s)
plt.plot(x,mylist)
plt.show()
```

9.15 *Solution*: Suppose, for example, we try to evaluate $\zeta(s)$ at $s = 1/100$, with the very modest error bound $1/10$, using (9.8). Recalling that, for an alternating-sign series, the incurred error is the first ignored term in the tail of the series. Here $1/n^s = 1/10$ means $n = 10^{1/s} = 10^{100}$, so we can give up on the whole idea! For the evaluation of $\zeta(s)$ when s is small we need to find out more on $\zeta(s)$, say further continuation into $\sigma > -1$, and show, for example, that $\zeta(s) = -\frac{1}{2} - \frac{s}{2}\log(2\pi) + O(|s|^2)$, as $s \to 0$.

Chapter 10

10.1 *Answer:* $15/\pi^2$.

10.2 *Hint:* Consider the Dirichlet series $\sum_{n\geq 1} \mu^2(n)\chi(n)/n^s$, with χ being one of the $\phi(Q)$ character (mod Q).

10.3 *Hint:* A multiple of 4 is not squarefree.

10.4 *Answer:*

$$C(a) = \begin{cases} 0 & a = 0, 4, 8, \\ \frac{3}{4\pi^2} & a = 1, 2, 5, 7, 10, 11, \\ \frac{1}{2\pi^2} & a = 3, 6, 9. \end{cases}$$

Denote the sums by $S_a(x)$. Then $S_a(x) = 0$ for $a = 0, 4, 8$, and we find that

$$S_a(10^8) = 7599045,\ 7599060,\ 7599077,\ 7599088,\ 7599113,\ 7599125,$$

for $a = 1, 2, 5, 7, 10, 11$, and

$$S_a(10^8) = 5066034,\ 5066061,\ 5066091, \qquad \text{for} \quad a = 3, 6, 9.$$

10.5 *Sketched solution:* A squarefull number has the unique representation a^2b^3, with b squarefree, so that the counting function for such numbers is $\sum_{a^2b^3 \le x} \mu^2(b)$. Making use of $\mu^2(b) = \sum_{c^2d=b} \mu(c)$, the double sum becomes the triple sum

$$\sum_{a^2d^3c^6 \le x} \mu(c) = \sum_{c^6 \le x} \mu(c) \sum_{a^2d^3 \le X} 1, \qquad X = x/c^6.$$

The inner sum has the asymptotic value $\zeta(\frac{3}{2})\sqrt{X}$, so that the counting function has the asymptotic value $C\sqrt{x}$, with $C = \zeta(\frac{3}{2})/\zeta(3)$.

10.6 *Answer:* $n = 234$, with

$$n^2 < 381^3 < 5^2 13^3 < 83^2 2^3 < 21^2 5^3 < (n+1)^2.$$

10.7 *Answer:* 293318625600.

10.8 *Hint:* Suppose that $\omega(n) \le k - 2$, where $k > 3$. Set $m = h(\lfloor n/h \rfloor + 1)$, the least multiple of h which exceeds m. By considering bounds for $\phi(n)/n$ and $\phi(m)/m$, deduce that $\phi(m) < \phi(n)$, which contradicts n being sparsely totient.

10.9 *Answer:* (i) 881790. *Hint:* (ii) $13{\cdot}17 > 11{\cdot}19$ and $\phi(11^2 19) > \phi(11{\cdot}13{\cdot}17)$. *Hint:* (iii) If $Q^2 < P$, or $Q^2 < R$, then set $m = nQ(\lfloor P/Q \rfloor + 1)/P$, or $m = nQ(\lfloor R/Q \rfloor + 1)/R$, accordingly, to derive a contradiction. If $P^4 | n$ then $P^3 \le R < Q^2$, and Q can be at most the smallest prime exceeding P, so that $Q \le 2P - 1$ by Bertrand's postulate (the simplest prime gap from Chebyshev's theorem).

10.11 *Answer:* (a) (i) $M(x) = 0$ for 5362 values of $x < 10^6$. The are three runs of 6 consecutive zeros, ending at $x = 46376,\ 265476,\ 973828$.
(ii) $(X_1, M(X_1) = (992998, +311)$ and $(X_2, M(X_2) = (926265, -368)$.

(b) For $1 \le x \le 10^6$, the function $L(x)$ attains the minimum at $L(925985) = -1253$; also $x = 586$ is the largest zero, that is $L(586) = 0$.
Remark: It is known that $L(x)$ changes signs infinitely often, and that $x = 906150257$ is the least $x > 1$ with $L(x) = 1$.

10.13 *Hint:* The normal order of $f(n)$ is an increasing function. Use it to estimate, for each $n \le x$, the number of $m \le n$ such that $100 f(n) \le 99 f(m)$.

Chapter 11

11.1 *Solution:* Making use of partial summation, we have, for $\sigma > 1$,

$$\begin{aligned}\log \zeta(s) &= -\sum_p \log\Big(1 - \frac{1}{p^s}\Big) = -\sum_{n\ge 2} \big(\pi(n) - \pi(n-1)\big) \log\Big(1 - \frac{1}{n^s}\Big) \\ &= -\sum_{n\ge 2} \pi(n) \Big(\log\Big(1 - \frac{1}{n^s}\Big) - \log\Big(1 - \frac{1}{(n+1)^s}\Big)\Big) \\ &= \sum_{n\ge 2} \pi(n) \int_n^{n+1} \frac{s}{x(x^s-1)}\, dx = s \int_2^\infty \frac{\pi(x)}{x(x^s-1)}\, dx.\end{aligned}$$

11.2 *Hint:* Apply the inverse to the Mellin transform of $\psi(x)$ given in Theorem 7.3.

11.3 *Answer/Hint:* From $5 + 8\cos\theta + 4\cos 2\theta + \cos 3\theta = (1 + \cos\theta)(1 + 2\cos\theta)^2 \ge 0$, it follows that $\zeta^5(\sigma)|\zeta(\sigma + it)|^8|\zeta(\sigma + 2it)|^4|\zeta(\sigma + 3it)| \ge 1$, for $\sigma > 1$.

11.5 *Hint:* Take $p = p_n \sim n \log n$ as $n \to \infty$.

11.6 *Answer:* $(p, p') = (396733, 396833), (838249, 838349)$.

11.7 *Solution:* Let $0 < a < b$ be fixed, and set $\epsilon = (b - a)/3$. By the prime number theorem for $\theta(x)$, if p is a large prime then

$$\theta(bp) \sim bp > (b - \epsilon)p, \qquad \text{and} \qquad \theta(ap) \sim ap < (a + \epsilon)p,$$

so that $\theta(bp) - \theta(ap) > (b - a - 2\epsilon)p > 0$, that is $\theta(bp) > \theta(ap)$. It follows that there is a prime p' in the interval $ap < p' < bp$, that is $a < p'/p < b$.

11.8 (i) *Remark:* The sequence of 5 successive composite numbers 24, 25, 26, 27, 28 enables 2^{28} to beat the product of primes for the last time.
(ii) *Solution:* We may set

$$M_{n+1} = \frac{(n+1)M_n}{\gcd(n+1, M_n)}, \qquad n = 1, 2, \ldots .$$

Proof: If $n+1|M_n$ then $\gcd(M_n, n+1) = n+1$ and $M_{n+1} = M_n$, which is correct. If $n+1 \nmid M_n$ then $n+1$ cannot have two coprime divisors, so that $n+1 = p^a$, a prime power, and $\gcd(M_n, n+1) = p^{a-1}$, so that $M_{n+1} = pM_n$, which is correct (See Fig. 2).
(iii) *Remark:* Note that

$$\log \prod_{p \le n} p = \sum_{p \le n} \log p = \theta(n),$$

and if $a = a(n, p)$ is the largest power of $p \le n$, that is $a = \lfloor \log n / \log p \rfloor = \sum_{p^a \le n} 1$, then we also have

$$\log \mathrm{LCM}(1, 2, \ldots, n) = \log \prod_{p \le n} p^a = \sum_{p^a \le n} \log p = \psi(n).$$

We now find, by the prime number theorem, that, as $n \to \infty$,

$$\prod_{p \le n} p = \exp(\theta(n)) = \exp(n + o(n)),$$

$$\mathrm{LCM}(1, 2, \ldots, n) = \exp(\psi(n)) = \exp(n + o(n)).$$

Since $2 < e < 3$, the required inequalities hold if n is large, and it is expected that $n \ge 29$ is sufficiently large. This can be accomplished using explicit estimates for $\pi(x)$, such as that given by Rosser and Schoenfeld in the text.
(iv) *Answer:*
That n_0 and n_1 do exist follows from $f(19) > e > f(118)$, and we find that $n_0 = 113$ and $n_1 = 222$, with

$$f(n_0) = 2{\cdot}82588\ldots \quad \text{and} \quad f(n_1) = 2{\cdot}59268\ldots .$$

It is expected (from the plot) that the values of n_0 and n_1 remain the same as $N \to \infty$, but explicit estimates for $\psi(x)$ are required to establish the claim.

11.9 *Hint:* Apply partial summation.

11.10 *Hint:* Consider

$$\frac{A(k) - A(k-1)}{k} = \frac{A(k)}{k} - \frac{A(k-1)}{k-1} + \frac{A(k-1)}{k-1} \cdot \frac{1}{k},$$

and note that $A(k) - A(k-1)$ is 1 or 0, depending on whether $k \in A$ or not.
(i) *Hint:* Take, for example, $A = \{k : 2^m < k \le 2^{m+1},\ m = 2, 4, 6, \ldots\}$, for which $A(n)$ fluctuates between essentially $n/3$ and $2n/3$, so that there is no asymptotic density. With the attached weight $1/n$ to members of A, its logarithmic density settles down to the value $\frac{1}{2}$, a number between $\frac{1}{3}$ and $\frac{2}{3}$.

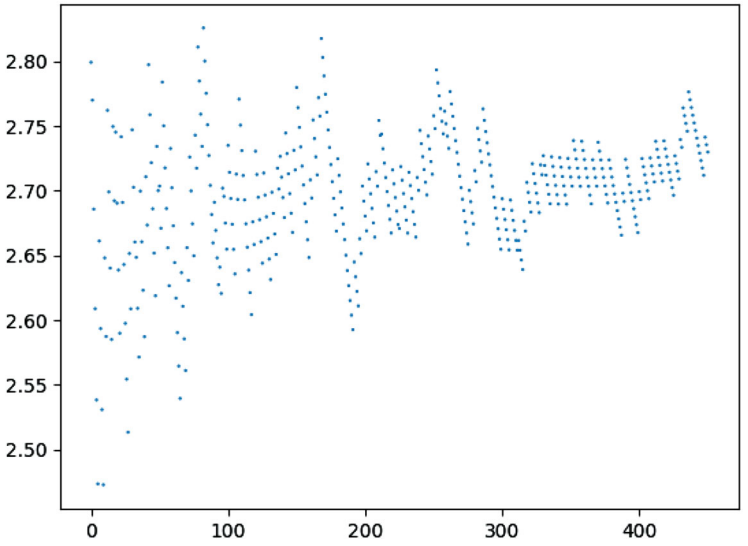


Fig. 2 Plot of $M_n^{1/n}$ for n in $0 < n - 30 < 470$

(ii) *Hint:* Try taking a set A with a much longer stretch of consecutive numbers following by an even longer gap, and repeat.

11.11 *Solution:* The gamma function has the integral representation

$$\Gamma(s) = \int_0^\infty e^{-x} x^{s-1} dx, \quad s = \sigma + iy, \quad \sigma > 0,$$

the Mellin transform of e^{-x}, and the required result is the inverse transform.

11.12 *Hint:* Take the Mellin transform of $f(x) = 1/(e^x - 1)$.

11.14 *Answers:*

$$f(x) = 0, 4, 33, 196, 1200, 8063, \qquad x = 10, 10^2, 10^3, 10^4, 10^5, 10^6.$$

(a) The prime number theorem (without an error term) can only deliver $f(x) = o(x/\log x)$, which gives no information on the sign of $f(x)$; (b) The lower bound for $f(x)$ can be deduced from the subtraction of the inequalities; (c) By de la Vallée Poussin's form of the prime number theorem,

$$f(x) = 2\int_2^x \frac{dt}{\log t} - \int_2^{2x} \frac{dt}{\log t} + O(x \exp(-200\sqrt{\log x}));$$

the required dominating term emerges from noting that

$$\int_2^{2x} \frac{dt}{\log t} = 2\int_2^x \frac{dt}{\log 2t} + O(1) = 2\int_2^x \frac{dt}{\log t + \log 2} + O(1).$$

11.15 *Hints/Answers/Remarks:* (i) Develop the asymptotic formula for $f(x)$ with an error term $O(x^2/\log^7 x)$.
(ii) Note that $x_0+1 = 2521\times 15215723$ is composite, and that $x_0/e = x_1+\eta$, where $x_1 = 14111427770$ and $0 < \eta < \frac{1}{2}$. Since $\eta + 1/e < 1$, it follows that $\lfloor x_0/e\rfloor = \lfloor (x_0+1)/e\rfloor = x_1$. Meissel's method gives

$$\pi(x_0) = \pi(x_0+1) = 1644673232, \qquad \pi(x_1) = 632207037,$$

which then deliver $f(x_0) < 0 < f(x_0+1)$.
Remark: Let X denote the largest integer such that $f(x) \le 0$. It has been shown that $X \le \exp(3361)$, and that, assuming the Riemann hypothesis, $X = x_0$; see [26].

11.16 *Answer:* $\max\limits_{1\le n\le 1000} \dfrac{\psi(n)}{n} = \dfrac{\psi(113)}{113} \approx 1{\cdot}03882.$
Remark: It is known (see [83]) that this maximum holds for all $n \in \mathbb{N}$.

Chapter 12

12.1 *Answer:* For $a = 1, 2, 3, 4, 5, 6$, we find that

$$\pi(10^6; 7, a) = 13063, 13065, 13105, 13069, 13105, 13090.$$

12.2 *Hint:* Take the partial sums of $\sum \chi(n)/n$ in blocks of 15 terms. The sum of the 8 non-zero terms $\chi(r)/(15m+r)$, where $\gcd(r, 15) = 1$, in a block is positive because ....
12.3 *Hint:* Consider the Euler product for $L(s, \chi_0)$.
12.4 *Hint:* Take blocks of q terms and apply ....
12.5 *Hint:* Apply partial summation to Theorem 8.1.
12.6 *Hint:* Take $a = 1000$ and proceed to show that there is a prime p such that $2^a | p-1$ and $3^a | p+1$. The Chinese remainder theorem and Dirichlet's theorem should be useful.
12.7 *First solution:* Let $M = \prod_{1\le k\le K} q_k$, a product of distinct primes exceeding $2K$. By the Chinese remainder theorem, the simultaneous system of $2K$ congruences

$$x \equiv +k \pmod{q_k}, \qquad x \equiv -k \pmod{K+k}, \qquad k = 1, 2, \ldots, K,$$

has a solution $x \equiv x_0 \pmod{M}$, where $\gcd(x_0, M) = 1$. Moreover, by Dirichlet's theorem, we may set x to be a prime p. It then follows that $p \pm k$ is divisible by either q_k or q_{K+k}, for $1 \le k \le K$, and is therefore composite. *Second solution:* Let $J = (2q-2)!/q$, where $q \ge K+2$ is a prime. Then $q \nmid J$, and $2 \le q - K < q + K \le 2q - 2$, so that $q \pm k | J$ for $1 \le k \le K$. By Dirichlet's theorem, there is a prime $p \equiv q \pmod{J}$, so that $p \pm k \equiv q \pm k \pmod{J}$, which implies that $q \pm k | p \pm k$, for $1 \le k \le K$.

12.8 *Answer:* For $a = 1, 2, 3, 4, 5, 6, 7$, we find that

$$D_a(10^6) = 1926814,\ 737672,\ 1226962,\ 736211,\ 1582304,\ 463305,\ 1711115,$$
$$E_a x \approx 1943596,\ 647865,\ 1110626,\ 647865,\ 1480835,\ 370208,\ 1627196.$$

Remarks: (i) The parity of $p - a$ is opposite that of a, which explains why the results for even a are significantly smaller than those for odd a. (ii) The computed results suggest that there is a secondary dominating term in the Titchmarsh-Linnik asymptotic formula, probably of the form $C_a x/\log x$, which is in agreement with the sharper result stated in Note 2.

12.9 *Answer:*

	(mod 3)		(mod 4)	
x	Team-1	Team-2	Team-1	Team-3
10^7	332194	332384	332180	332398
10^8	2880517	2880937	2880504	2880950

(mod 8) :

x	Team-1	Team-3	Team-5	Team-7
10^7	165976	166161	166204	166237
10^8	1439970	1440544	1440534	1440406

(mod 10) :

x	Team-1	Team-3	Team-7	Team-9
10^7	166104	166230	166211	166032
10^8	1440298	1440474	1440495	1440186

12.10 *Answer:*

(mod 11) :

Team	10^6	10^7	10^8
1	7858	66386	576103
2	7843	66541	576332
3	7814	66480	575872
4	7839	66452	575818
5	7853	66376	576332
6	7876	66448	576056
7	7874	66490	576487
8	7873	66507	576172
9	7828	66425	575927
10	7839	66473	576355

Chapter 13

13.3 *Hint:* Let $(A, B, C, X) = (\log 2, \log 3, \log 5, \log x)$, so that $\Psi(x, 5)$ is given by the triple sum $\sum_{0 \le aA+bB+cC \le X} 1$ over a, b, c.

13.4 *Hint:* Use numerical quadrature to evaluate the integral when $2 < u \le 3$.

13.5 *Answer:* (i) $\rho(5) = 3{\cdot}5472470045\ldots$, (ii) $\rho(10) = 2{\cdot}7701718\ldots \times 10^{-11}$.

13.6 *Hint:* Apply induction on u to

$$\rho(u) = \frac{1}{u}\int_{u-1}^{u} \rho(t)\,dt \le \frac{\rho(u-1)}{u}.$$

13.7 *Hint:* Consider $\Psi(x, y) \le \sum_{n \le x^\sigma} 1 + x^{-\delta} \sum_{n \le x,\ P(n) \le y} n^\delta$, for $0 < \delta < \sigma < 1$. The sum concerned is bounded by a product over primes $p \le y$. Fix σ, and set $\delta = \delta(y)$, accordingly to deliver the required results.

13.8 *Hint:* The function $k(n)$ is multiplicative, and $N(x, y) \le \sum_{n \le x} y/k(n)$. Also

$$N(x, y) \le \sum_{n \le x,\, k(n) \le y} \Big(\frac{x}{n}\Big)^\delta \Big(\frac{y}{k(n)}\Big)^{1-\delta} \le y e^{\delta v} \sum_{n \le x,\ P(n) \le y} \frac{1}{n^\delta k(n)^{1-\delta}},$$

and the last sum can be expressed as a product over primes $p \le y$. Set $\delta = \sqrt{2/v}$.

13.11 *Solution:* Let $q \le y$ be any prime. If $q^a | n$ then $a \le b$, so that $q^a | r^b$. This means that any such prime power of n divides s, and hence it will divide t. Therefore t is the y-smooth part of n, which is itself y-smooth if, and only if, $t = n$.

Chapter 14

14.1 *Answer:* $m = 243, 586, 1272$.

14.2 *Answer:* $m = 6671, 6673, 7831$.

14.3 *Hints:* For the lower bound, consider partitions of n using numbers from a subset of $\{1, 2, \ldots, \sqrt{n}\}$, together with one more number. For the upper bound, consider $p(n)x^n$ being a term of the generating function.

14.5 *Hints:* (i) (a) Observe that $a^3 + b^3 \leq (a + b)^3 \leq 4(a^3 + b^3)$ for $a, b \geq 0$.
(b) Consider $x^3 \pmod 7$ and $\pmod 9$.
(ii) *Answer:* $4104 = 2^3 + 16^3 = 9^3 + 15^3$.
(iii) *Answer:* $(167, 436)$, $(228, 423)$, $(255, 414)$.

14.9 *Answer:* 6^3.

14.13 *Hint:* Compute the relevant singular series for $k = 4$, $s = 21$, and for $n \equiv 2, 10 \pmod{16}$.
Remark: My calculation indicates that 'about 200 times more ways' can be replaced by 'nearly 300 times more ways'.

14.15 *Hint:* $a^6 \equiv 0$ or 1, depending on whether $3|a$ or not. Consequently a sum of 8 sixth powers must $\cdots$.

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Index

A
Agrawal, M., 75
Algorithm
  Brillhart, 160
  complexity, 29
  Euclid, 10, 11, 29, 160, 167
Ankeny, N.C., 146
Apéry, R., 227
Approximations, 207
  accurate, 208
  bad, 208
  best, 210
  Diophantine, 207, 384
  good, 210
  level κ, 210
  to π, 210
Arithmetic
  function, 33
    additive, multiplicative, 33
    Carmichael $\lambda(n)$, 133
    Euler totient $\phi(n)$, 38
    generating, 235, 265, 372, 398
    Liouville $\lambda(n)$, 49
    Mangoldt $\Lambda(n)$, 55, 300
    Möbius $\mu(n)$, 35
    $\omega(n)$, $\Omega(n)$, $d(n)$, 8
    one-way, 83
    partition, 371
    singular series $\mathfrak{S}(n)$, 382
  modular, 10
    evaluation of $a^b \pmod n$, 27
Artin, E., 125
Aryabhata, 169

B
Baker, A., 223
Baker, R.C., 312
Bambah, R.P., 152
Bays, C., 315
Bernstein, D., 368
Bertrand, J.L.F., postulate, 312
Beukers, F., 227
Bézout identity (for gcd), 11
Bombelli, R., 169
Borozdin, K., 405
Brahmagupta, 183
Brillhart, J., 160, 186
Brouncker, Lord, 183
Brüdern, J., 379
Brun, V., sieve, 91
Buchstab, A.A., iterative formula, 351

C
Cantor, G., 217
Cauchy, A.-L., 248
  criterion, 260
  inequality, 394, 404
  integral formula, 302, 374
  sequence, 171
Chebyshev, P.L., 66
Chen, J.-R., 91
Chowla, S.D., 152
Cohen, H., 224, 227
Congruence
  binomial, 137
  classes, 9

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exponential, 137
linear, 13
quadratic, 95
Conjecture/hypothesis
Artin, 125
Bambah-Chowla, 152
Carmichael, 279
Cramér, 312
Dirichlet/Gauss circle problems, 165
Elliott-Halberstam, 330, 346
Gauss, 309
Goldbach, 162, 396
Hardy-Landau, 53
Hardy-Littlewood (H), 162
Knapowski-Turán, 335
Legendre, 309
Mertens, 287
$\phi(n) = \phi(n+1)$, 64
prime gaps, 313
prime twins, 313
Riemann (RH), 152, 312
generalised (GRH), 328–331, 346, 405
Shanks-Rényi, 339
Sierpiński', 279
Continued fraction
complete quotient, 168
computation, 190
continuant, 174
convergent, 168
for e, 224
partial quotient, 168
periodic, 176
Cows come home (CCH), 9, 77, 187
Cramér, H., 312
Critical strip, 310
Cryptography, 83

D
Davenport, H., 91, 218, 380, 382
de Bruijn, N.G., 351, 354
Dedekind, R., 375
de la Vallée Poussin, C., 66
Density
asymptotic, 315
logarithmic, 315
of primes, 309
Dickman, K., 350
Dickson, L.E., 381
Diffie, W., 85
Diophantus, 6, 153, 207, 377
Dirichlet, P.G.L., 66
character, 248, 249
convolution, 54
series, 236
Discrete logarithm (DL), 136
Division
divisor, quotient, remainder, 7
greatest common divisor gcd, 10
trial, 8
Dolan, S., 148
Dyson, F., 218

E
Eclipse, nodes, months, Saros, 202
Eisenstein, F.G., 100
Elliott, P.D.T.A., 330
Equation
algebraic, 207
differential-difference, 350
Diophantine, 218
linear indeterminate, 14
Pell, 183
quadratic, 95
Equivalence relation, 10
Eratosthenes, sieve, 67
Erdős, P., 122, 275, 282
Estermann, T., 28, 332, 371
Euclid, 8
Euler, L., 66
criterion, 97
product, 254
summation formula, 238

F
Factorisation
continued fraction method, 186
Fermat's method, 76
Pollard's ρ-method, 76
Farey, J., arcs, 384
major $\mathfrak{M}$, minor $\mathfrak{m}$, 385, 398
Farey, J., sequence, 374
Fermat, P., 3
descent, 3, 5, 145, 185, 186
factorisation method, 80, 102
Fermat-Euler theorem, 42
little theorem, 25, 26
PSP, 108
two-squares theorem, 146
Fibonacci sequence, 12
Binet's formula, 13
Ford, K., 312
Function
Dedekind $\eta(\tau)$, 375
Dickman $\rho(u)$, 350

Dirichlet $L(s, \chi)$, 256
logarithmic integral $\mathrm{li}(x)$, 309
Mertens $M(x)$, 287
Riemann zeta $\zeta(s)$, 235, 255
symmetric, 224

G
Galois, É., 180
Gauss, C.F., 10, 20, 66, 75, 98, 145, 191, 192, 309
dream, 75
elimination, 190
lemma, 99
Gelfond, A., 223
Girard, A., 145
Goldbach, C., 162, 396
Green, B., 312

H
Hadamard, J., 66
Halberstam, H., v, 330
Hall, M., 105
Hardy, G.H., 301, 350, 353, 379
Harman, G., 312
Heath-Brown, D.R., 64, 163, 276
Helfgott, H., 405
Hellman, M., 85
Hermite, C., 219
Hilbert, D., 377
Hildebrand, A., 352, 354
Hoheisel, G., 312
Homomorphism, 249
Hua, L.-K., 382
lemma, 395
Hudson, R.H., 315
Huygens, C., 169

I
Ikehara, S., 301
Inclusion-exclusion principle, 69
Index, 136
Induction, 4, 21
Inequality
Cauchy, 394
Mertens cosine, 305
Pólya-Vinogradov, 342
triangle, 38
Weyl, 394
Ingham, A.E., 301
Involution, 175
Ivory, J., 63

J
Jacobi, C.G.J., symbol, 103

K
Kayal, N., 75
Key-exchange protocol, 138
Khinchin, A.Y., 190
Knapowski, S., 335
Konyagin, S., 312
Kraitchik, M., 81
Kuzmin, R., 191

L
Lambert, J.H., 219
Landau, E., 151, 301, 380
Laplace, P.S., 192
transform, 301, 303, 304
Legendre, A.M., 66
symbol, 96
Lehmer, D.H., 71, 186, 195
Lévy, P., 191
Linnik, Y.V., 329, 380
Littlewood, J.E., 301, 379
Liu, M.C., 405
Lucas, E., 79

M
Mahler, K., 233, 378
Marsaglia, G., 357
Marsaglia, J.C.W., 357
Masser, D.W., 63, 278
Maynard, J., 312
Measure, Lebesgue, 190
Mellin, H.
transform, 265, 300, 302, 307, 310
Menon, P.K., 279
Merkle, R., 85
Mertens, F., conjecture, 287
Method
circle, 371
continued fraction, 186
Floyd cycle-finding, 82, 140
Hermite, 220
Meissel-Lehmer, 71
Monte Carlo, 83
Newton, 80, 194
Pollard ρ for DL, 140
Minkowski, H., 146
Mirsky, L., 275
Möbius, A.F., 36
inversion, 36

Modular form, 374
Mordell, L.J., 371
Moree, P., 143, 417
Morrison, M.A., 186
Multiple, least common (LCM), 10

N
Nair, M., 308
Newman, D.J., 301
Nicolas, J.L., 275
Niven, I., 219
Numbers
  algebraic, 20, 190, 193, 216, 217
  Bernoulli, 318
  biquadrate, 377, 380
  Carmichael, 133
  composite, 8
  coprime, 10
  decimal (representation), 43
  digit (missing), 48
  e, 173, 219
  Euler's constant γ, 52, 239
  factorial (prime power), 21
  Fermat, 30, 130, 186
  highly composite, 272
  irrational (criterion), 23
  Khinchin's constant K, 190
  pentagonal, 373
  π, 173, 219
  prime, 7
  quadratic irrationals, 197
  Skewes, 315
  smooth, 349
    discrete logarithm, 361
  sparsely totient, 278
  squarefree, 9
  squarefull, 202, 296
  standard factorisation, 8
  transcendental, 216–218

O
Odlyzko, A., 287
One-time pad, 84
Order
  of a (mod n), 43
  normal, 159, 289

P
Pan, C.T., 329
Pell, J., 183
Periodic
  continued fractions, 173
  decimals, 45
Perron, O., formula, 265
Pigeonhole principle, 4, 22, 28
Pillai, S.S., 353
Pinch, R.G.E., 135
Pintz, J., 312
Pocklington, H.C., 79
Poisson, S.D., formula, 262
Pollard, J.M., 81, 138
Pomerance, C., 122
Power, R.E., 186
Prime
  counting, 69
  probable, 108
  relatively, 10
Primitive root, 125
Problem
  birthday, 82, 91
  chicken nugget, 14
  circle, 149
  Dirichlet divisor, 51
  Waring, 377
Pseudoprime, 77, 78
  Euler, 107
  strong, 107
Public-key code, RSA, 85
Pythagorean triples, 30
Python, 7

Q
Quadratic irrational
  continued fraction, 177
  reduced, 180
Quadratic residues, 95
  non-residues, 95
    least, 113
  reciprocity law, 98

R
Rademacher, H., 375
Ramanujan, S., 272, 350
Rankin, R.A., 358, 367
  trick, 358
Residue classes, representation, 56
Residue system
  complete, 9
  reduced, 13
Richter, F.K., 267
Riemann, B., 66
  hypothesis, 310

Rosser, J.B., 310
Roth, K.F., 218
Rubinstein, M., 315

S
Sarnak, P., 315
Saxena, N., 75
Scheinder, T., 223
Scoenfeld, L., 310
Segal, S.L., 291
Selberg, A., sieve, 91
Shanks, D., 111, 373
Siegel, C.L., 218, 329
Sierpiński, W., 279
Siksek, S., 381
Singular series/integral, 386
Smith, H.J.S., 175
Stirling, J., formula, 241
Sums
  of cubes, 380
  Dedekind, 375, 376
  exponential, 384
  Gauss, 116, 263
  Kloosterman, 376
  Legendre, 71
  moment, 271

T
Tao, T., 312
Tauber, A., 301
  Tauberian theory, 301
te Reile, H., 287
Test
  AKS, 67, 87
  Baillie-PSW, 122
  divisibility, 46
  Fermat, 77
  Millar-Rabin, 122
  primality, 75, 76
  probabilistic primality, 108
Theorem
  Abel's continuity, 261
  Apéry, 230
  Aubry, 22
  Bambah-Chowla, 152
  Bombieri, 330
  Brun-Titchmarsh, 329
  Cauchy, 300, 384, 397
  Chebyshev, 215, 236
  Chinese remainder, 19
  Dirichlet's approximation, 22, 147, 160, 210, 215, 232, 394, 399, 404
  Erdős-Mirsky, 275
  Estermann-Roth, 285
  Euclid, 65
  fundamental (algebra), 25
  fundamental (arithmetic), 20
  Galois, 180
  Gauss, 150
  Gauss-Kuzmin, 191
  Gelfond-Scheinder, 223
  Hardy-Littlewood, 382
  Hardy-Ramanujan, 289
  Hermite, 221
  Hurwitz, 214
  Ikehara-Wiener, 301
  Jarnik, 164
  Khinchin, 191
  Korselt, 135
  Lagrange, 25, 182, 185
  Lagrange, four-squares, 153
  Lamé, 12
  Landau, 380
  Legendre, 212
  Lévy, 191
  Lindermann, 222
  Liouville, 216
  Lucas-Lehmer, 79
  Menon, 280
  Mertens, 244, 354
  Pocklington-Lehmer, 79
  prime number, 50, 66, 235
  primitive roots, 128
  Rademacher, 375
  Rubinstein-Sarnak, 337
  Segal, 293
  Siegel-Walfisz, 329, 397
  symmetric polynomials, 224
  Thue, 22
  Thue-Siegel-Roth, 218, 379
  Titchmarsh-Linnik, 331
  Vinogradov, 405
  Wilson, 26
Thue, A., 23, 218
Tonelli, A., 111
Tournament, round-robin, 18
Turán, P., 289, 291, 335

V
van der Poorten, A.J., 202
Vaughan, R.C., 372, 382, 397
  identity, 402
Vinogradov, I.M., 379, 382, 397
  lemma, 403

W
Walfisz, A., 329
Wallis, J., 183
  formula, 241
Wang, T.Z., 405
Waring, E., 377
Watson, G.L., 380
Well ordering, 4, 7, 30, 411
Weyl, H., 394
Wieferich, A., 380
Wiener, N., 301
Wintner, A., 315
Wooley, T.D., 372

X
Xylouris, T., 329

Z
Zagier, D., 163, 227
Zaman, A., 357
Zhang, Y., 313
Zudilin, W., 233